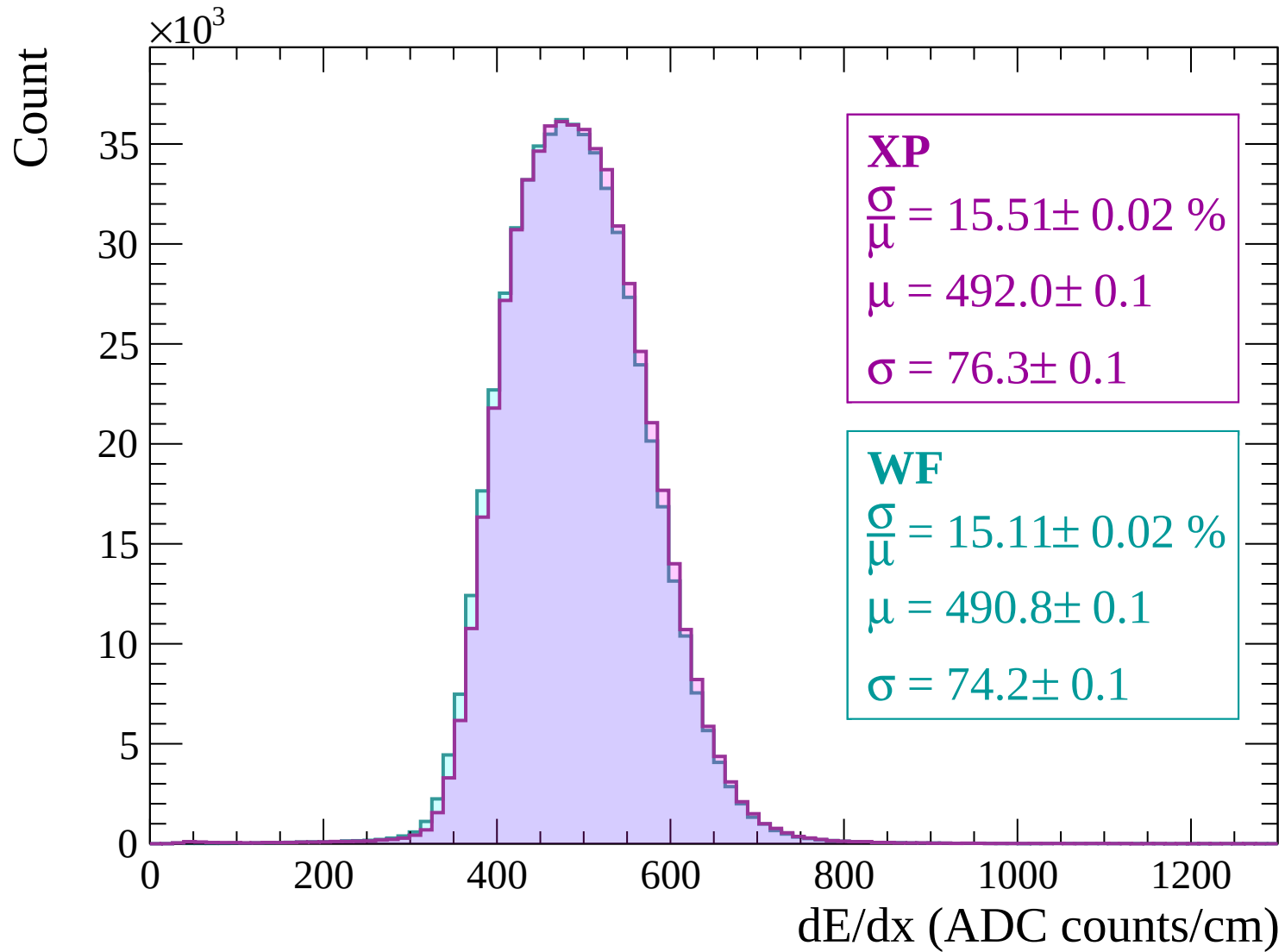
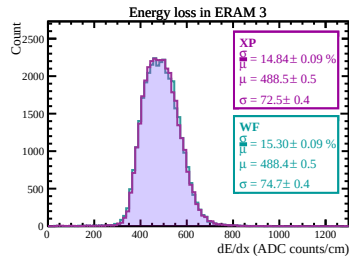
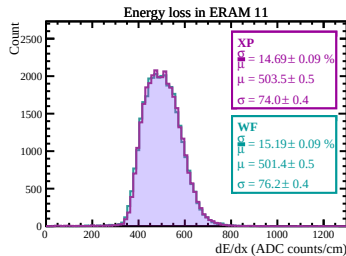
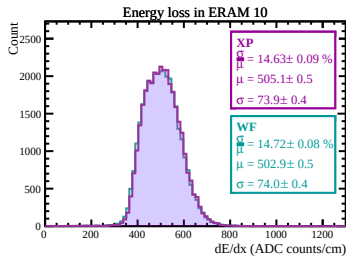
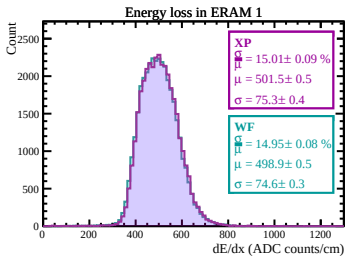
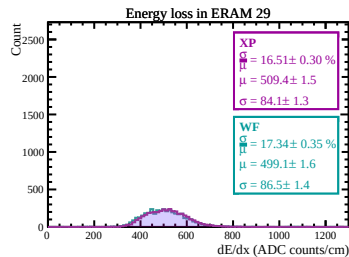
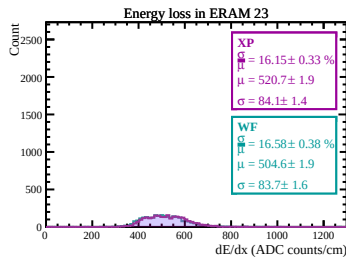
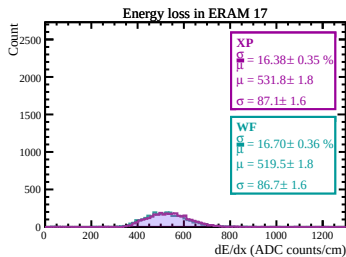
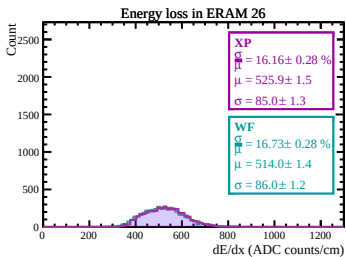
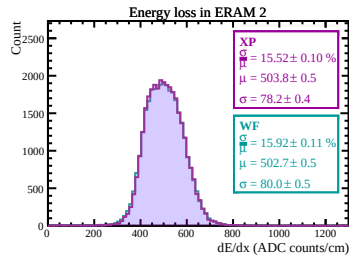
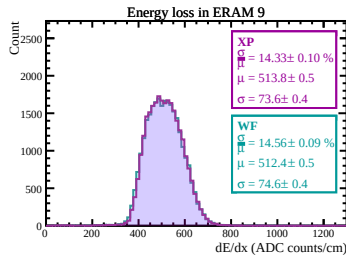
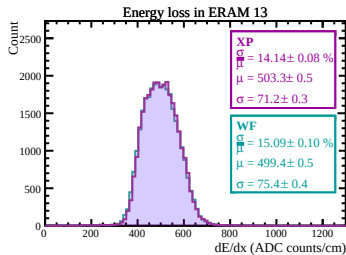
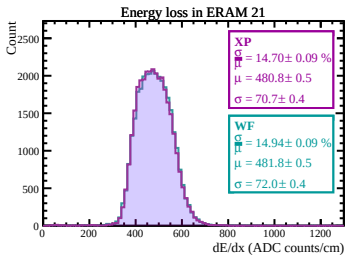
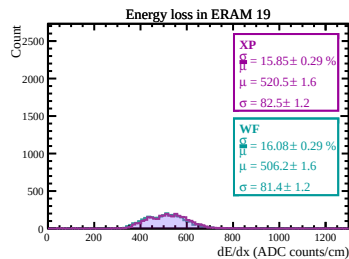
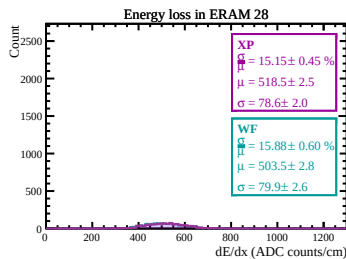
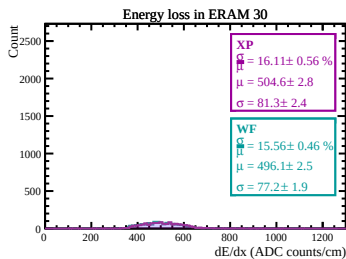
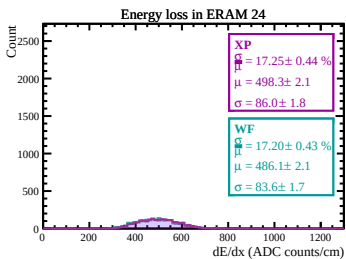
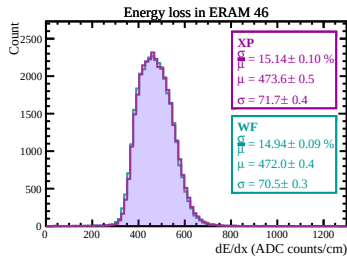
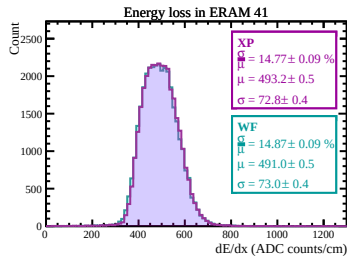
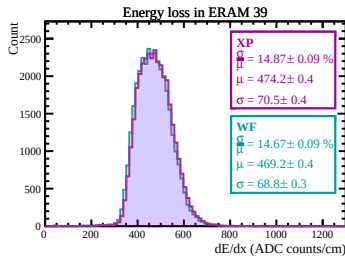
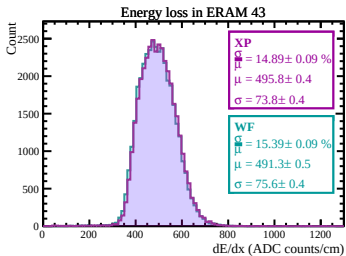
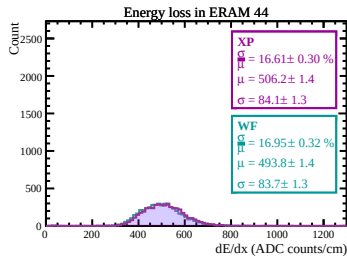
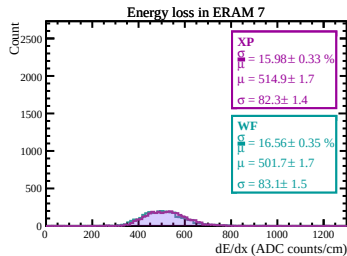
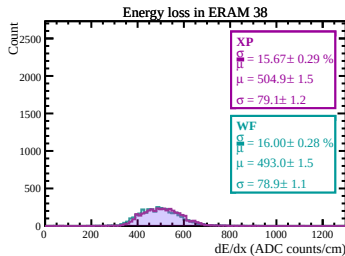
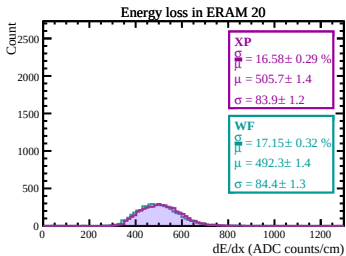
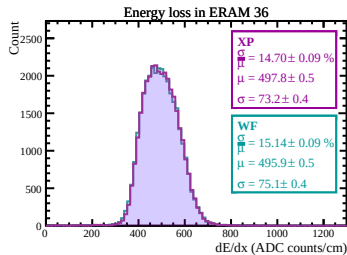
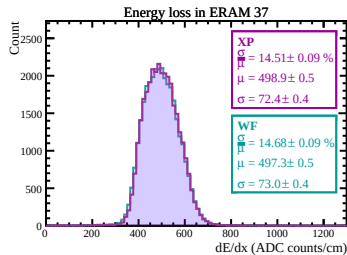
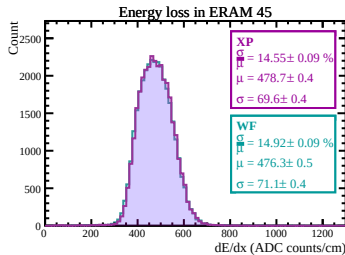
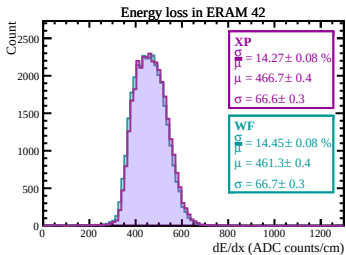
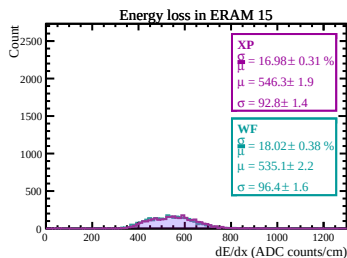
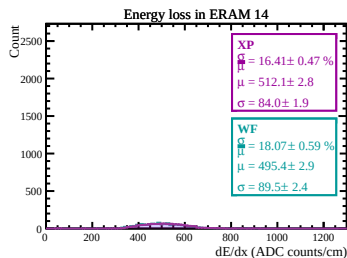
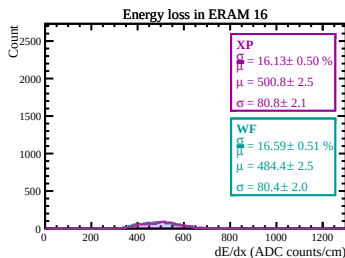
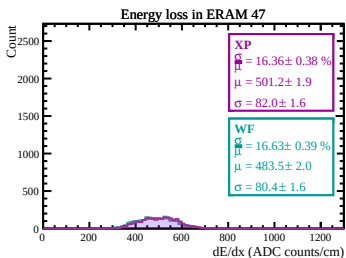
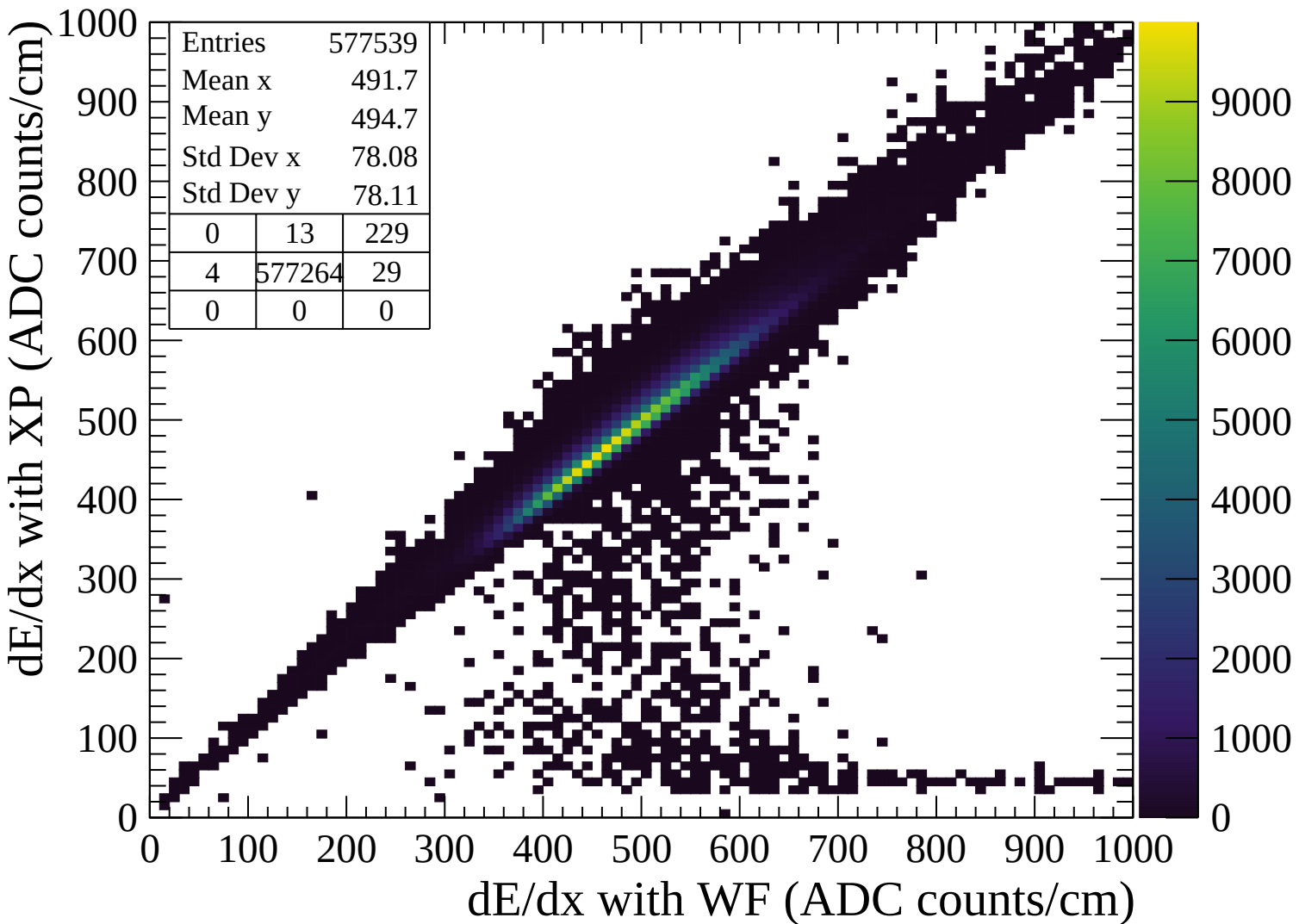
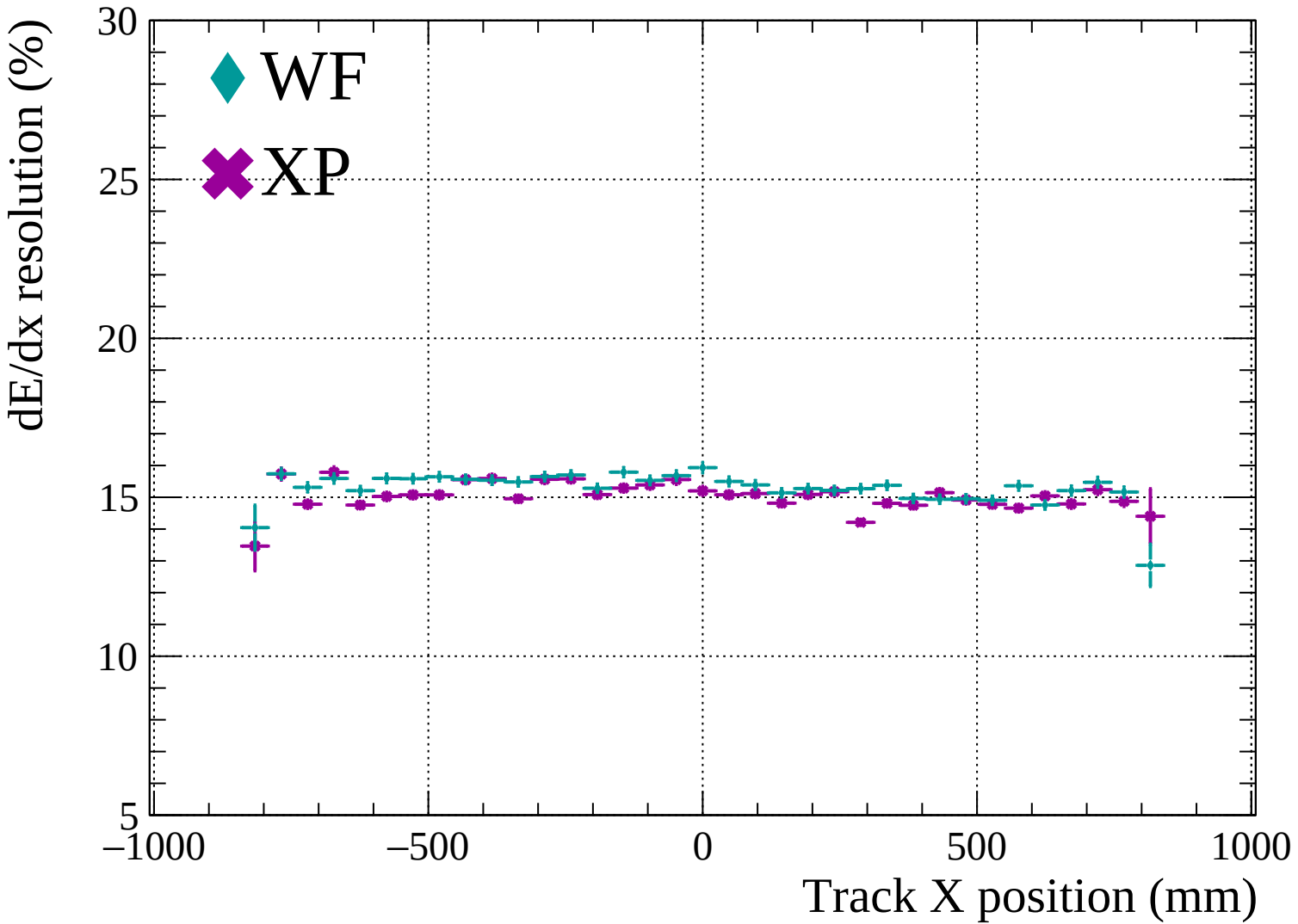


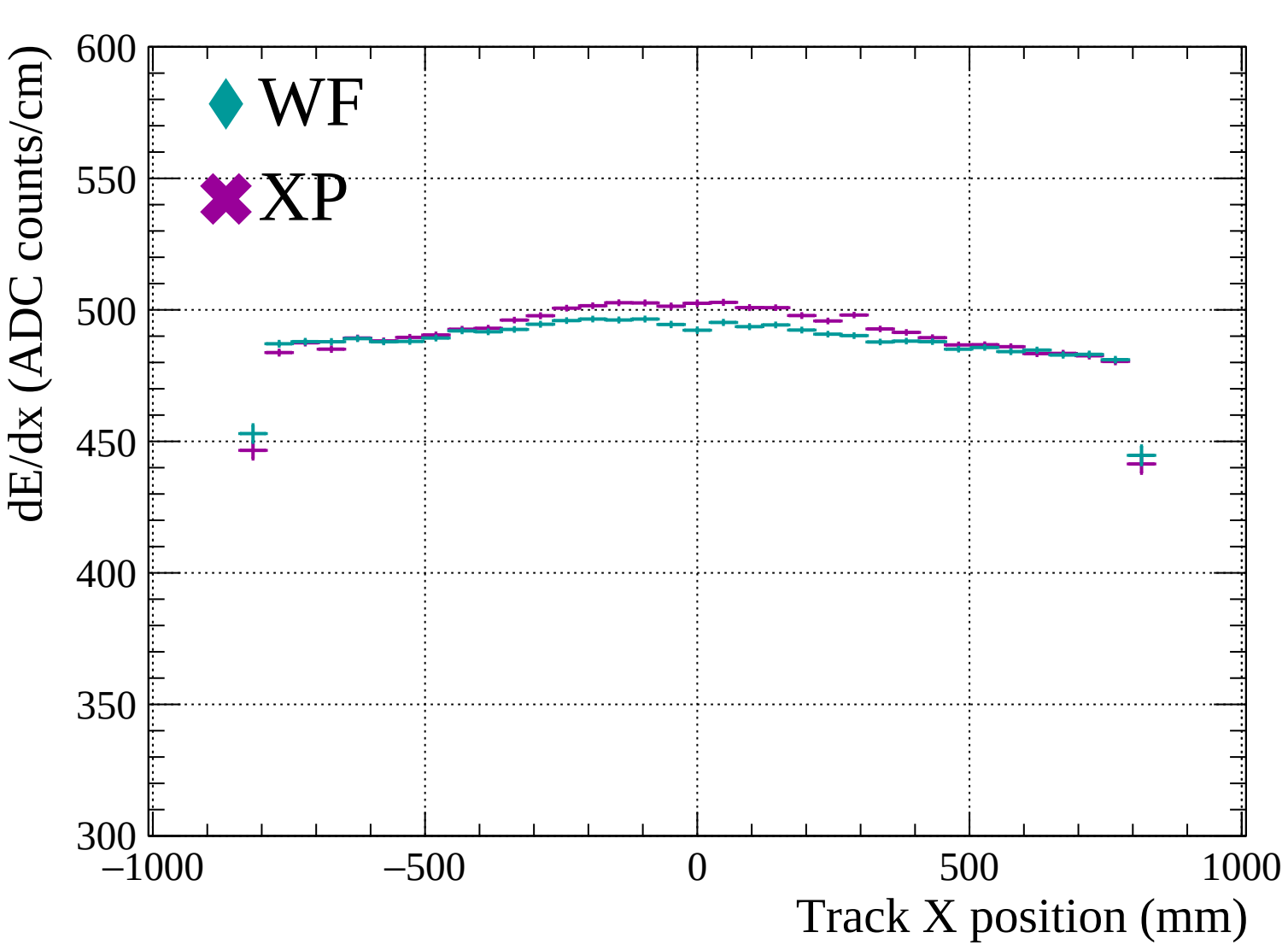


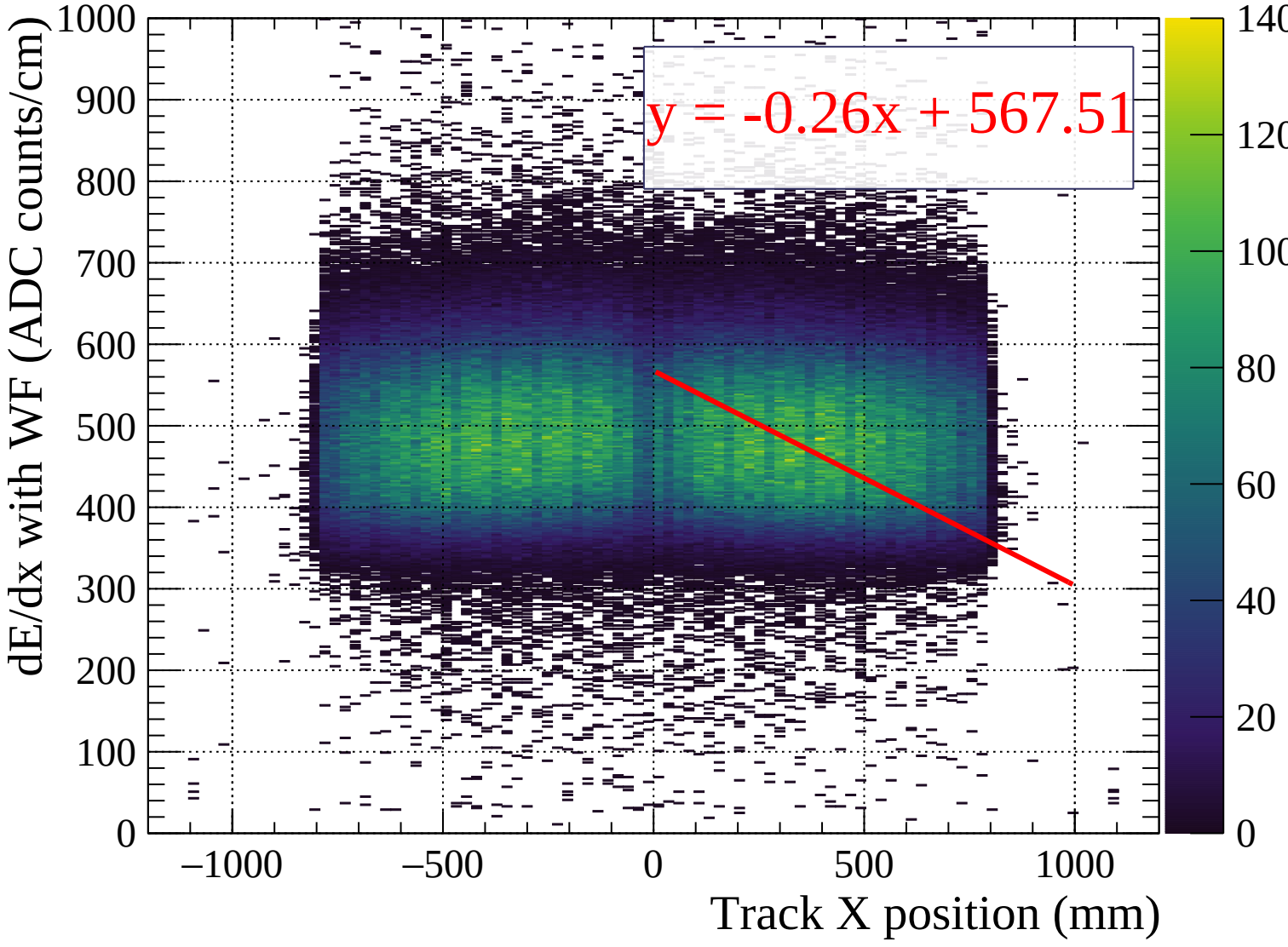


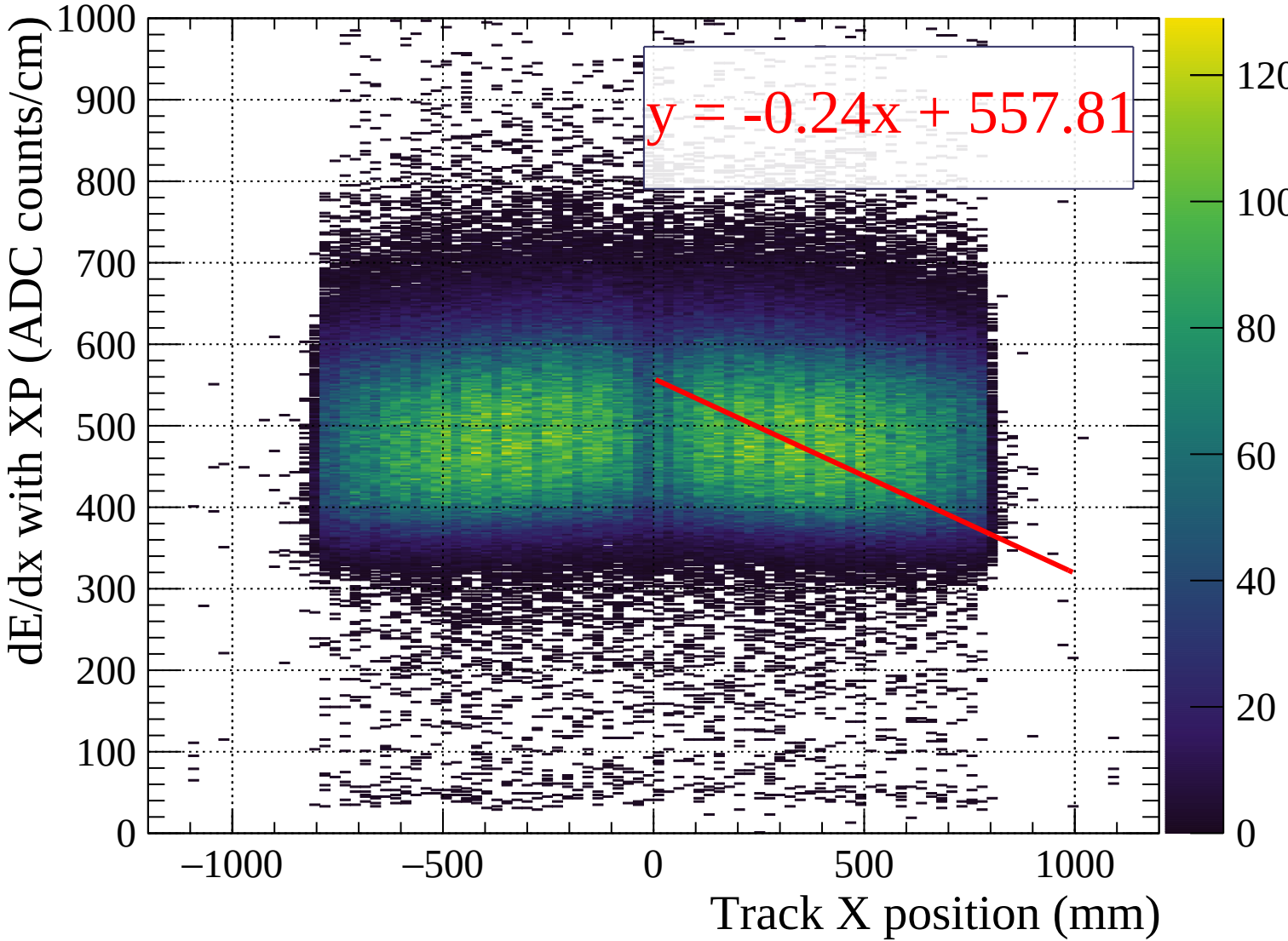


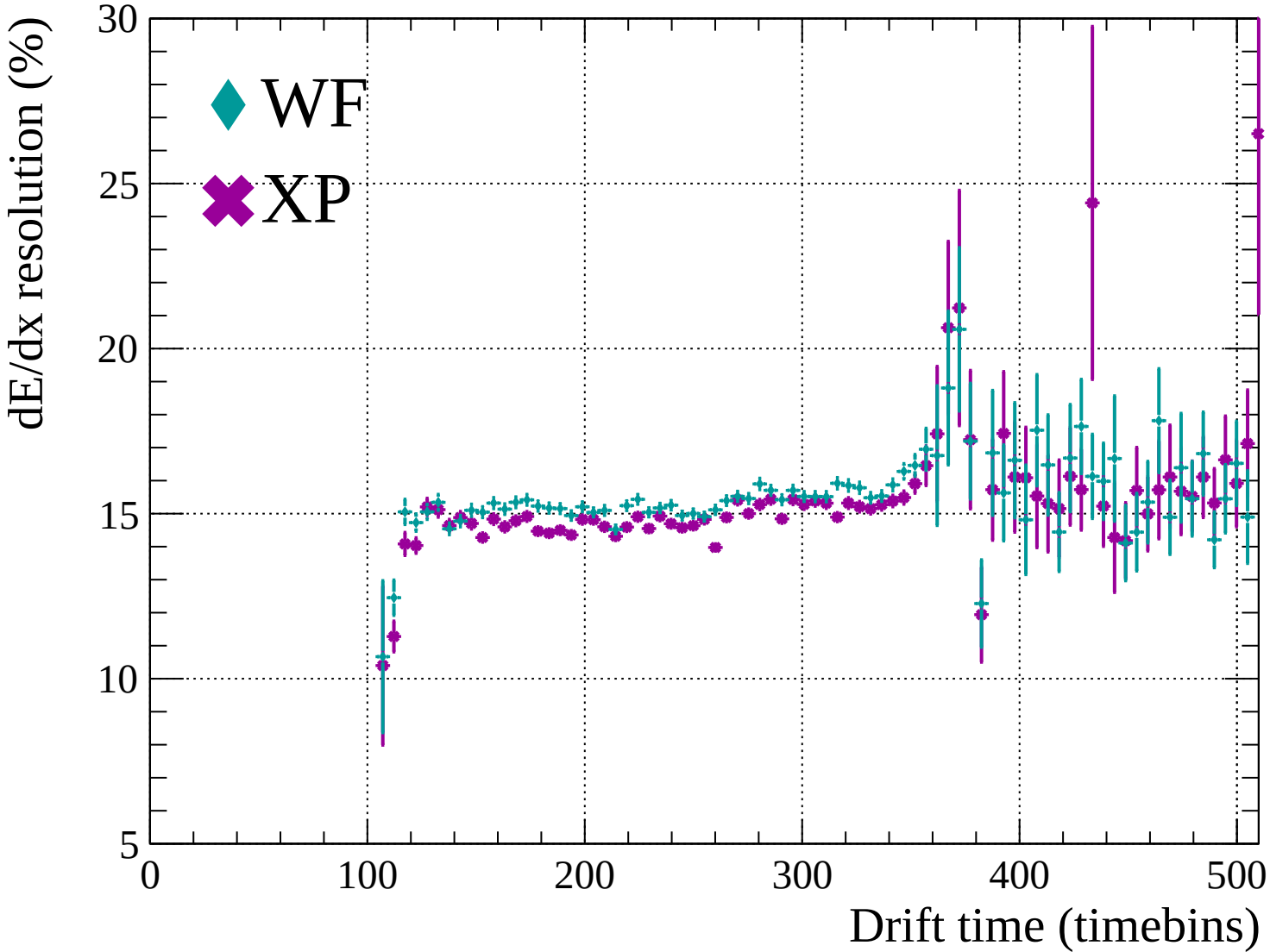


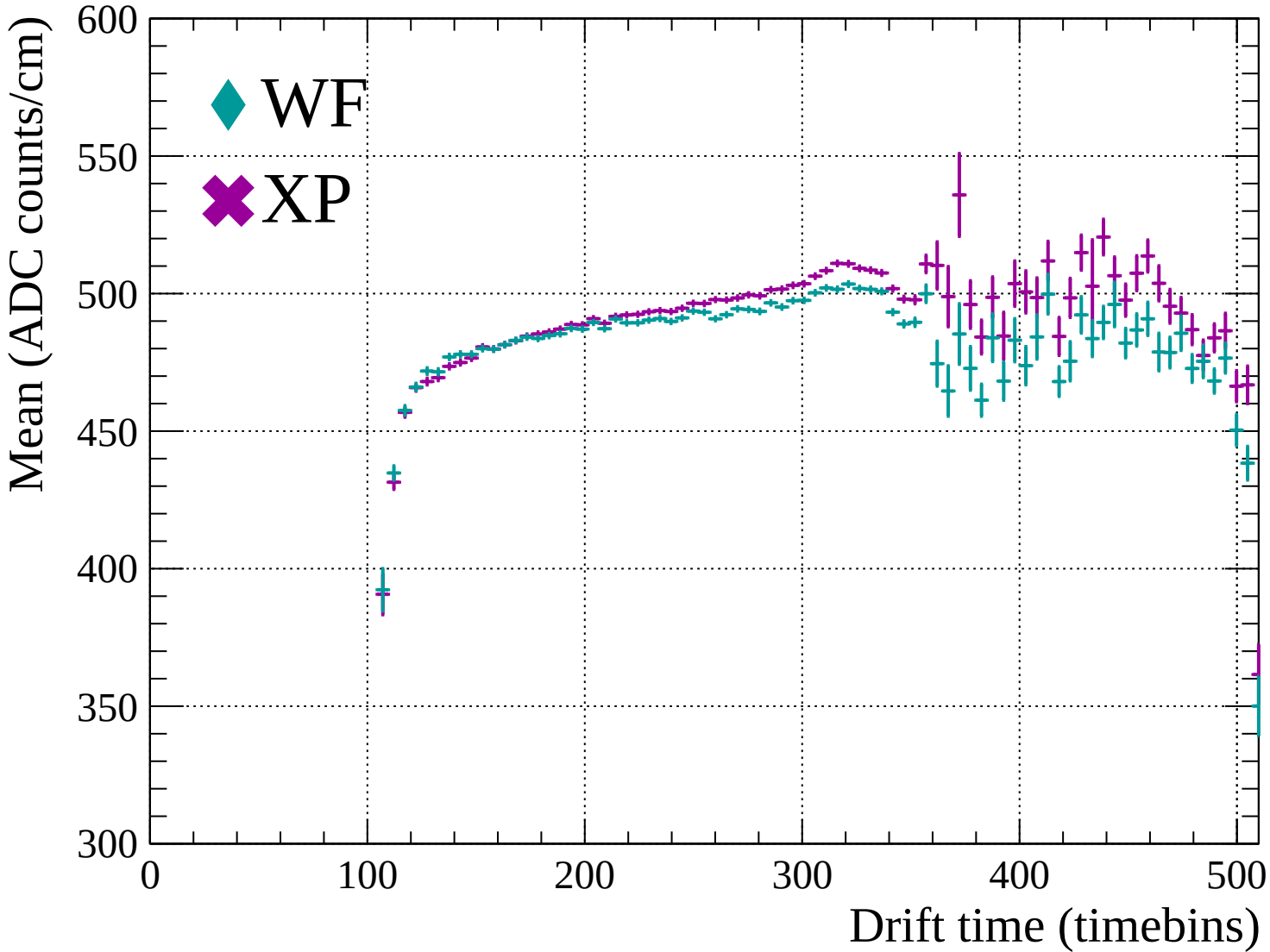


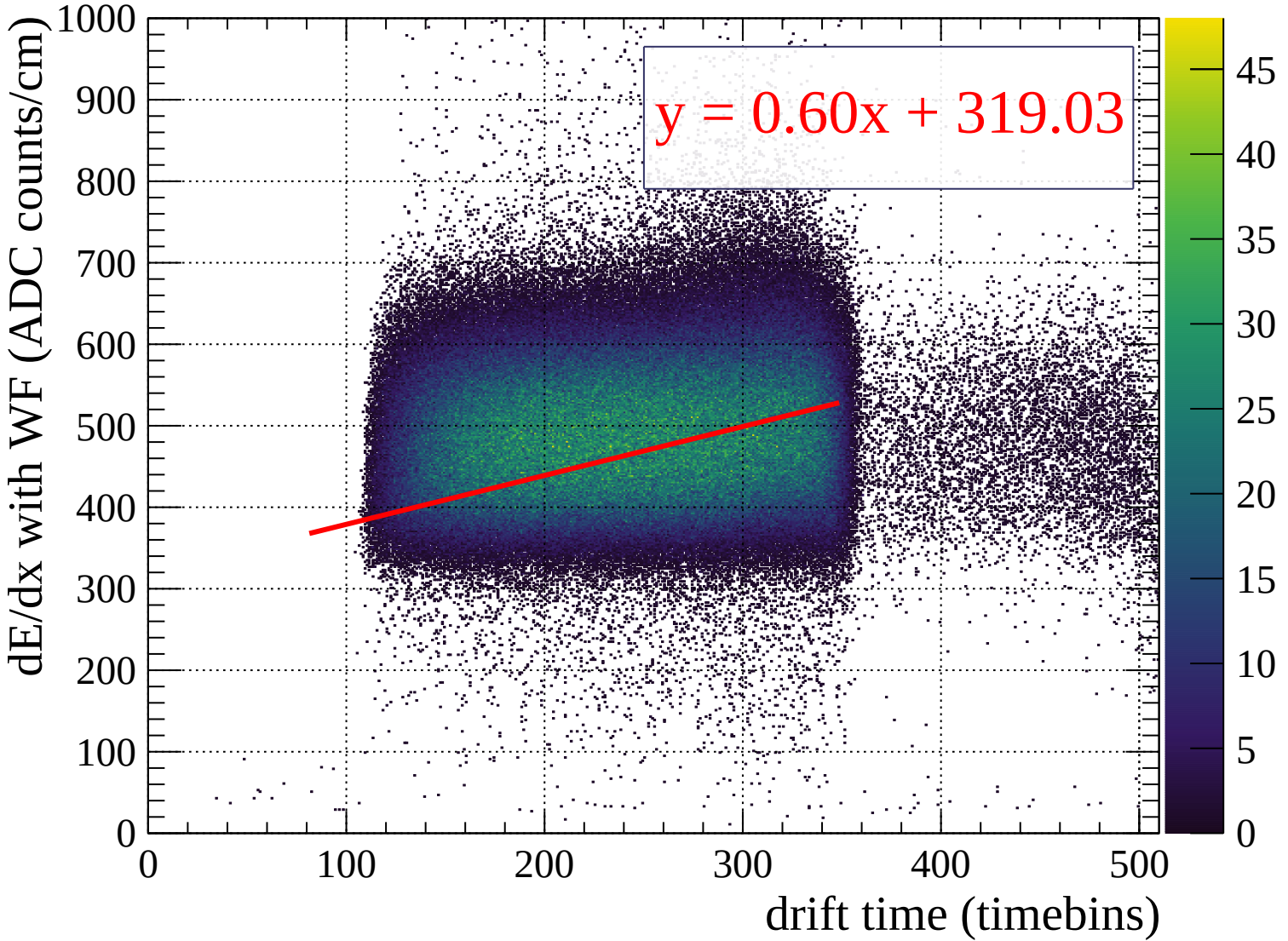


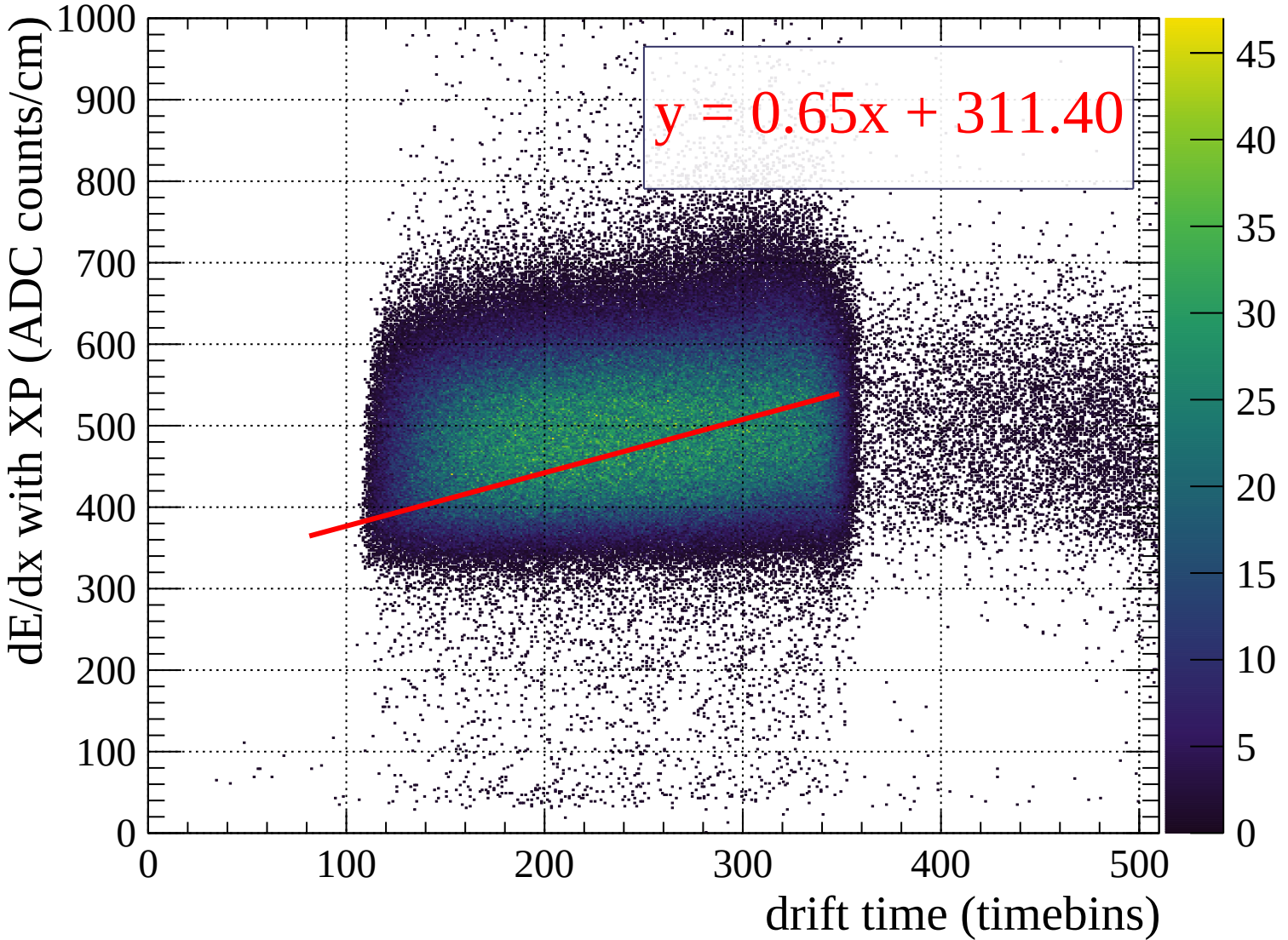


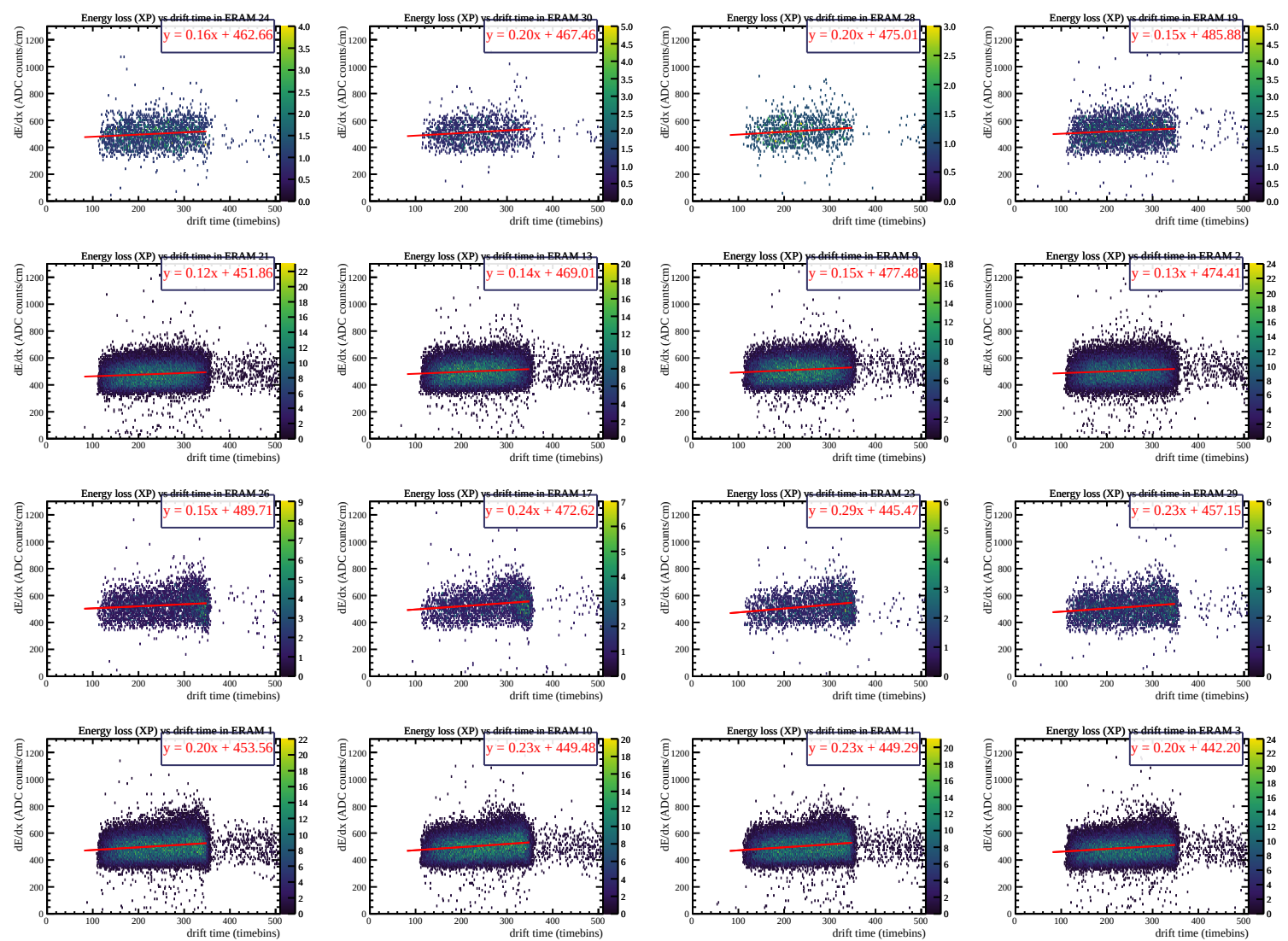


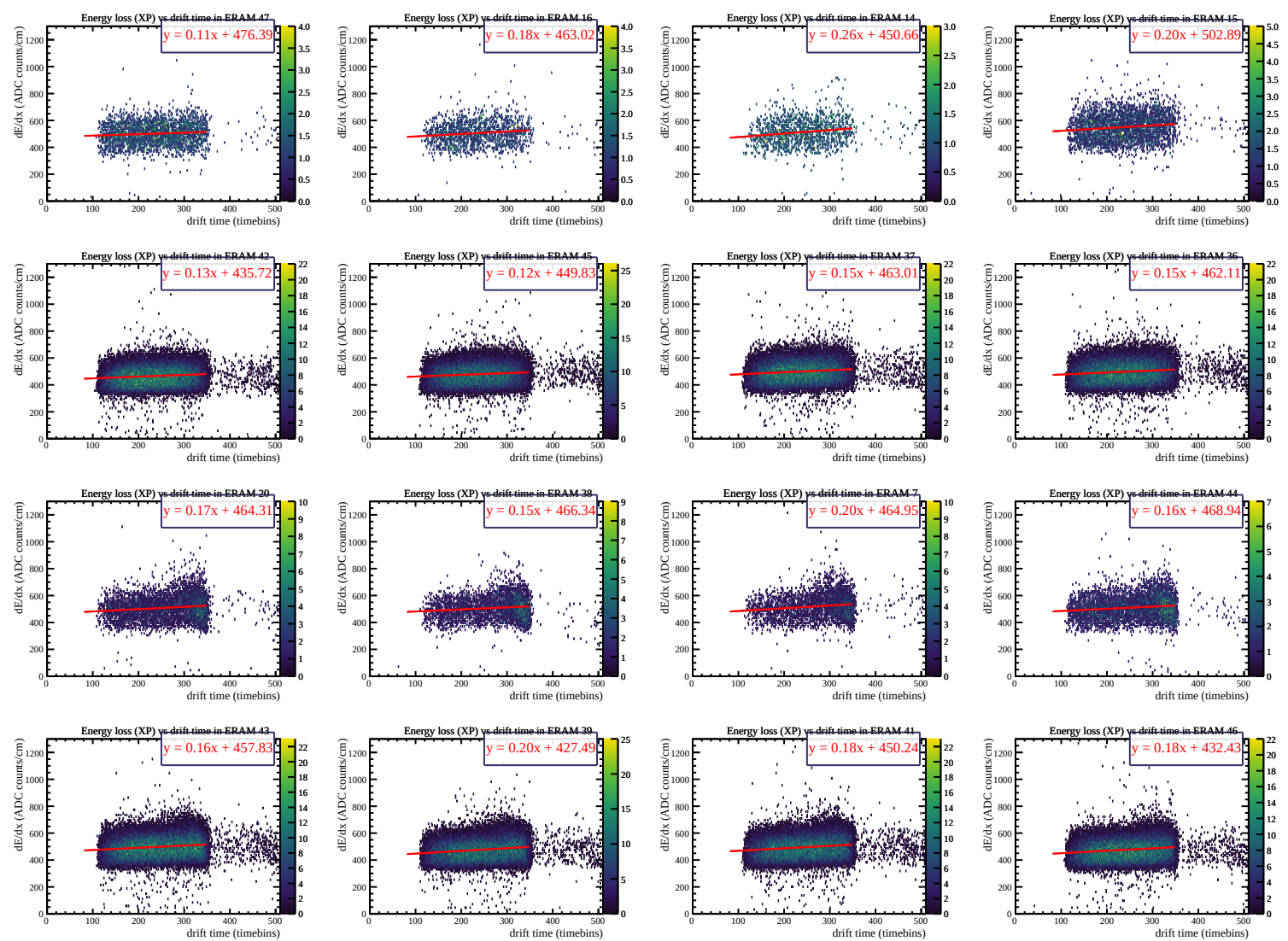


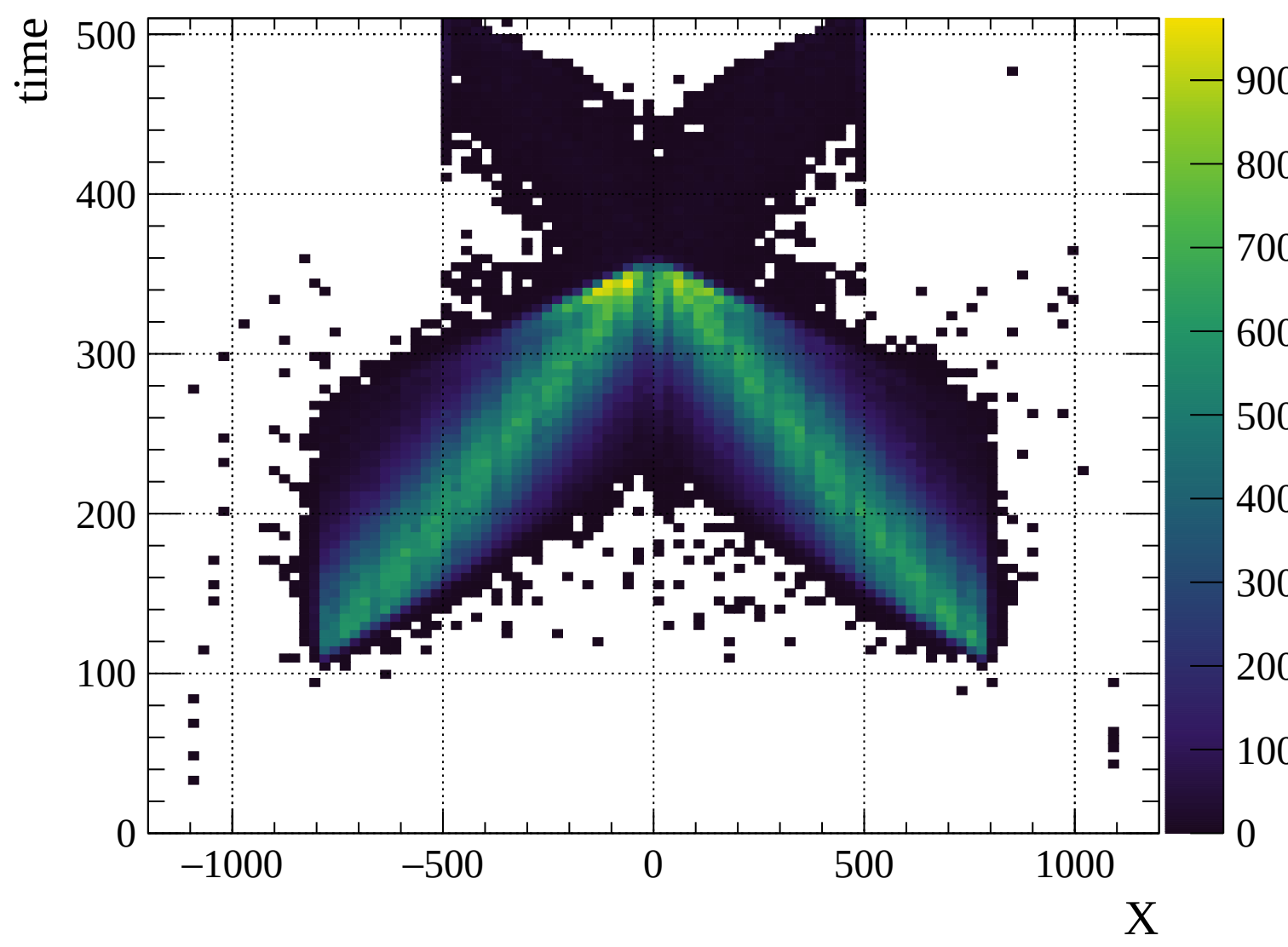


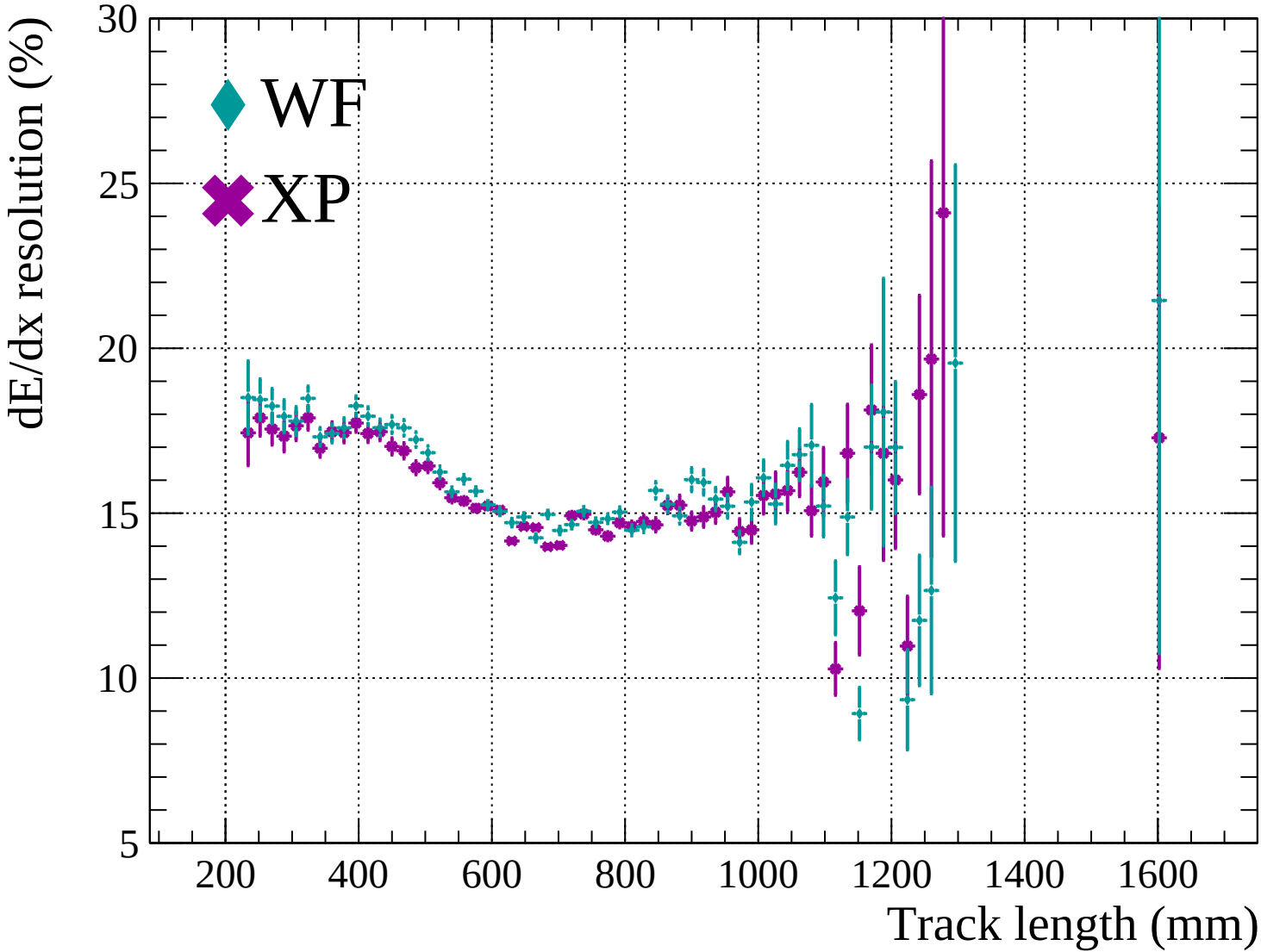


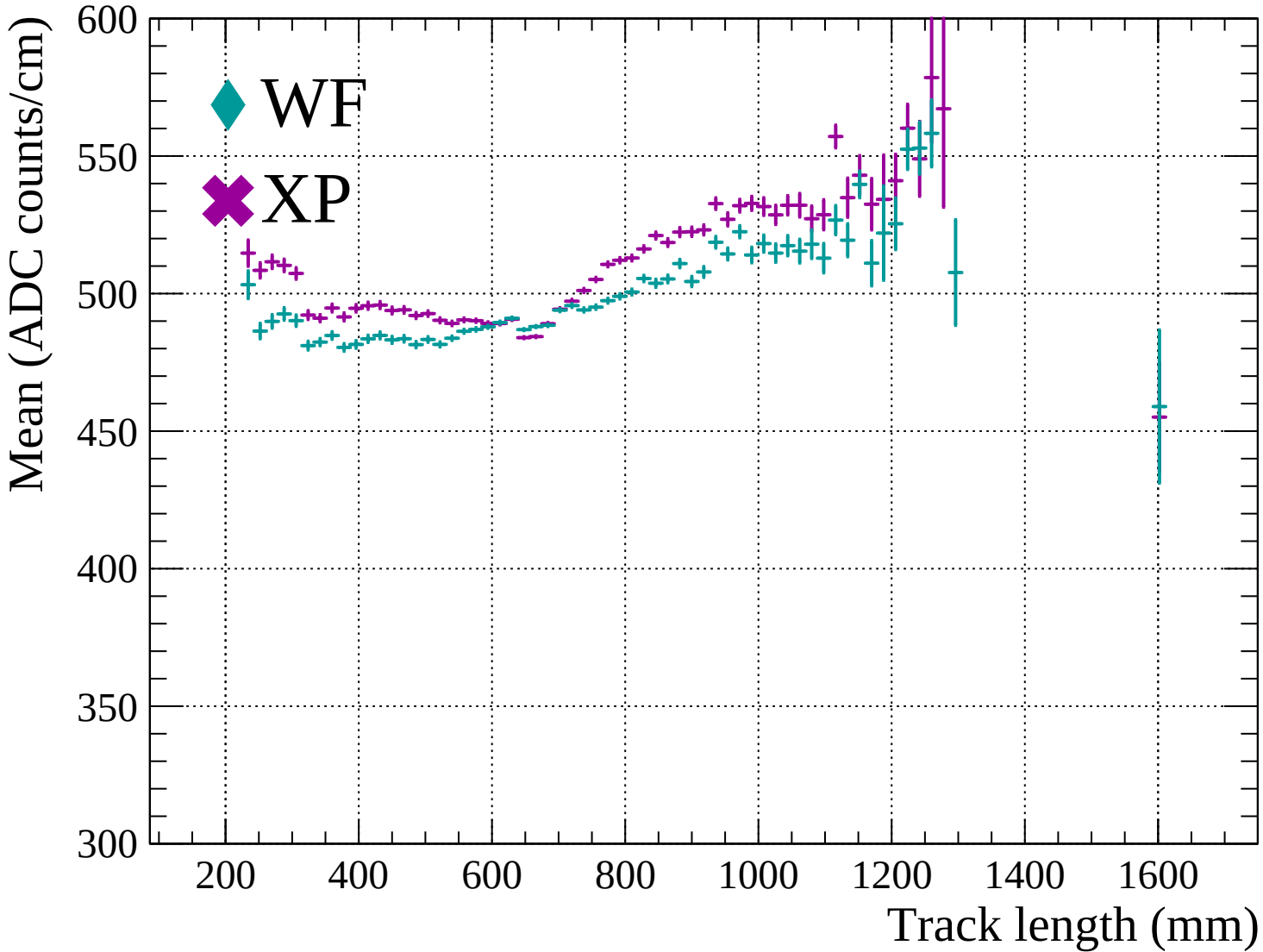


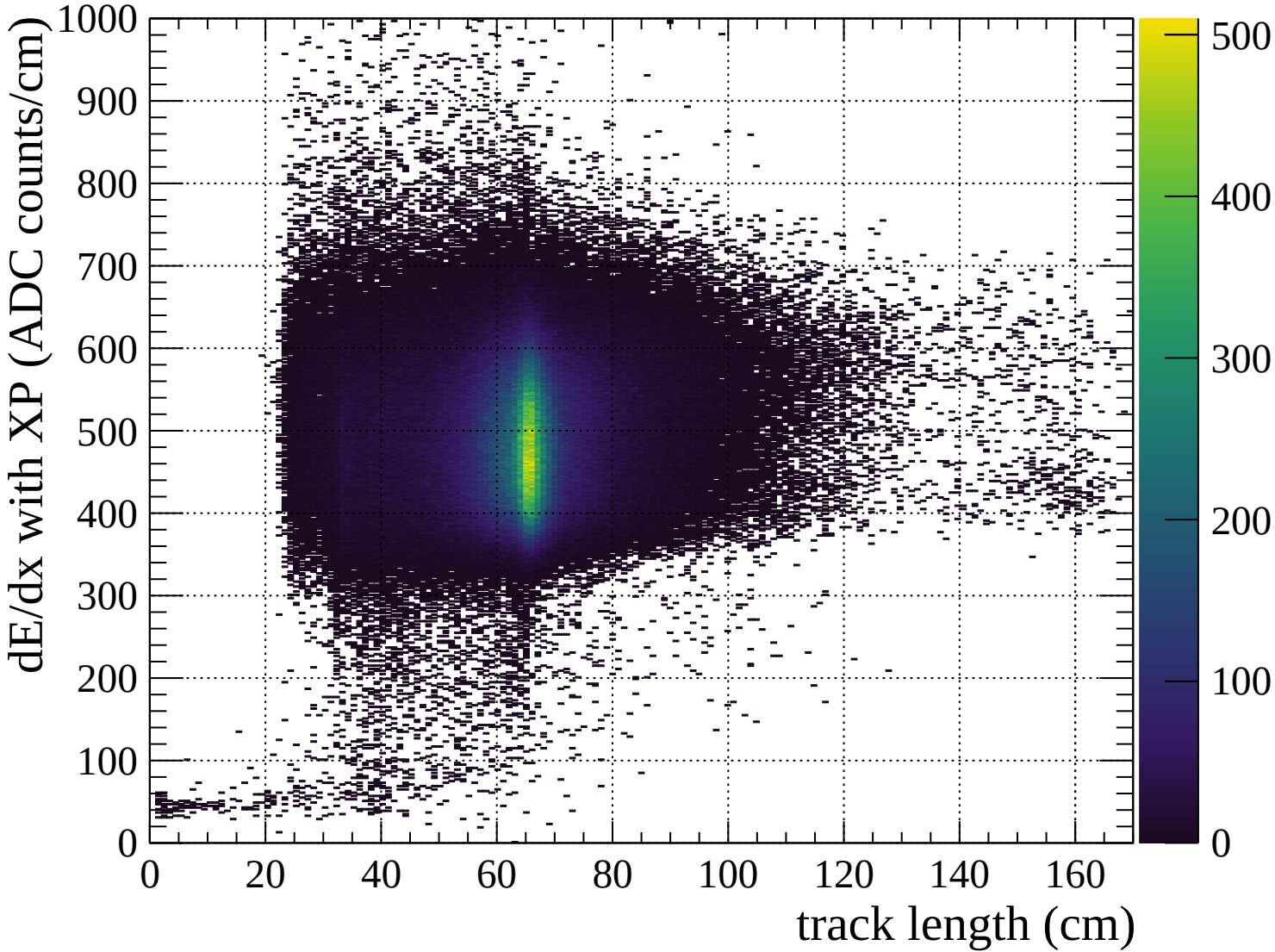


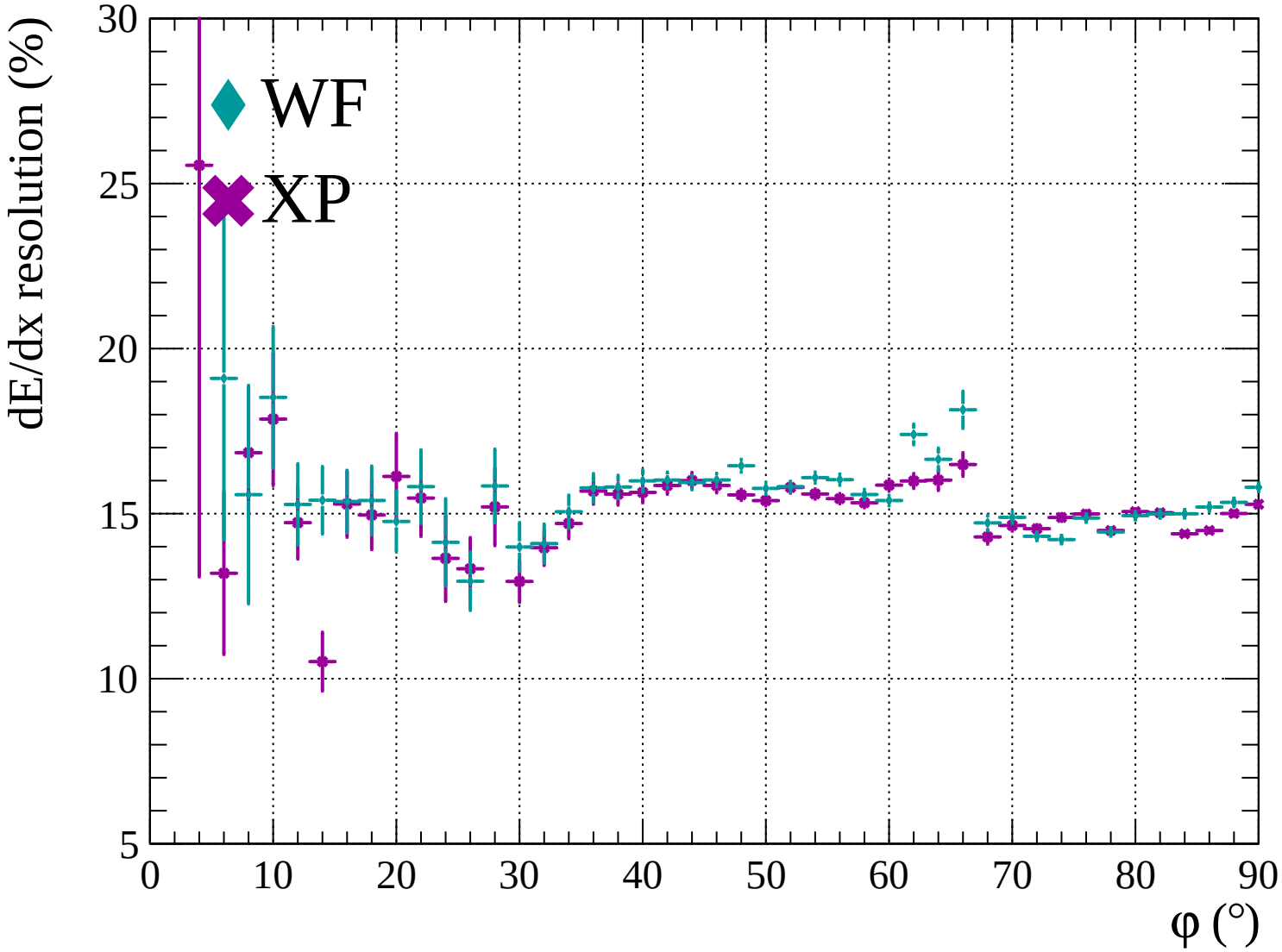


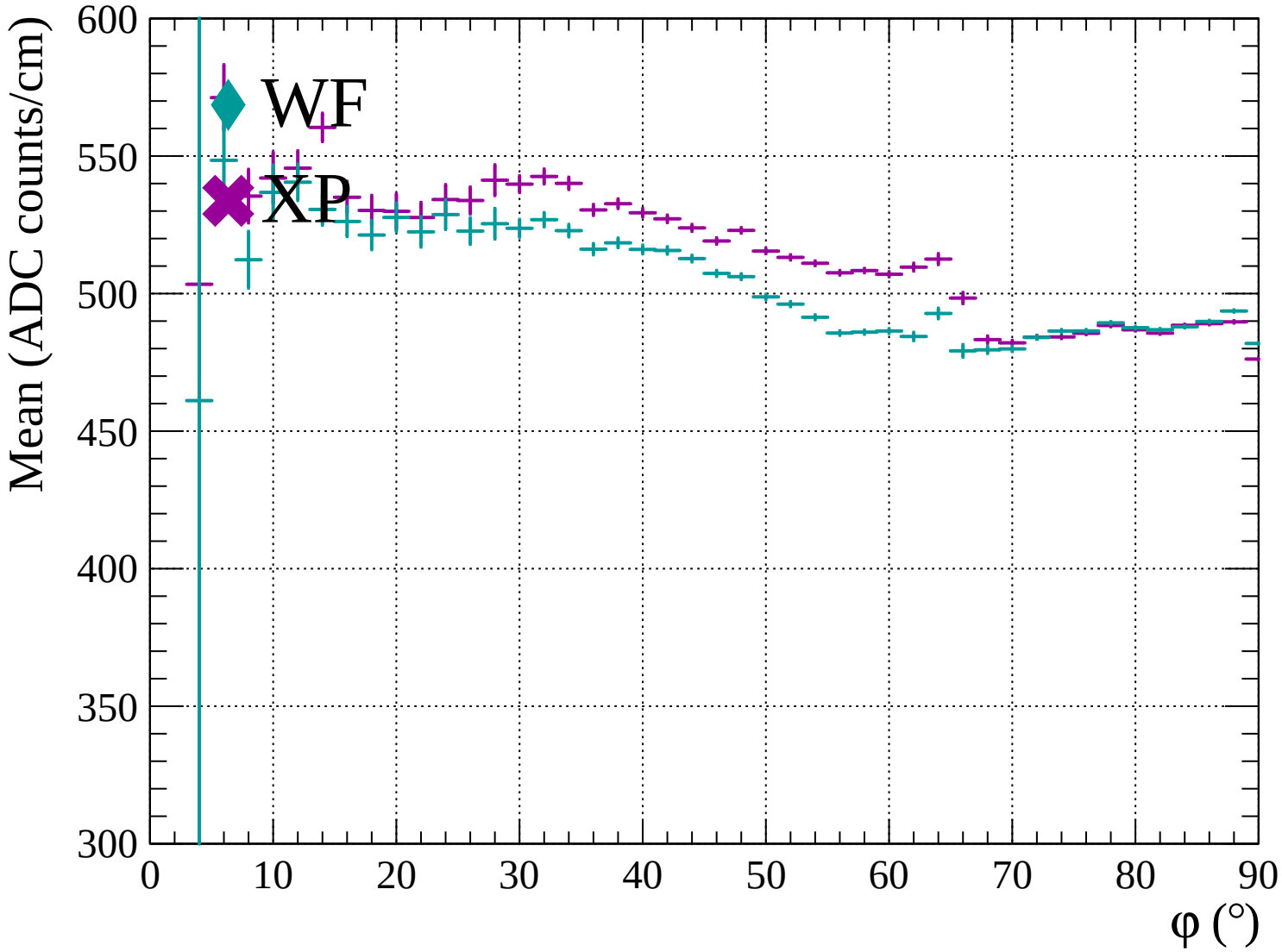


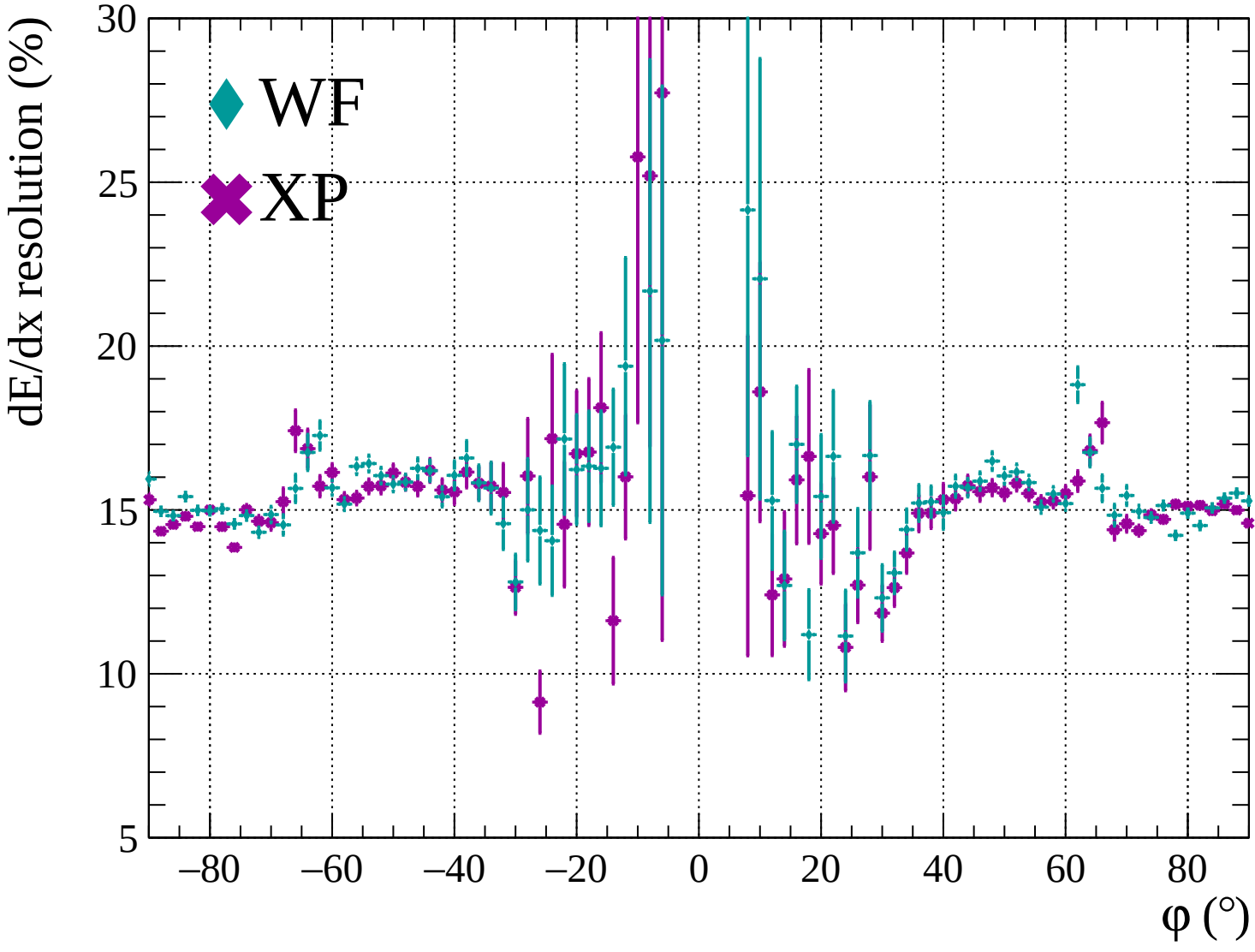


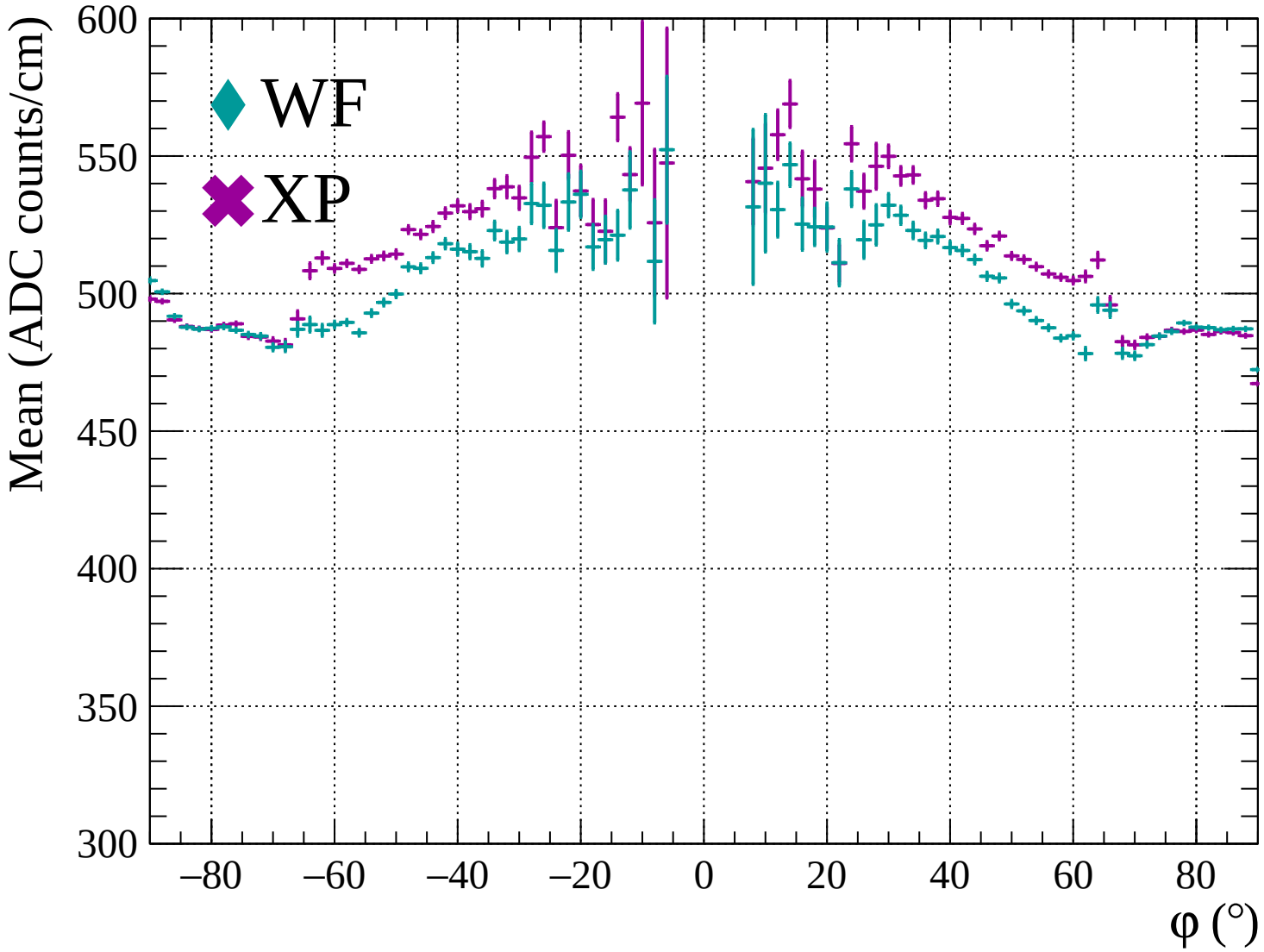


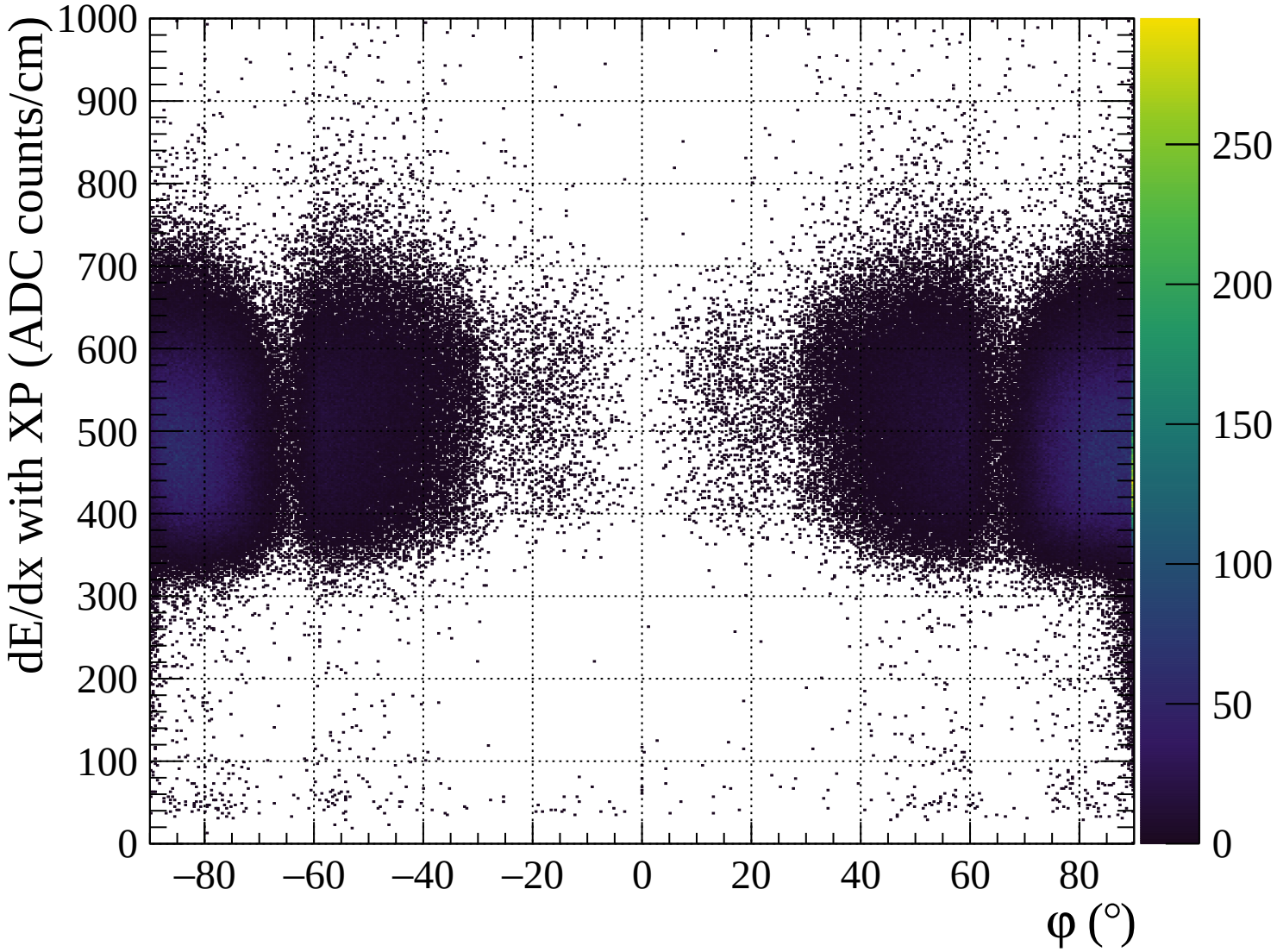


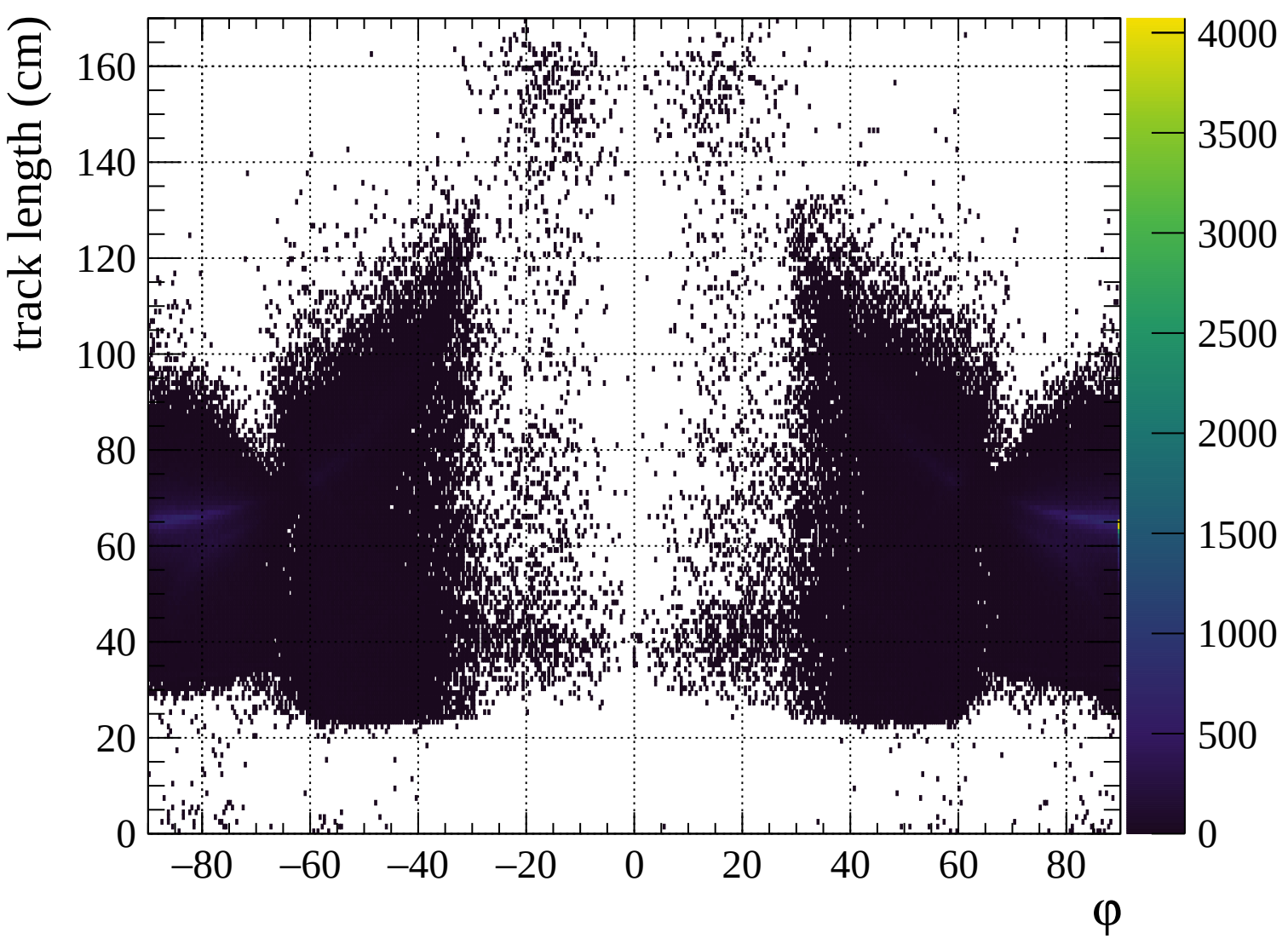


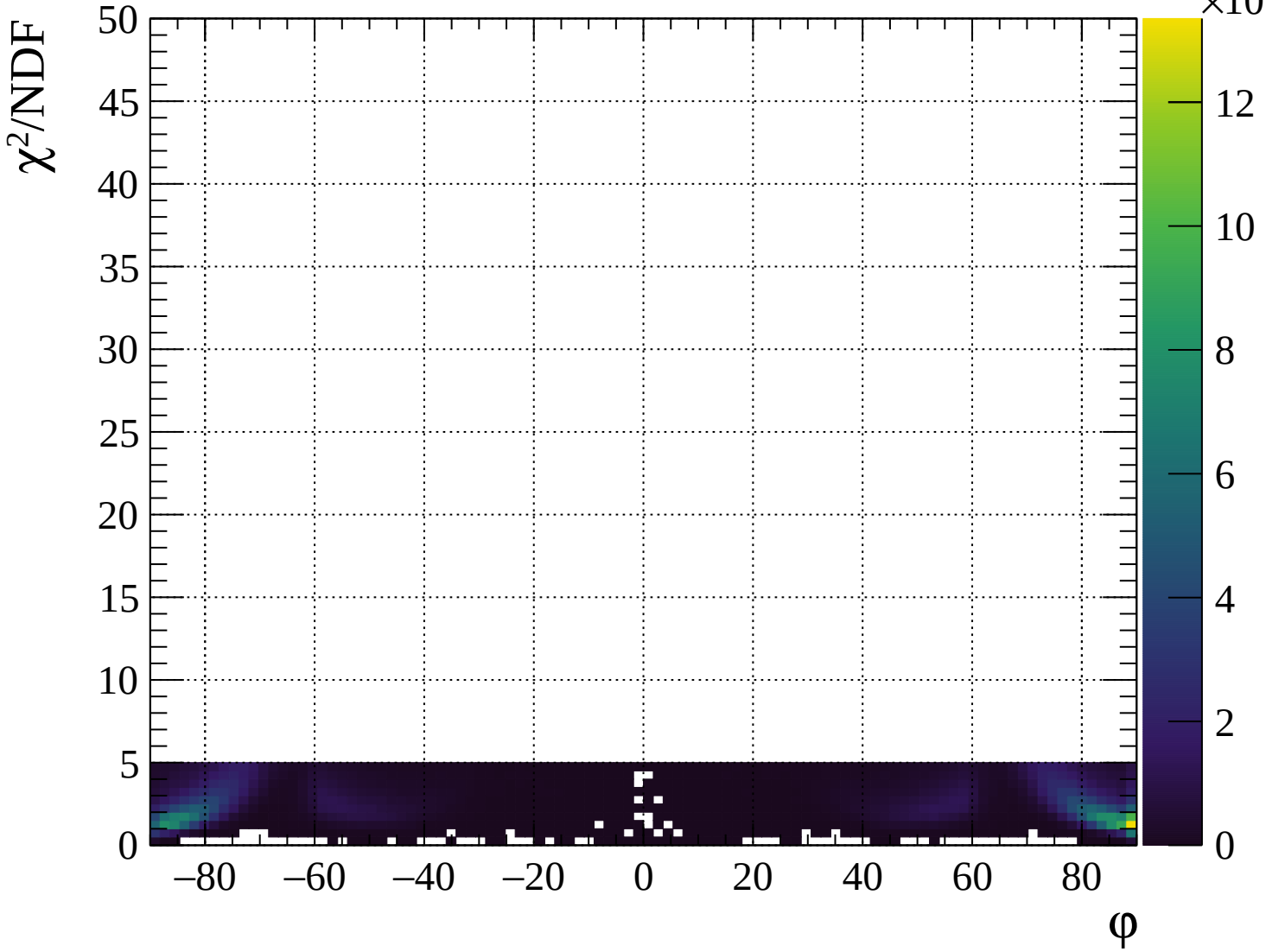


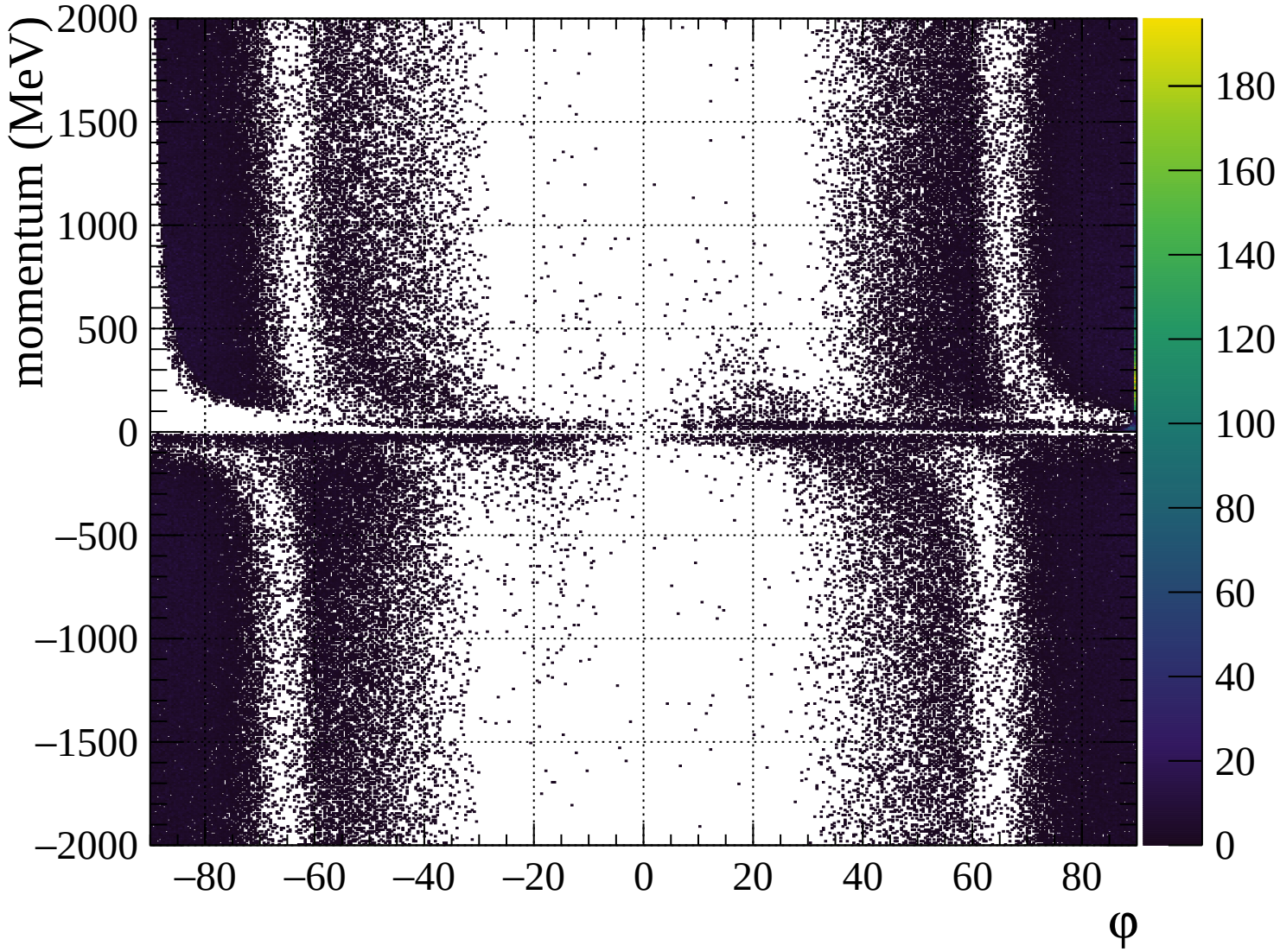


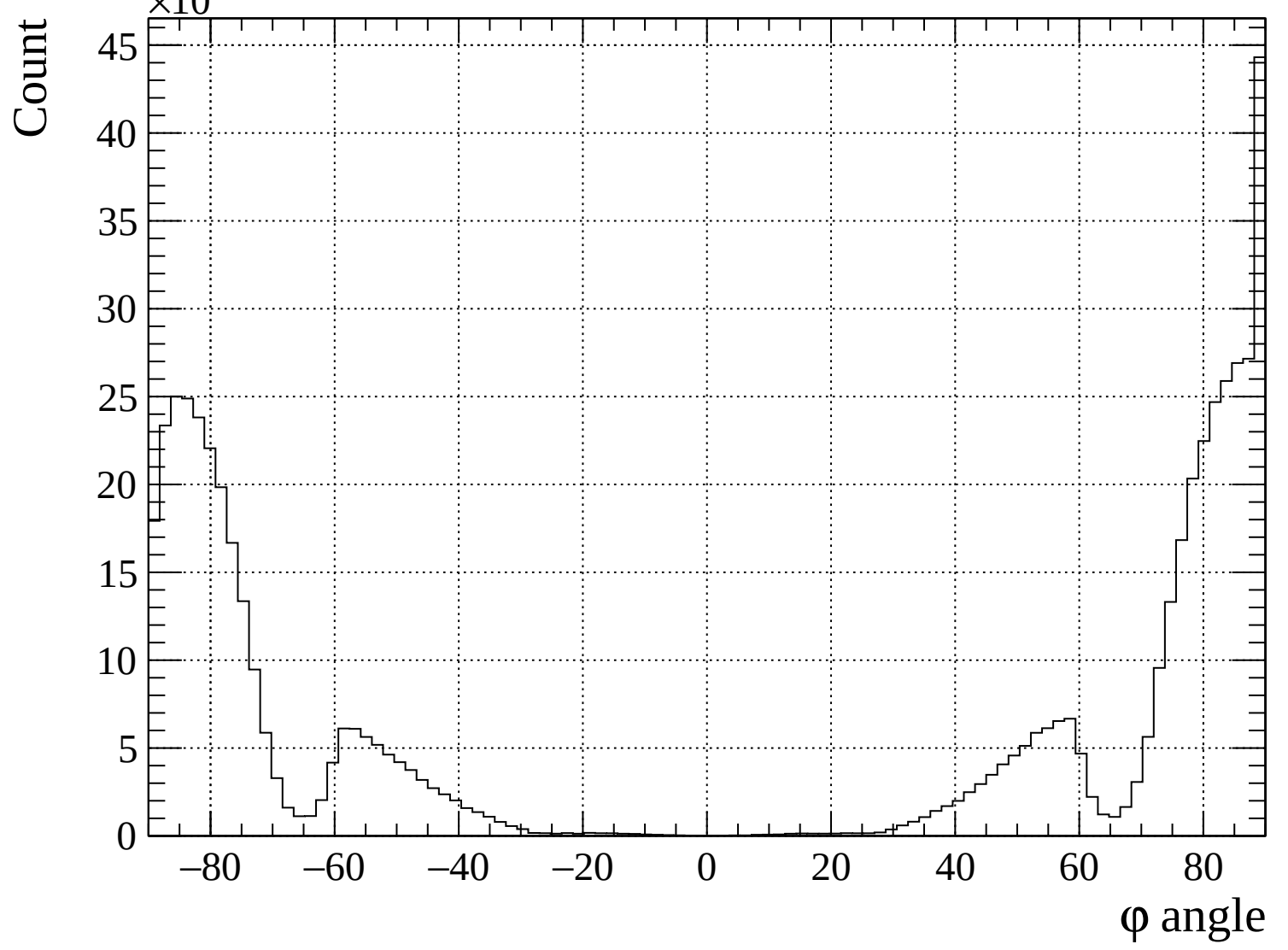


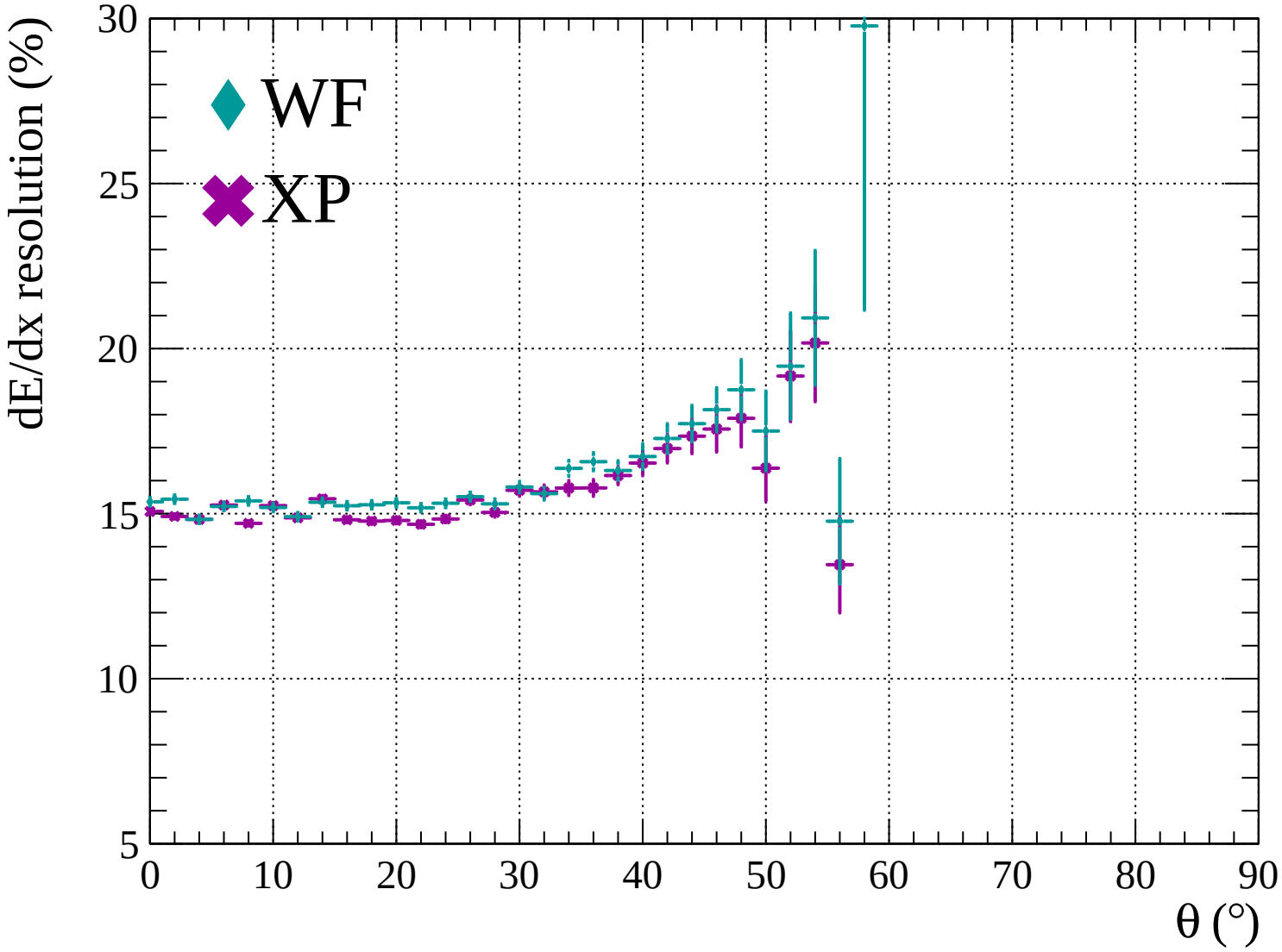


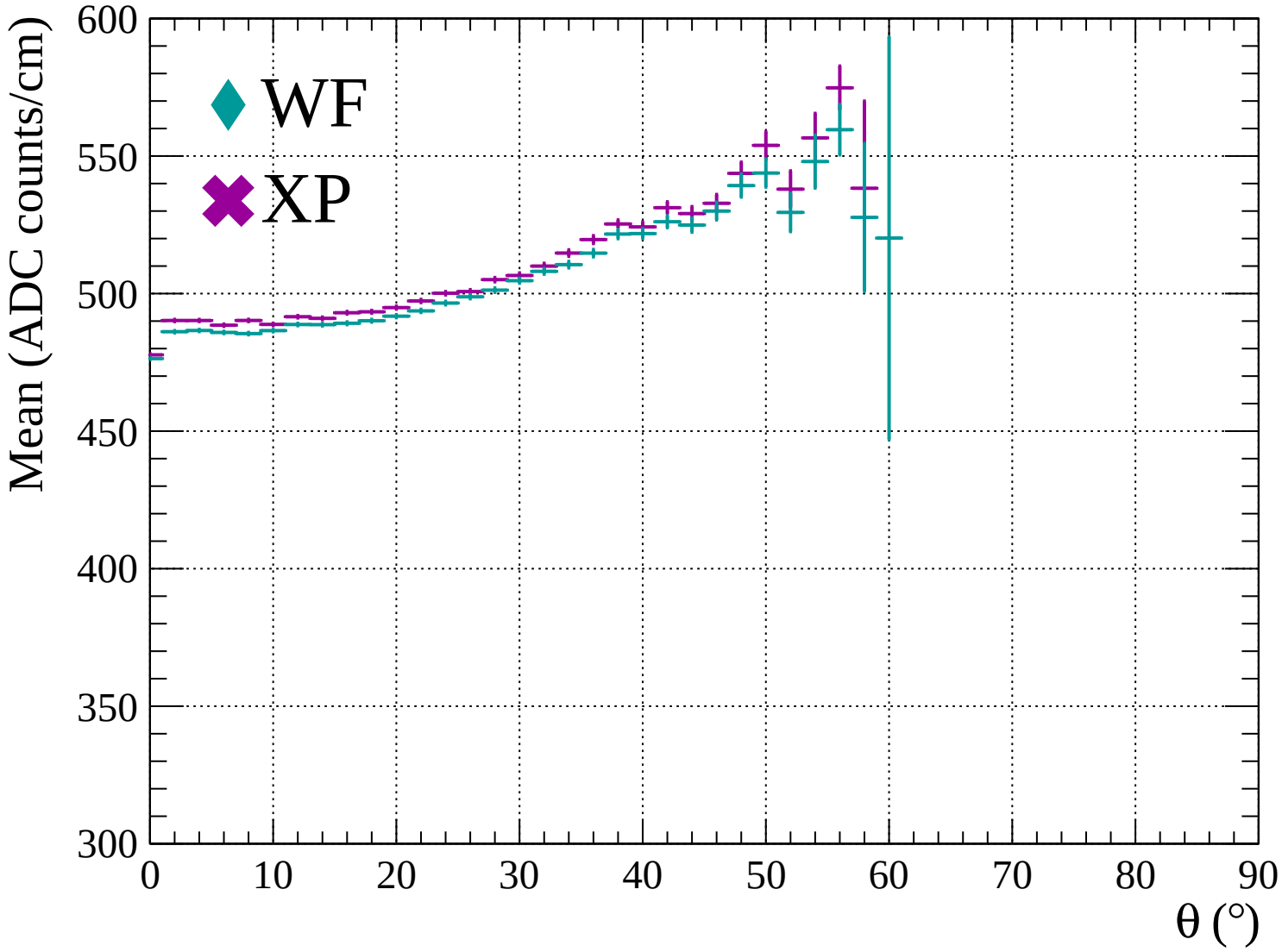


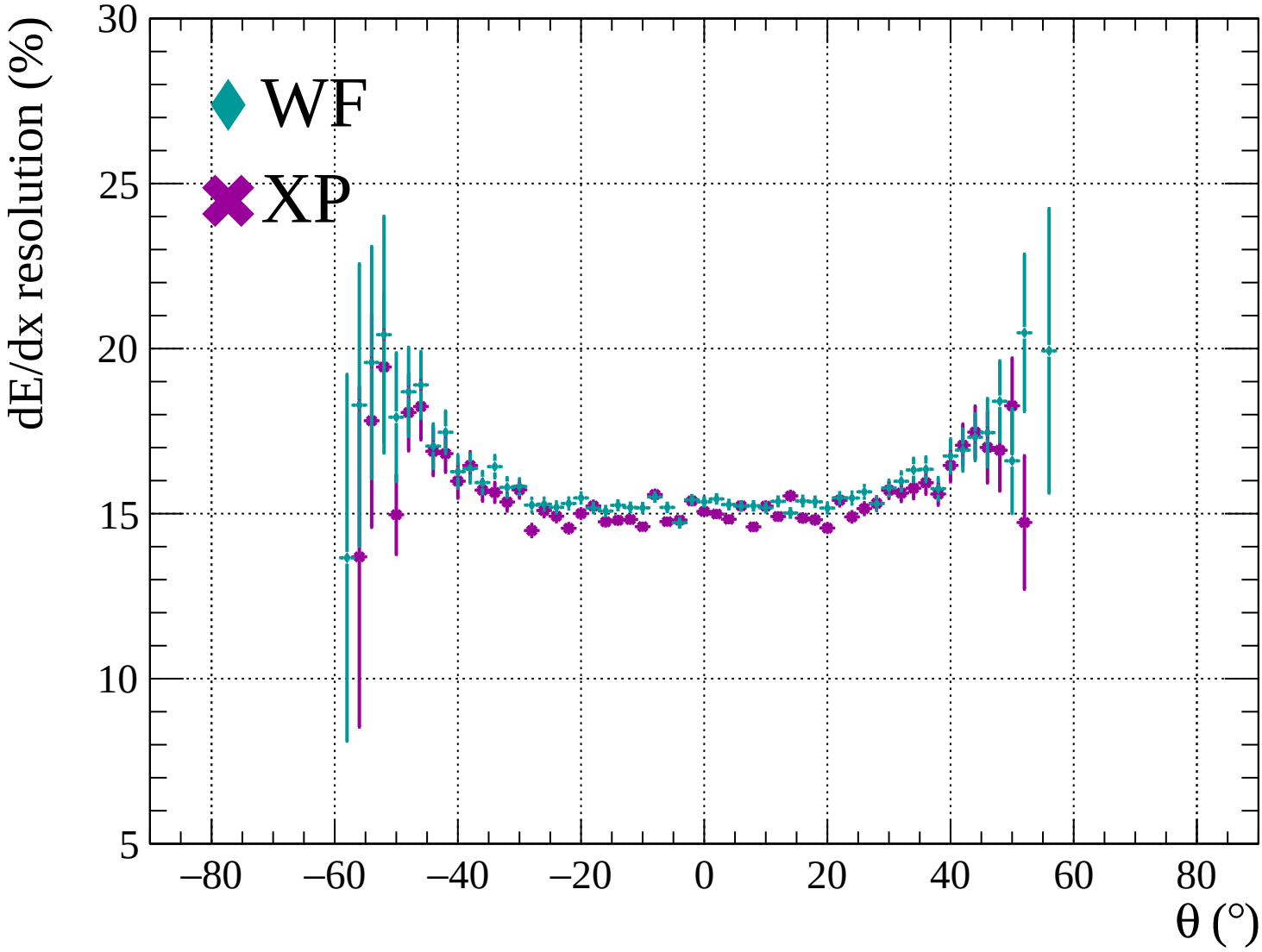


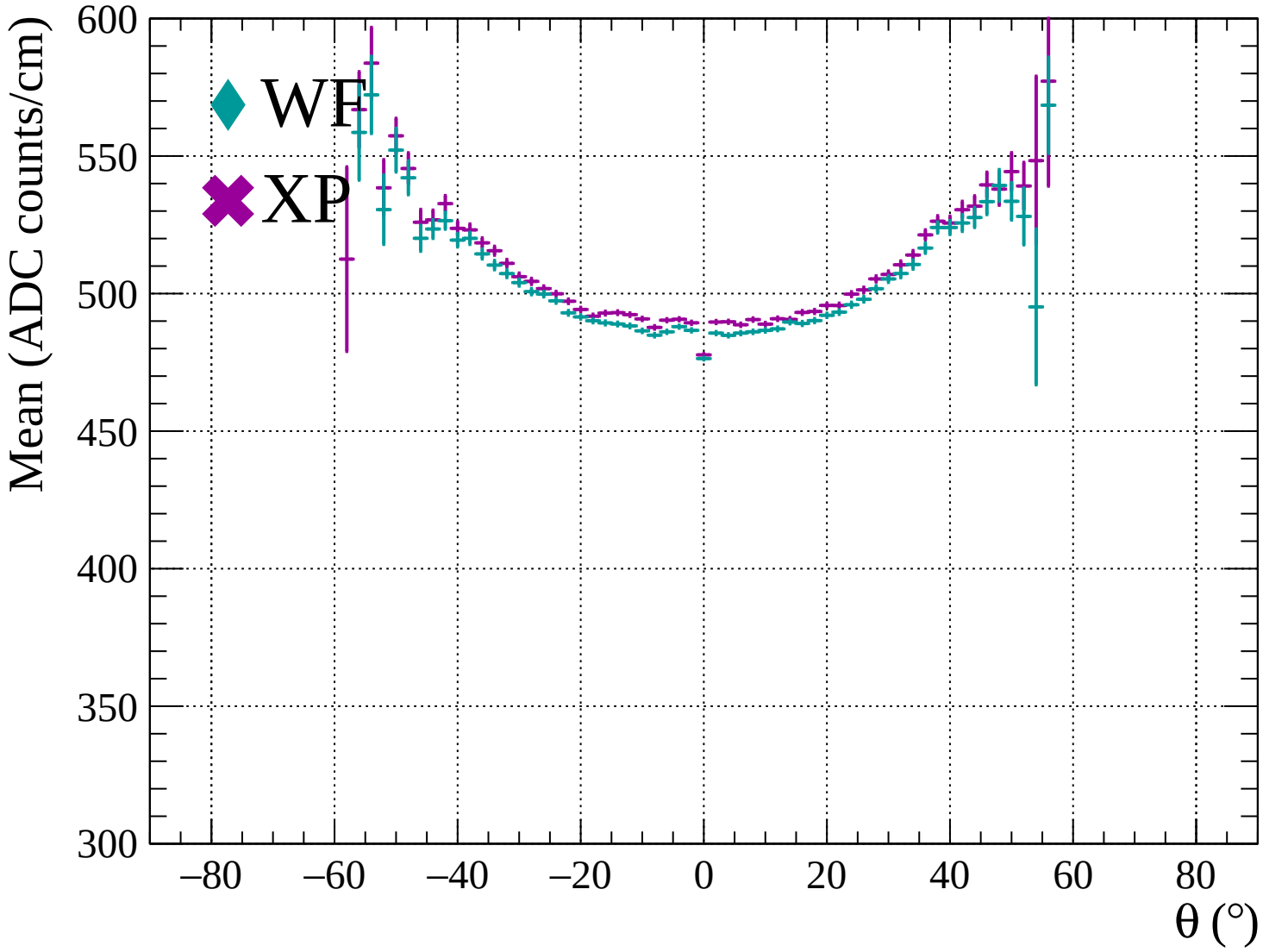


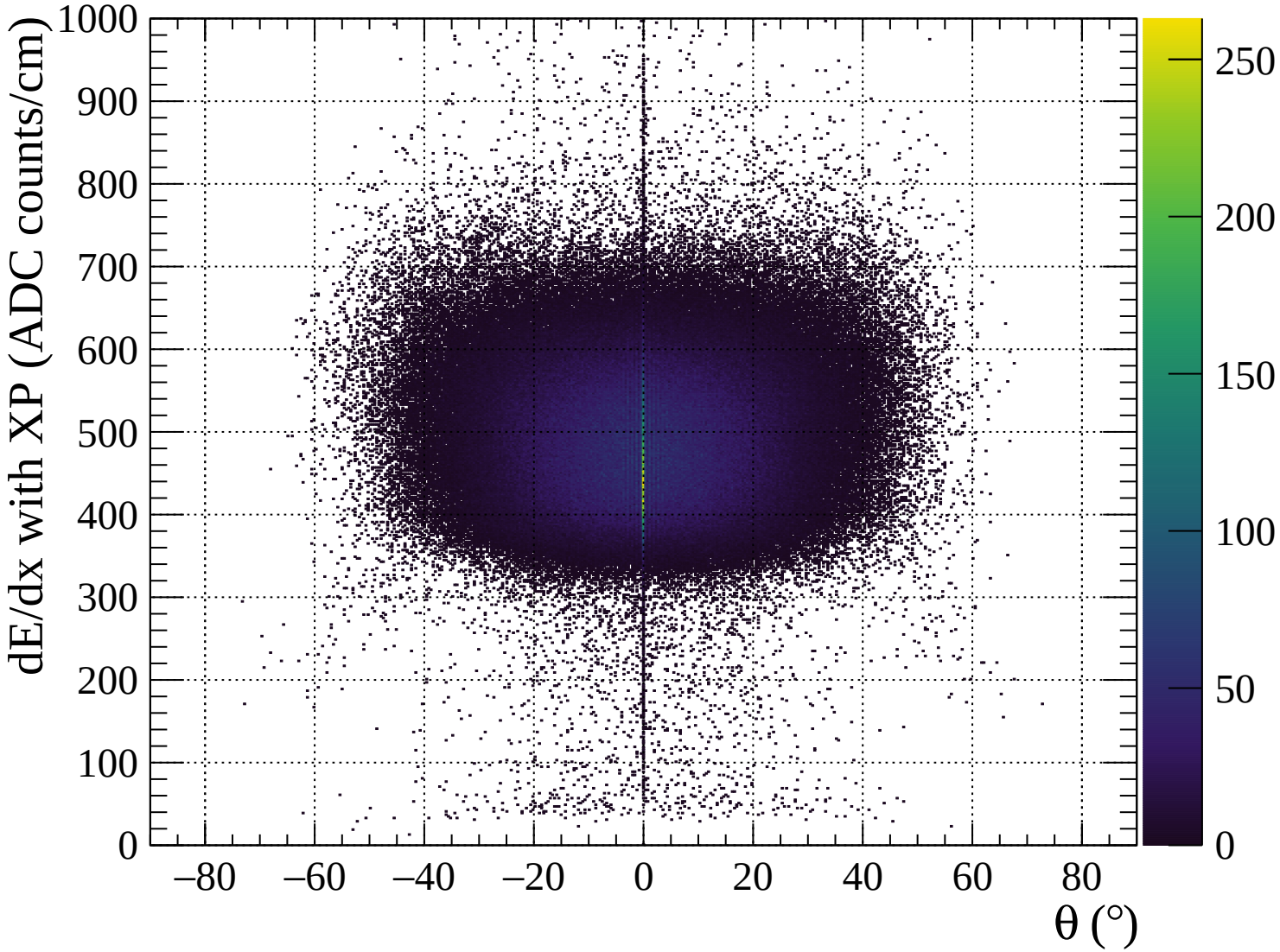


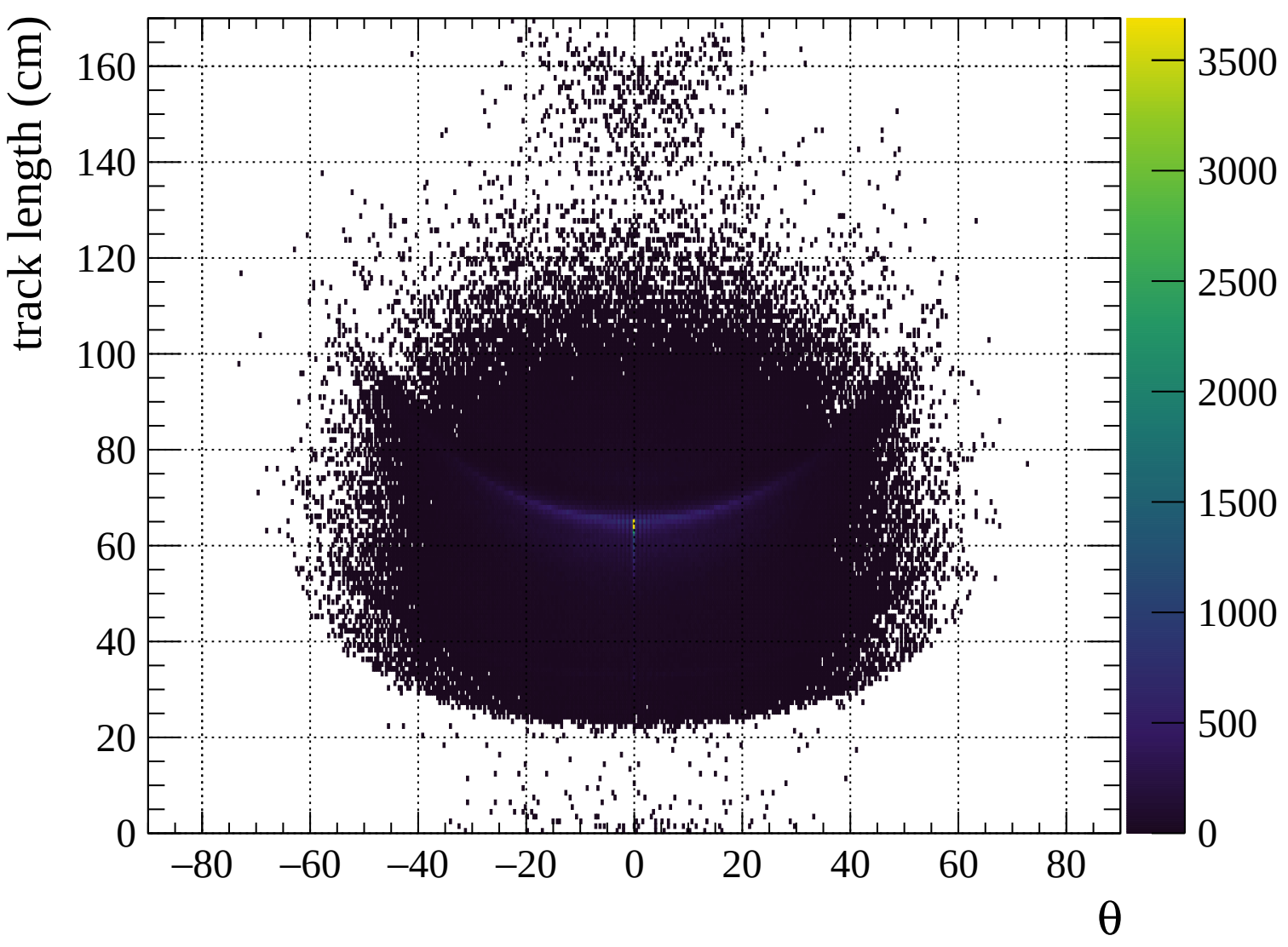


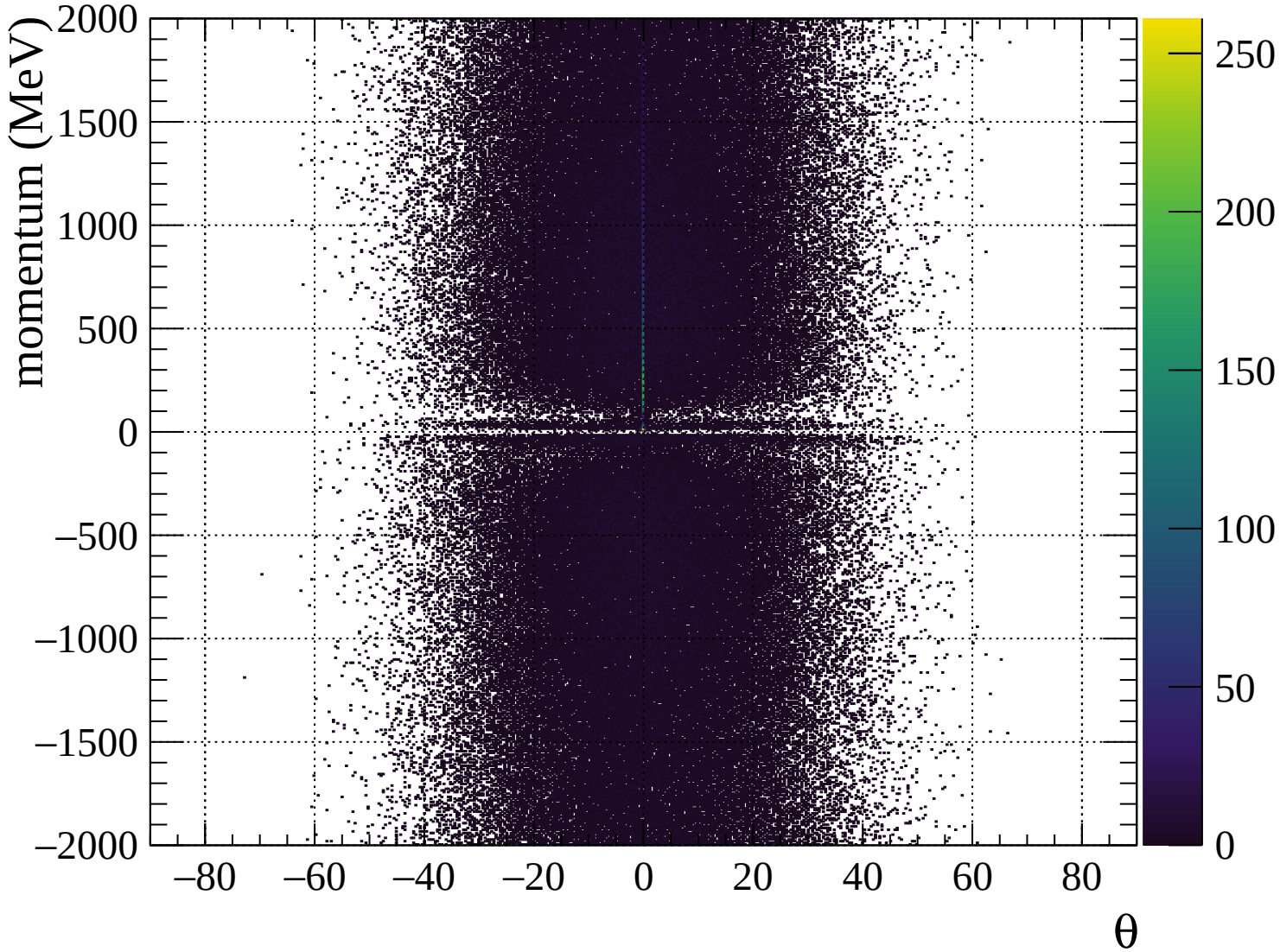


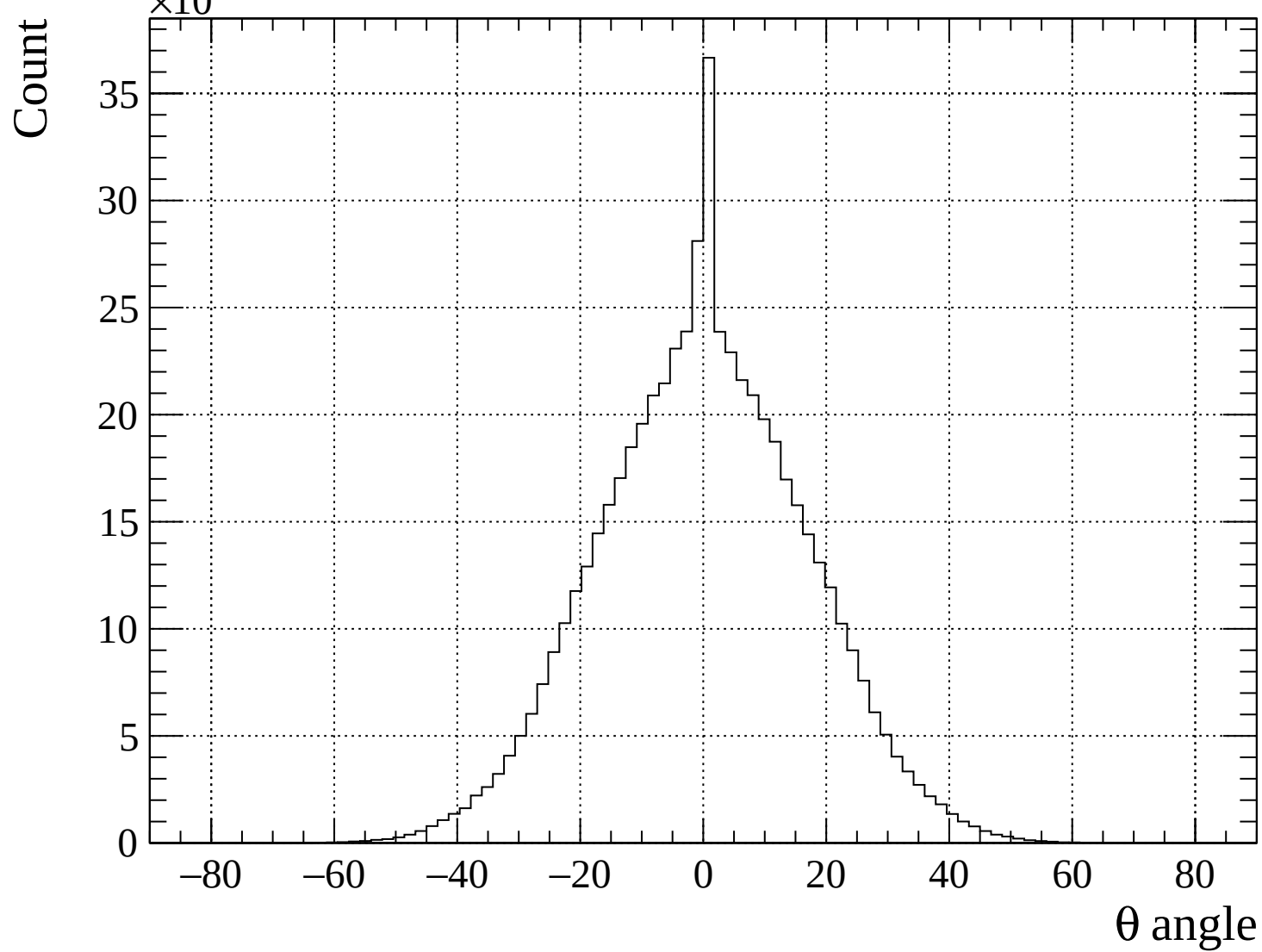


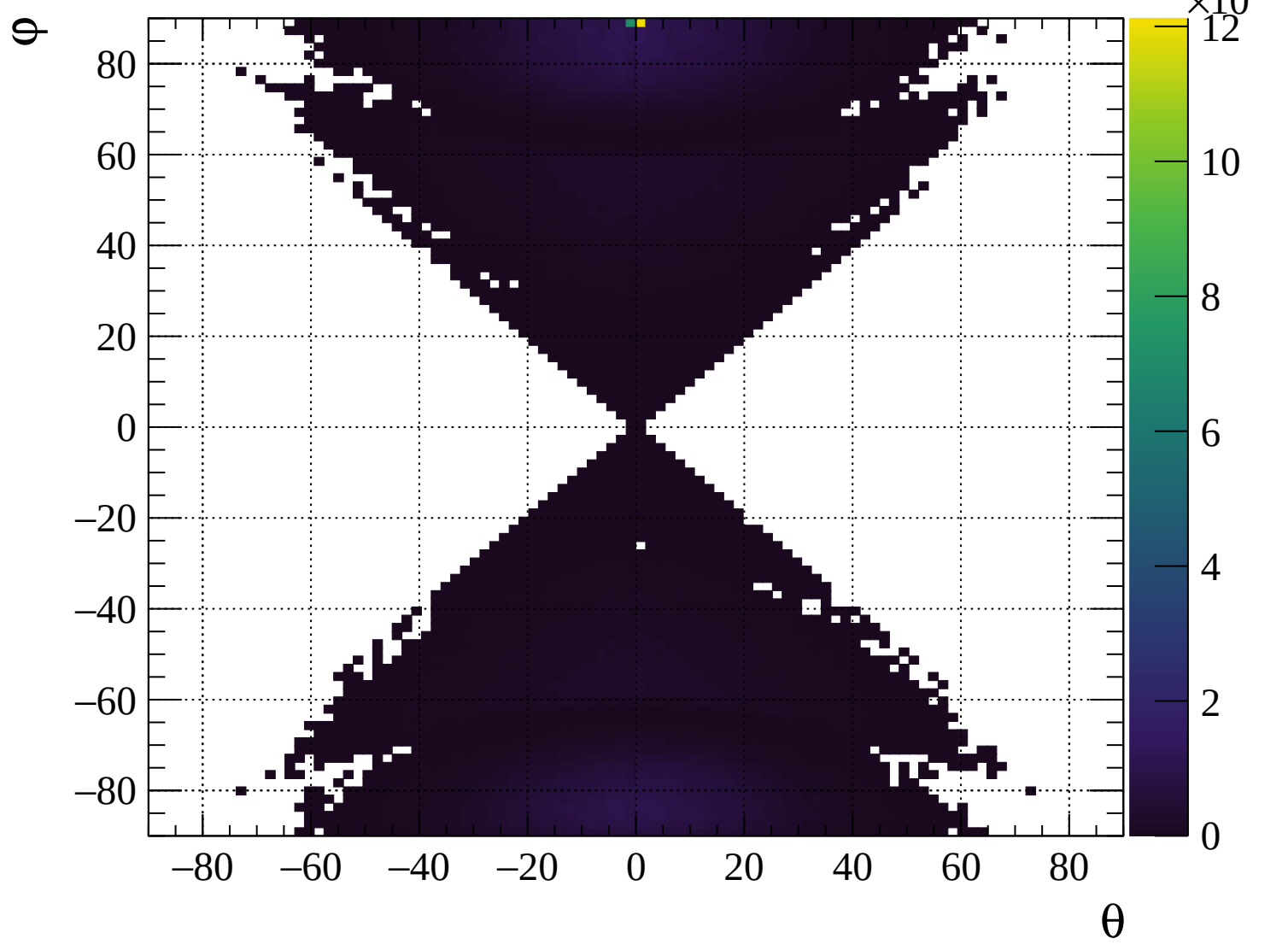


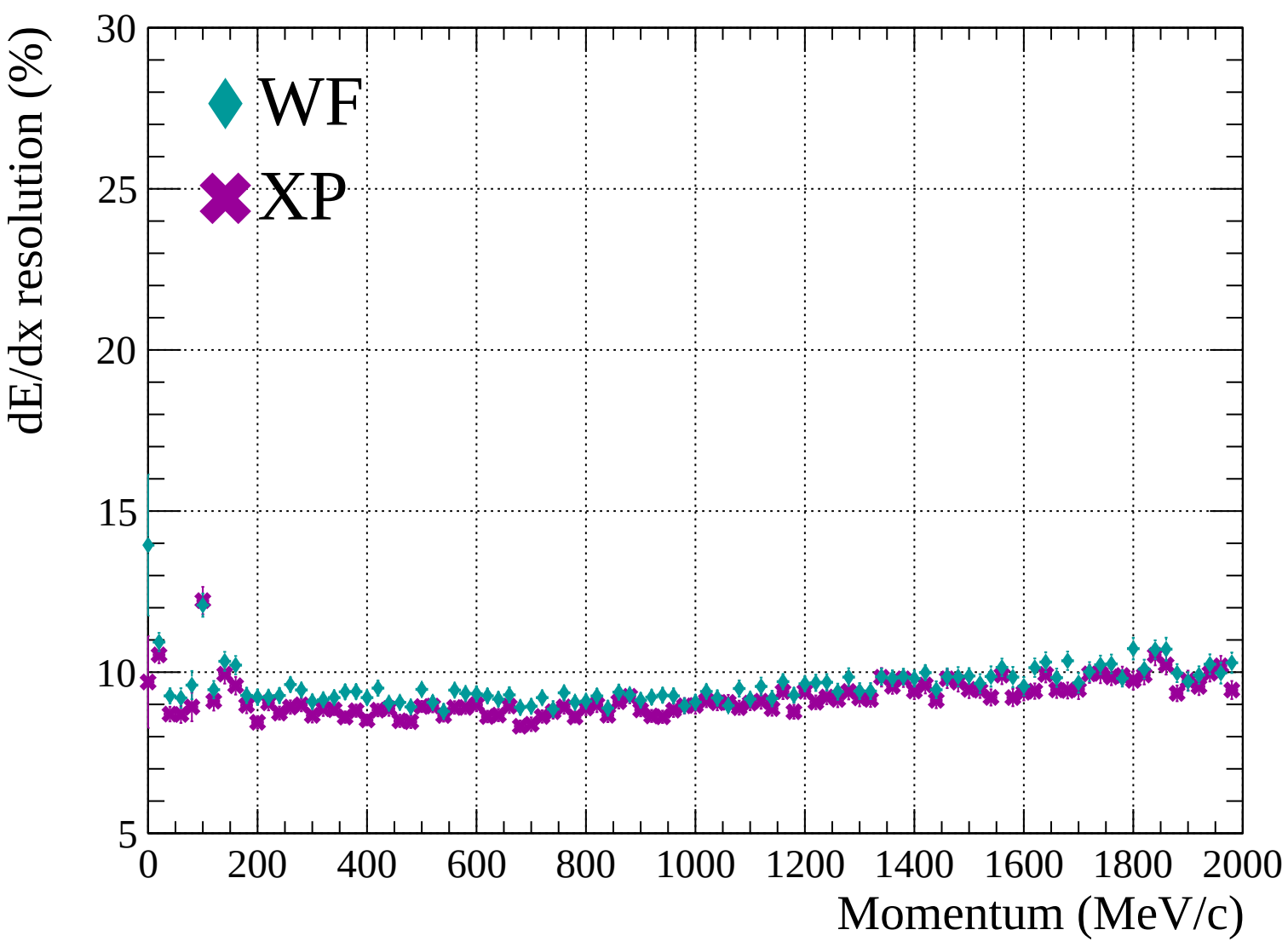


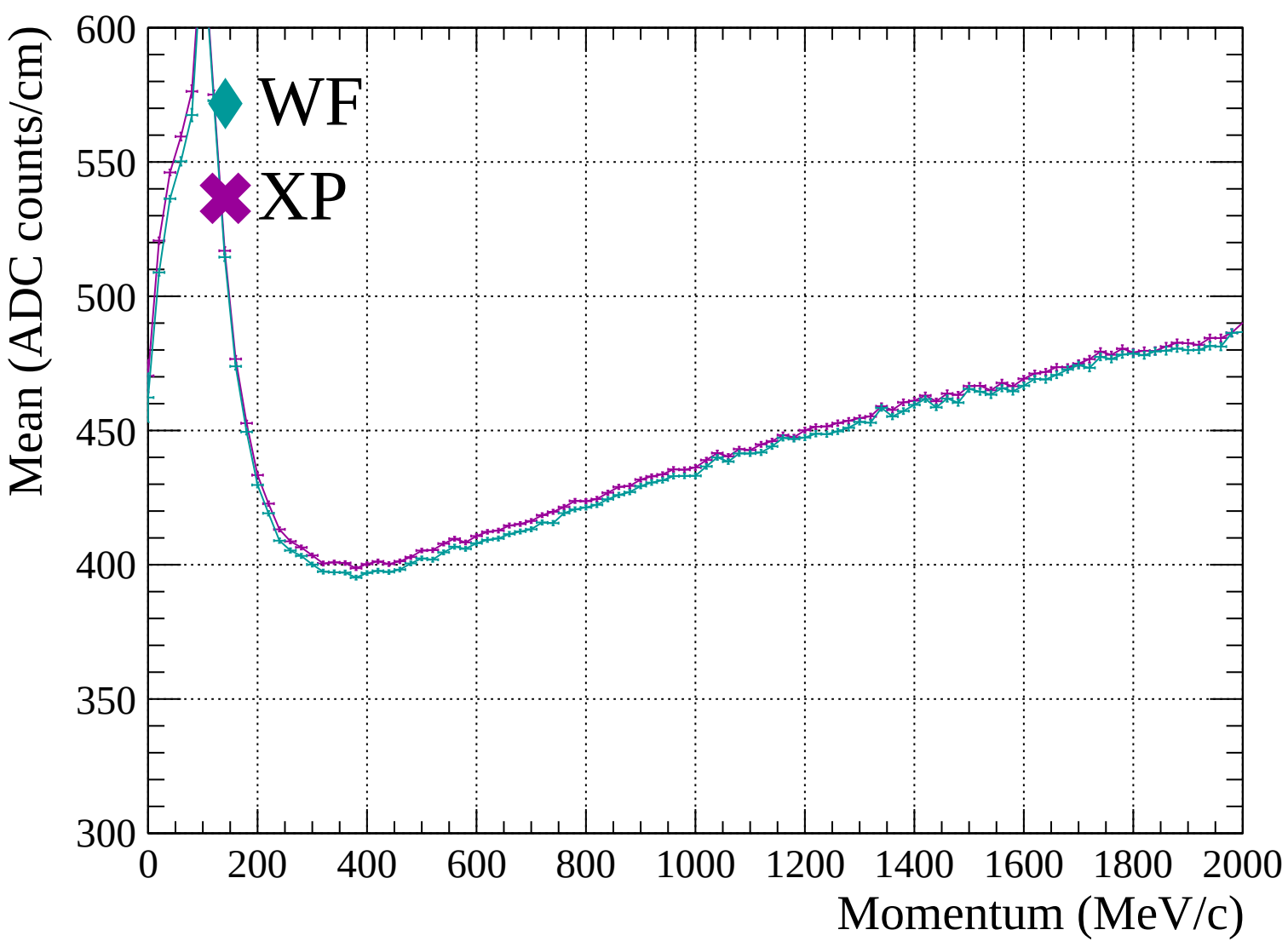


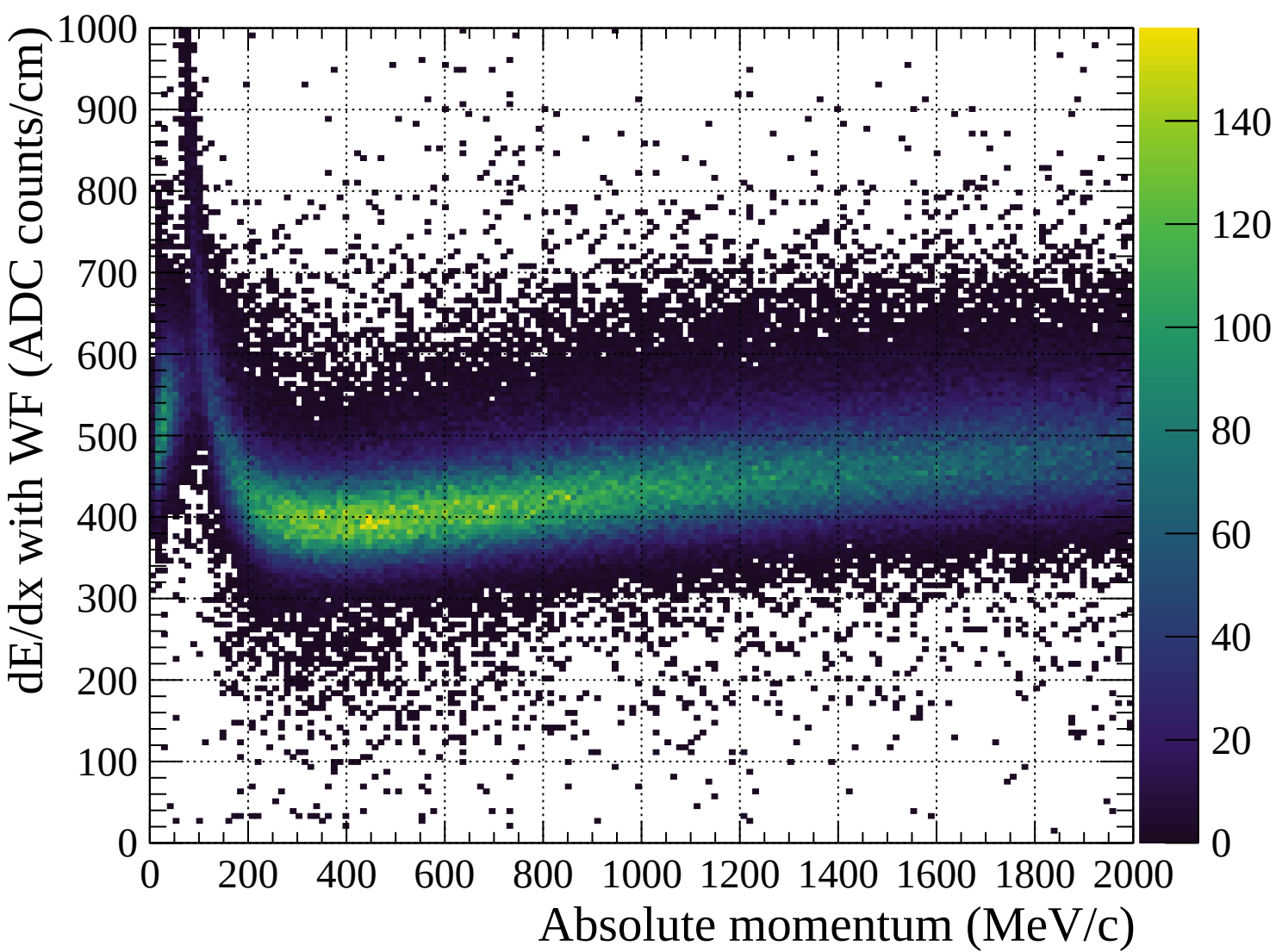


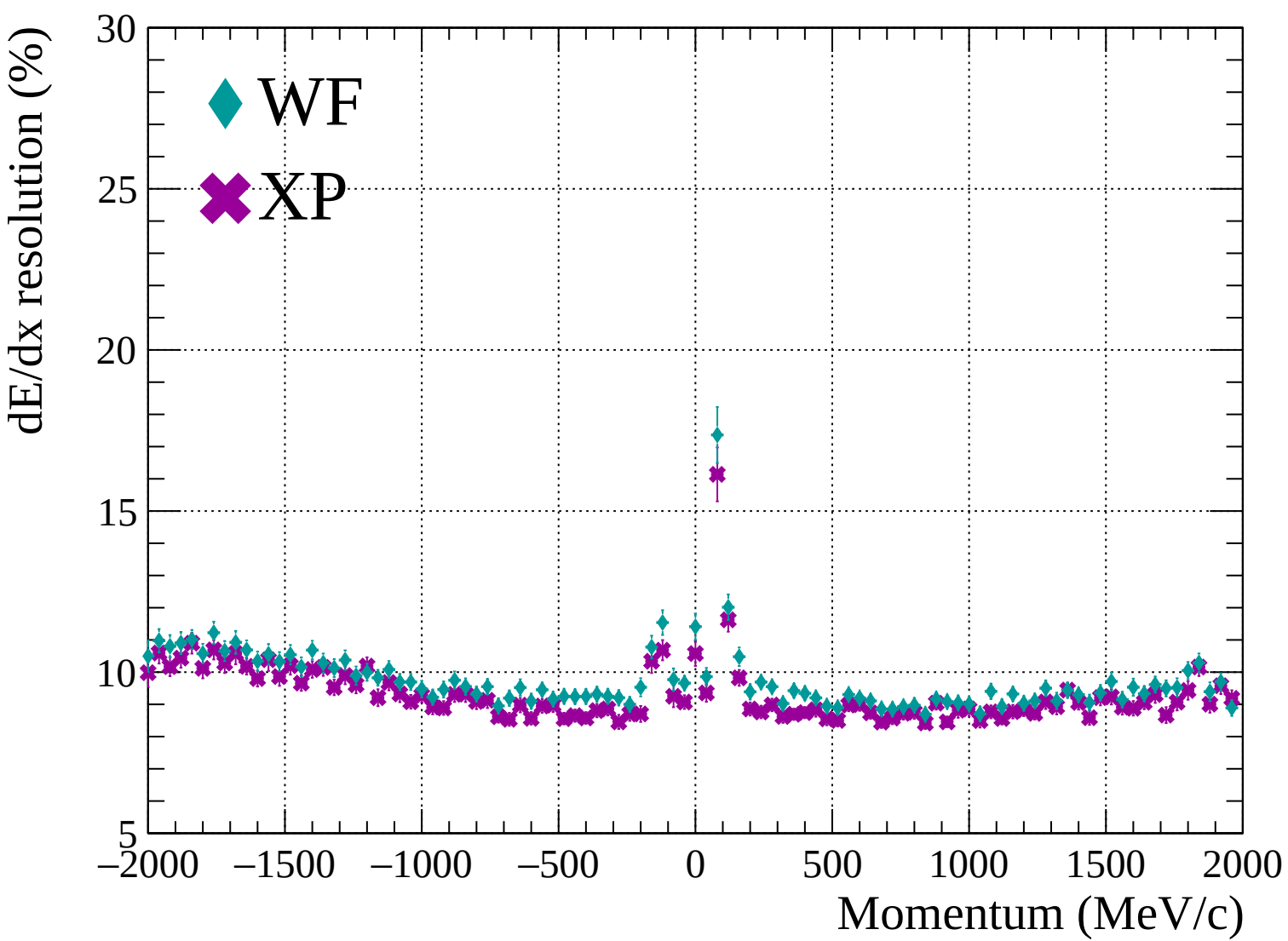


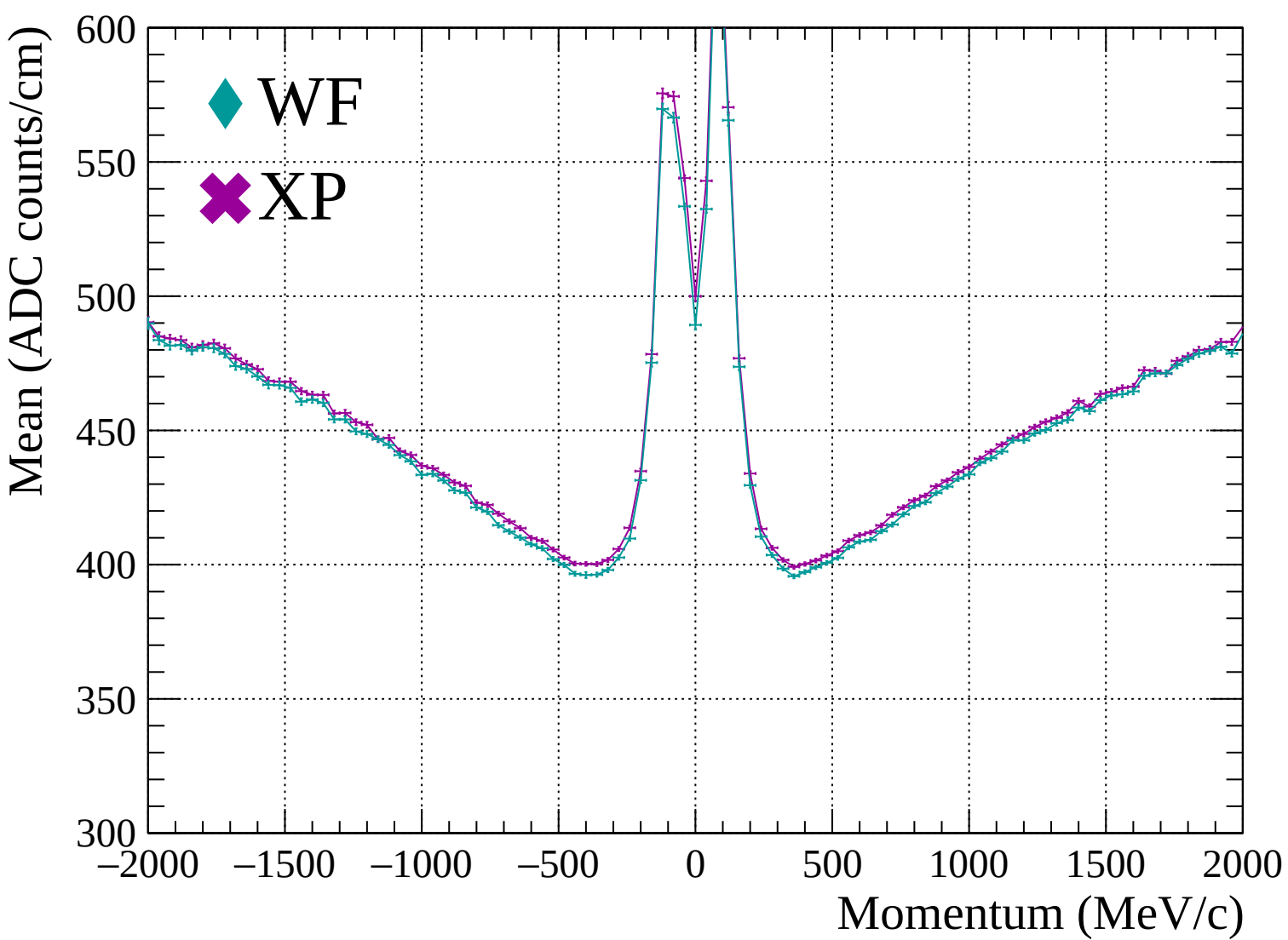


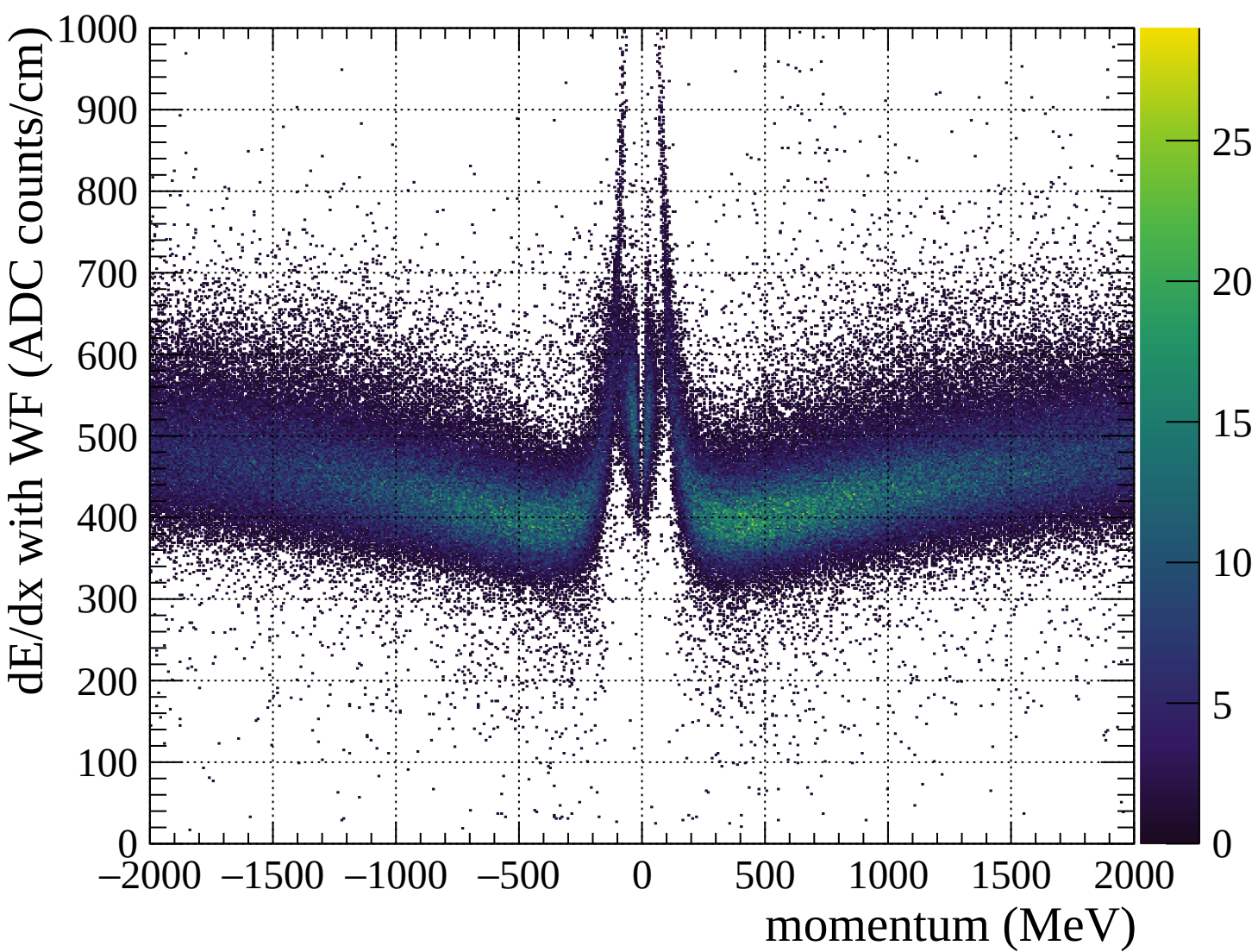


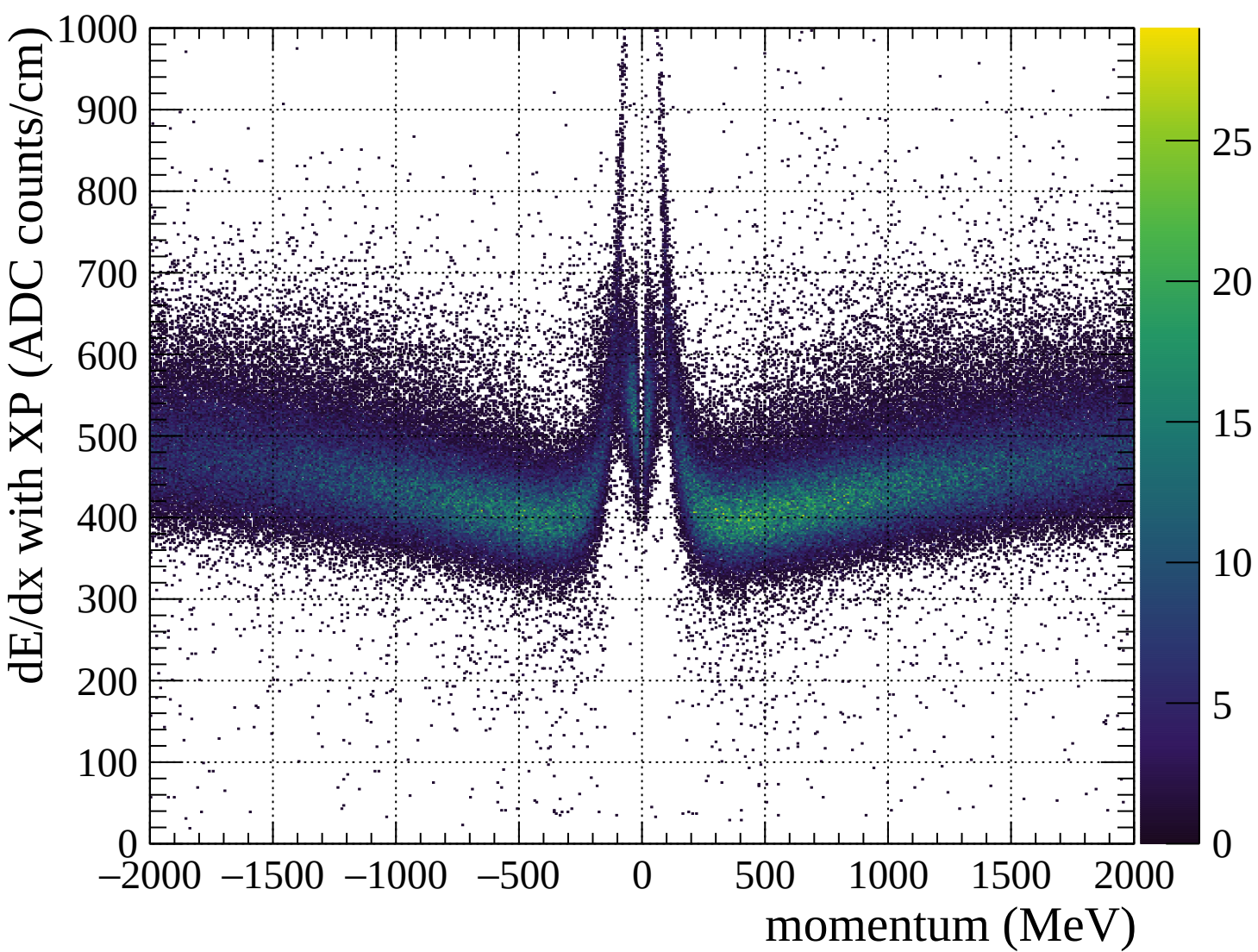


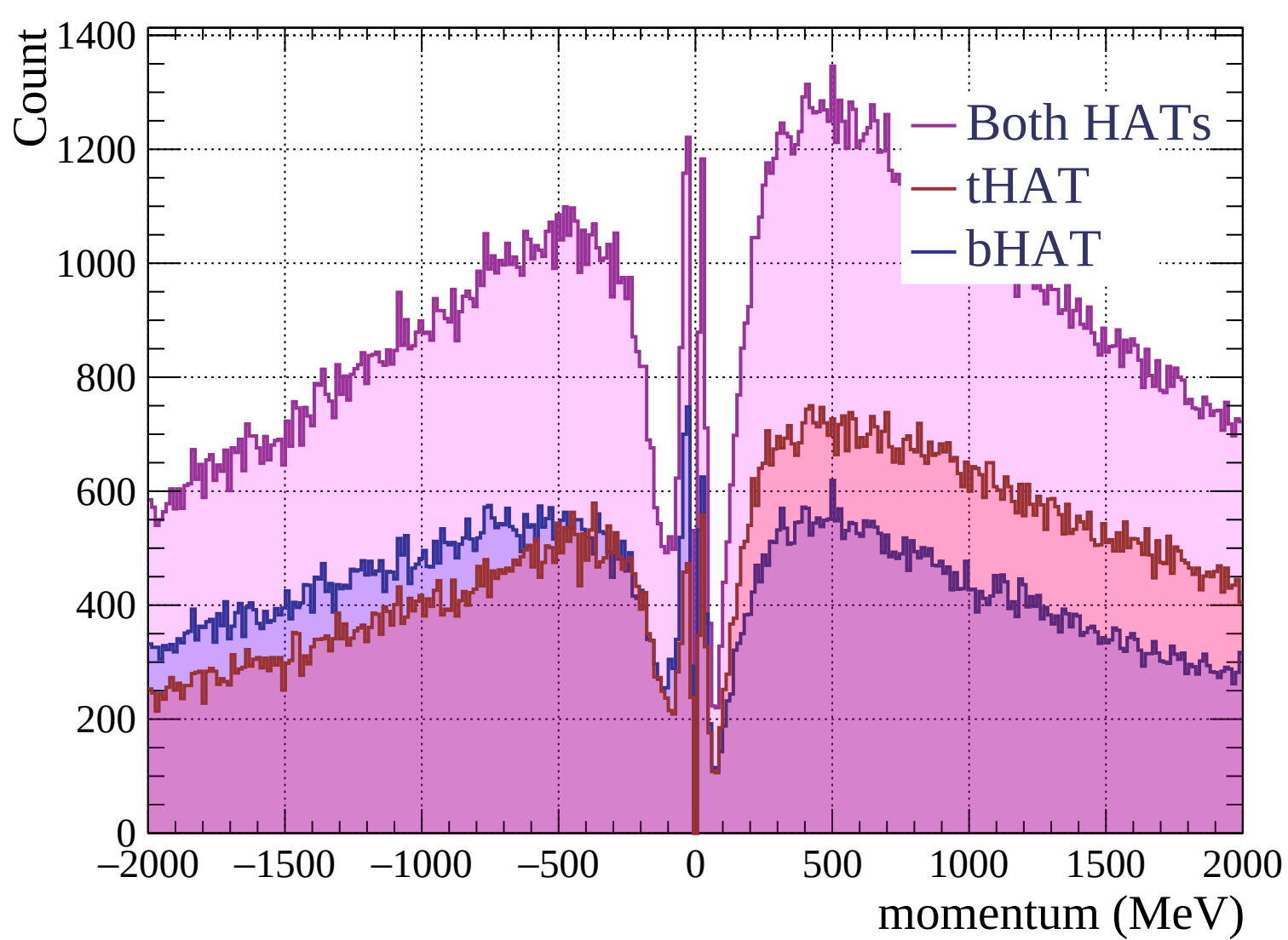


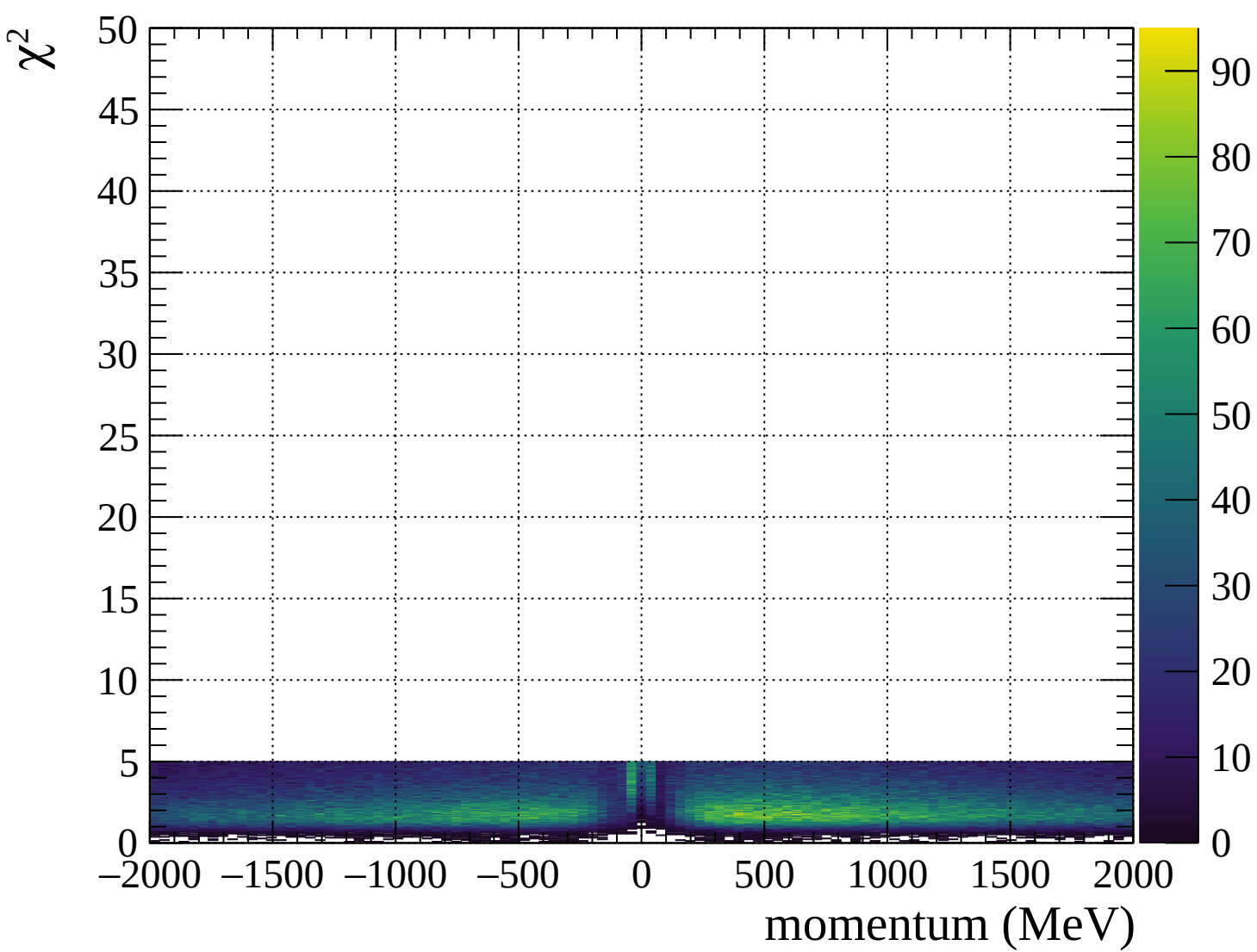




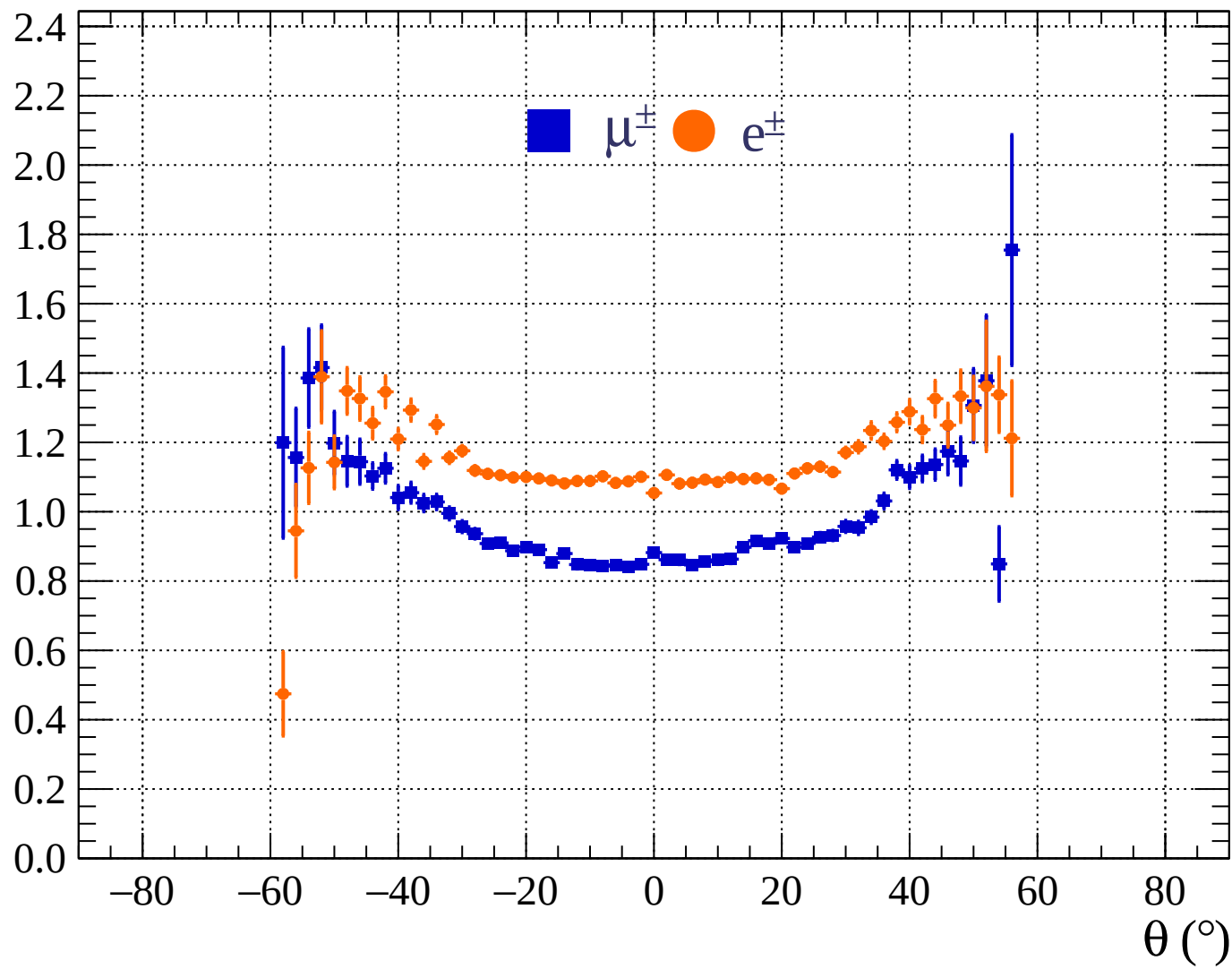




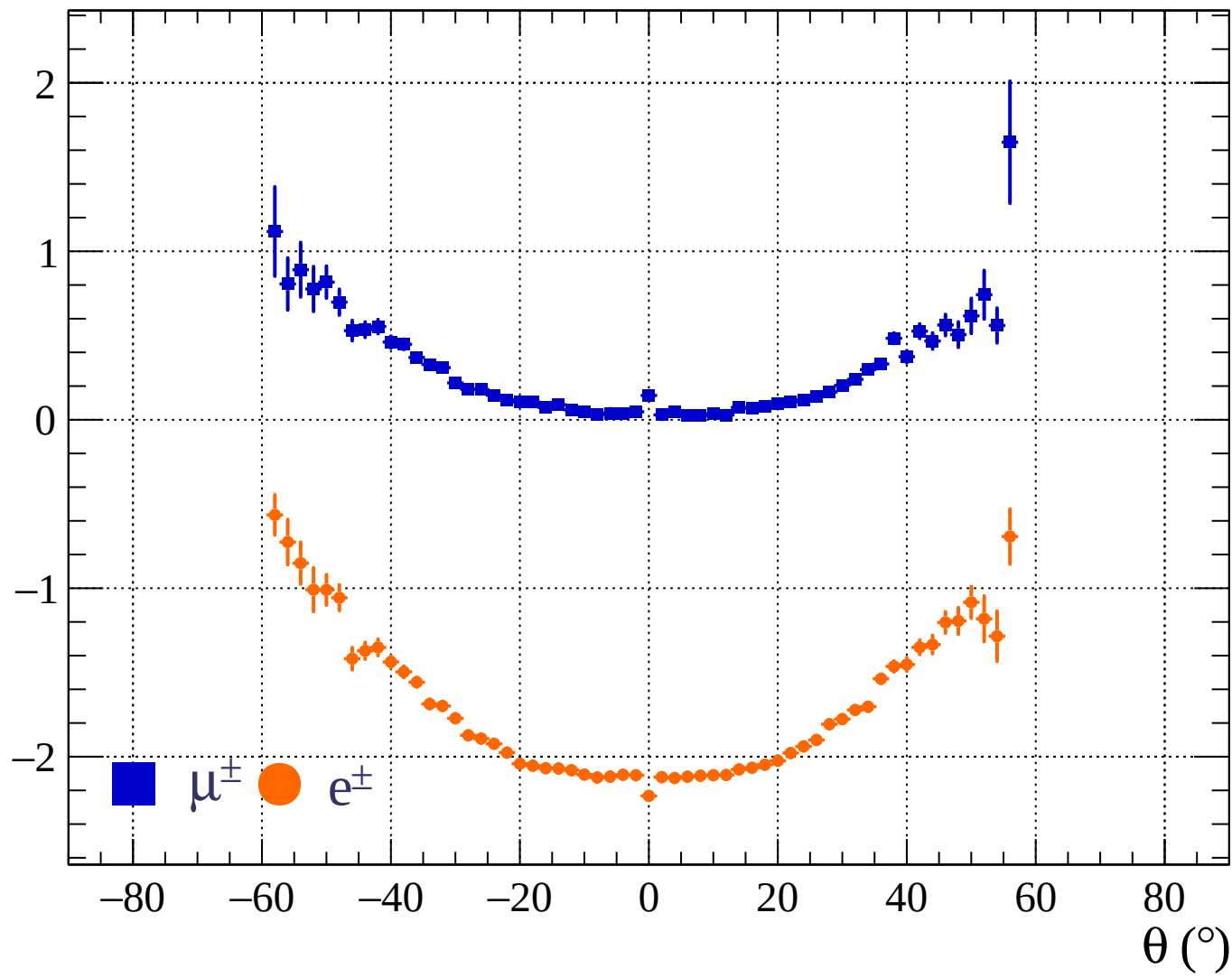




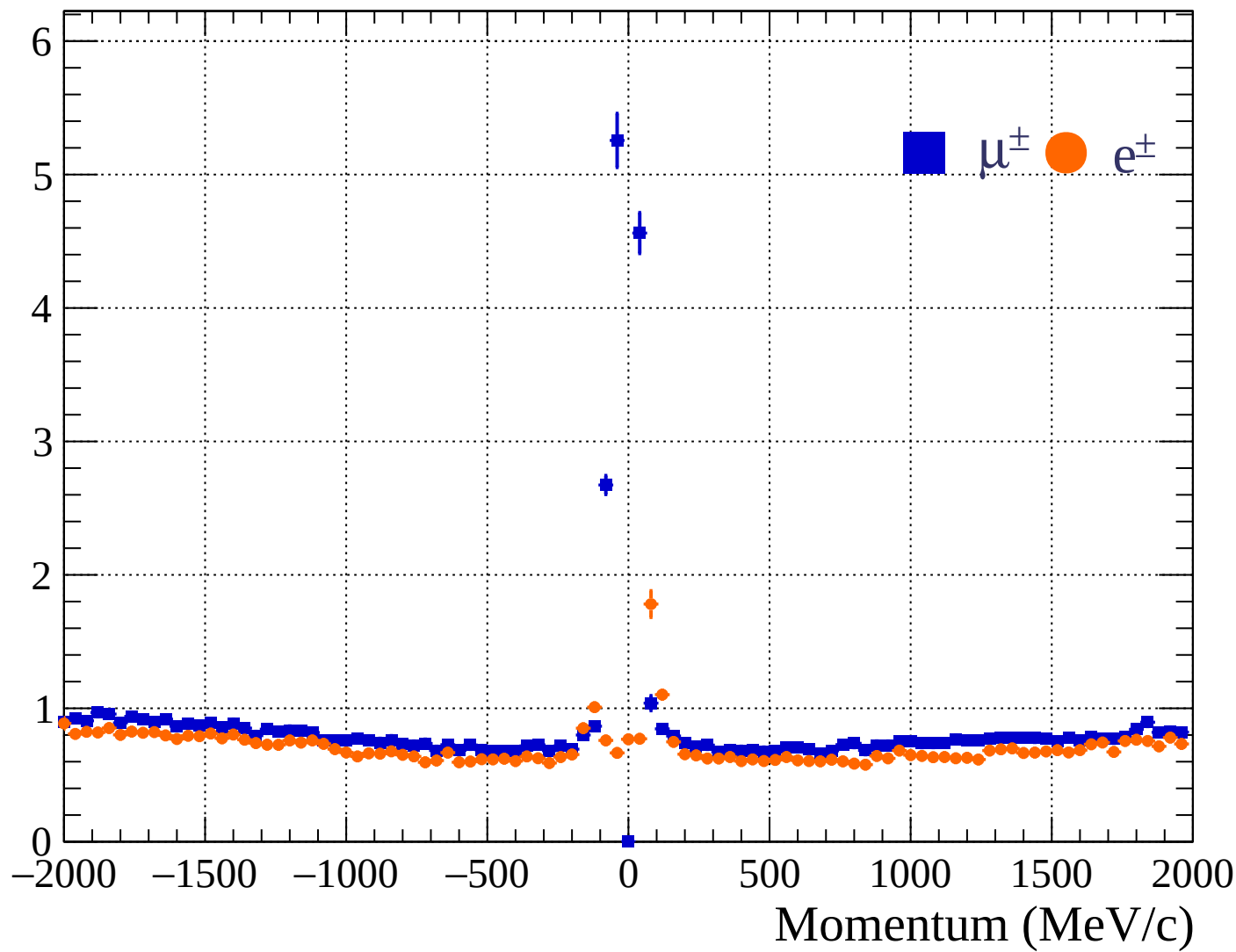
Pulls standard deviation



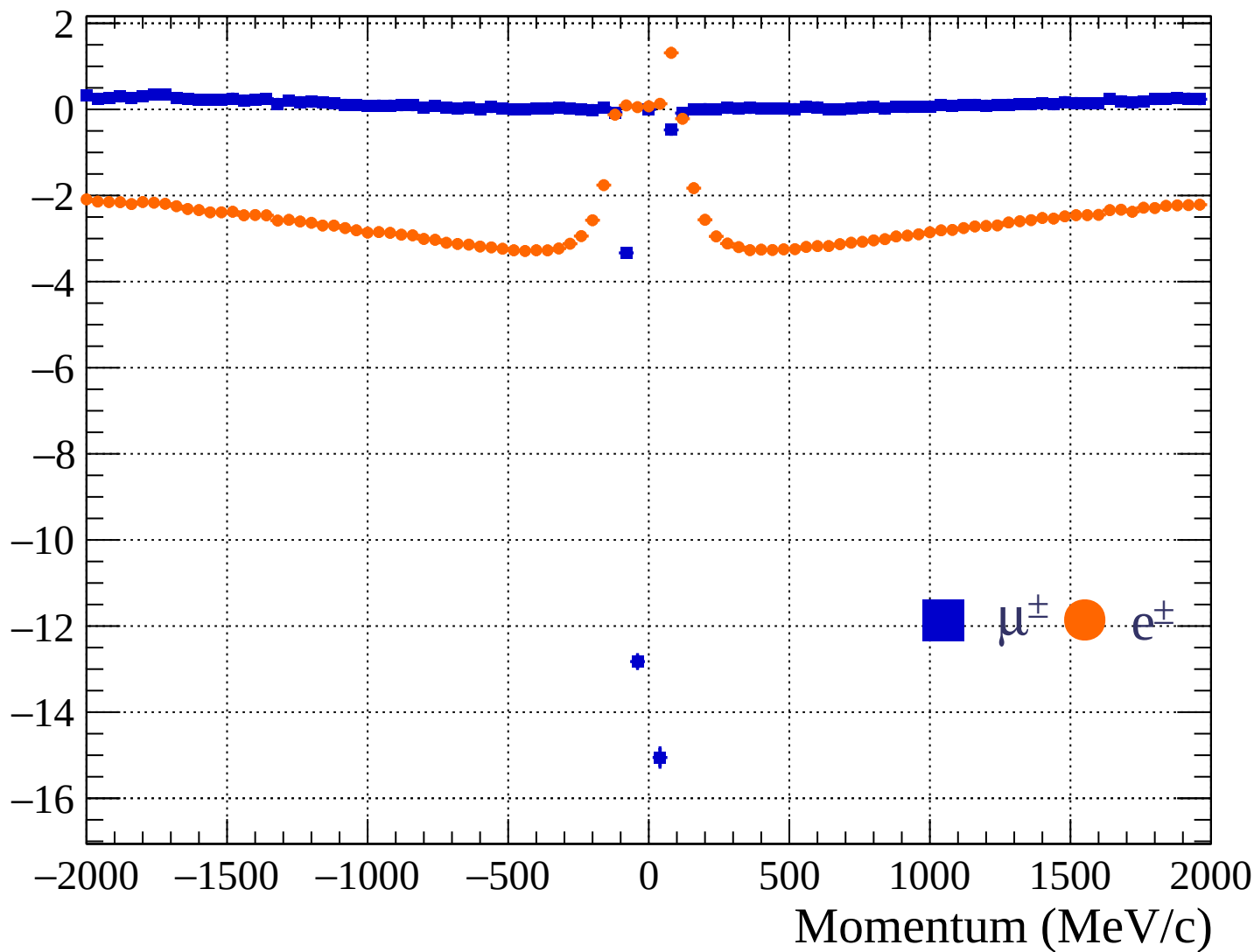
Pulls mean

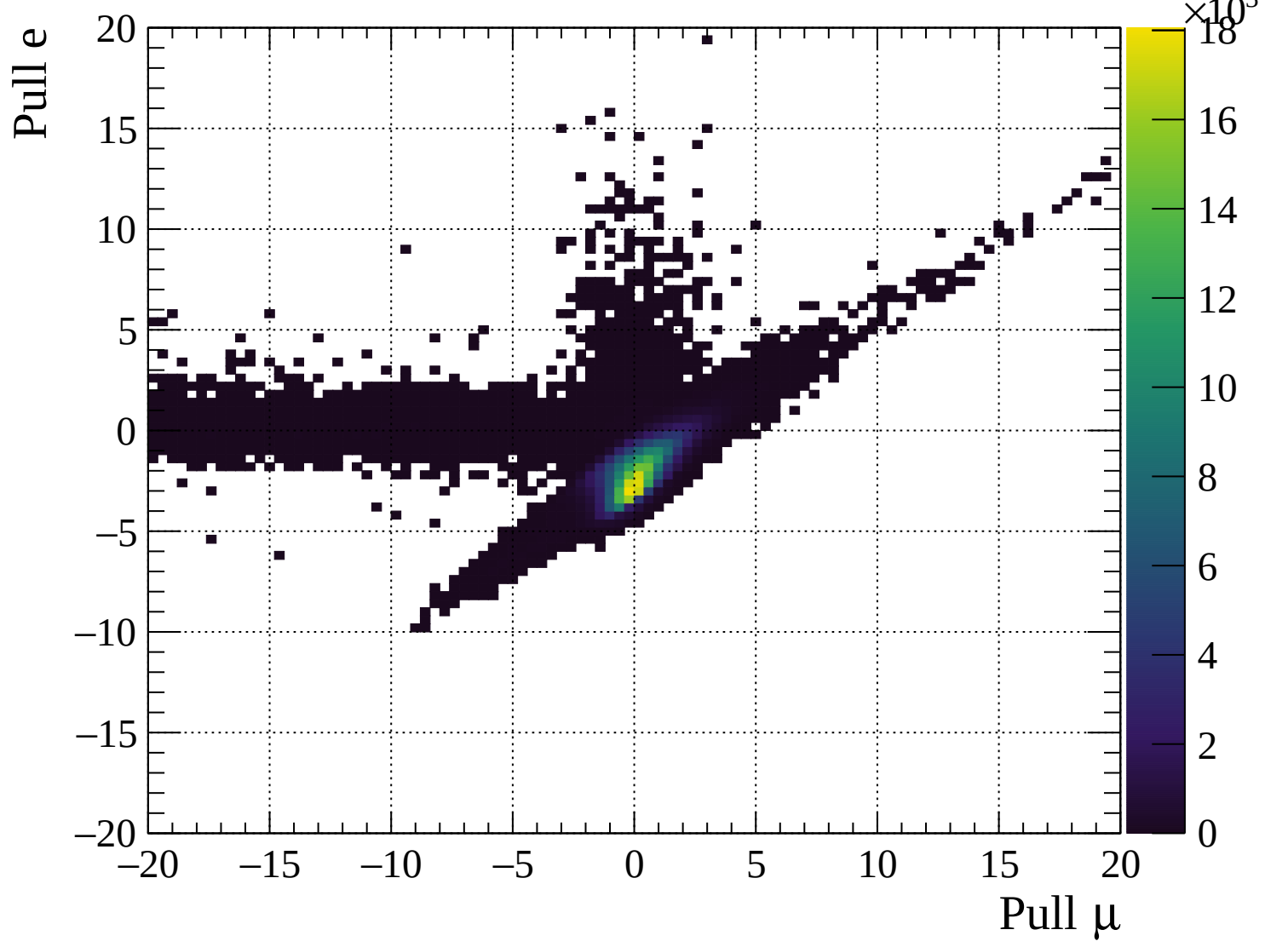


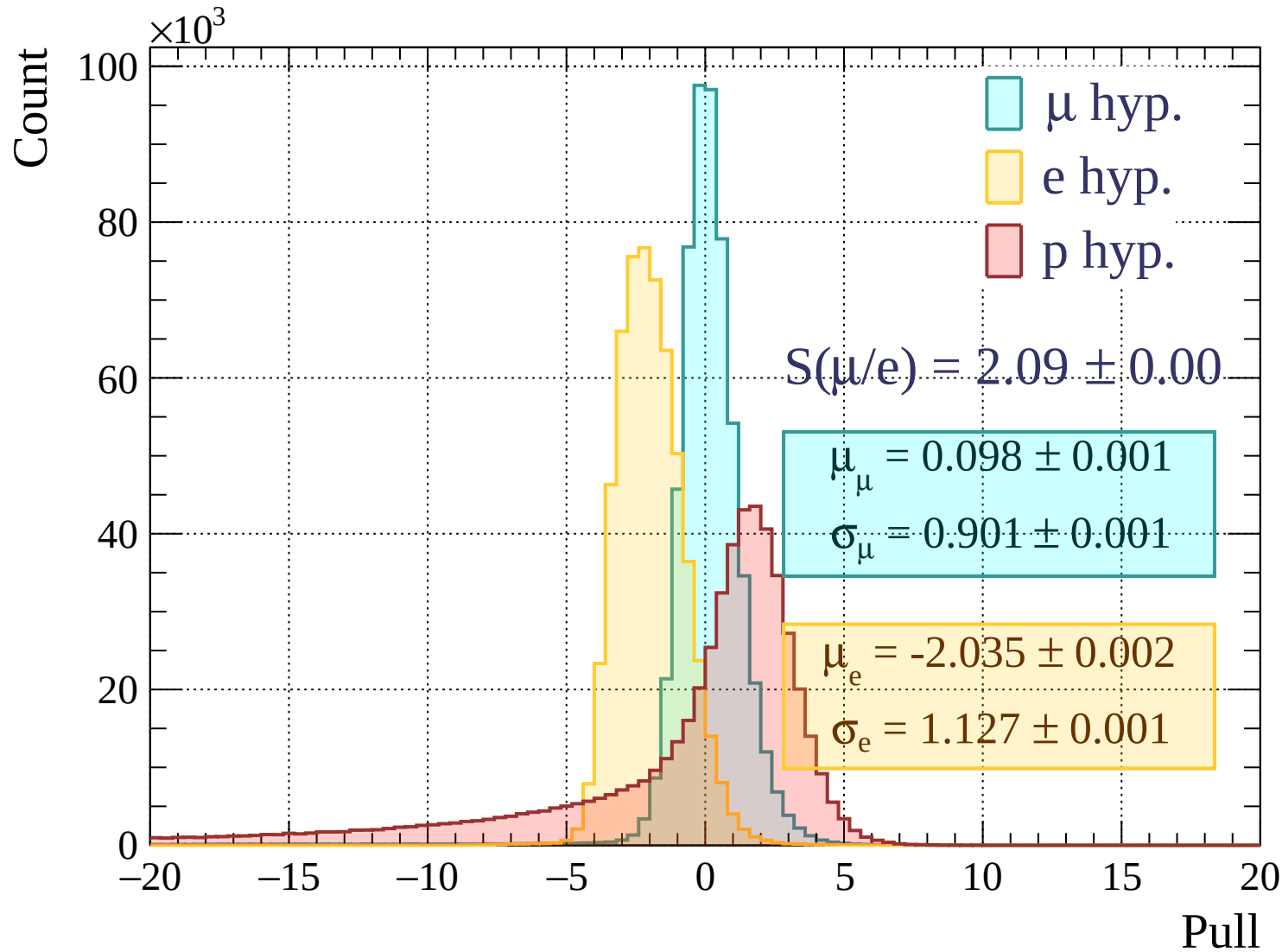
Pulls standard deviation

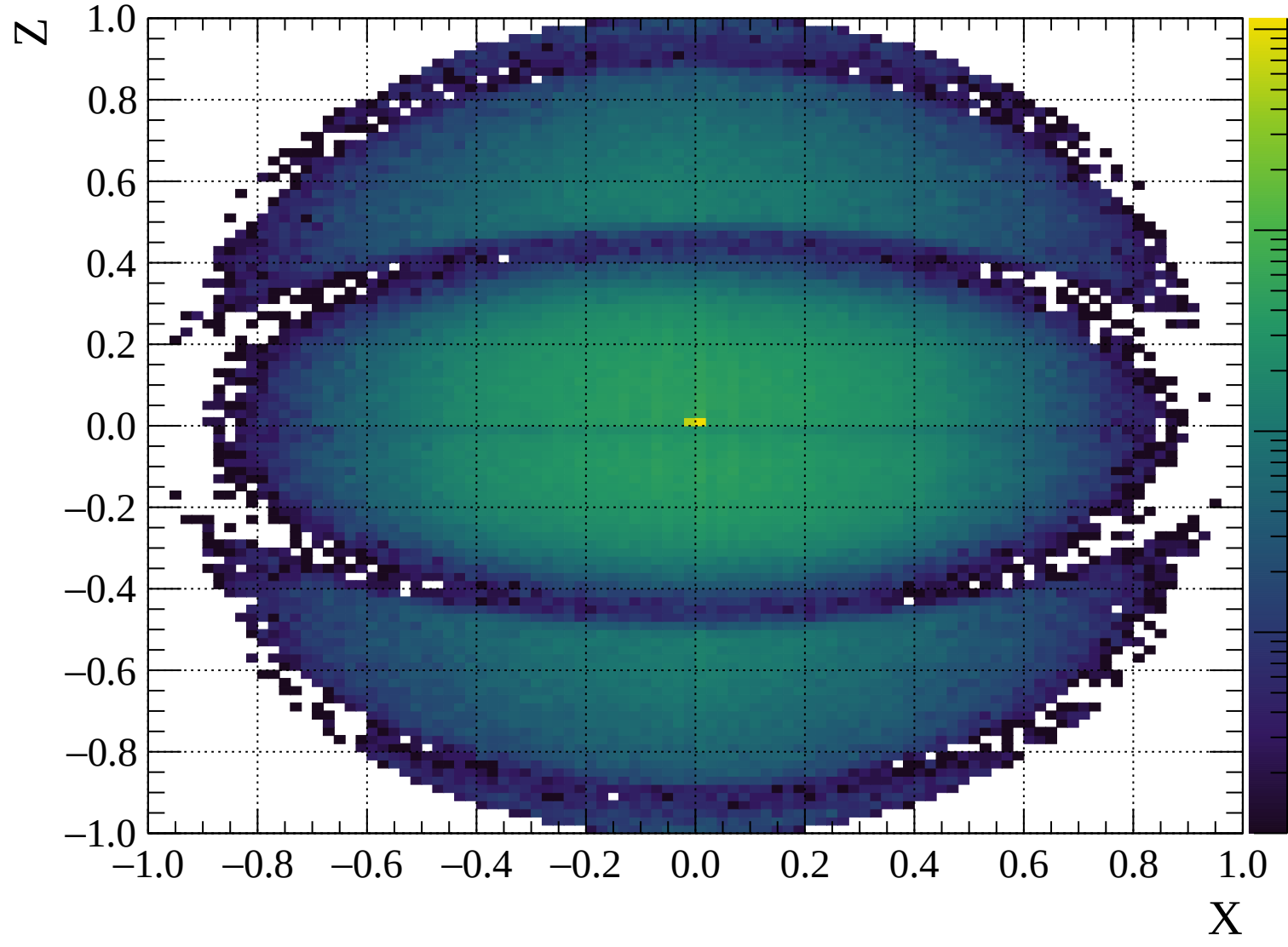


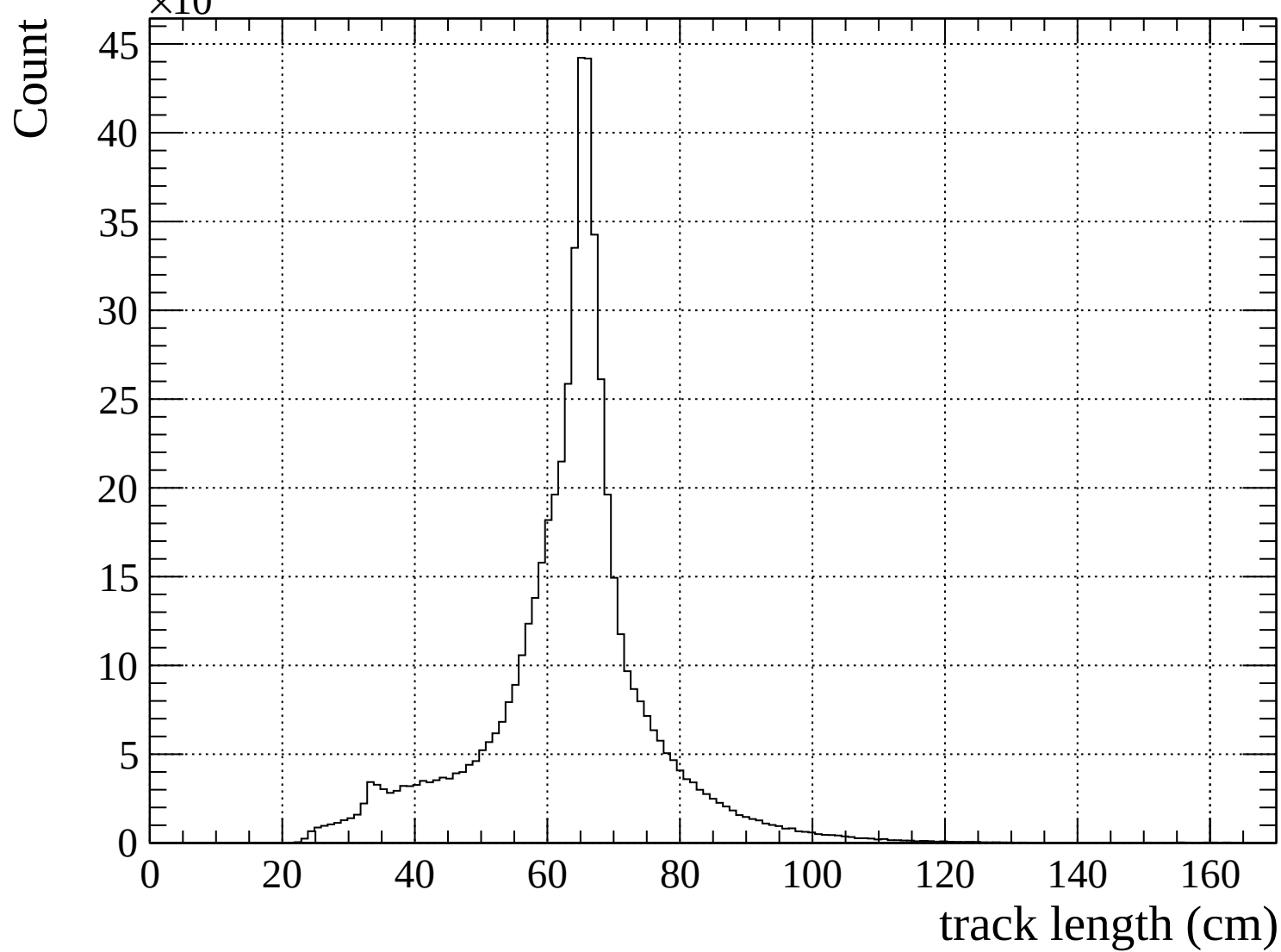
Pulls mean

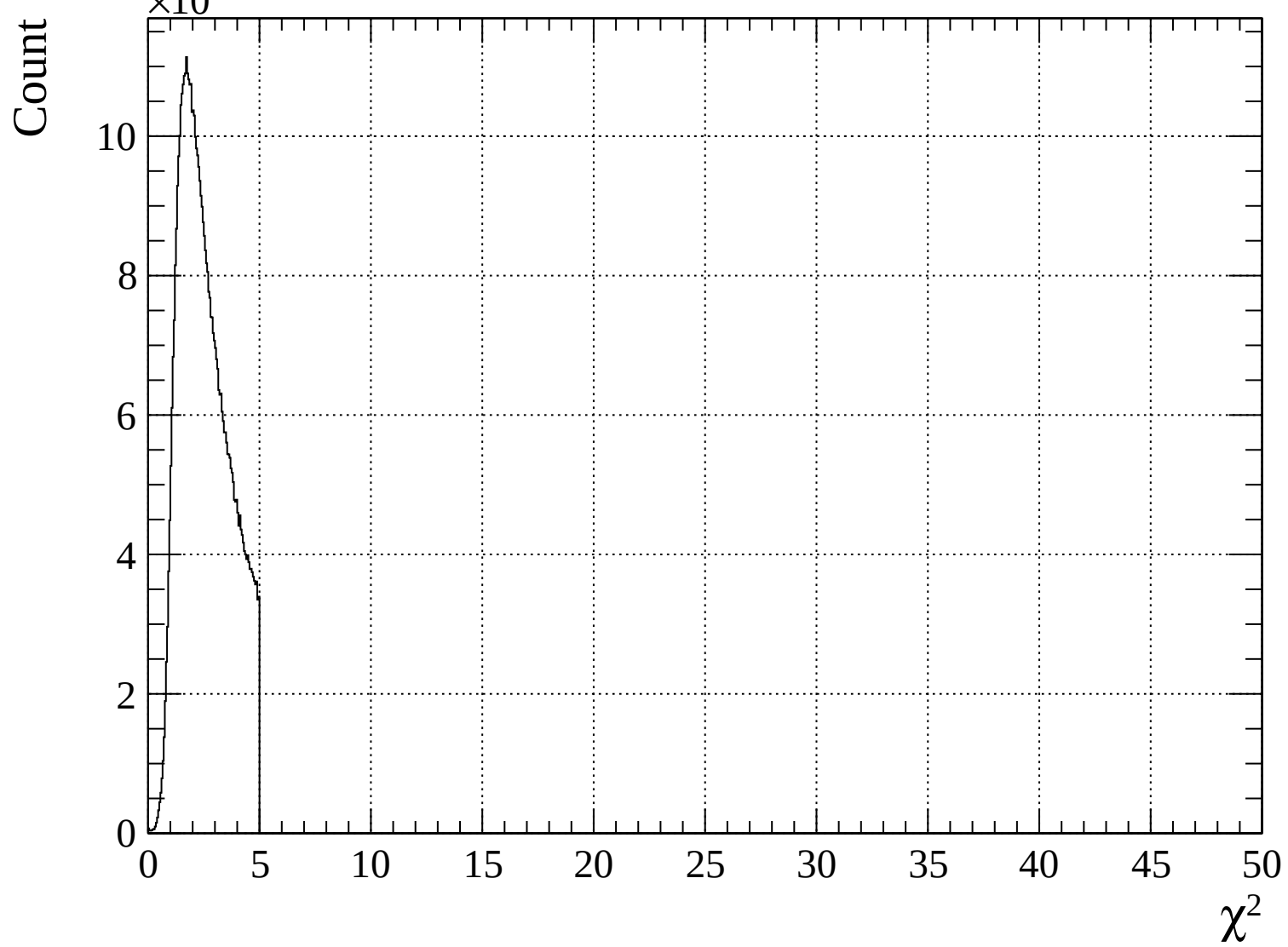


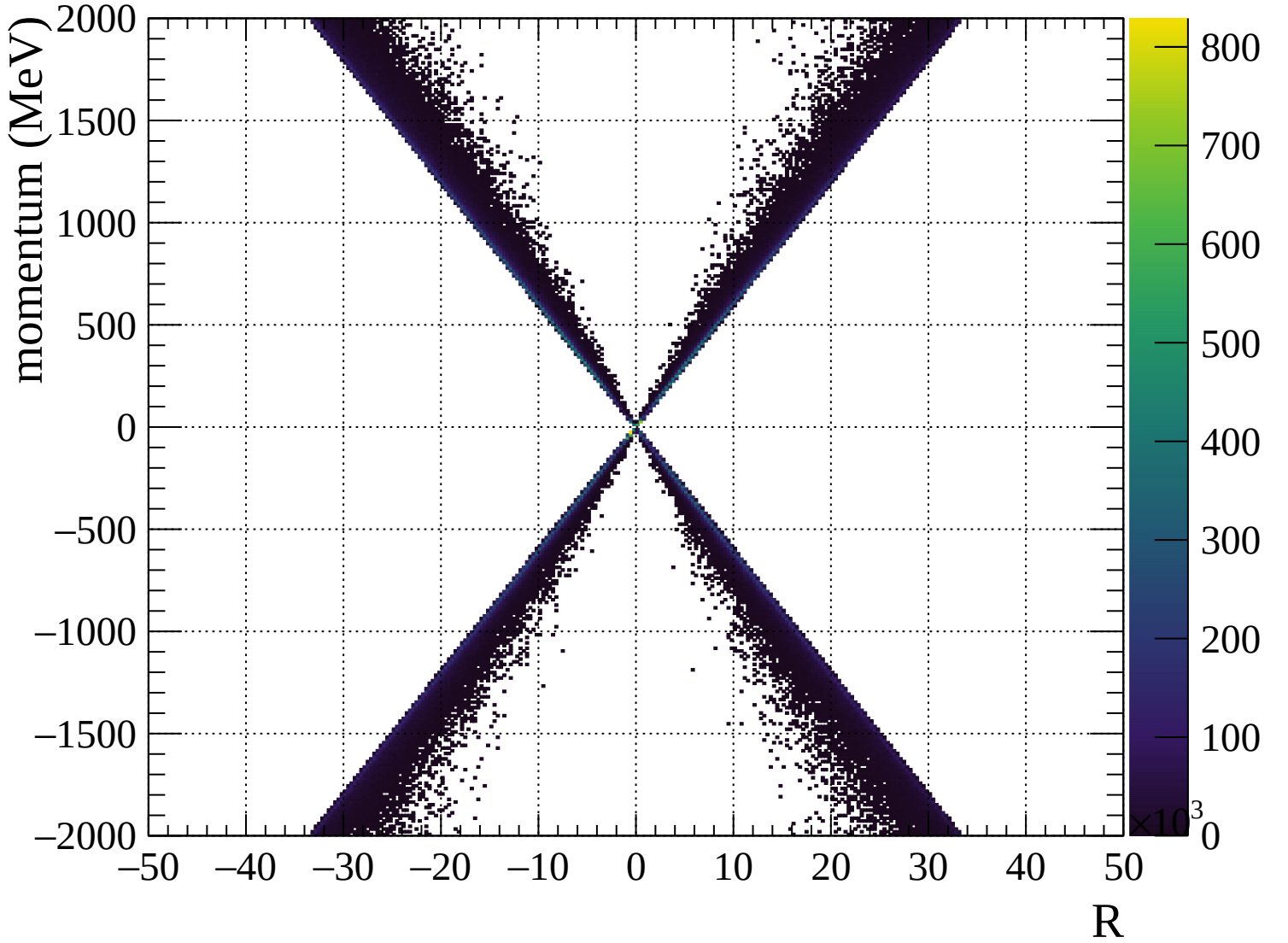


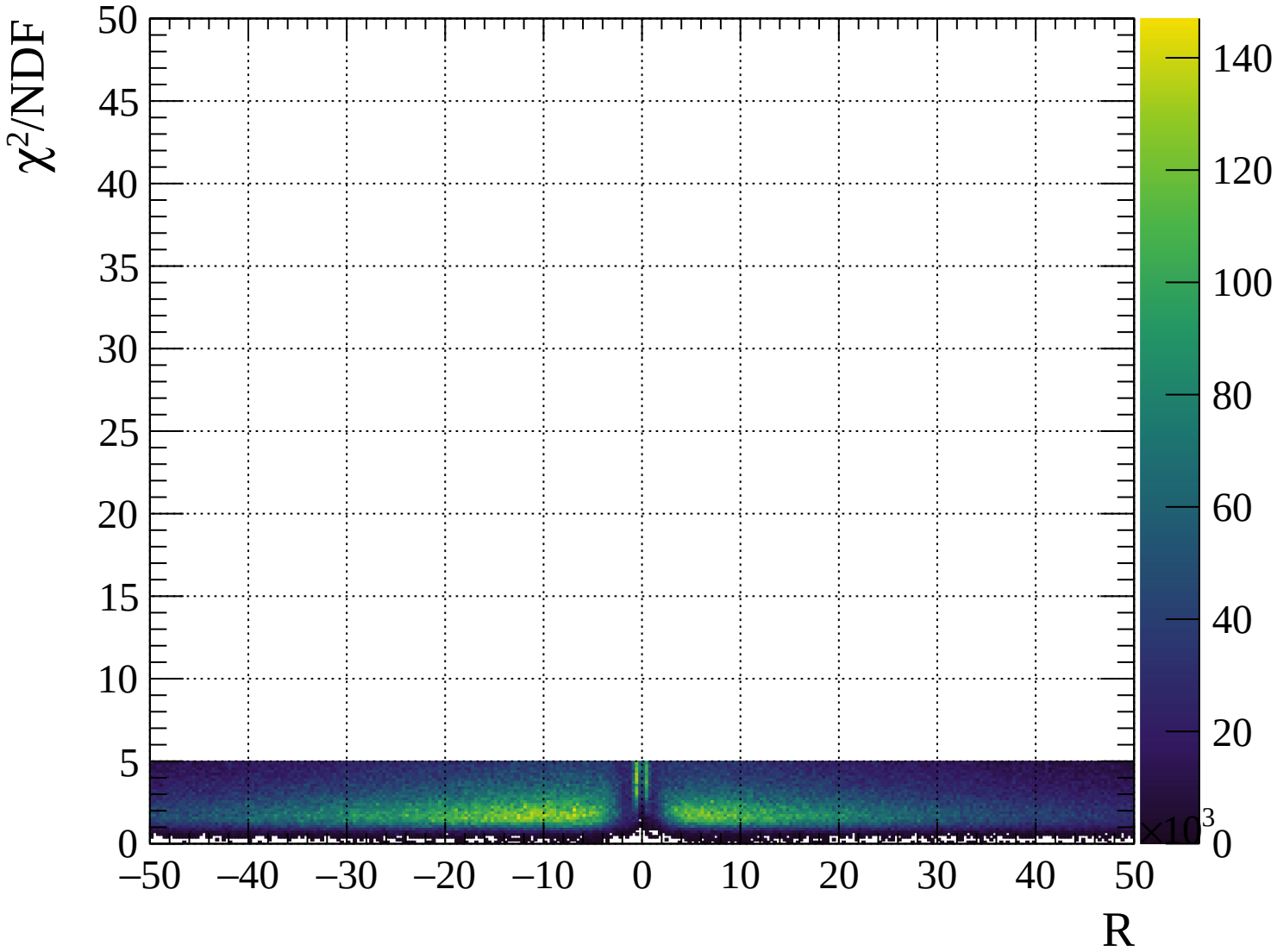






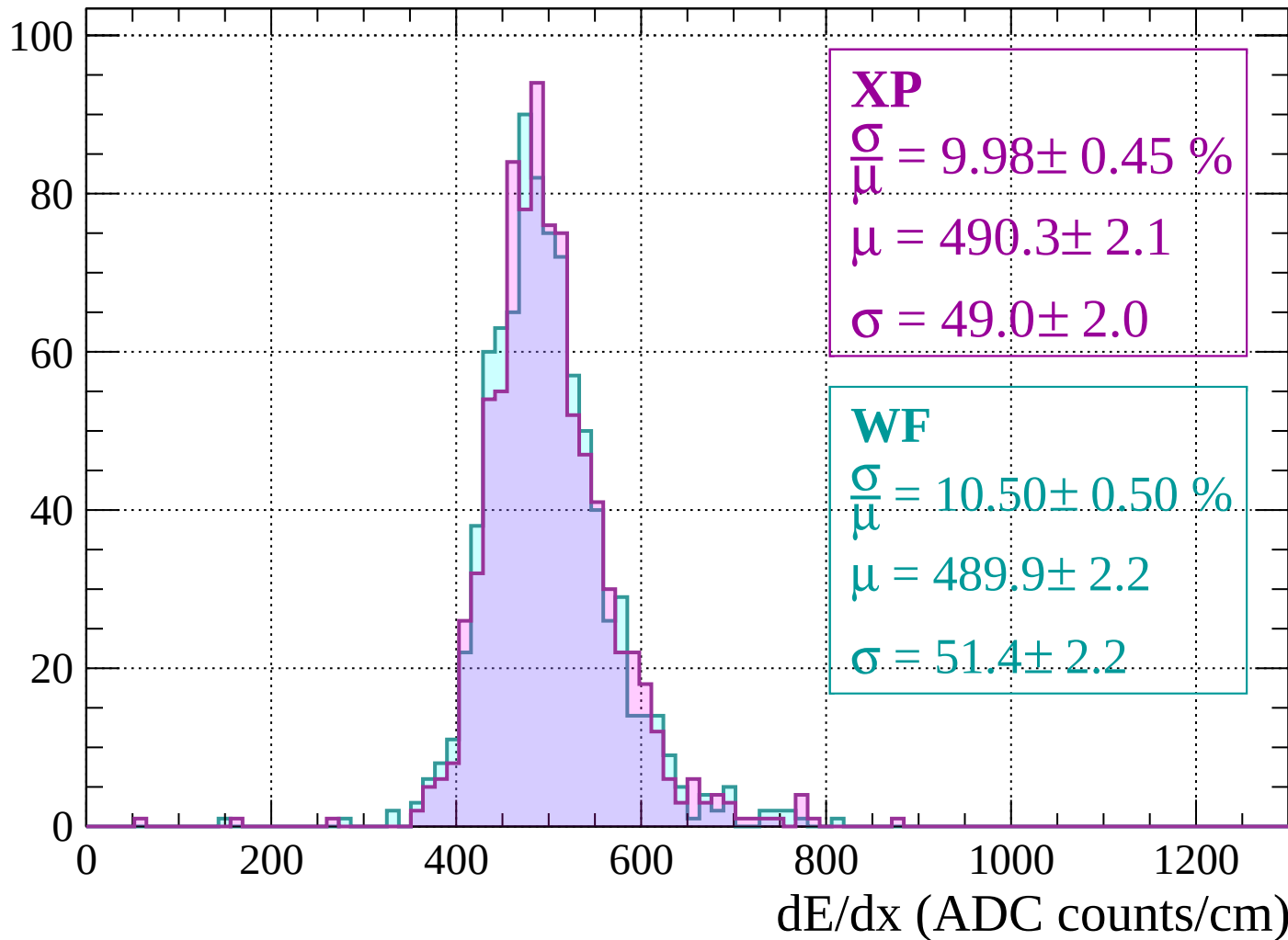






Energy loss | $-2000 < p < -1960$

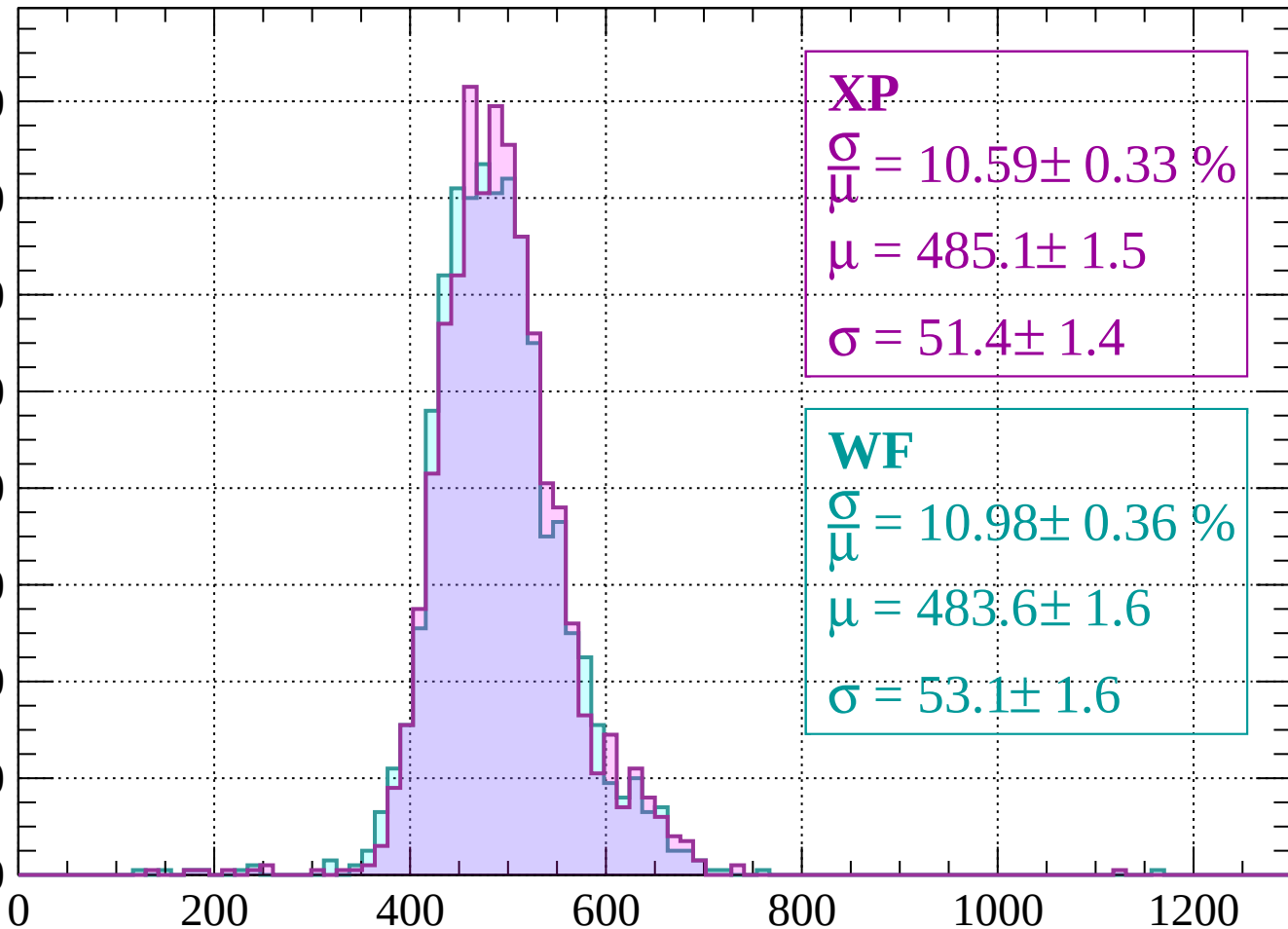
Count



Energy loss | -1960 < p < -1920

Count

160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 10.59 \pm 0.33 \%$$

$$\mu = 485.1 \pm 1.5$$

$$\sigma = 51.4 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 10.98 \pm 0.36 \%$$

$$\mu = 483.6 \pm 1.6$$

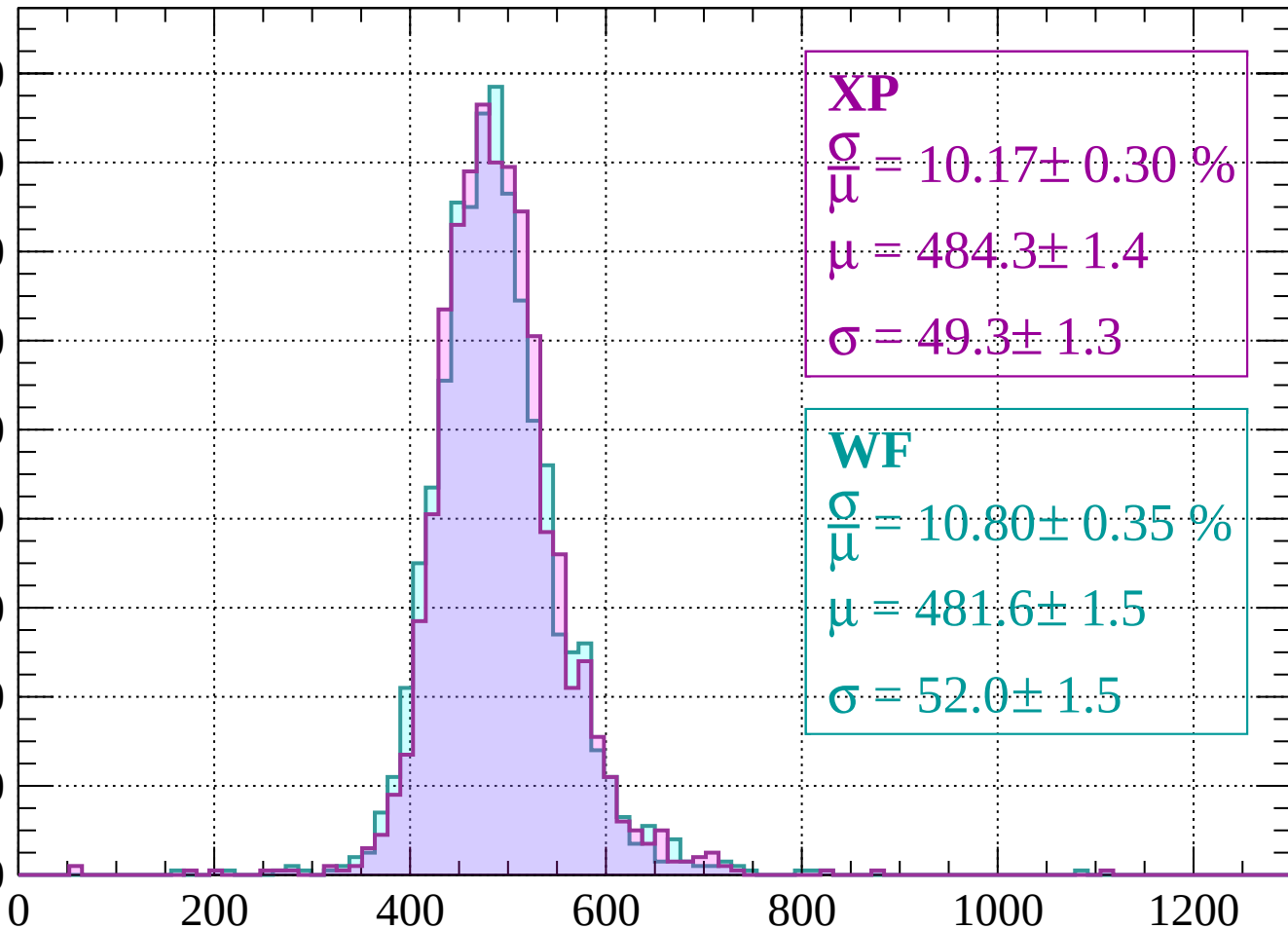
$$\sigma = 53.1 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1880

Count

180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\bar{\mu}} = 10.17 \pm 0.30 \%$$

$$\mu = 484.3 \pm 1.4$$

$$\sigma = 49.3 \pm 1.3$$

WF

$$\frac{\sigma}{\bar{\mu}} = 10.80 \pm 0.35 \%$$

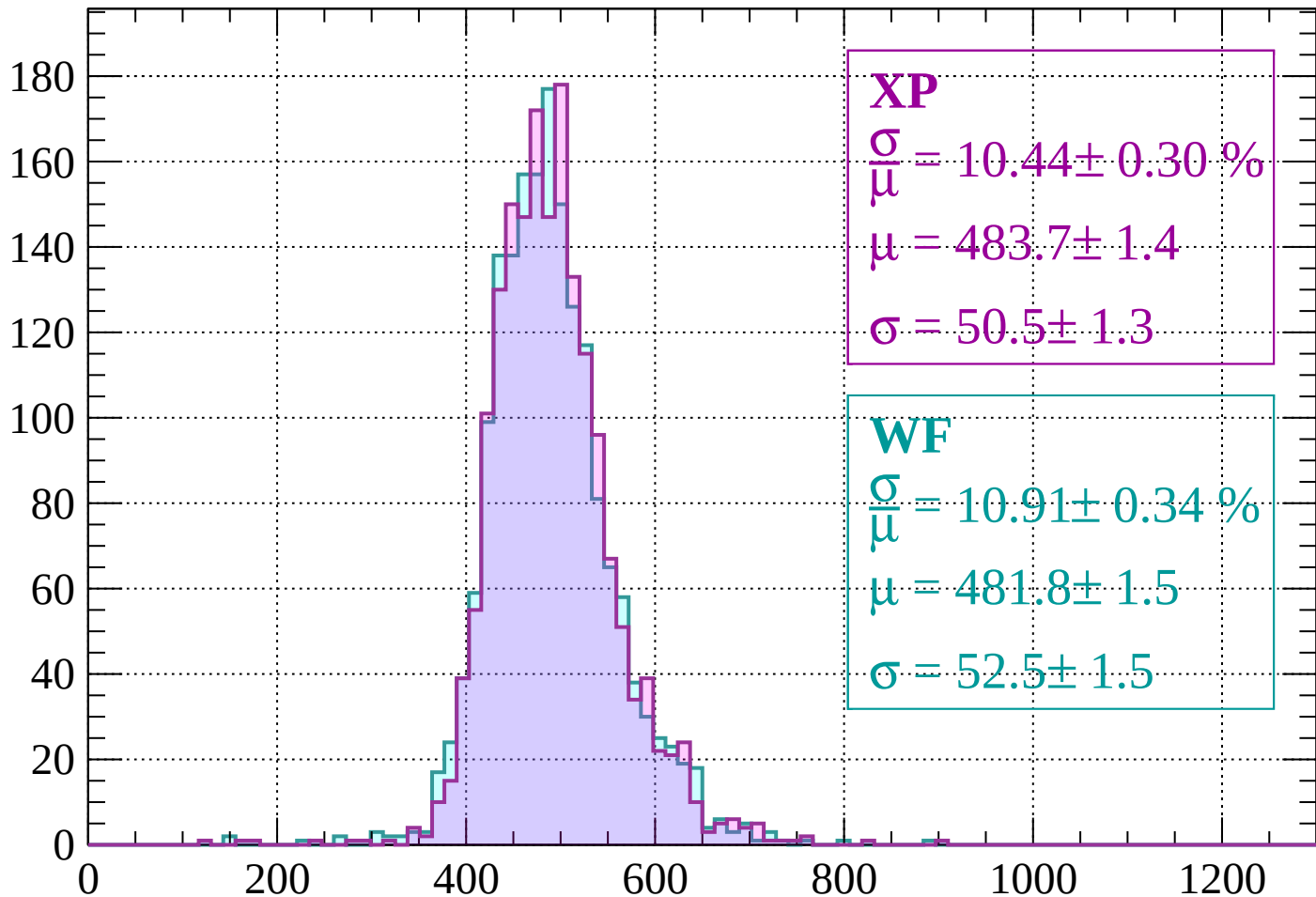
$$\mu = 481.6 \pm 1.5$$

$$\sigma = 52.0 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | -1880 < p < -1840

Count



XP

$$\frac{\sigma}{\mu} = 10.44 \pm 0.30 \%$$

$$\mu = 483.7 \pm 1.4$$

$$\sigma = 50.5 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.91 \pm 0.34 \%$$

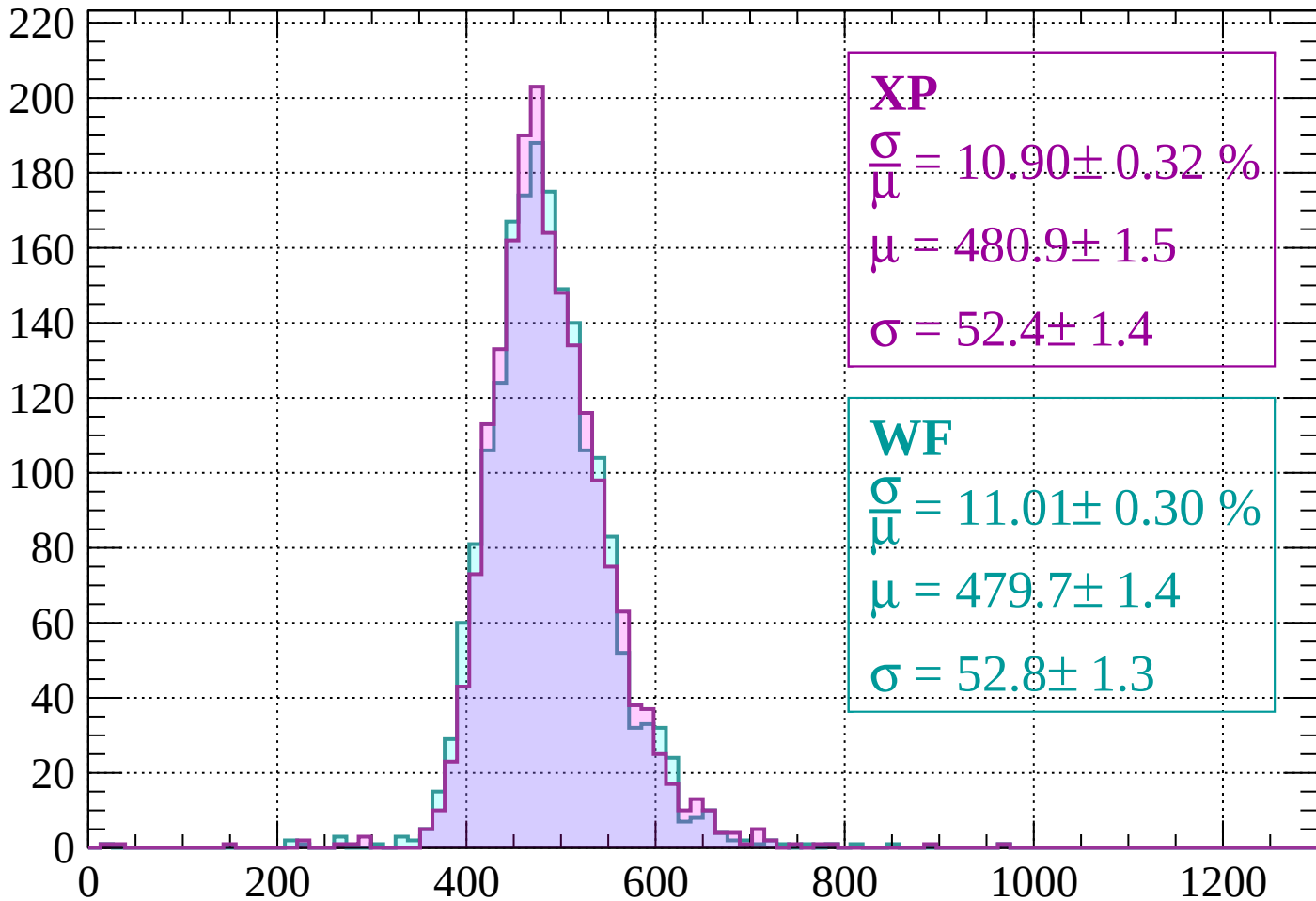
$$\mu = 481.8 \pm 1.5$$

$$\sigma = 52.5 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1840 < p < -1800$

Count



XP

$$\frac{\sigma}{\mu} = 10.90 \pm 0.32 \%$$

$$\mu = 480.9 \pm 1.5$$

$$\sigma = 52.4 \pm 1.4$$

WF

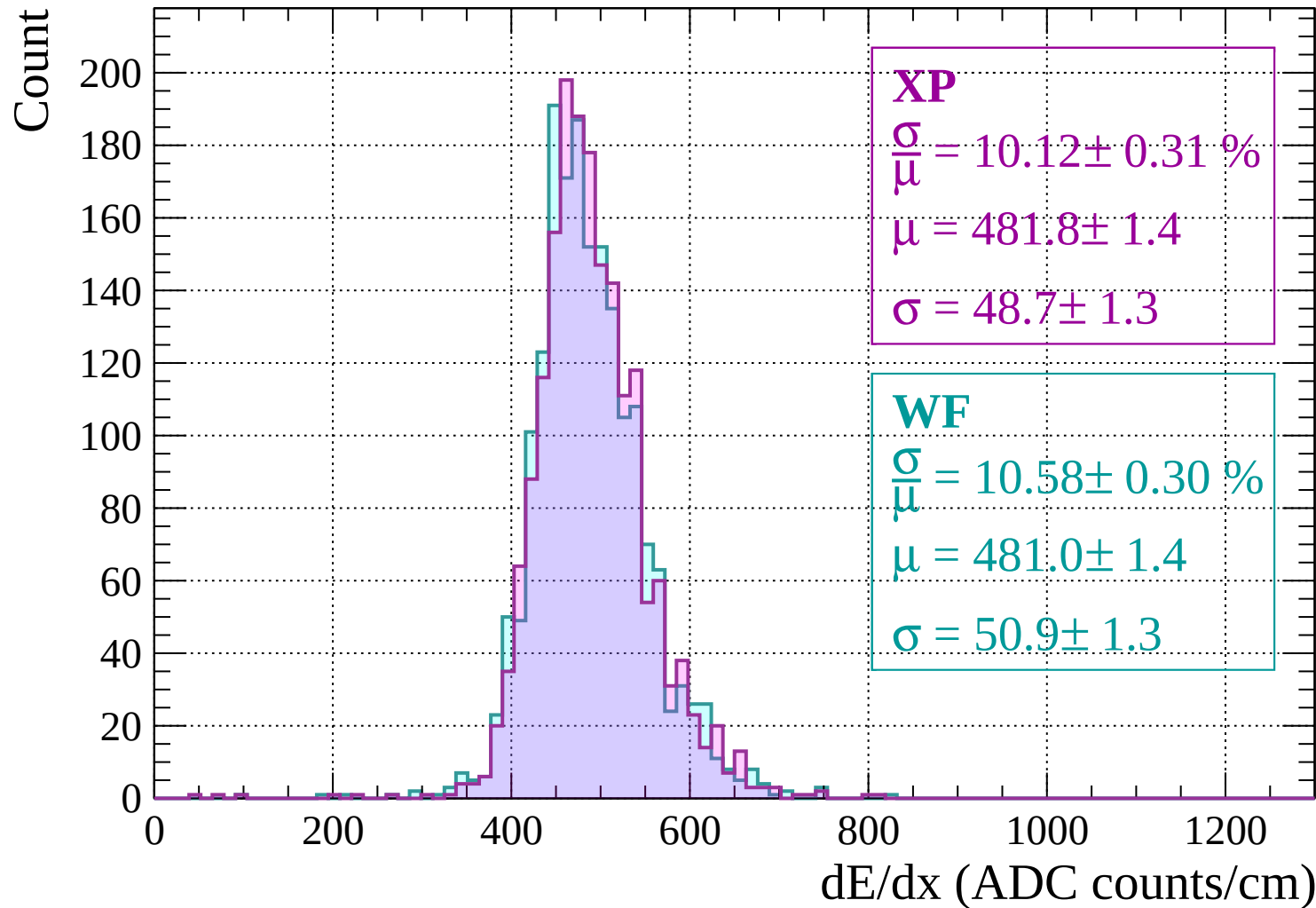
$$\frac{\sigma}{\mu} = 11.01 \pm 0.30 \%$$

$$\mu = 479.7 \pm 1.4$$

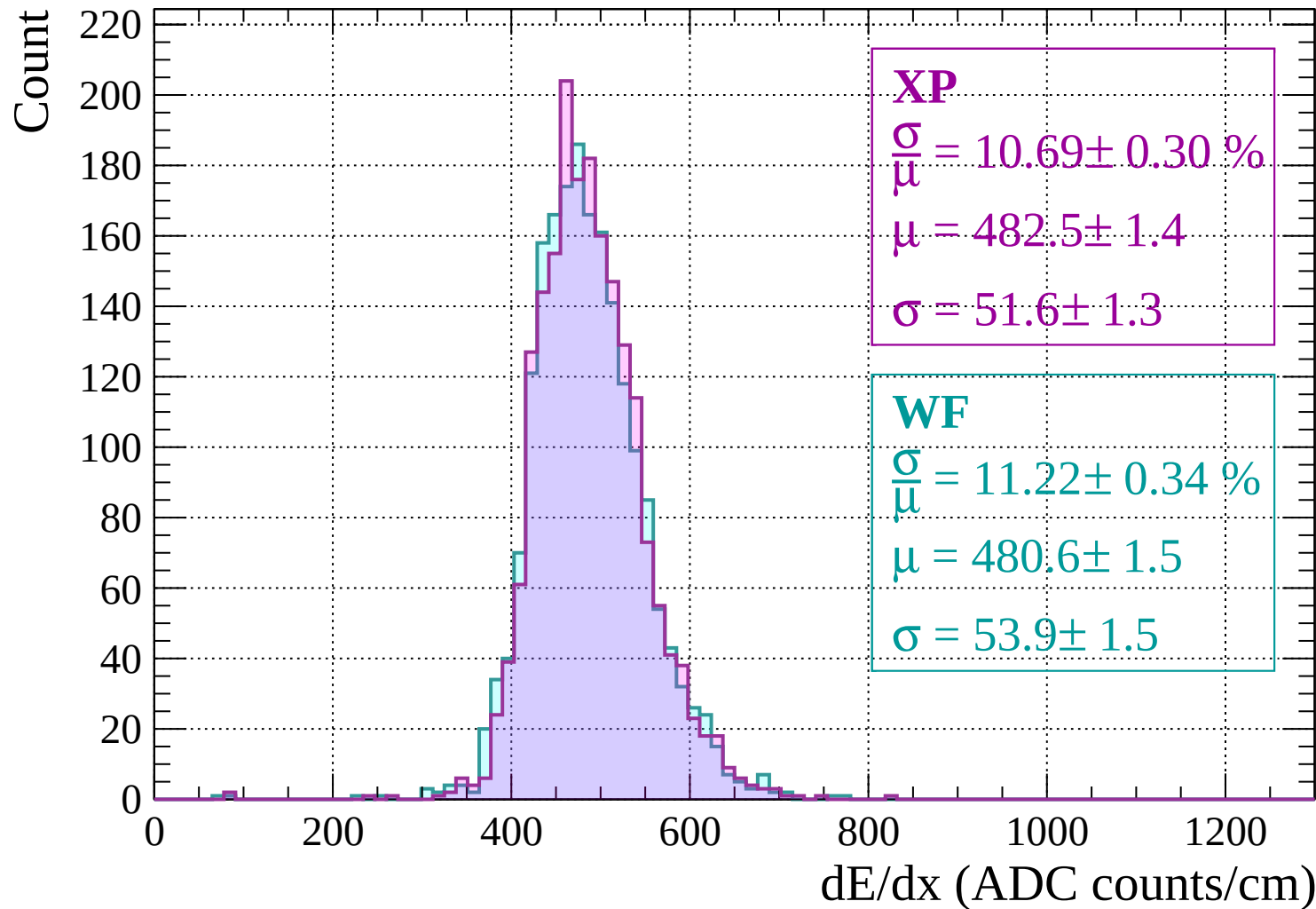
$$\sigma = 52.8 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | -1800 < p < -1760

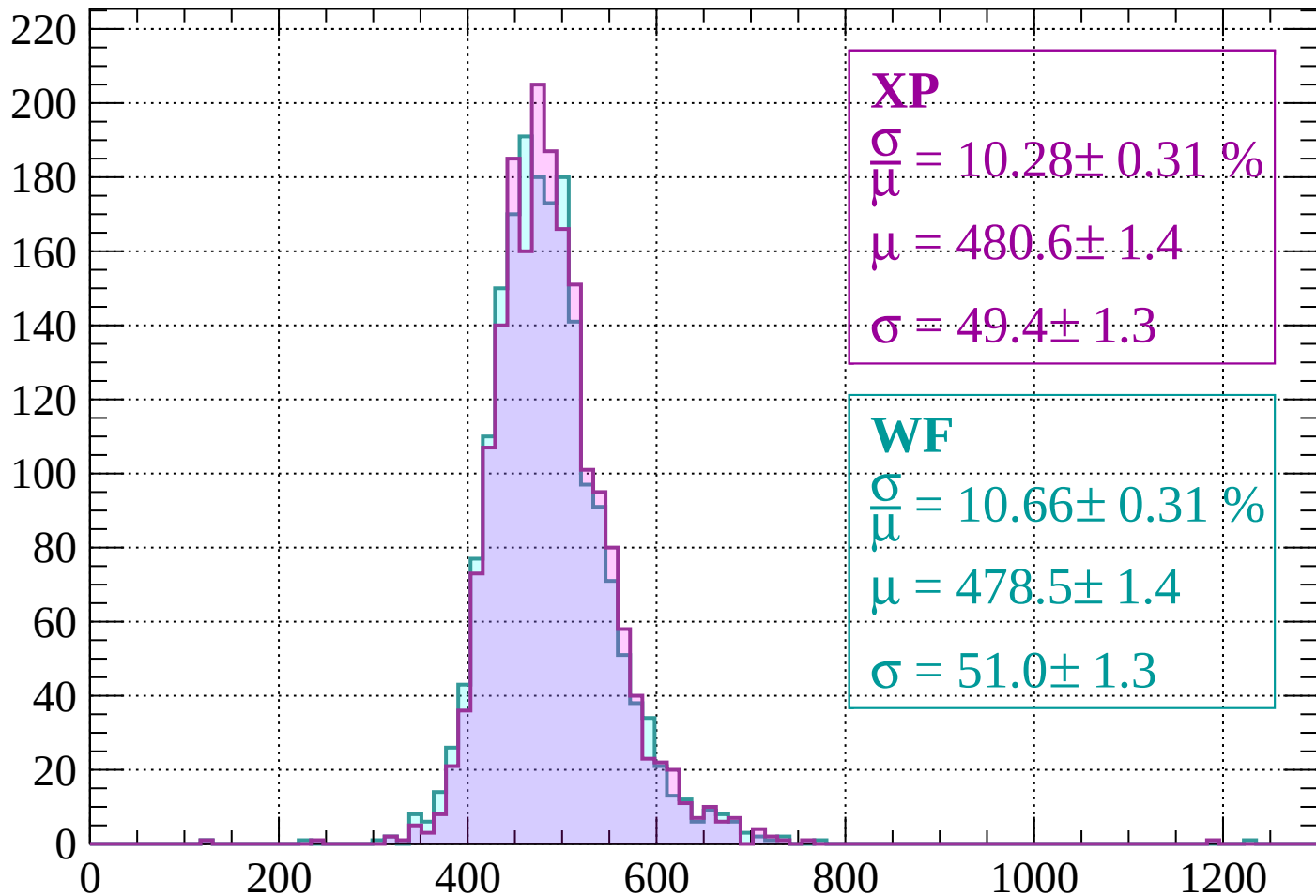


Energy loss | -1760 < p < -1720



Energy loss | -1720 < p < -1680

Count



XP

$$\frac{\sigma}{\mu} = 10.28 \pm 0.31 \%$$

$$\mu = 480.6 \pm 1.4$$

$$\sigma = 49.4 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.66 \pm 0.31 \%$$

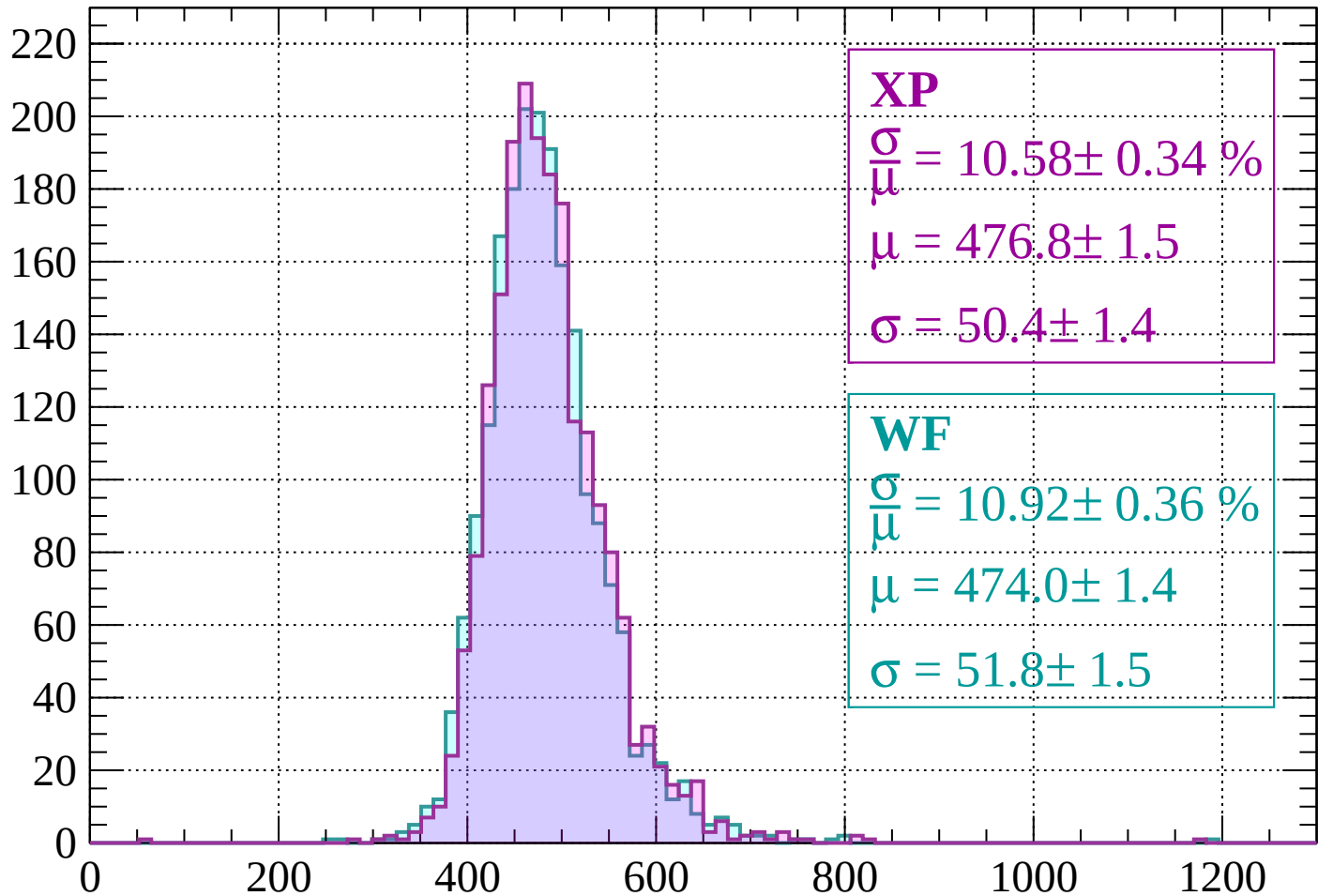
$$\mu = 478.5 \pm 1.4$$

$$\sigma = 51.0 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1640$

Count



XP

$$\frac{\sigma}{\mu} = 10.58 \pm 0.34 \%$$

$$\mu = 476.8 \pm 1.5$$

$$\sigma = 50.4 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 10.92 \pm 0.36 \%$$

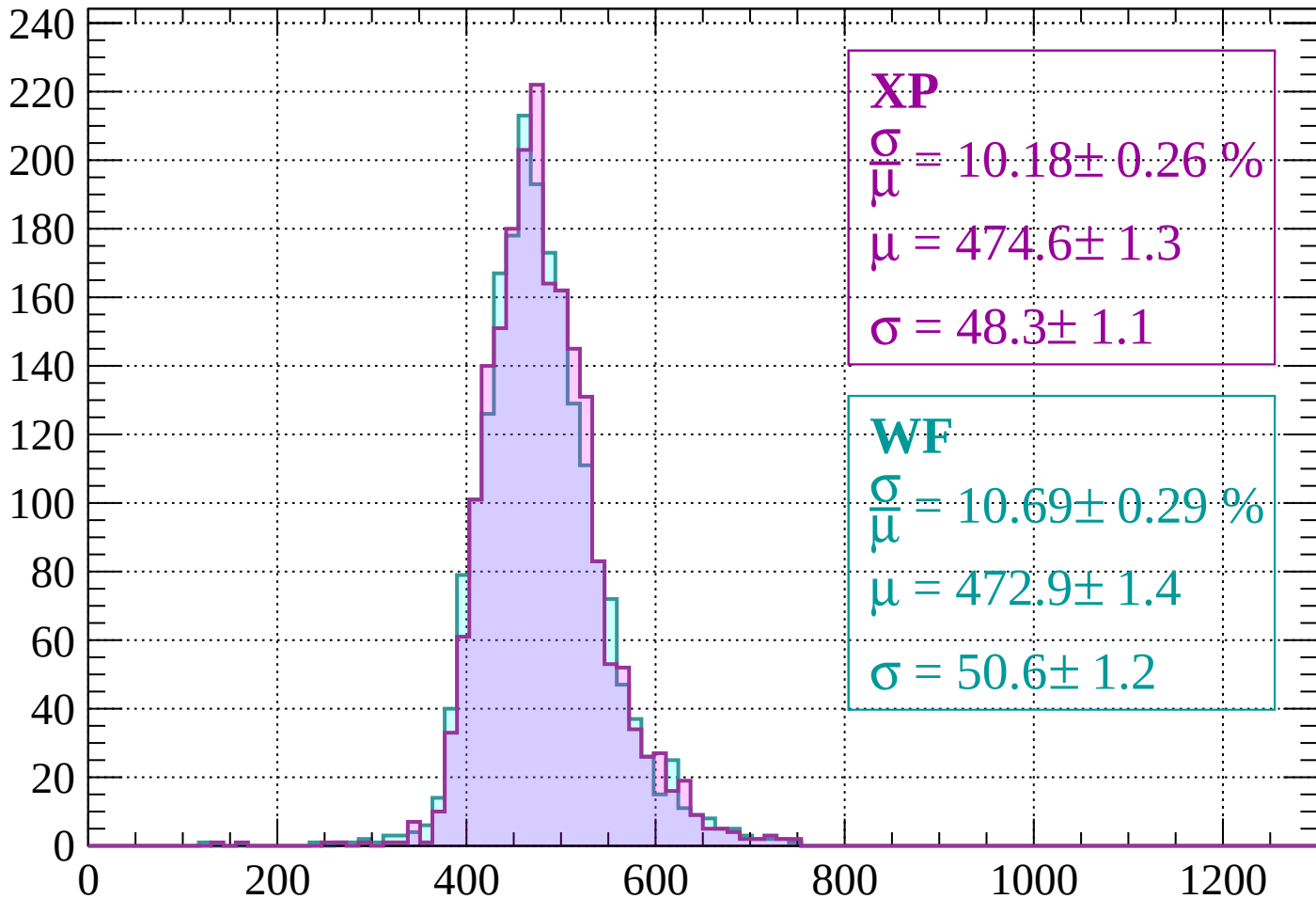
$$\mu = 474.0 \pm 1.4$$

$$\sigma = 51.8 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1640 < p < -1600$

Count



XP

$$\frac{\sigma}{\mu} = 10.18 \pm 0.26 \%$$

$$\mu = 474.6 \pm 1.3$$

$$\sigma = 48.3 \pm 1.1$$

WF

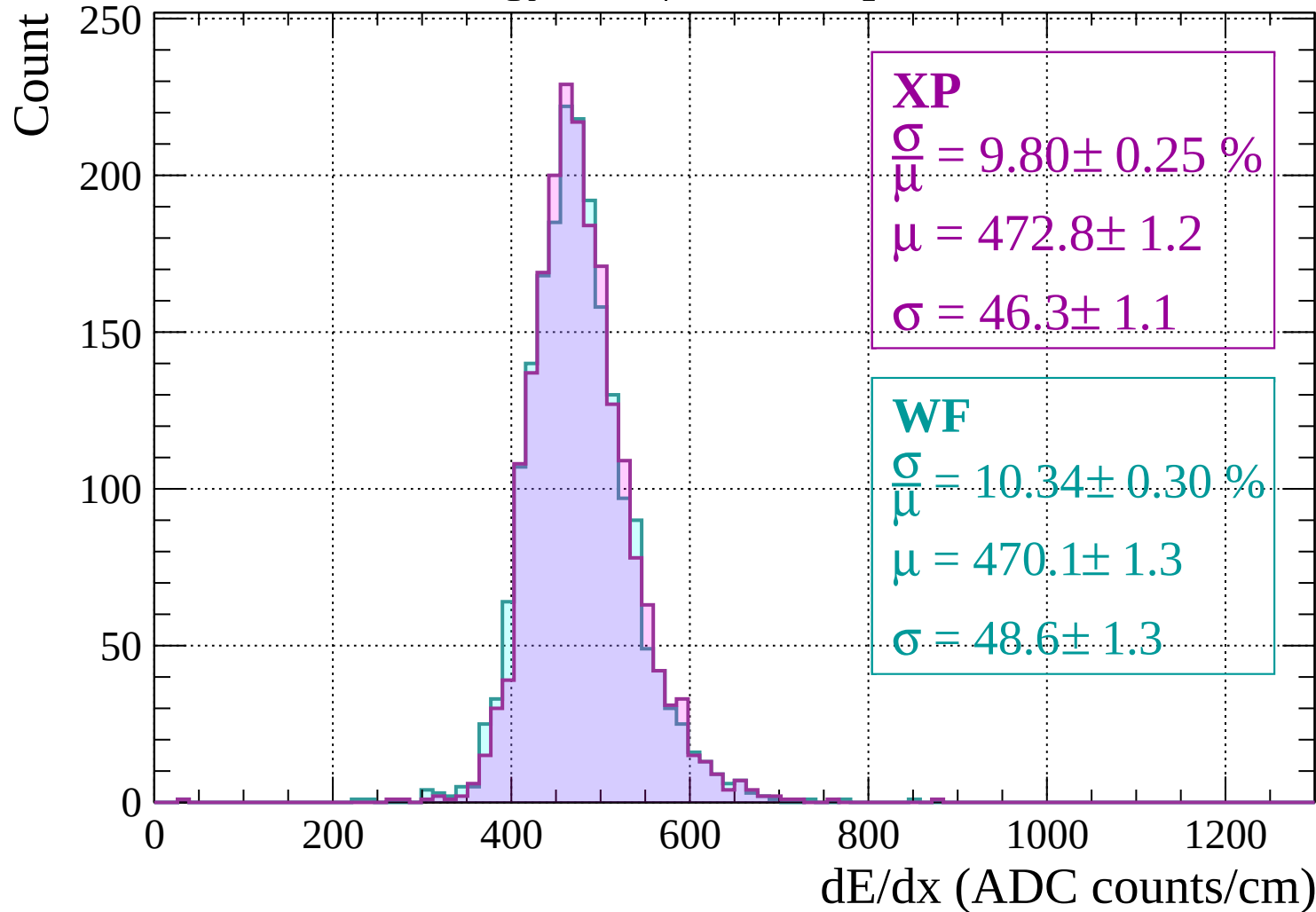
$$\frac{\sigma}{\mu} = 10.69 \pm 0.29 \%$$

$$\mu = 472.9 \pm 1.4$$

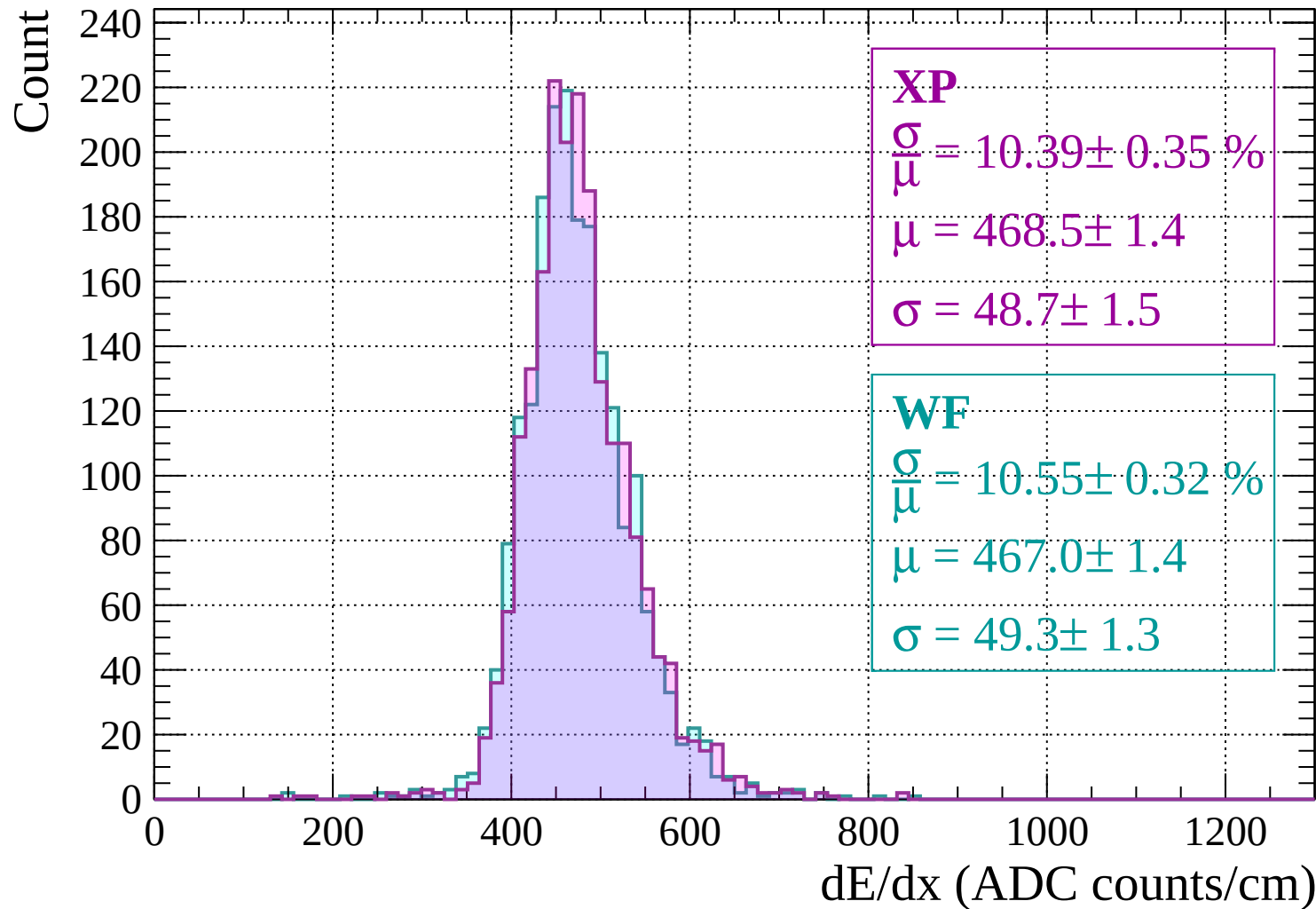
$$\sigma = 50.6 \pm 1.2$$

dE/dx (ADC counts/cm)

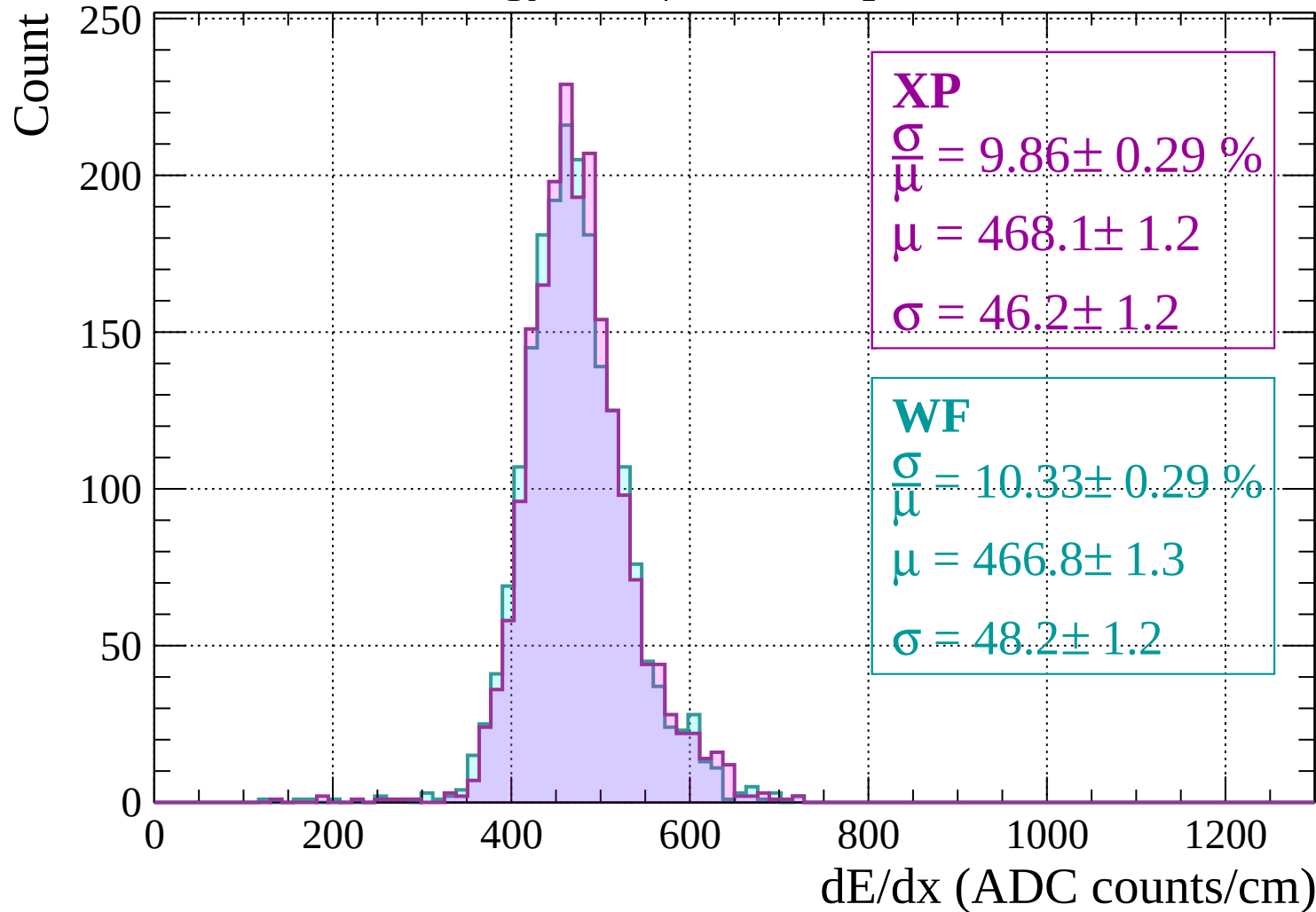
Energy loss | -1600 < p < -1560



Energy loss | -1560 < p < -1520

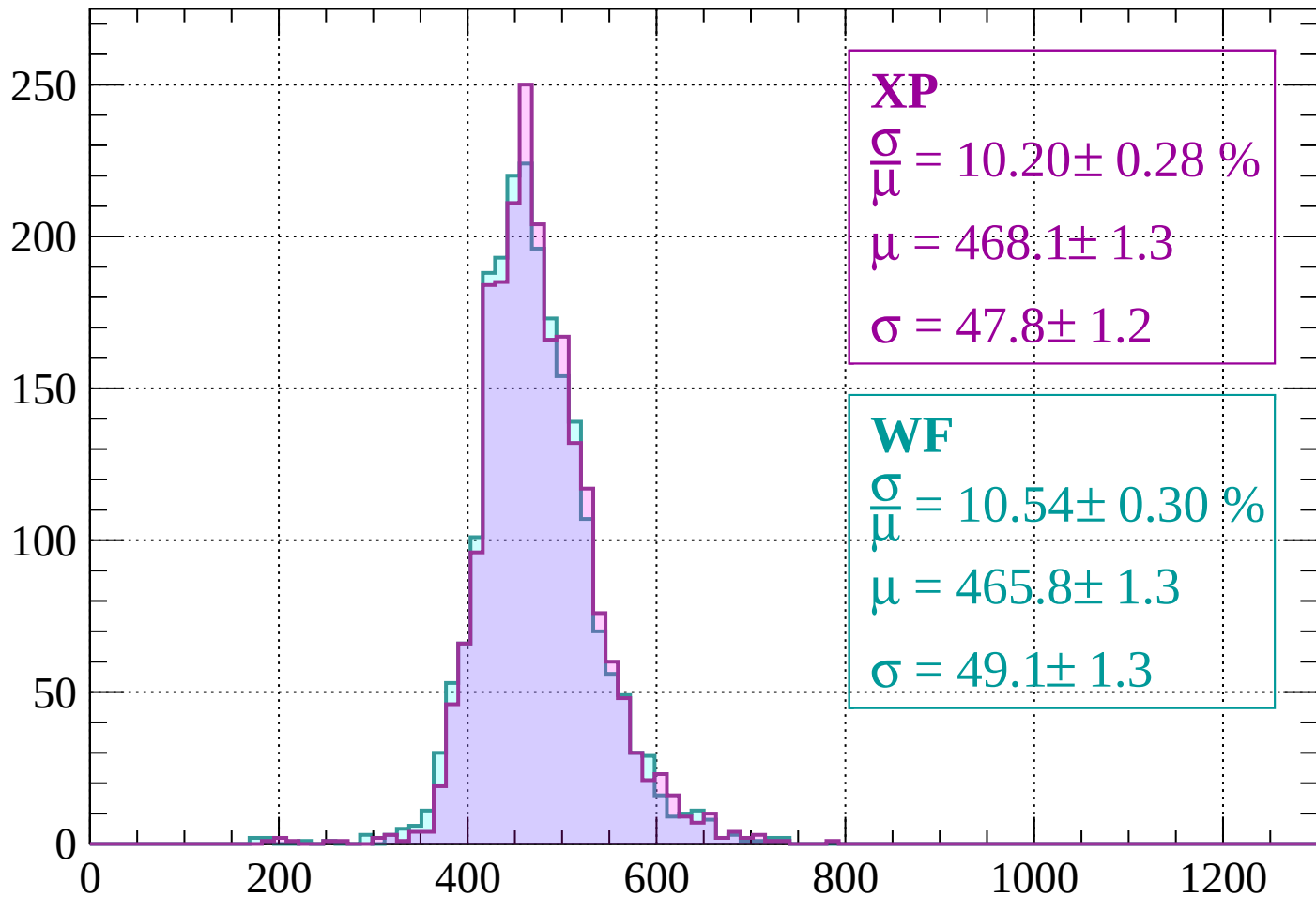


Energy loss | $-1520 < p < -1480$



Energy loss | $-1480 < p < -1440$

Count



XP

$$\frac{\sigma}{\mu} = 10.20 \pm 0.28 \%$$

$$\mu = 468.1 \pm 1.3$$

$$\sigma = 47.8 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.54 \pm 0.30 \%$$

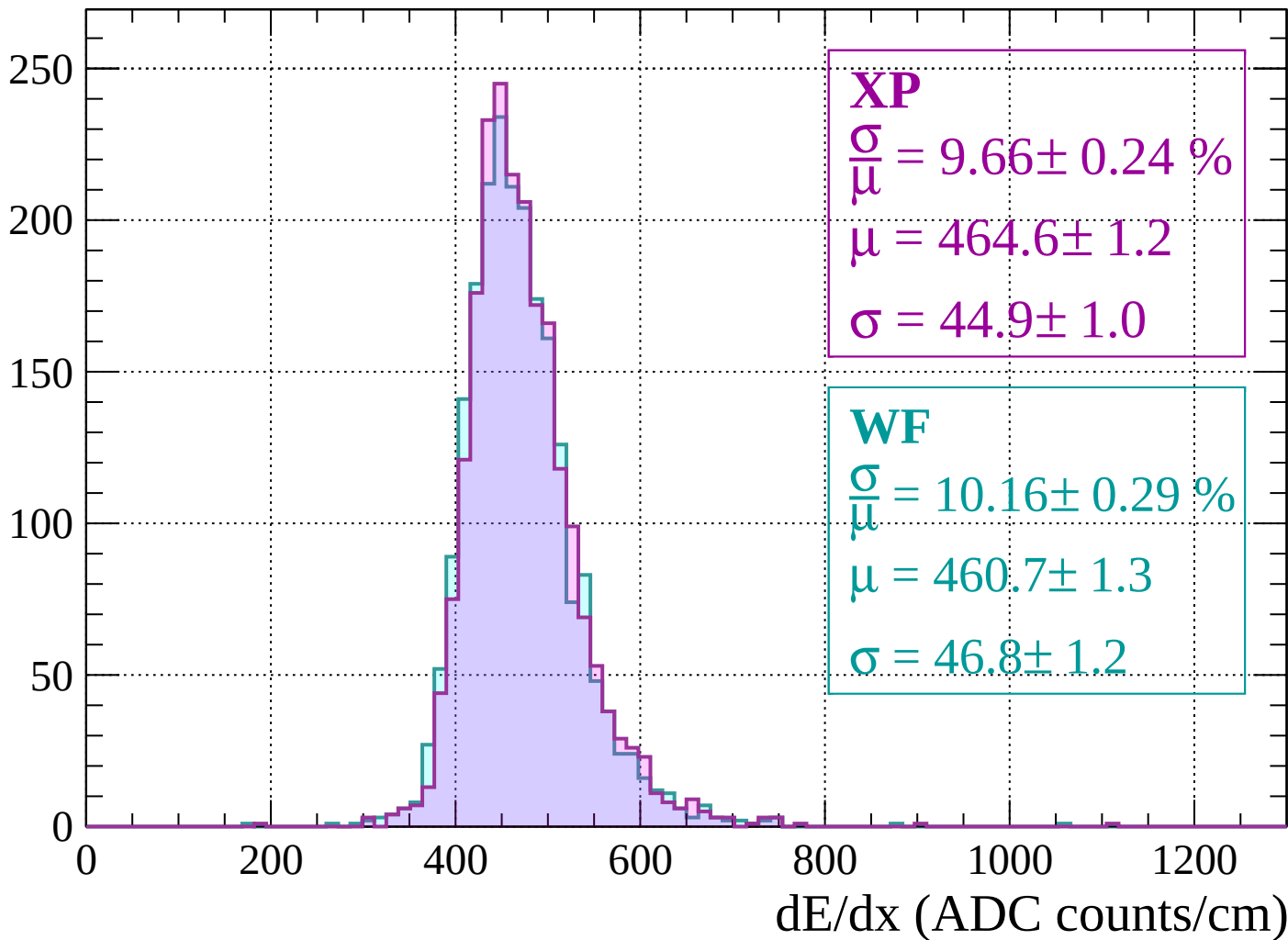
$$\mu = 465.8 \pm 1.3$$

$$\sigma = 49.1 \pm 1.3$$

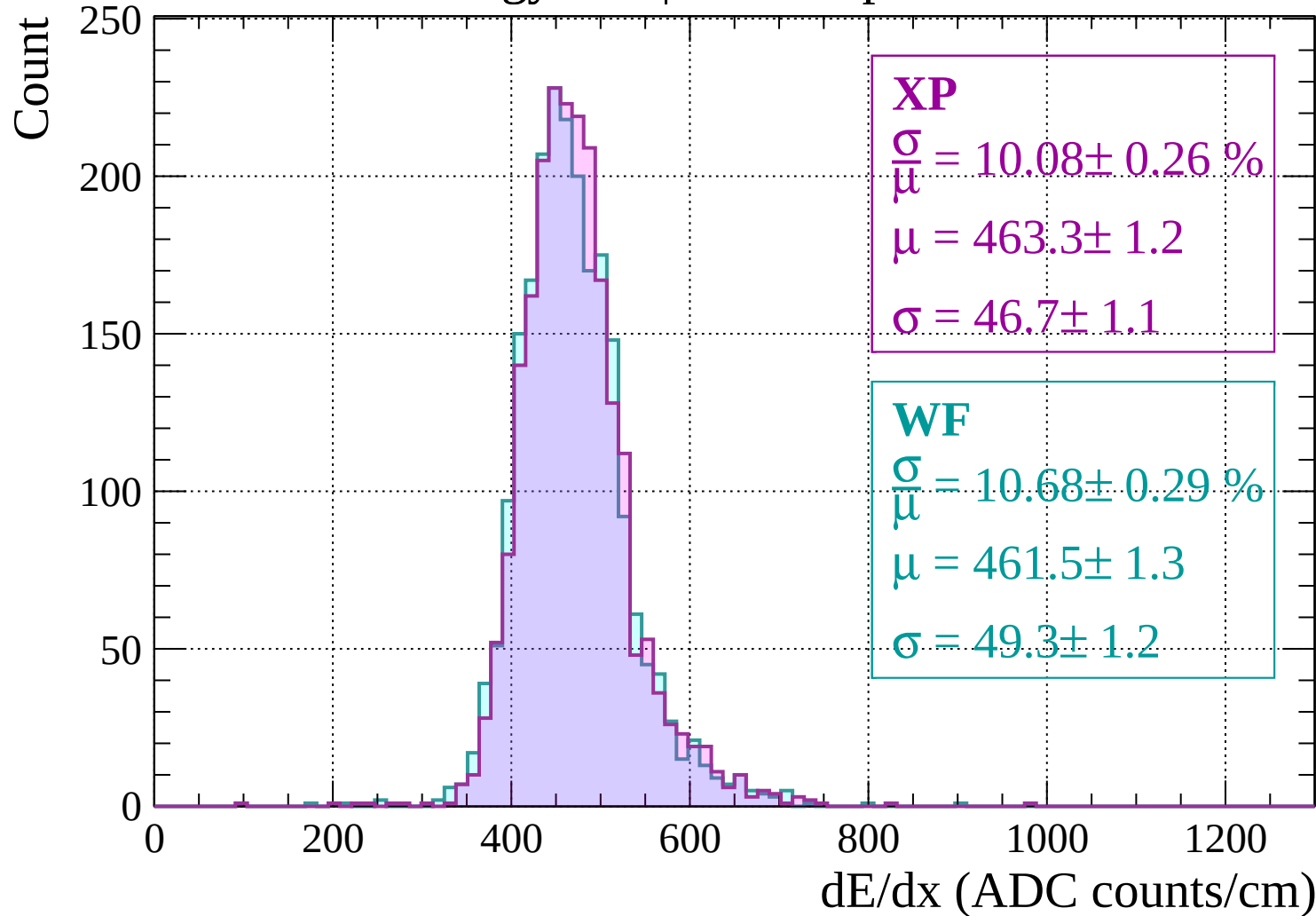
dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1400$

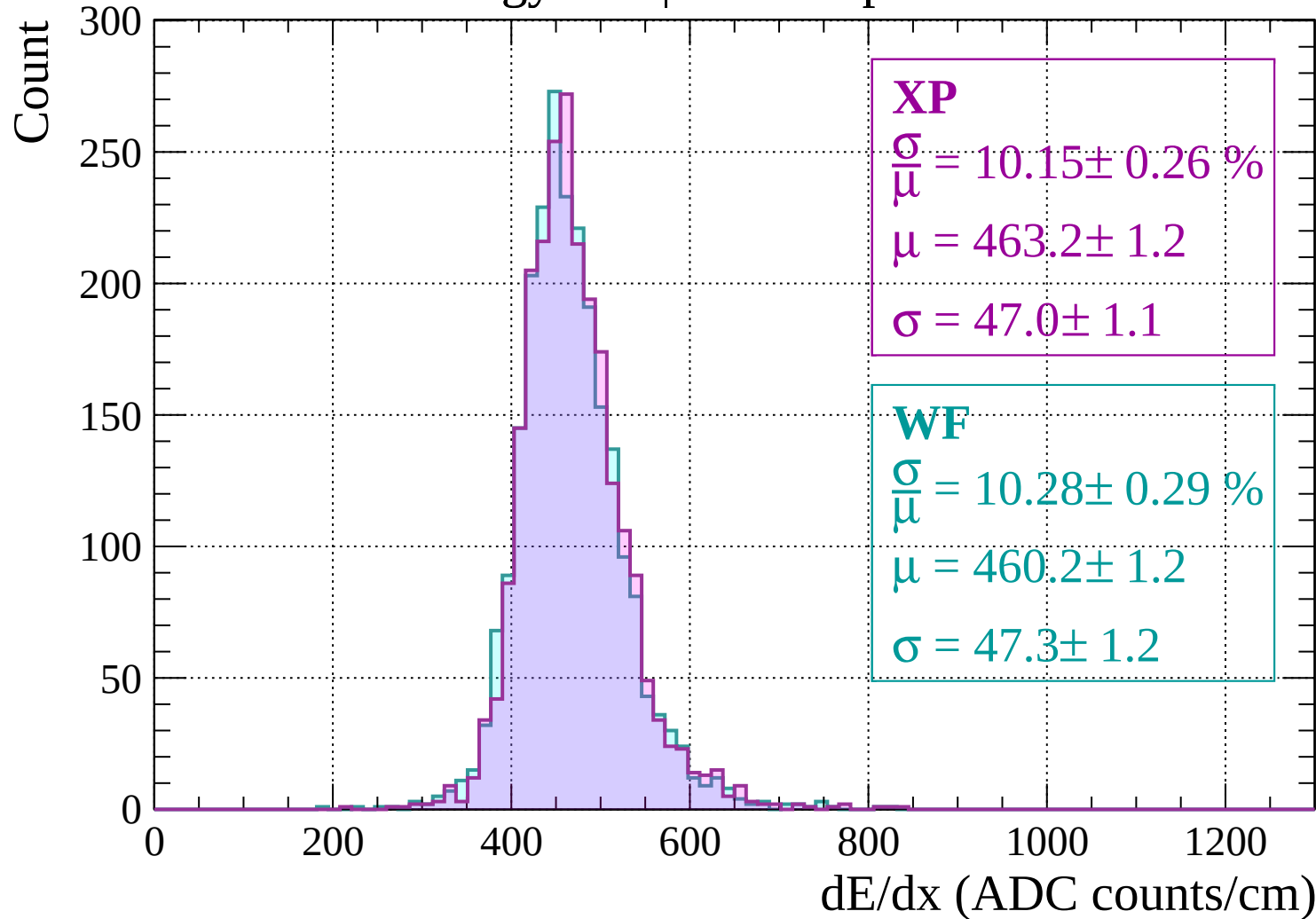
Count



Energy loss | -1400 < p < -1360

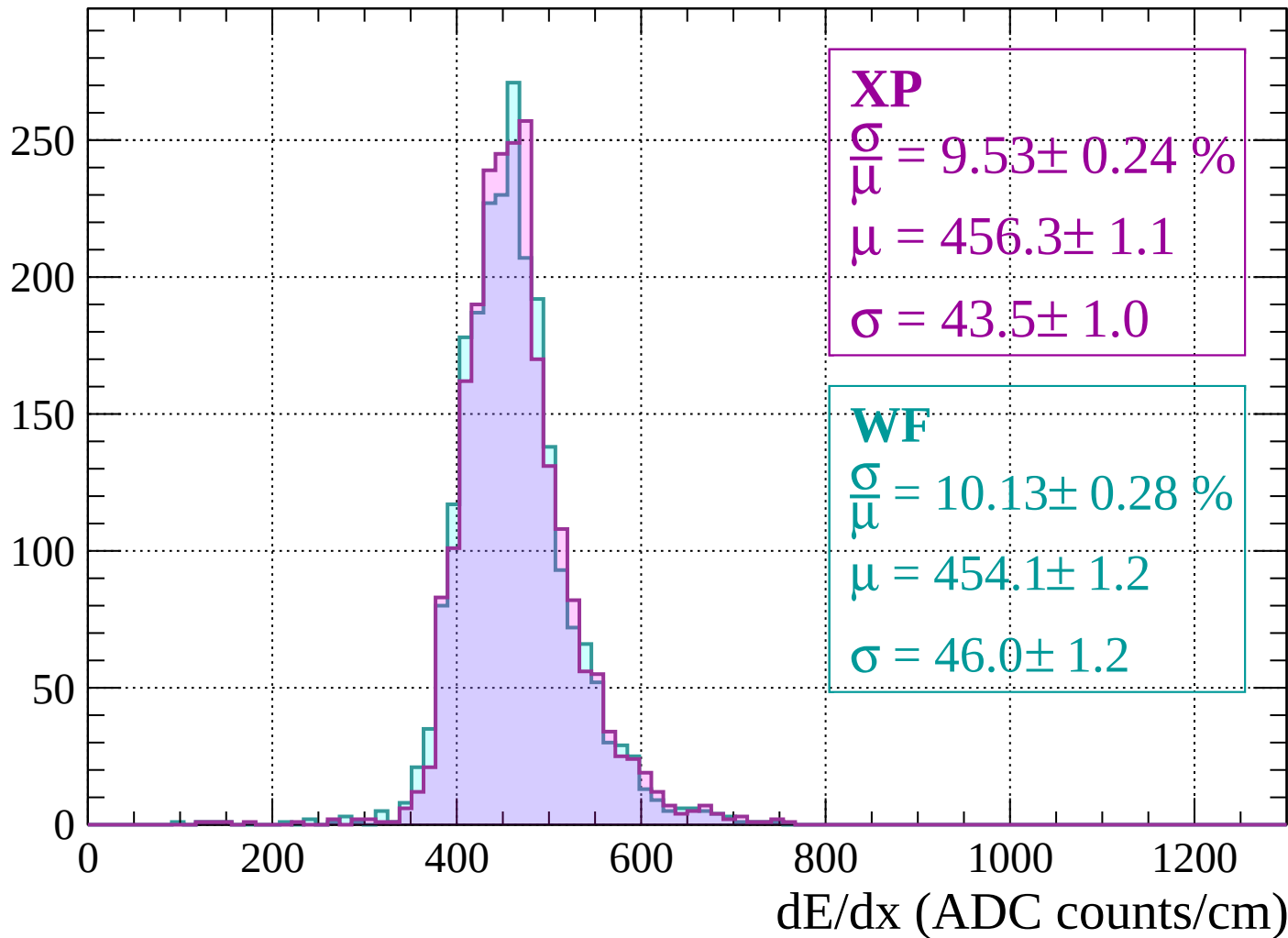


Energy loss | -1360 < p < -1320



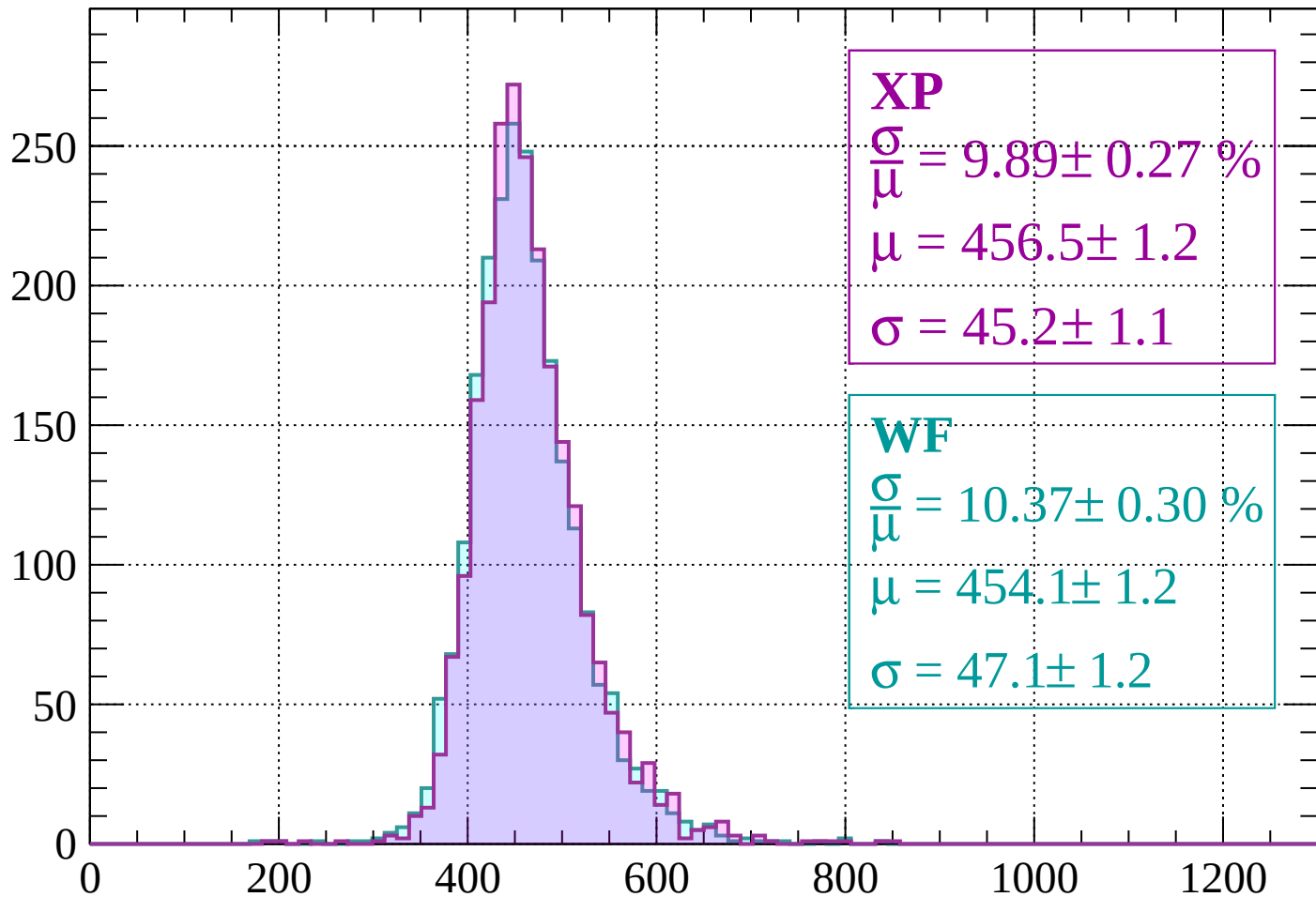
Energy loss | $-1320 < p < -1280$

Count



Energy loss | $-1280 < p < -1240$

Count



XP

$$\frac{\sigma}{\mu} = 9.89 \pm 0.27 \%$$

$$\mu = 456.5 \pm 1.2$$

$$\sigma = 45.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.37 \pm 0.30 \%$$

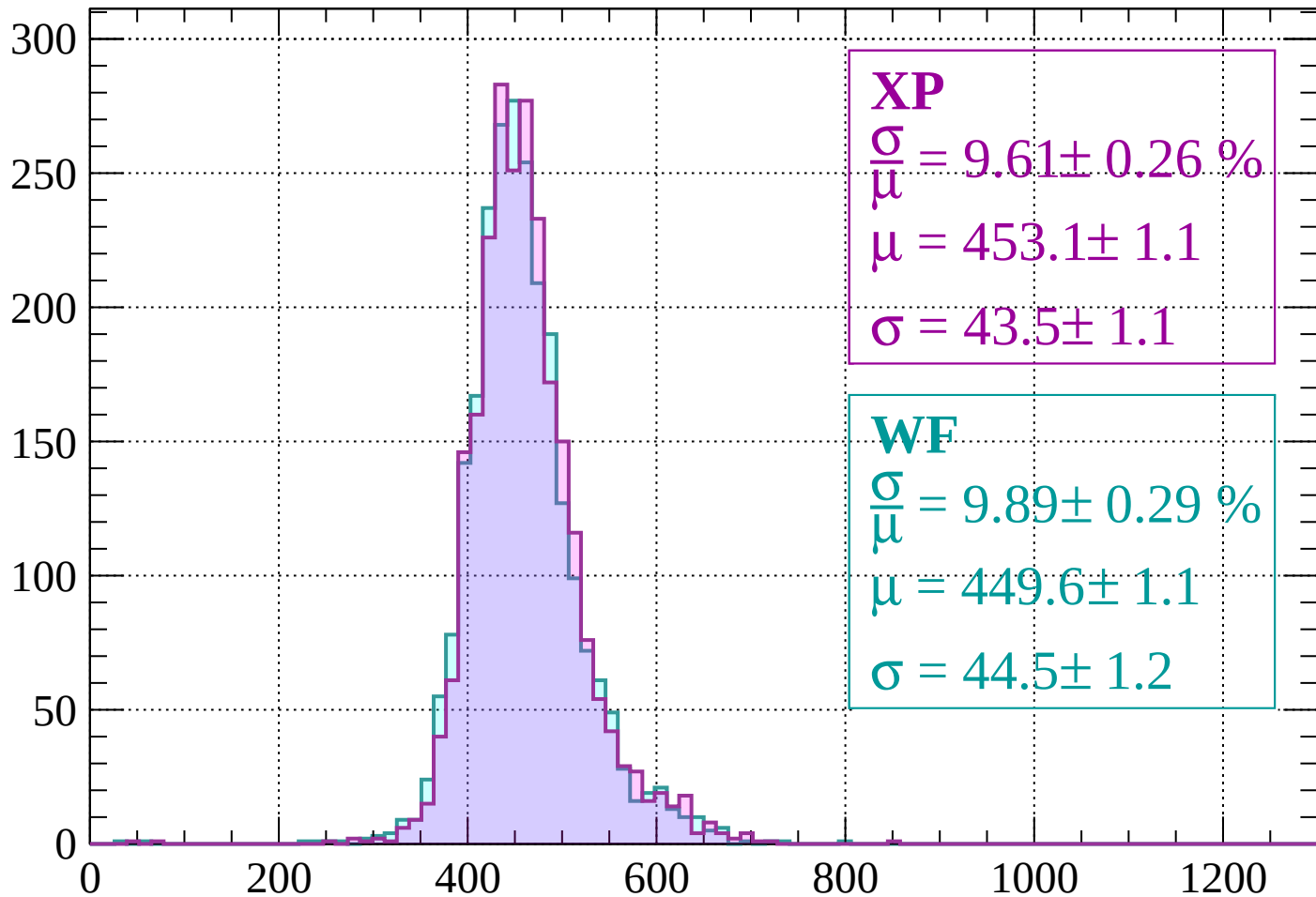
$$\mu = 454.1 \pm 1.2$$

$$\sigma = 47.1 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-1240 < p < -1200$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.61 \pm 0.26 \%$$

$$\mu = 453.1 \pm 1.1$$

$$\sigma = 43.5 \pm 1.1$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.89 \pm 0.29 \%$$

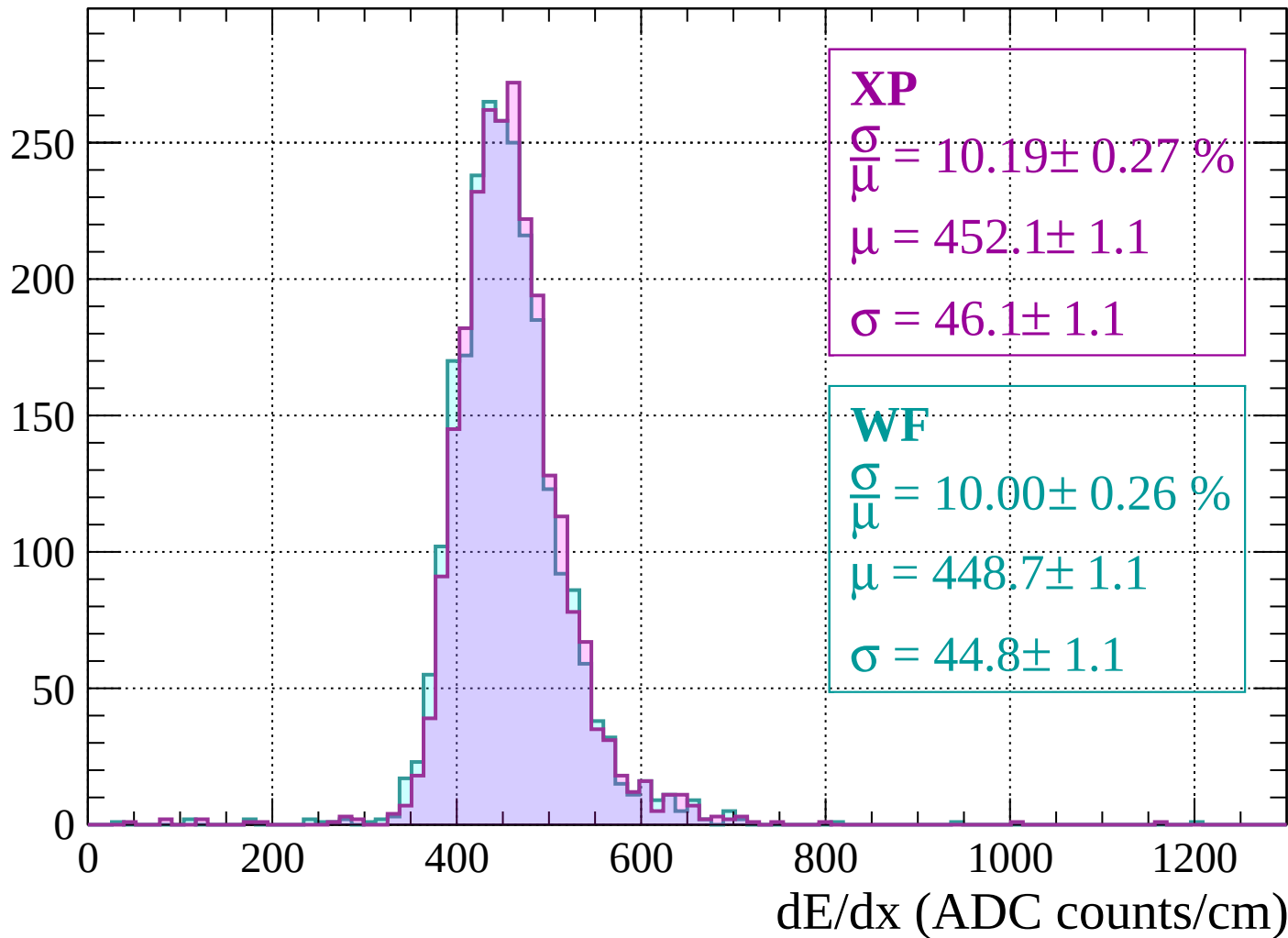
$$\mu = 449.6 \pm 1.1$$

$$\sigma = 44.5 \pm 1.2$$

dE/dx (ADC counts/cm)

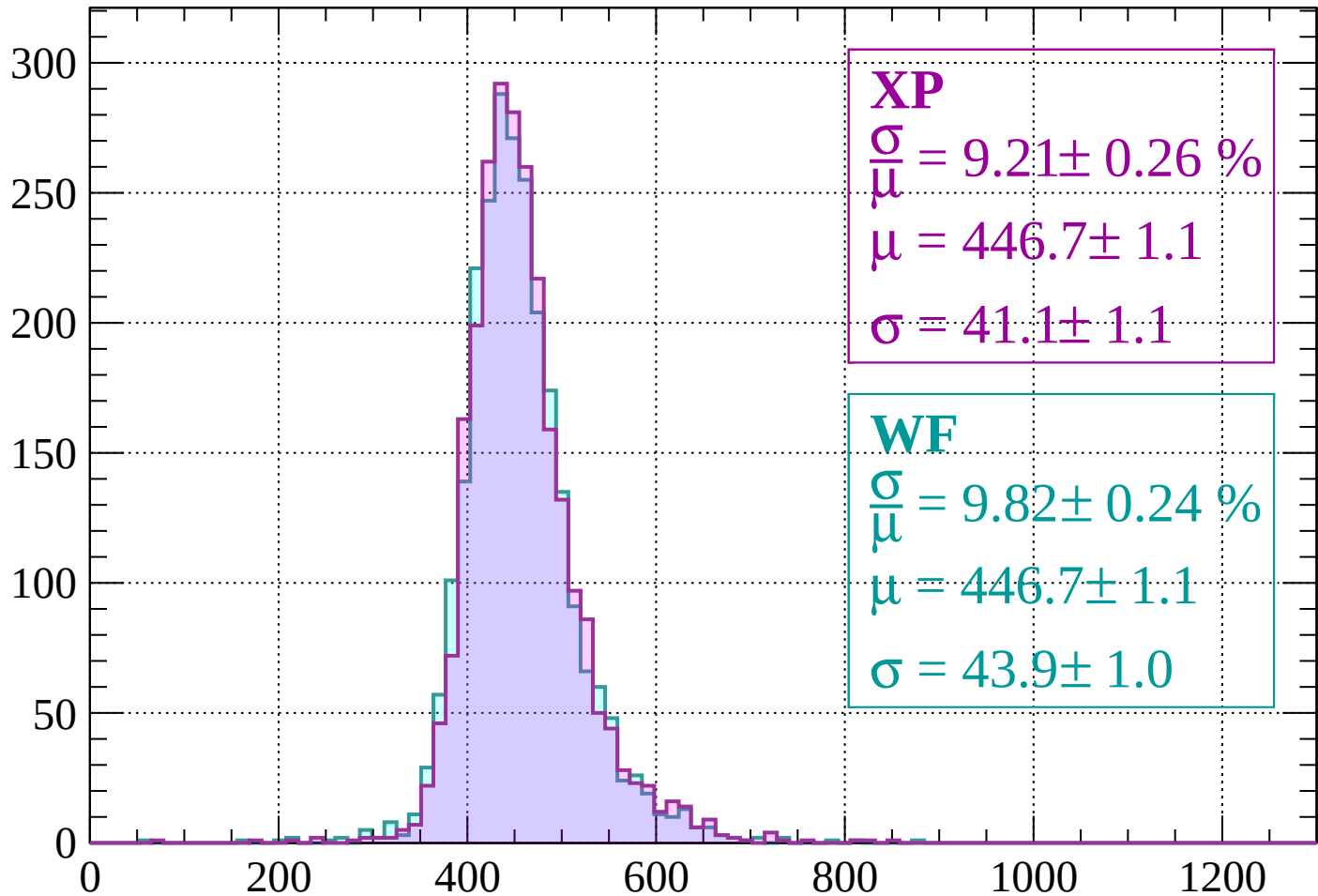
Energy loss | $-1200 < p < -1160$

Count



Energy loss | $-1160 < p < -1120$

Count



XP

$$\frac{\sigma}{\mu} = 9.21 \pm 0.26 \%$$

$$\mu = 446.7 \pm 1.1$$

$$\sigma = 41.1 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.82 \pm 0.24 \%$$

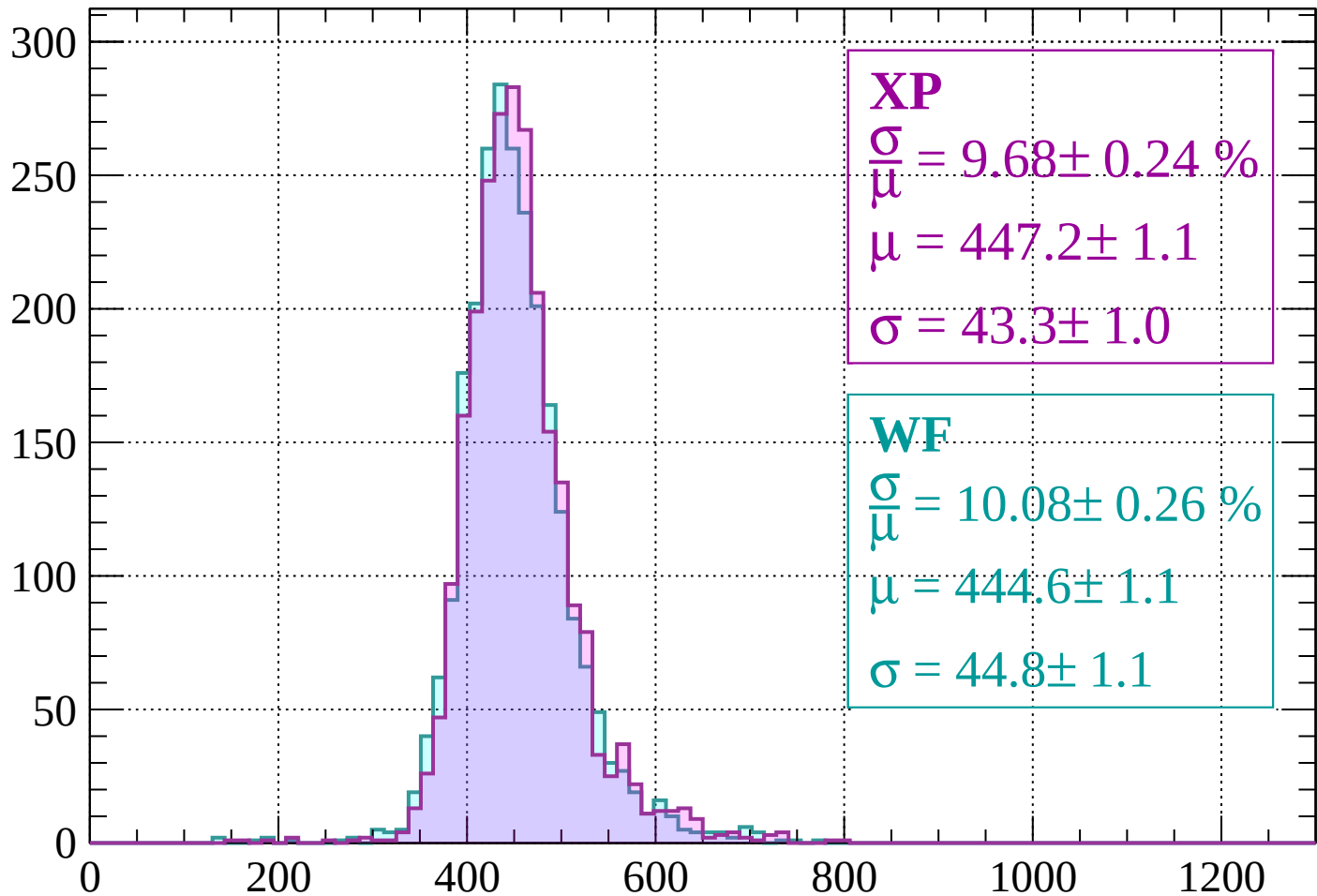
$$\mu = 446.7 \pm 1.1$$

$$\sigma = 43.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1120 < p < -1080$

Count



XP

$$\frac{\sigma}{\mu} = 9.68 \pm 0.24 \%$$

$$\mu = 447.2 \pm 1.1$$

$$\sigma = 43.3 \pm 1.0$$

WF

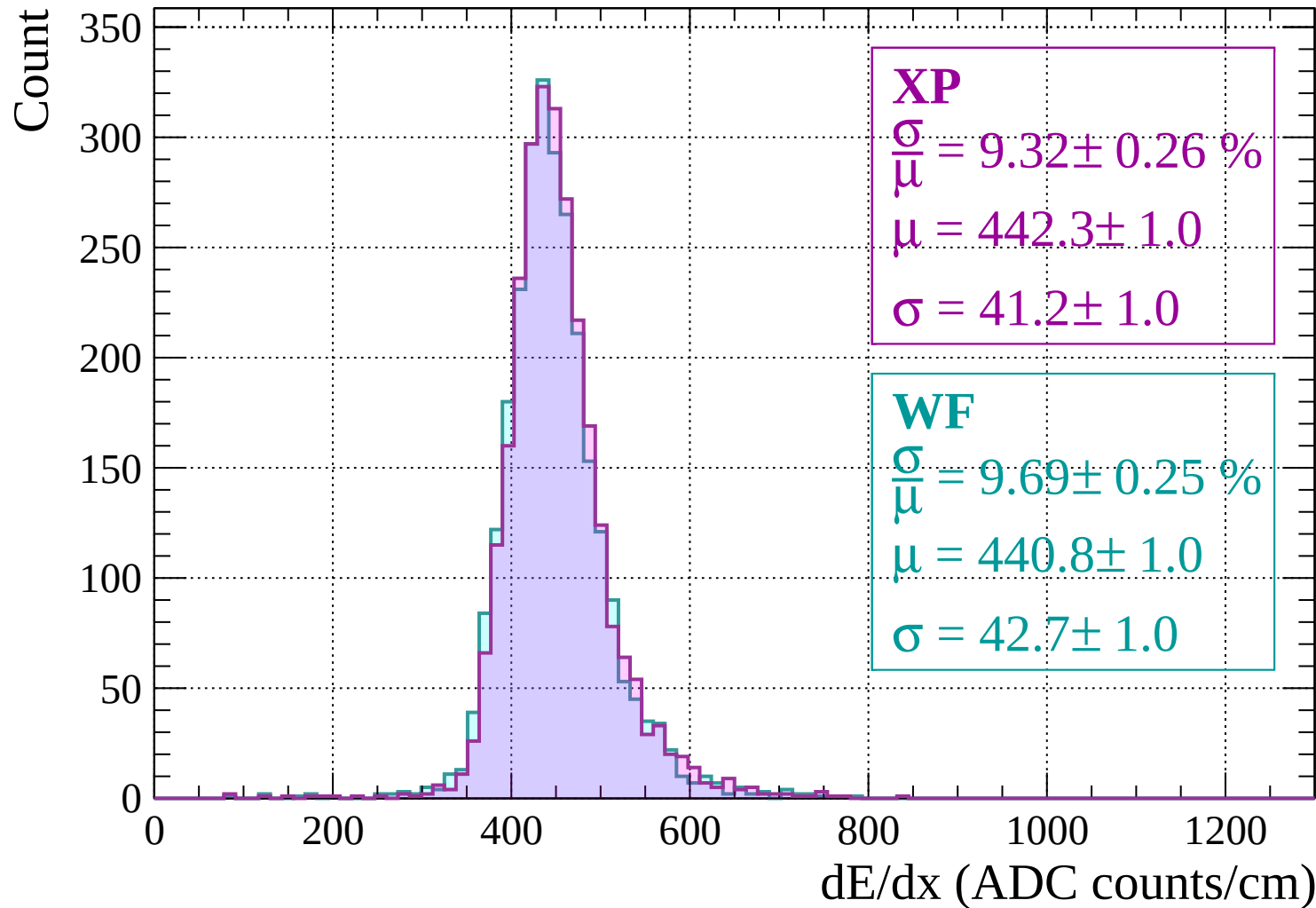
$$\frac{\sigma}{\mu} = 10.08 \pm 0.26 \%$$

$$\mu = 444.6 \pm 1.1$$

$$\sigma = 44.8 \pm 1.1$$

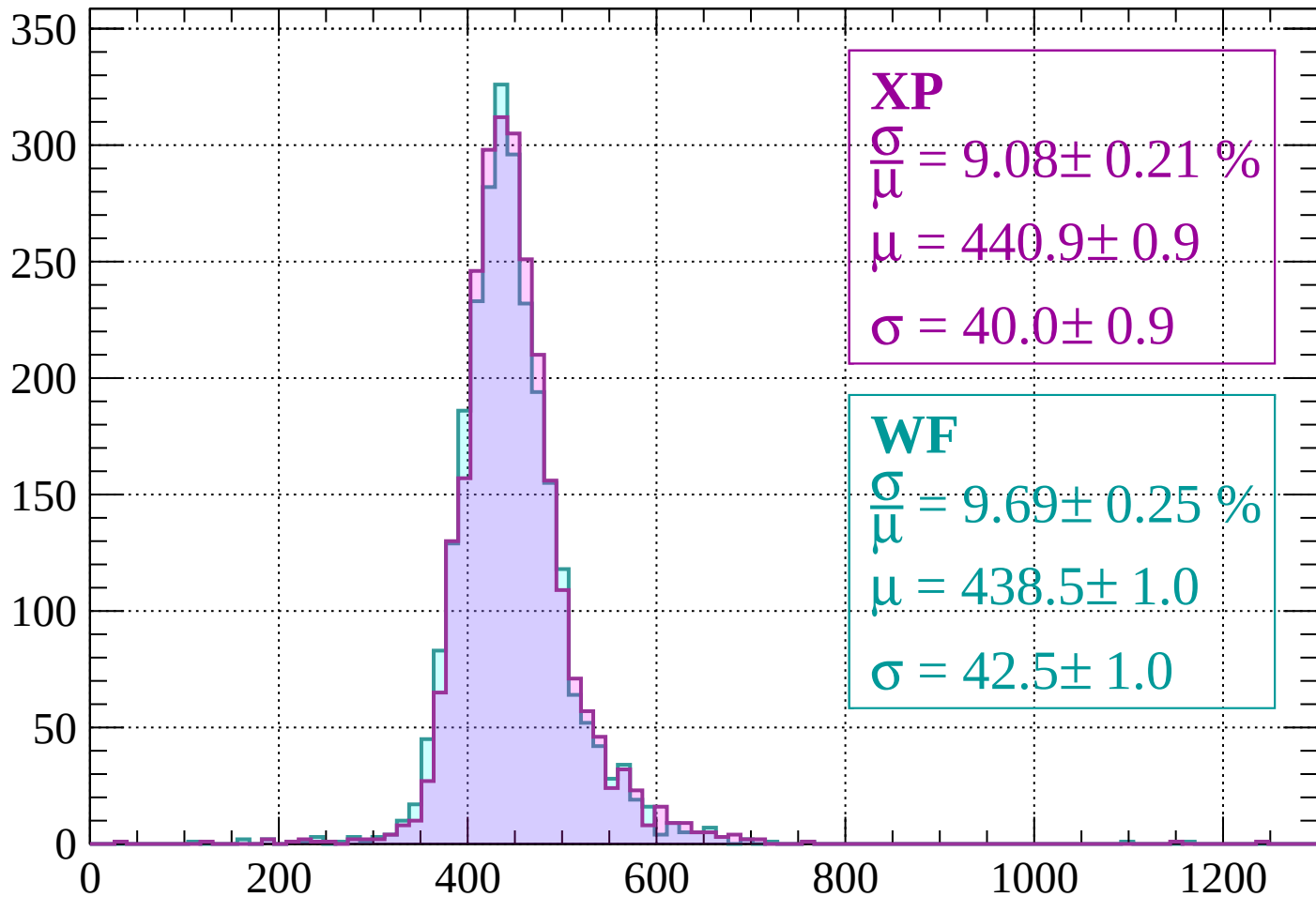
dE/dx (ADC counts/cm)

Energy loss | -1080 < p < -1040



Energy loss | $-1040 < p < -1000$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.08 \pm 0.21 \%$$

$$\mu = 440.9 \pm 0.9$$

$$\sigma = 40.0 \pm 0.9$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.69 \pm 0.25 \%$$

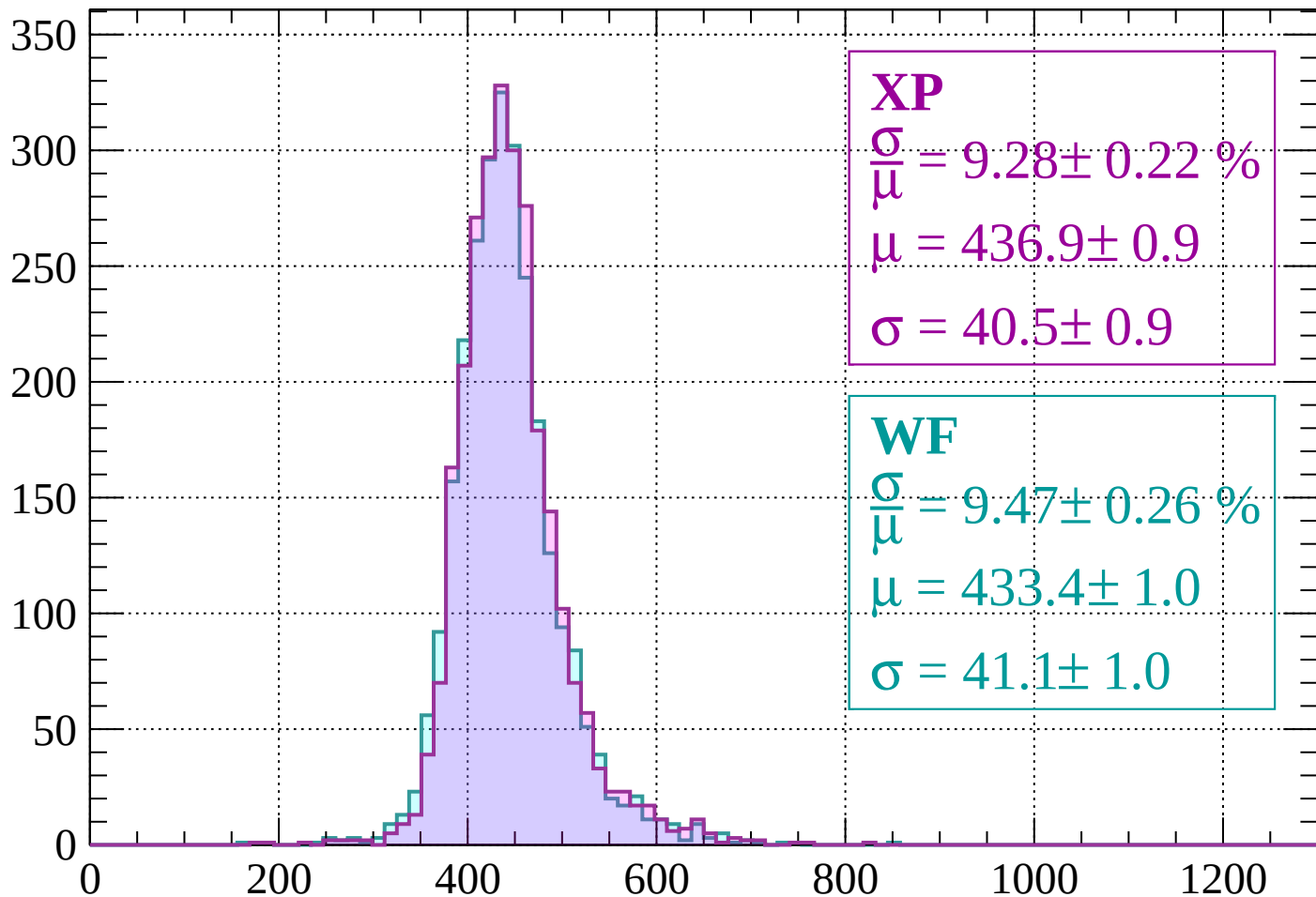
$$\mu = 438.5 \pm 1.0$$

$$\sigma = 42.5 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1000 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 9.28 \pm 0.22 \%$$

$$\mu = 436.9 \pm 0.9$$

$$\sigma = 40.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.47 \pm 0.26 \%$$

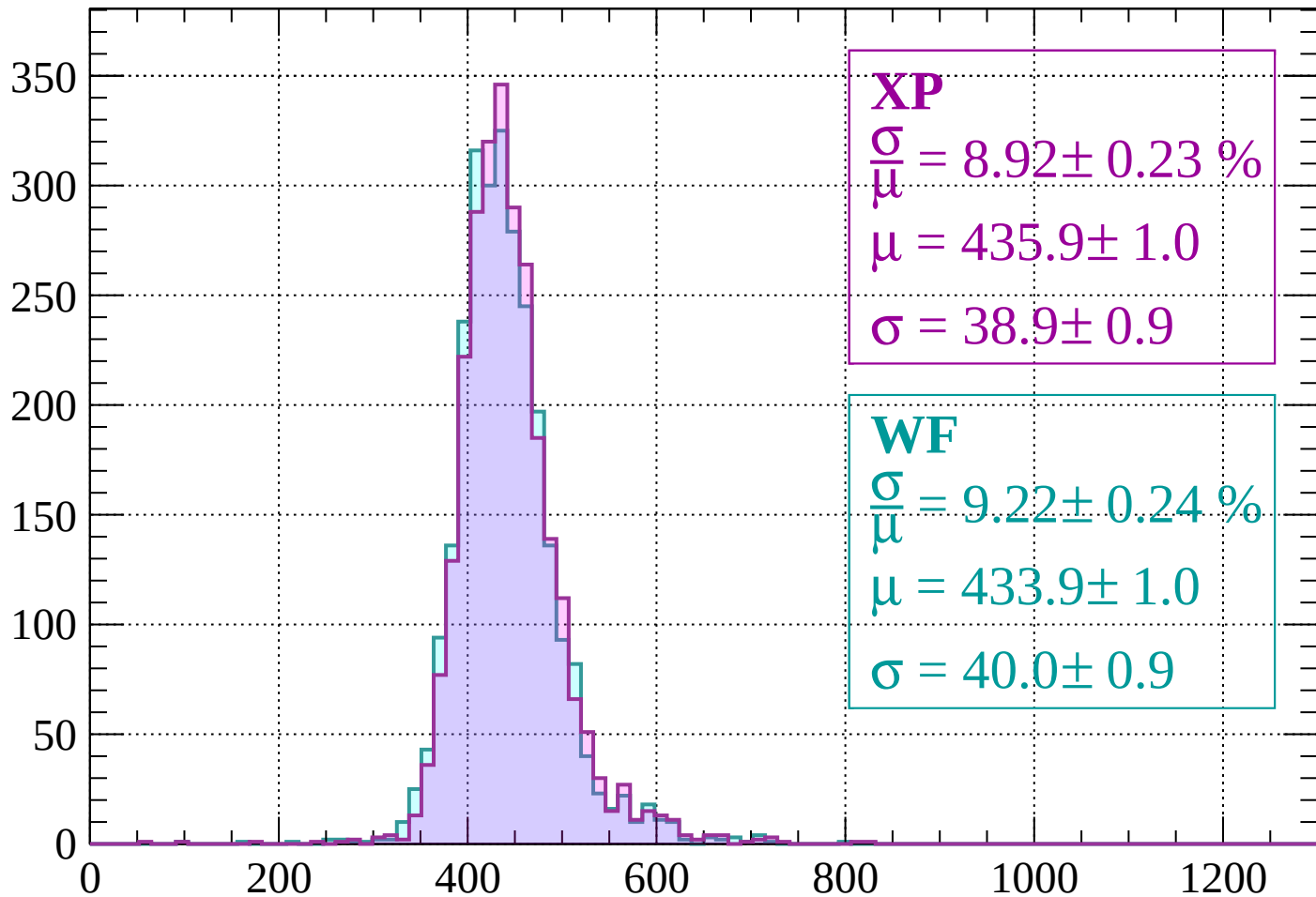
$$\mu = 433.4 \pm 1.0$$

$$\sigma = 41.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 8.92 \pm 0.23 \%$$

$$\mu = 435.9 \pm 1.0$$

$$\sigma = 38.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.22 \pm 0.24 \%$$

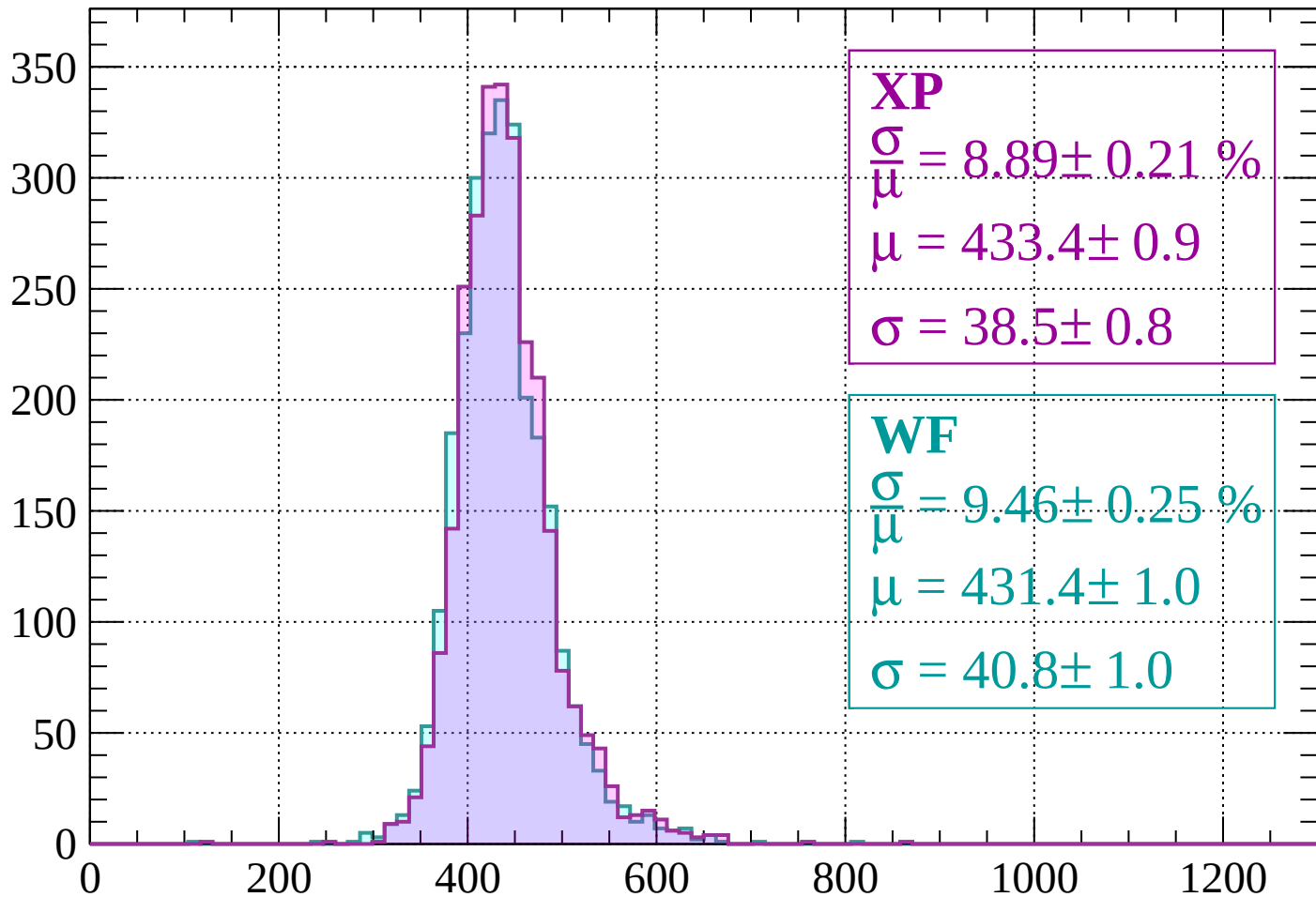
$$\mu = 433.9 \pm 1.0$$

$$\sigma = 40.0 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 8.89 \pm 0.21 \%$$

$$\mu = 433.4 \pm 0.9$$

$$\sigma = 38.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.46 \pm 0.25 \%$$

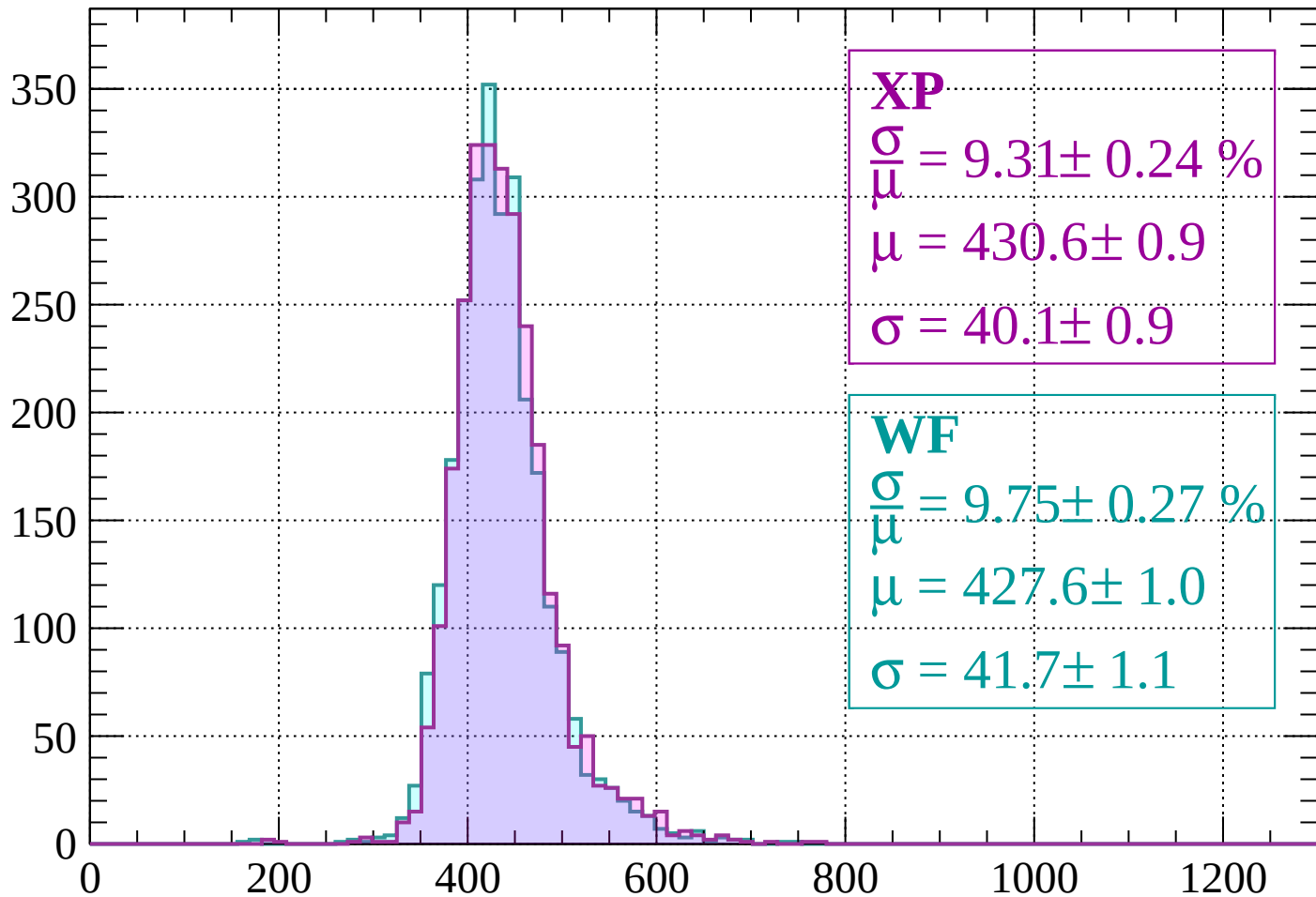
$$\mu = 431.4 \pm 1.0$$

$$\sigma = 40.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 9.31 \pm 0.24 \%$$

$$\mu = 430.6 \pm 0.9$$

$$\sigma = 40.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.75 \pm 0.27 \%$$

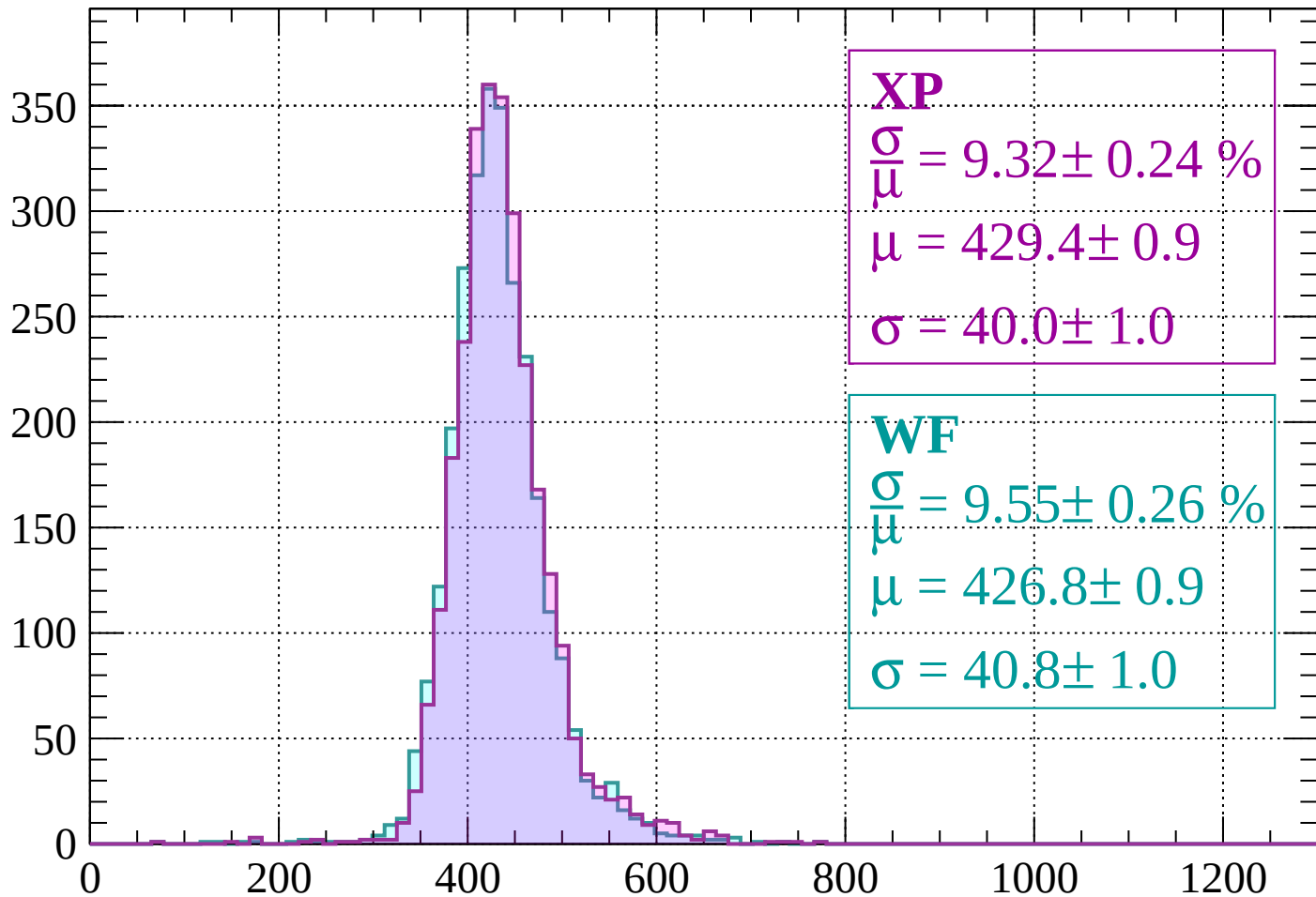
$$\mu = 427.6 \pm 1.0$$

$$\sigma = 41.7 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 9.32 \pm 0.24 \%$$

$$\mu = 429.4 \pm 0.9$$

$$\sigma = 40.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.55 \pm 0.26 \%$$

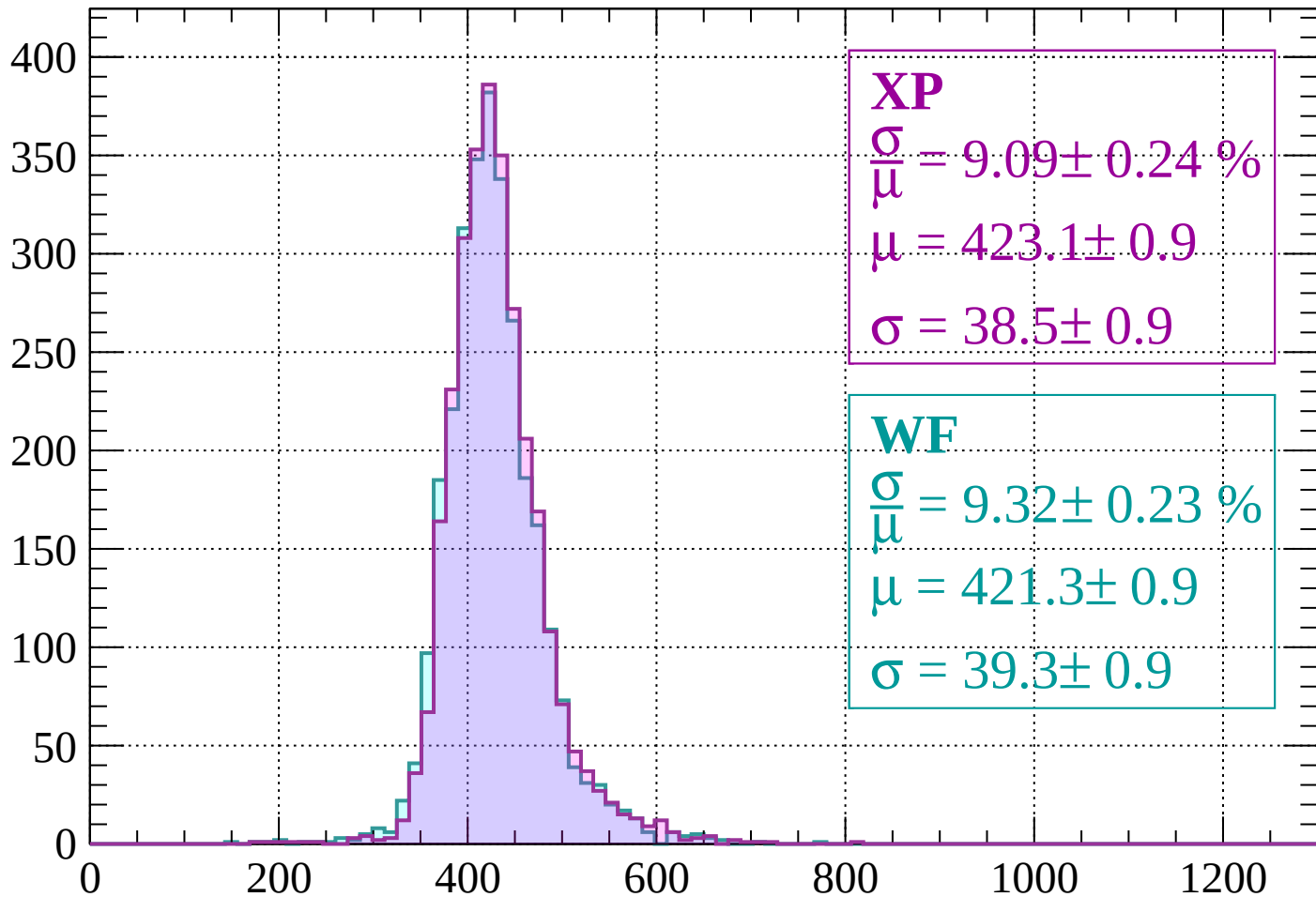
$$\mu = 426.8 \pm 0.9$$

$$\sigma = 40.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 9.09 \pm 0.24 \%$$

$$\mu = 423.1 \pm 0.9$$

$$\sigma = 38.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.32 \pm 0.23 \%$$

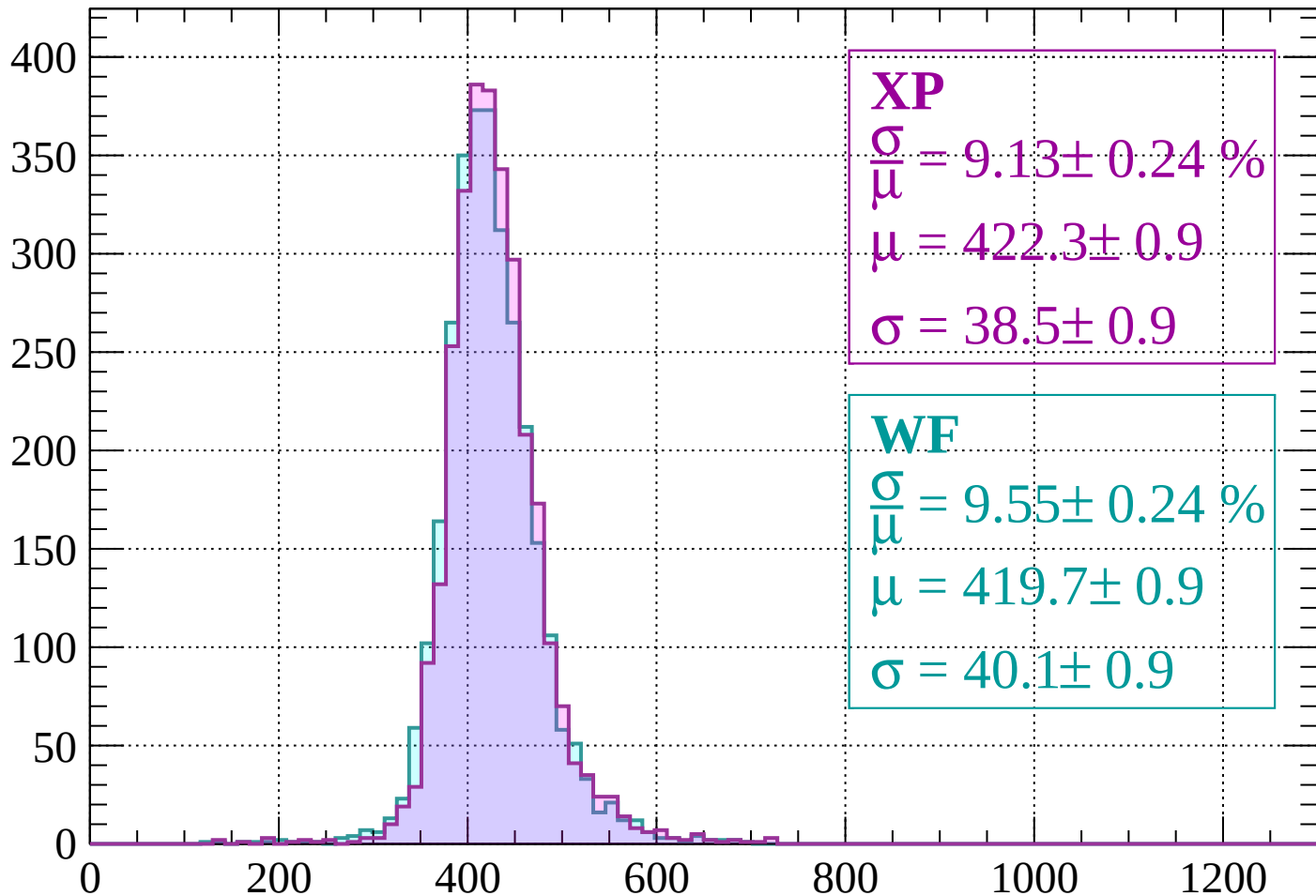
$$\mu = 421.3 \pm 0.9$$

$$\sigma = 39.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 9.13 \pm 0.24 \%$$

$$\mu = 422.3 \pm 0.9$$

$$\sigma = 38.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.55 \pm 0.24 \%$$

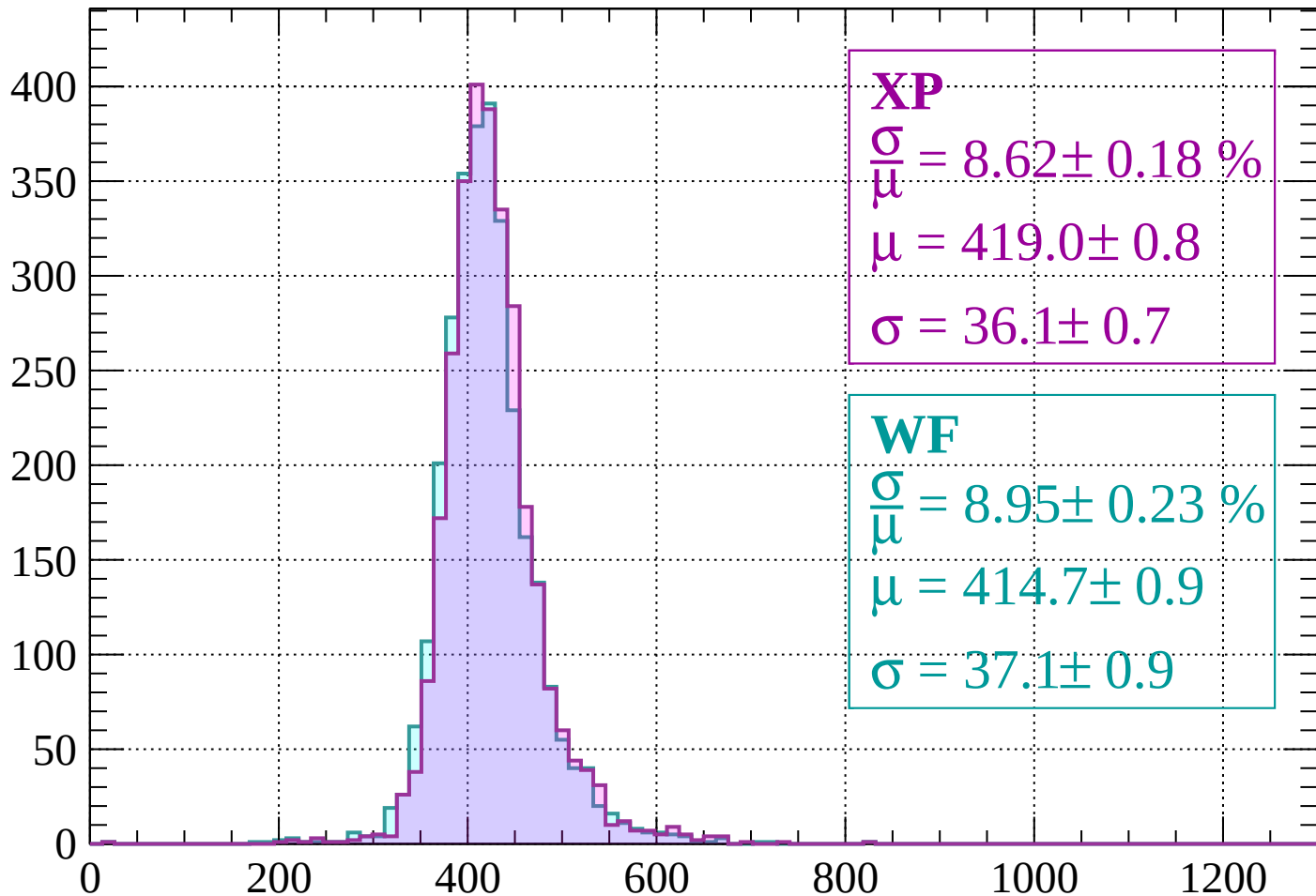
$$\mu = 419.7 \pm 0.9$$

$$\sigma = 40.1 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 8.62 \pm 0.18 \%$$

$$\mu = 419.0 \pm 0.8$$

$$\sigma = 36.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.95 \pm 0.23 \%$$

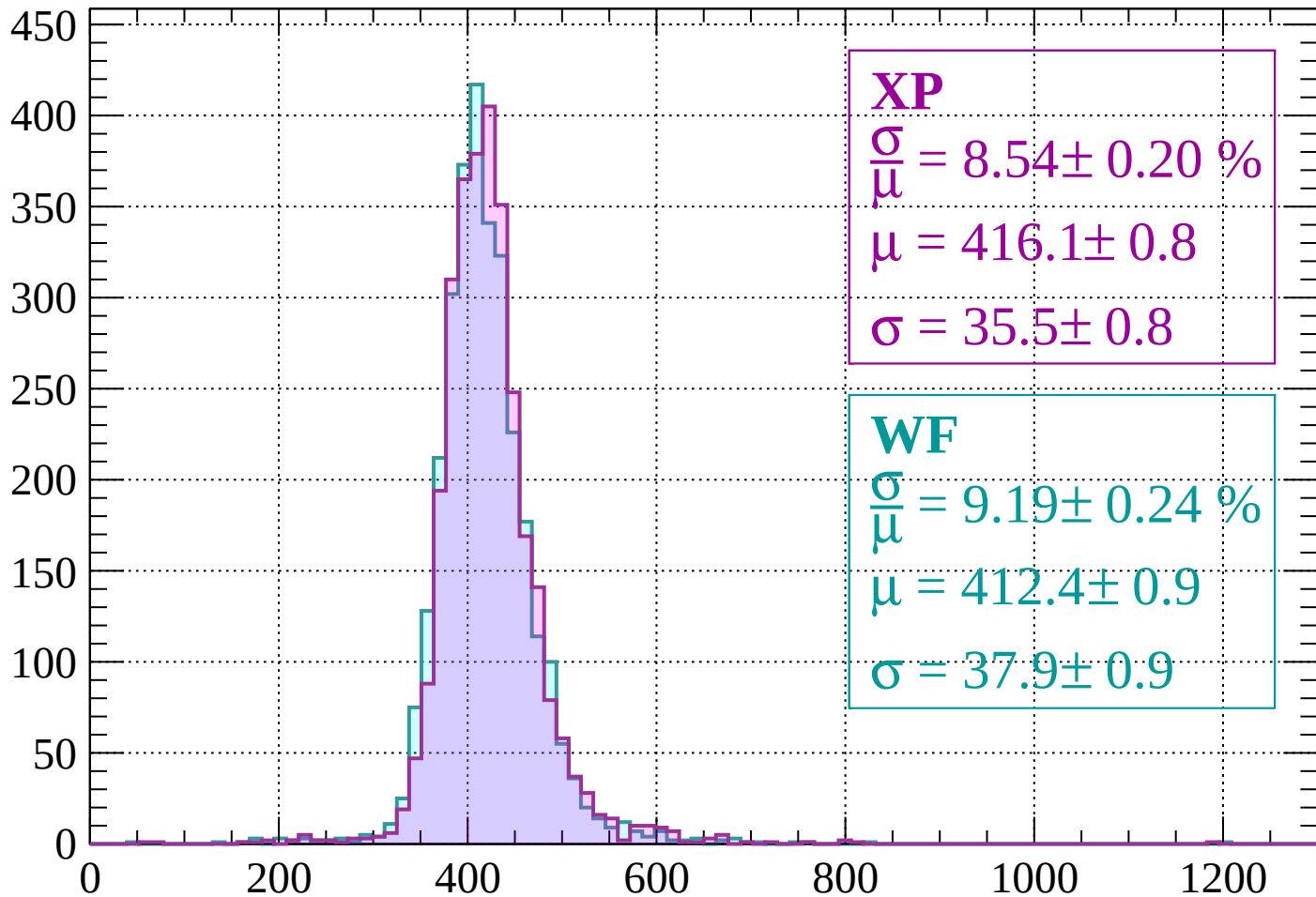
$$\mu = 414.7 \pm 0.9$$

$$\sigma = 37.1 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$

Count



XP

$$\frac{\sigma}{\mu} = 8.54 \pm 0.20 \%$$

$$\mu = 416.1 \pm 0.8$$

$$\sigma = 35.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.19 \pm 0.24 \%$$

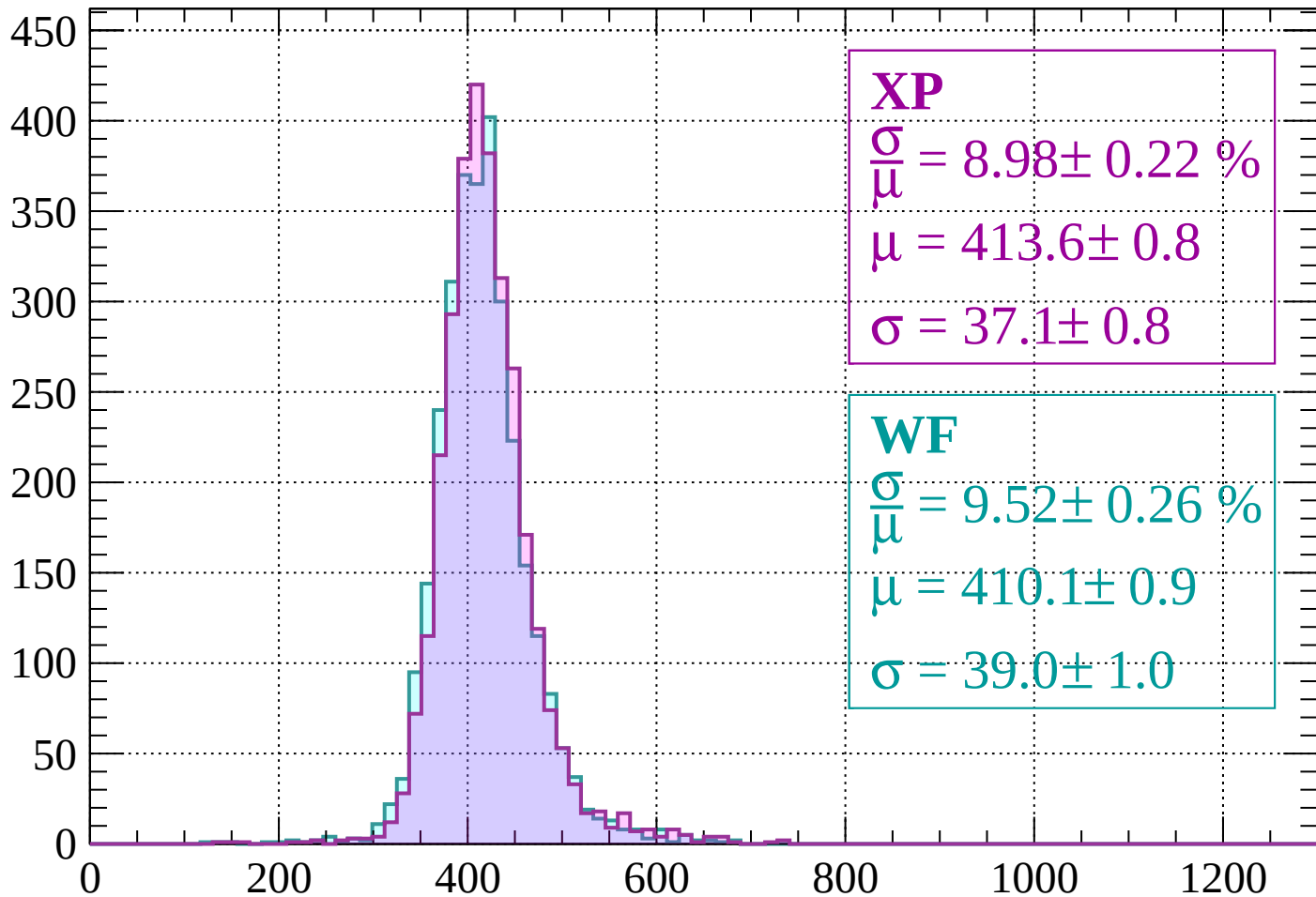
$$\mu = 412.4 \pm 0.9$$

$$\sigma = 37.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 8.98 \pm 0.22 \%$$

$$\mu = 413.6 \pm 0.8$$

$$\sigma = 37.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.52 \pm 0.26 \%$$

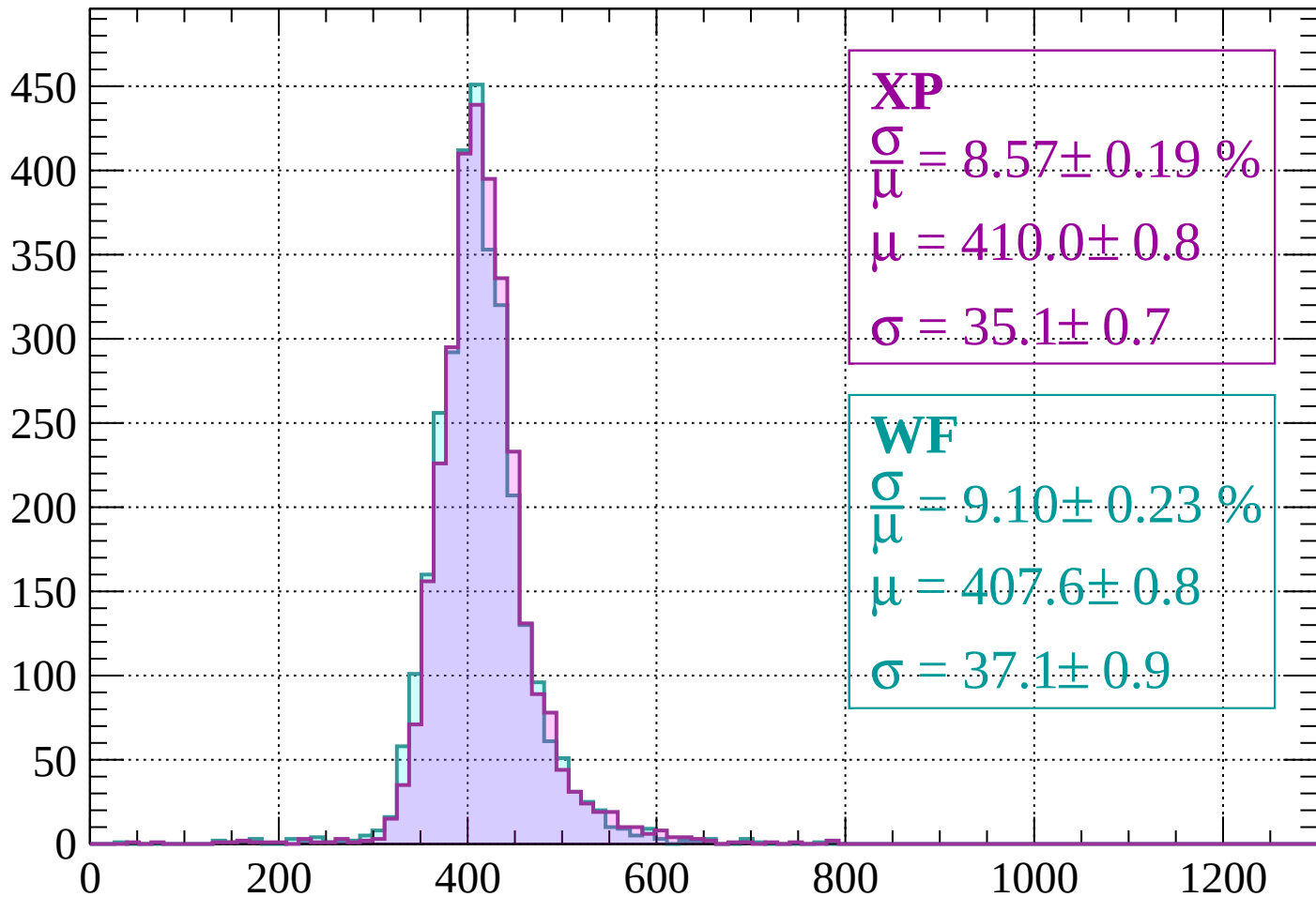
$$\mu = 410.1 \pm 0.9$$

$$\sigma = 39.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 8.57 \pm 0.19 \%$$

$$\mu = 410.0 \pm 0.8$$

$$\sigma = 35.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.10 \pm 0.23 \%$$

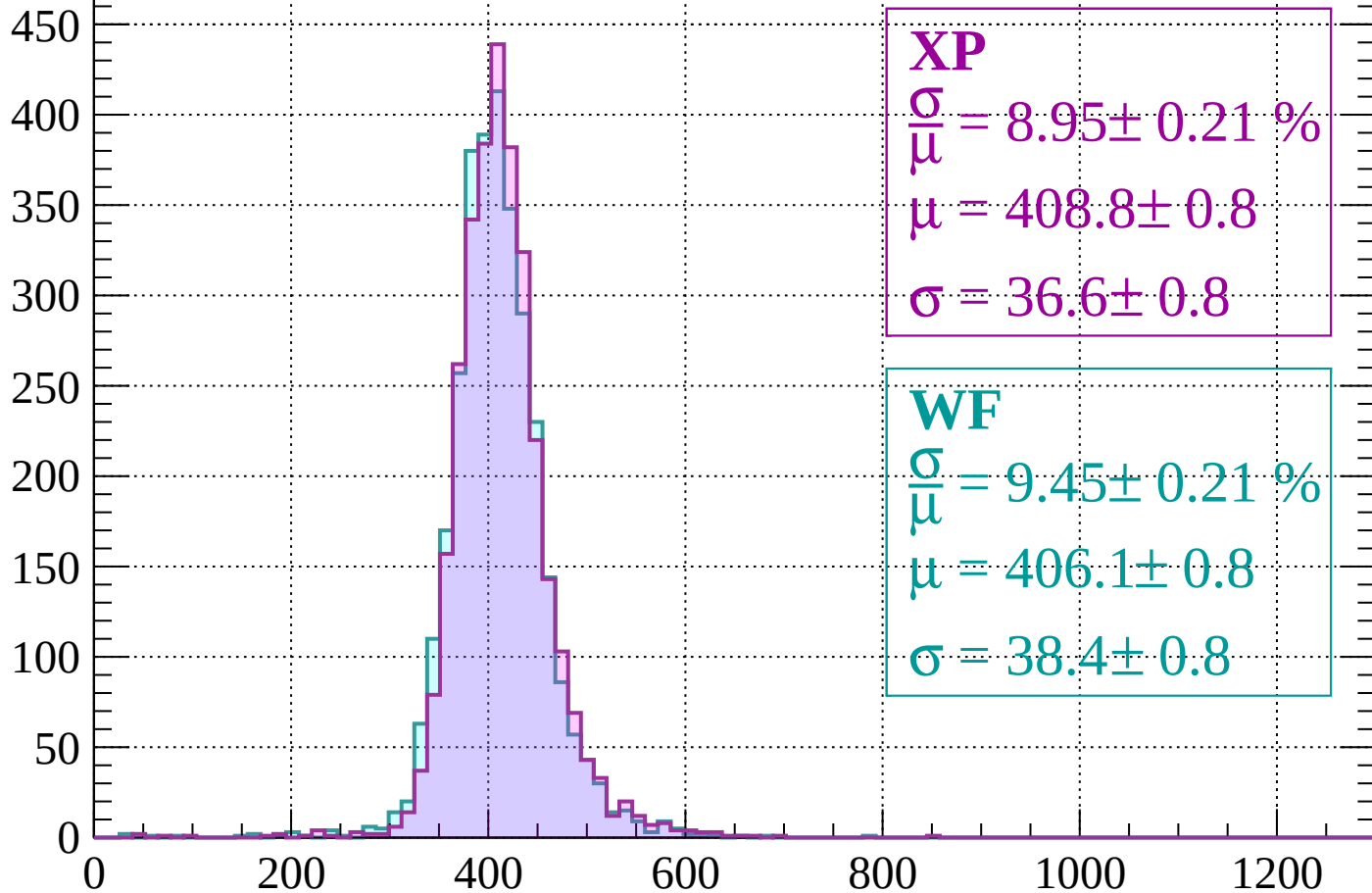
$$\mu = 407.6 \pm 0.8$$

$$\sigma = 37.1 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 8.95 \pm 0.21 \%$$

$$\mu = 408.8 \pm 0.8$$

$$\sigma = 36.6 \pm 0.8$$

WF

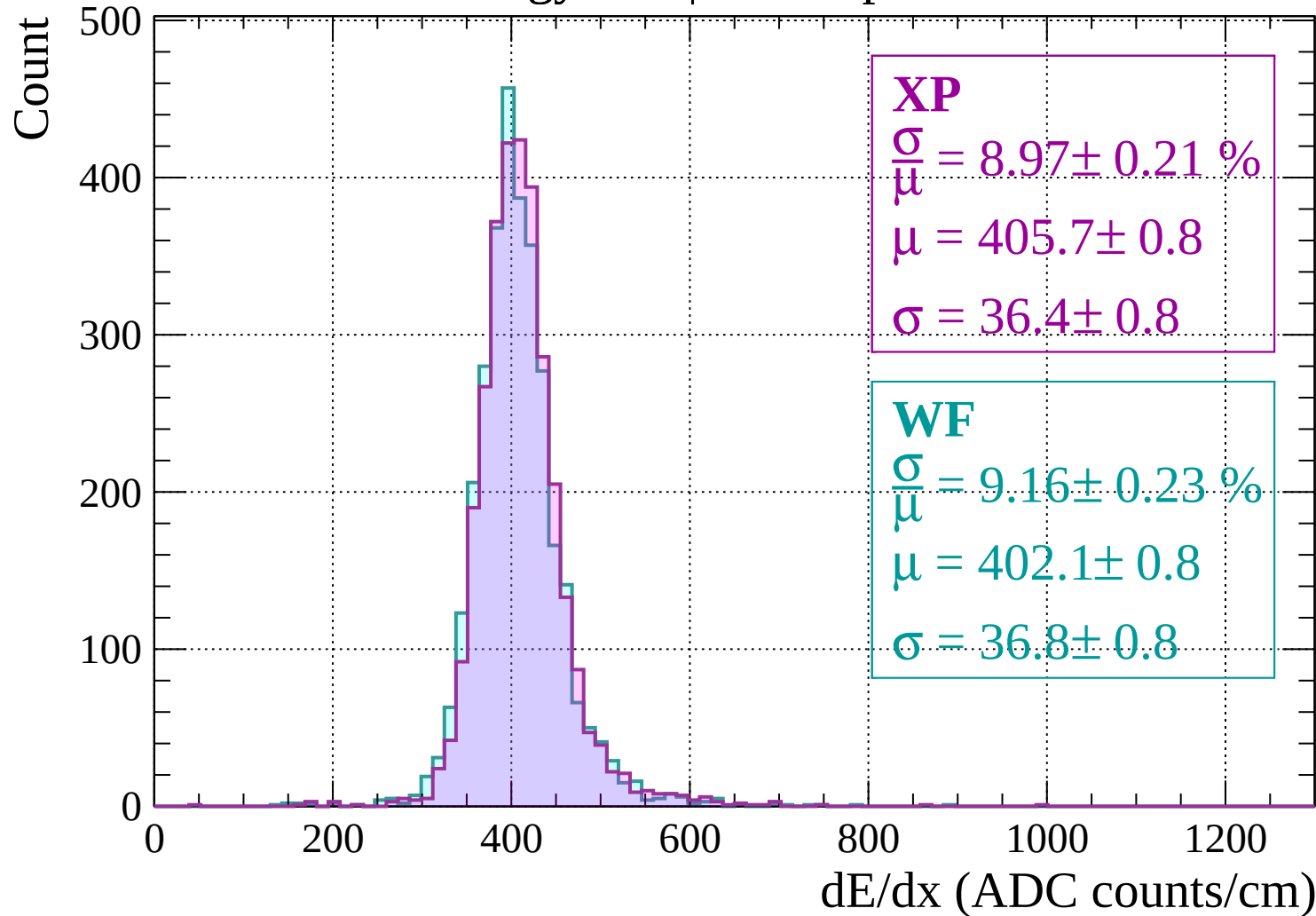
$$\frac{\sigma}{\mu} = 9.45 \pm 0.21 \%$$

$$\mu = 406.1 \pm 0.8$$

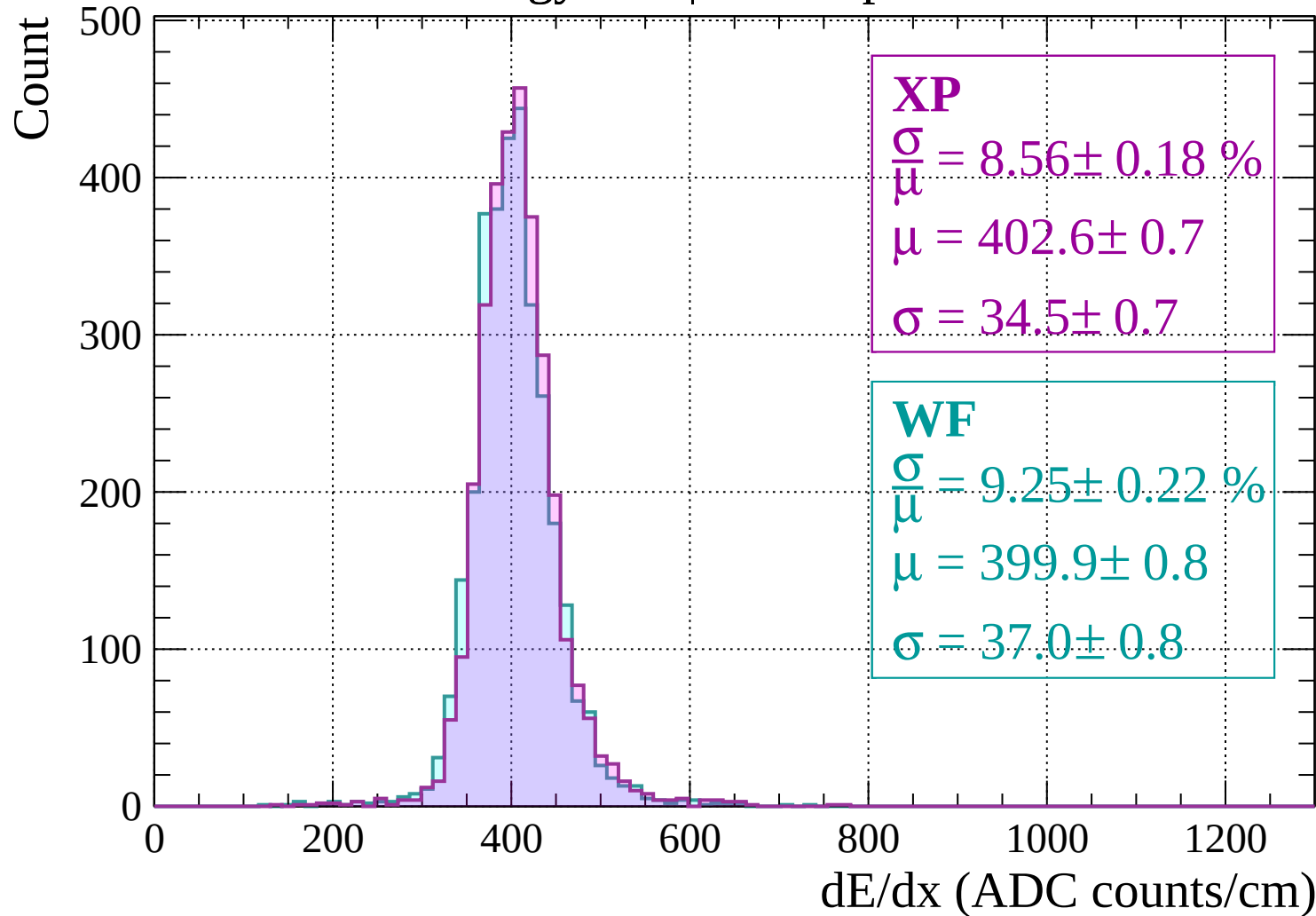
$$\sigma = 38.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

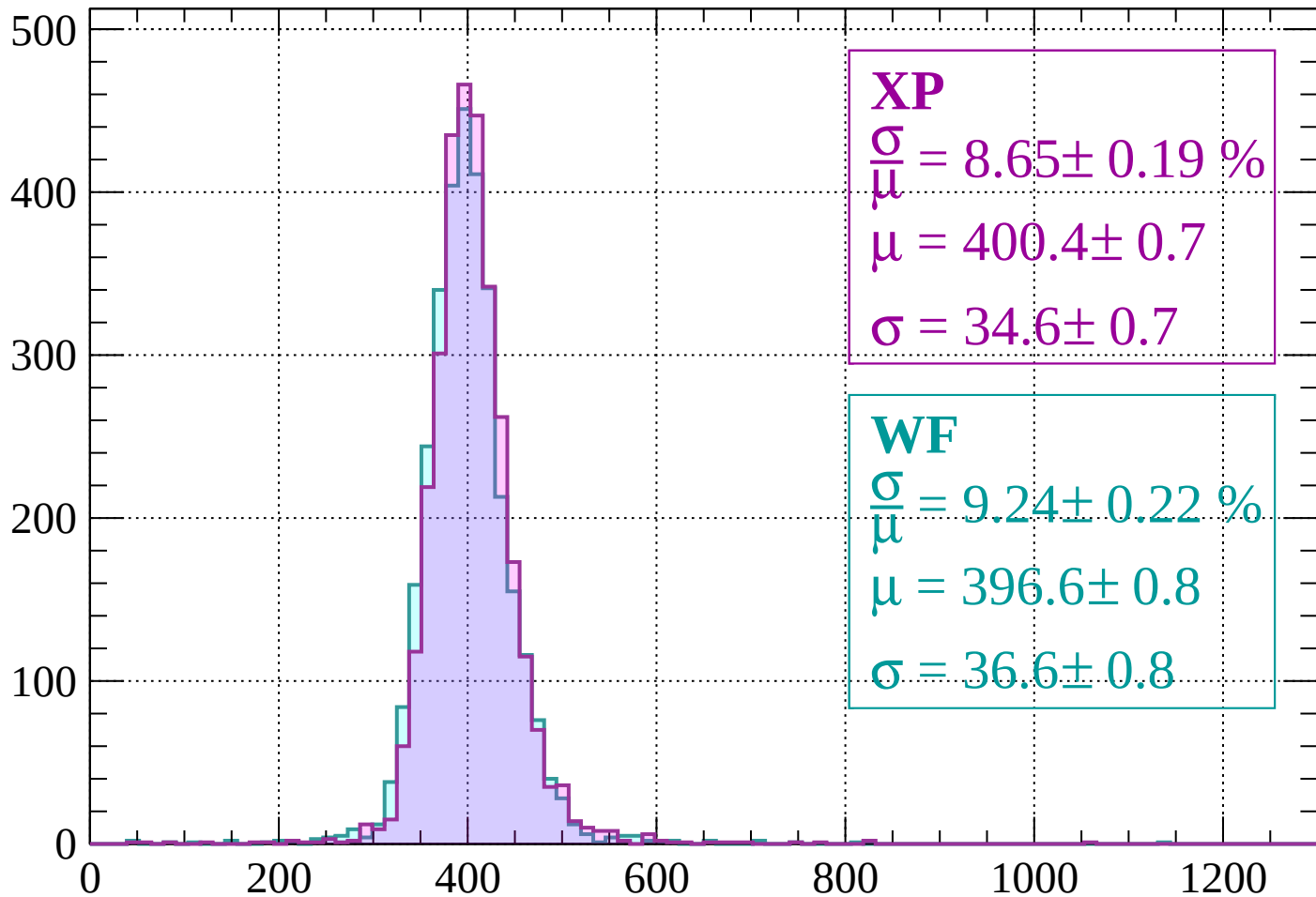


Energy loss | $-480 < p < -440$



Energy loss | $-440 < p < -400$

Count



XP

$$\frac{\sigma}{\mu} = 8.65 \pm 0.19 \%$$

$$\mu = 400.4 \pm 0.7$$

$$\sigma = 34.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.24 \pm 0.22 \%$$

$$\mu = 396.6 \pm 0.8$$

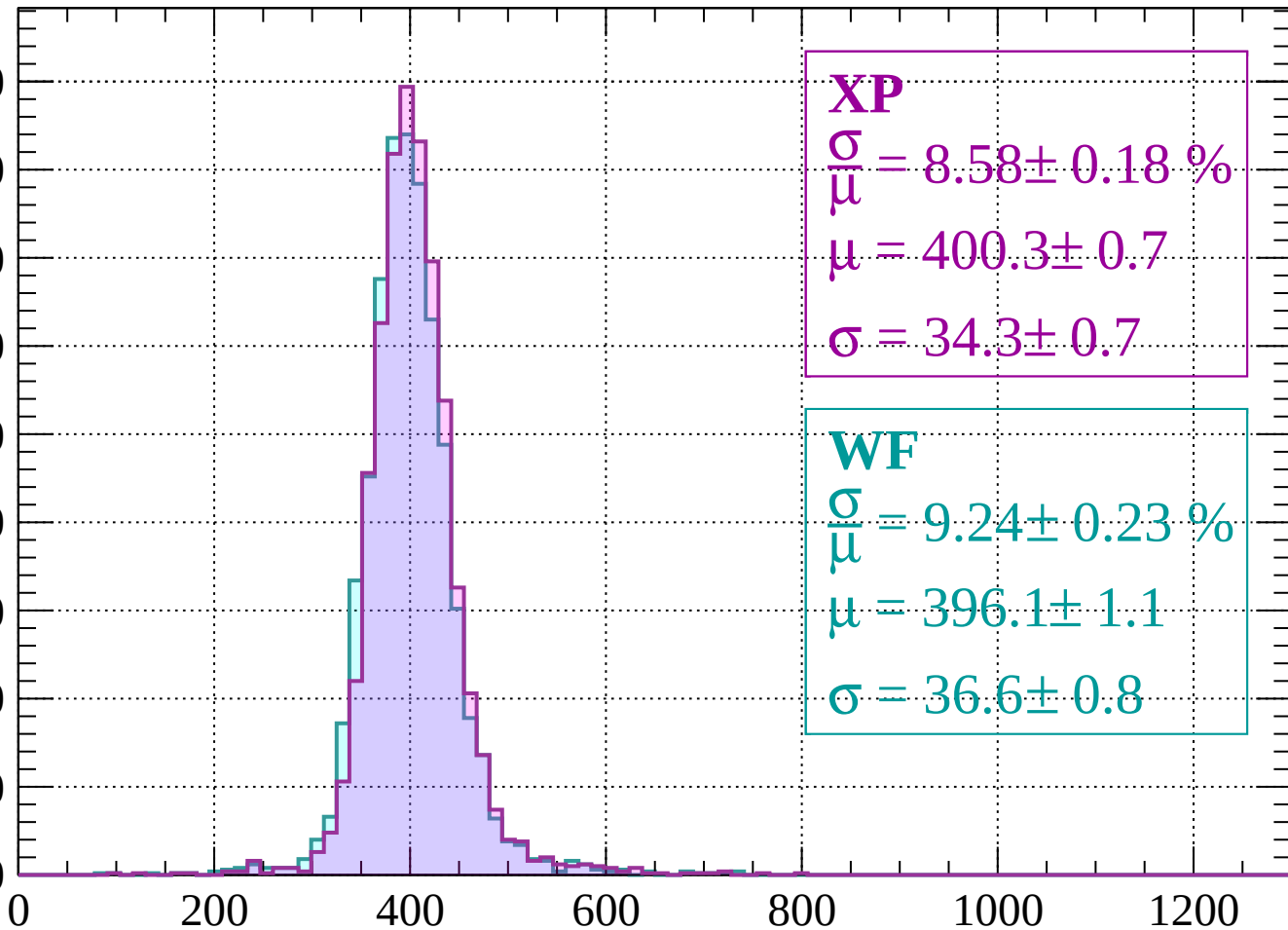
$$\sigma = 36.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 8.58 \pm 0.18 \%$$

$$\mu = 400.3 \pm 0.7$$

$$\sigma = 34.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.24 \pm 0.23 \%$$

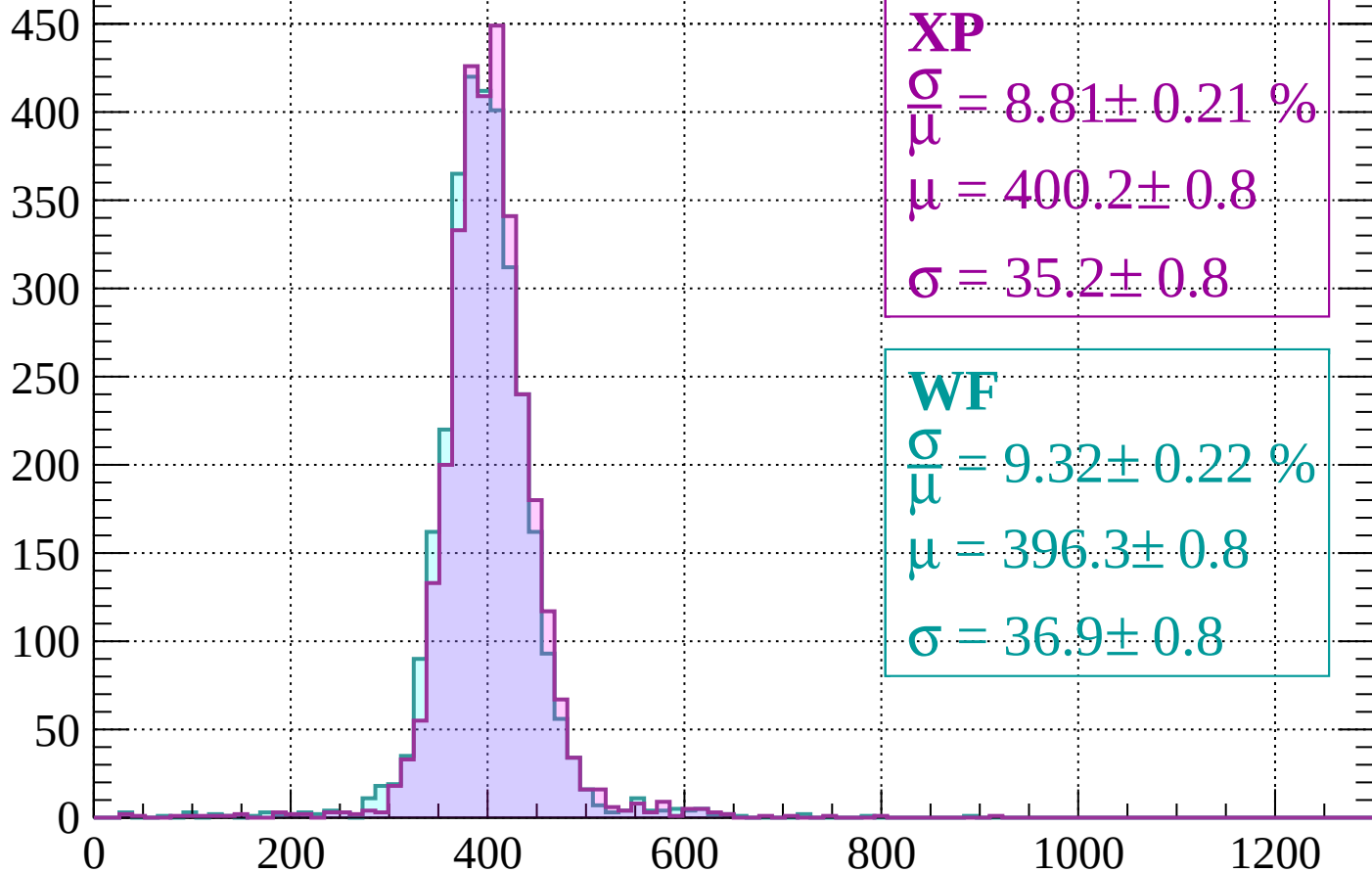
$$\mu = 396.1 \pm 1.1$$

$$\sigma = 36.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$

Count



XP

$$\frac{\sigma}{\mu} = 8.81 \pm 0.21 \%$$

$$\mu = 400.2 \pm 0.8$$

$$\sigma = 35.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.32 \pm 0.22 \%$$

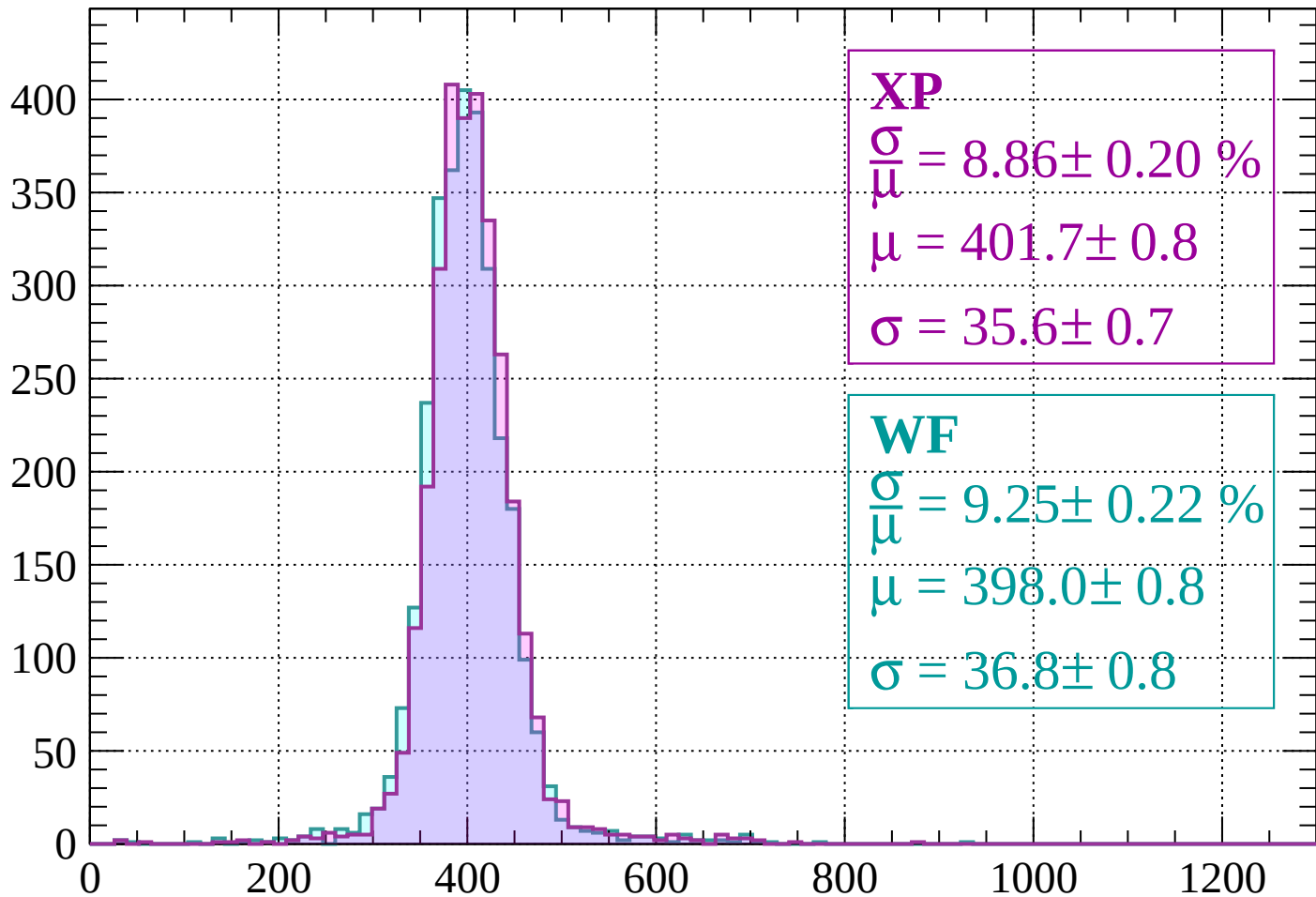
$$\mu = 396.3 \pm 0.8$$

$$\sigma = 36.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\mu} = 8.86 \pm 0.20 \%$$

$$\mu = 401.7 \pm 0.8$$

$$\sigma = 35.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.25 \pm 0.22 \%$$

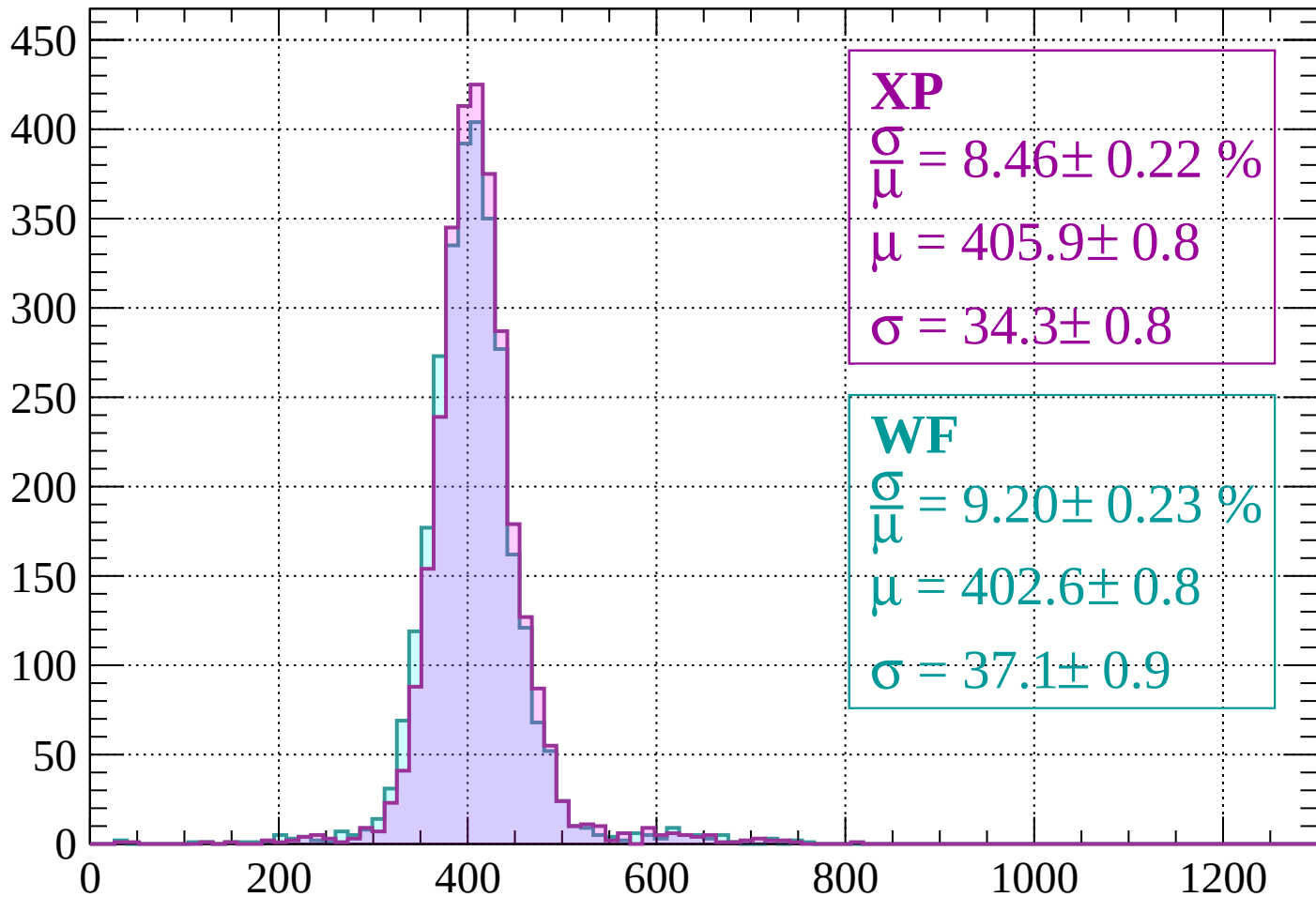
$$\mu = 398.0 \pm 0.8$$

$$\sigma = 36.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 8.46 \pm 0.22 \%$$

$$\mu = 405.9 \pm 0.8$$

$$\sigma = 34.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.20 \pm 0.23 \%$$

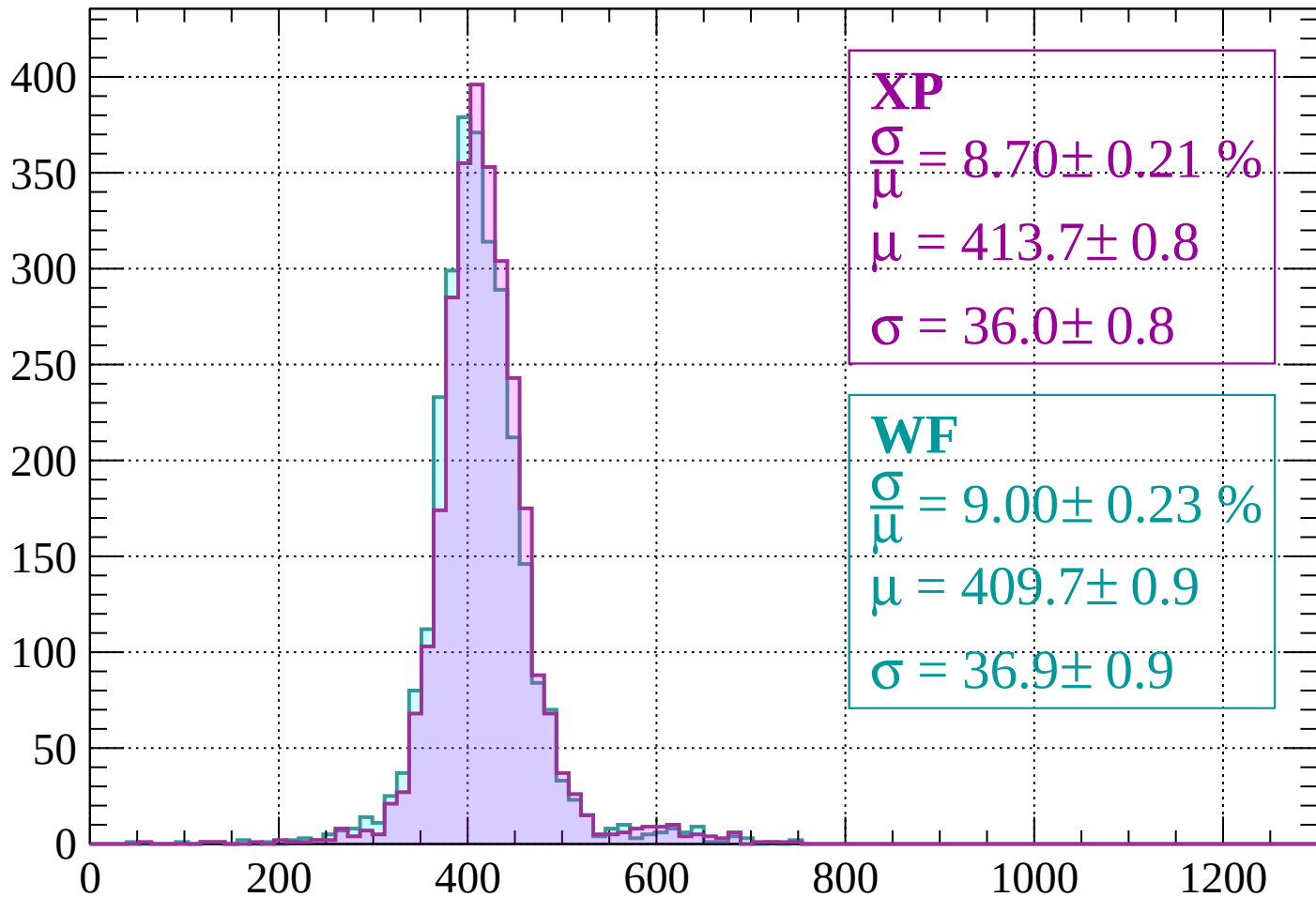
$$\mu = 402.6 \pm 0.8$$

$$\sigma = 37.1 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

Count



XP

$$\frac{\sigma}{\mu} = 8.70 \pm 0.21 \%$$

$$\mu = 413.7 \pm 0.8$$

$$\sigma = 36.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.00 \pm 0.23 \%$$

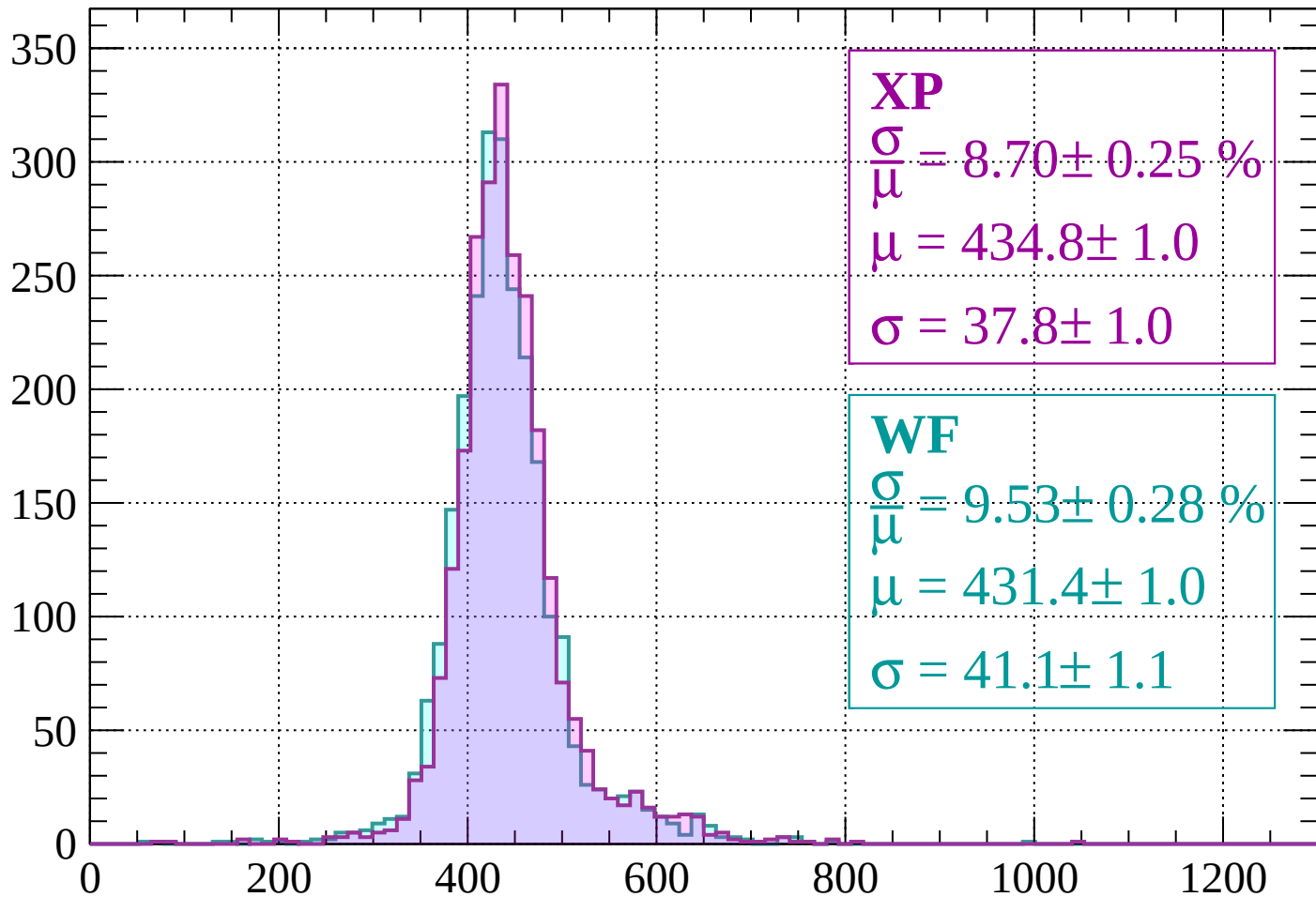
$$\mu = 409.7 \pm 0.9$$

$$\sigma = 36.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$

Count



XP

$$\frac{\sigma}{\mu} = 8.70 \pm 0.25 \%$$

$$\mu = 434.8 \pm 1.0$$

$$\sigma = 37.8 \pm 1.0$$

WF

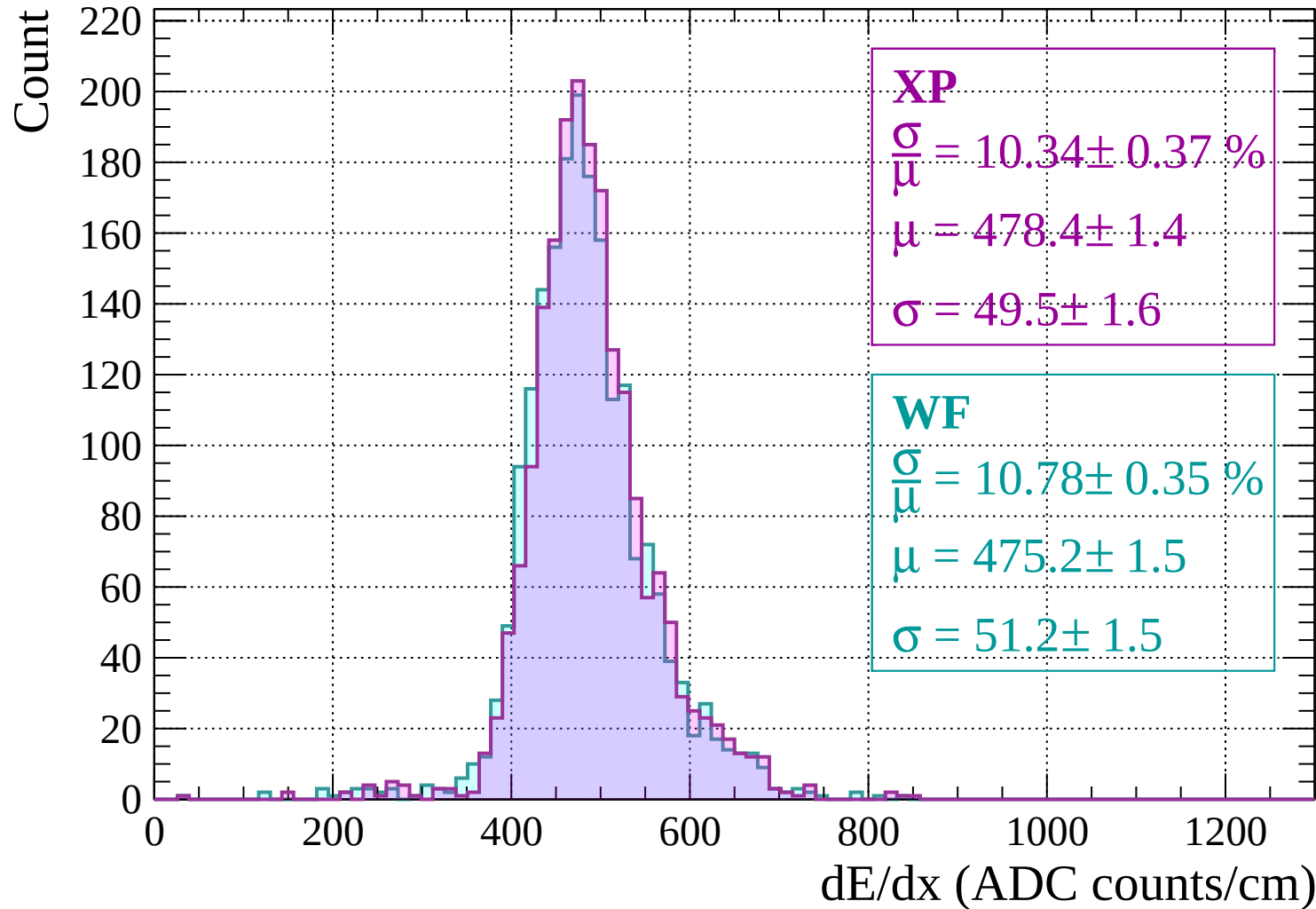
$$\frac{\sigma}{\mu} = 9.53 \pm 0.28 \%$$

$$\mu = 431.4 \pm 1.0$$

$$\sigma = 41.1 \pm 1.1$$

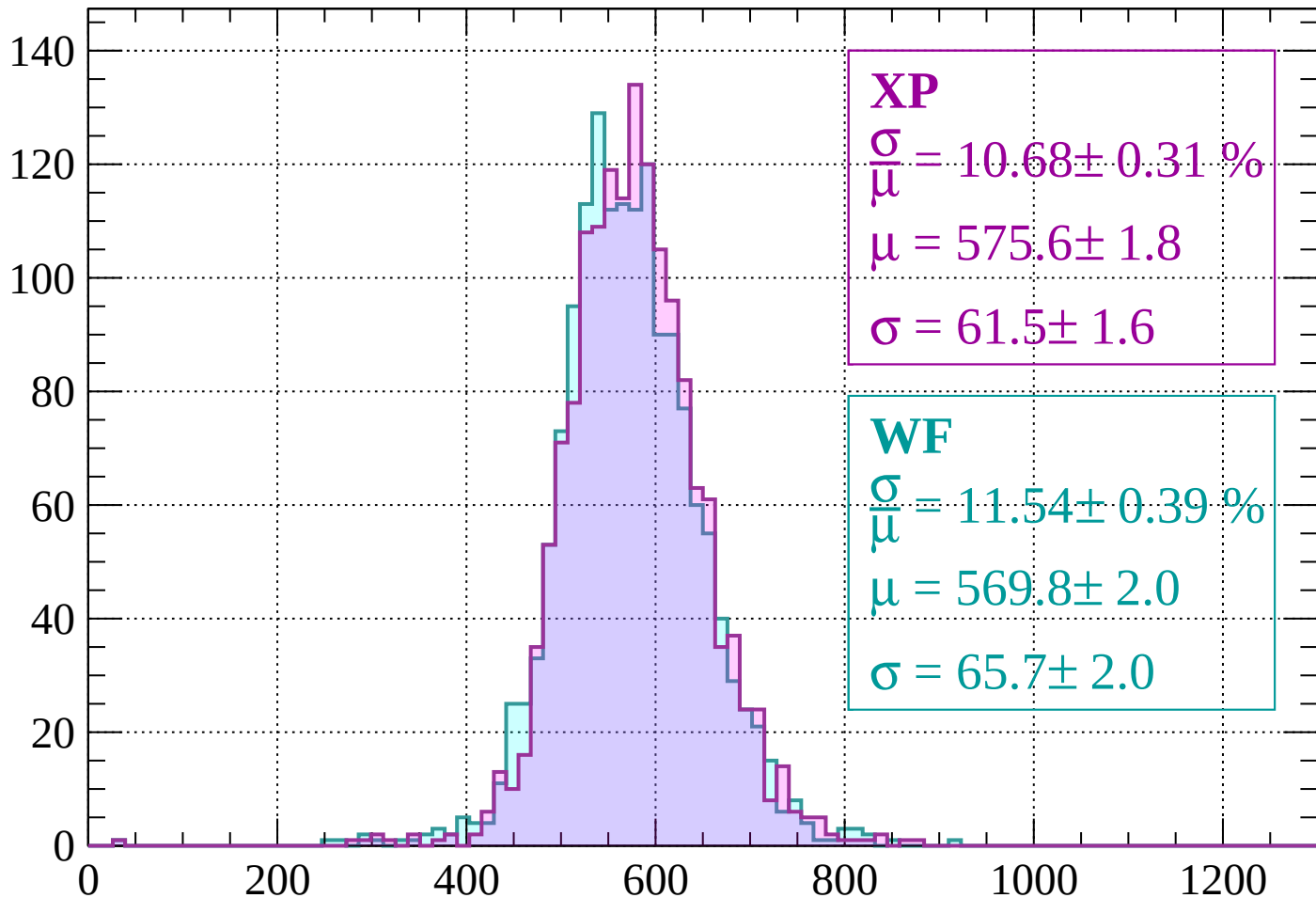
dE/dx (ADC counts/cm)

Energy loss | $-160 < p < -120$



Energy loss | $-120 < p < -80$

Count



XP

$$\frac{\sigma}{\mu} = 10.68 \pm 0.31 \%$$

$$\mu = 575.6 \pm 1.8$$

$$\sigma = 61.5 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 11.54 \pm 0.39 \%$$

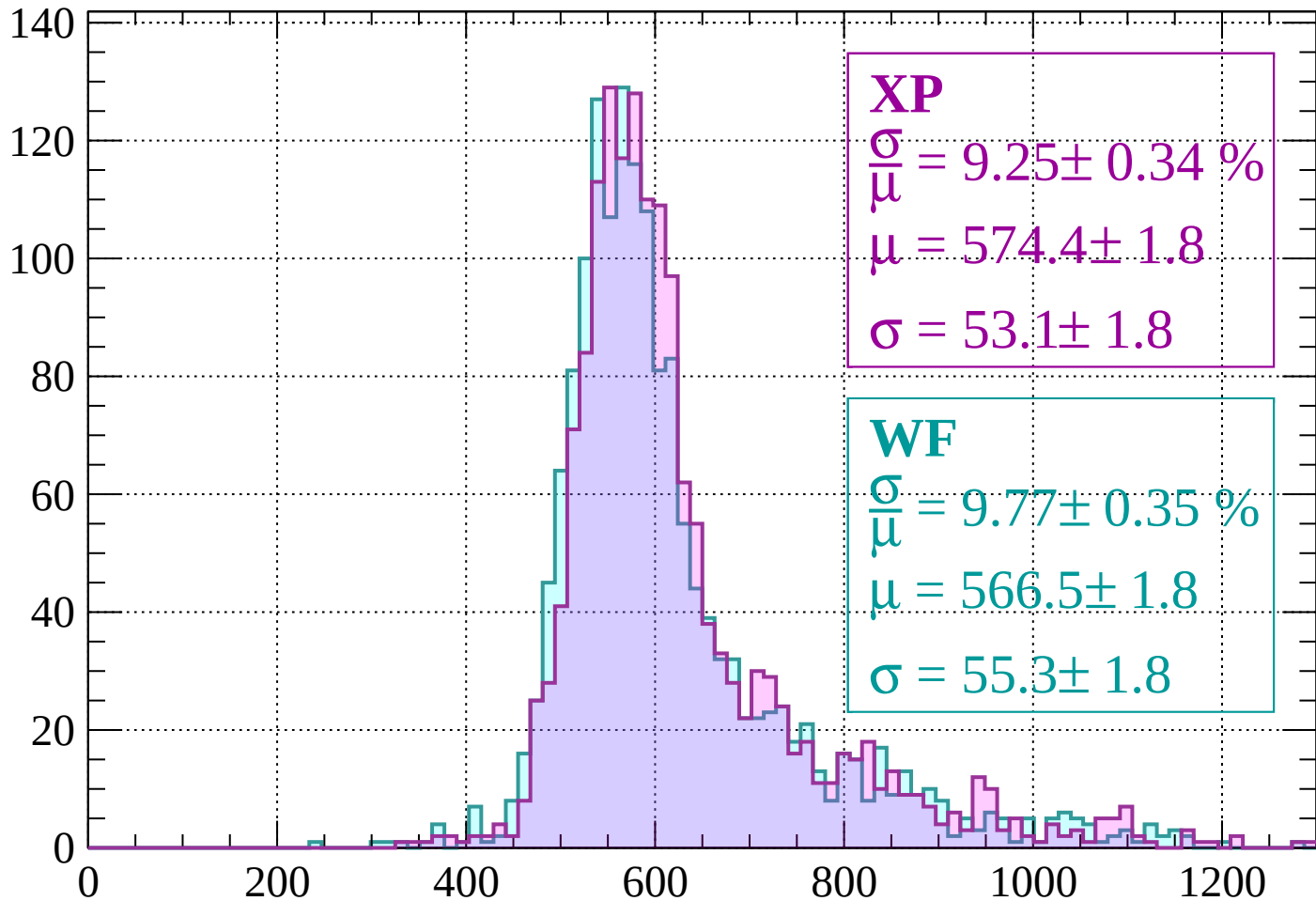
$$\mu = 569.8 \pm 2.0$$

$$\sigma = 65.7 \pm 2.0$$

dE/dx (ADC counts/cm)

Energy loss | $-80 < p < -40$

Count



XP

$$\frac{\sigma}{\mu} = 9.25 \pm 0.34 \%$$

$$\mu = 574.4 \pm 1.8$$

$$\sigma = 53.1 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 9.77 \pm 0.35 \%$$

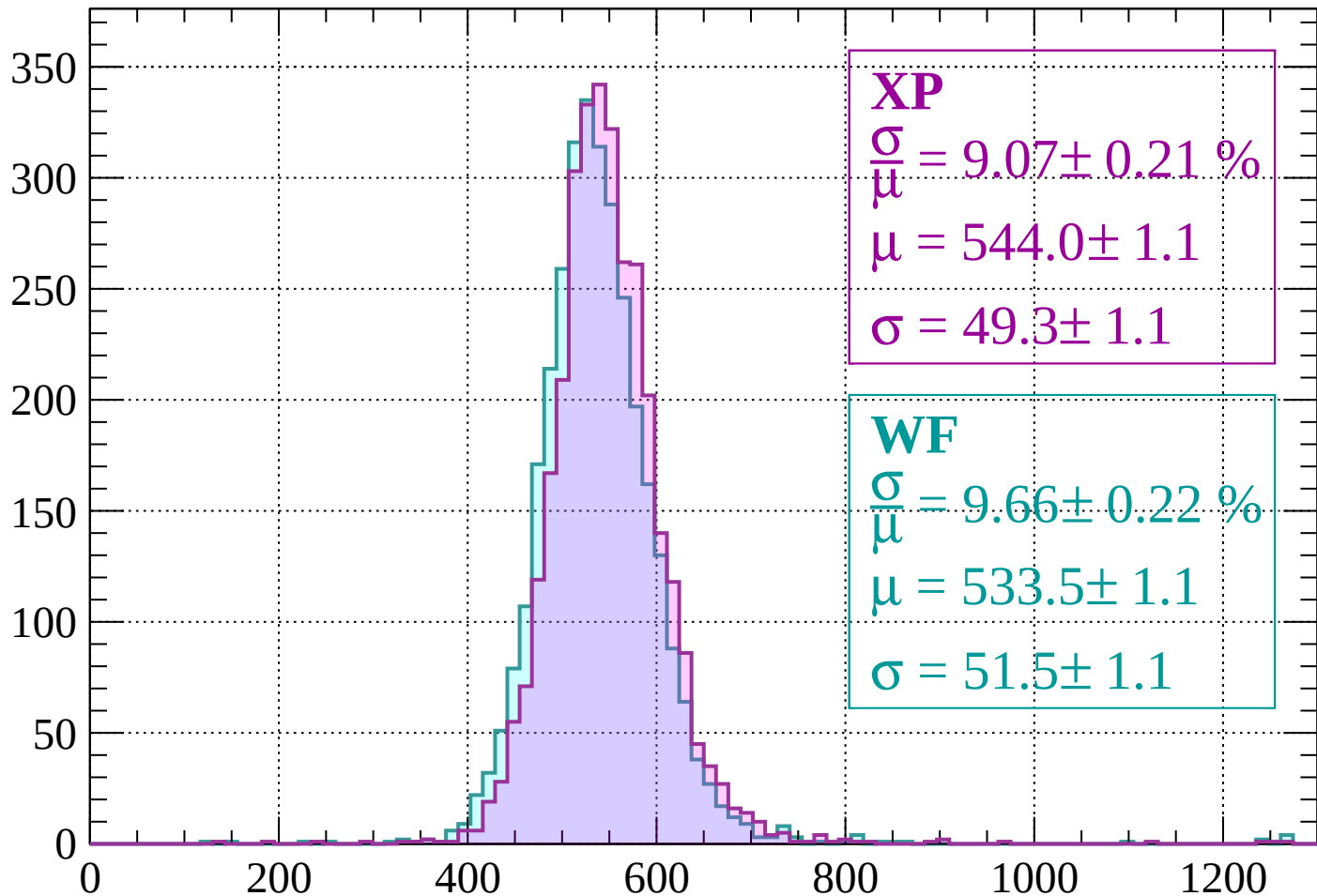
$$\mu = 566.5 \pm 1.8$$

$$\sigma = 55.3 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $-40 < p < 0$

Count



XP

$$\frac{\sigma}{\mu} = 9.07 \pm 0.21 \%$$

$$\mu = 544.0 \pm 1.1$$

$$\sigma = 49.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.66 \pm 0.22 \%$$

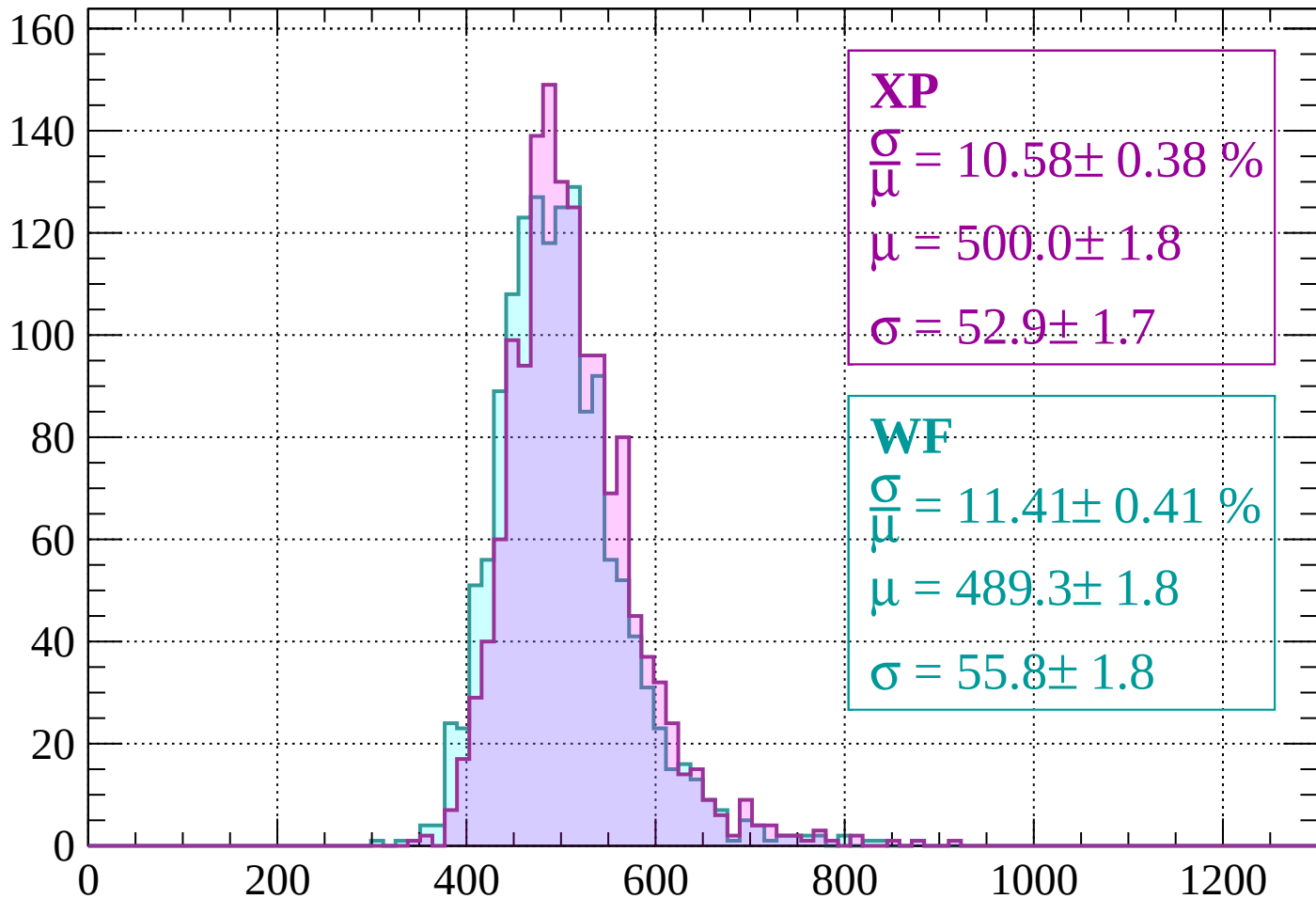
$$\mu = 533.5 \pm 1.1$$

$$\sigma = 51.5 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $0 < p < 40$

Count



XP

$$\frac{\sigma}{\mu} = 10.58 \pm 0.38 \%$$

$$\mu = 500.0 \pm 1.8$$

$$\sigma = 52.9 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 11.41 \pm 0.41 \%$$

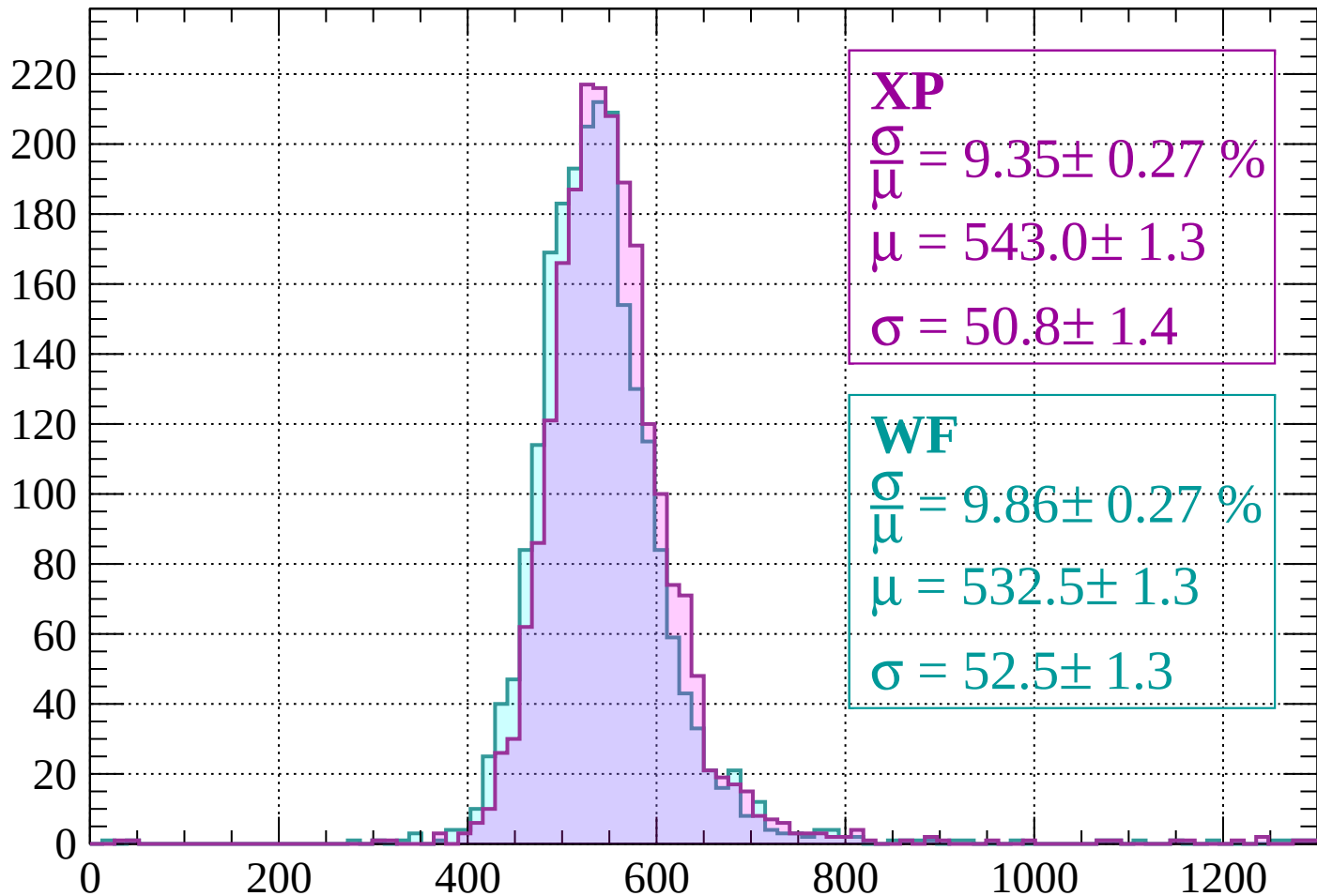
$$\mu = 489.3 \pm 1.8$$

$$\sigma = 55.8 \pm 1.8$$

dE/dx (ADC counts/cm)

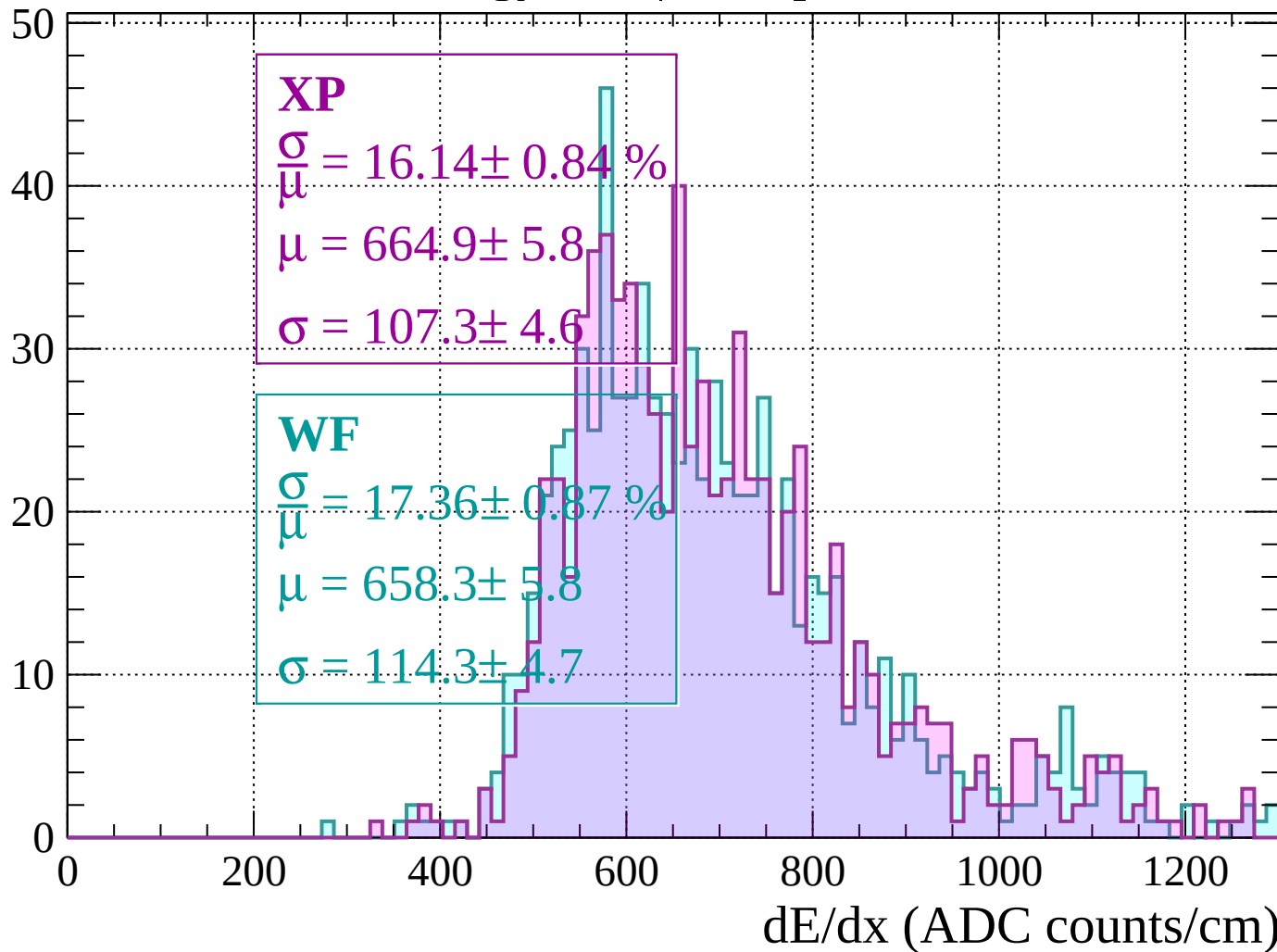
Energy loss | $40 < p < 80$

Count



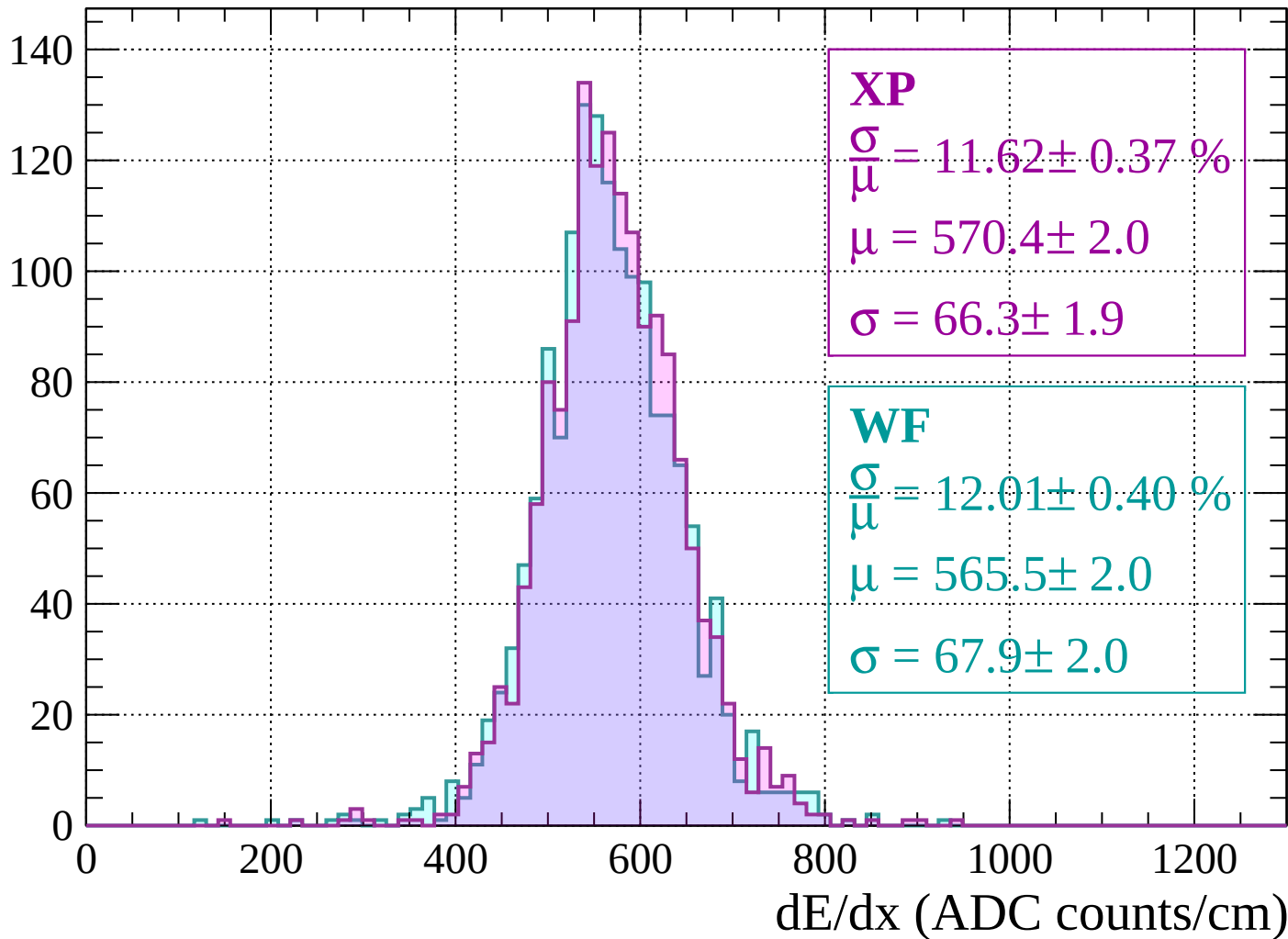
Energy loss | $80 < p < 120$

Count



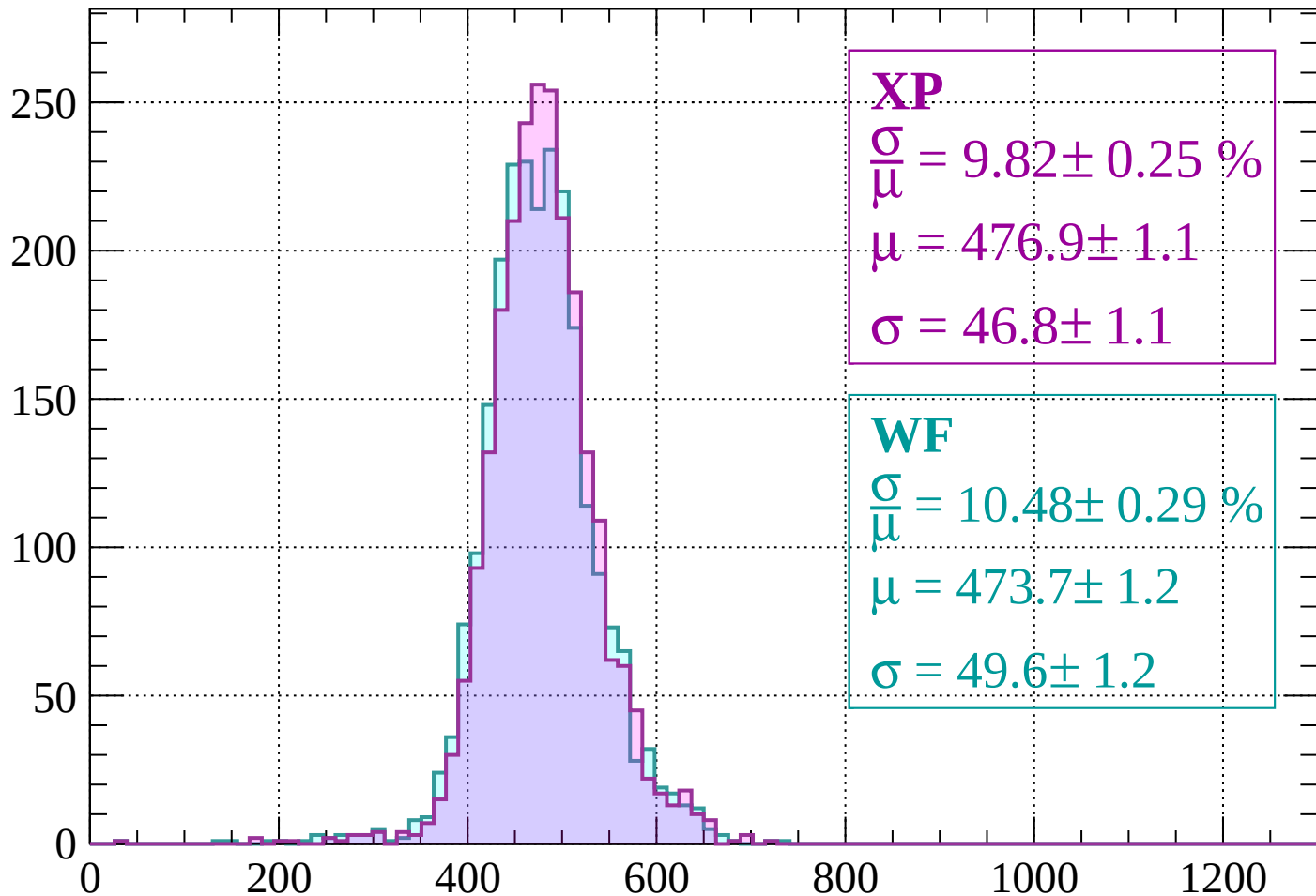
Energy loss | $120 < p < 160$

Count



Energy loss | $160 < p < 200$

Count



XP

$$\frac{\sigma}{\mu} = 9.82 \pm 0.25 \%$$

$$\mu = 476.9 \pm 1.1$$

$$\sigma = 46.8 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.48 \pm 0.29 \%$$

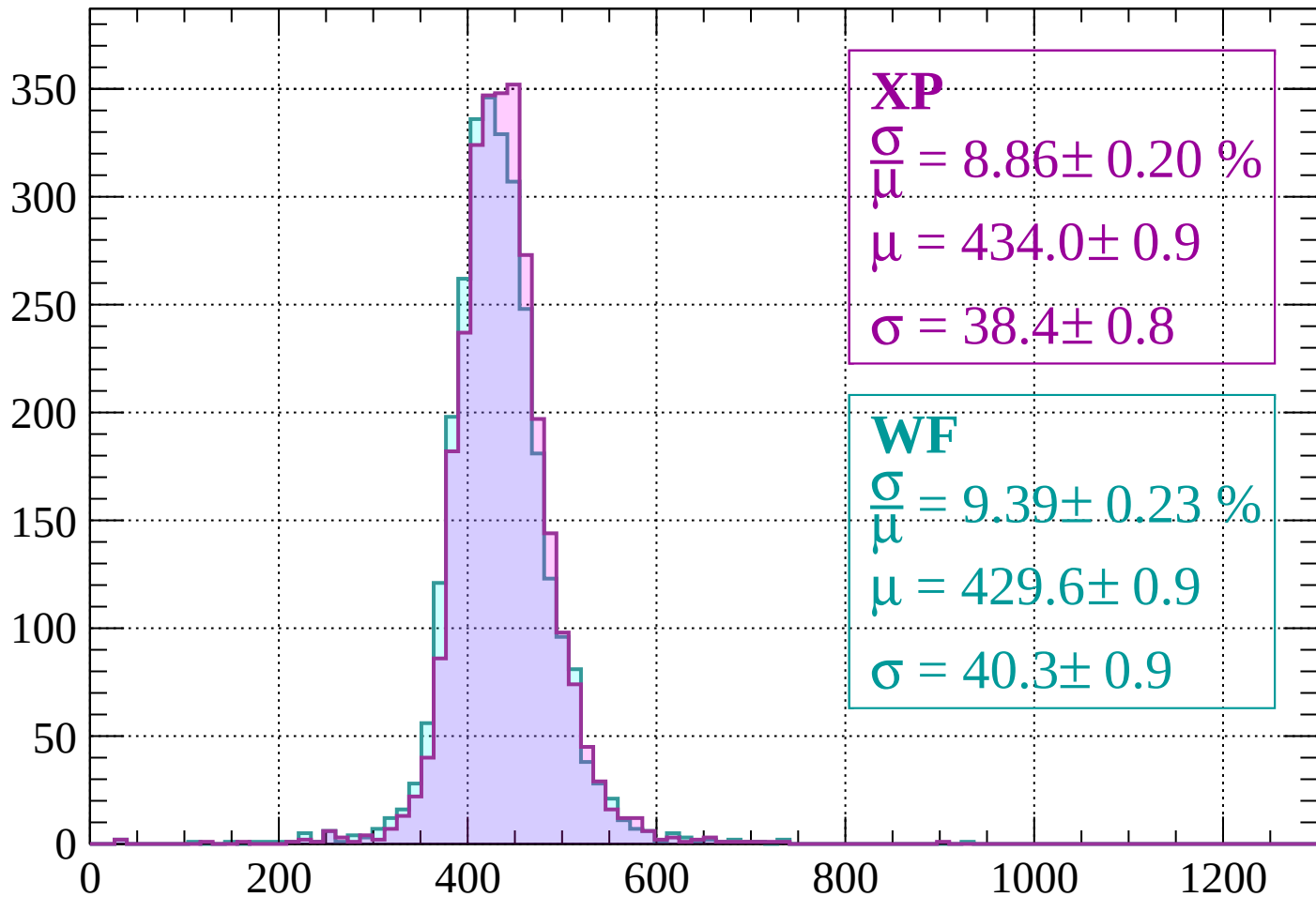
$$\mu = 473.7 \pm 1.2$$

$$\sigma = 49.6 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $200 < p < 240$

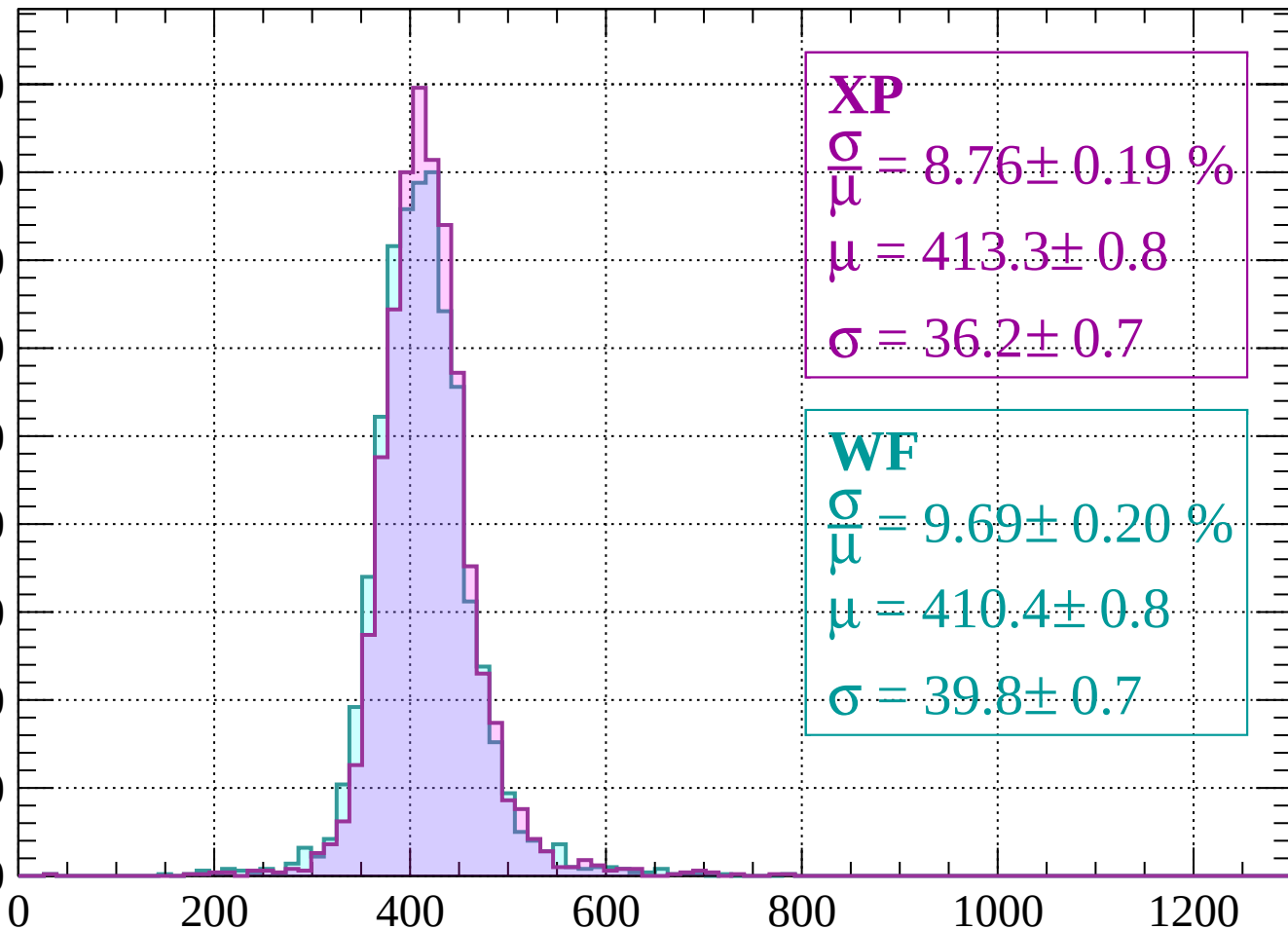
Count



Energy loss | $240 < p < 280$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 8.76 \pm 0.19 \%$$

$$\mu = 413.3 \pm 0.8$$

$$\sigma = 36.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.69 \pm 0.20 \%$$

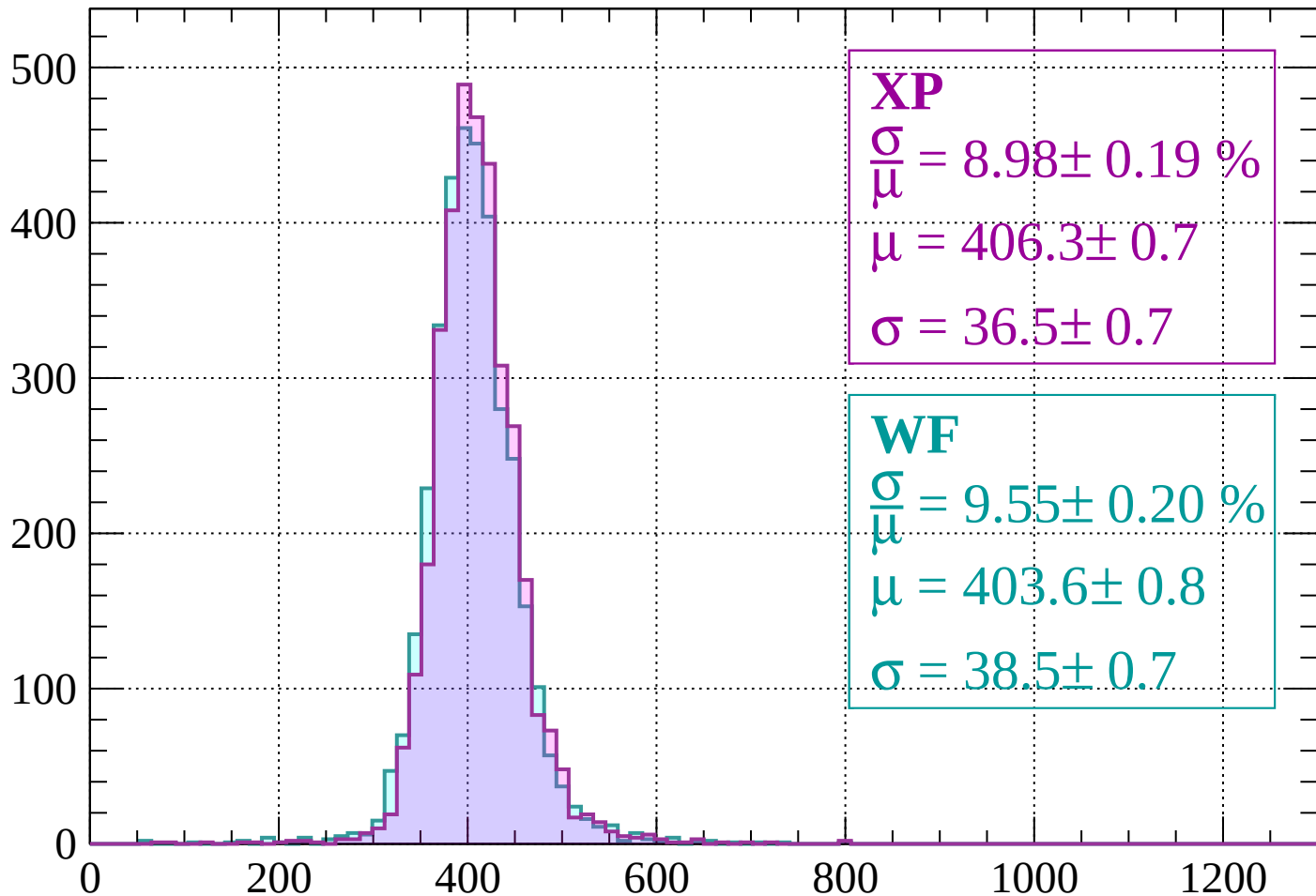
$$\mu = 410.4 \pm 0.8$$

$$\sigma = 39.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 8.98 \pm 0.19 \%$$

$$\mu = 406.3 \pm 0.7$$

$$\sigma = 36.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.55 \pm 0.20 \%$$

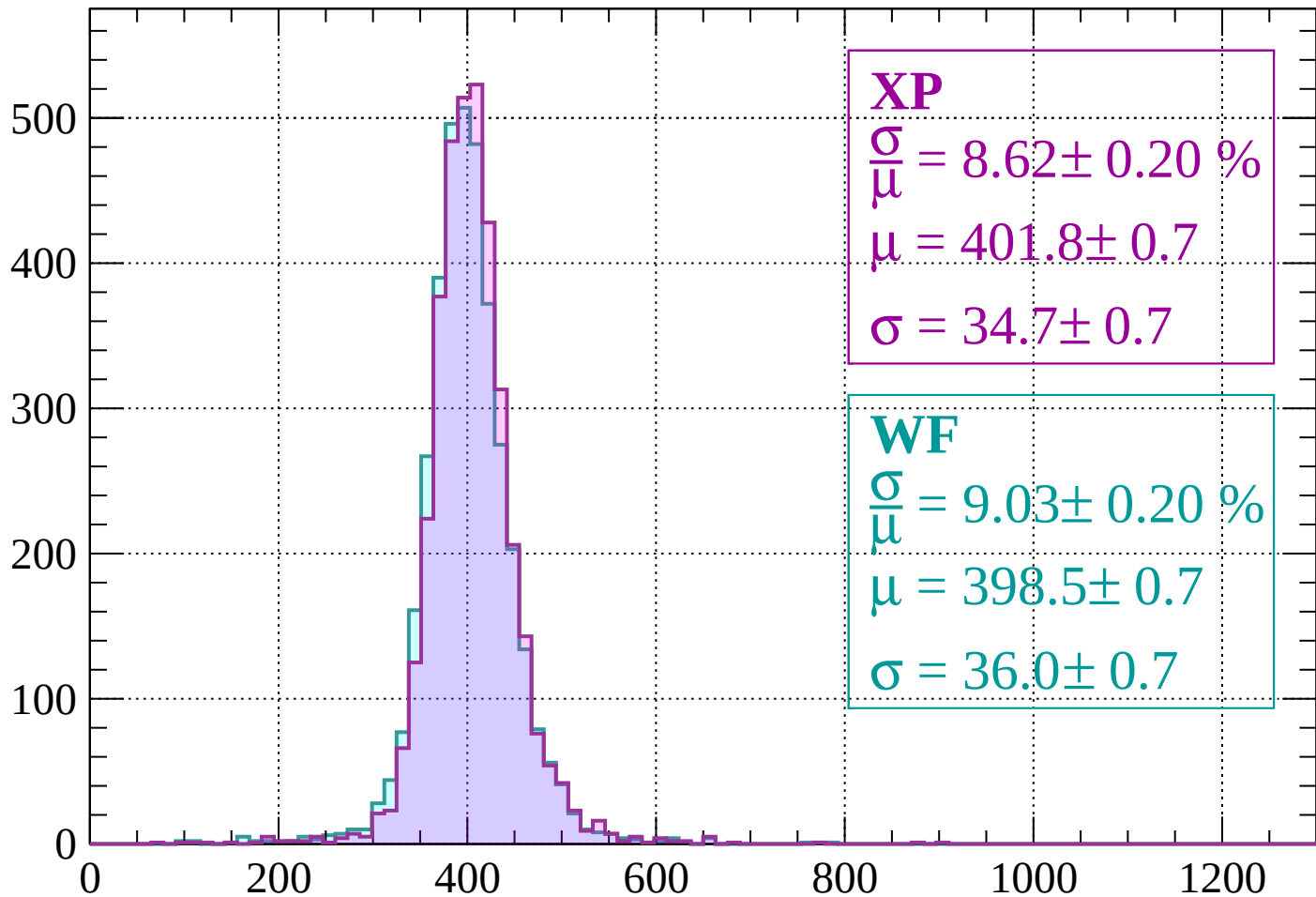
$$\mu = 403.6 \pm 0.8$$

$$\sigma = 38.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

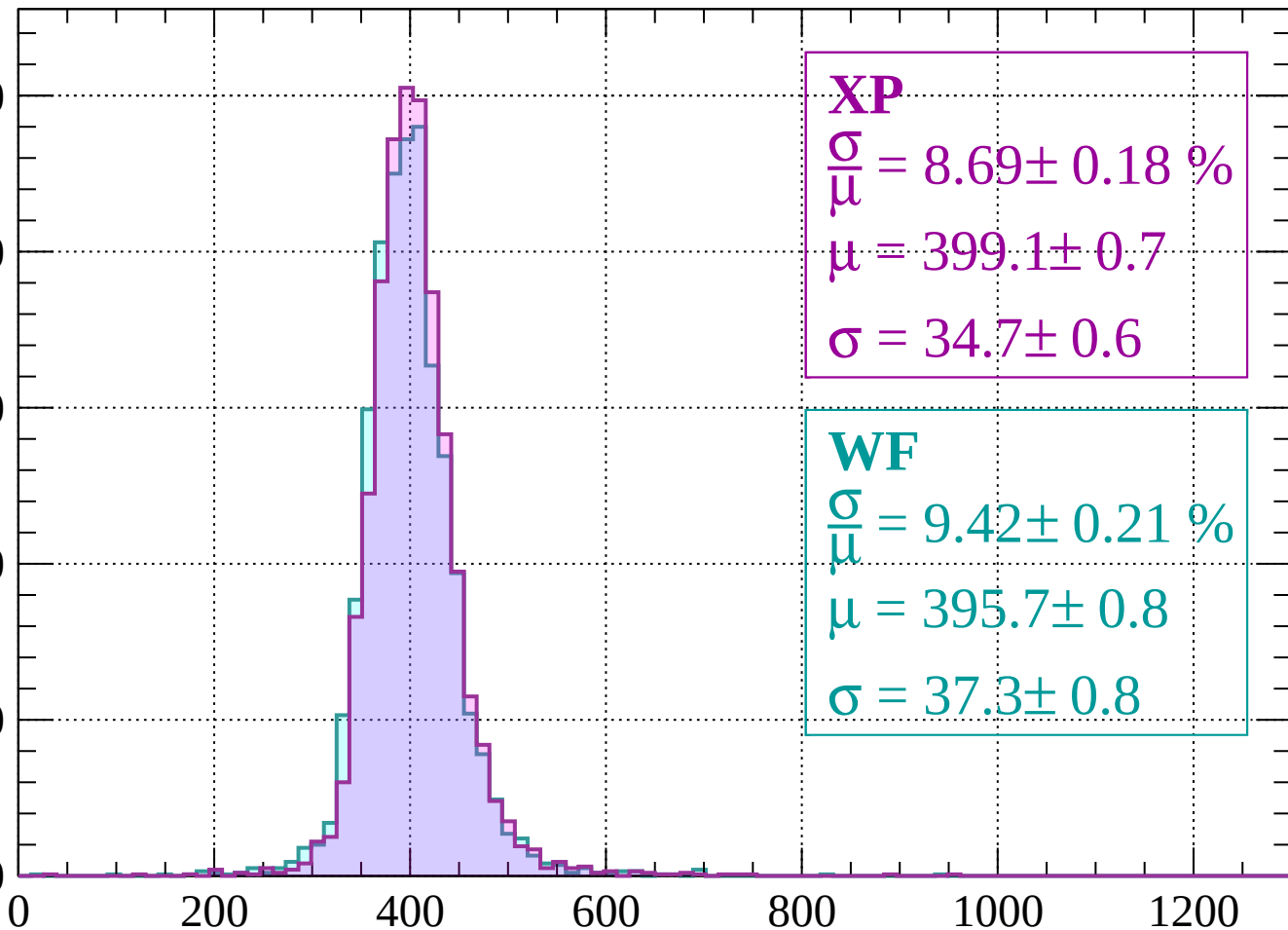
Count



Energy loss | $360 < p < 400$

Count

500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 8.69 \pm 0.18 \%$$

$$\mu = 399.1 \pm 0.7$$

$$\sigma = 34.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.42 \pm 0.21 \%$$

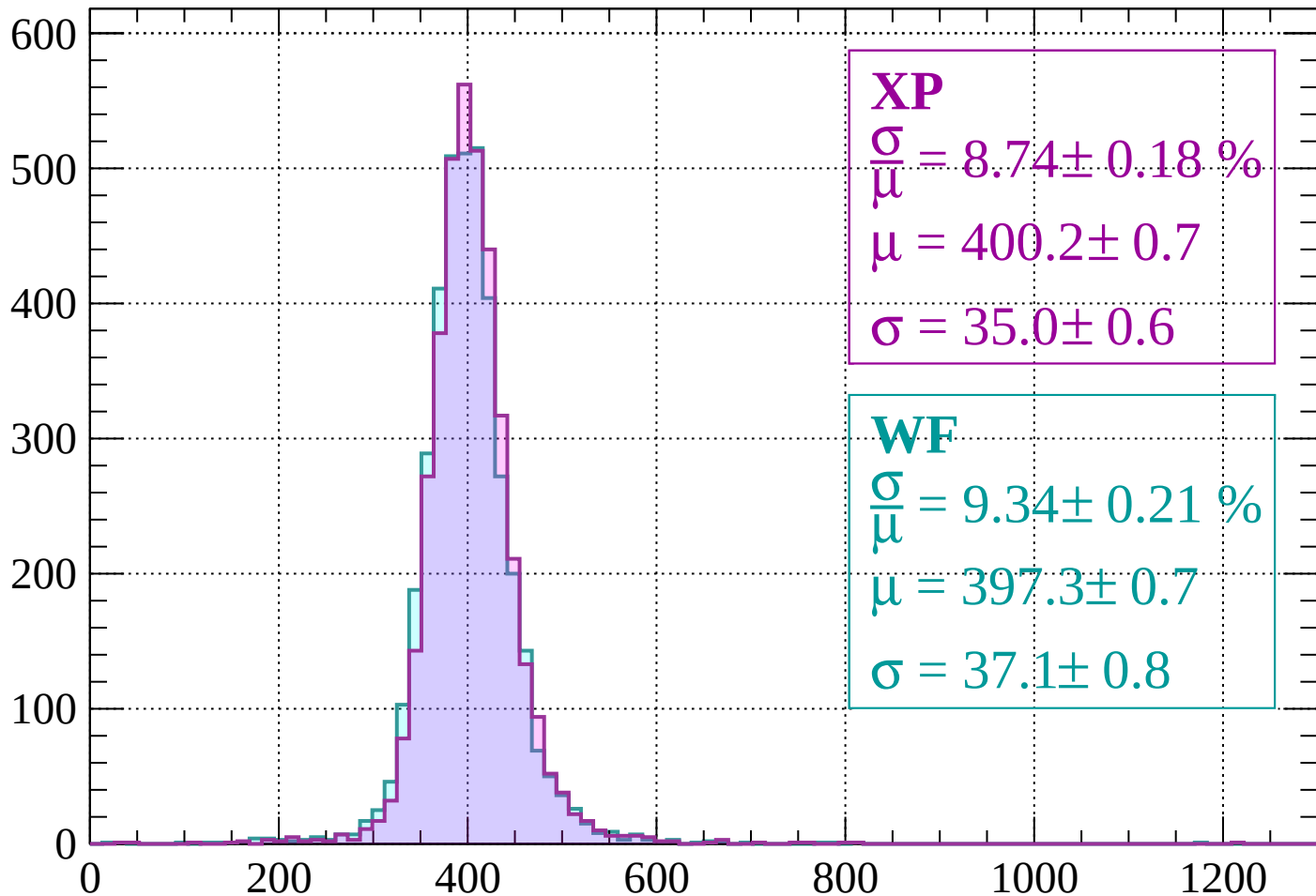
$$\mu = 395.7 \pm 0.8$$

$$\sigma = 37.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $400 < p < 440$

Count



XP

$$\frac{\sigma}{\mu} = 8.74 \pm 0.18 \%$$

$$\mu = 400.2 \pm 0.7$$

$$\sigma = 35.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.34 \pm 0.21 \%$$

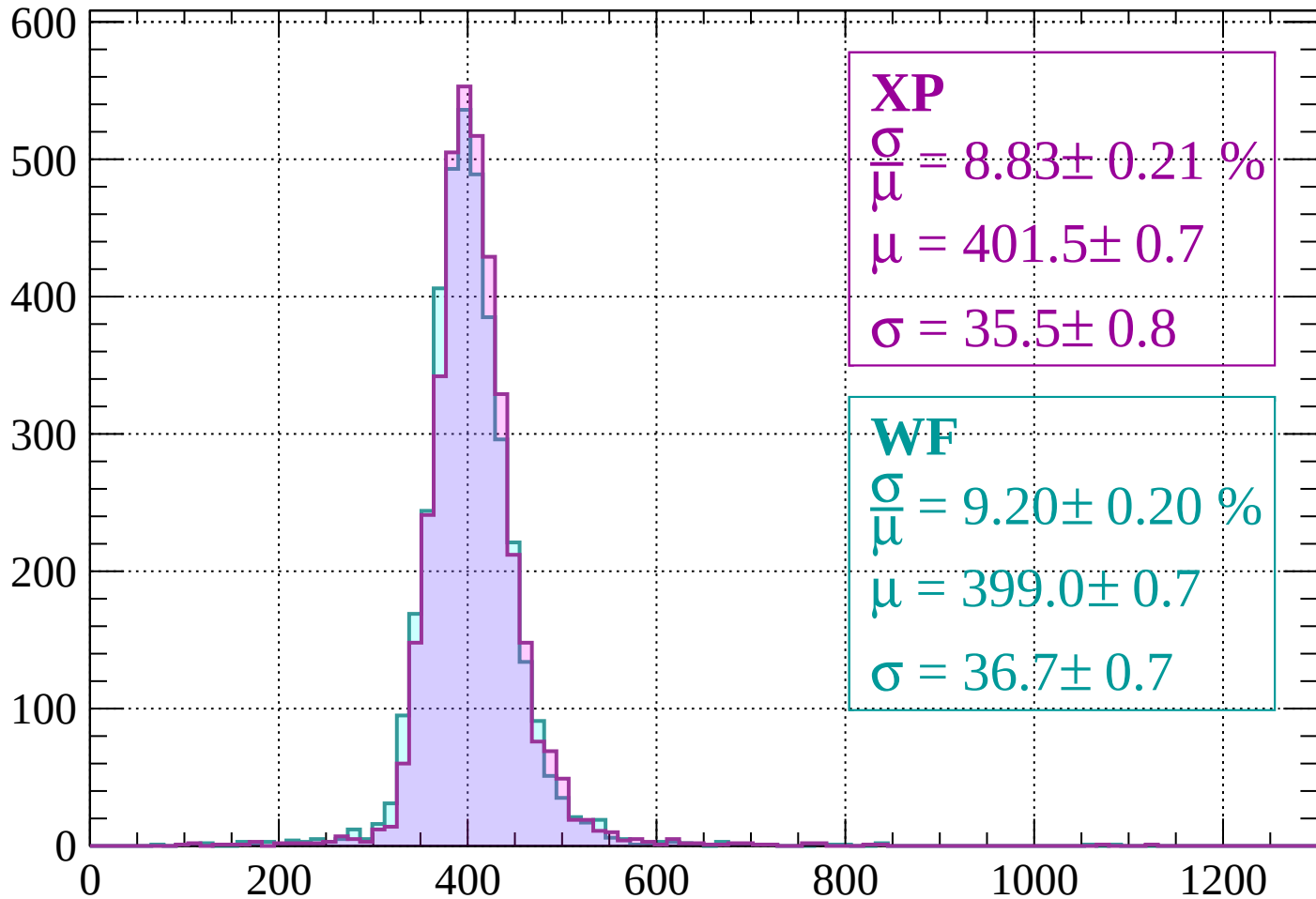
$$\mu = 397.3 \pm 0.7$$

$$\sigma = 37.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $440 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 8.83 \pm 0.21 \%$$

$$\mu = 401.5 \pm 0.7$$

$$\sigma = 35.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.20 \pm 0.20 \%$$

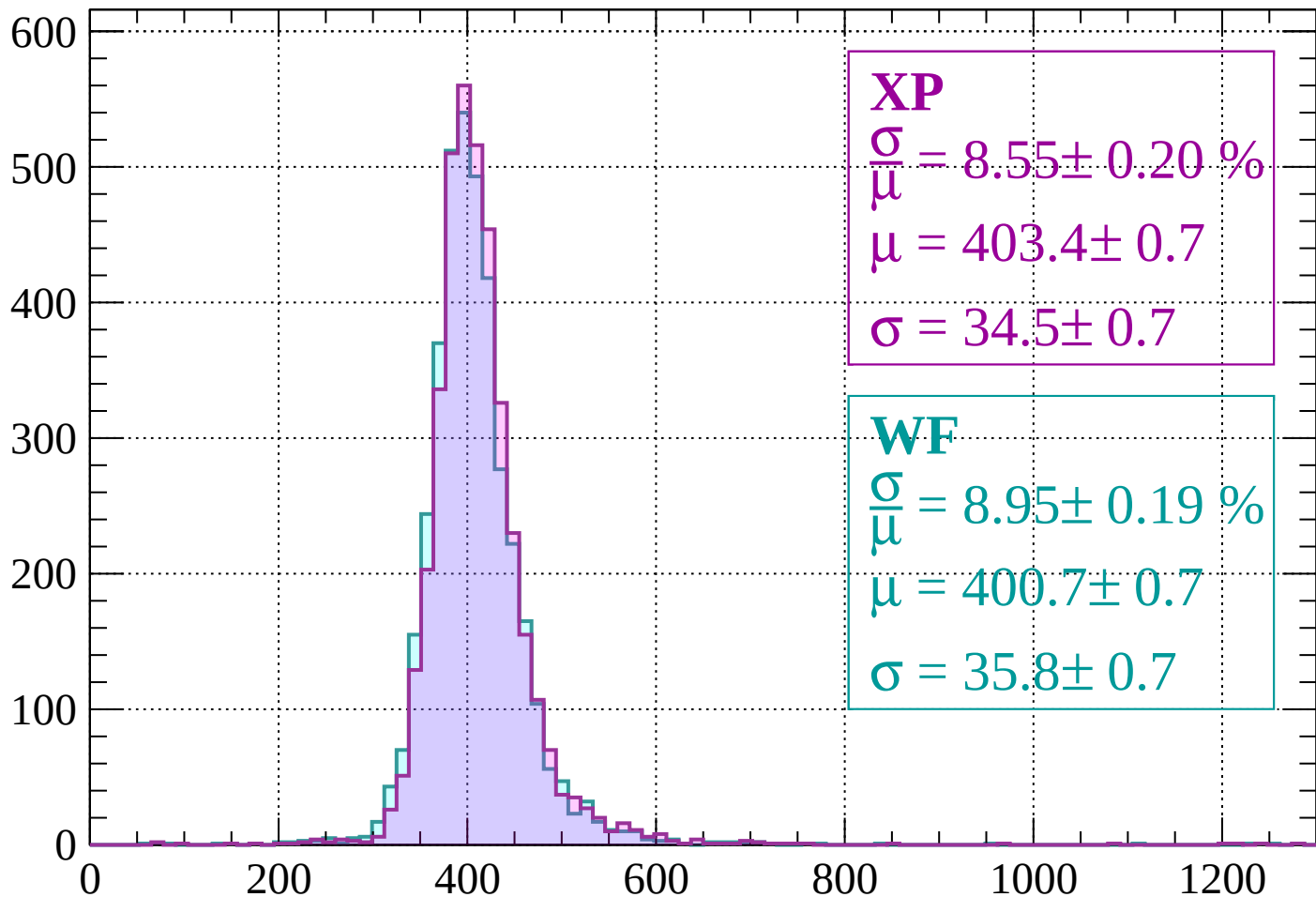
$$\mu = 399.0 \pm 0.7$$

$$\sigma = 36.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 520$

Count



XP

$$\frac{\sigma}{\mu} = 8.55 \pm 0.20 \%$$

$$\mu = 403.4 \pm 0.7$$

$$\sigma = 34.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.95 \pm 0.19 \%$$

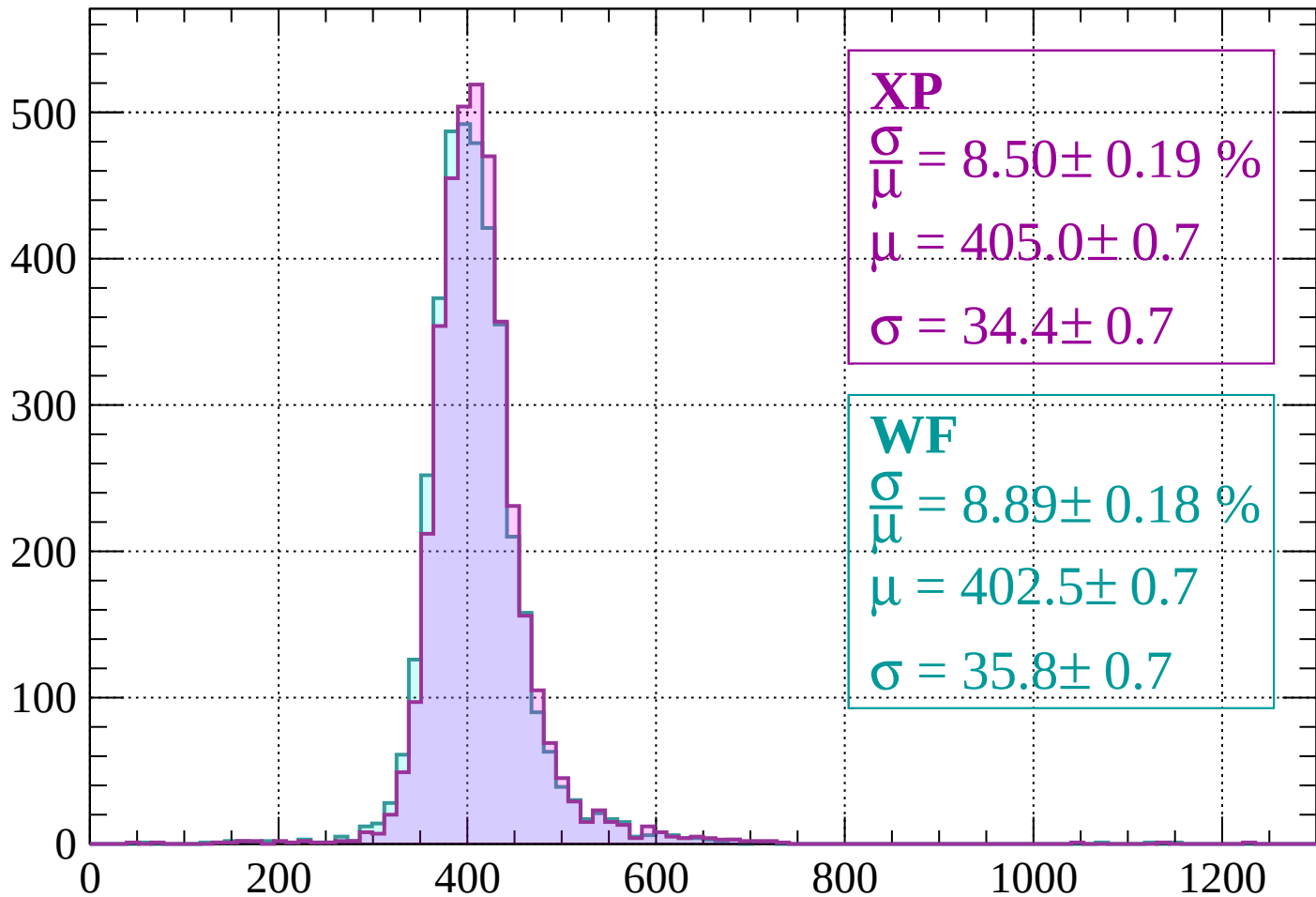
$$\mu = 400.7 \pm 0.7$$

$$\sigma = 35.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $520 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 8.50 \pm 0.19 \%$$

$$\mu = 405.0 \pm 0.7$$

$$\sigma = 34.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.89 \pm 0.18 \%$$

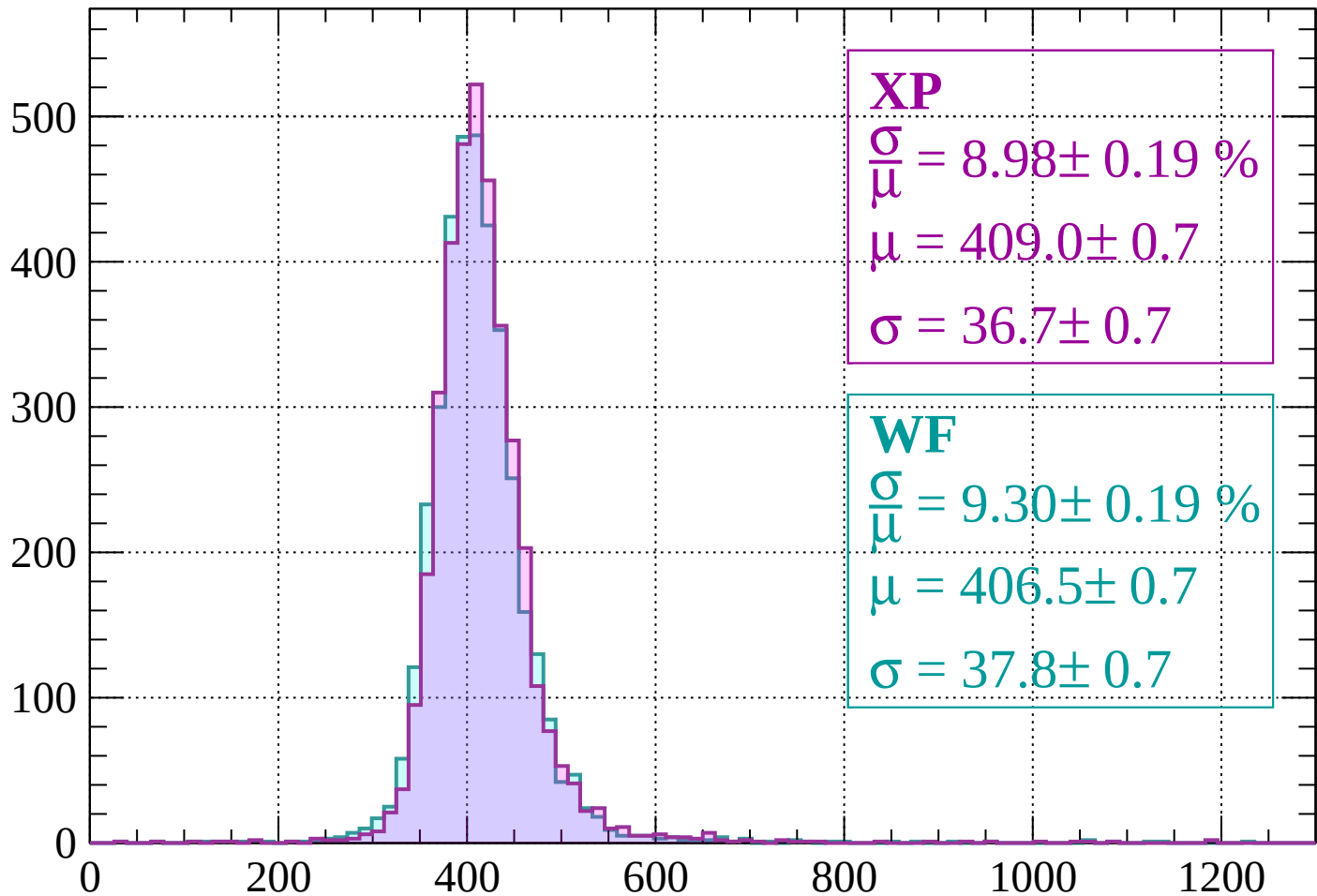
$$\mu = 402.5 \pm 0.7$$

$$\sigma = 35.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 600$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.98 \pm 0.19 \%$$

$$\mu = 409.0 \pm 0.7$$

$$\sigma = 36.7 \pm 0.7$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.30 \pm 0.19 \%$$

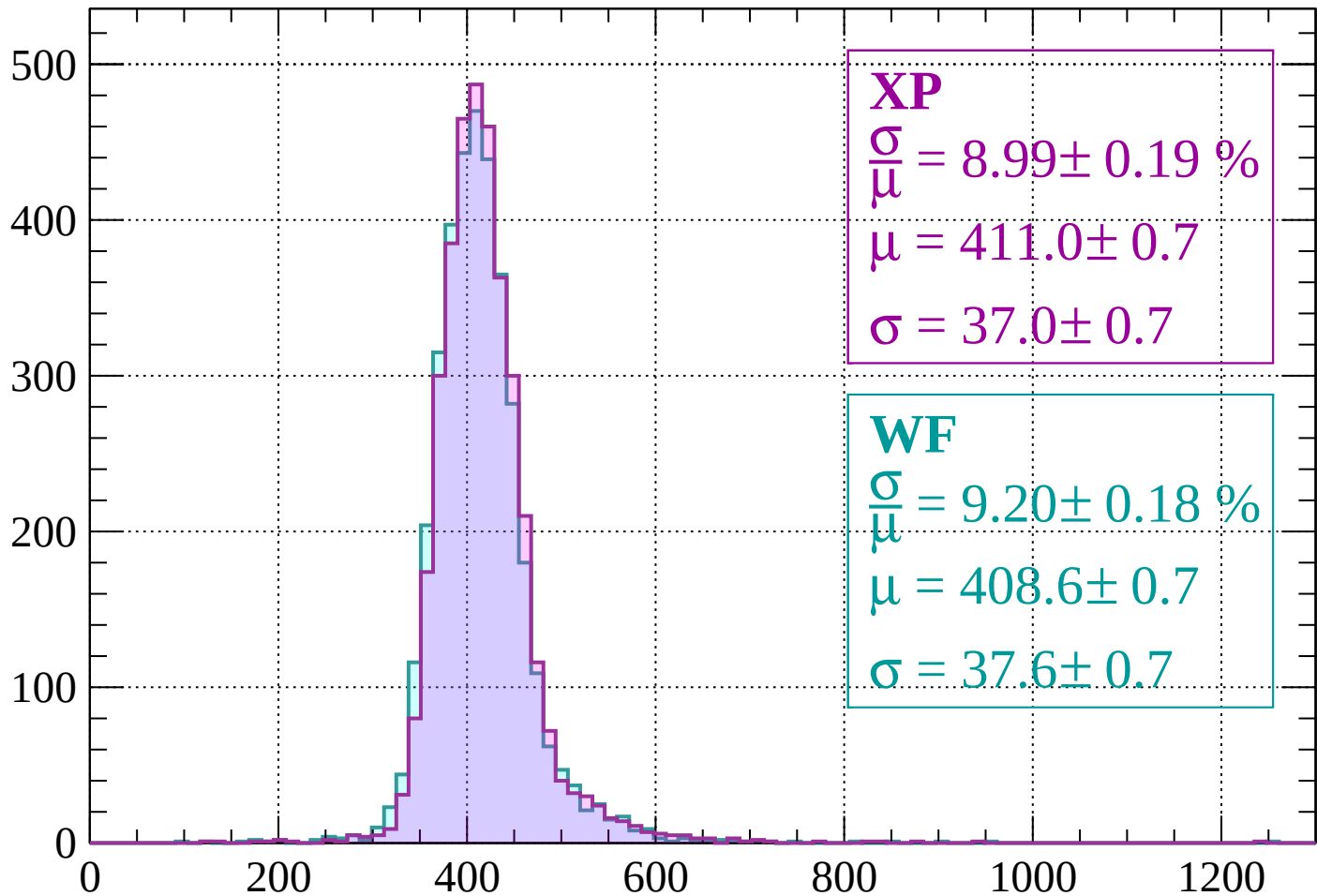
$$\mu = 406.5 \pm 0.7$$

$$\sigma = 37.8 \pm 0.7$$

dE/dx (ADC counts/cm)

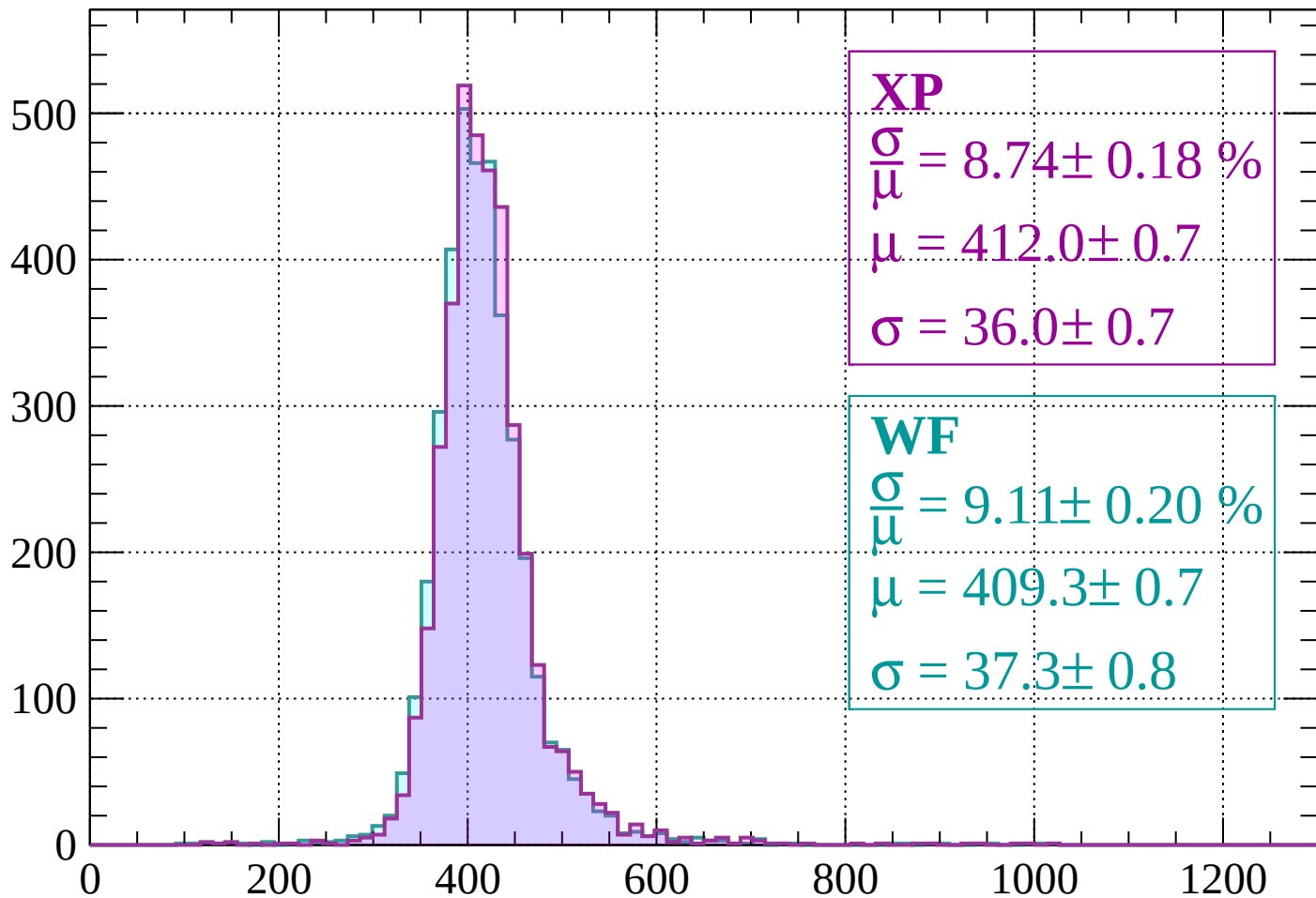
Energy loss | $600 < p < 640$

Count



Energy loss | $640 < p < 680$

Count



XP

$$\frac{\sigma}{\mu} = 8.74 \pm 0.18 \%$$

$$\mu = 412.0 \pm 0.7$$

$$\sigma = 36.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.11 \pm 0.20 \%$$

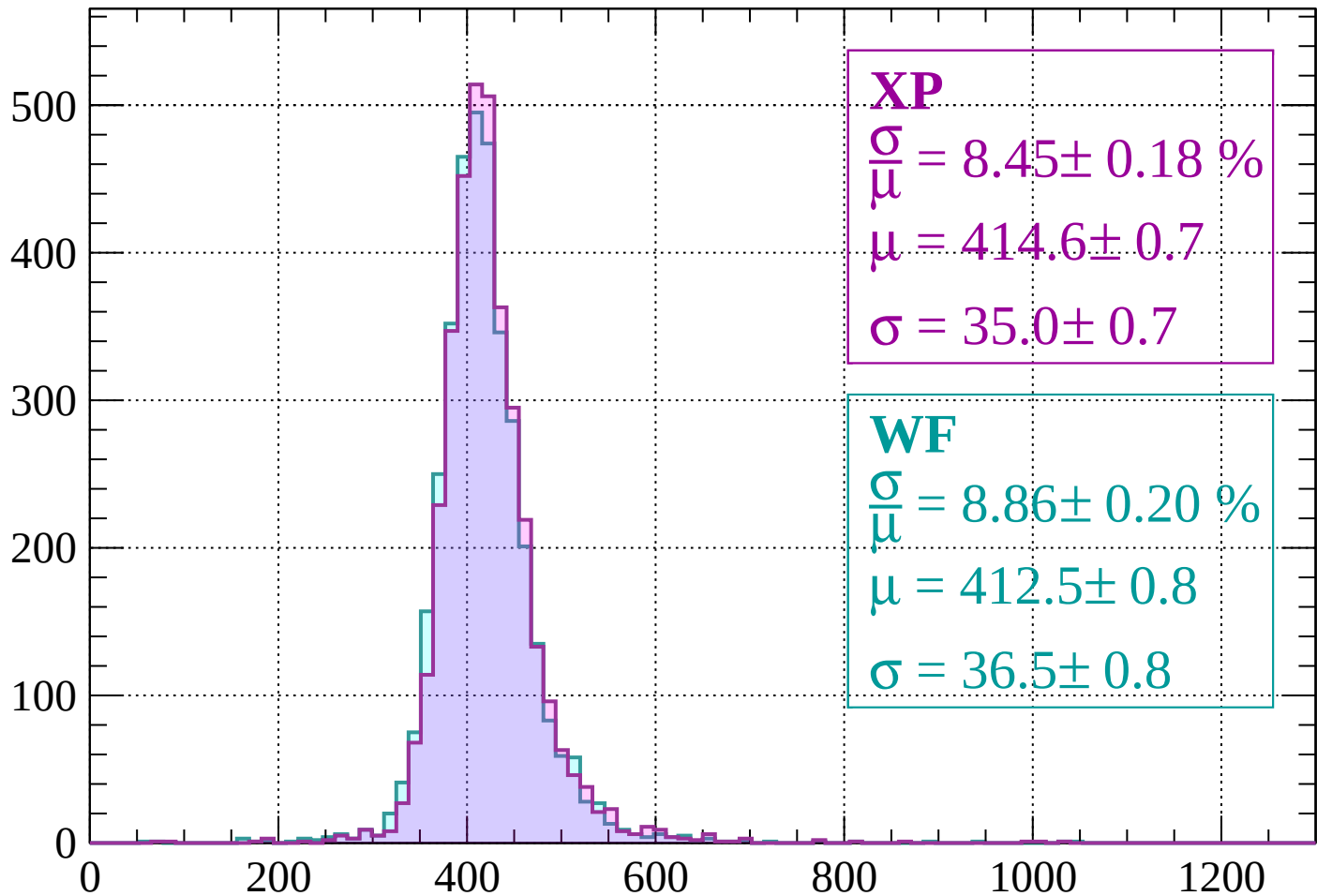
$$\mu = 409.3 \pm 0.7$$

$$\sigma = 37.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $680 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 8.45 \pm 0.18 \%$$

$$\mu = 414.6 \pm 0.7$$

$$\sigma = 35.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.86 \pm 0.20 \%$$

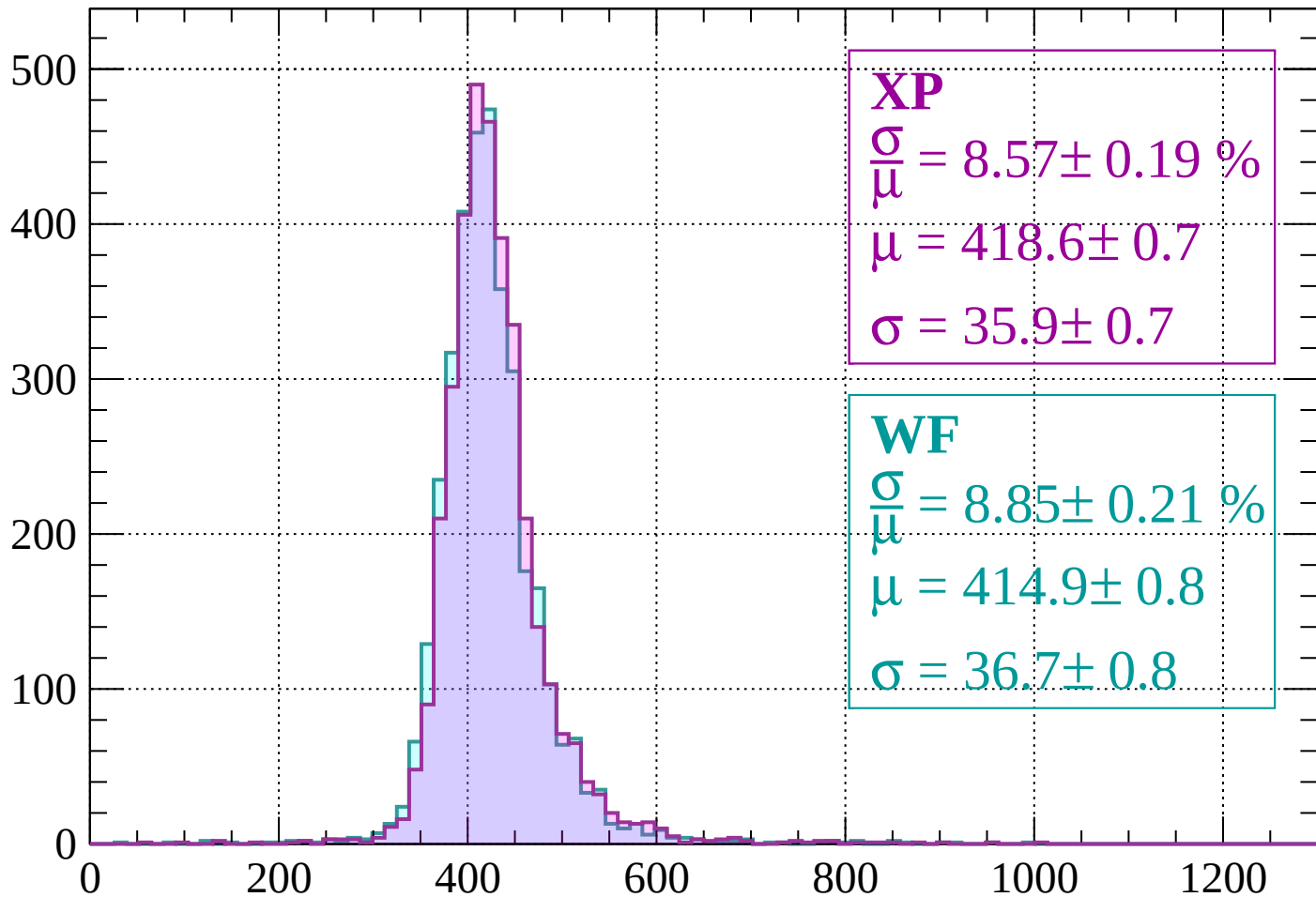
$$\mu = 412.5 \pm 0.8$$

$$\sigma = 36.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 760$

Count



XP

$$\frac{\sigma}{\mu} = 8.57 \pm 0.19 \%$$

$$\mu = 418.6 \pm 0.7$$

$$\sigma = 35.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.85 \pm 0.21 \%$$

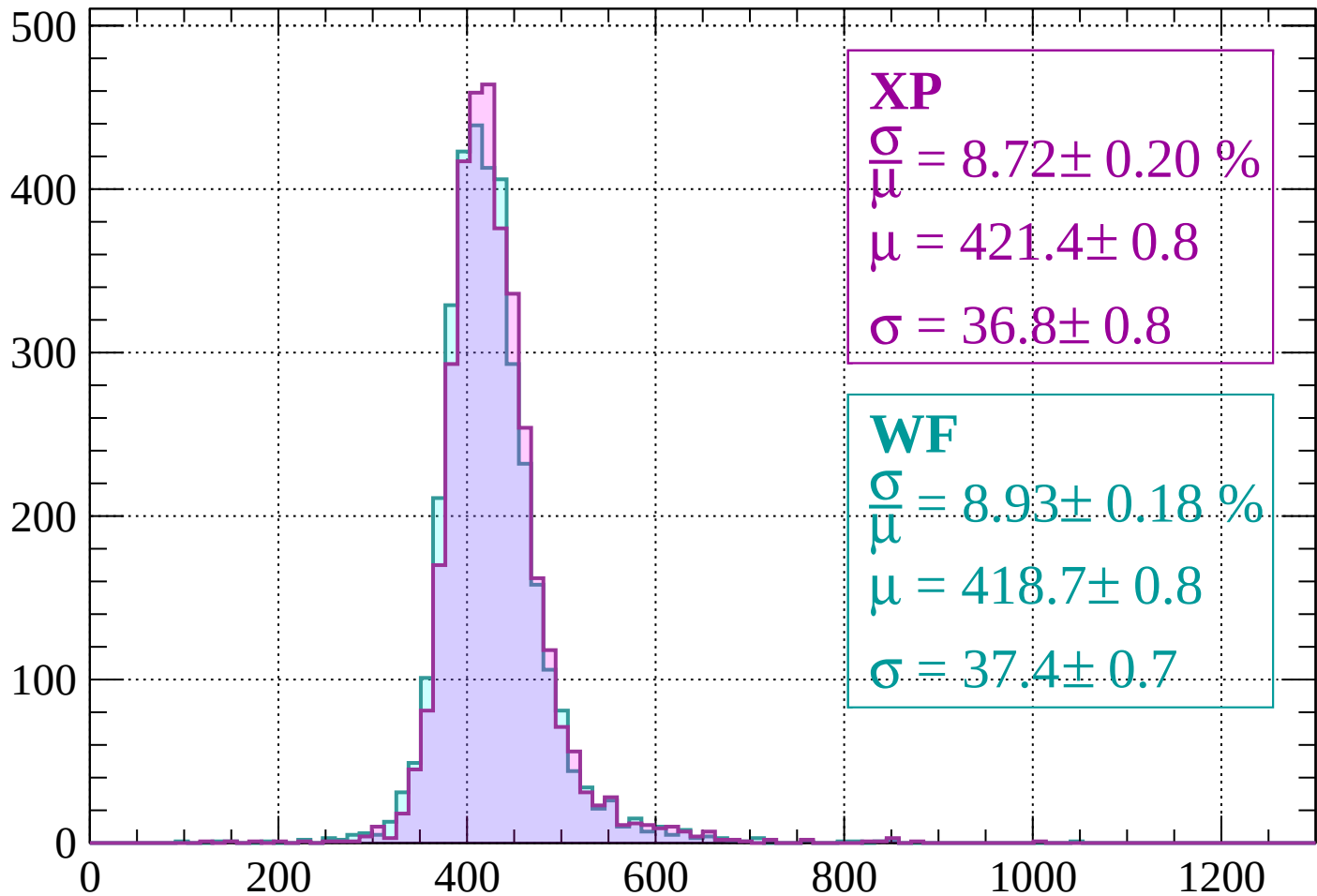
$$\mu = 414.9 \pm 0.8$$

$$\sigma = 36.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $760 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 8.72 \pm 0.20 \%$$

$$\mu = 421.4 \pm 0.8$$

$$\sigma = 36.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.93 \pm 0.18 \%$$

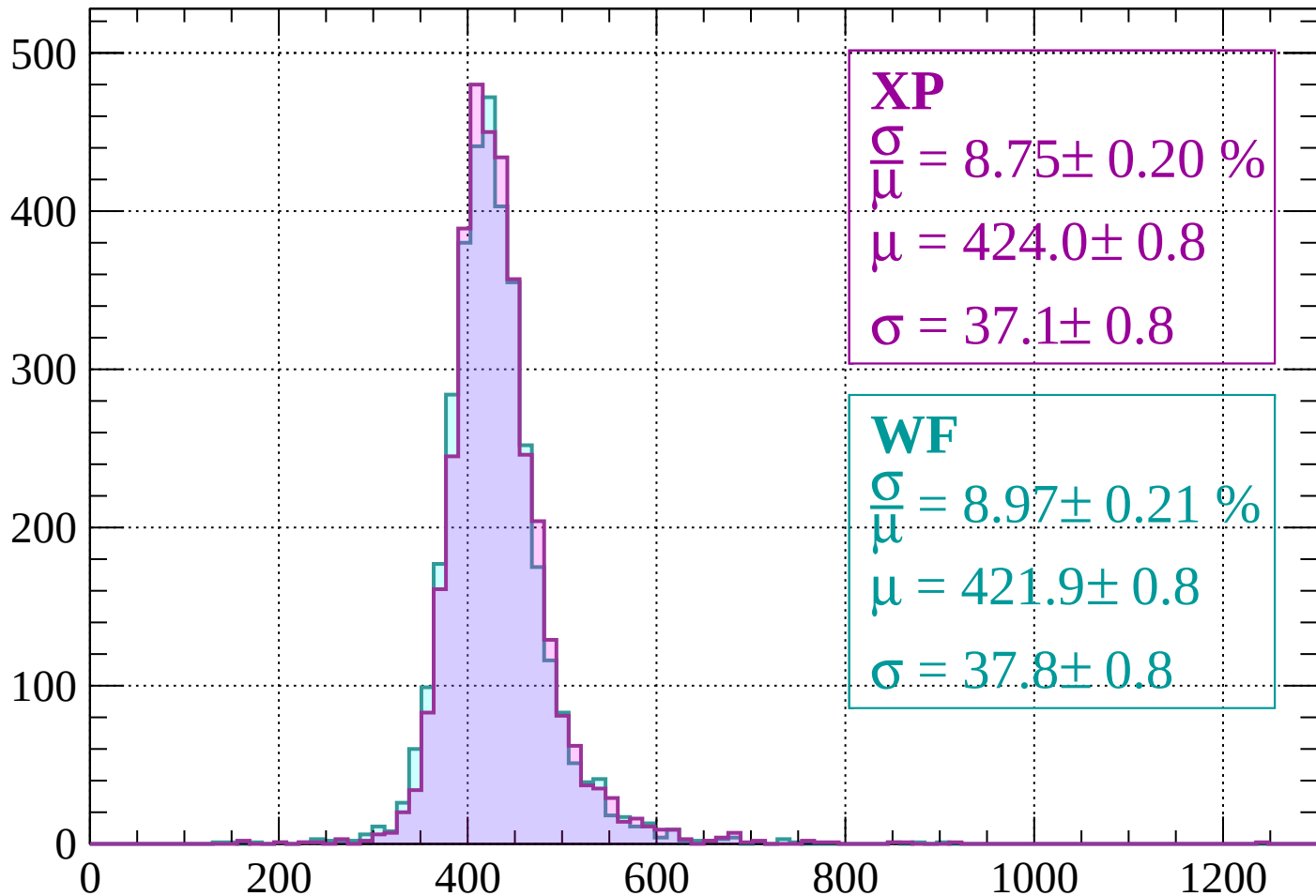
$$\mu = 418.7 \pm 0.8$$

$$\sigma = 37.4 \pm 0.7$$

dE/dx (ADC counts/cm)

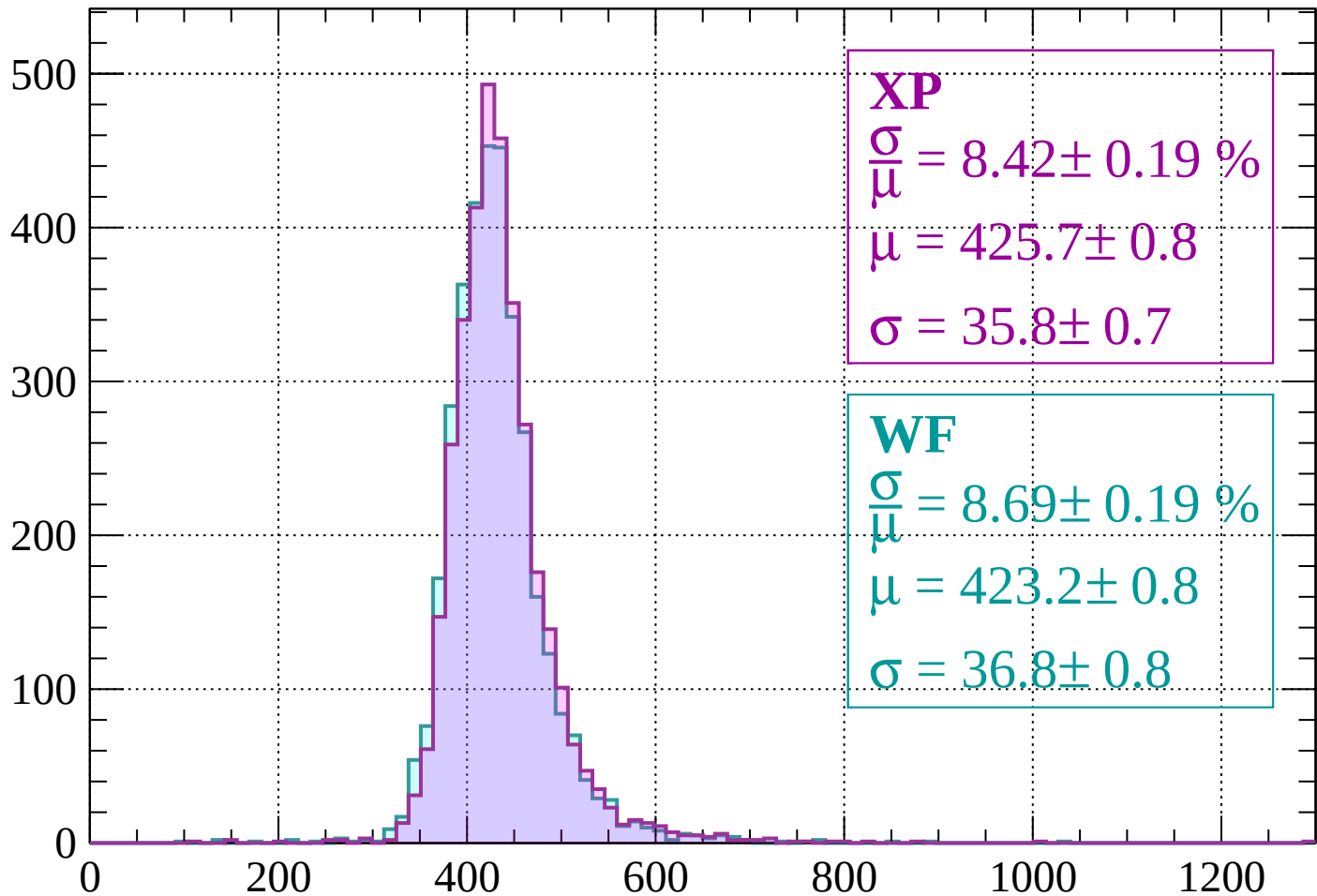
Energy loss | $800 < p < 840$

Count



Energy loss | $840 < p < 880$

Count



XP

$$\frac{\sigma}{\mu} = 8.42 \pm 0.19 \%$$

$$\mu = 425.7 \pm 0.8$$

$$\sigma = 35.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.69 \pm 0.19 \%$$

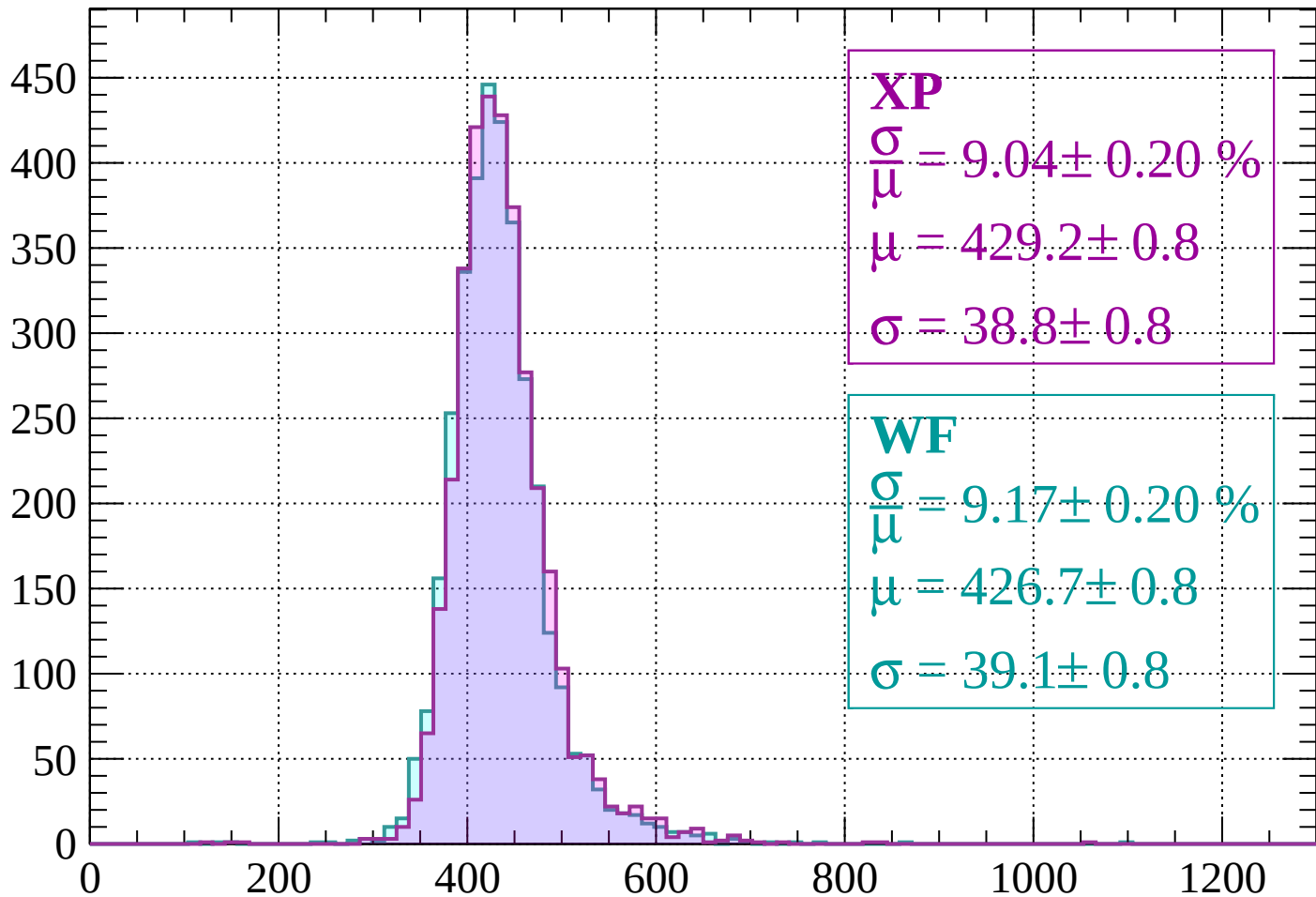
$$\mu = 423.2 \pm 0.8$$

$$\sigma = 36.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $880 < p < 920$

Count



XP

$$\frac{\sigma}{\mu} = 9.04 \pm 0.20 \%$$

$$\mu = 429.2 \pm 0.8$$

$$\sigma = 38.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.17 \pm 0.20 \%$$

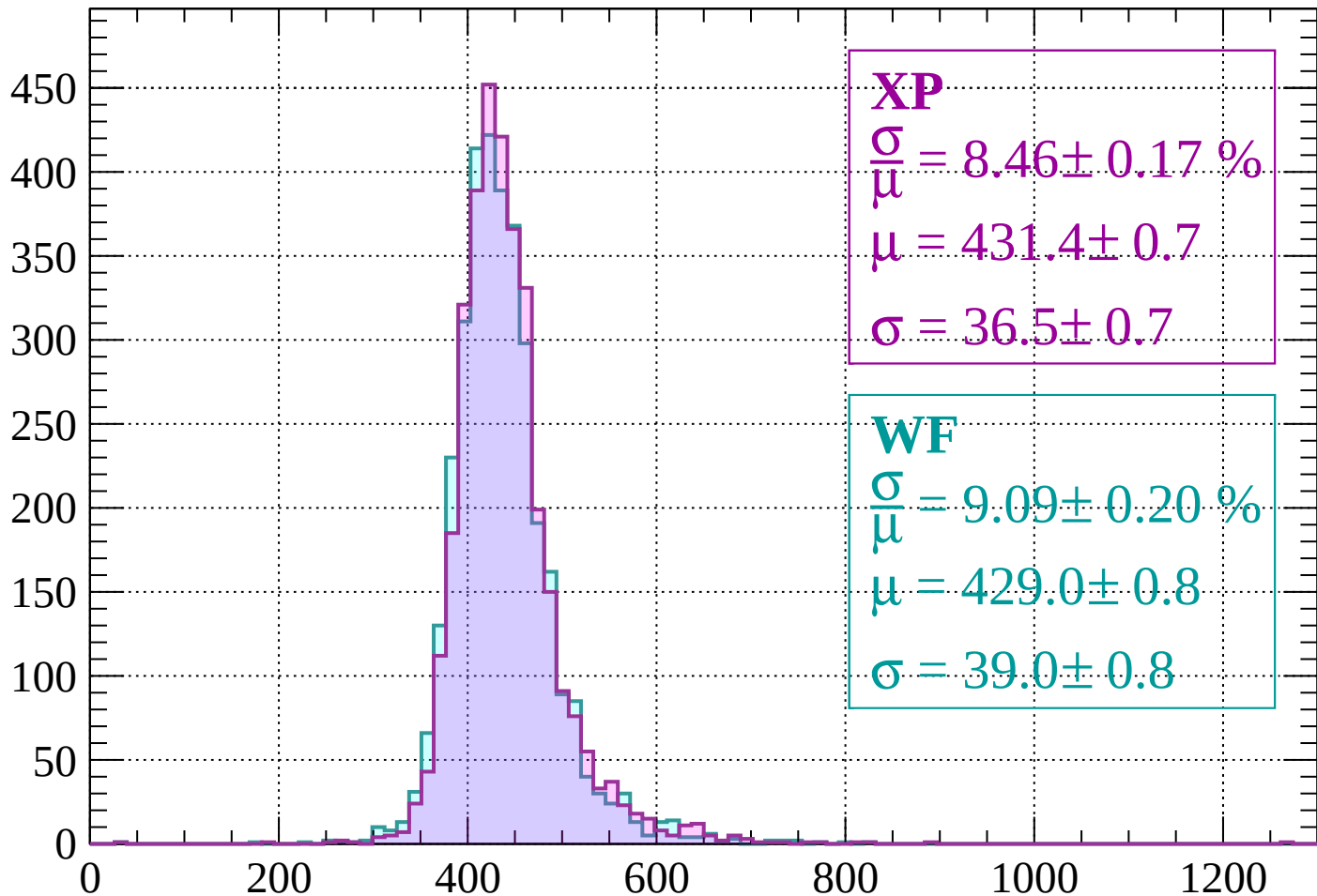
$$\mu = 426.7 \pm 0.8$$

$$\sigma = 39.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $920 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 8.46 \pm 0.17 \%$$

$$\mu = 431.4 \pm 0.7$$

$$\sigma = 36.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.09 \pm 0.20 \%$$

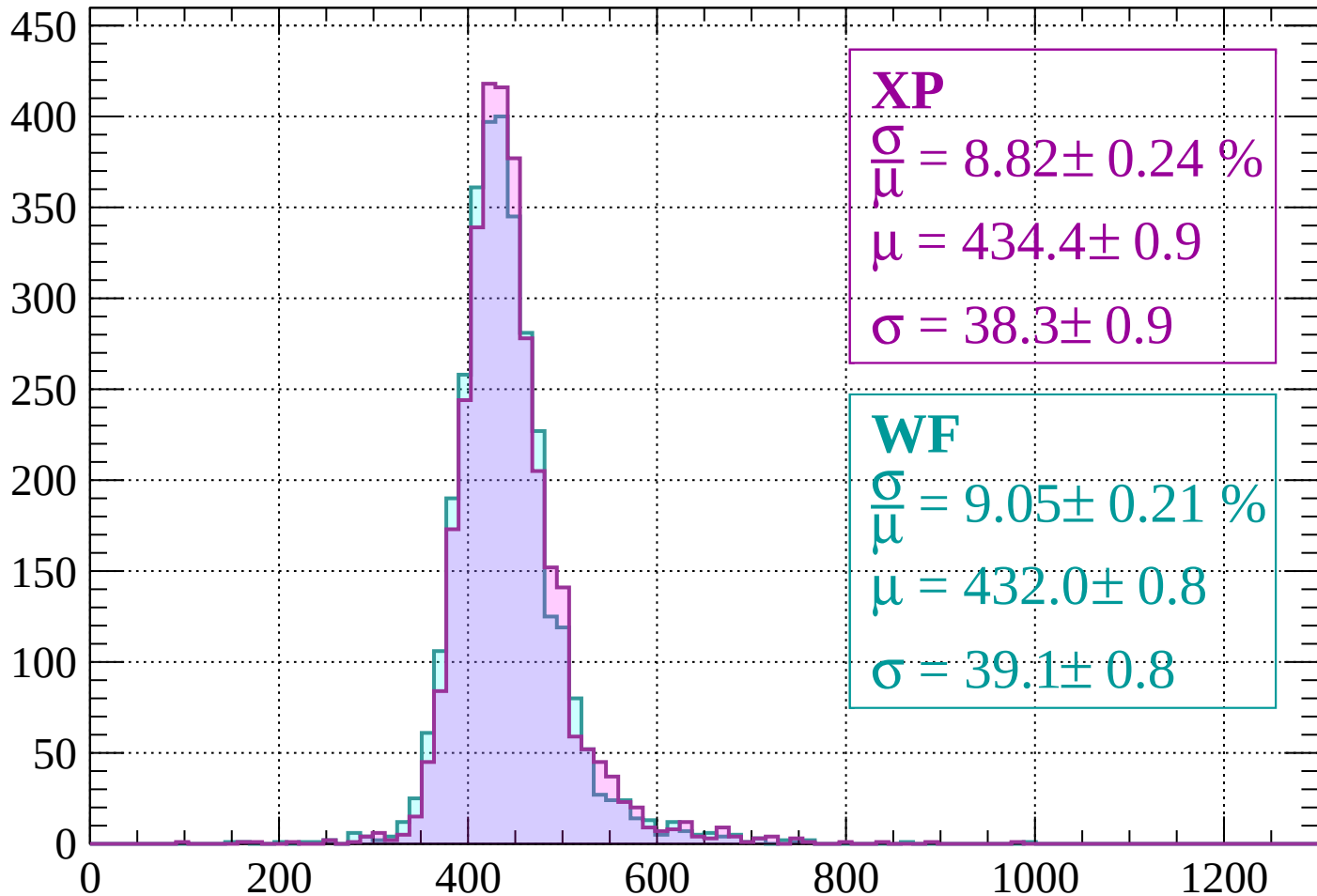
$$\mu = 429.0 \pm 0.8$$

$$\sigma = 39.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1000$

Count



XP

$$\frac{\sigma}{\mu} = 8.82 \pm 0.24 \%$$

$$\mu = 434.4 \pm 0.9$$

$$\sigma = 38.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.05 \pm 0.21 \%$$

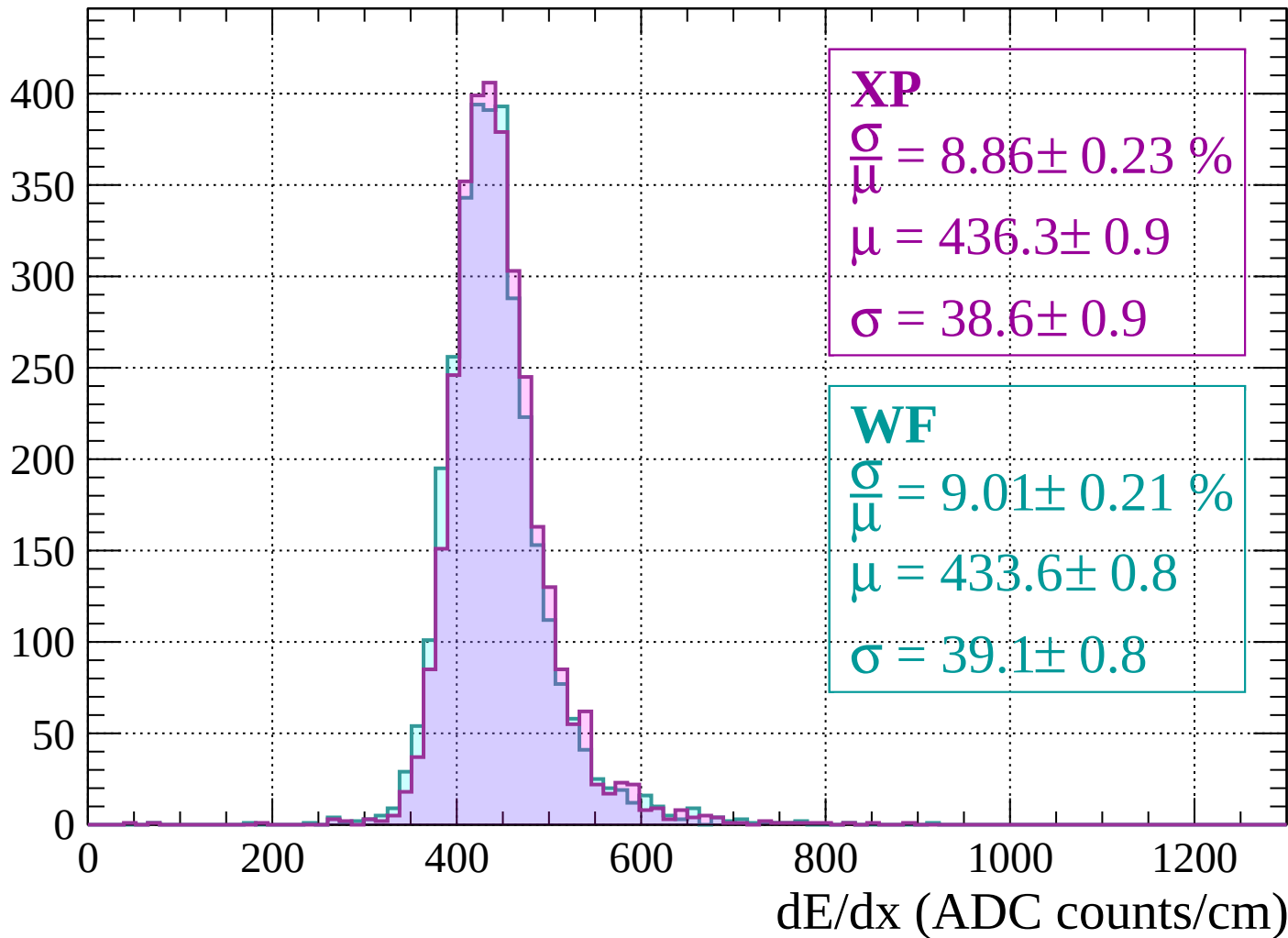
$$\mu = 432.0 \pm 0.8$$

$$\sigma = 39.1 \pm 0.8$$

dE/dx (ADC counts/cm)

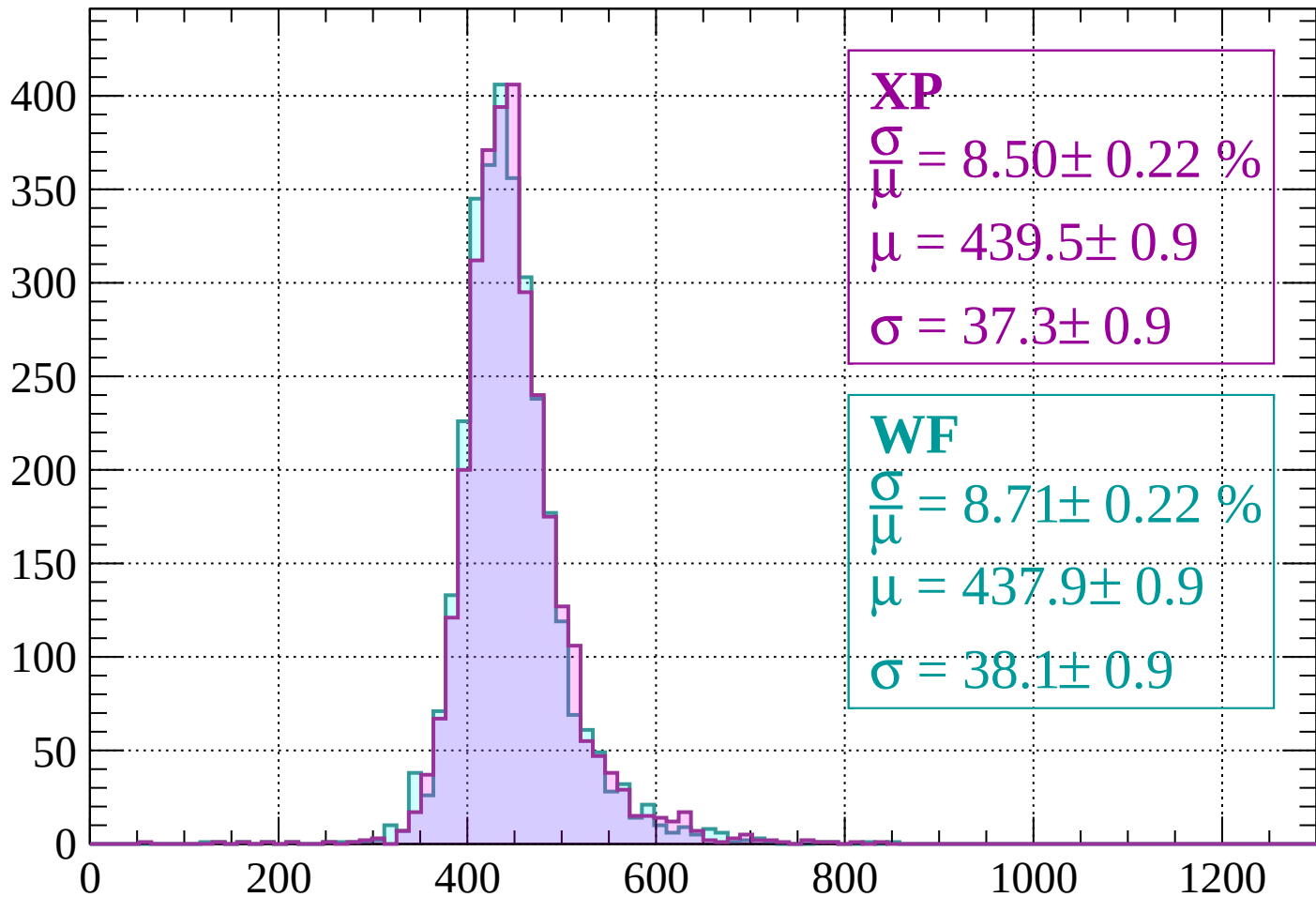
Energy loss | $1000 < p < 1040$

Count



Energy loss | $1040 < p < 1080$

Count



XP

$$\frac{\sigma}{\mu} = 8.50 \pm 0.22 \%$$

$$\mu = 439.5 \pm 0.9$$

$$\sigma = 37.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.71 \pm 0.22 \%$$

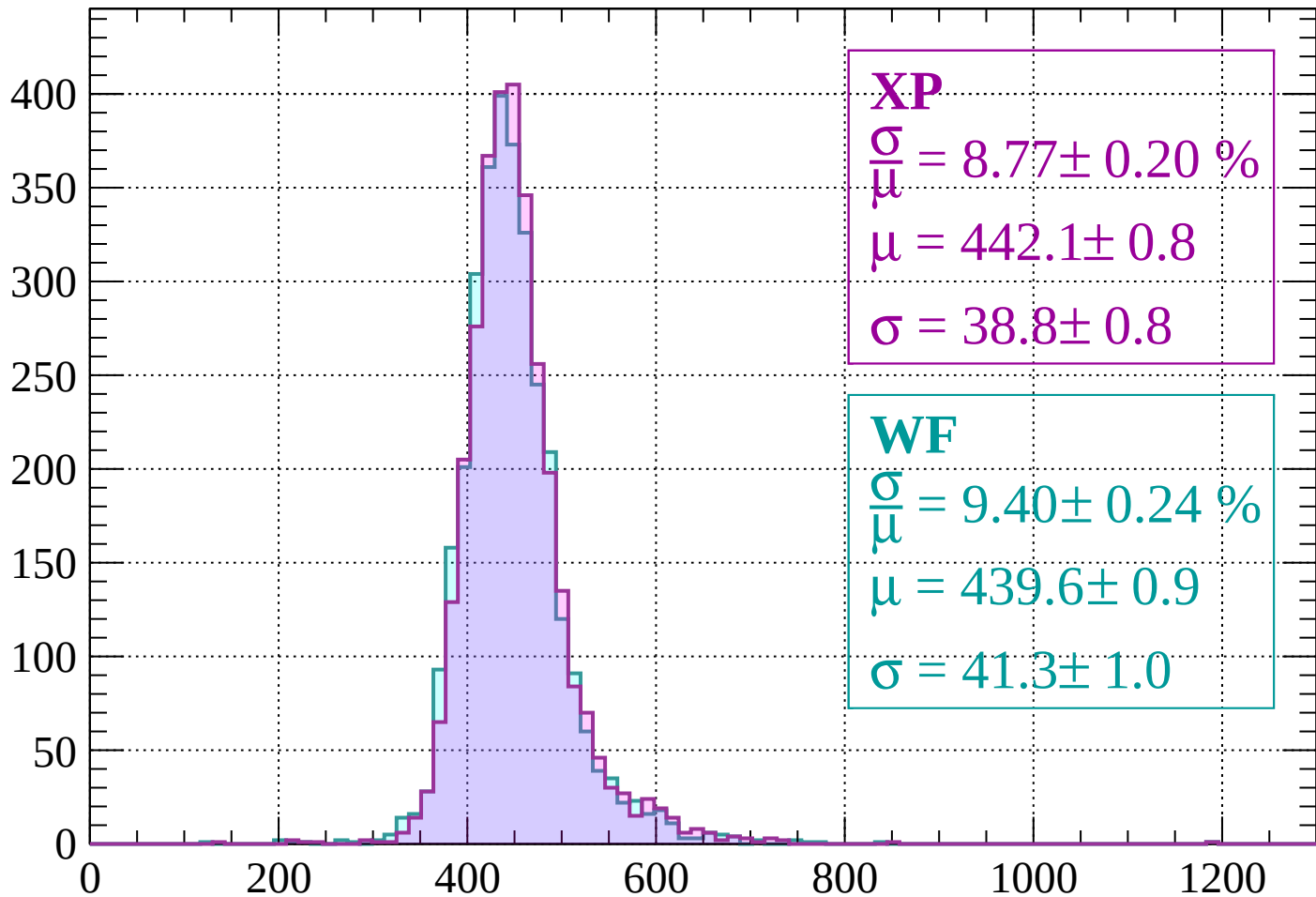
$$\mu = 437.9 \pm 0.9$$

$$\sigma = 38.1 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 8.77 \pm 0.20 \%$$

$$\mu = 442.1 \pm 0.8$$

$$\sigma = 38.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.40 \pm 0.24 \%$$

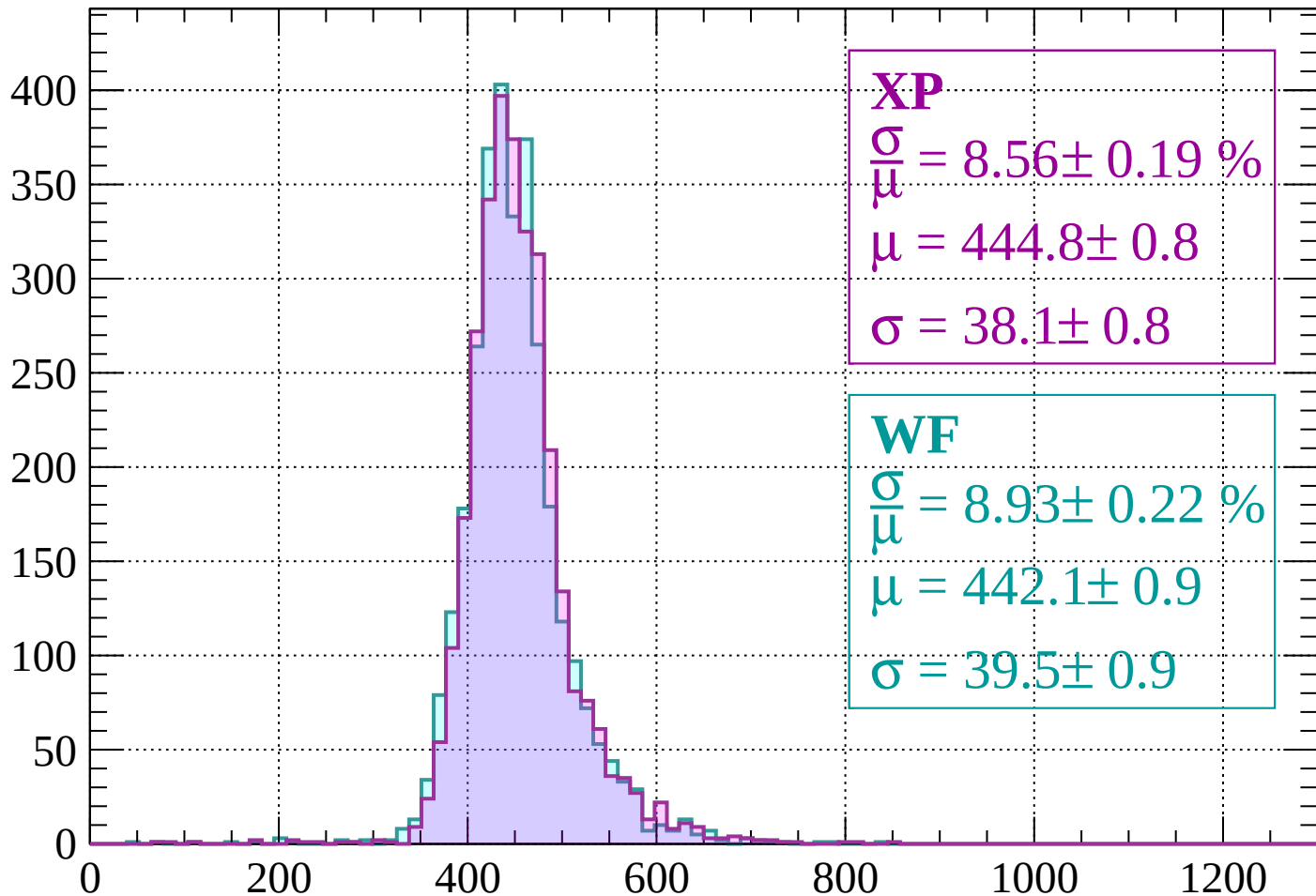
$$\mu = 439.6 \pm 0.9$$

$$\sigma = 41.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1160$

Count



XP

$$\frac{\sigma}{\mu} = 8.56 \pm 0.19 \%$$

$$\mu = 444.8 \pm 0.8$$

$$\sigma = 38.1 \pm 0.8$$

WF

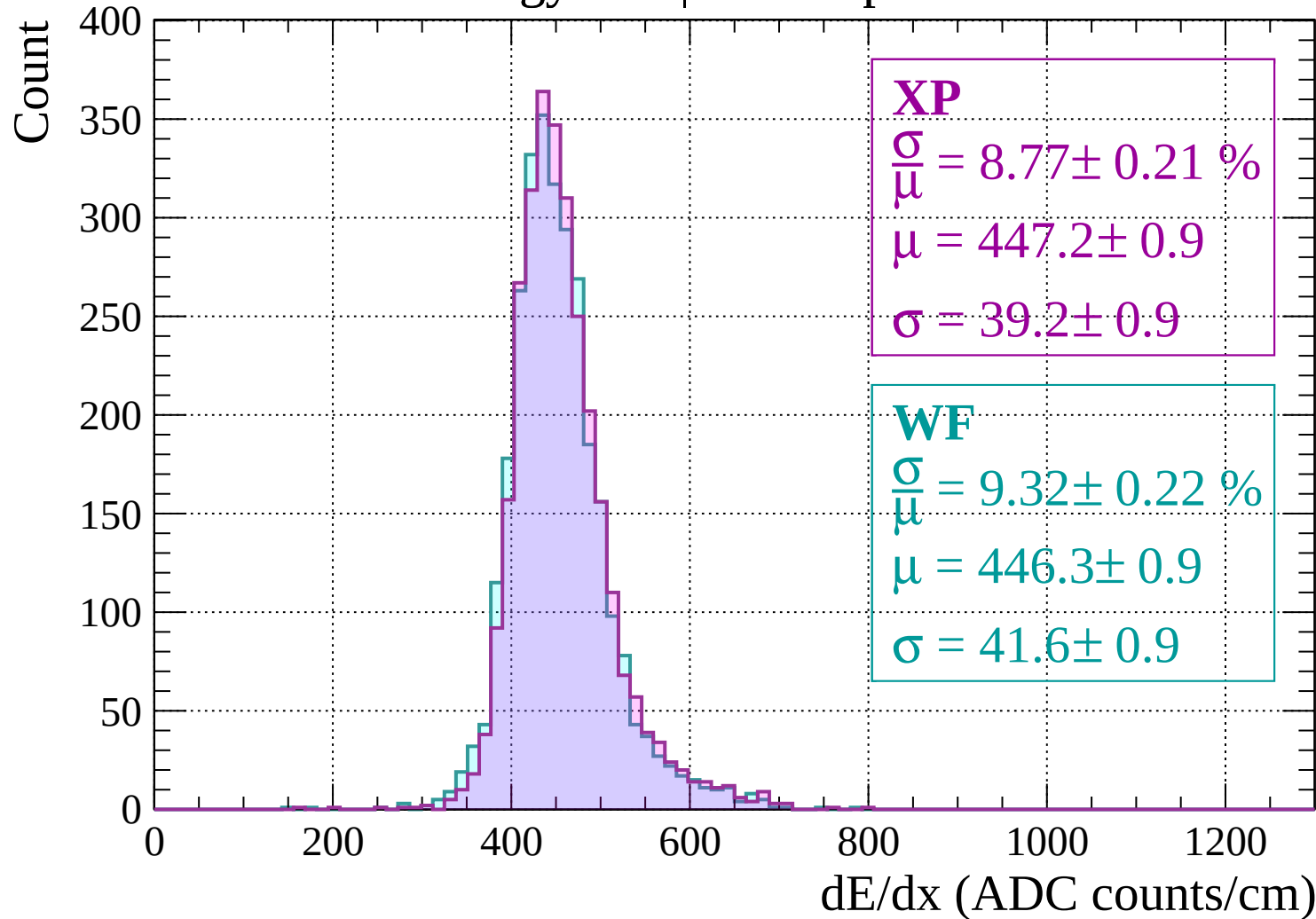
$$\frac{\sigma}{\mu} = 8.93 \pm 0.22 \%$$

$$\mu = 442.1 \pm 0.9$$

$$\sigma = 39.5 \pm 0.9$$

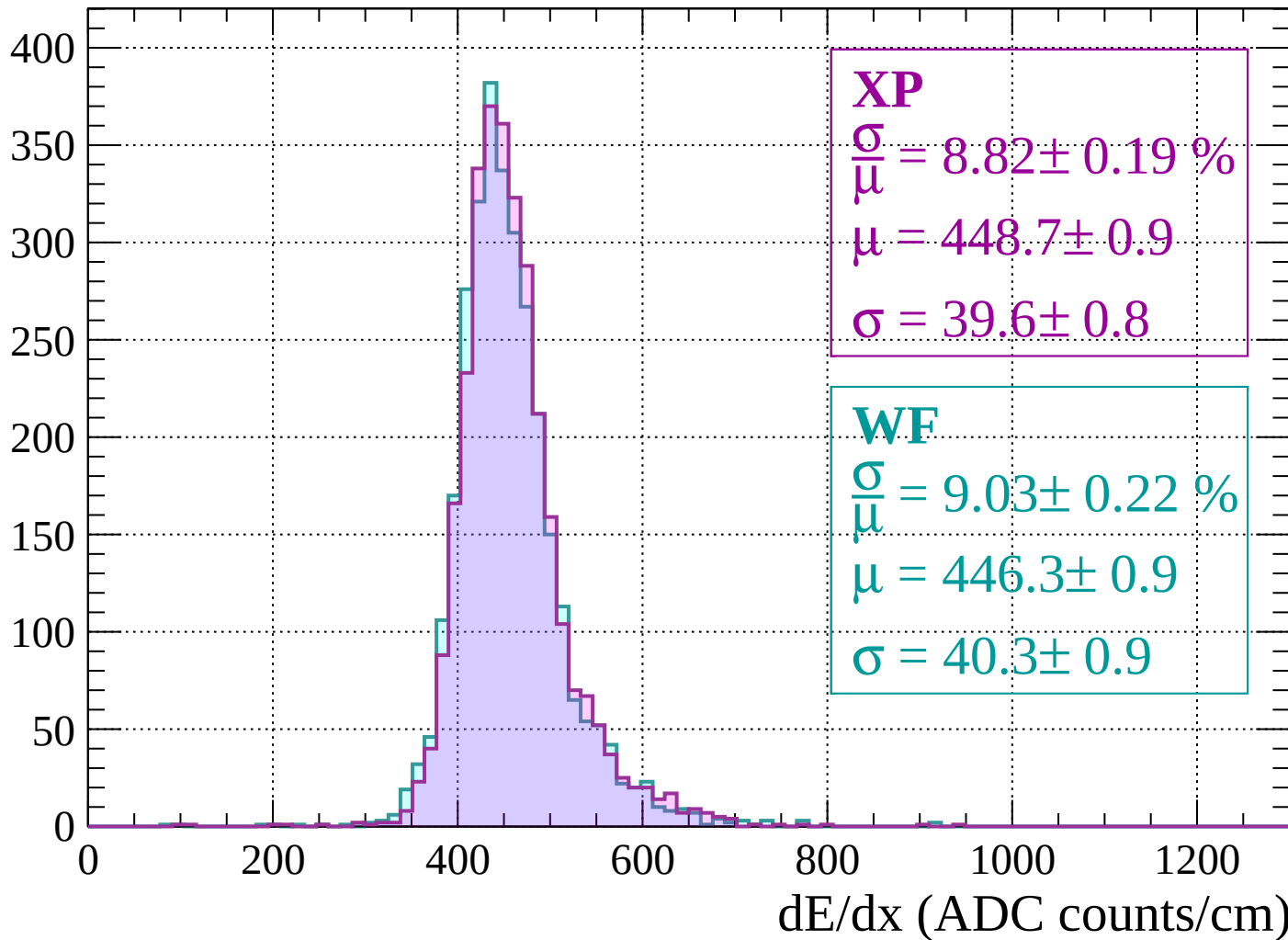
dE/dx (ADC counts/cm)

Energy loss | $1160 < p < 1200$



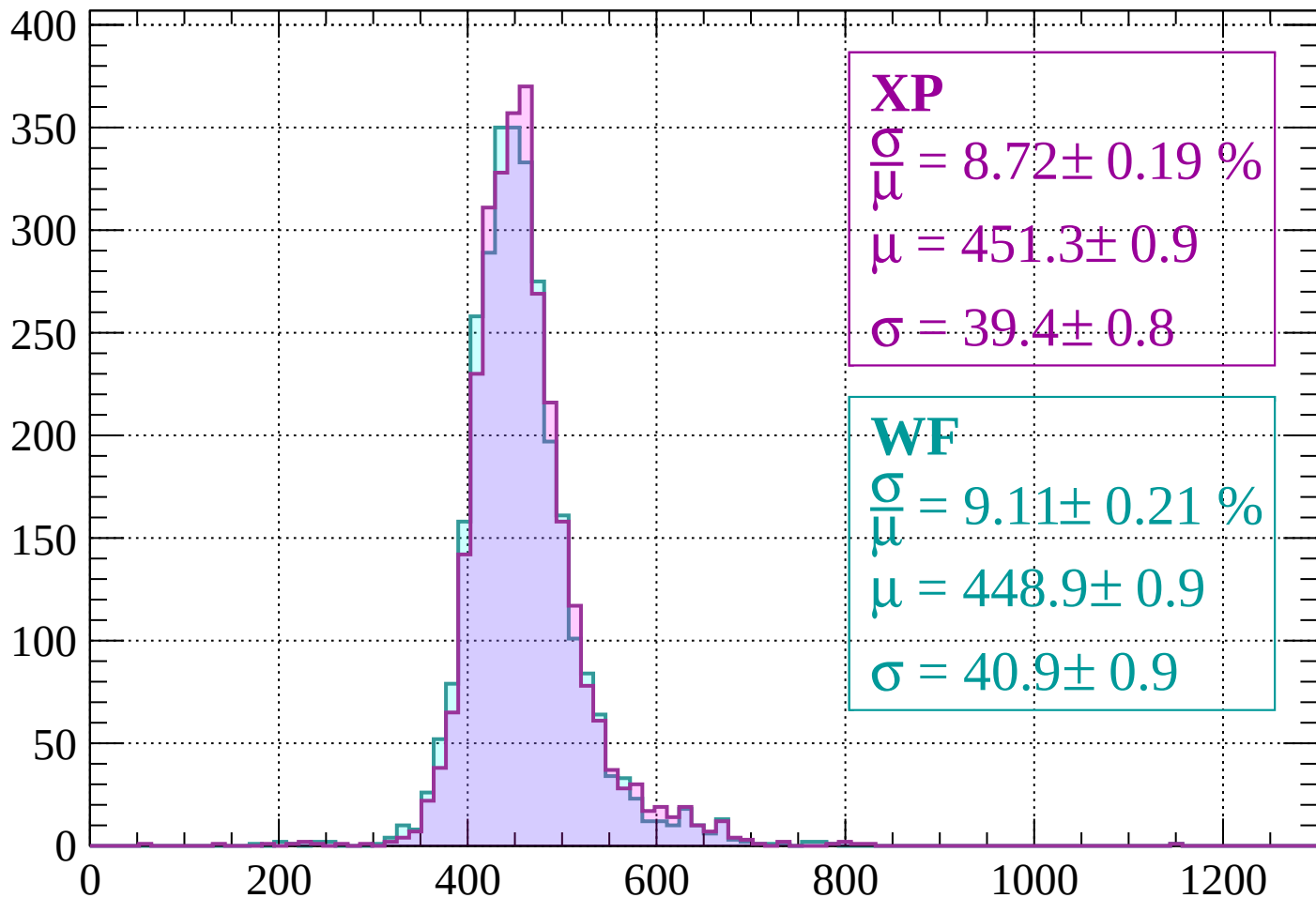
Energy loss | $1200 < p < 1240$

Count



Energy loss | $1240 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 8.72 \pm 0.19 \%$$

$$\mu = 451.3 \pm 0.9$$

$$\sigma = 39.4 \pm 0.8$$

WF

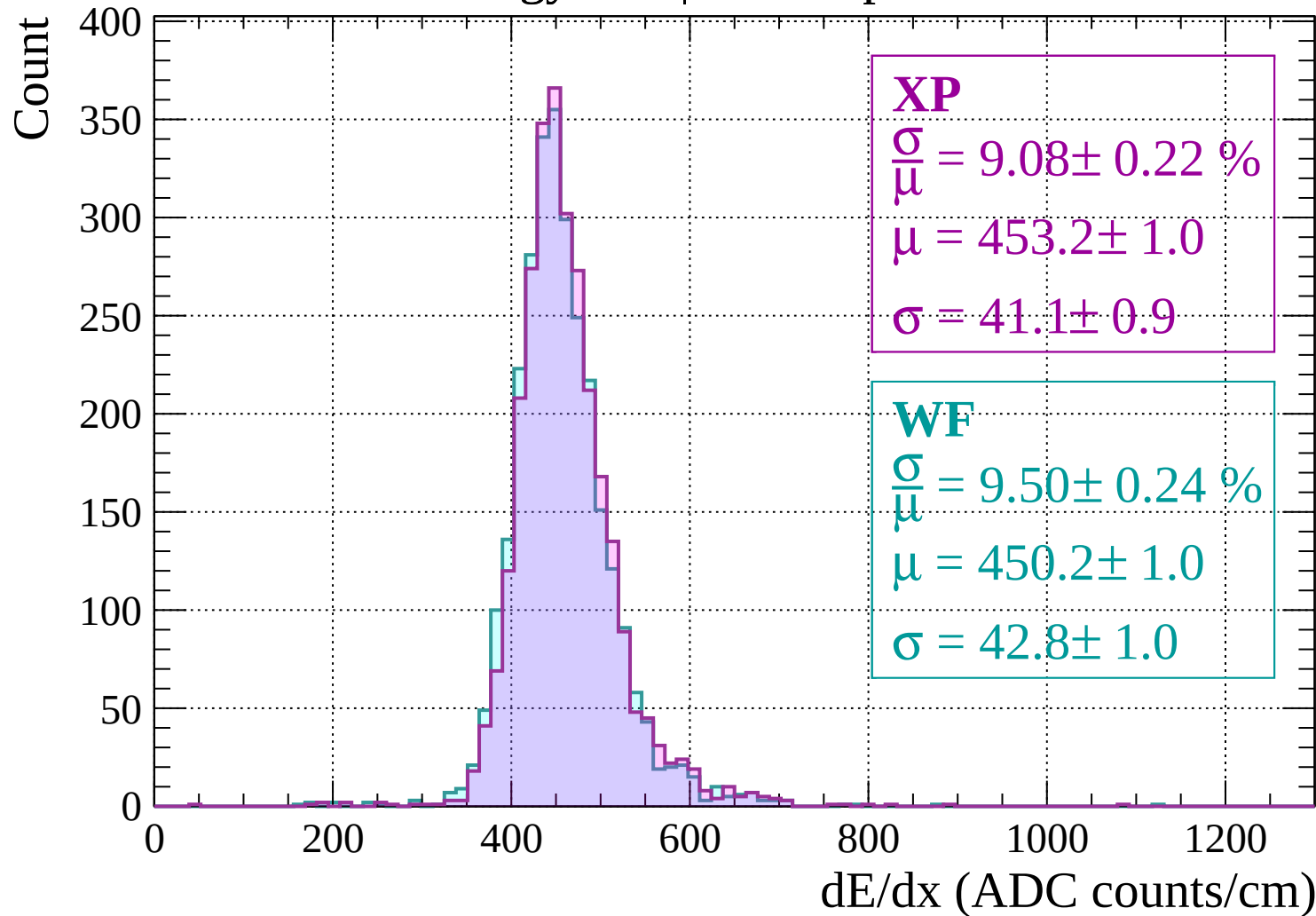
$$\frac{\sigma}{\mu} = 9.11 \pm 0.21 \%$$

$$\mu = 448.9 \pm 0.9$$

$$\sigma = 40.9 \pm 0.9$$

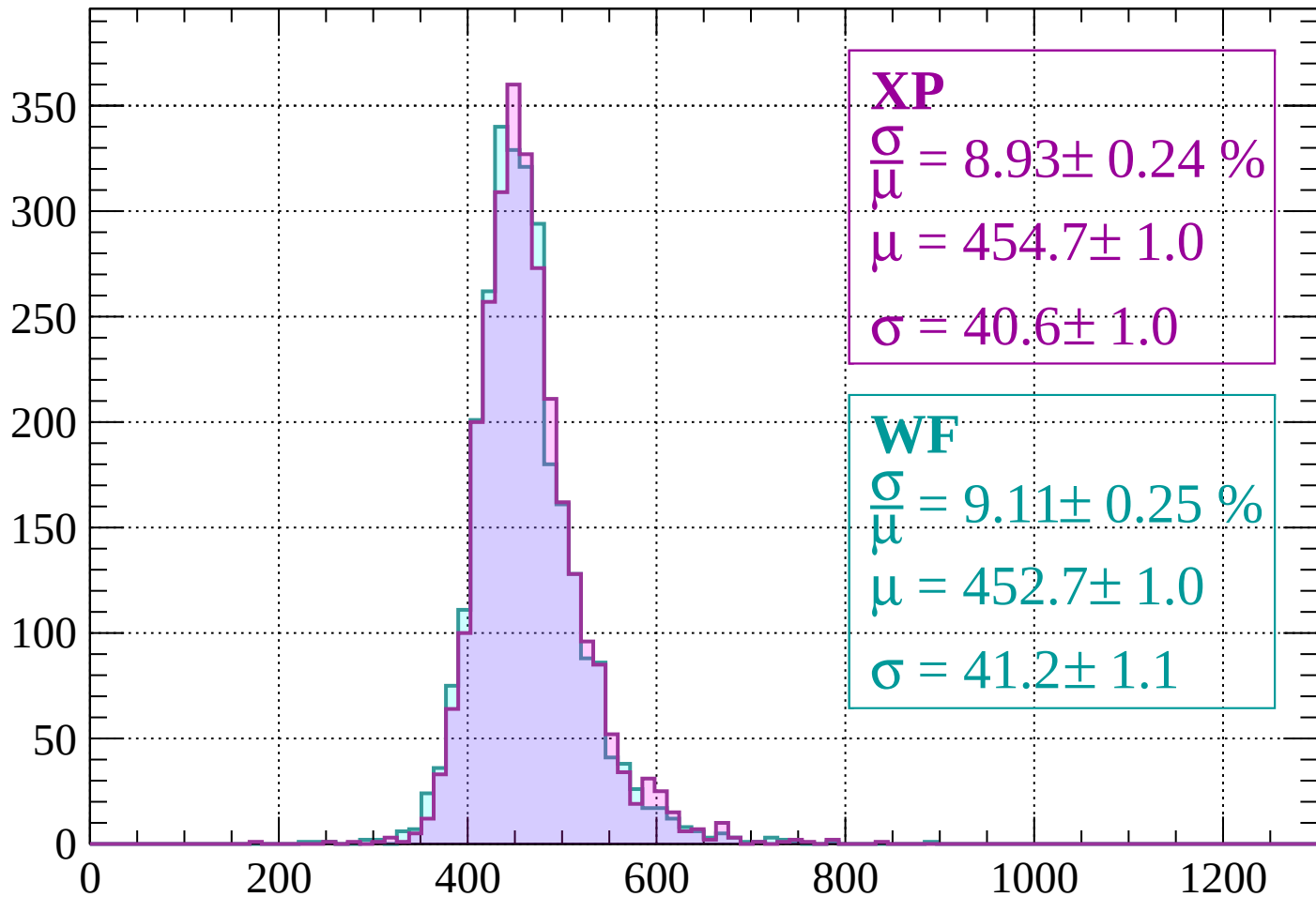
dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1320$



Energy loss | $1320 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 8.93 \pm 0.24 \%$$

$$\mu = 454.7 \pm 1.0$$

$$\sigma = 40.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.11 \pm 0.25 \%$$

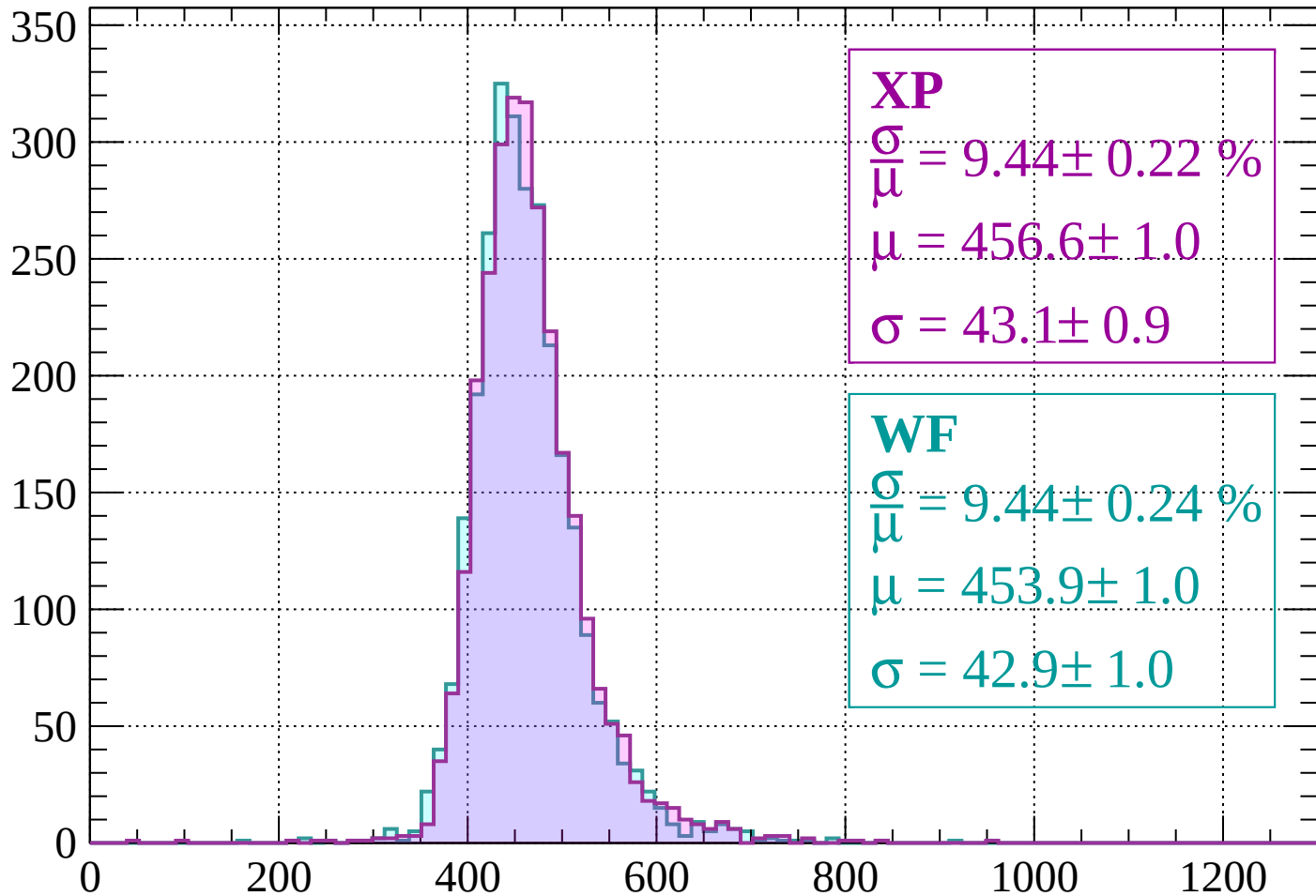
$$\mu = 452.7 \pm 1.0$$

$$\sigma = 41.2 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1400$

Count



XP

$$\frac{\sigma}{\mu} = 9.44 \pm 0.22 \%$$

$$\mu = 456.6 \pm 1.0$$

$$\sigma = 43.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.44 \pm 0.24 \%$$

$$\mu = 453.9 \pm 1.0$$

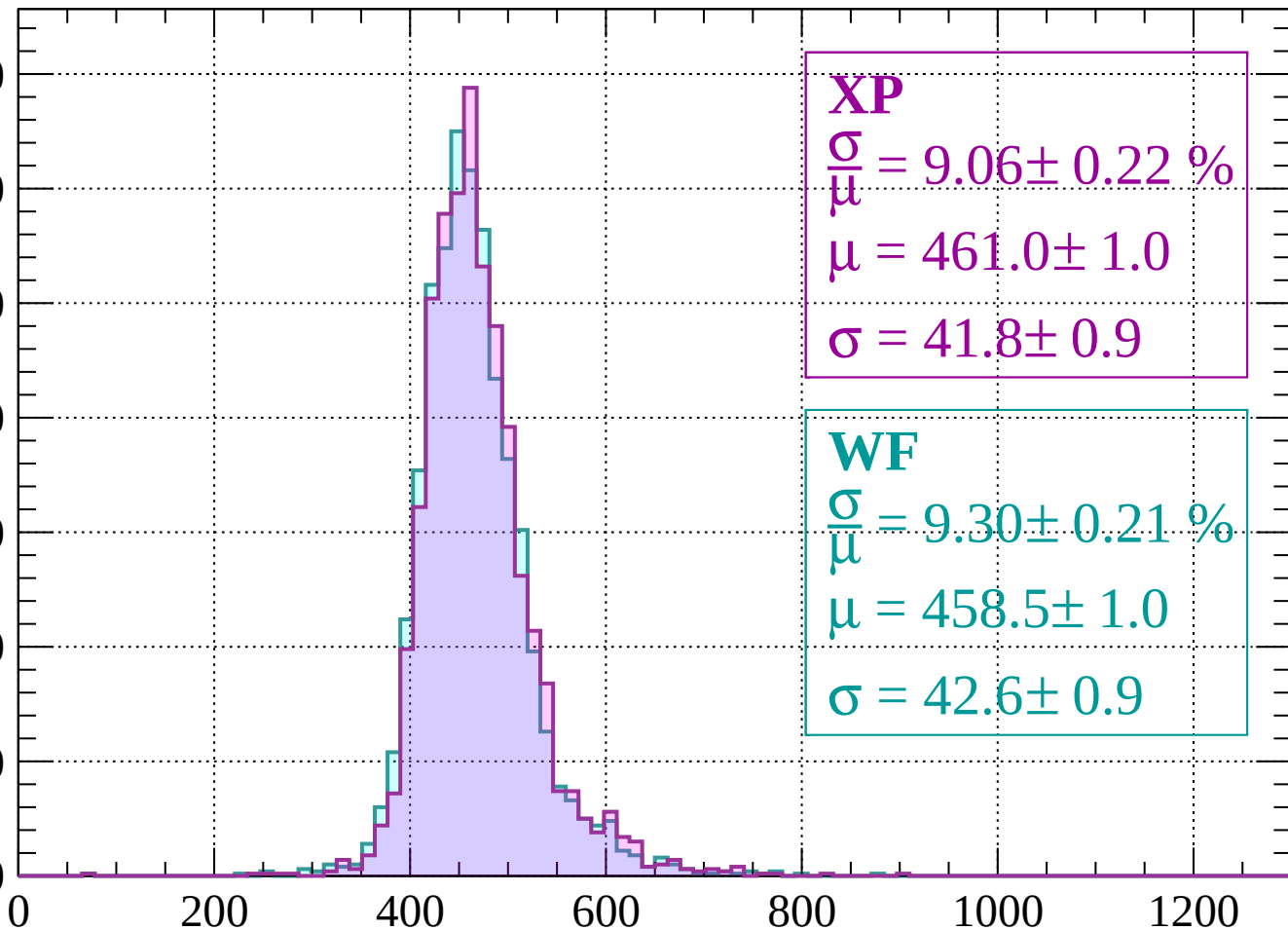
$$\sigma = 42.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1400 < p < 1440$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.06 \pm 0.22 \%$$

$$\mu = 461.0 \pm 1.0$$

$$\sigma = 41.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.30 \pm 0.21 \%$$

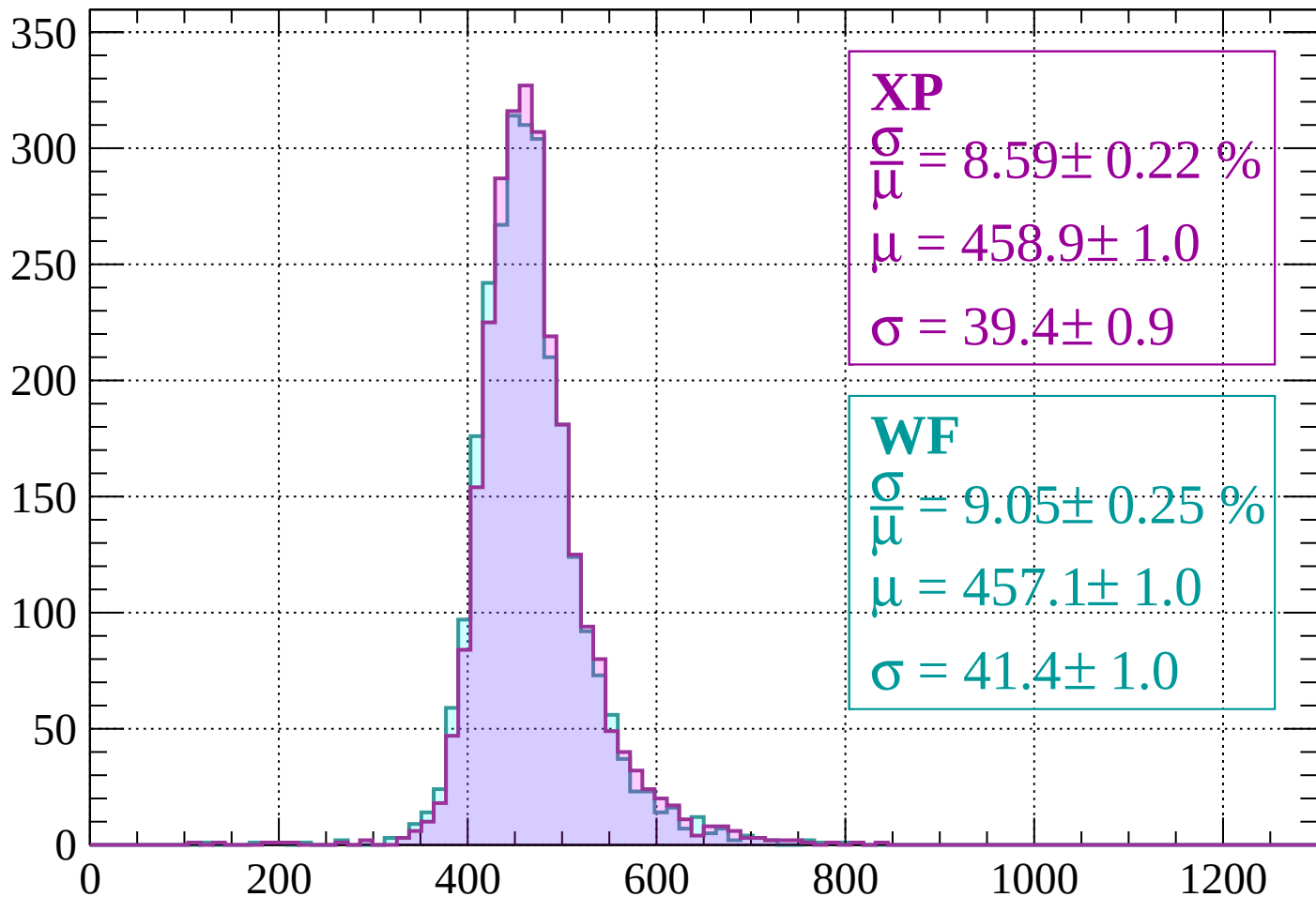
$$\mu = 458.5 \pm 1.0$$

$$\sigma = 42.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1480$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.59 \pm 0.22 \%$$

$$\mu = 458.9 \pm 1.0$$

$$\sigma = 39.4 \pm 0.9$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.05 \pm 0.25 \%$$

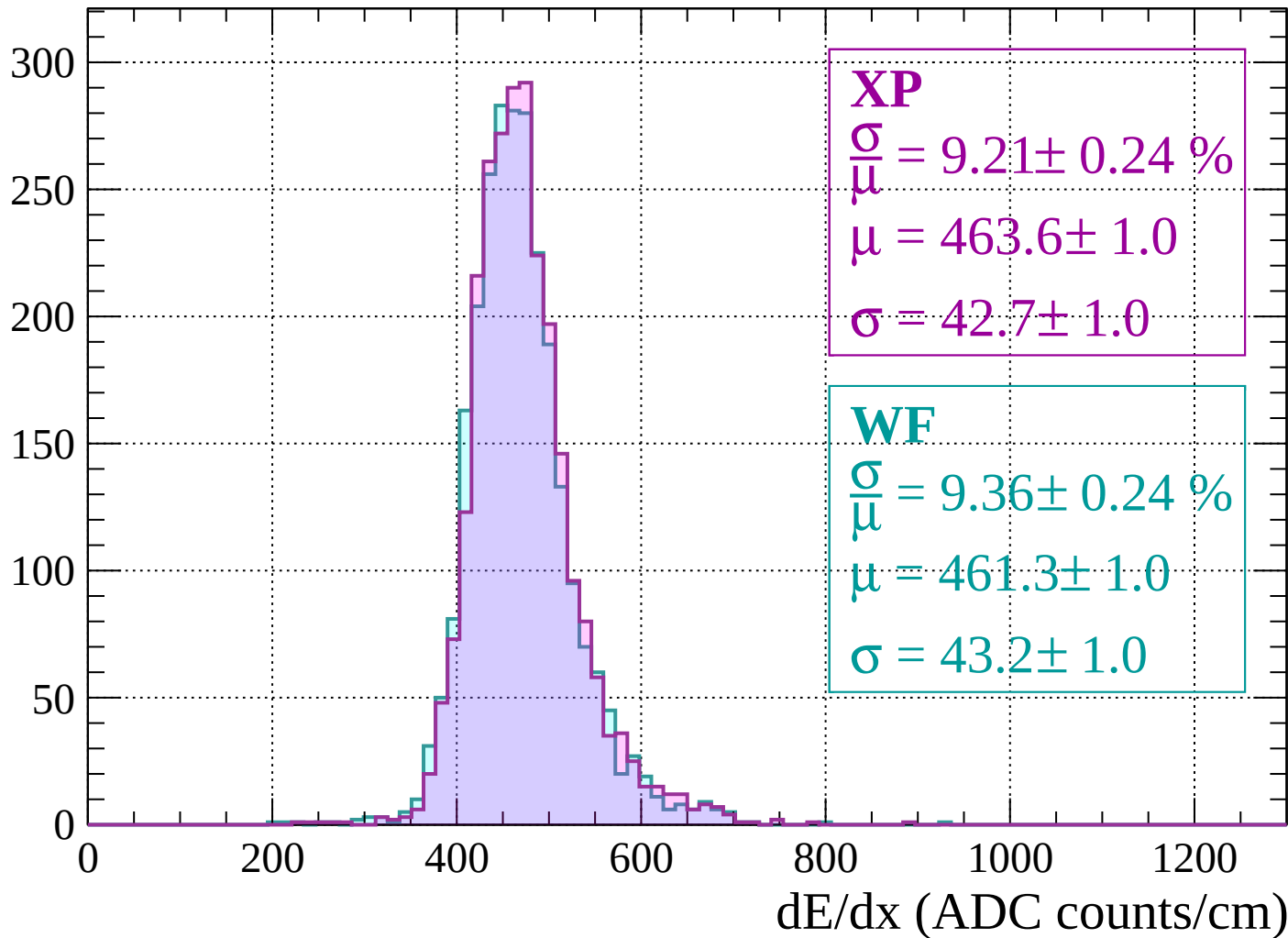
$$\mu = 457.1 \pm 1.0$$

$$\sigma = 41.4 \pm 1.0$$

dE/dx (ADC counts/cm)

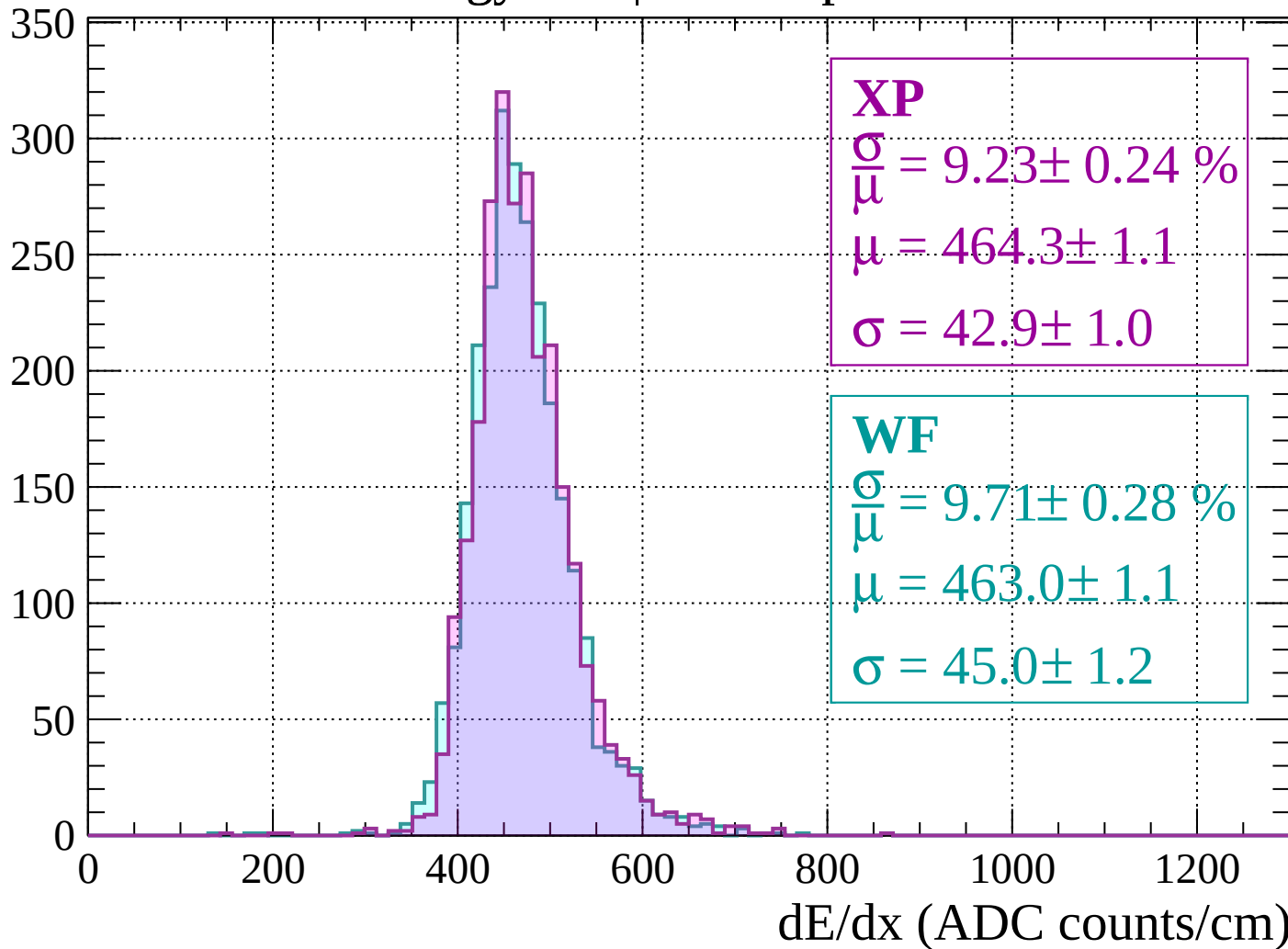
Energy loss | $1480 < p < 1520$

Count



Energy loss | $1520 < p < 1560$

Count



Energy loss | $1560 < p < 1600$

Count

300

250

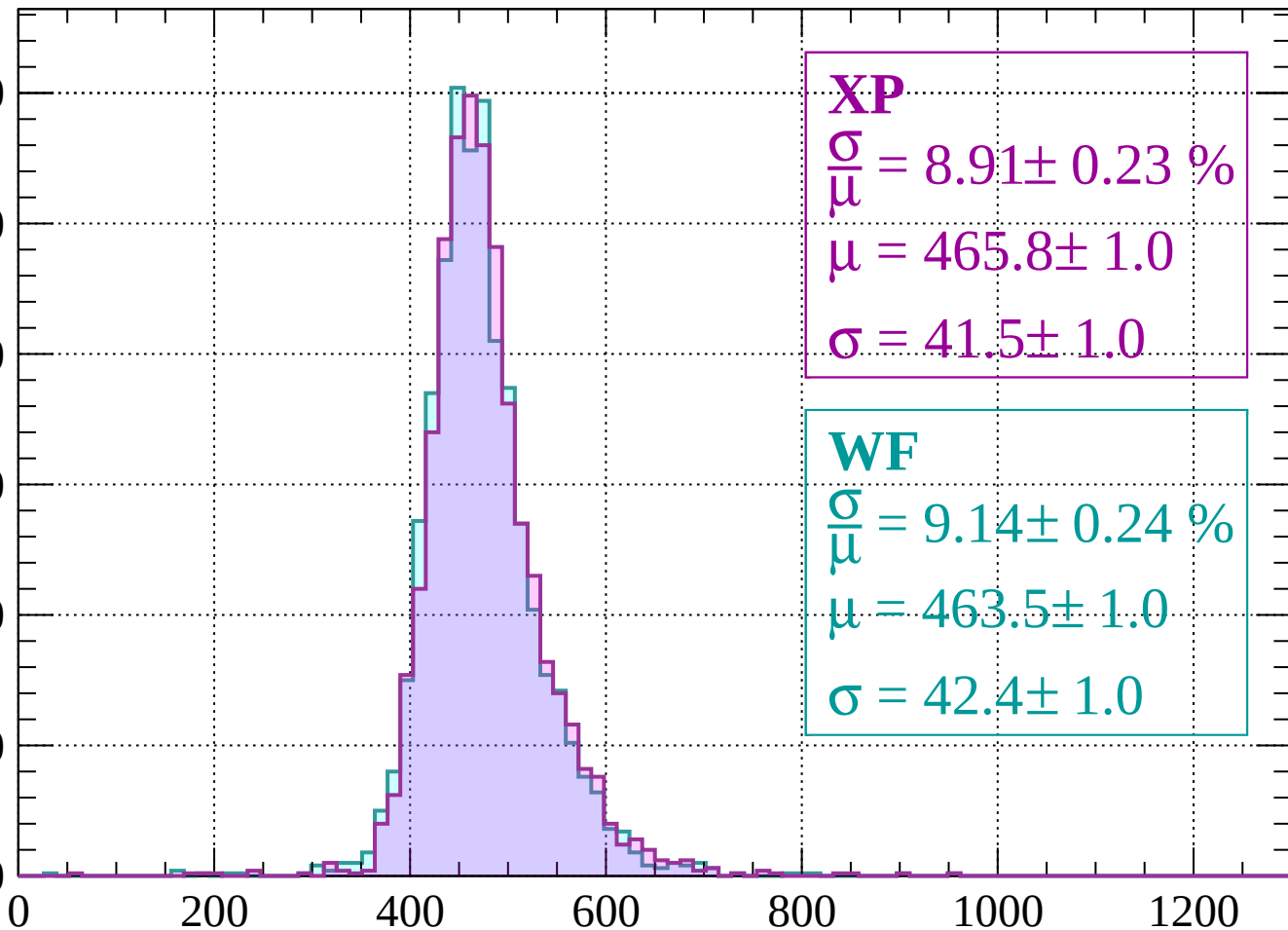
200

150

100

50

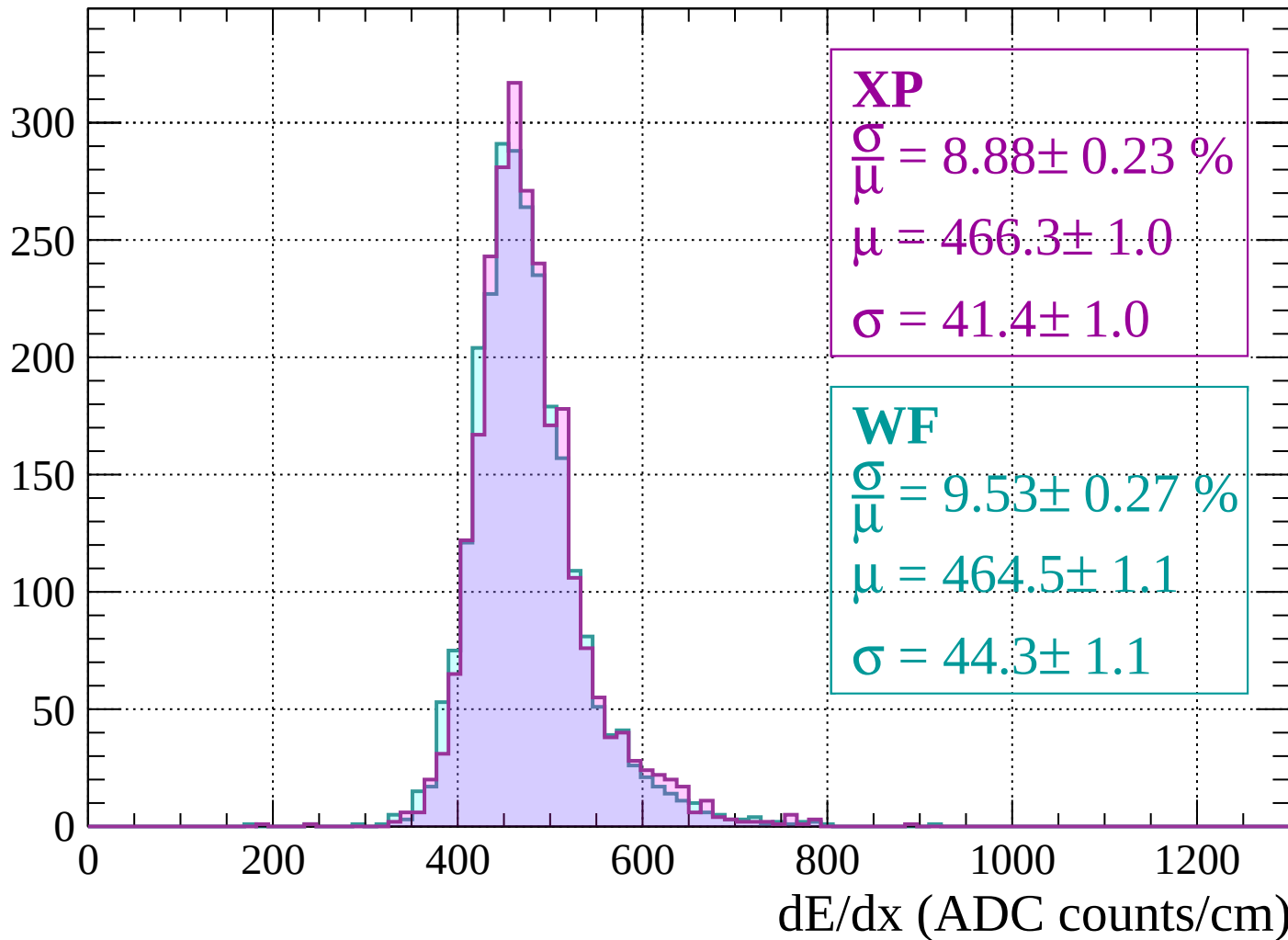
0



dE/dx (ADC counts/cm)

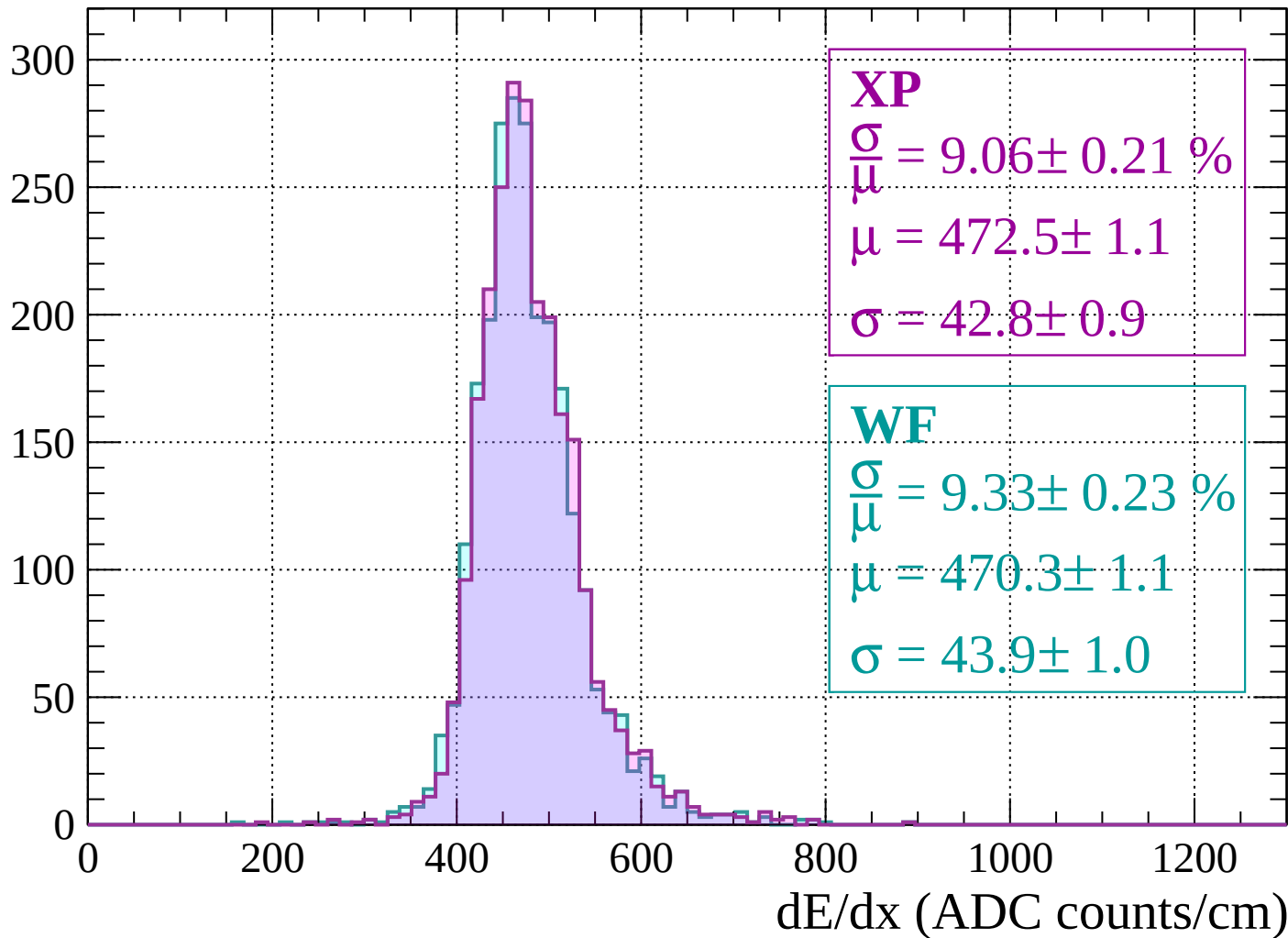
Energy loss | $1600 < p < 1640$

Count



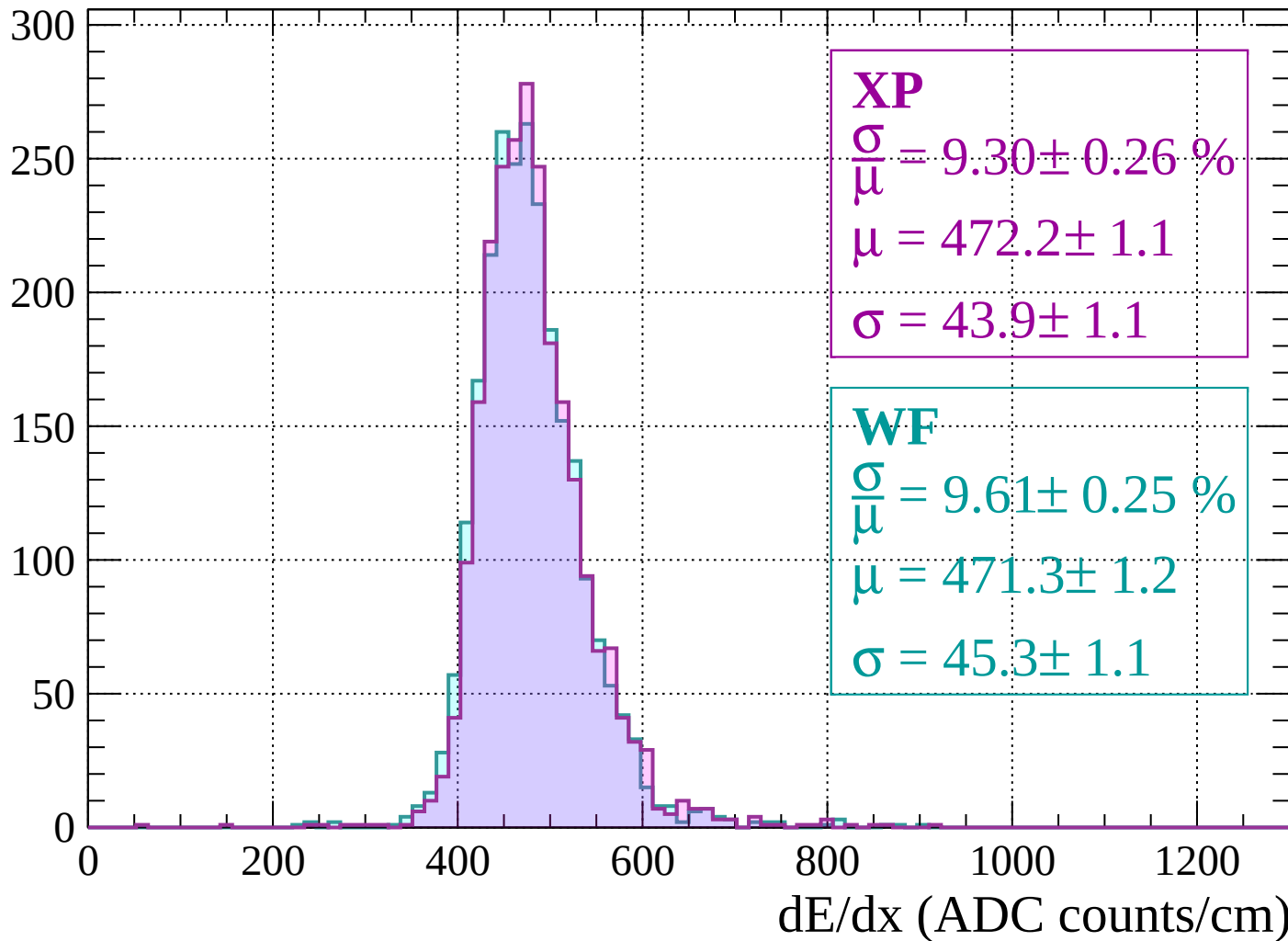
Energy loss | $1640 < p < 1680$

Count



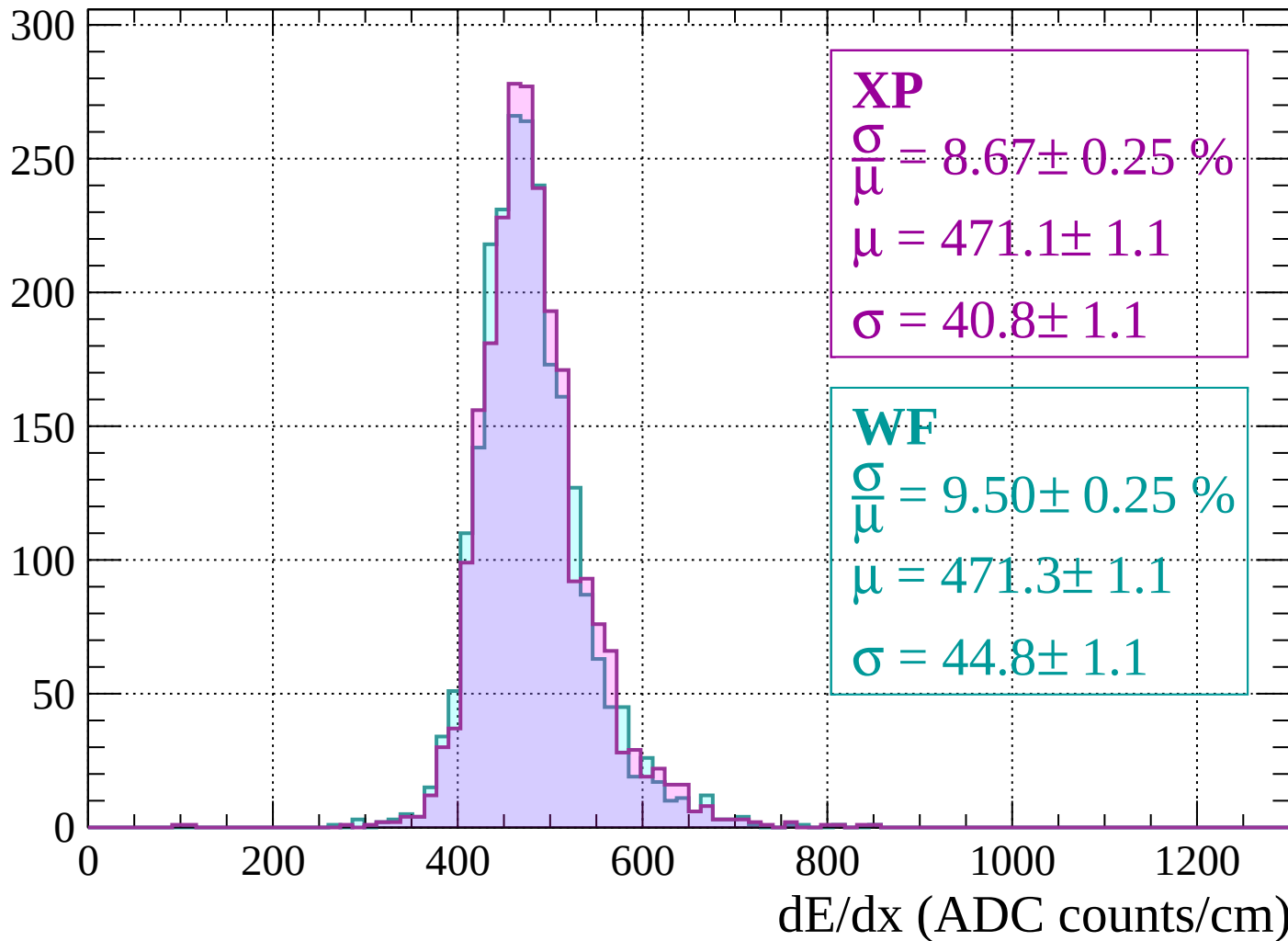
Energy loss | $1680 < p < 1720$

Count



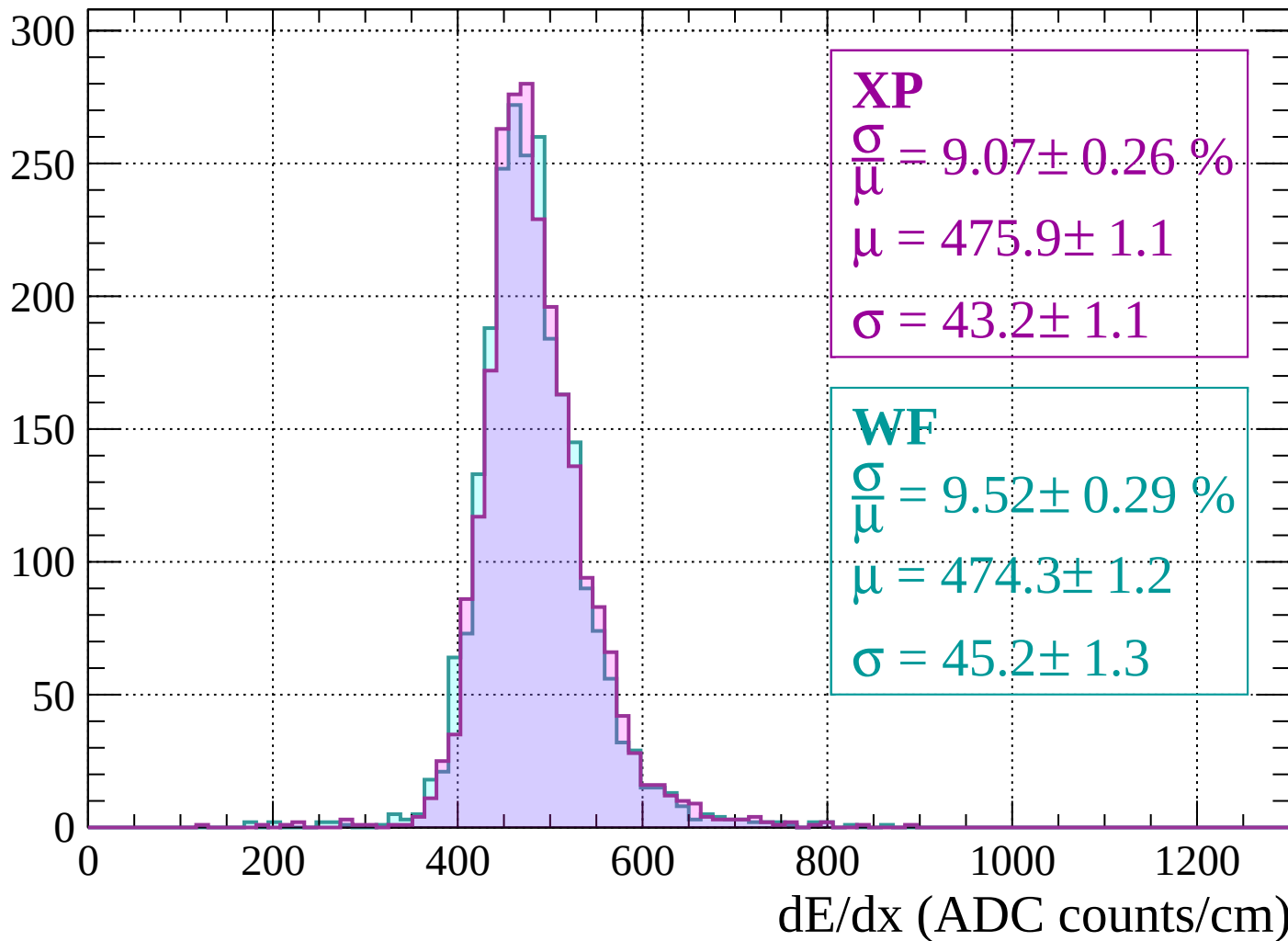
Energy loss | $1720 < p < 1760$

Count



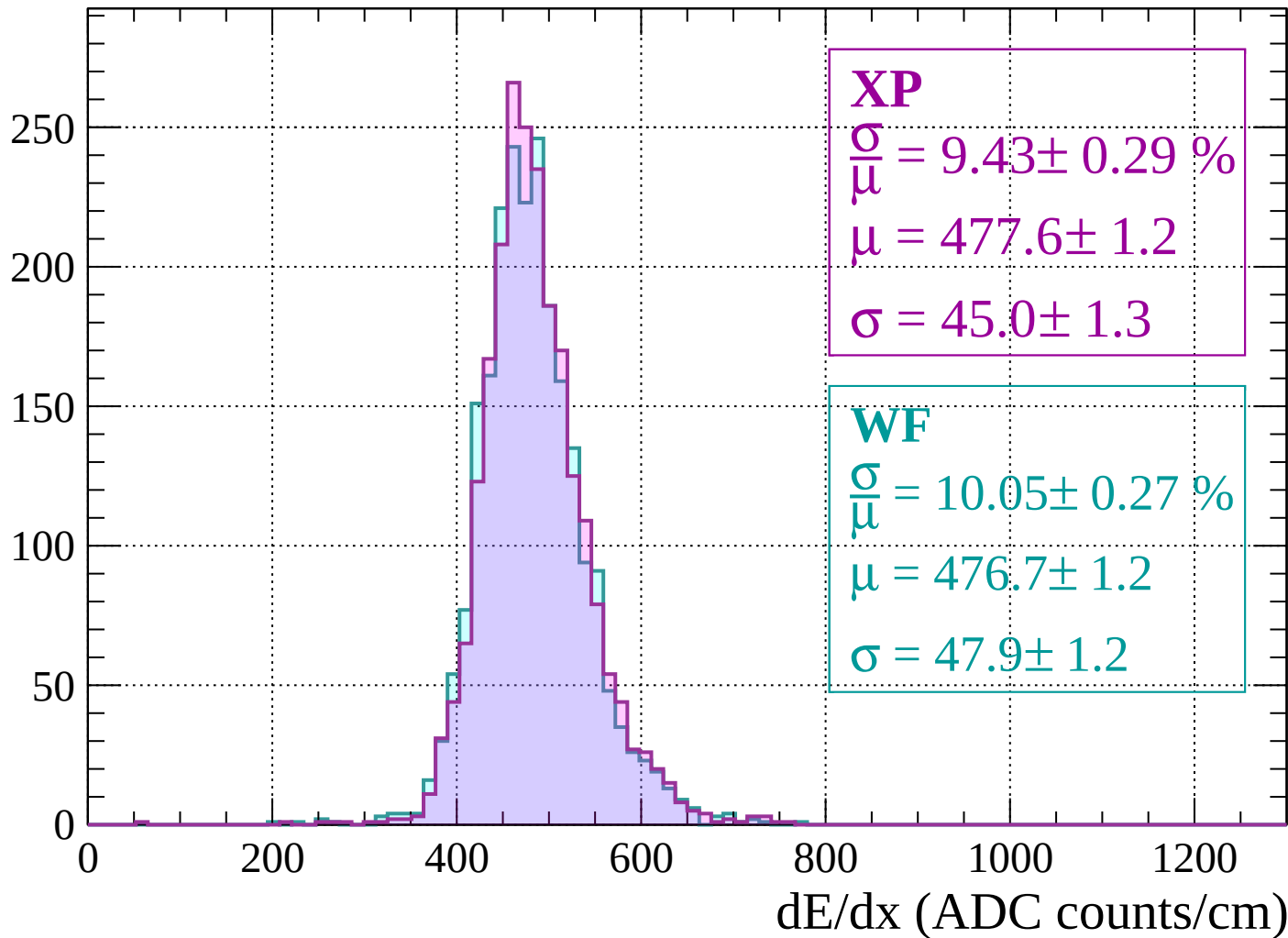
Energy loss | $1760 < p < 1800$

Count



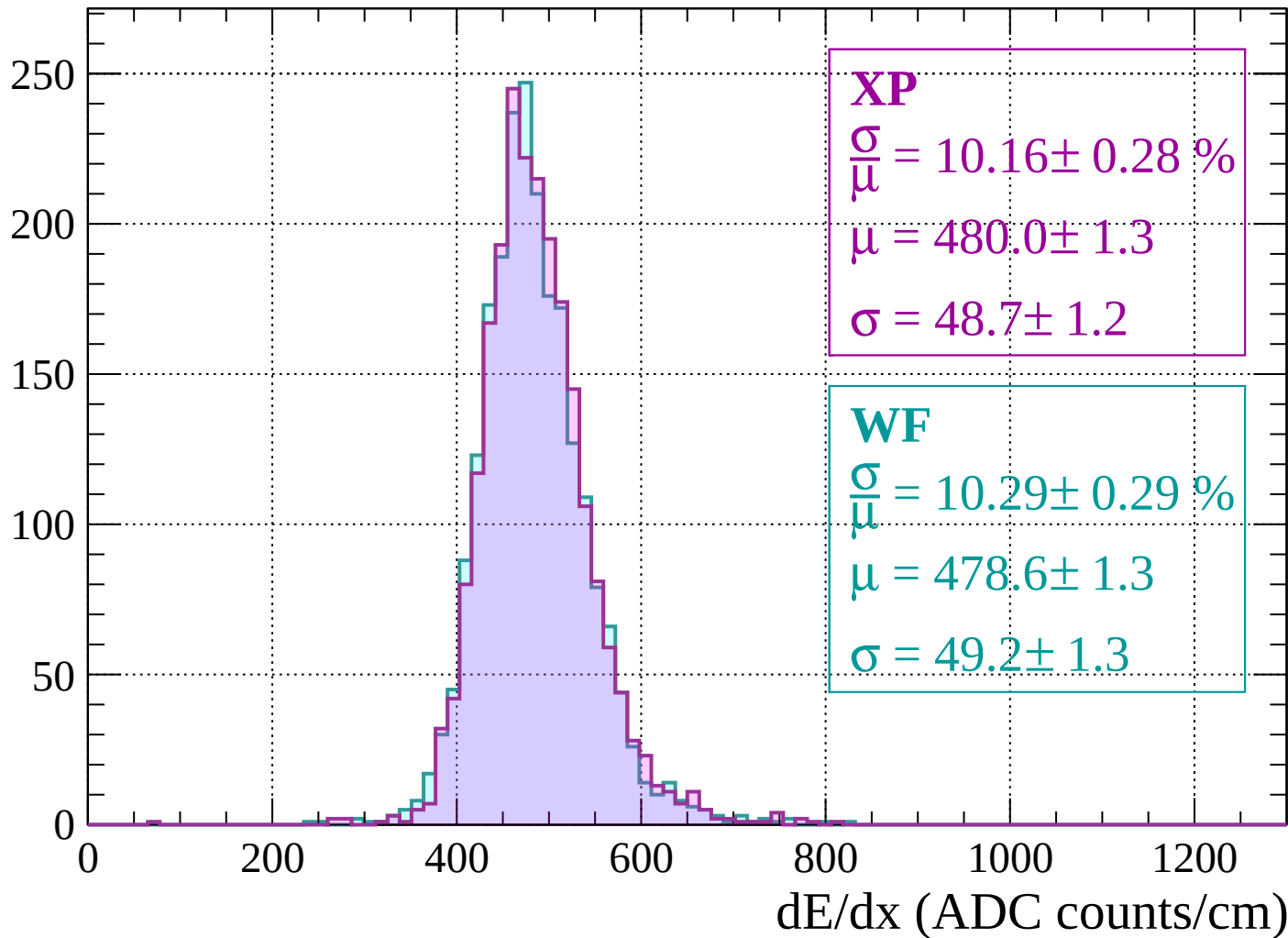
Energy loss | $1800 < p < 1840$

Count



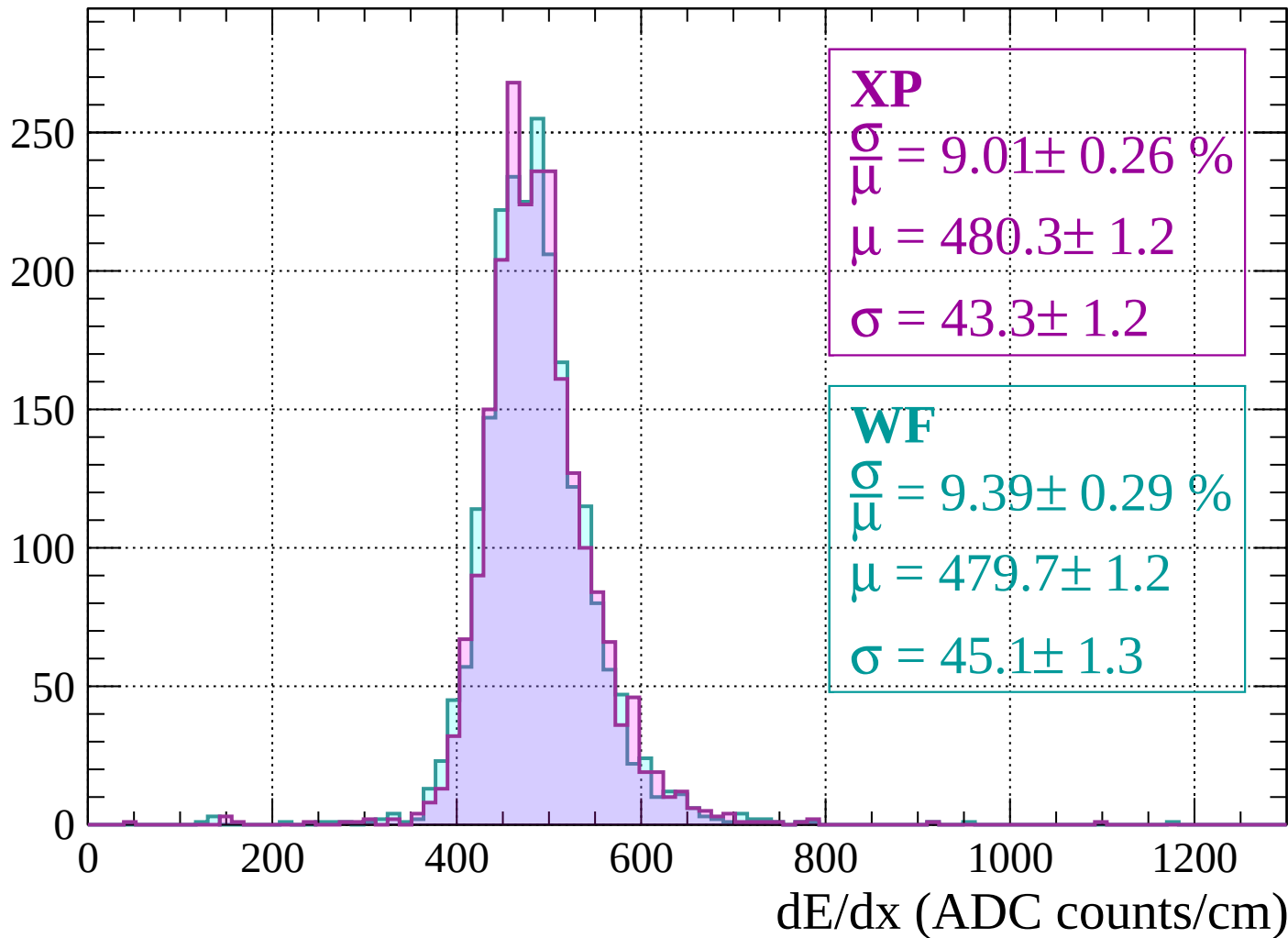
Energy loss | $1840 < p < 1880$

Count



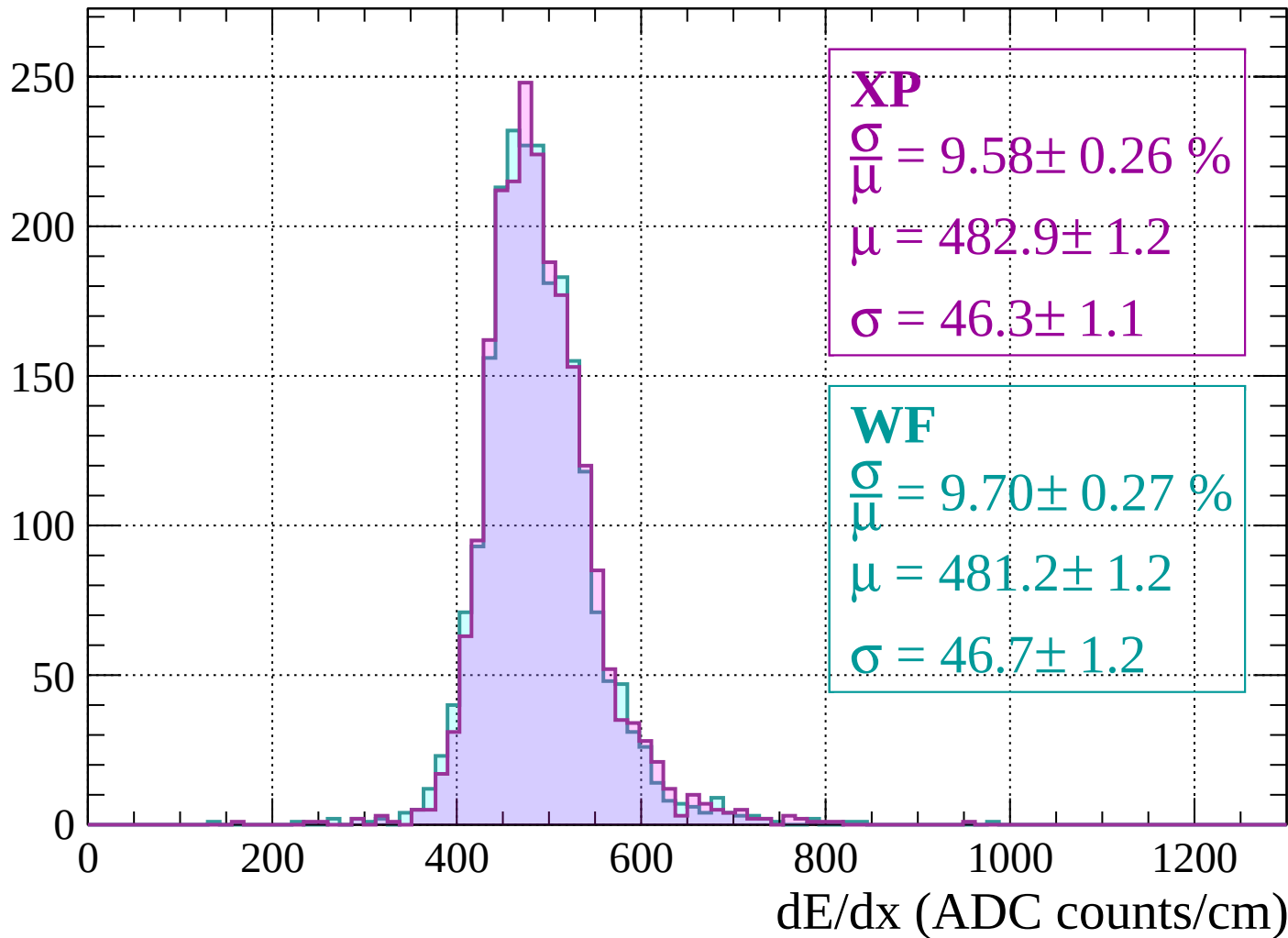
Energy loss | $1880 < p < 1920$

Count



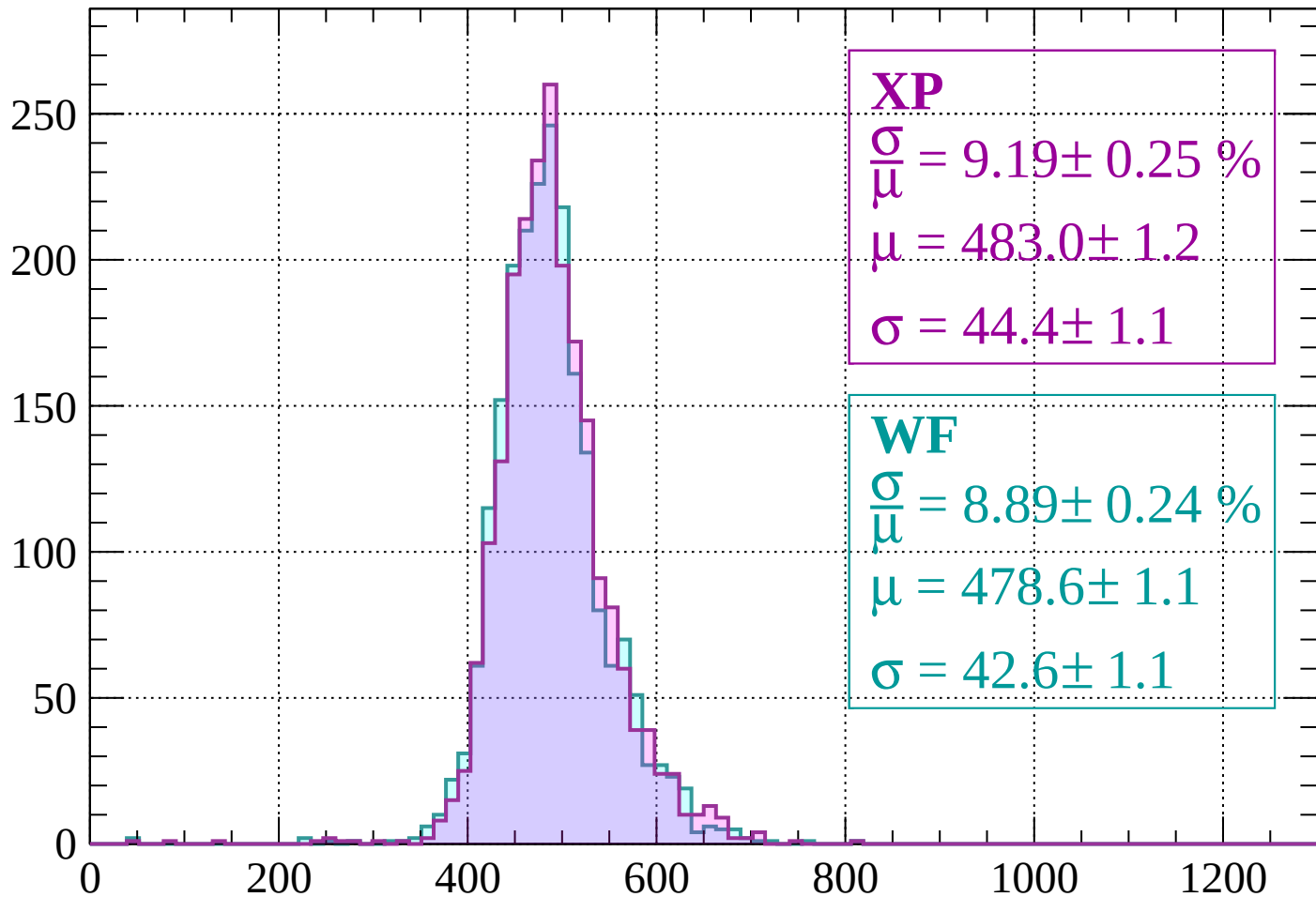
Energy loss | $1920 < p < 1960$

Count



Energy loss | $1960 < p < 2000$

Count



XP

$$\frac{\sigma}{\mu} = 9.19 \pm 0.25 \%$$

$$\mu = 483.0 \pm 1.2$$

$$\sigma = 44.4 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.89 \pm 0.24 \%$$

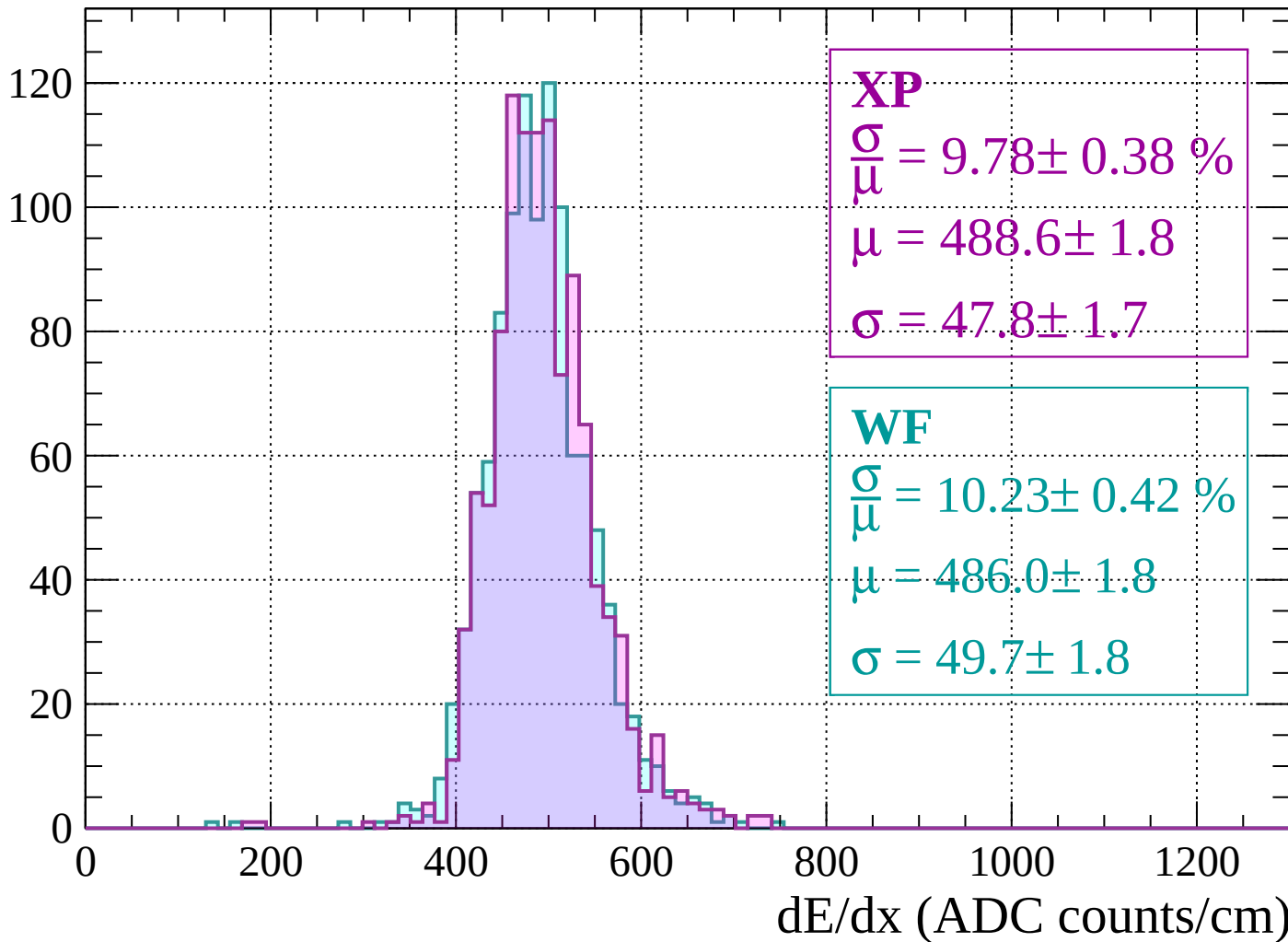
$$\mu = 478.6 \pm 1.1$$

$$\sigma = 42.6 \pm 1.1$$

dE/dx (ADC counts/cm)

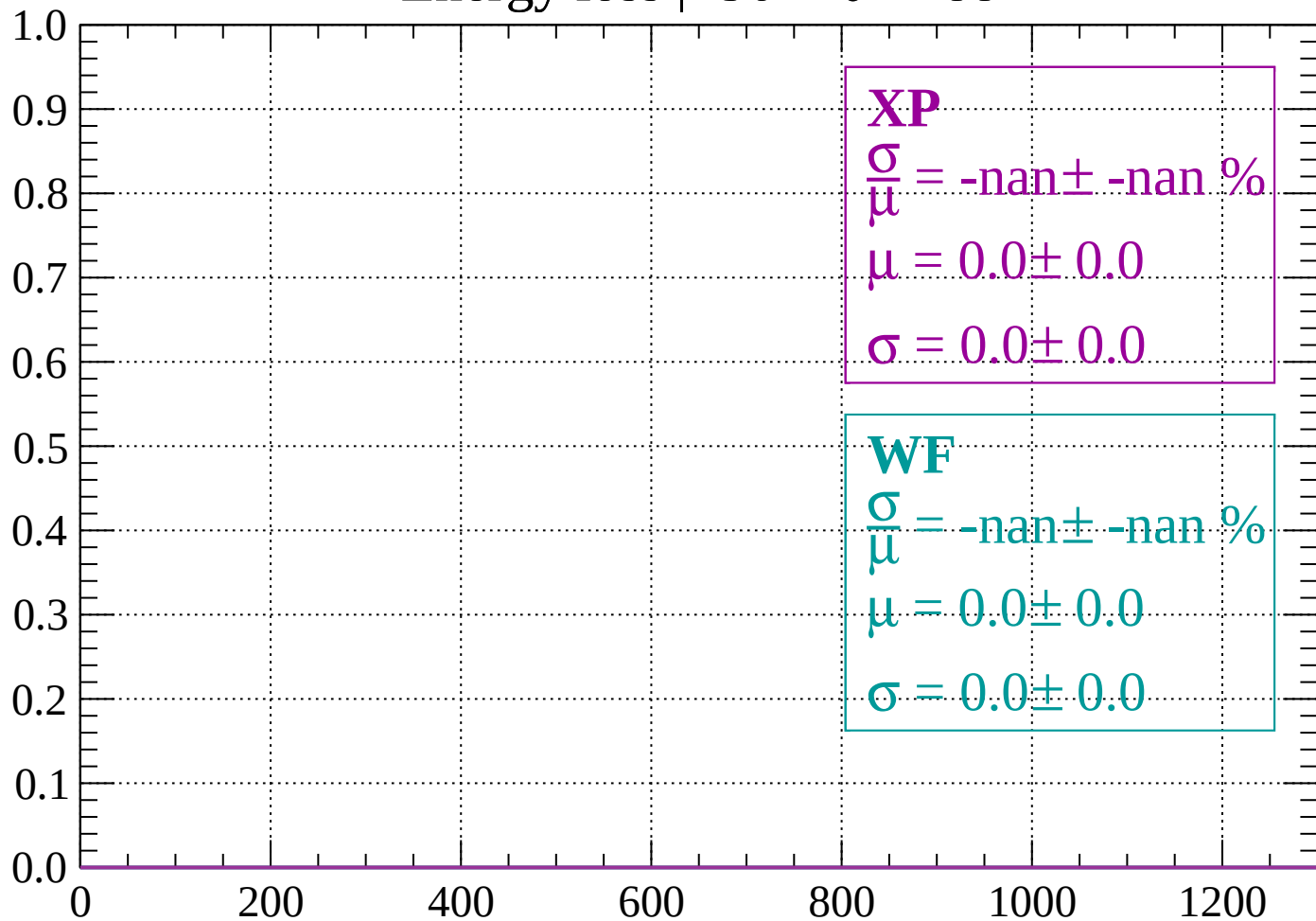
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -88$

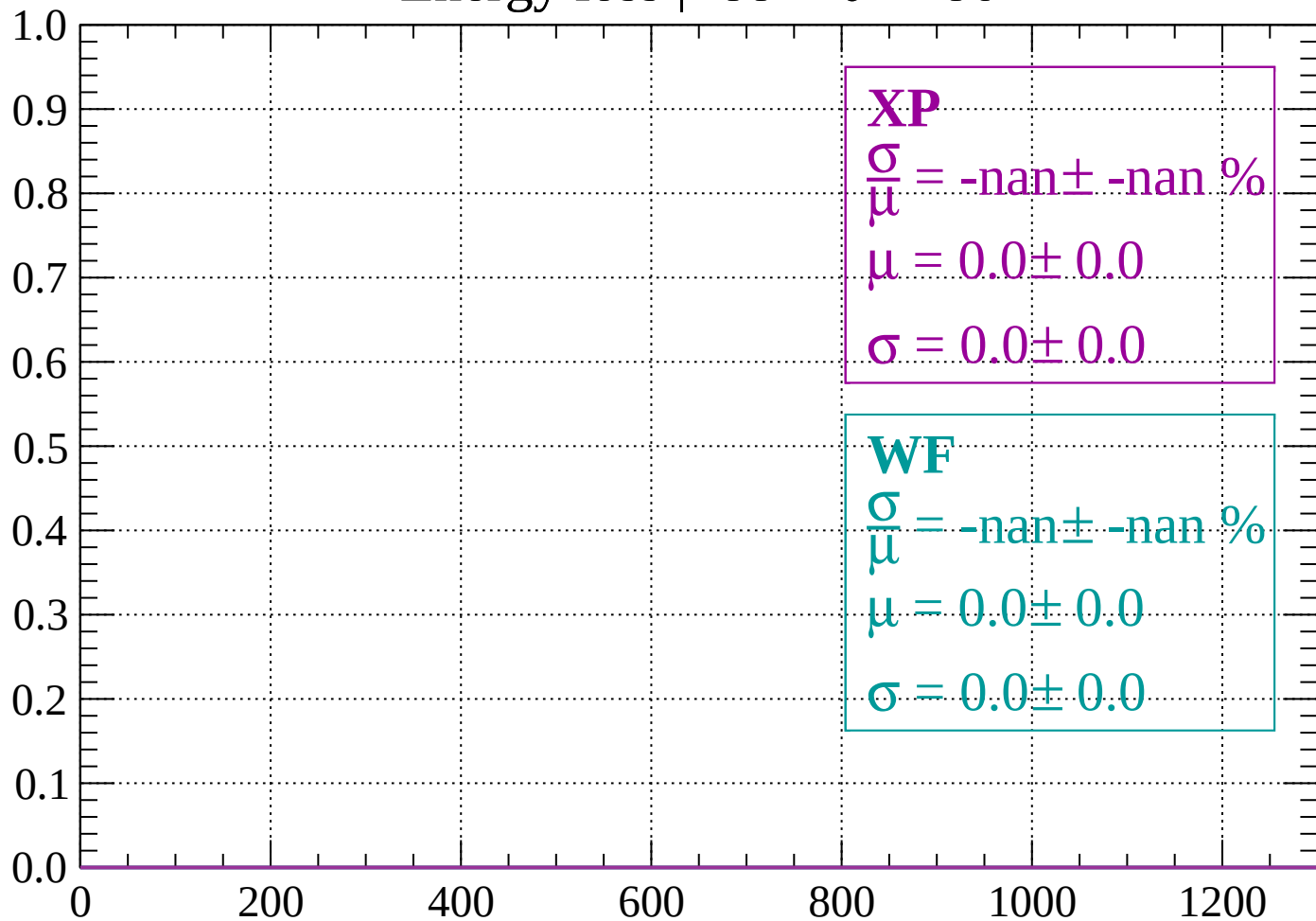
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

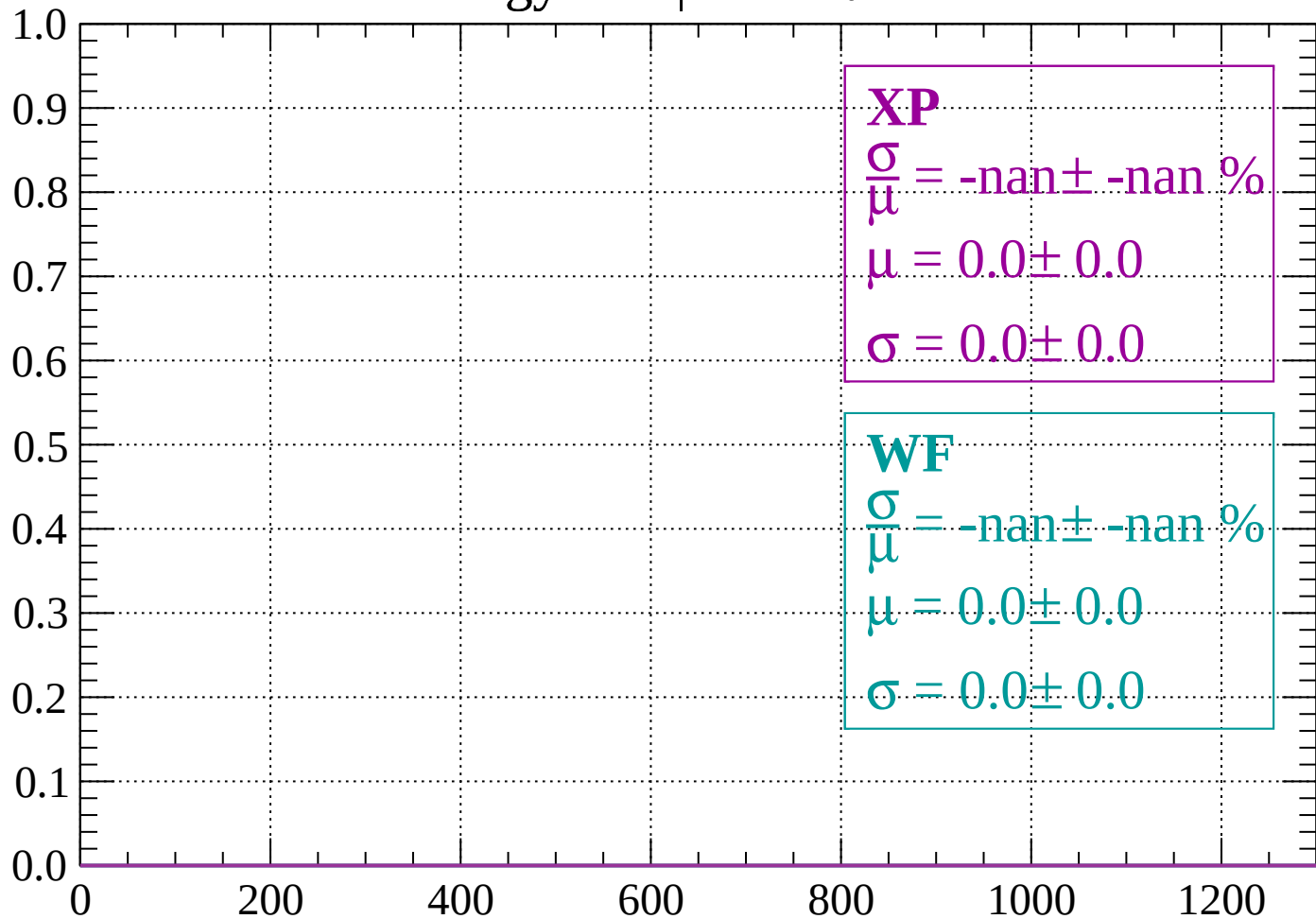
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

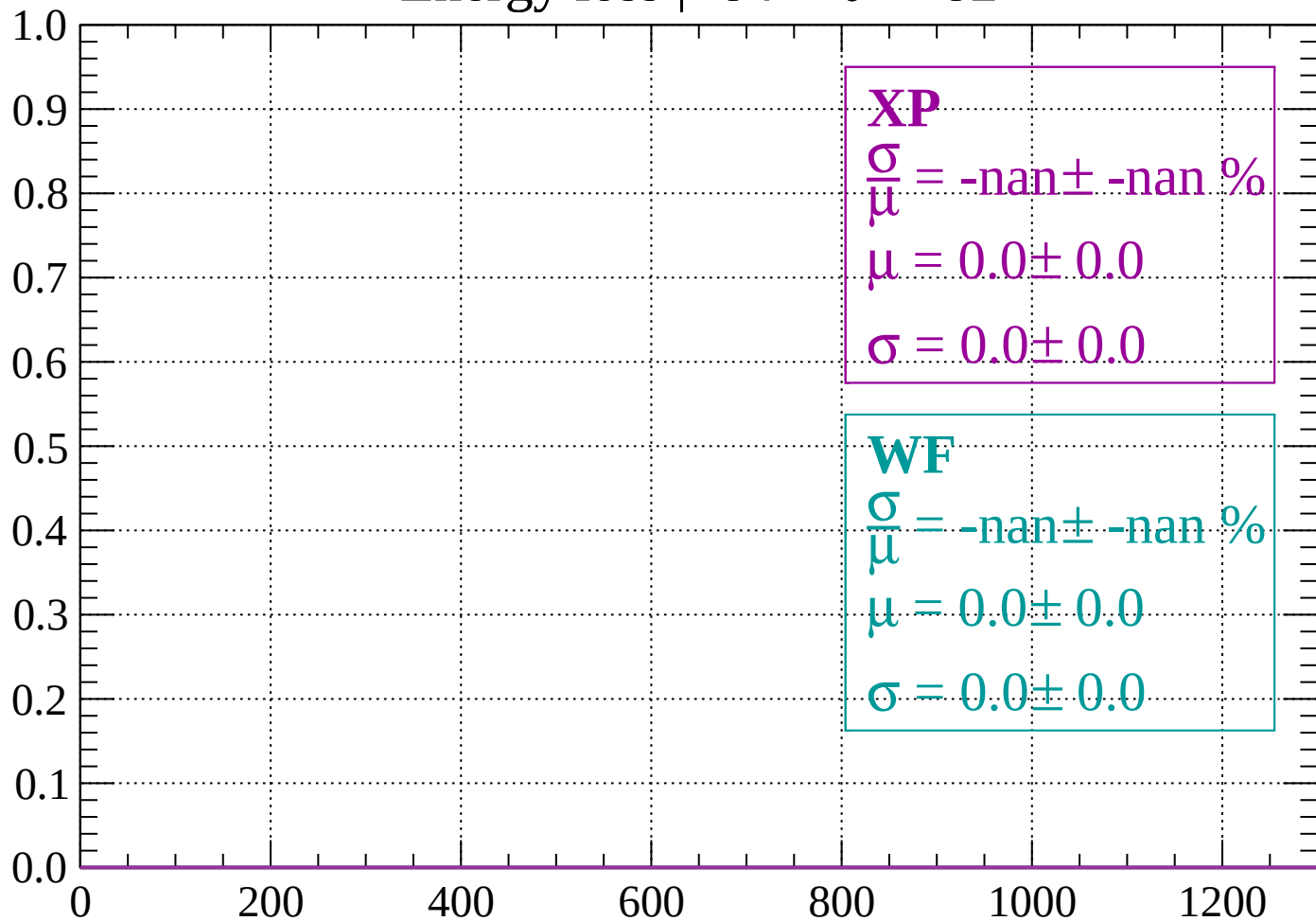
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

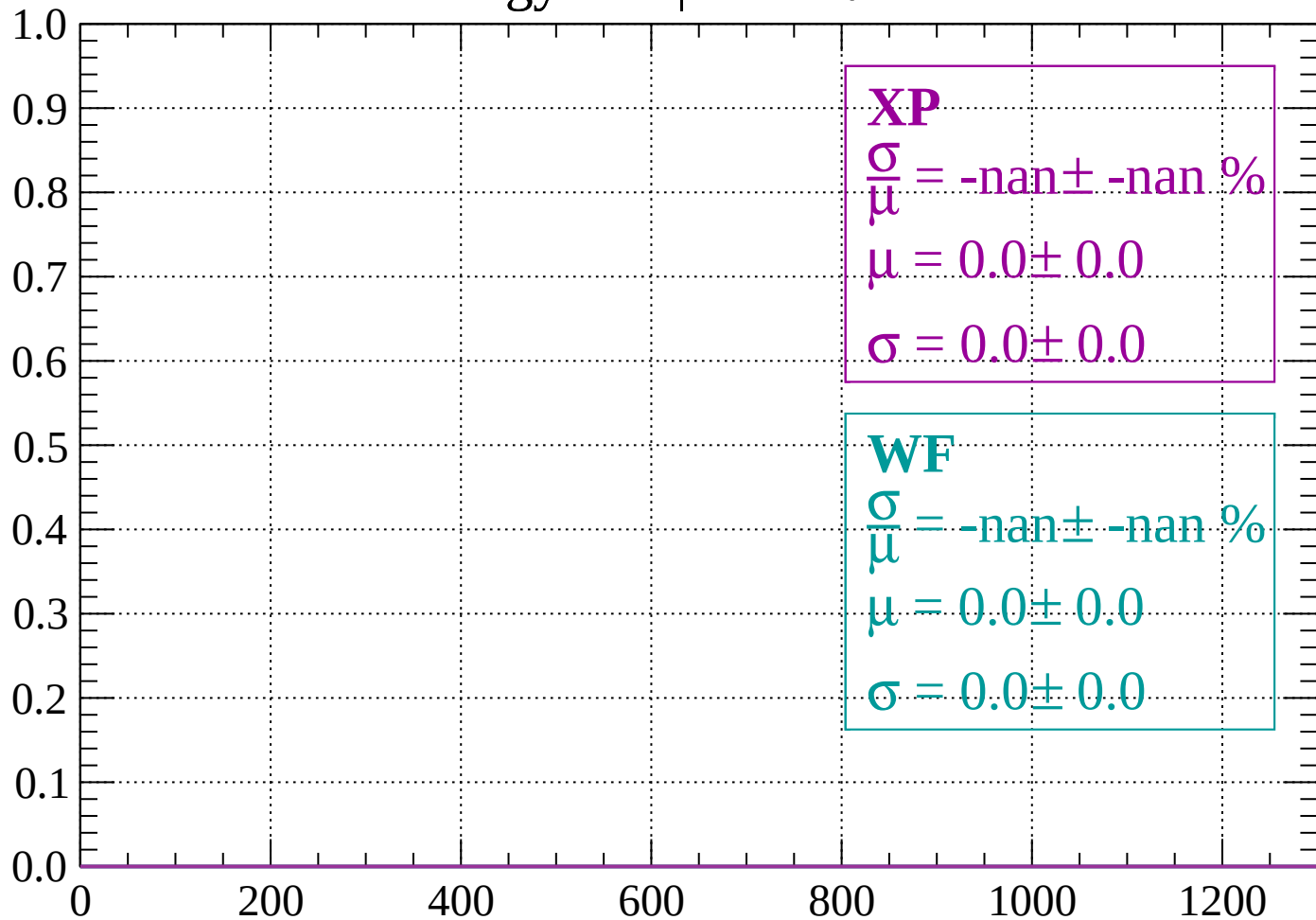
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

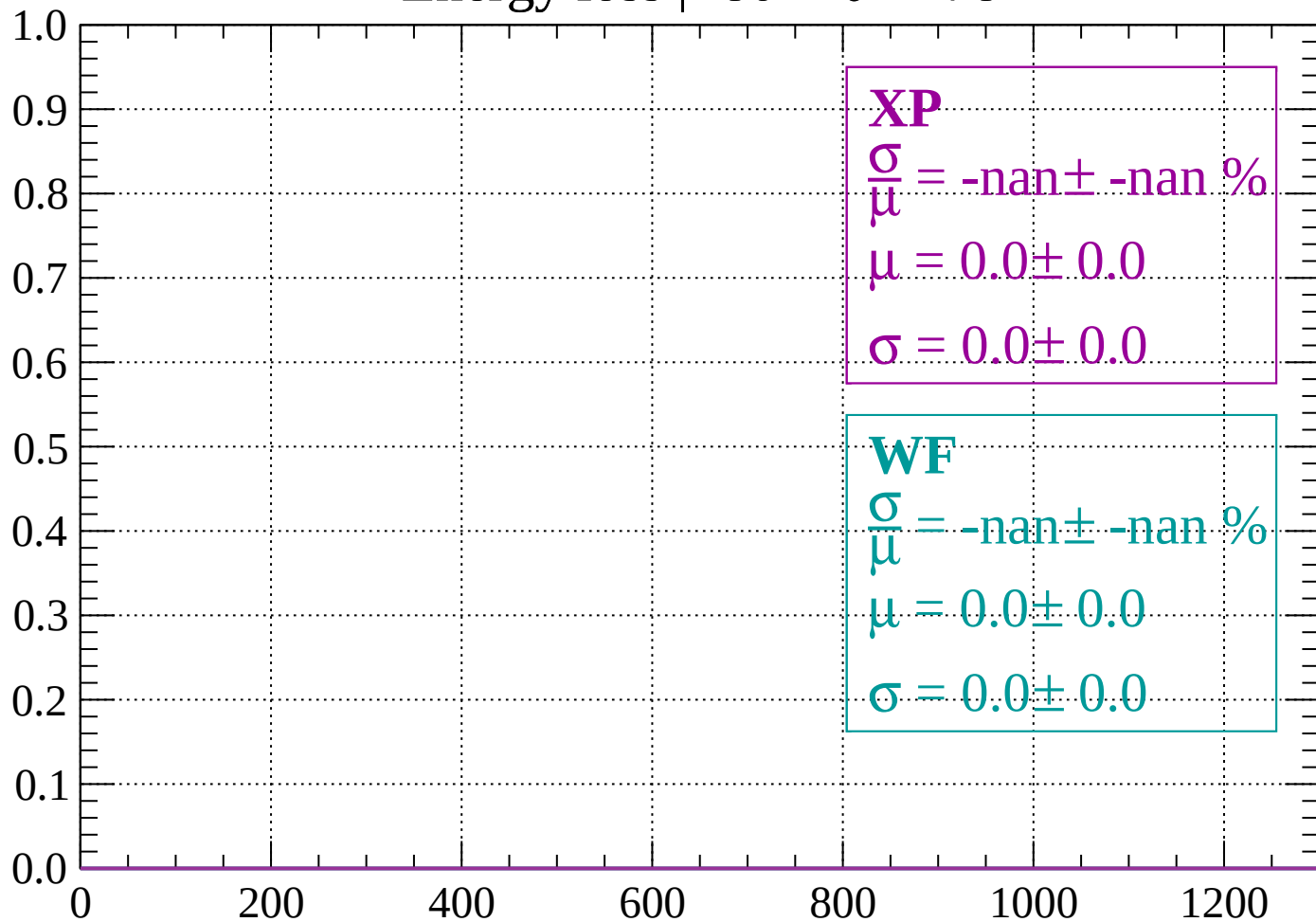
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

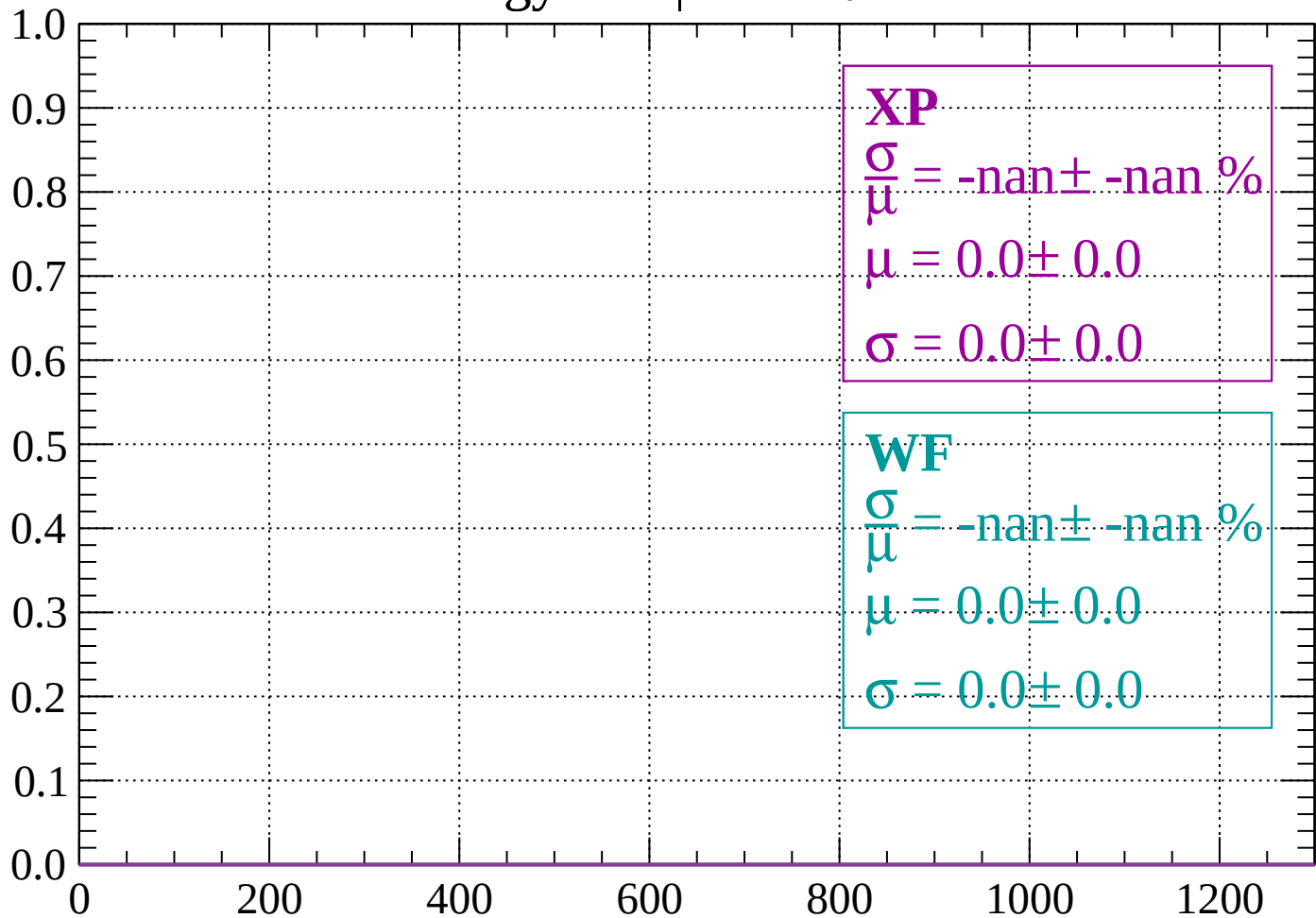
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

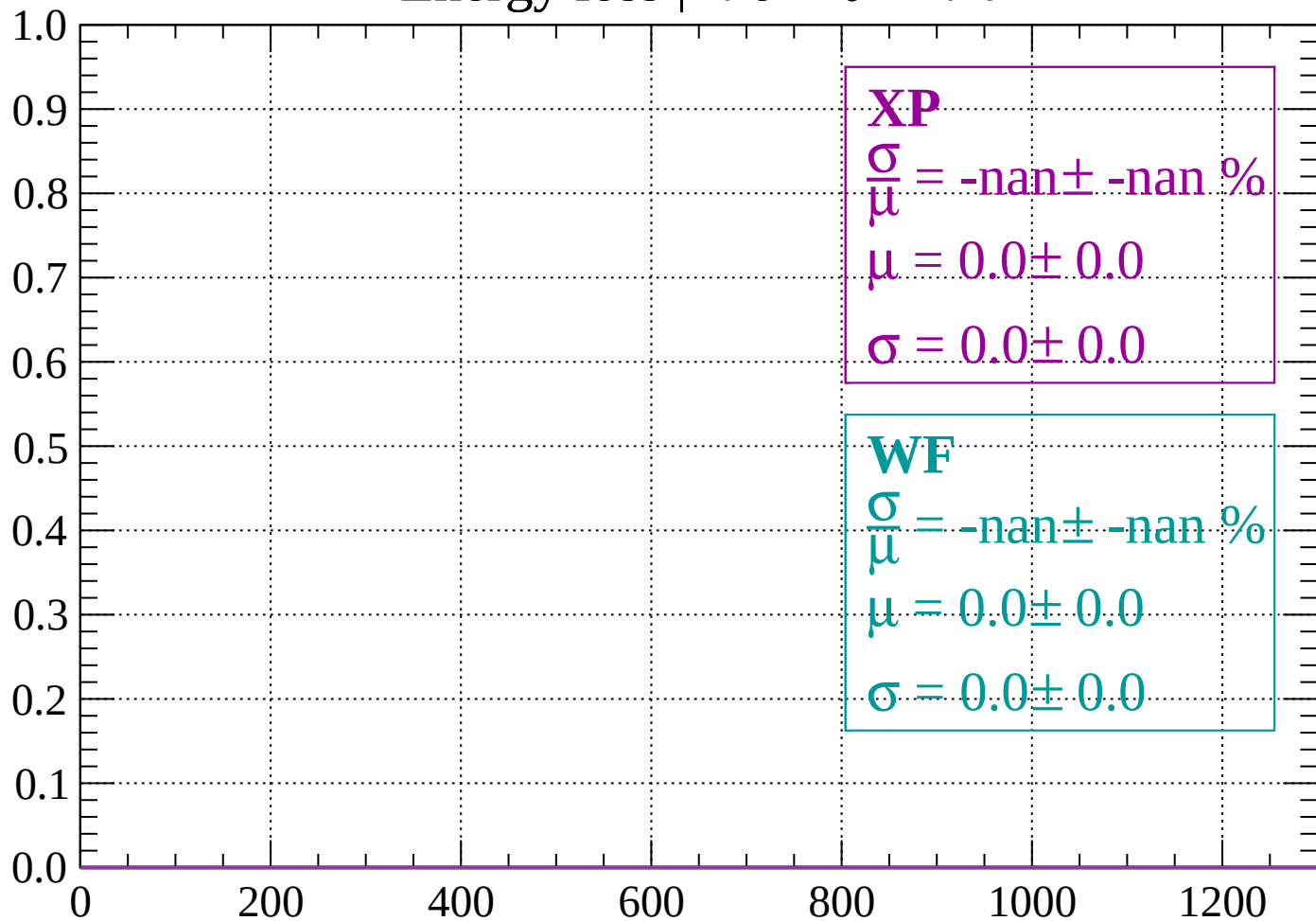
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

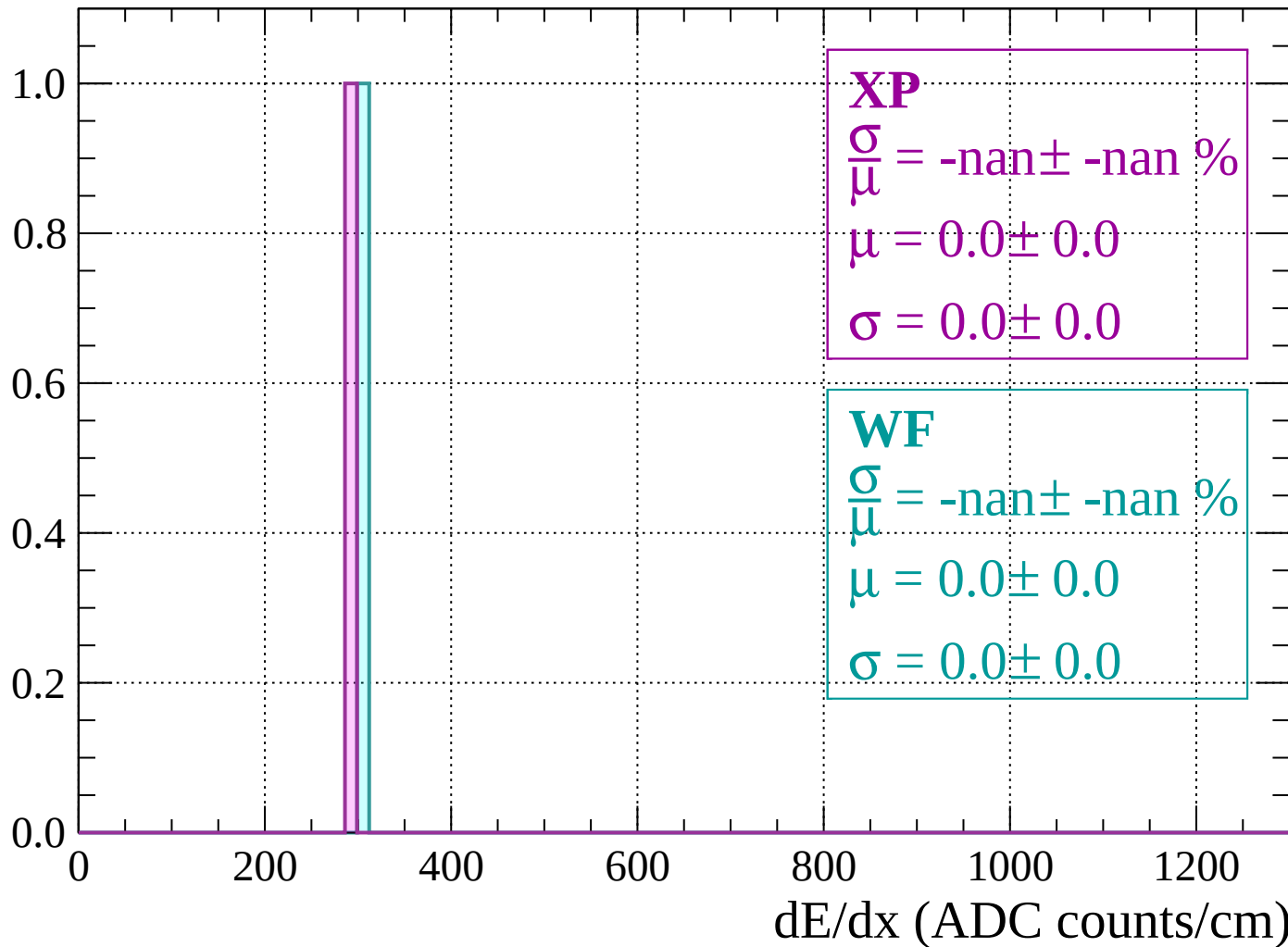
Count



dE/dx (ADC counts/cm)

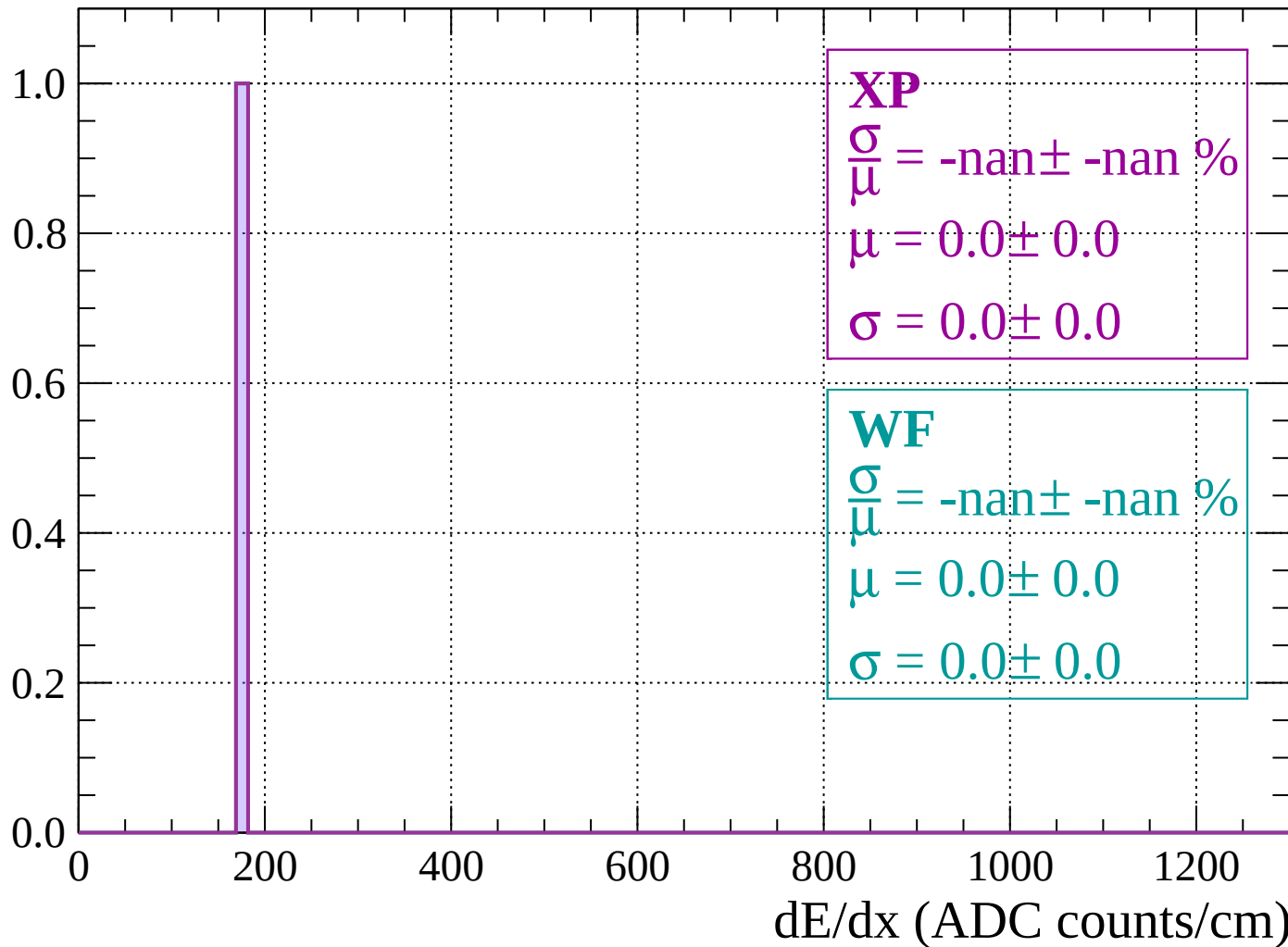
Energy loss | $-74 < \theta < -72$

Count



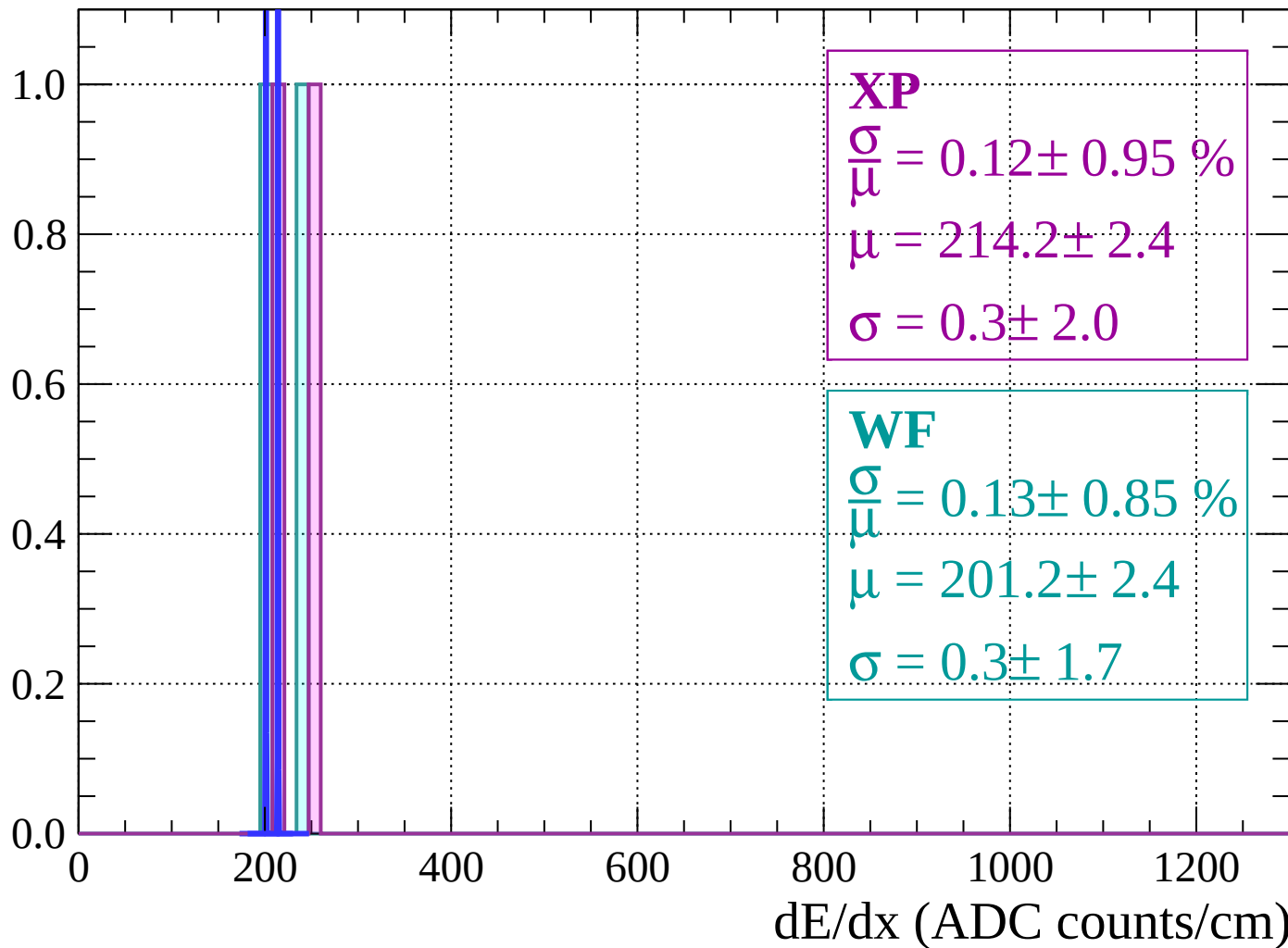
Energy loss | $-72 < \theta < -70$

Count



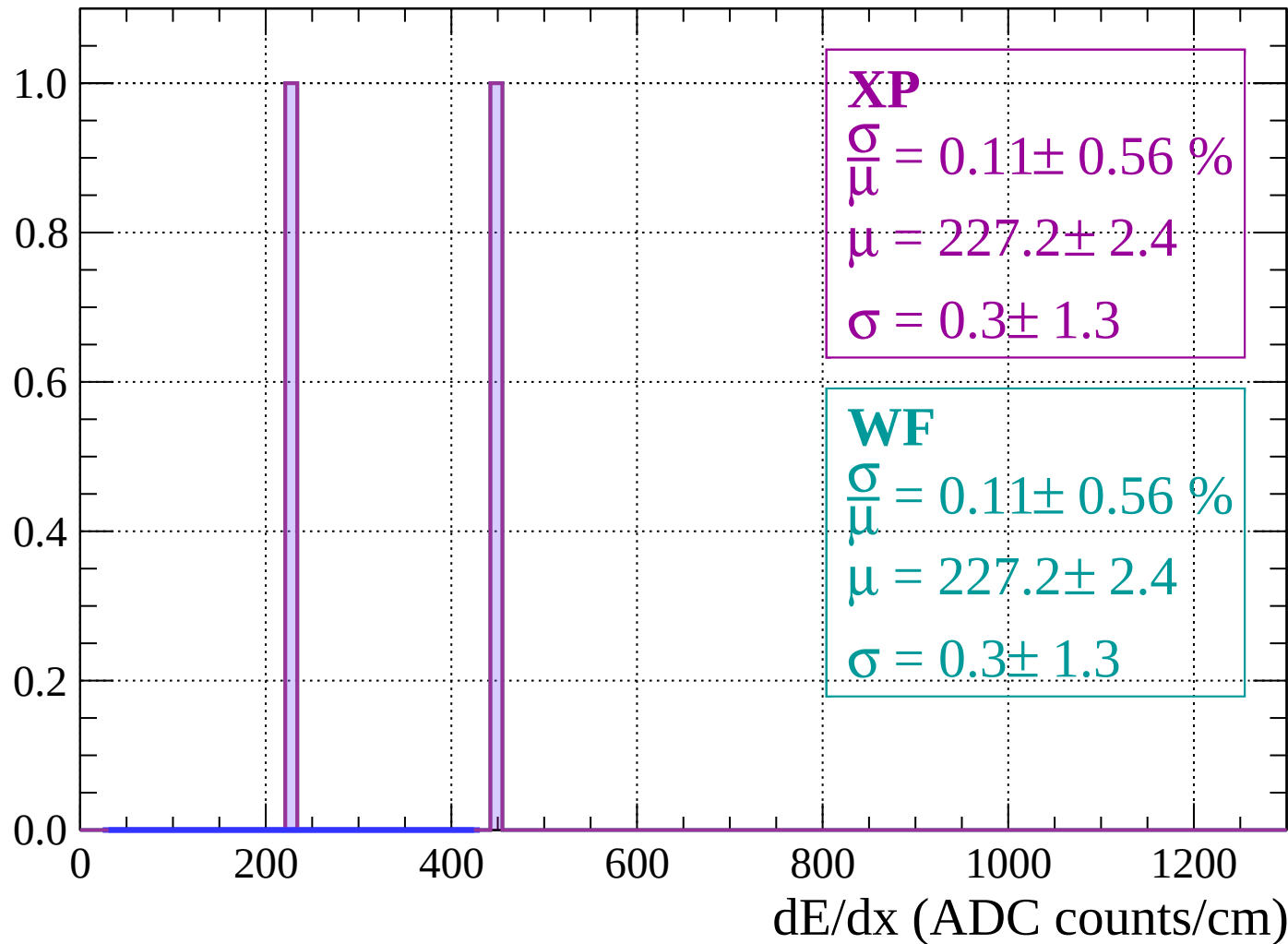
Energy loss | $-70 < \theta < -68$

Count



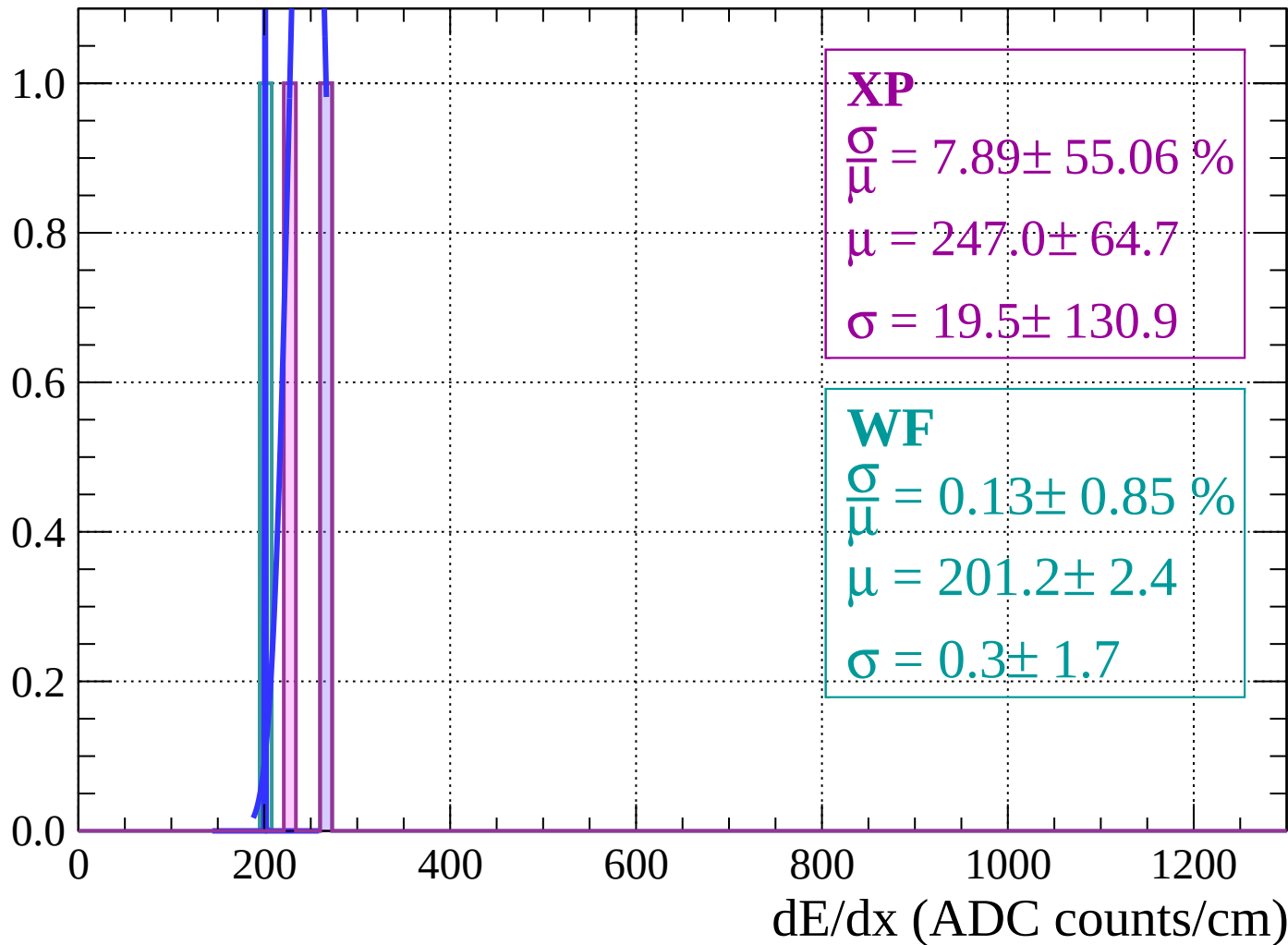
Energy loss | $-68 < \theta < -66$

Count



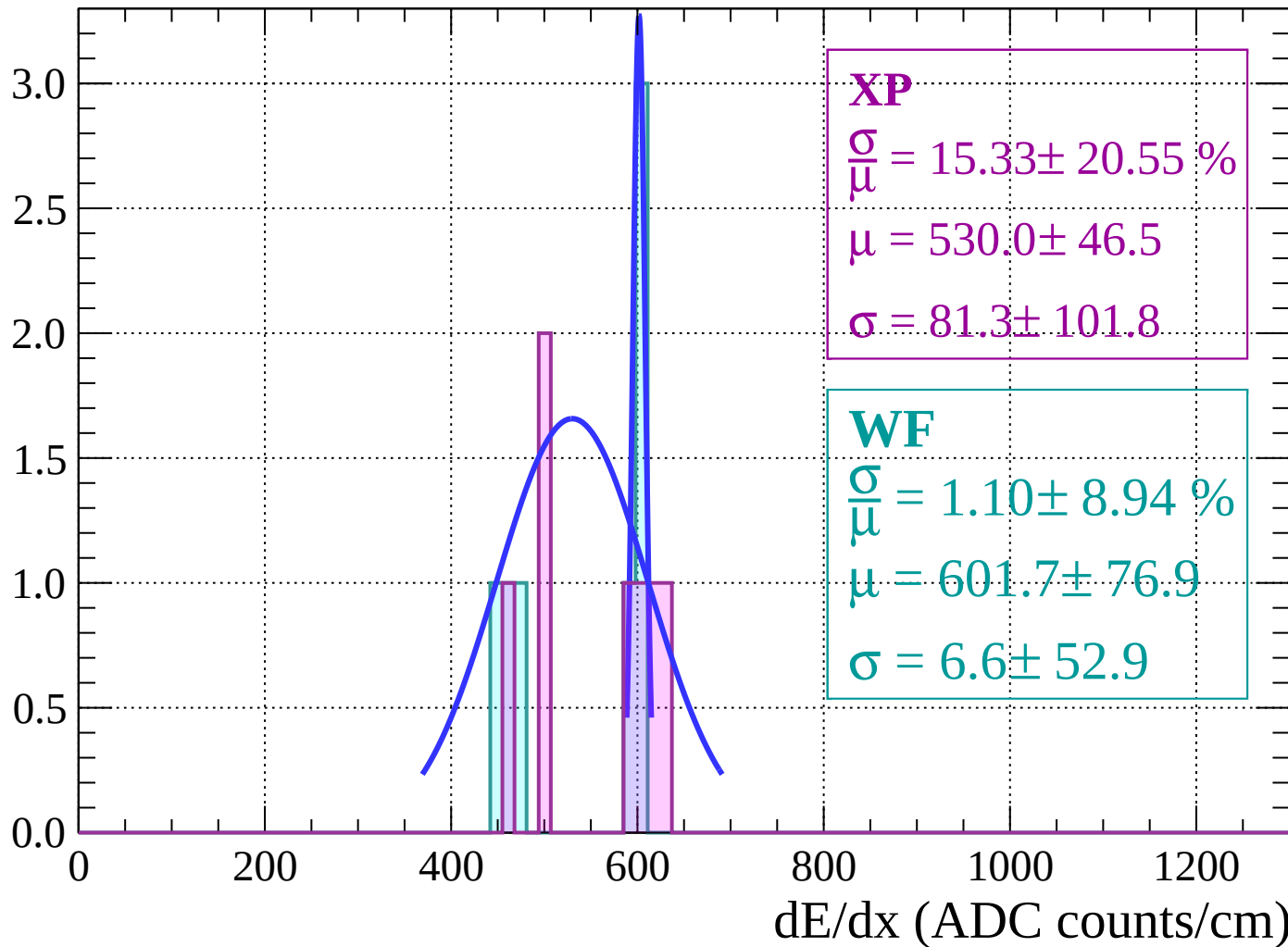
Energy loss | $-66 < \theta < -64$

Count



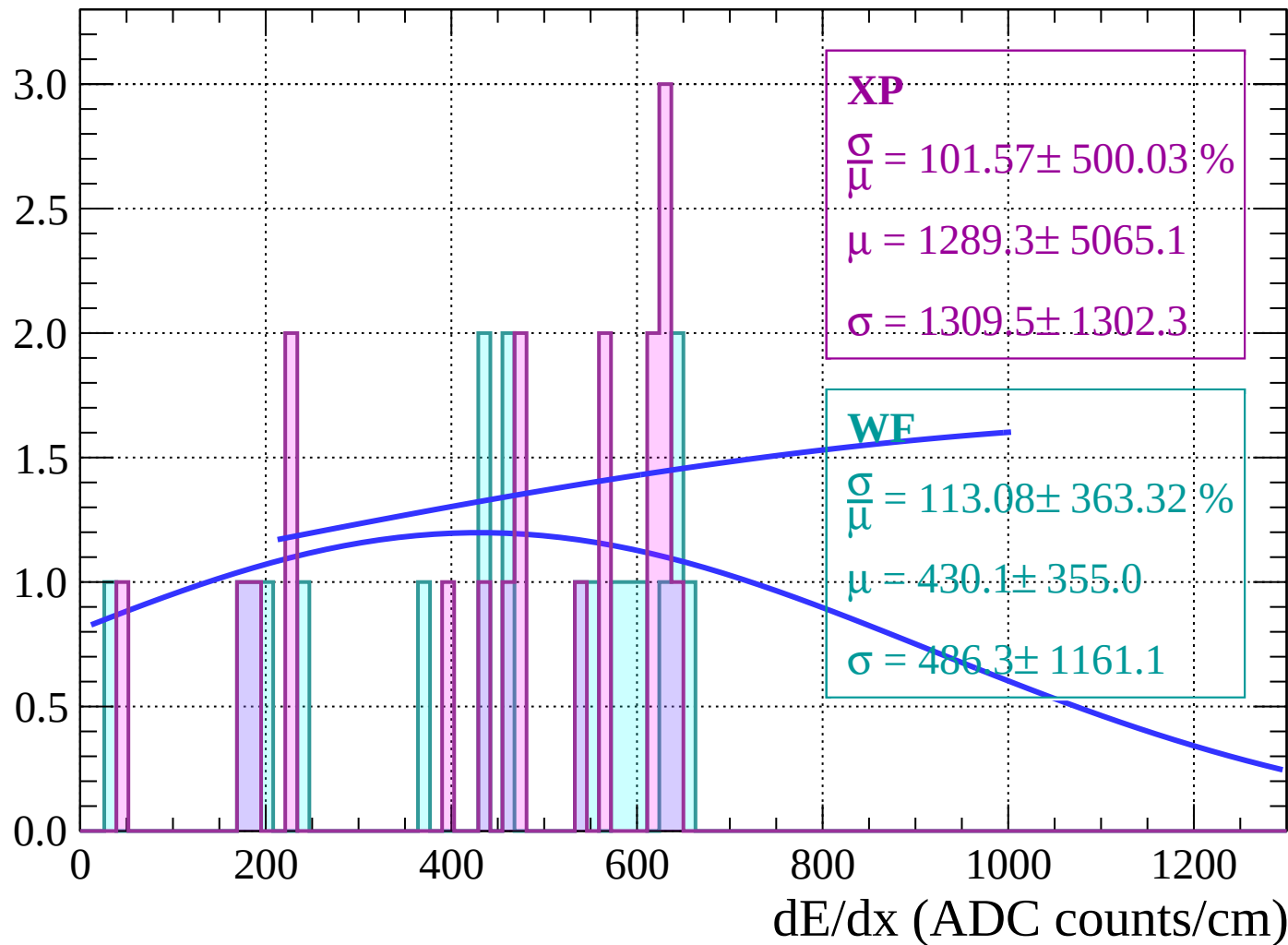
Energy loss | $-64 < \theta < -62$

Count



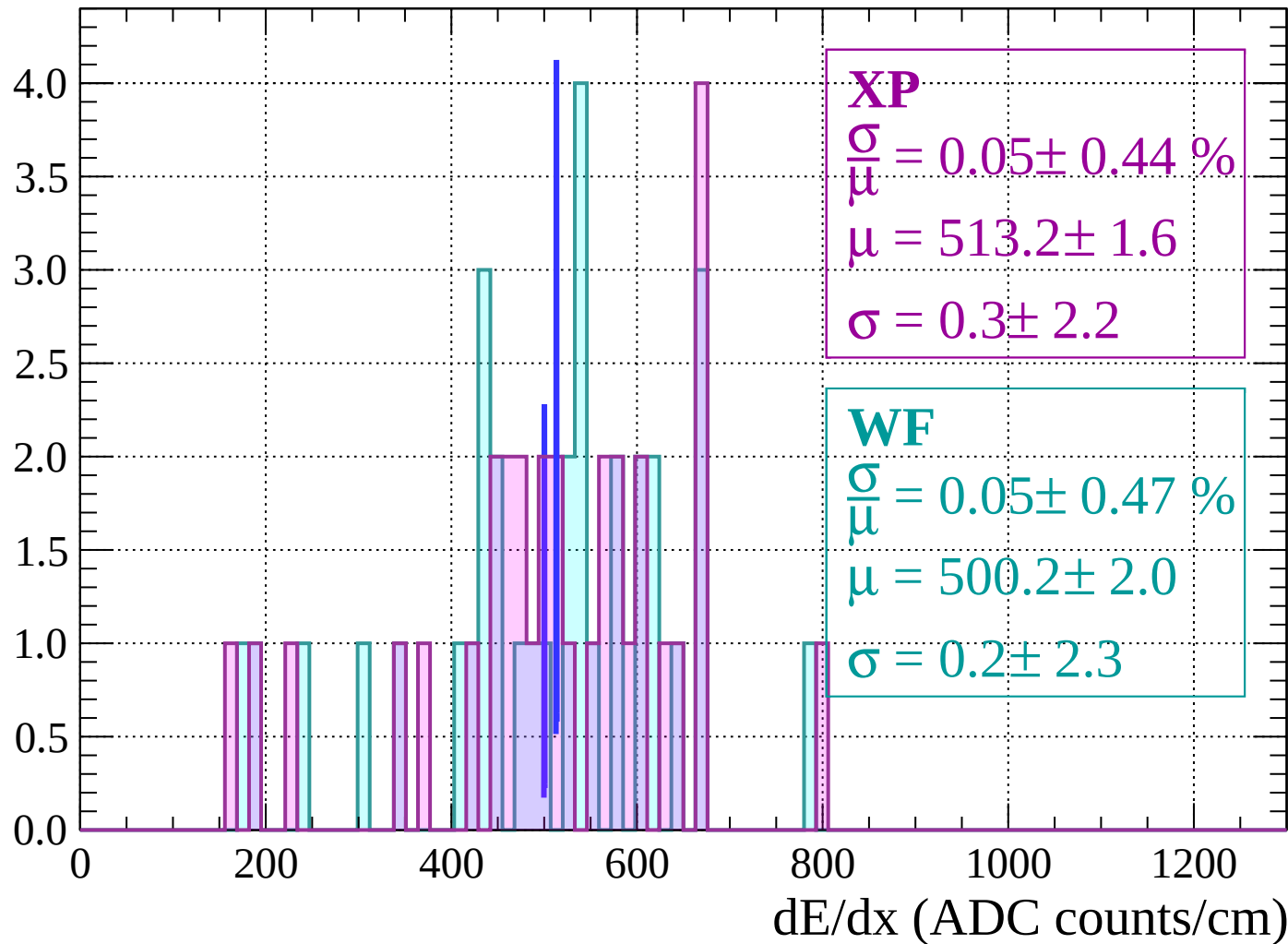
Energy loss | $-62 < \theta < -60$

Count



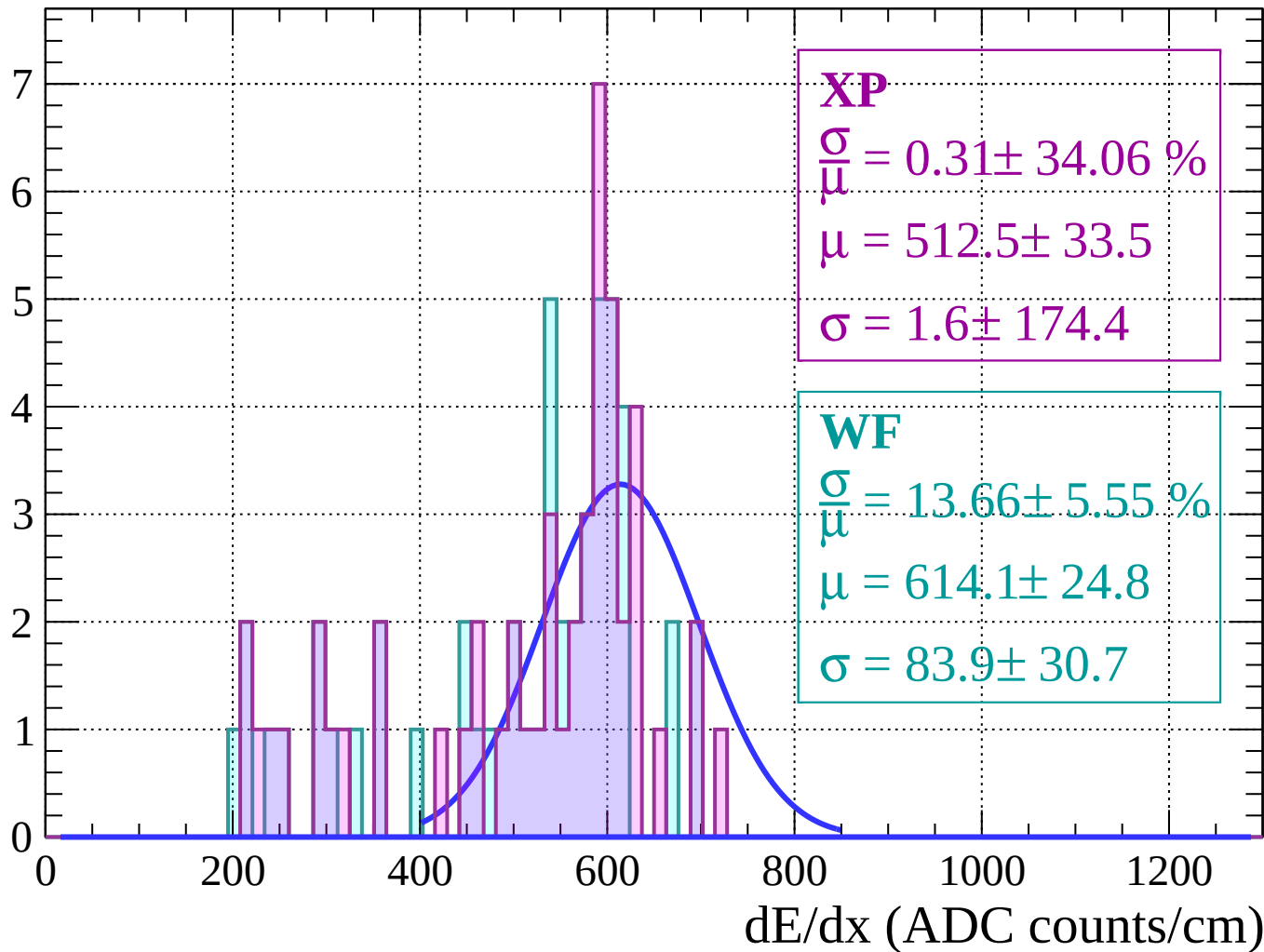
Energy loss | $-60 < \theta < -58$

Count



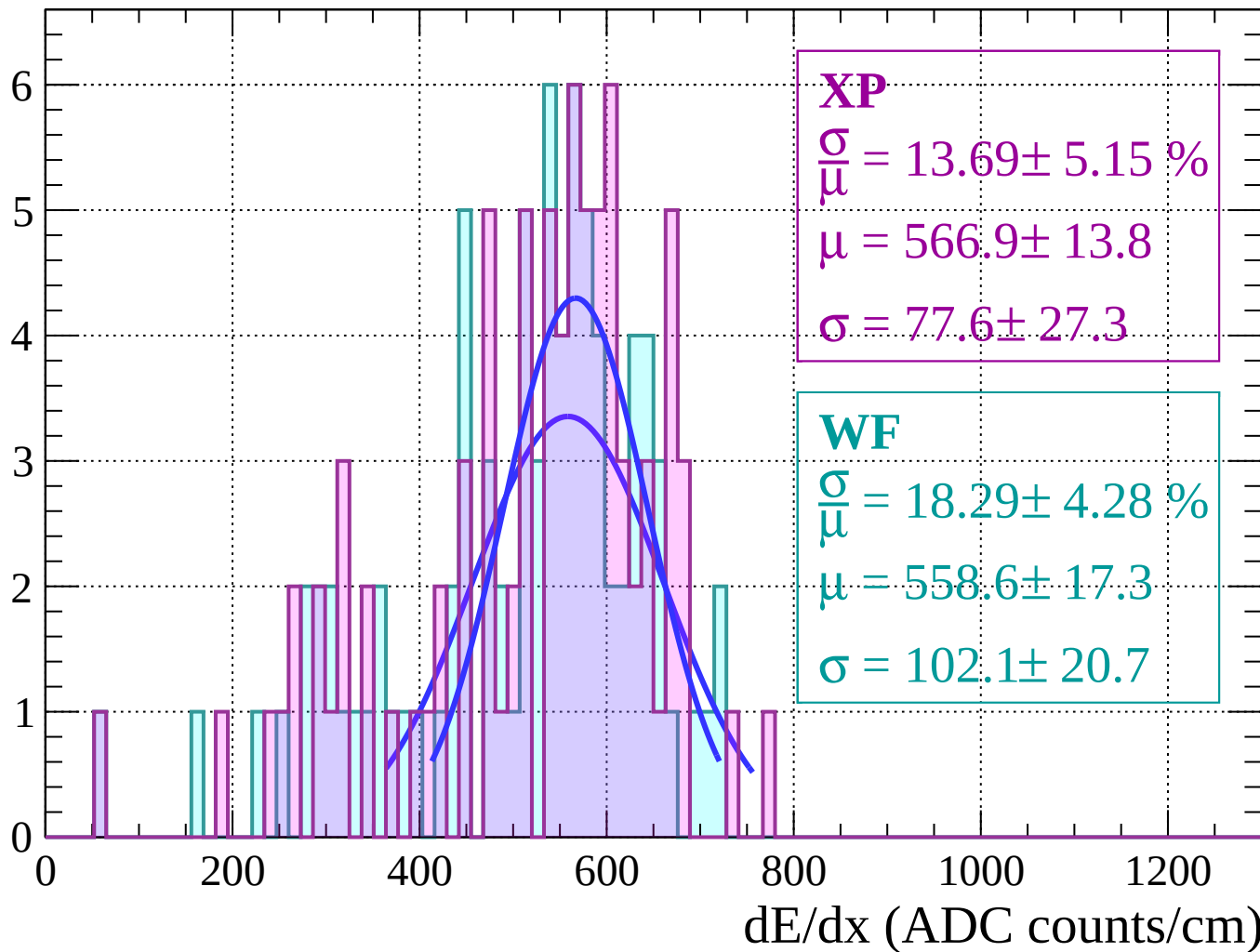
Energy loss | $-58 < \theta < -56$

Count



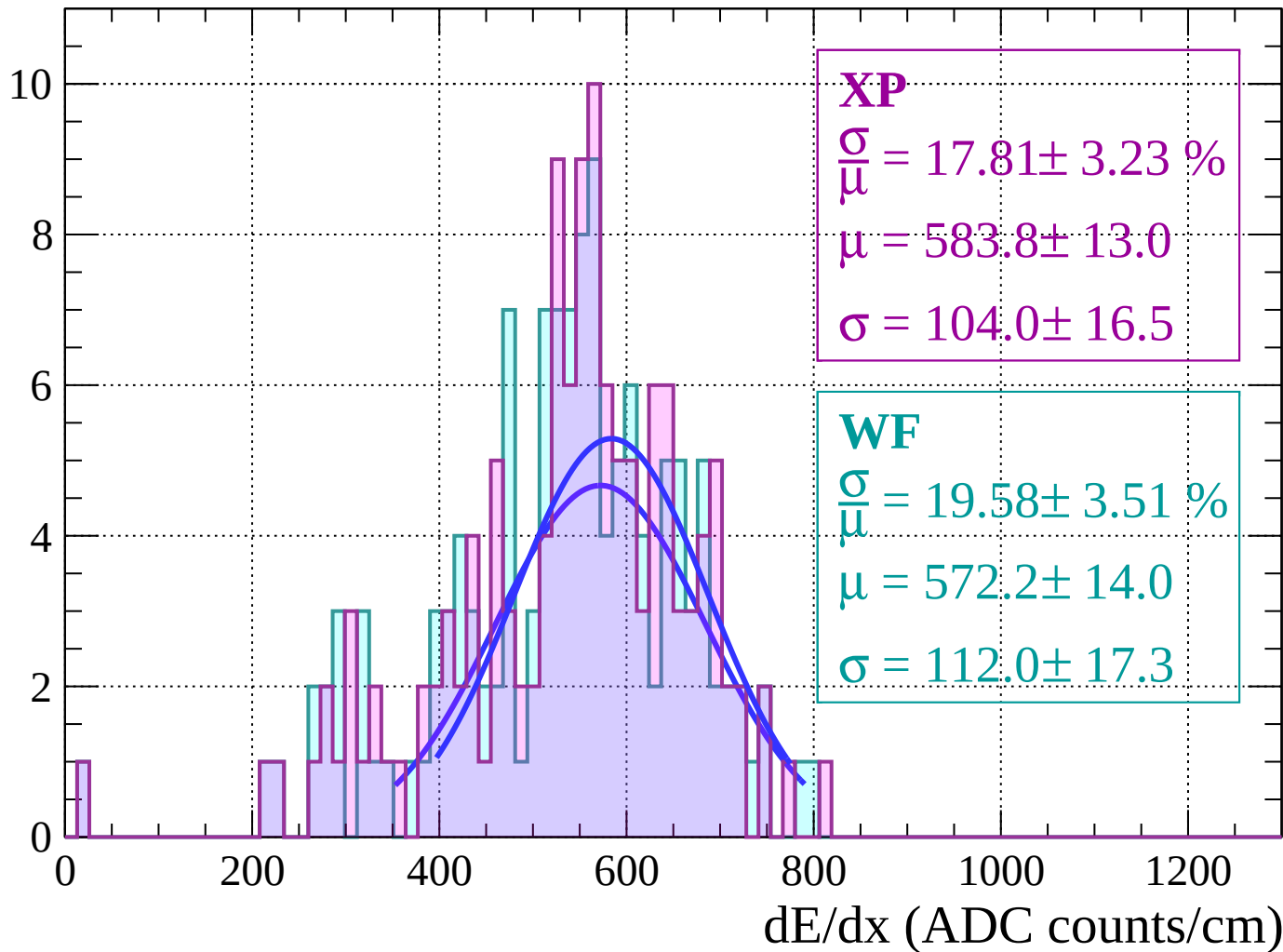
Energy loss $|-56 < \theta < -54$

Count



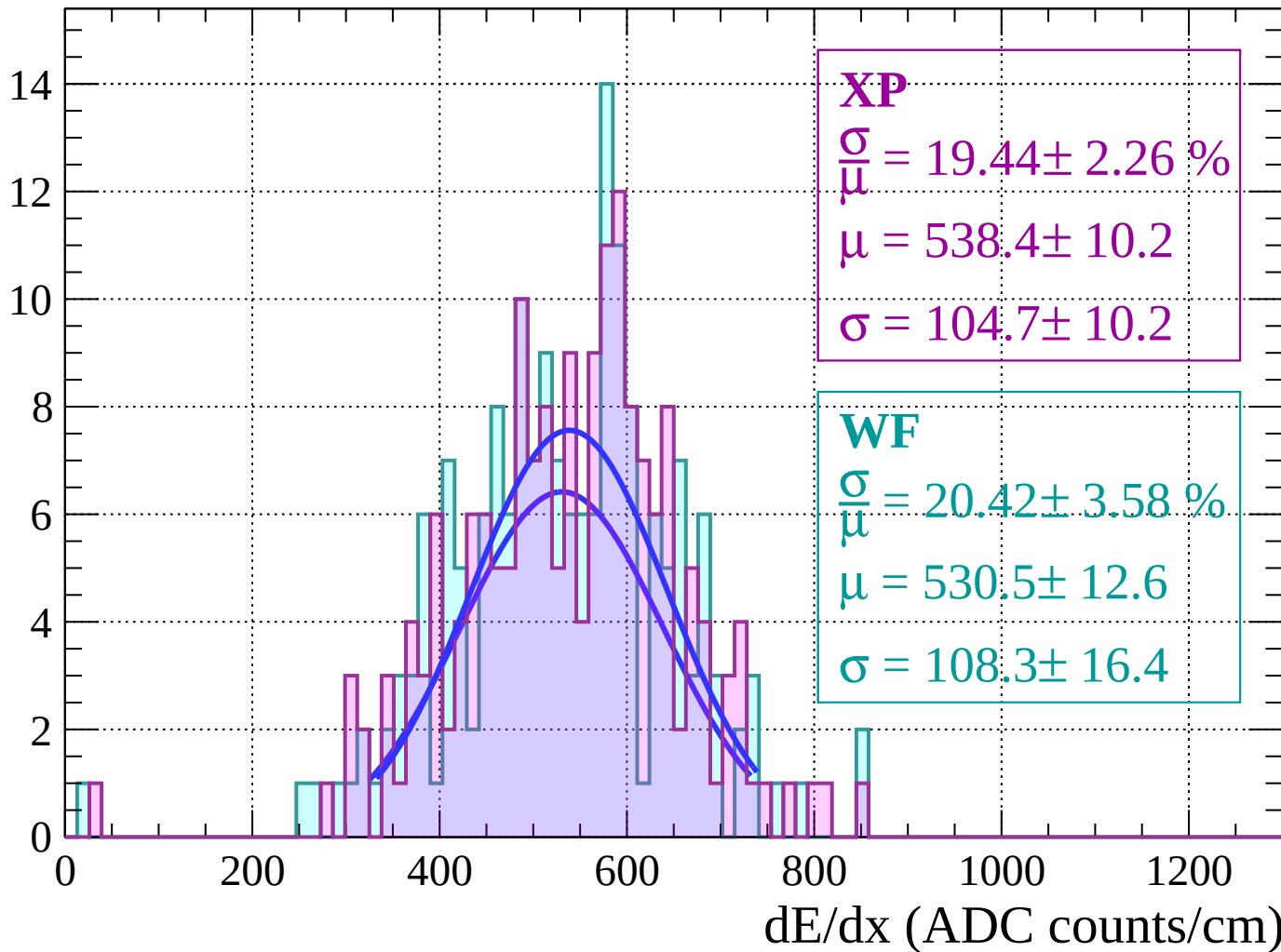
Energy loss $|-54 < \theta < -52$

Count



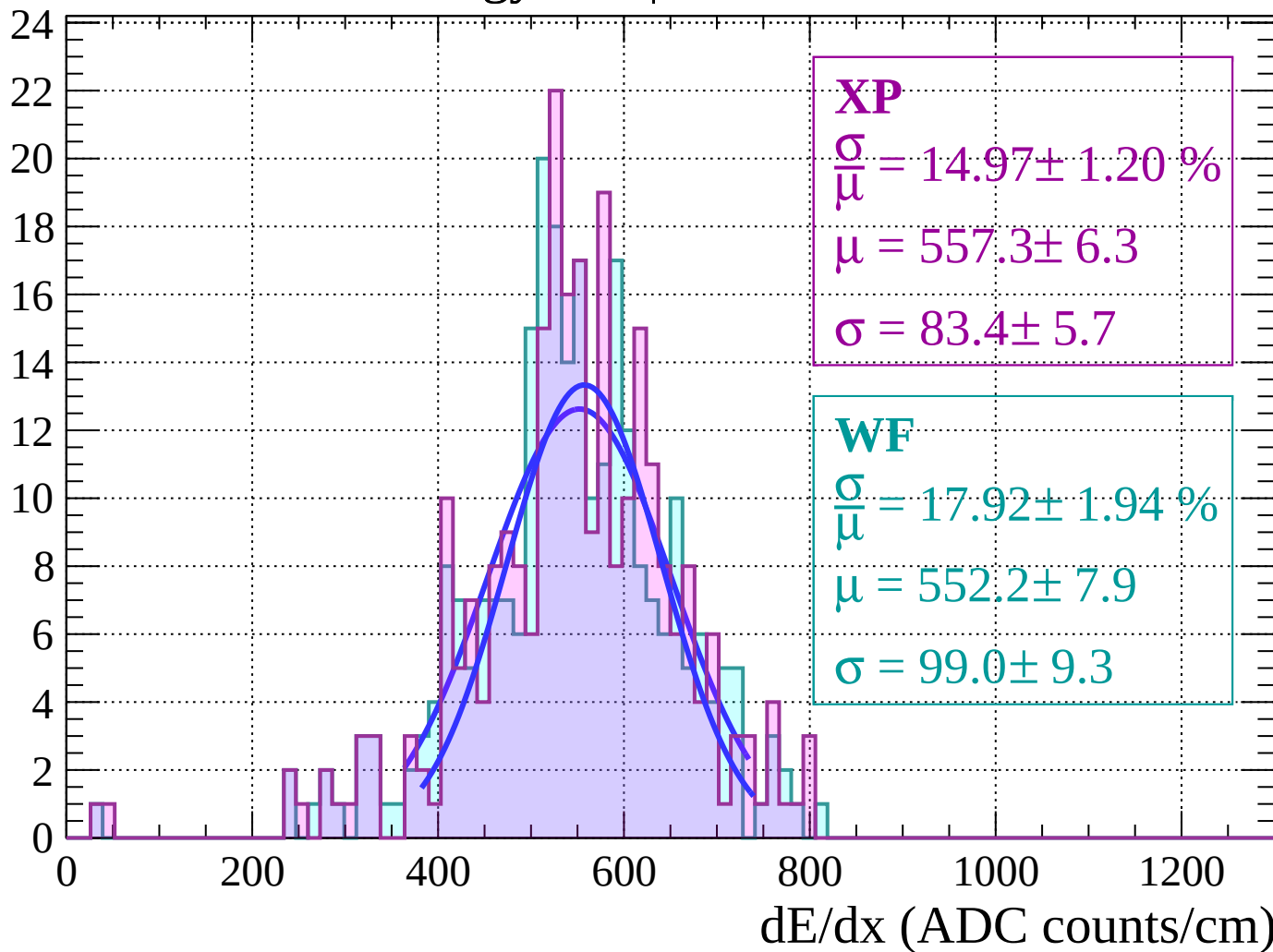
Energy loss | $-52 < \theta < -50$

Count



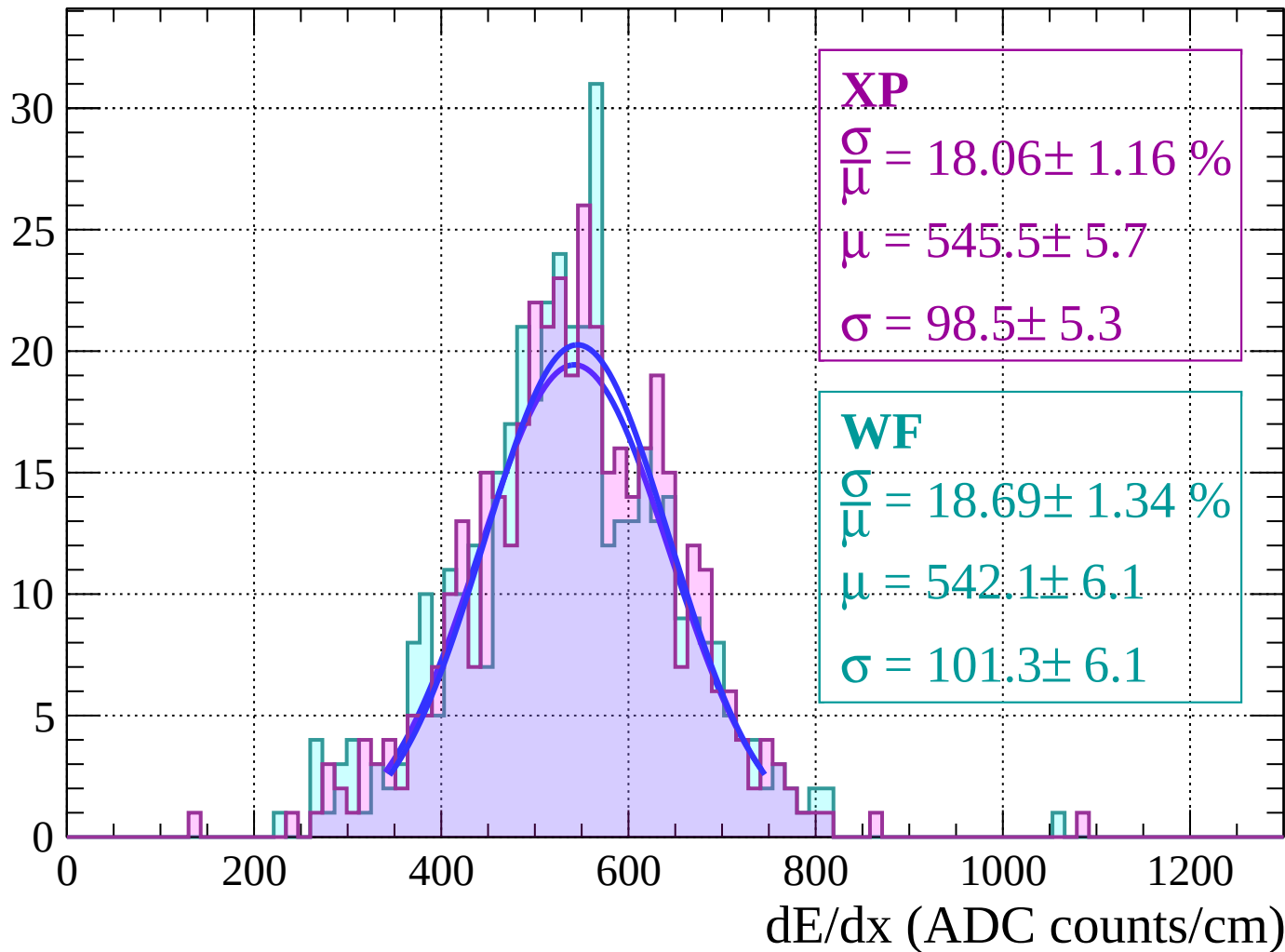
Energy loss | $-50 < \theta < -48$

Count



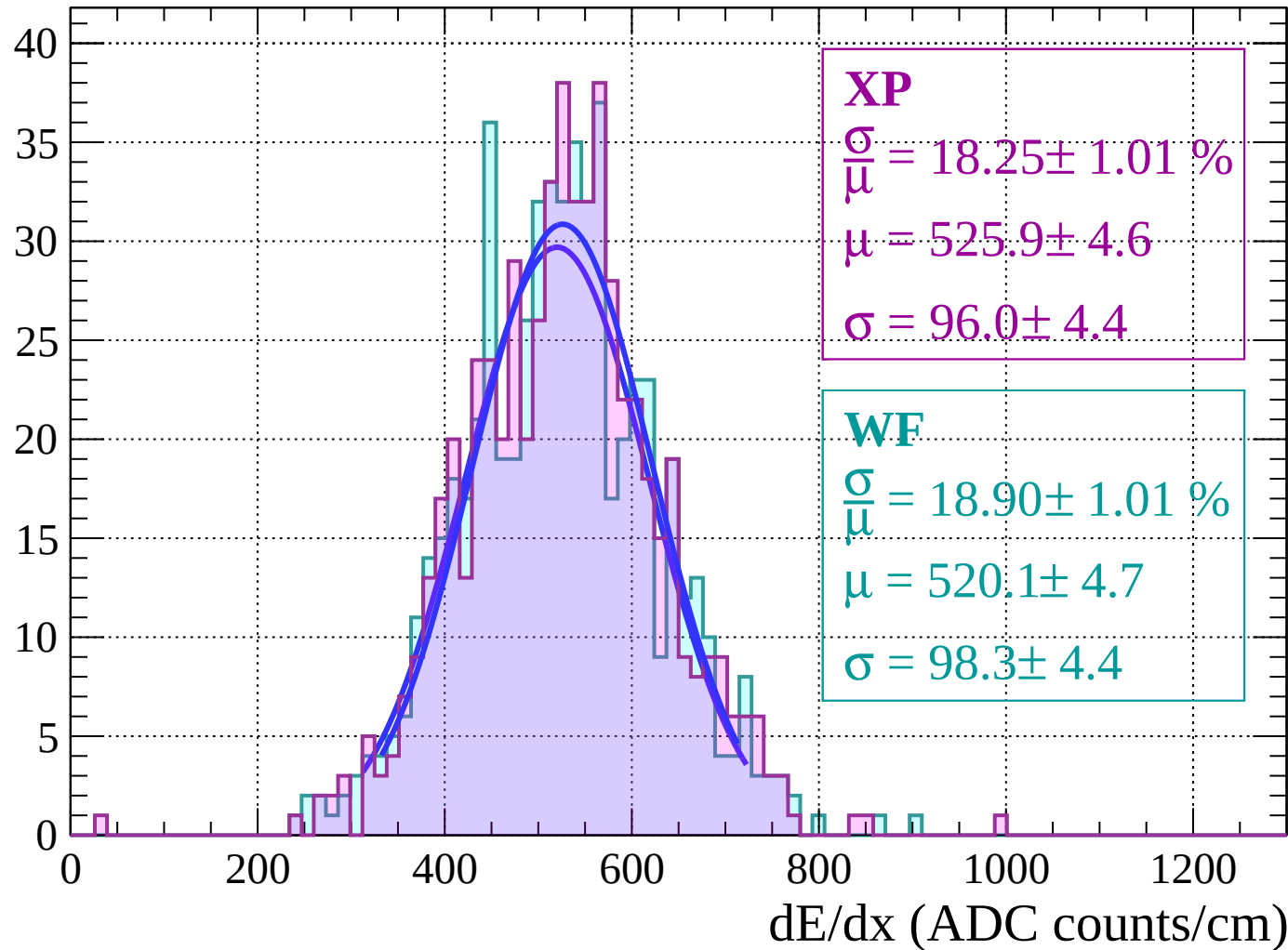
Energy loss | $-48 < \theta < -46$

Count



Energy loss $|-46 < \theta < -44$

Count



Energy loss | $-44 < \theta < -42$

Count

XP

$$\frac{\sigma}{\mu} = 16.89 \pm 0.73 \%$$

$$\mu = 526.8 \pm 3.5$$

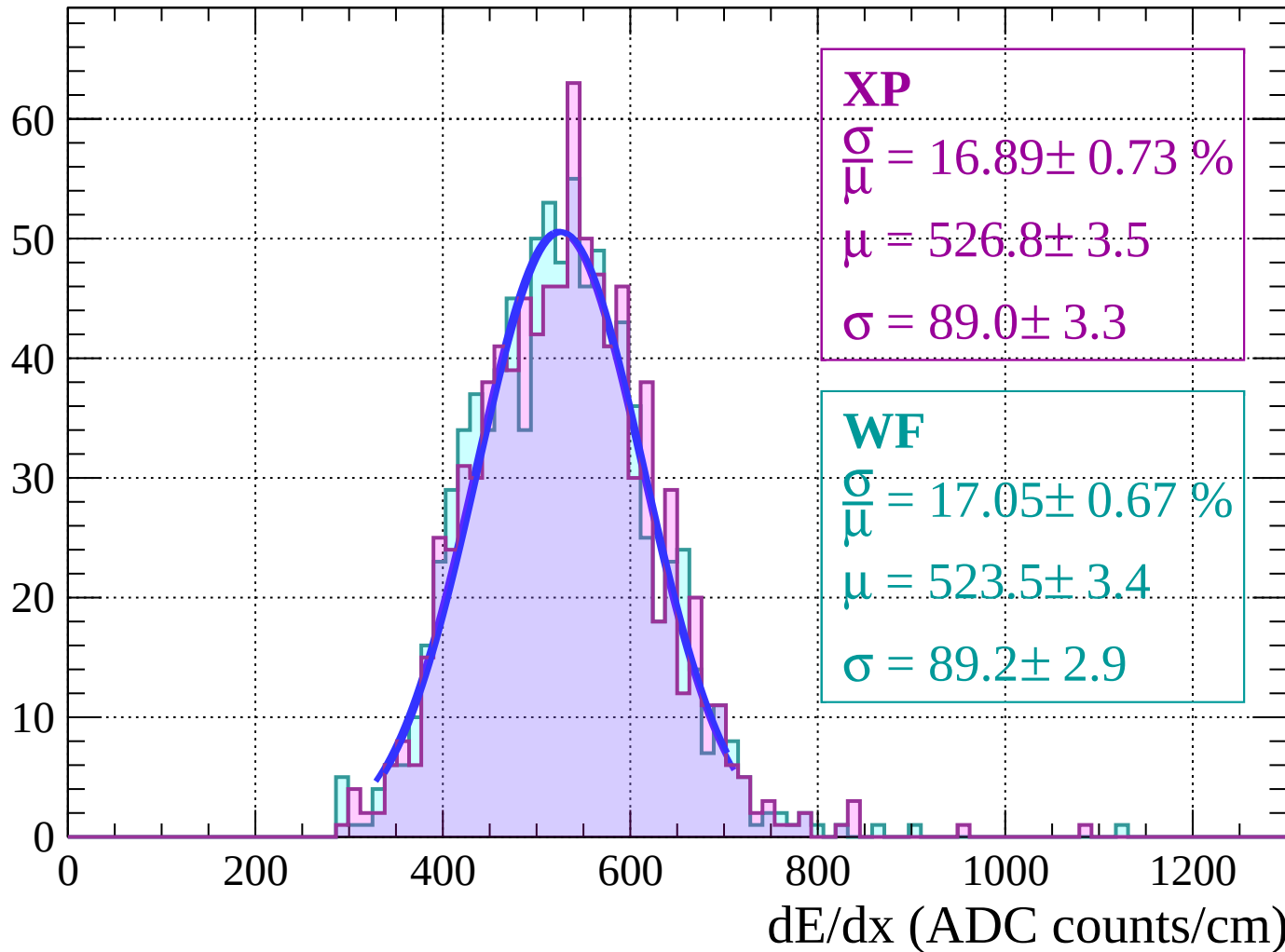
$$\sigma = 89.0 \pm 3.3$$

WF

$$\frac{\sigma}{\mu} = 17.05 \pm 0.67 \%$$

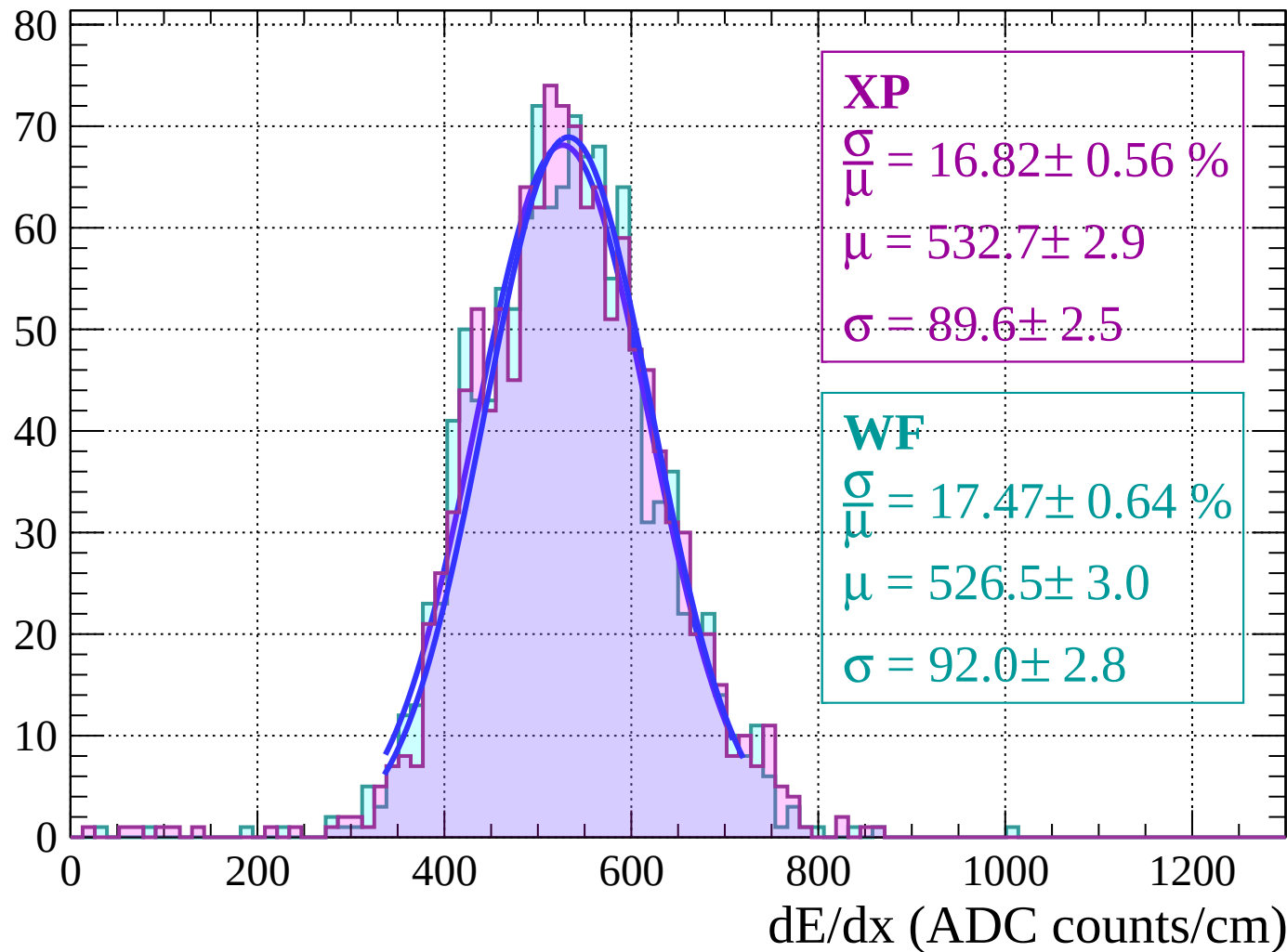
$$\mu = 523.5 \pm 3.4$$

$$\sigma = 89.2 \pm 2.9$$



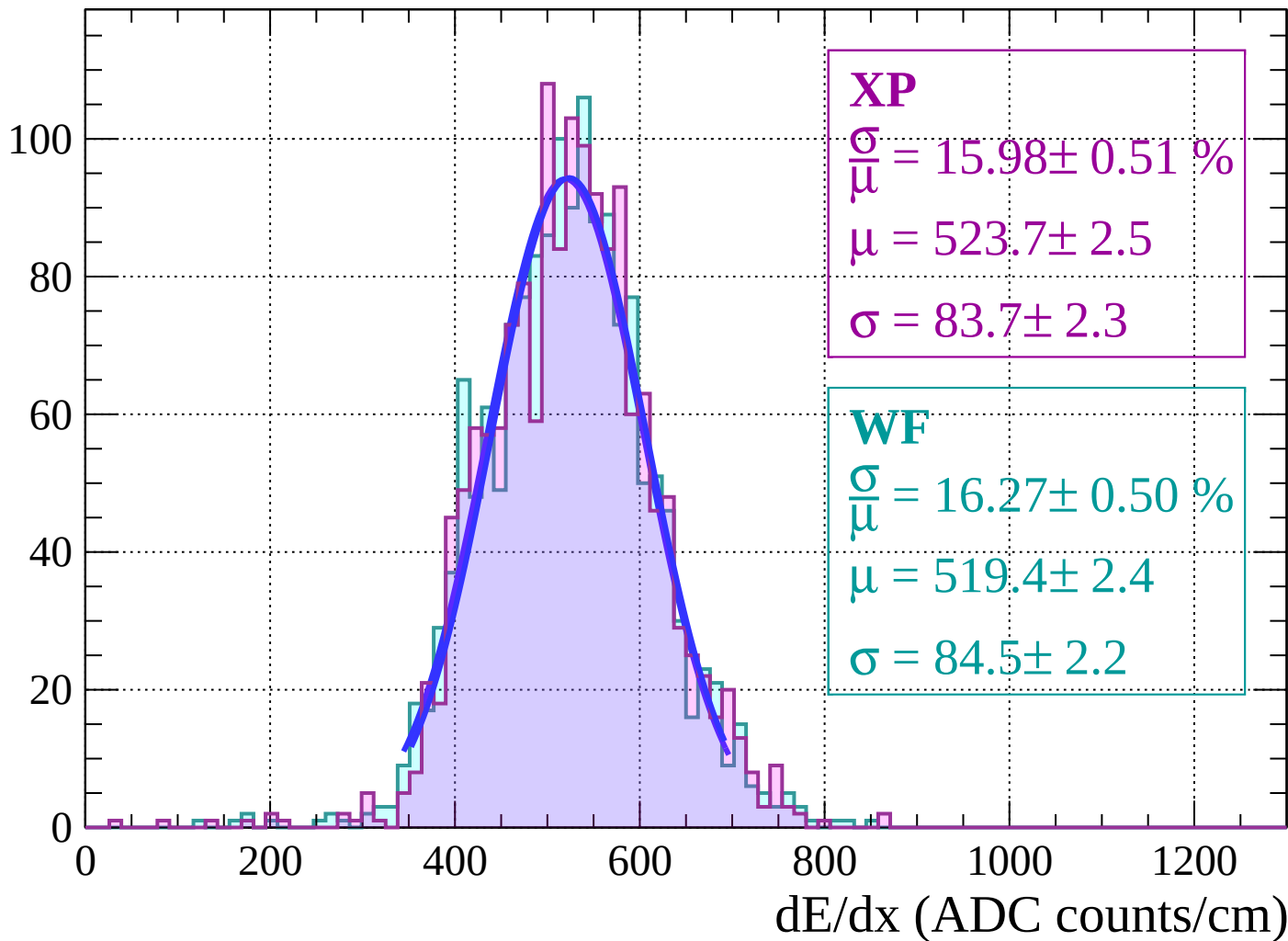
Energy loss | $-42 < \theta < -40$

Count



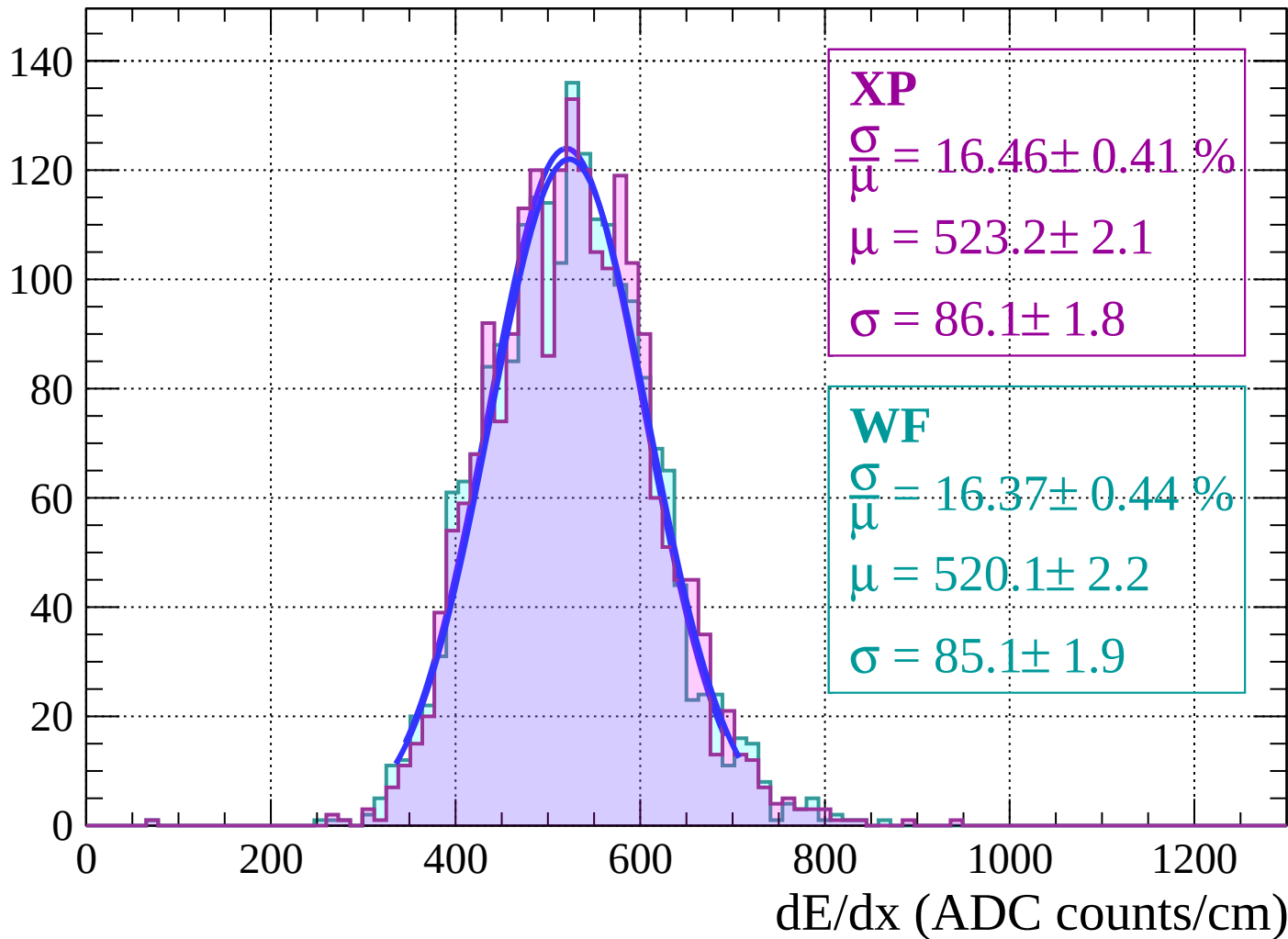
Energy loss | $-40 < \theta < -38$

Count



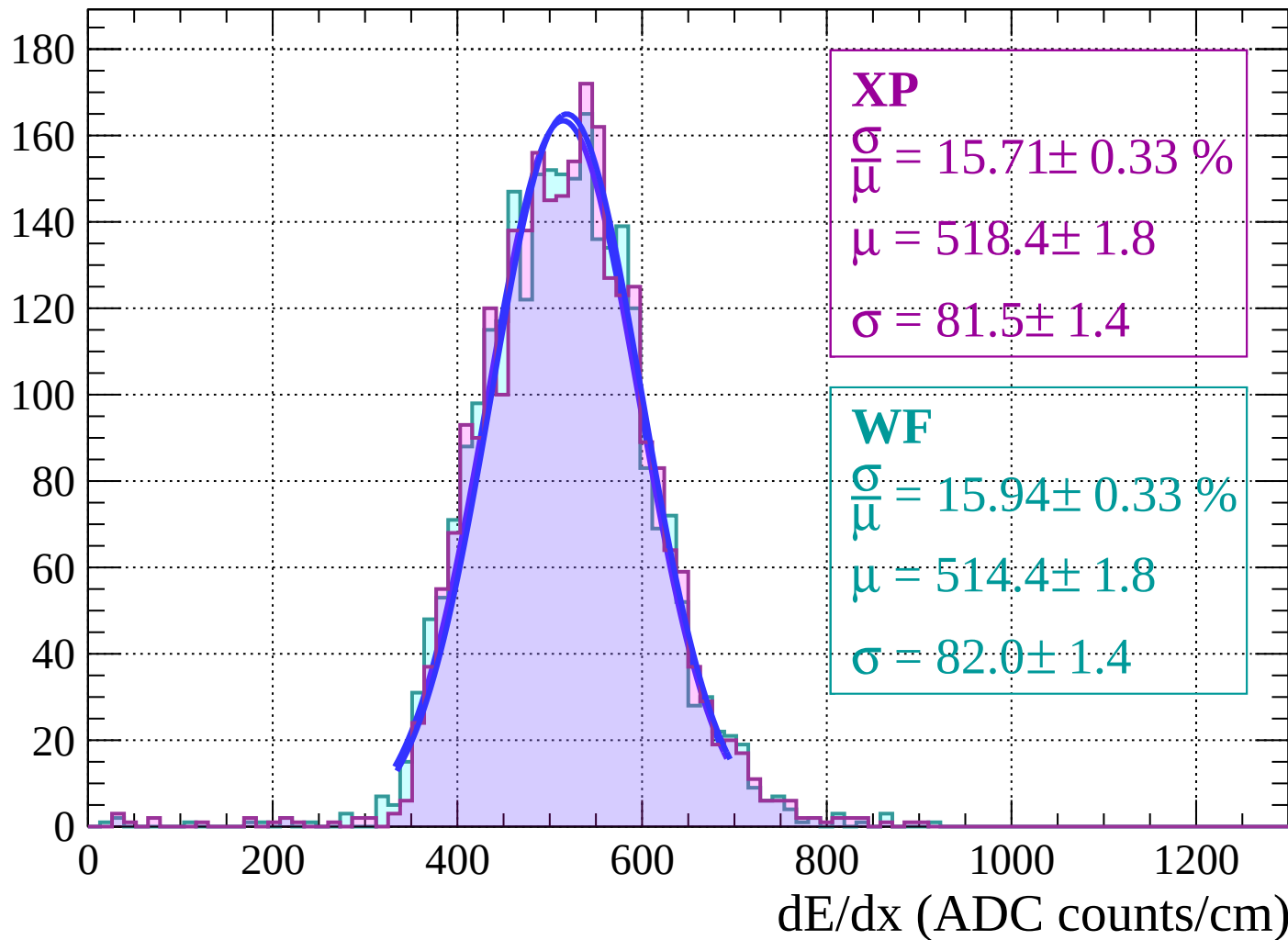
Energy loss $|-38 < \theta < -36$

Count



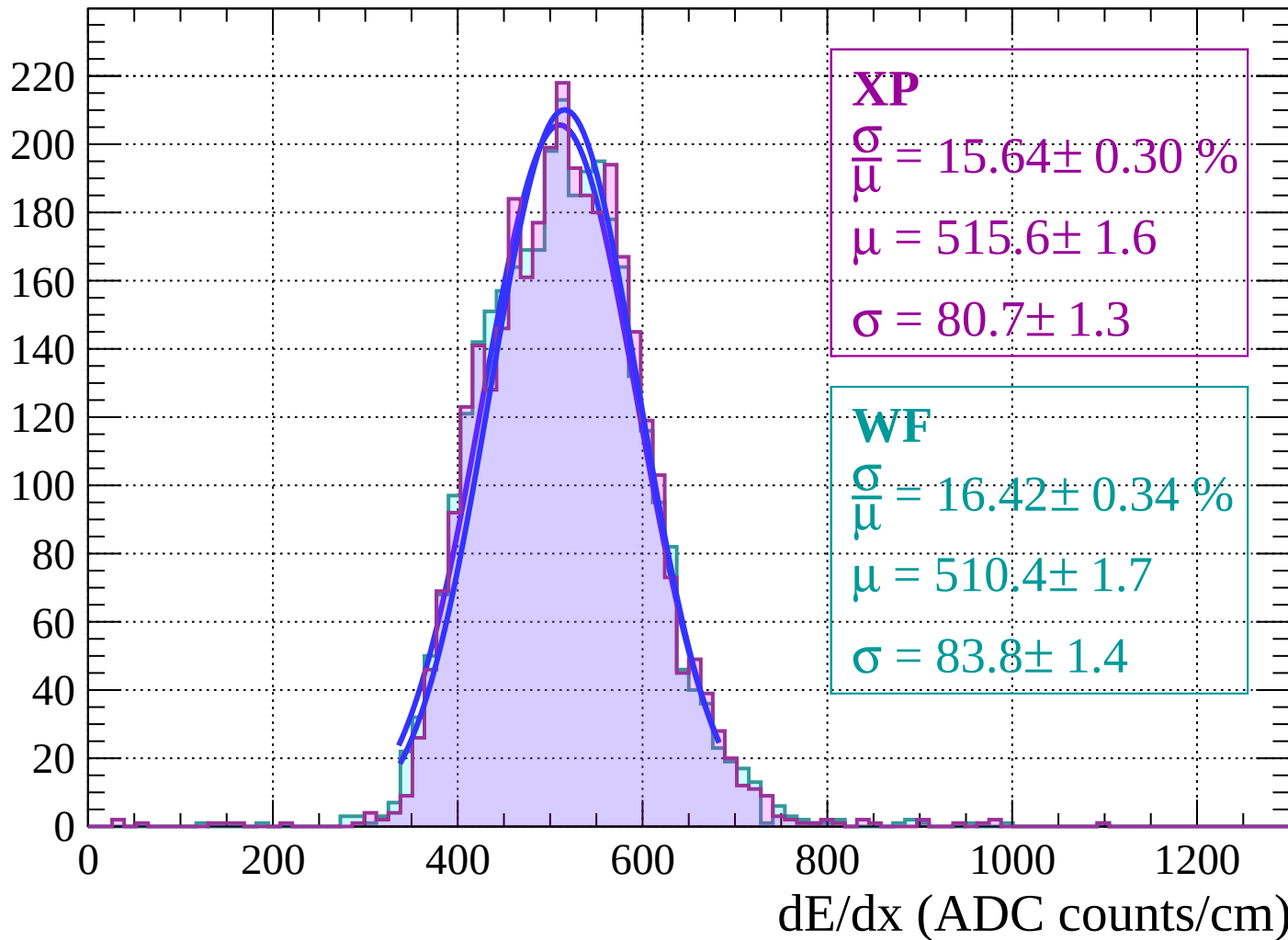
Energy loss $|-36 < \theta < -34$

Count

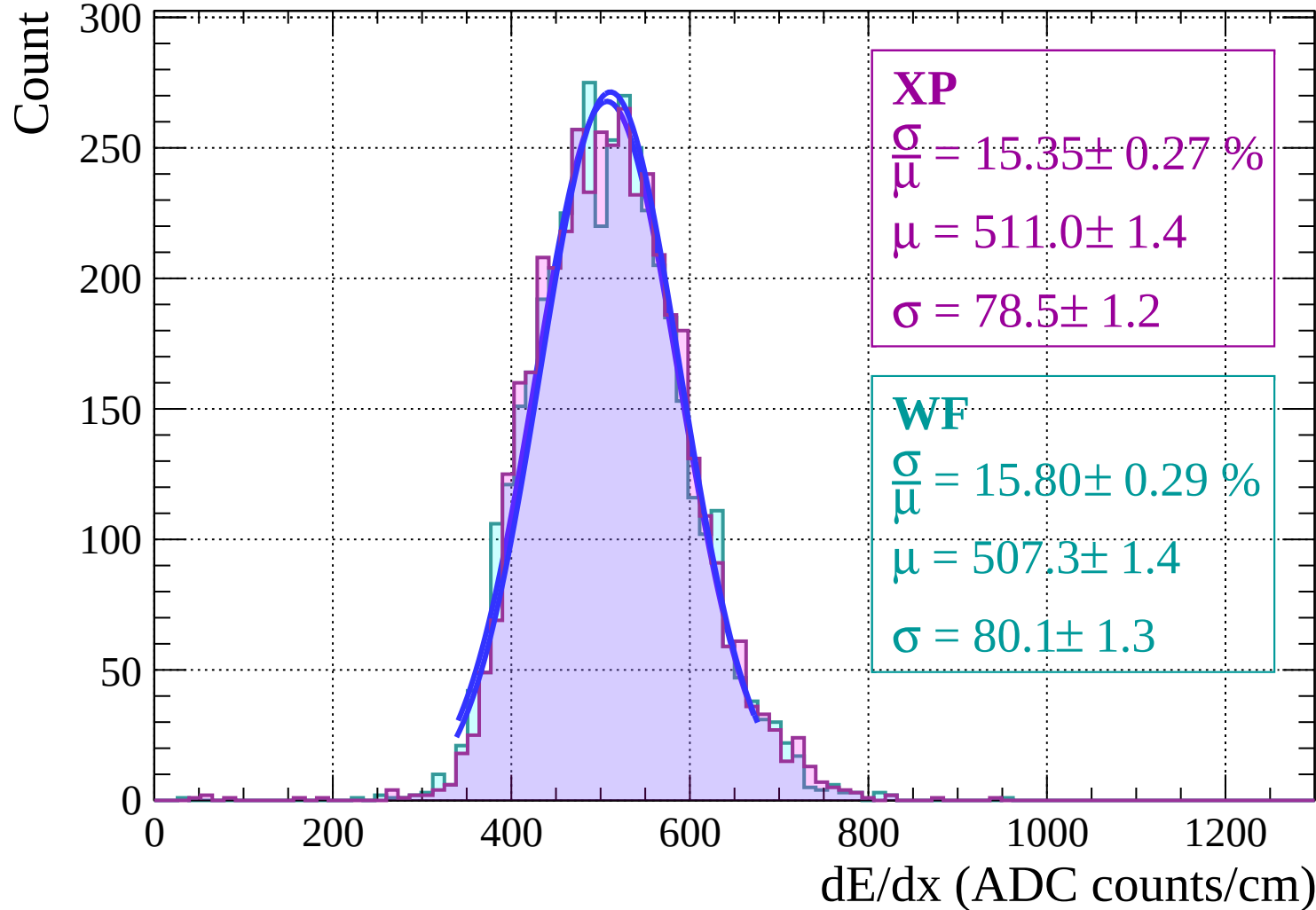


Energy loss $|-34 < \theta < -32$

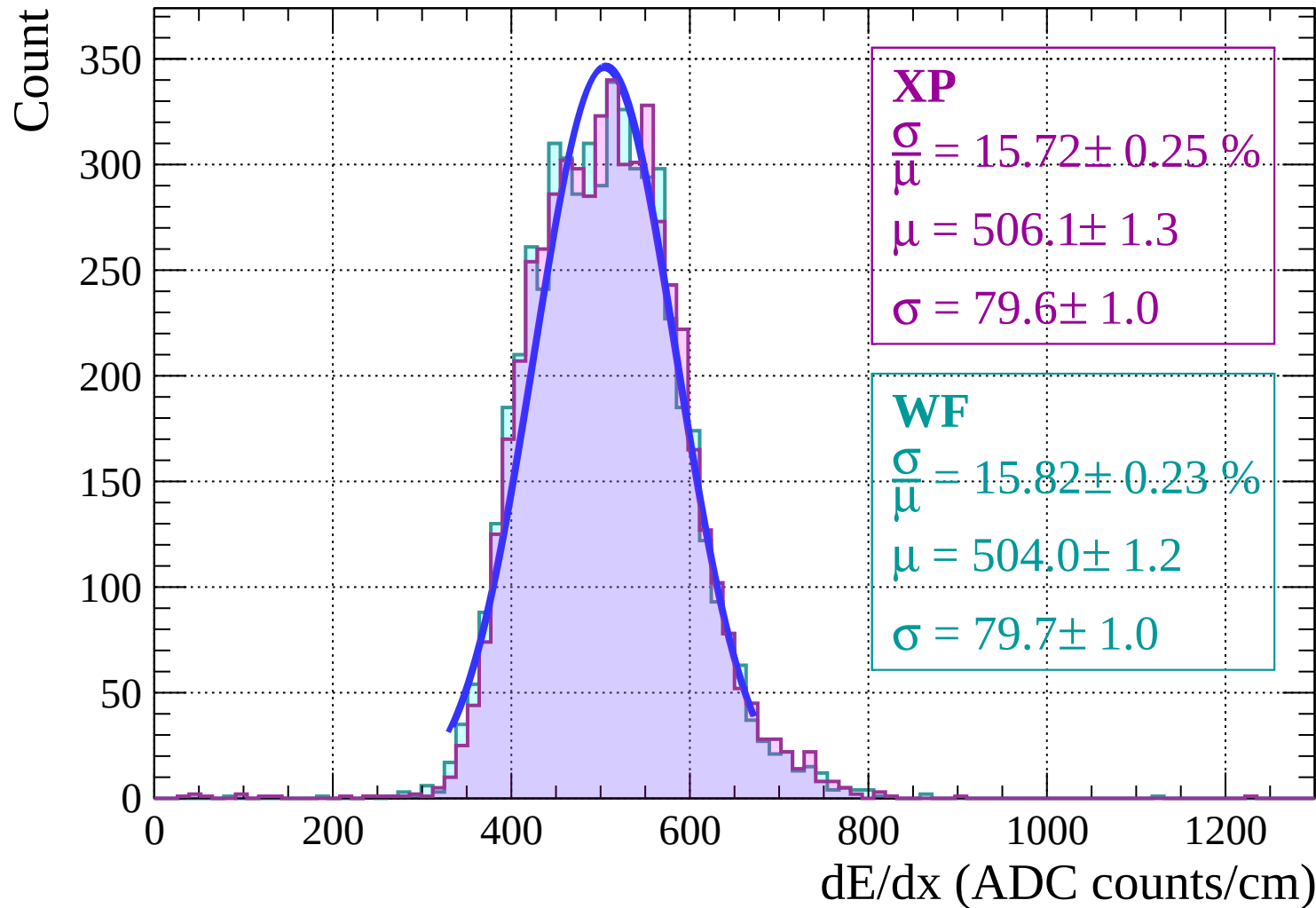
Count



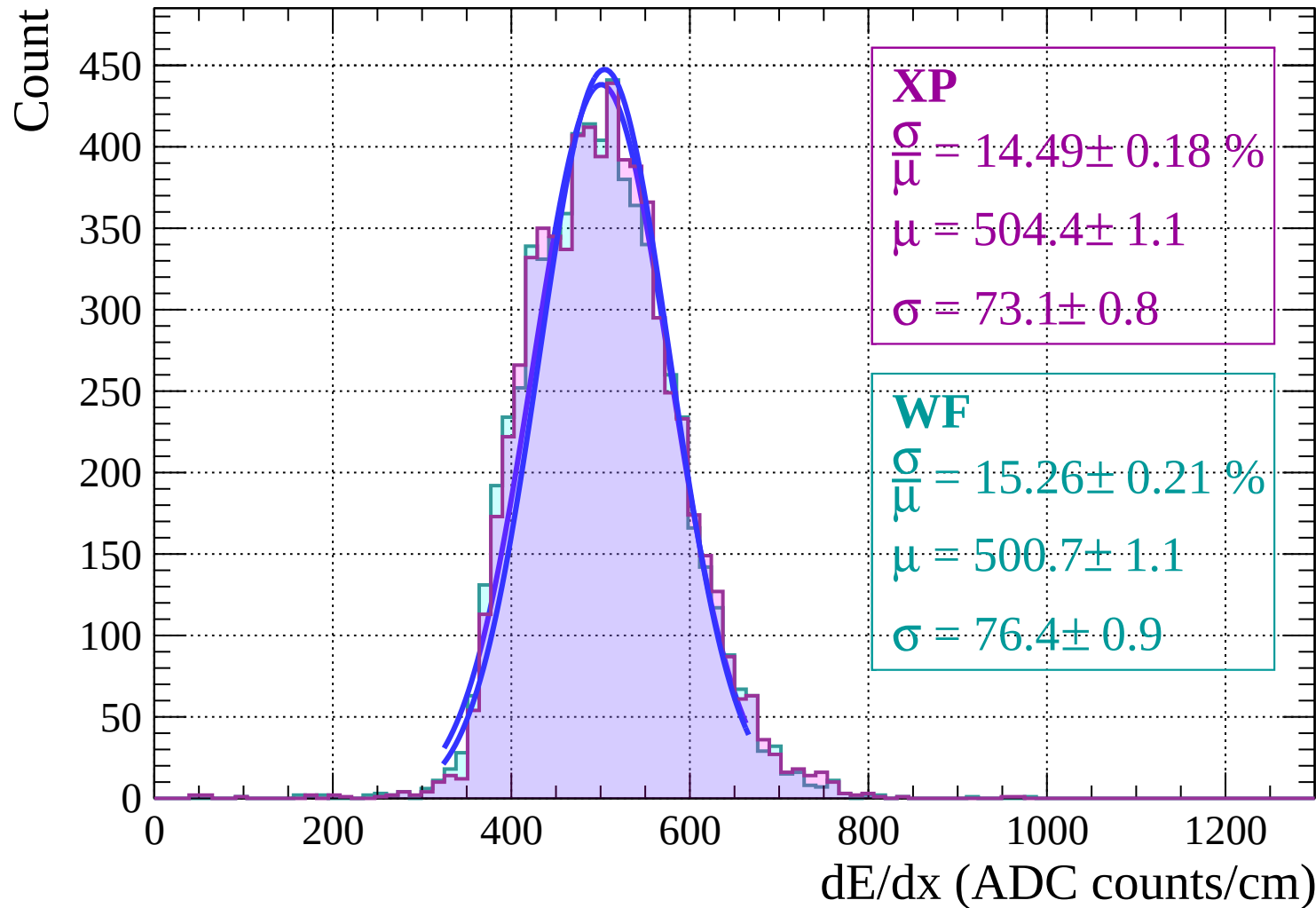
Energy loss $|-32 < \theta < -30$



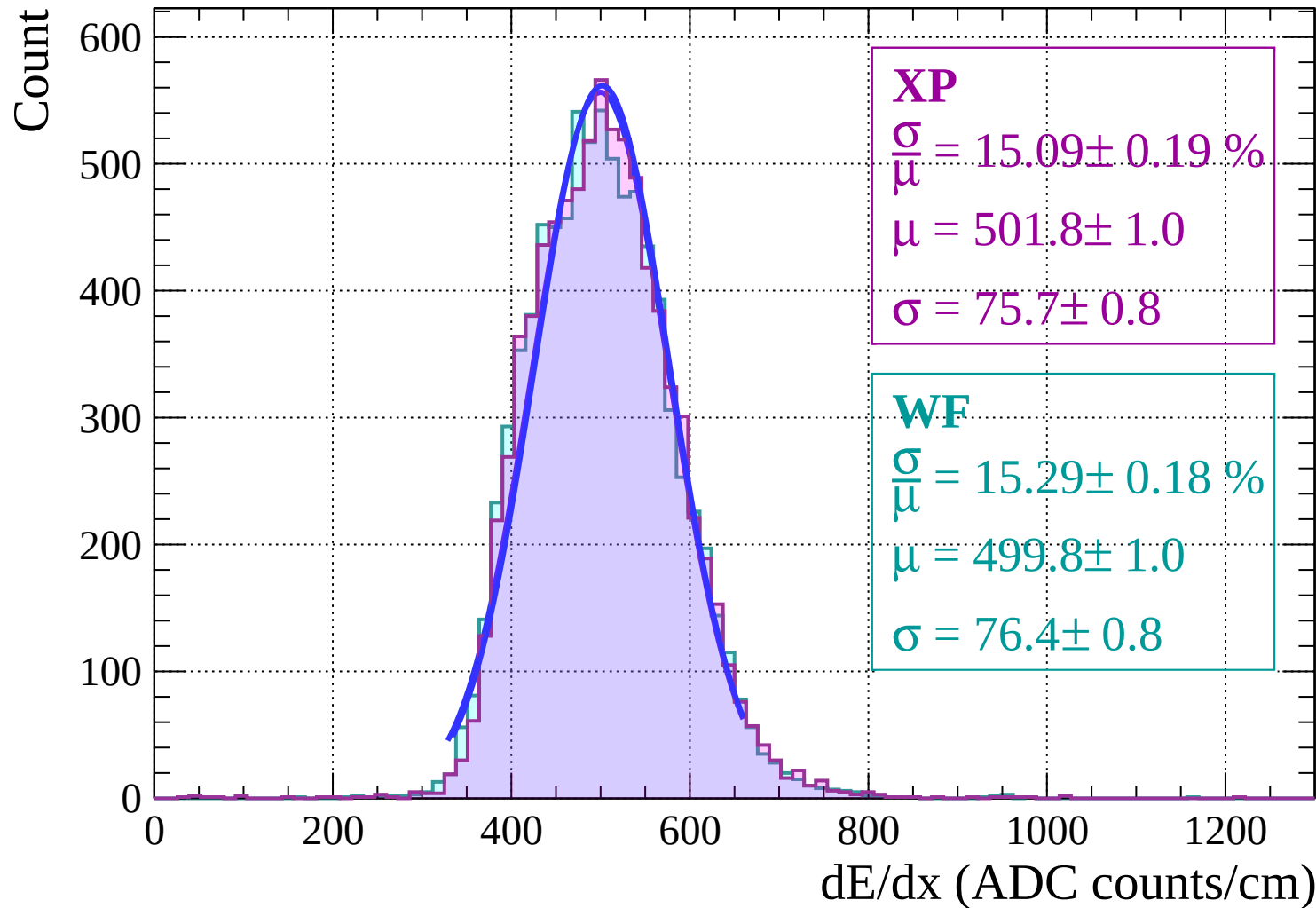
Energy loss | $-30 < \theta < -28$



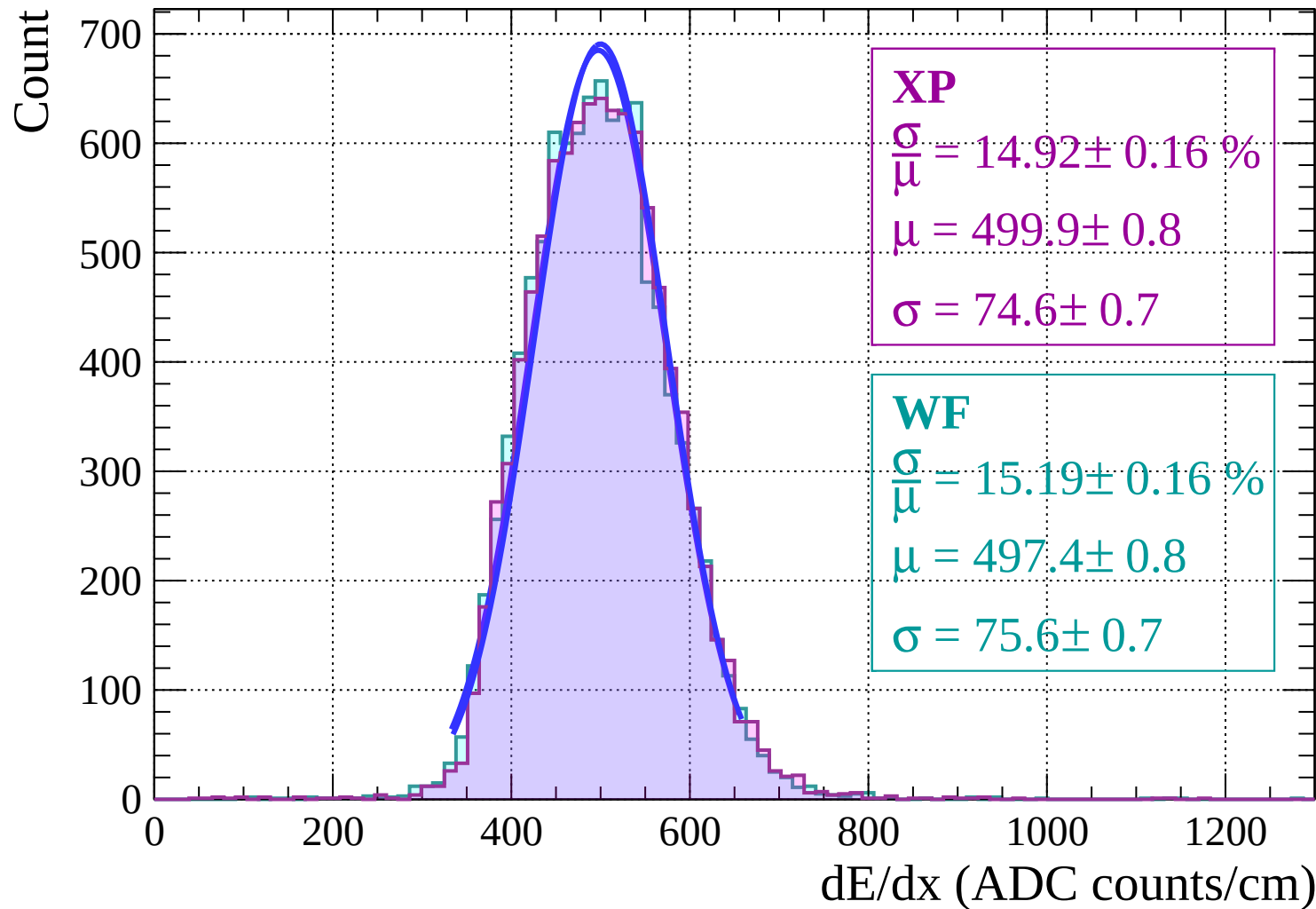
Energy loss | $-28 < \theta < -26$



Energy loss | $-26 < \theta < -24$

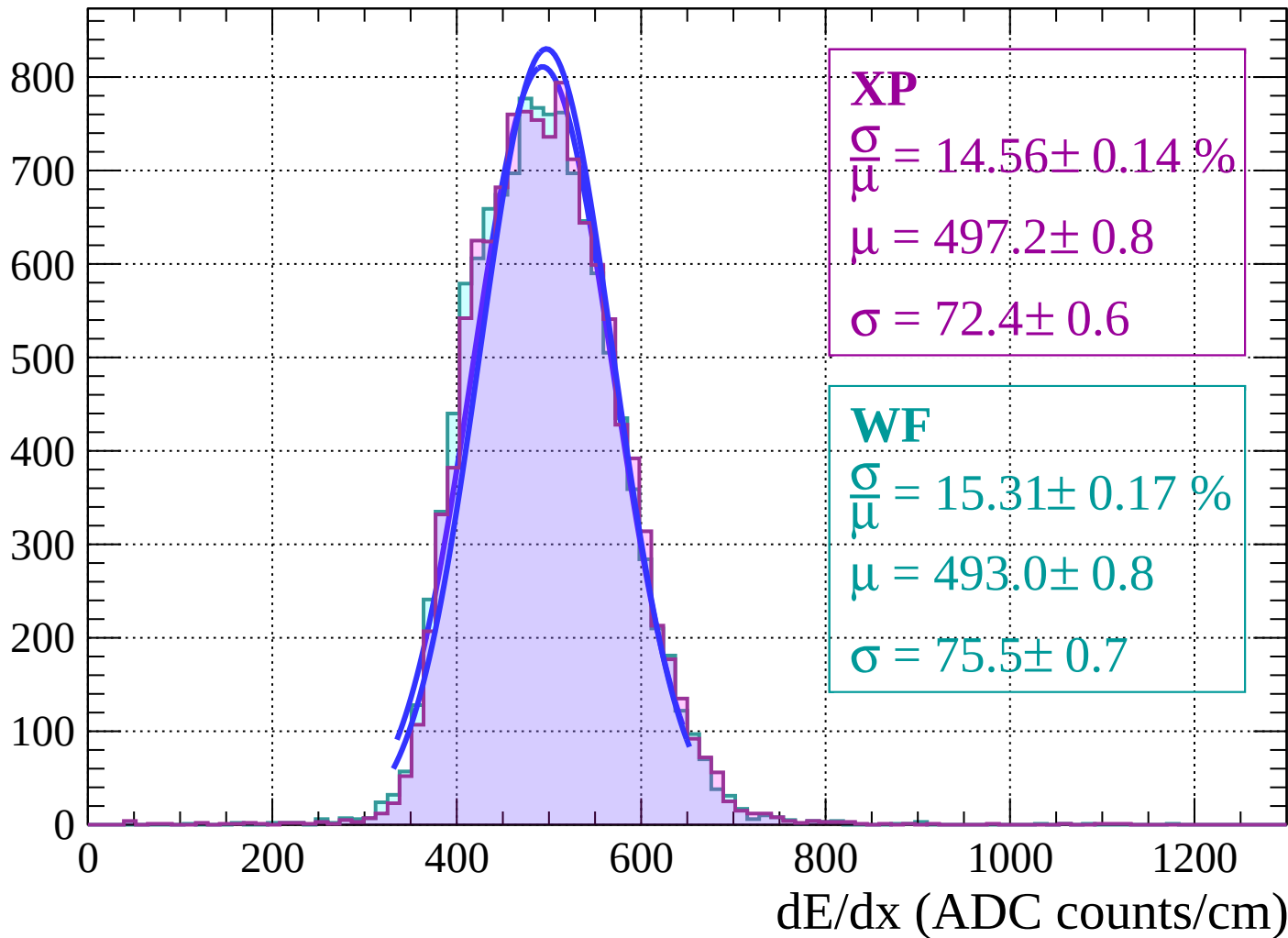


Energy loss | $-24 < \theta < -22$

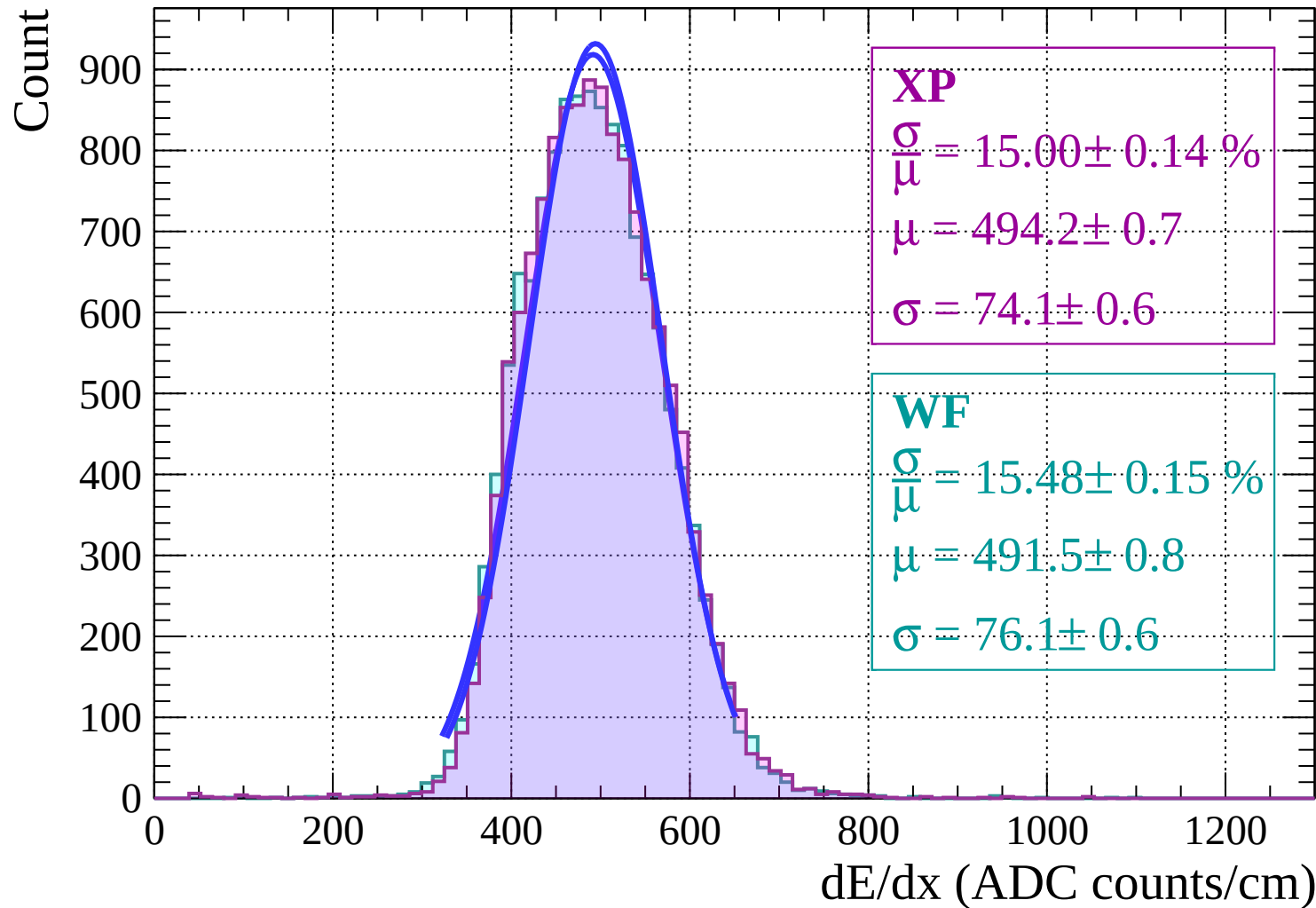


Energy loss | $-22 < \theta < -20$

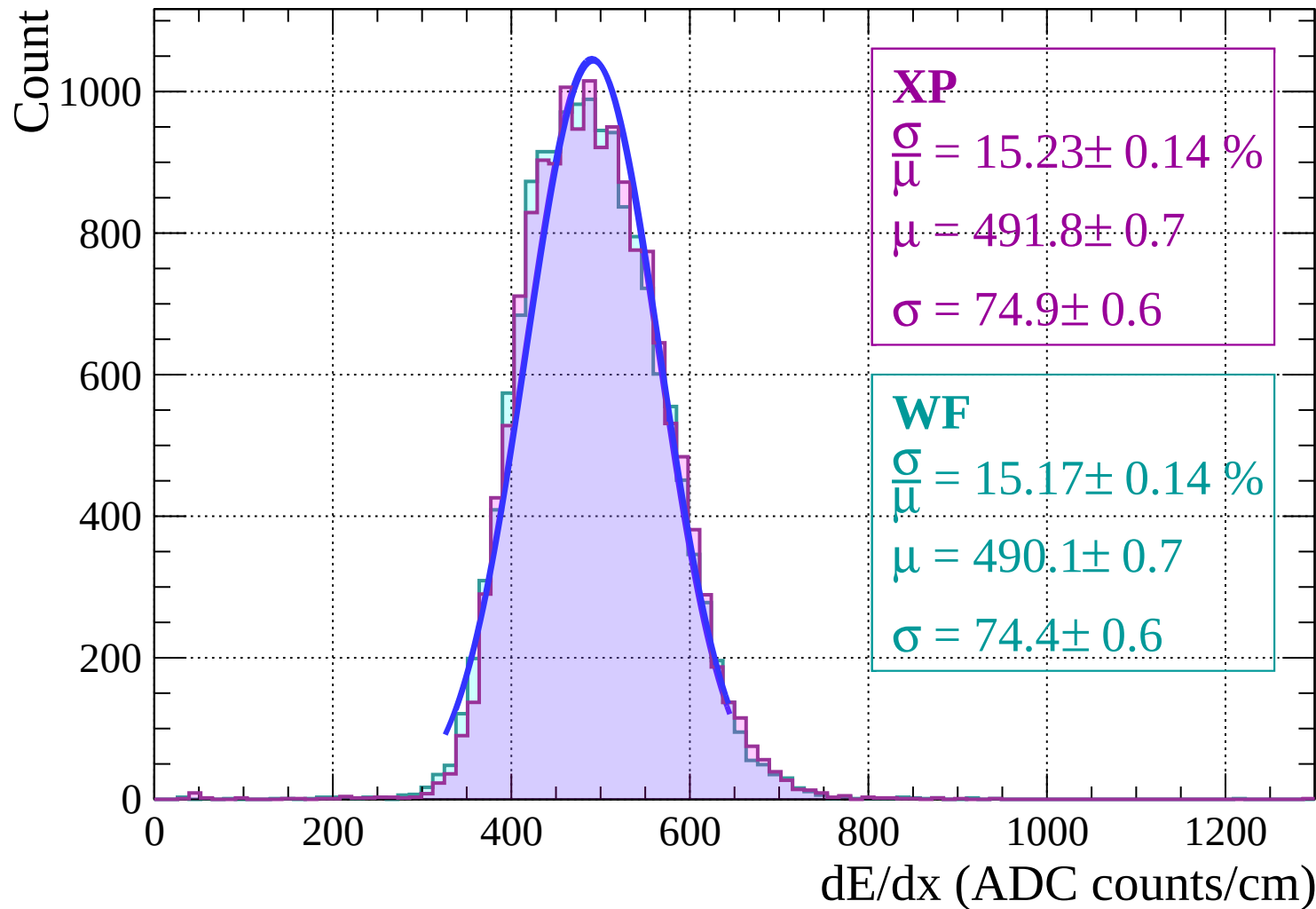
Count



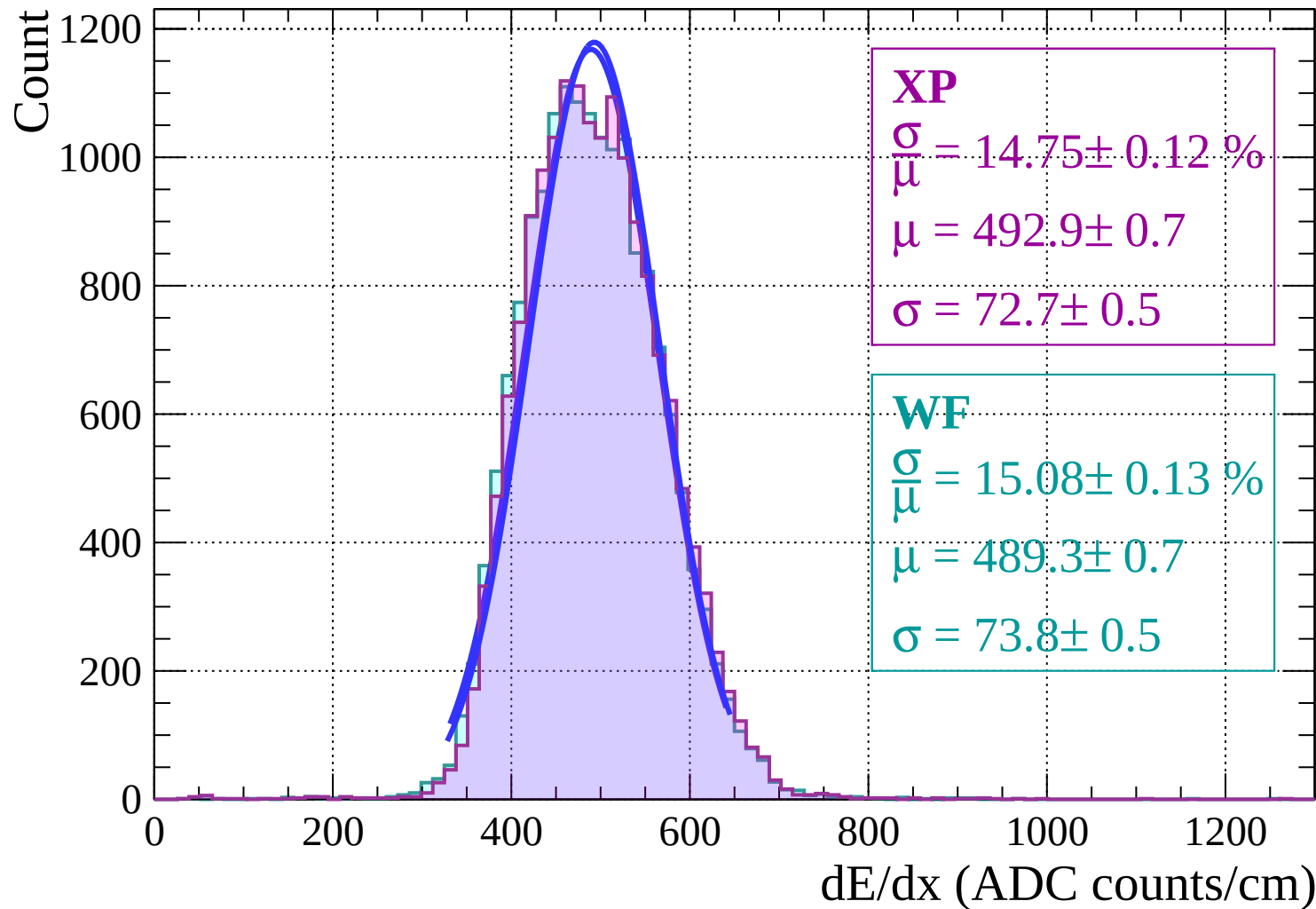
Energy loss | $-20 < \theta < -18$



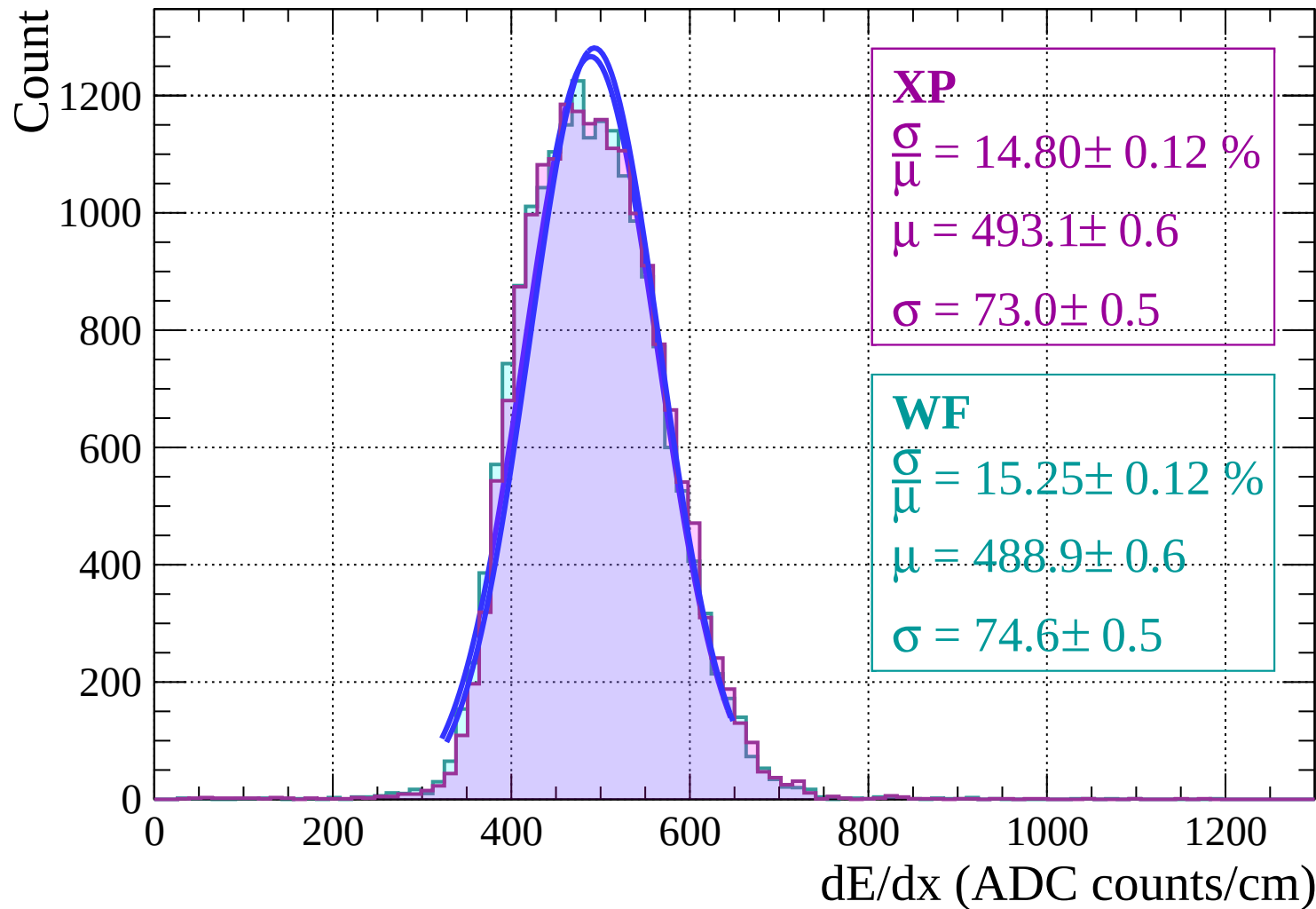
Energy loss | $-18 < \theta < -16$



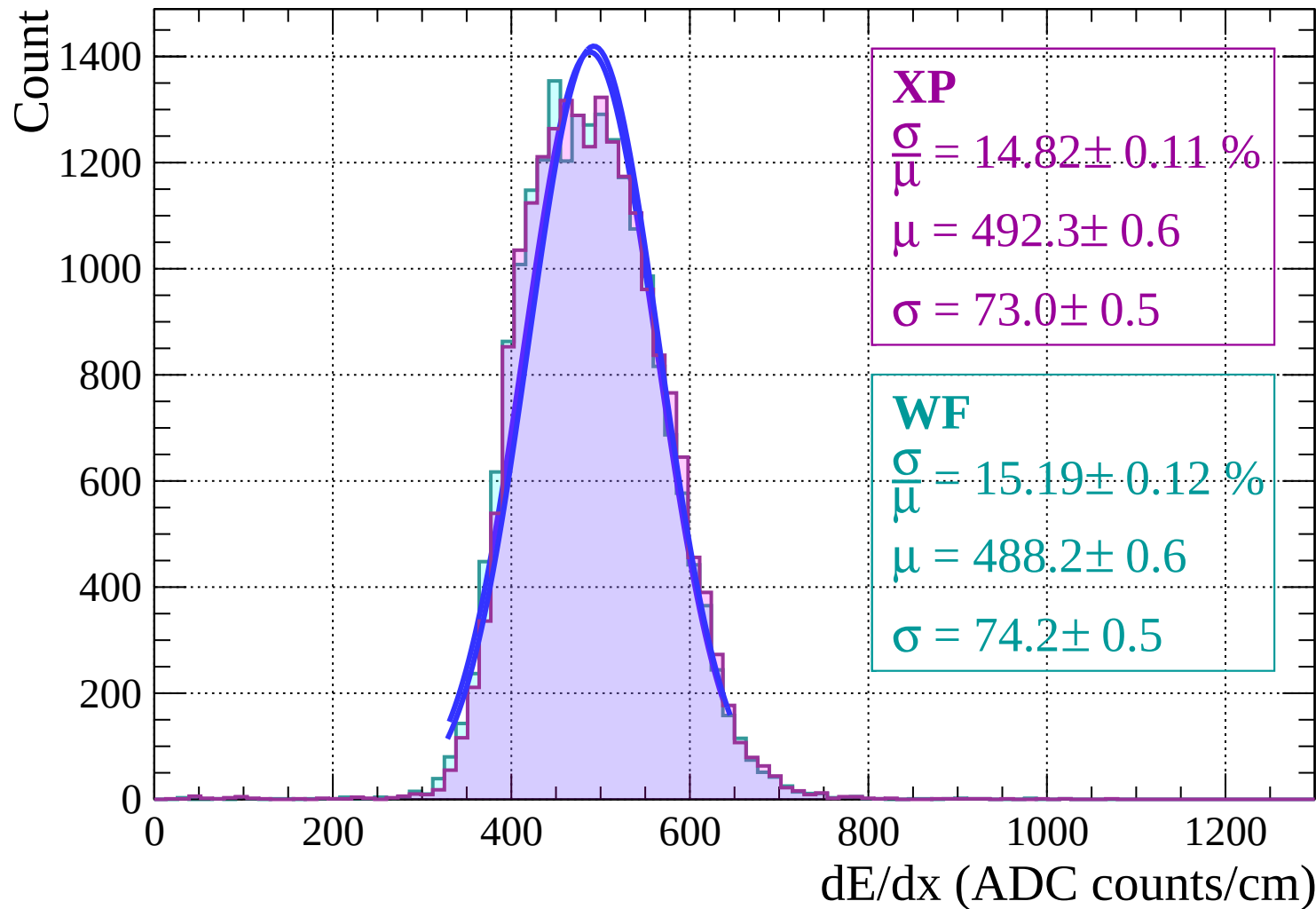
Energy loss $|-16 < \theta < -14$



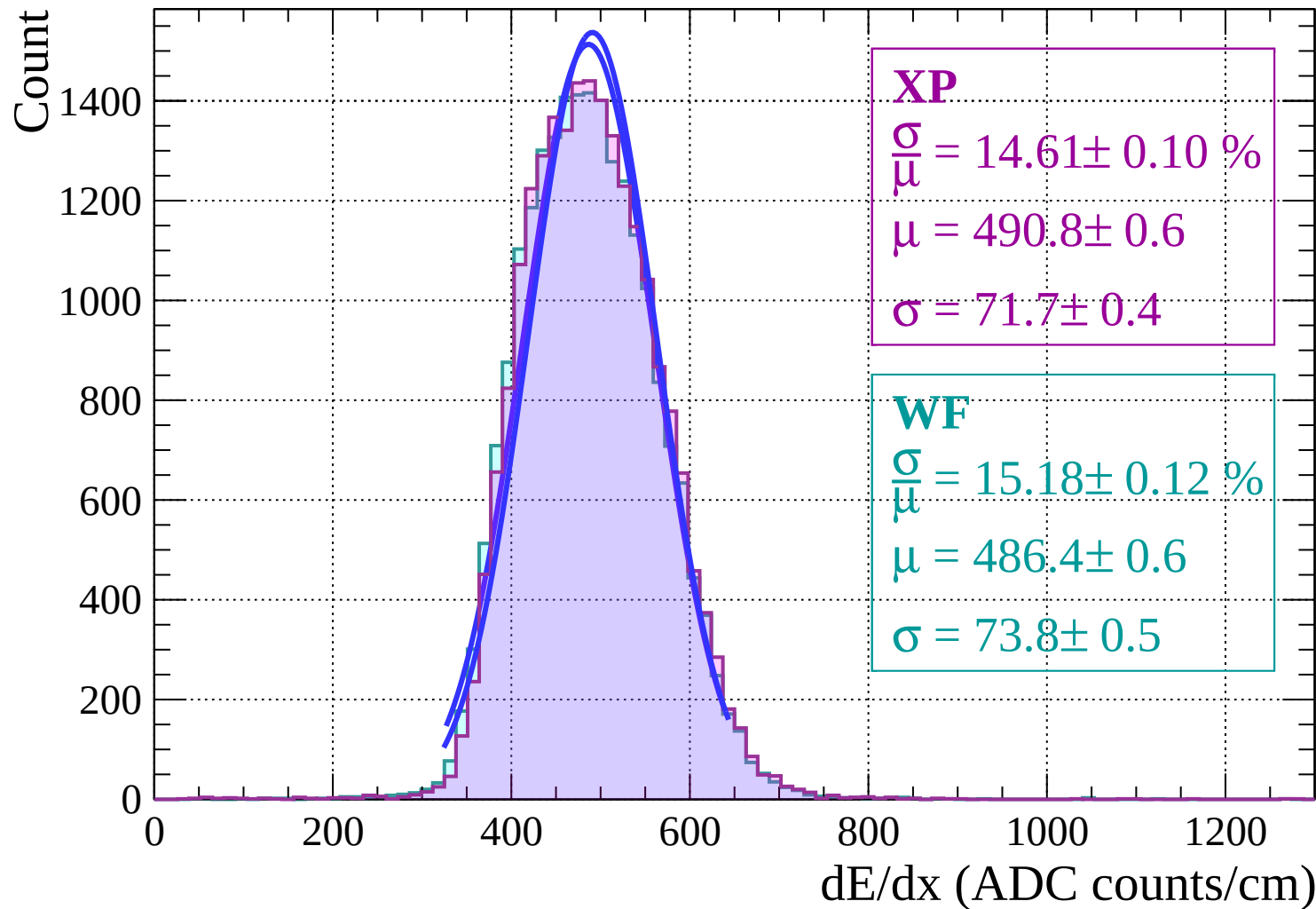
Energy loss | $-14 < \theta < -12$



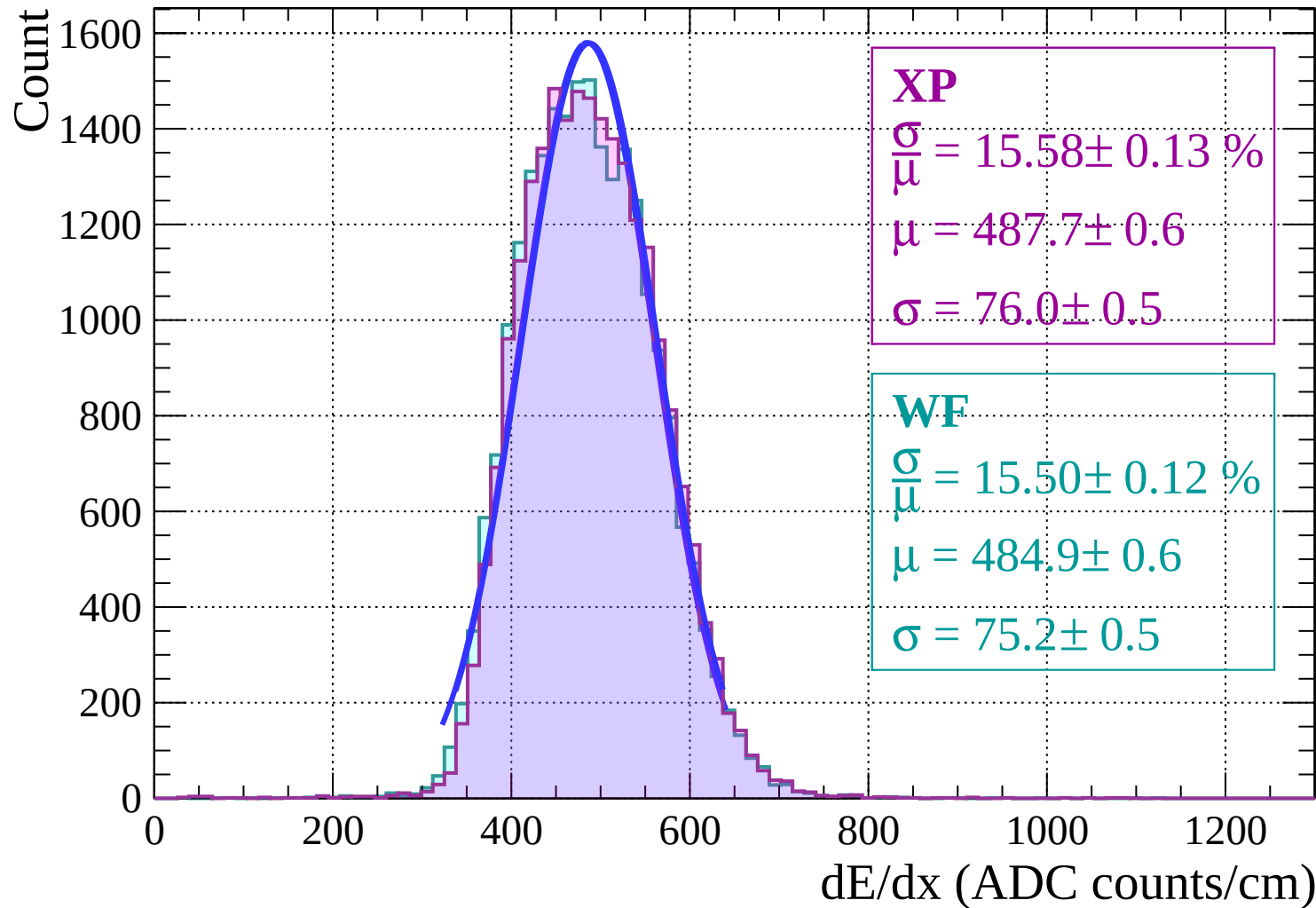
Energy loss | $-12 < \theta < -10$



Energy loss | $-10 < \theta < -8$

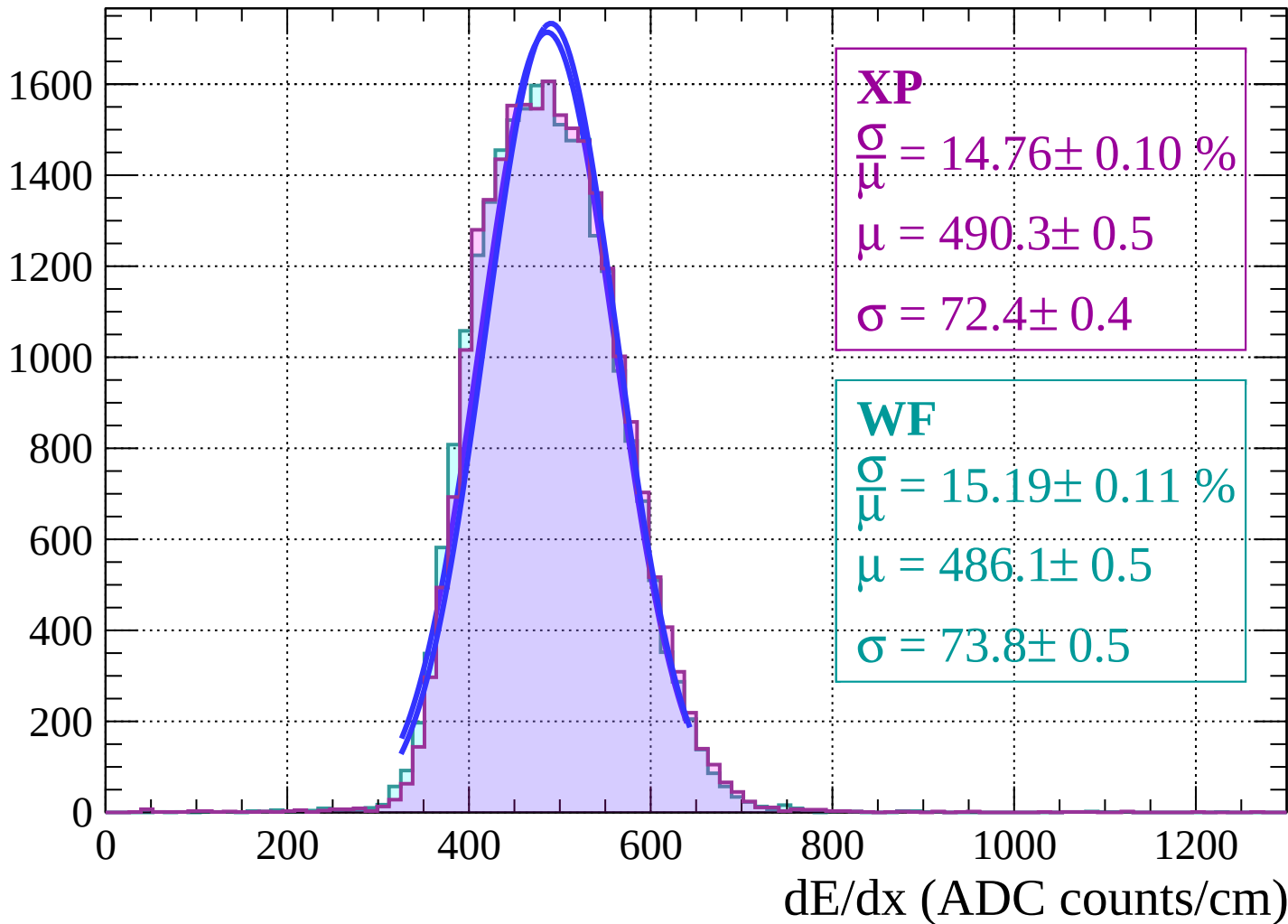


Energy loss | $-8 < \theta < -6$

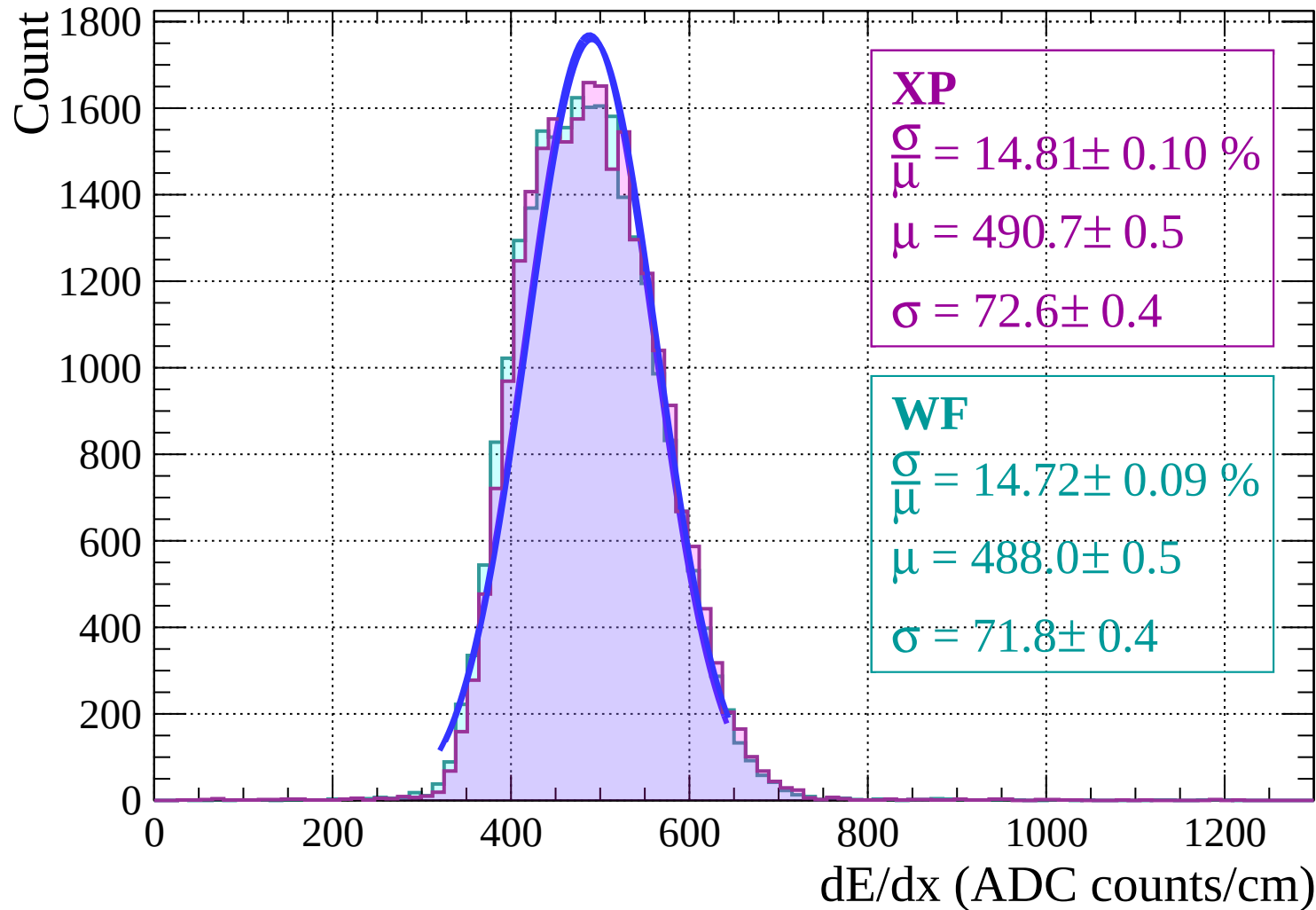


Energy loss | $-6 < \theta < -4$

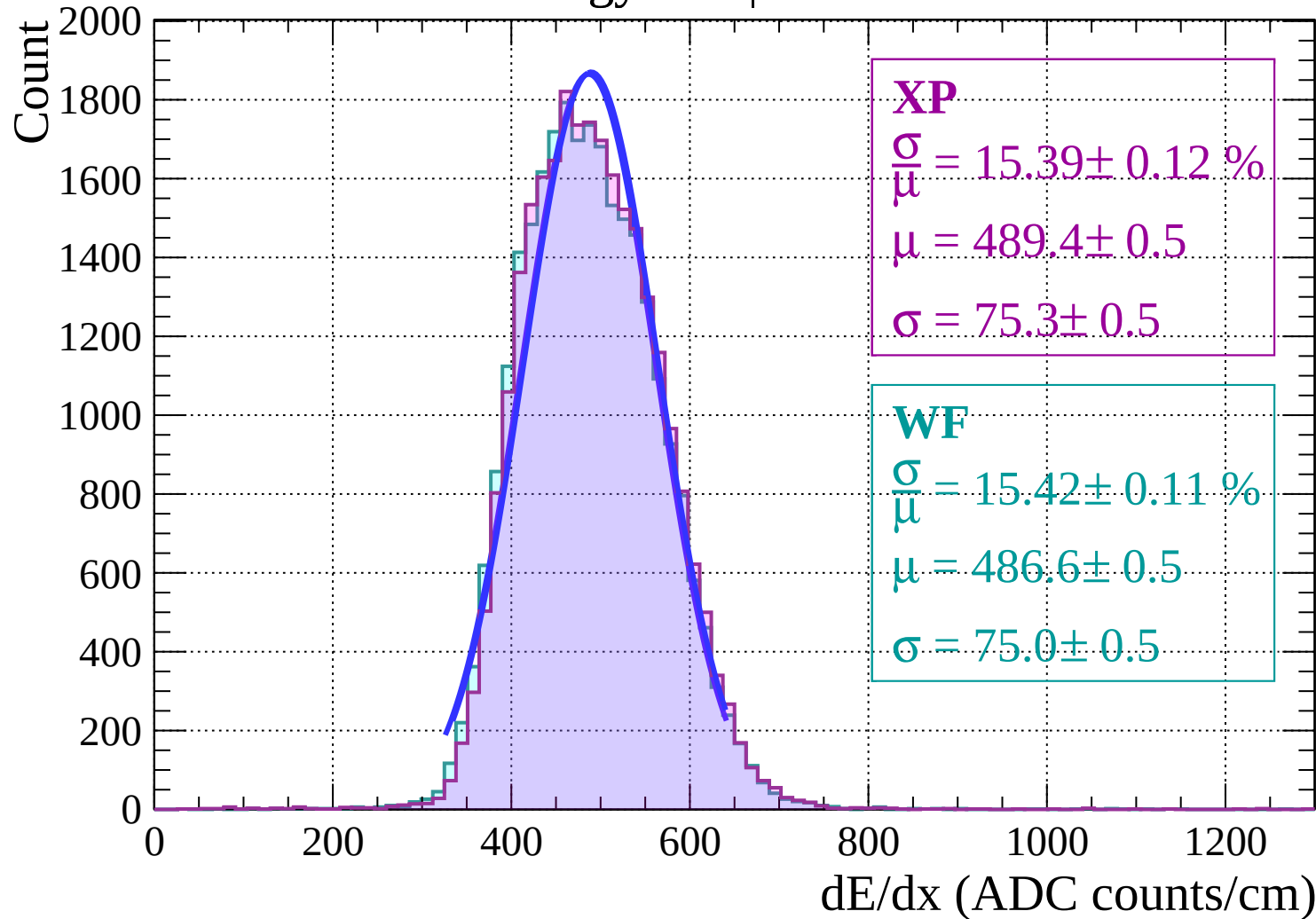
Count



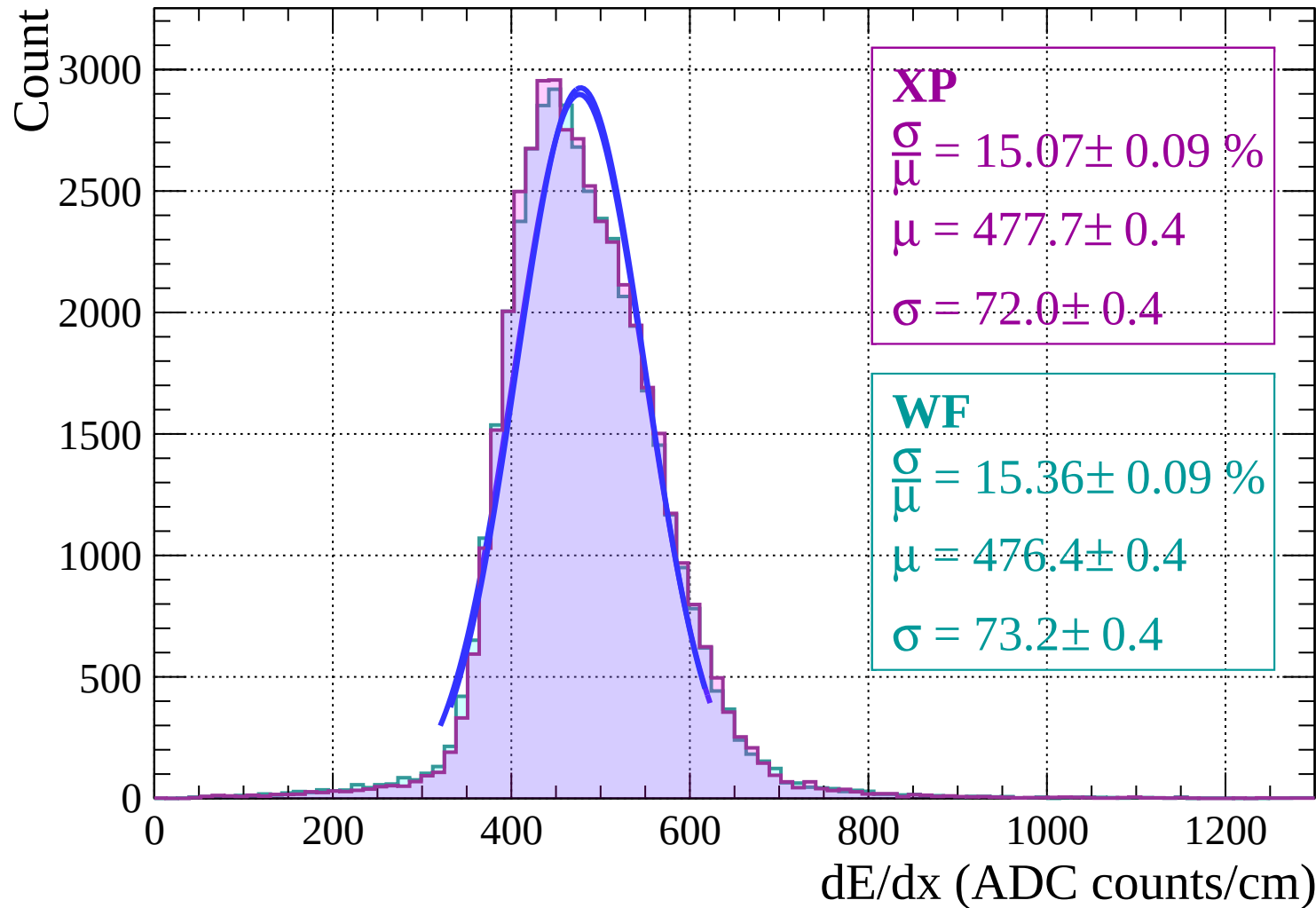
Energy loss | $-4 < \theta < -2$



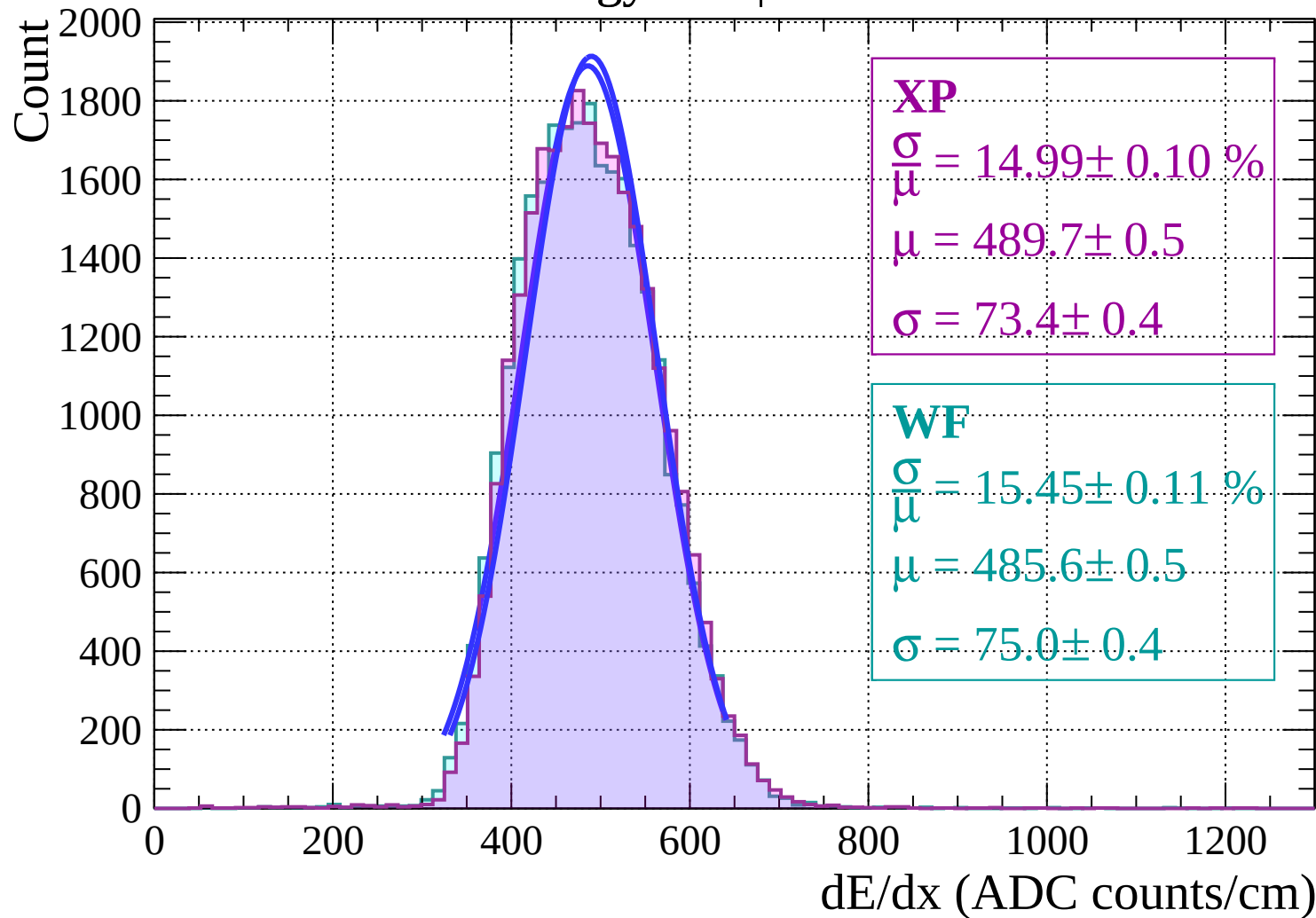
Energy loss | $-2 < \theta < 0$



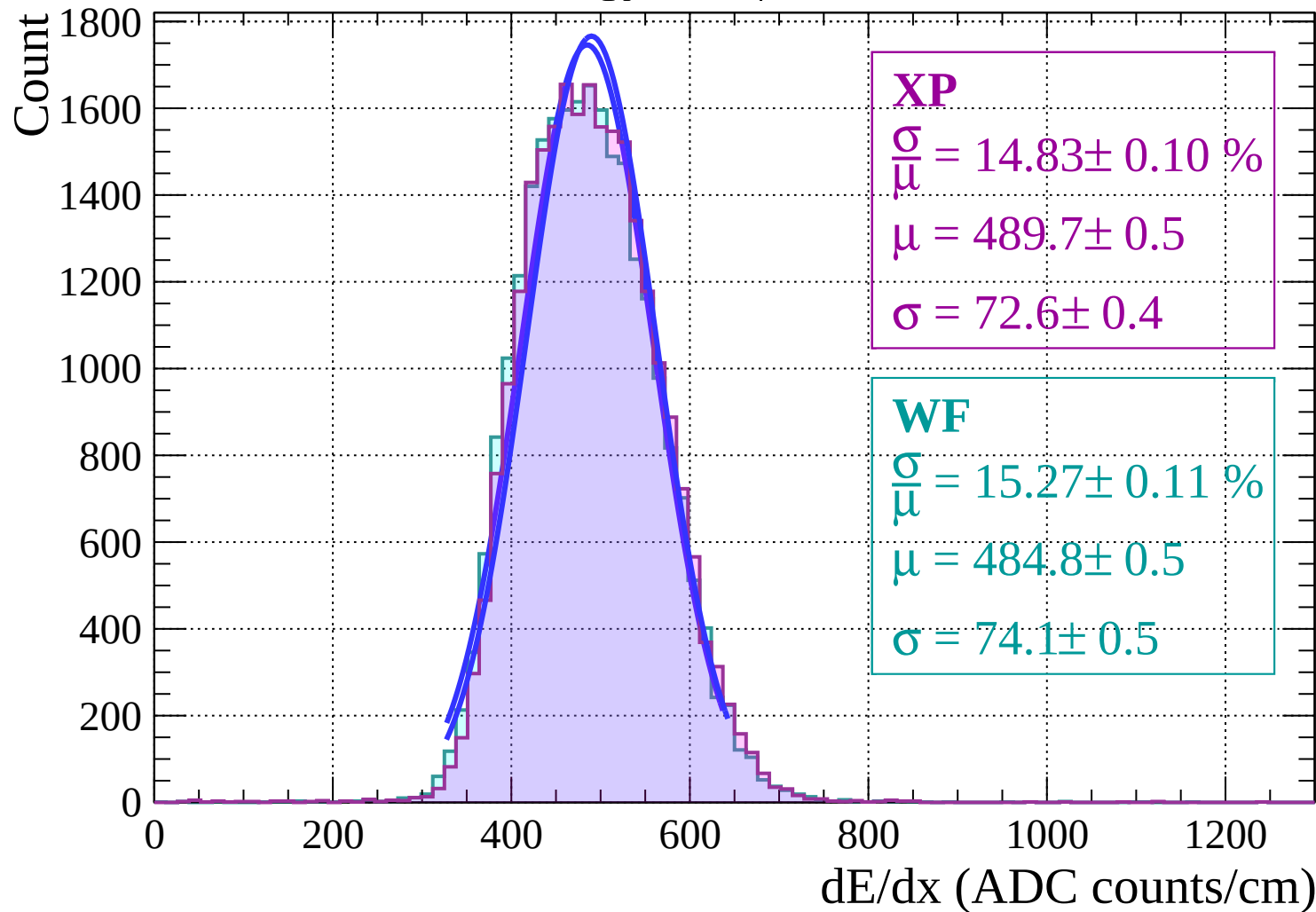
Energy loss | $0 < \theta < 2$



Energy loss | $2 < \theta < 4$

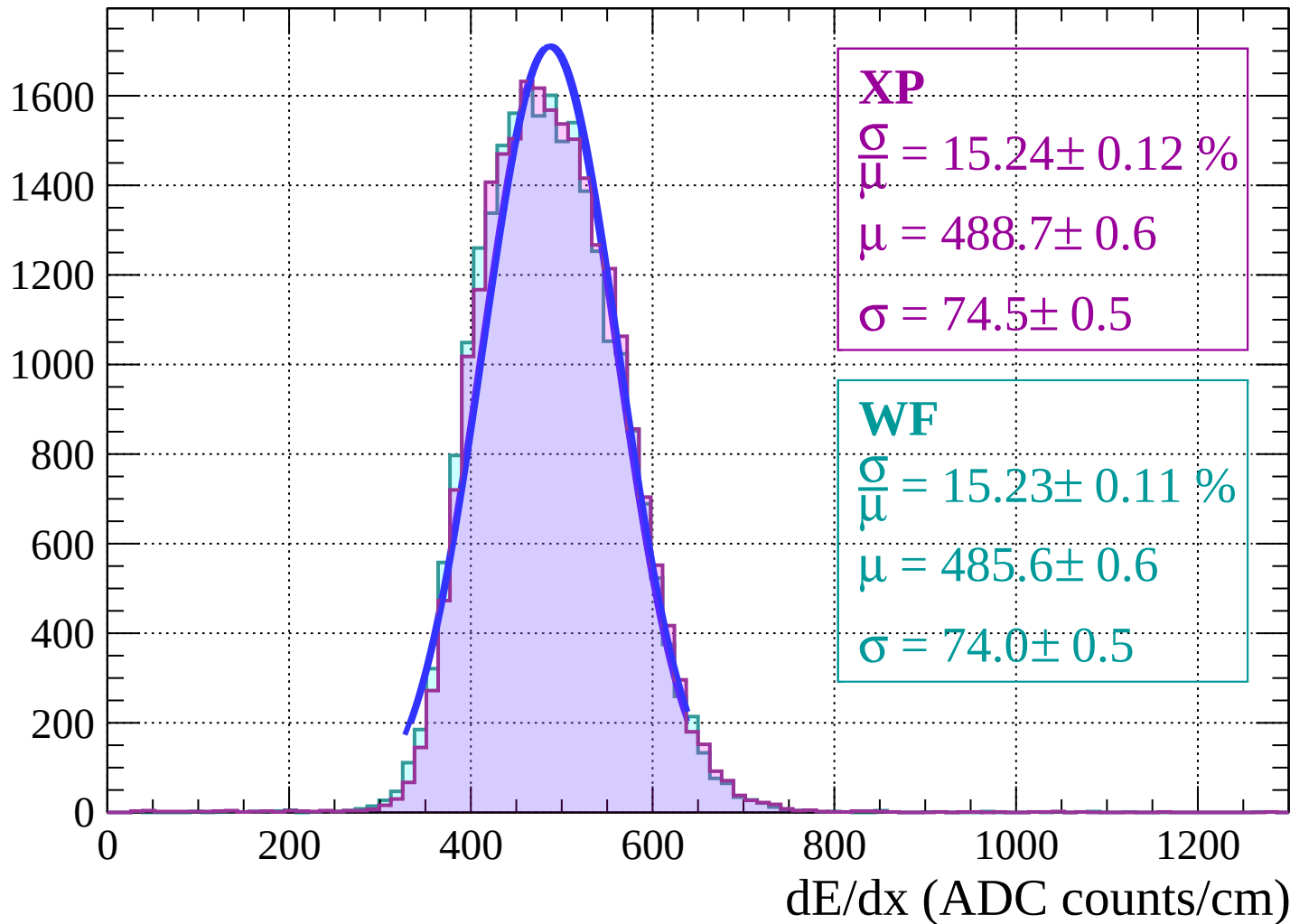


Energy loss | $4 < \theta < 6$

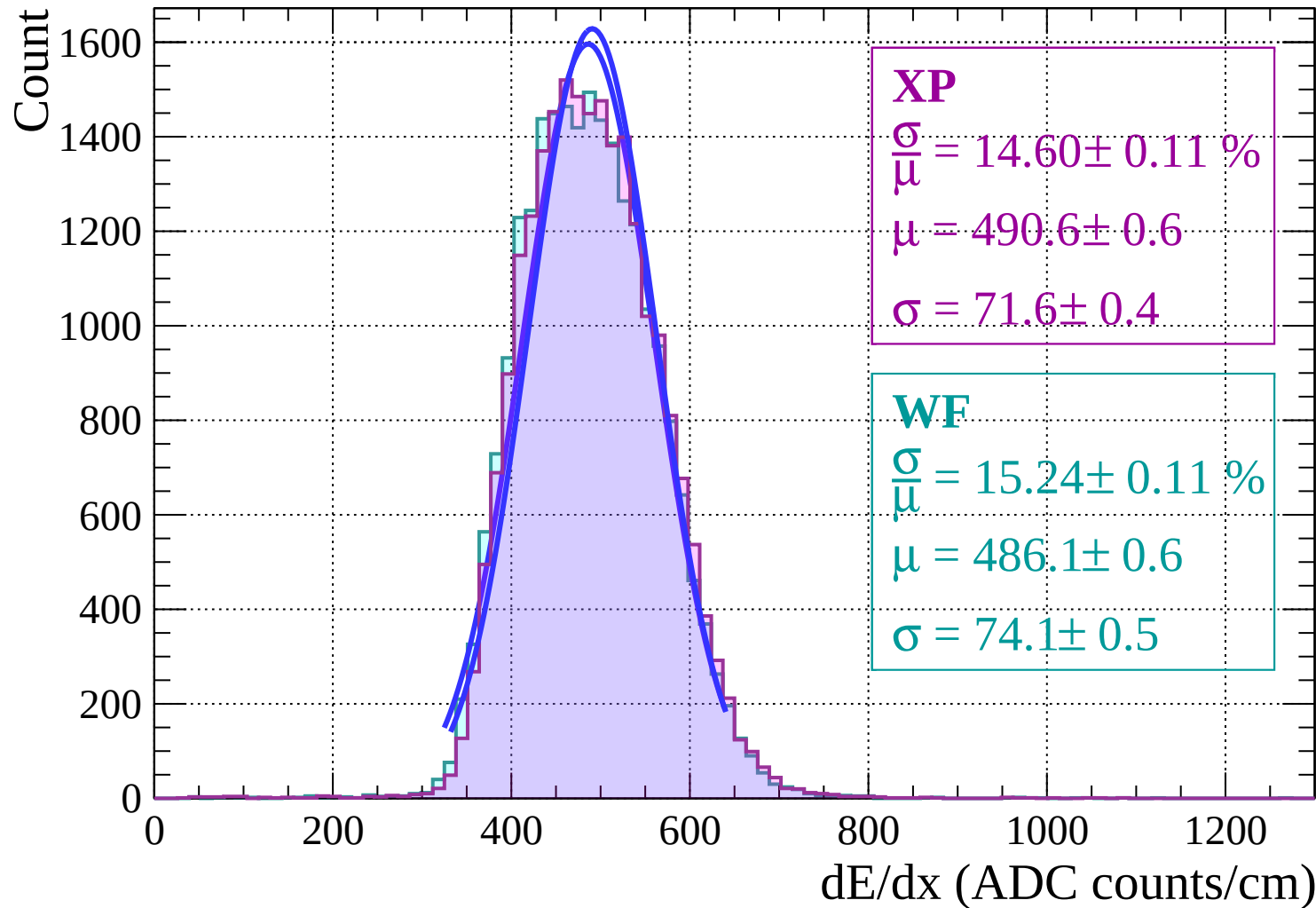


Energy loss | $6 < \theta < 8$

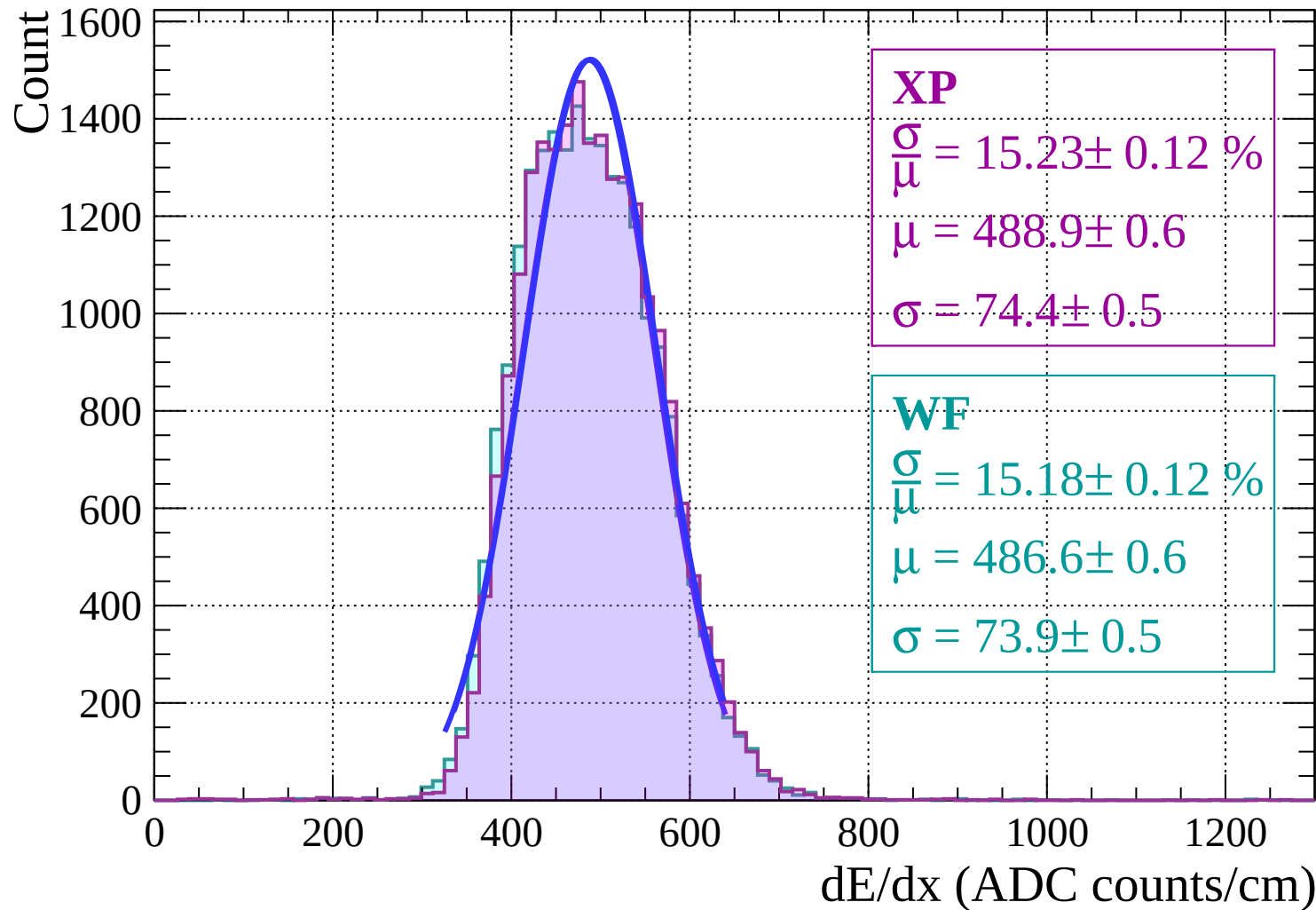
Count



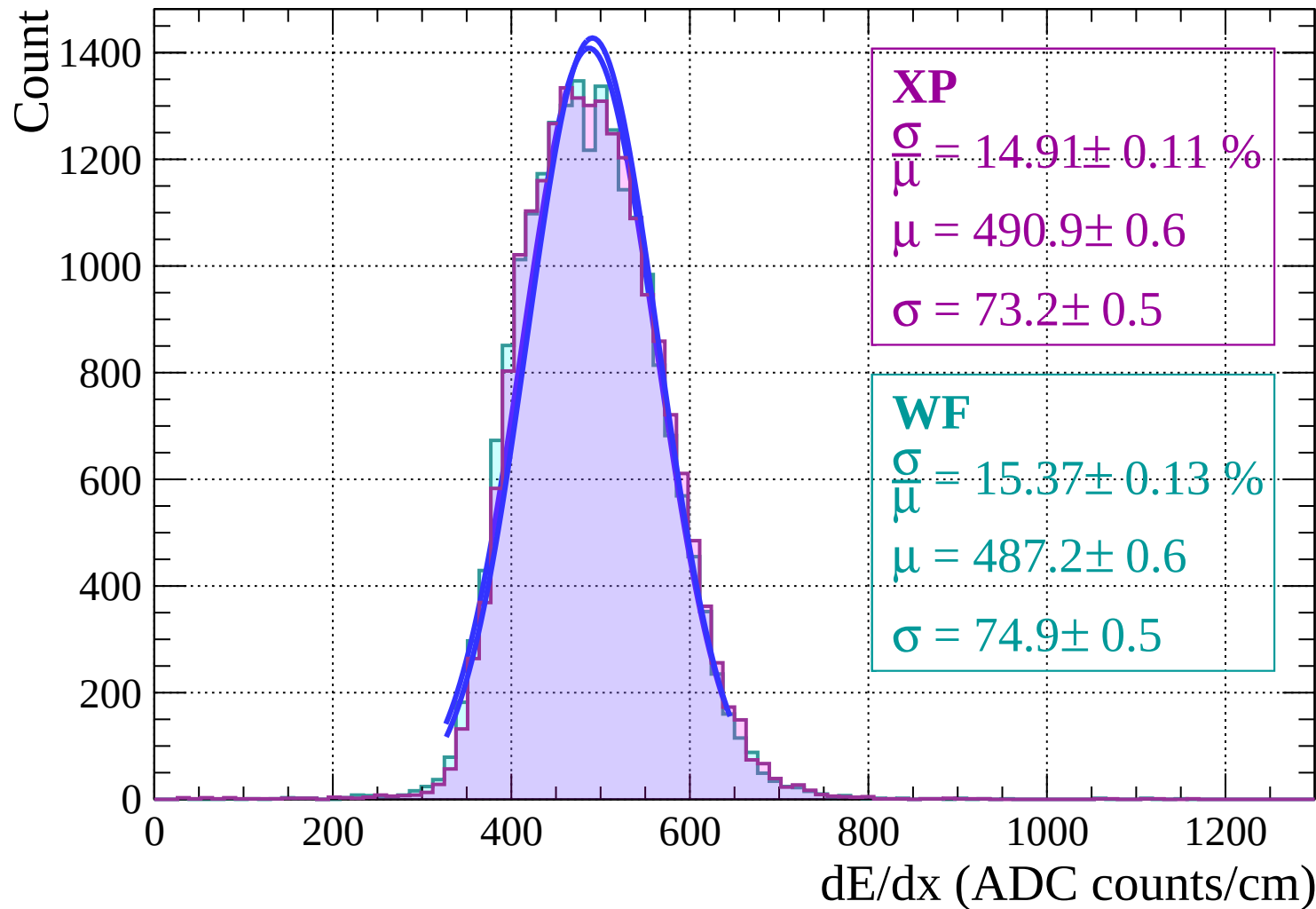
Energy loss | $8 < \theta < 10$



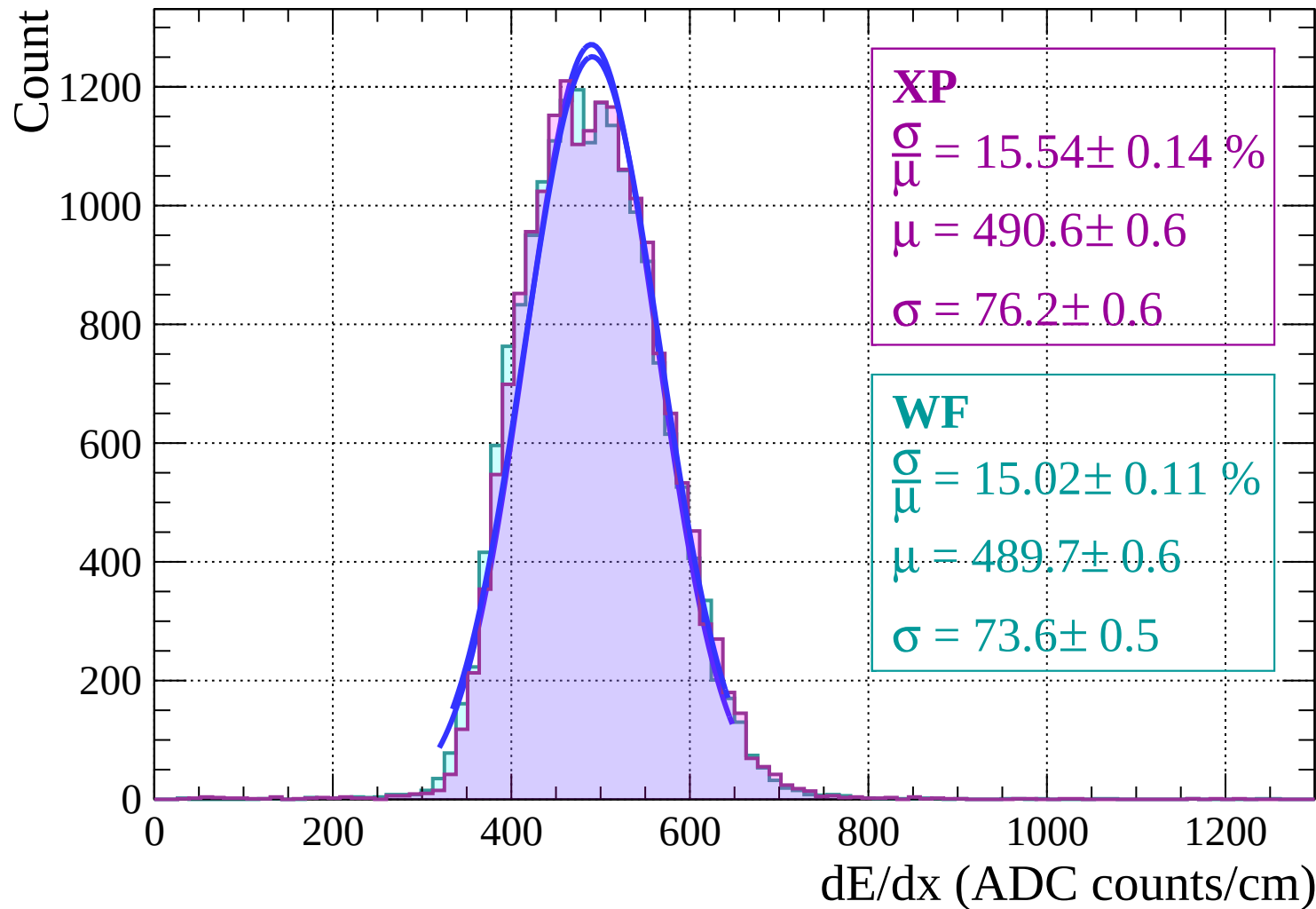
Energy loss | $10 < \theta < 12$



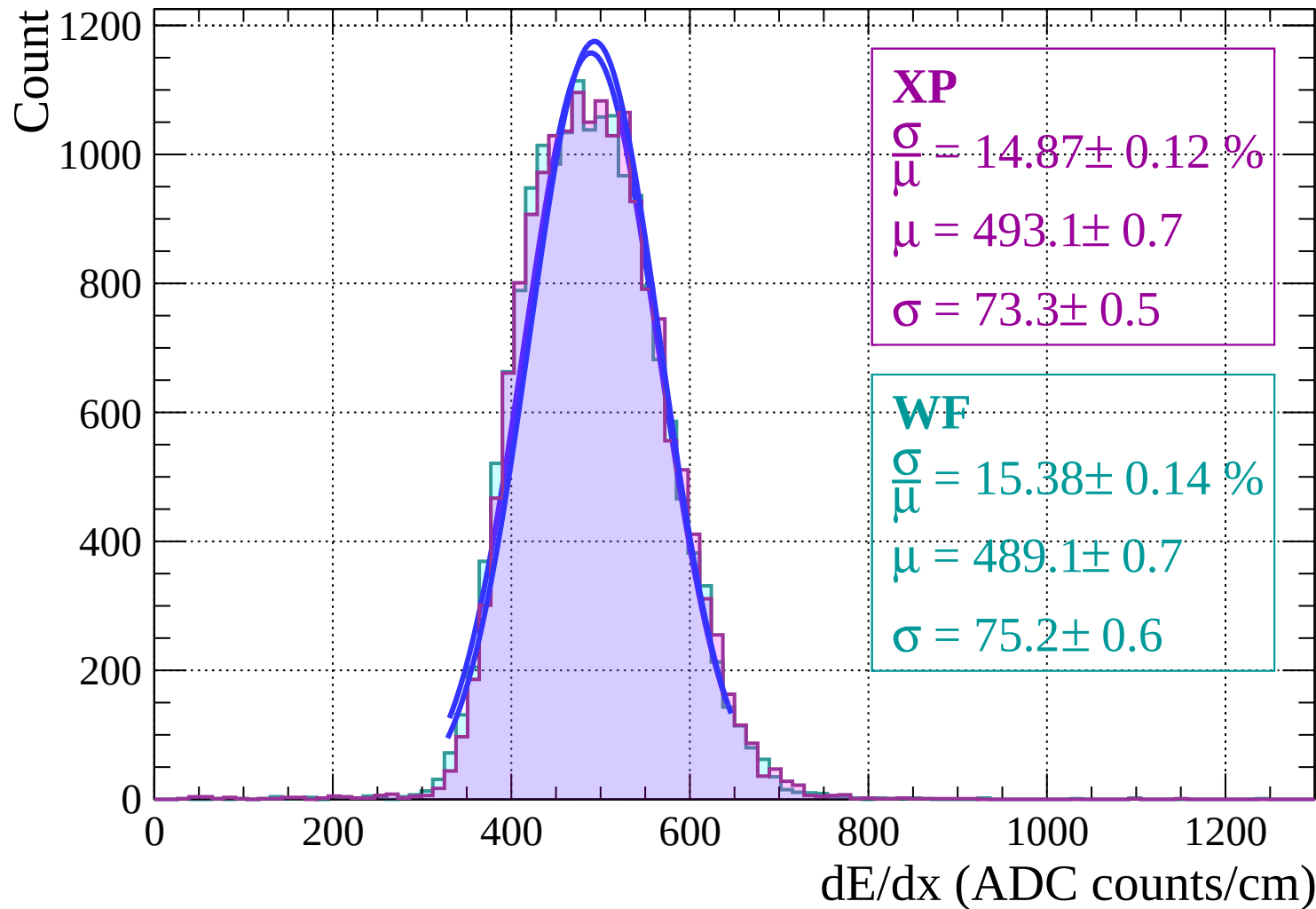
Energy loss | $12 < \theta < 14$



Energy loss | $14 < \theta < 16$

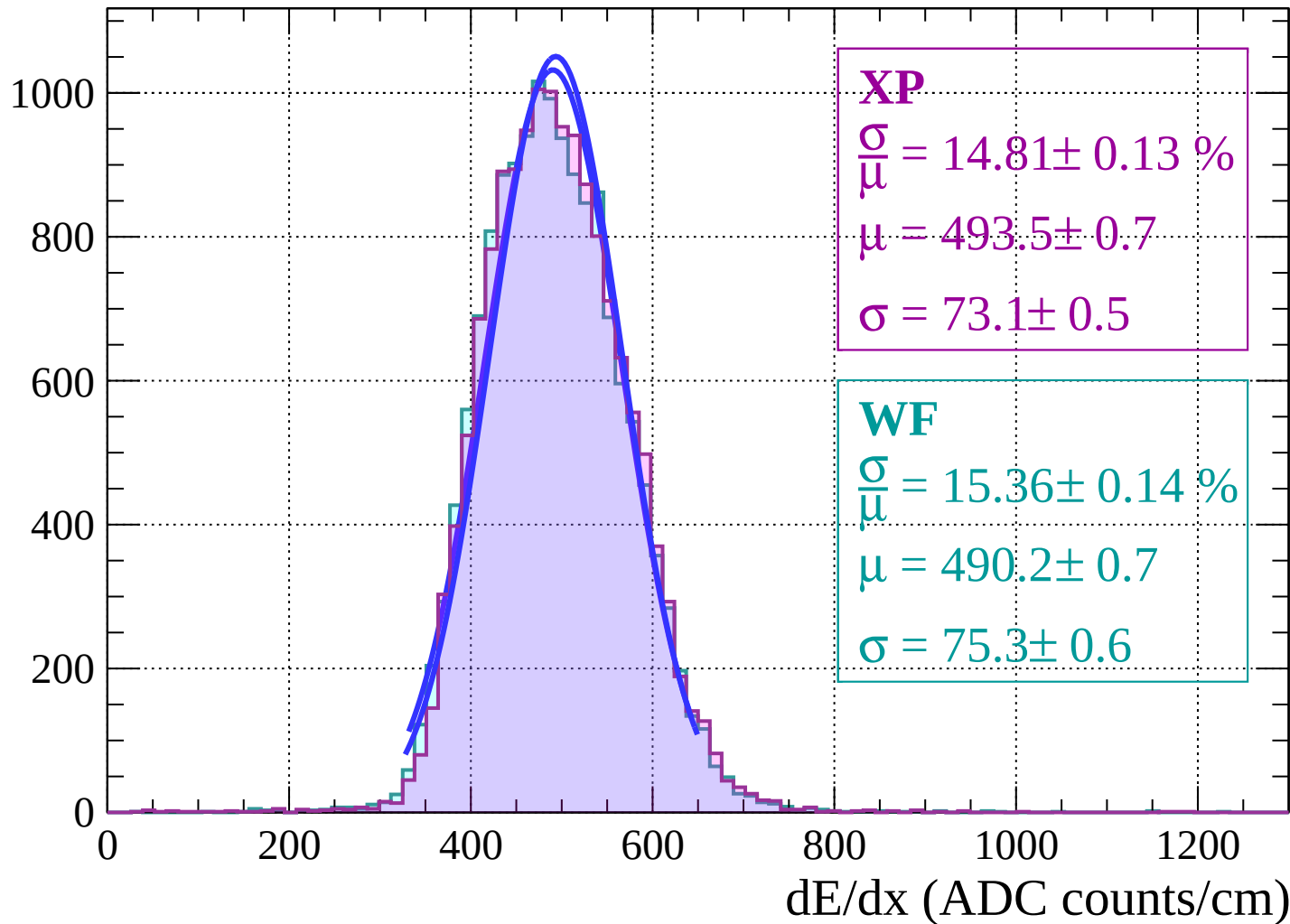


Energy loss | $16 < \theta < 18$



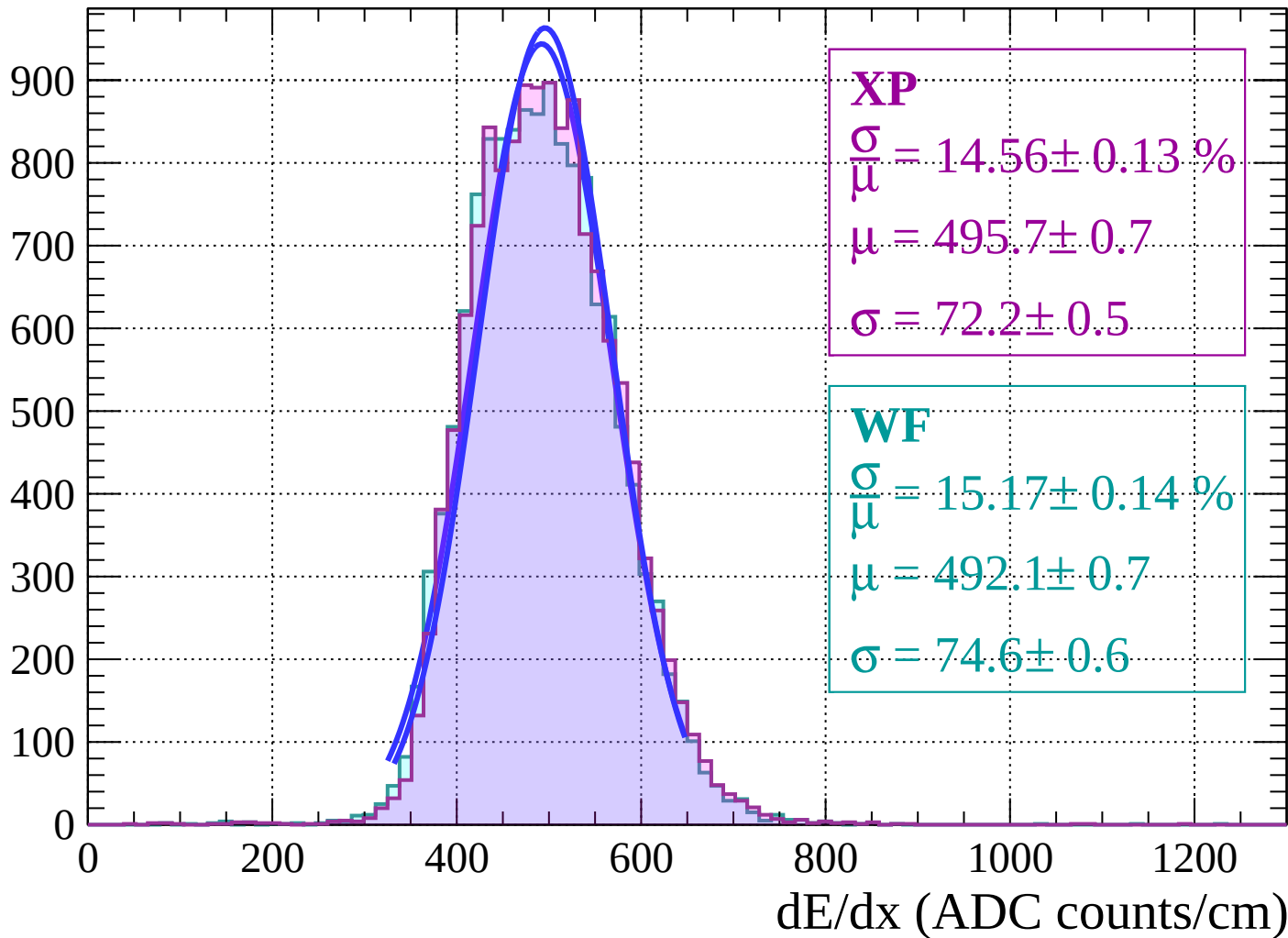
Energy loss | $18 < \theta < 20$

Count

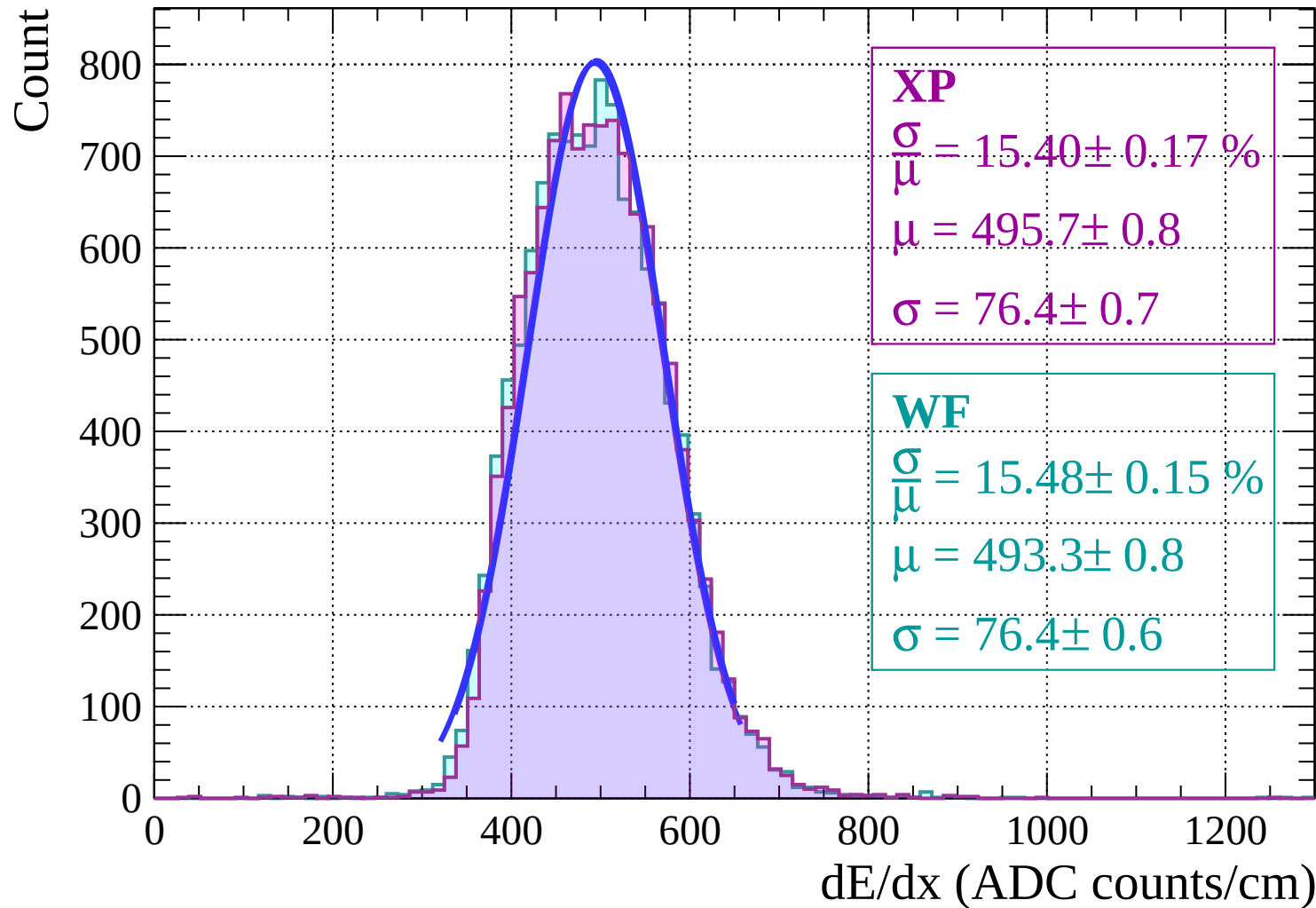


Energy loss | $20 < \theta < 22$

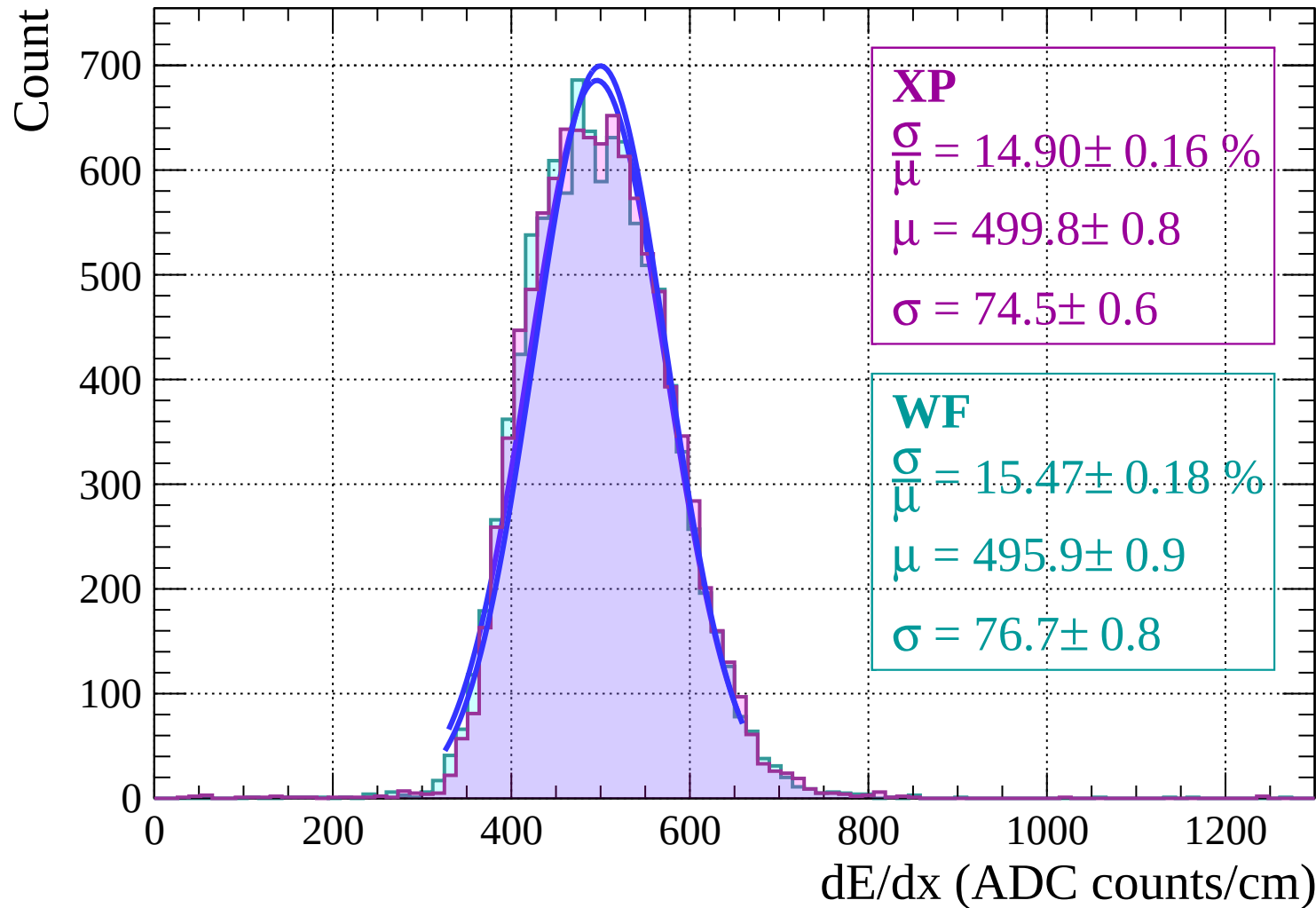
Count



Energy loss | $22 < \theta < 24$

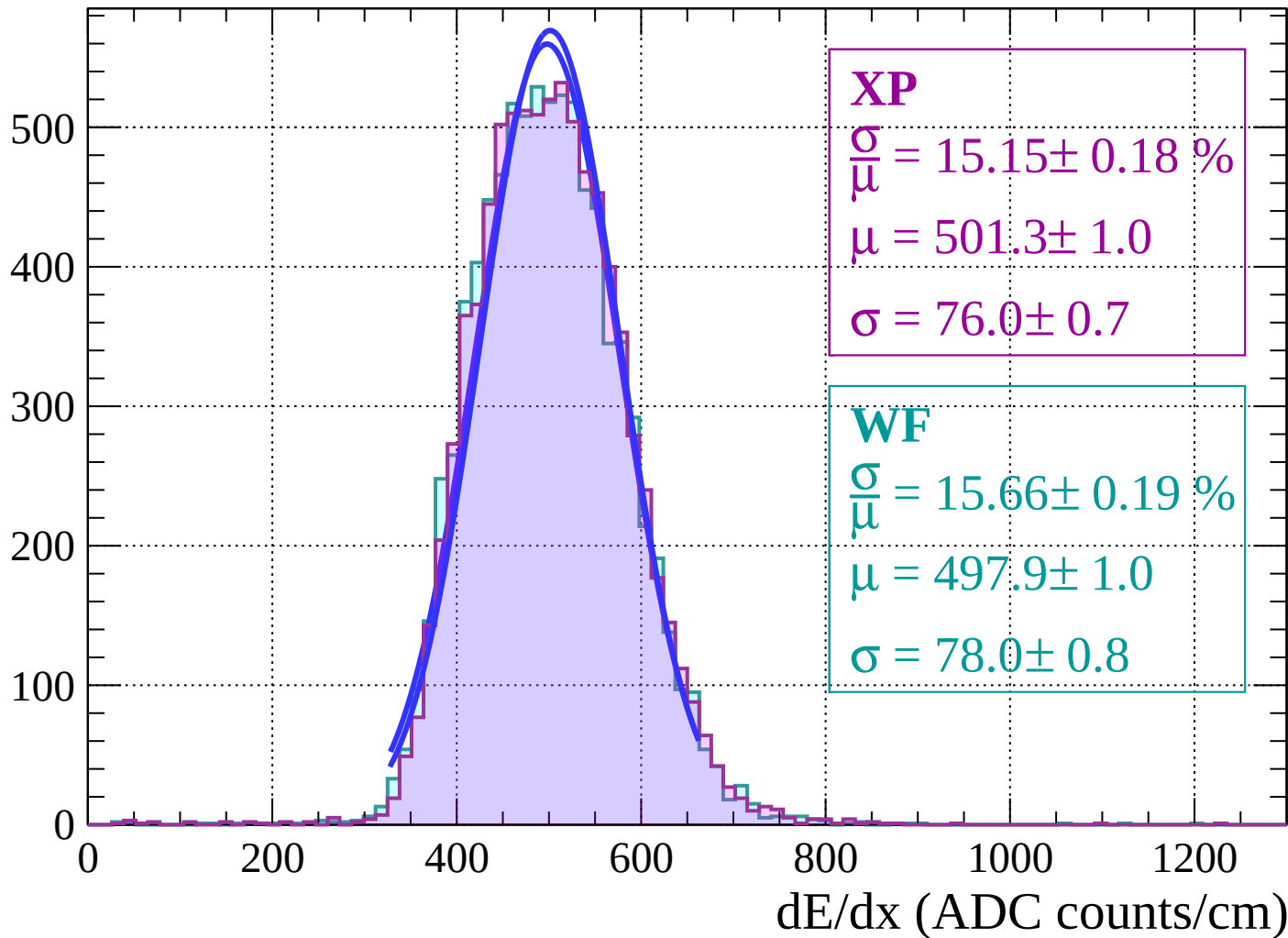


Energy loss | $24 < \theta < 26$



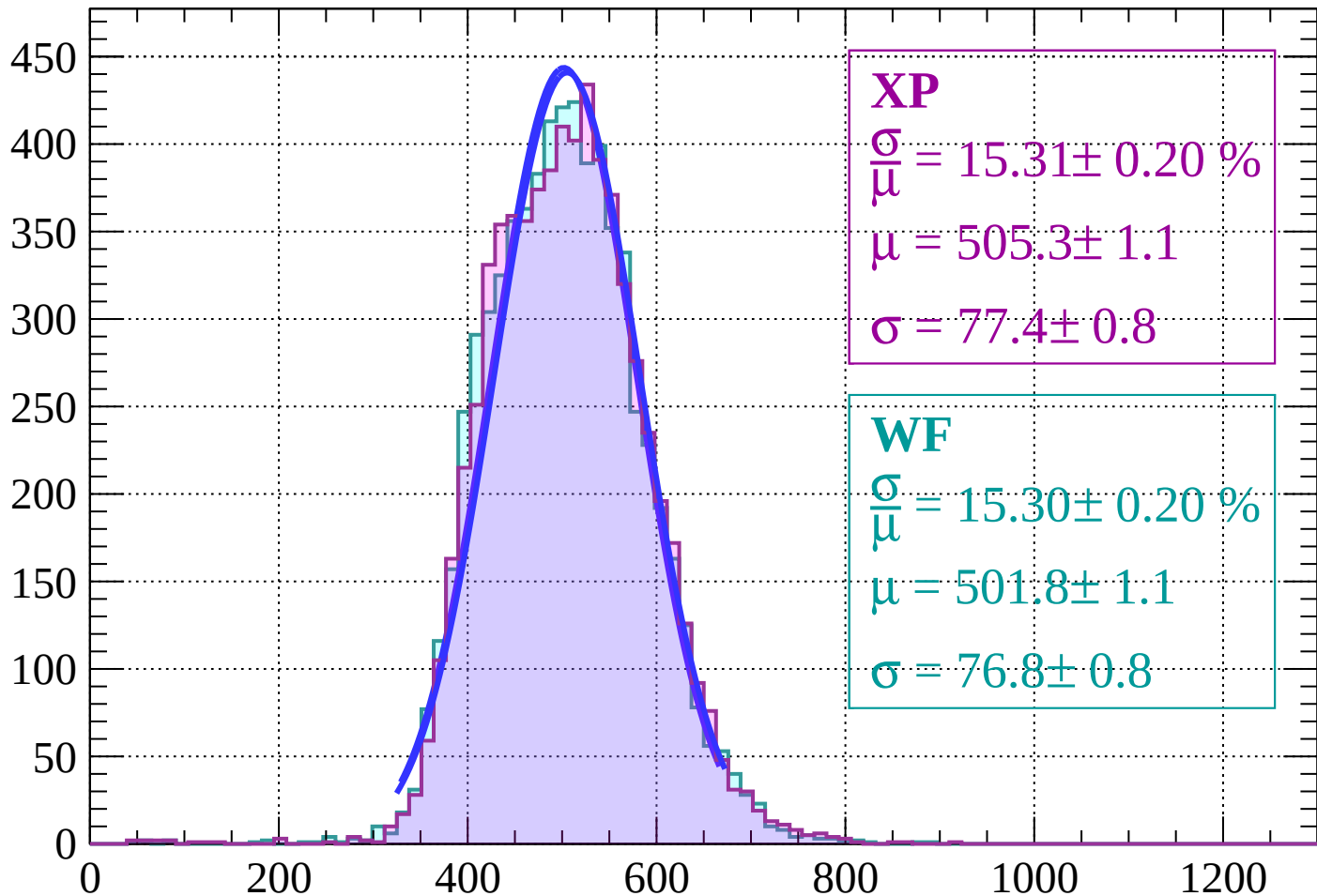
Energy loss | $26 < \theta < 28$

Count



Energy loss | $28 < \theta < 30$

Count



XP

$$\frac{\sigma}{\mu} = 15.31 \pm 0.20 \%$$

$$\mu = 505.3 \pm 1.1$$

$$\sigma = 77.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 15.30 \pm 0.20 \%$$

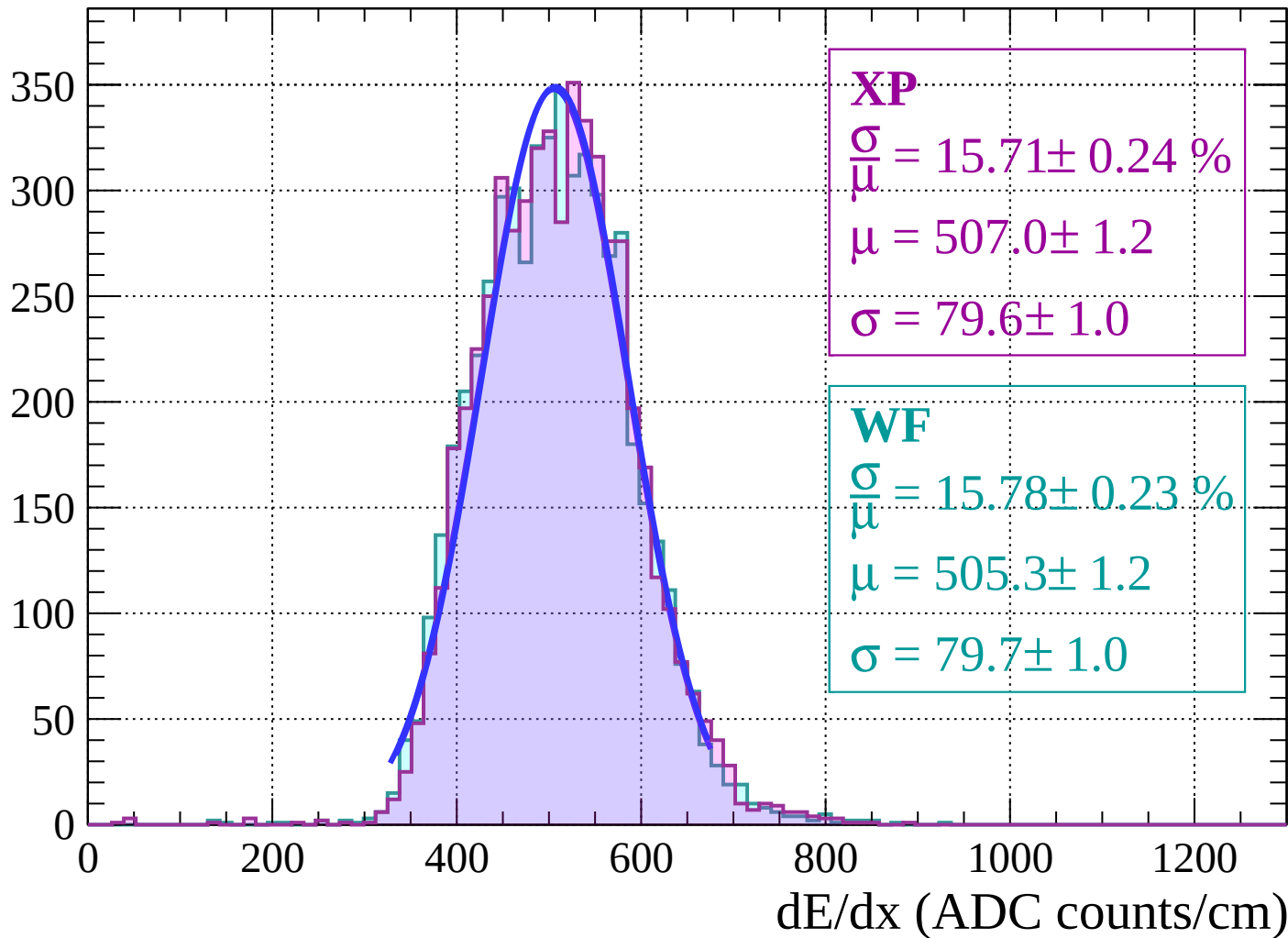
$$\mu = 501.8 \pm 1.1$$

$$\sigma = 76.8 \pm 0.8$$

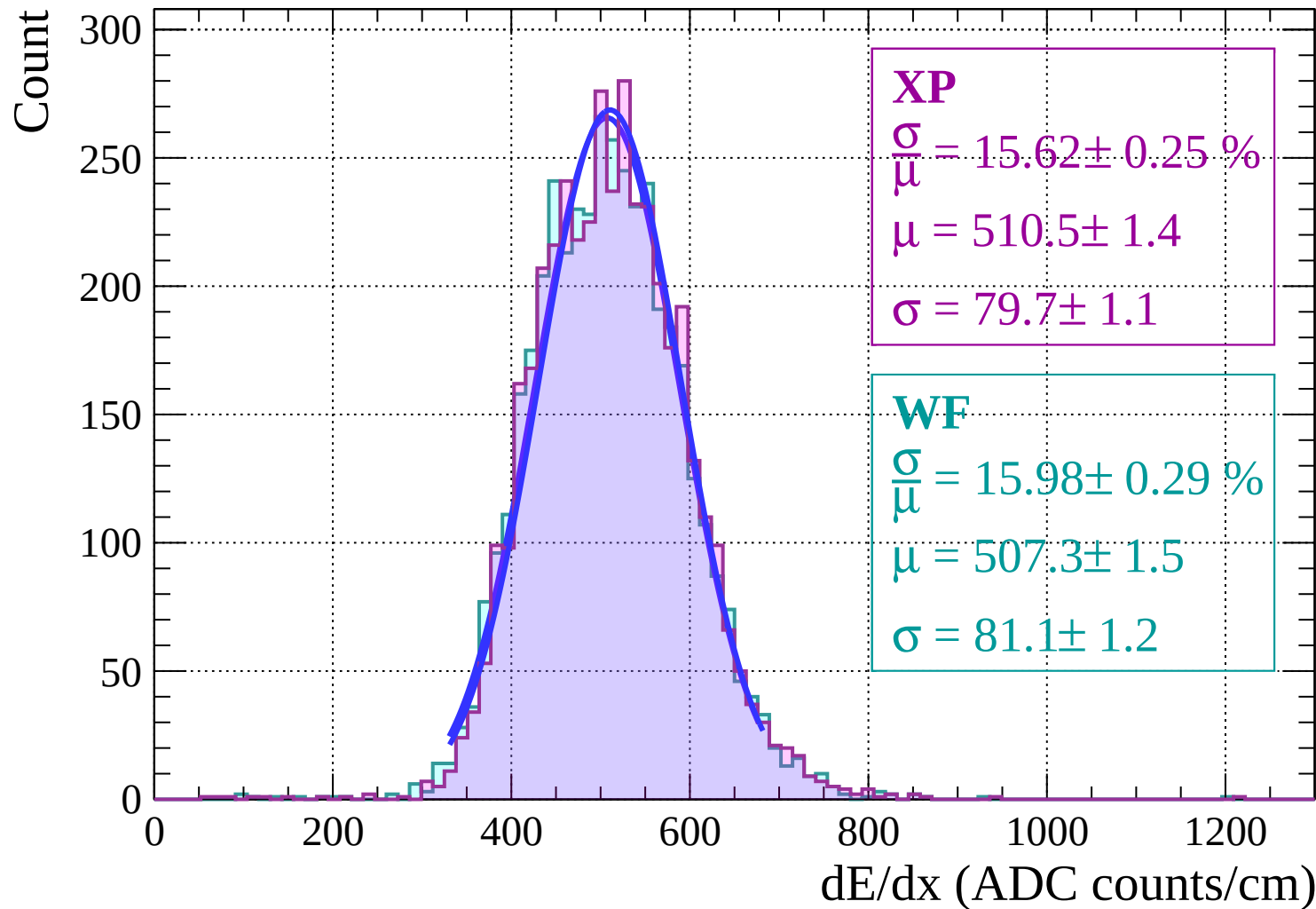
dE/dx (ADC counts/cm)

Energy loss | $30 < \theta < 32$

Count

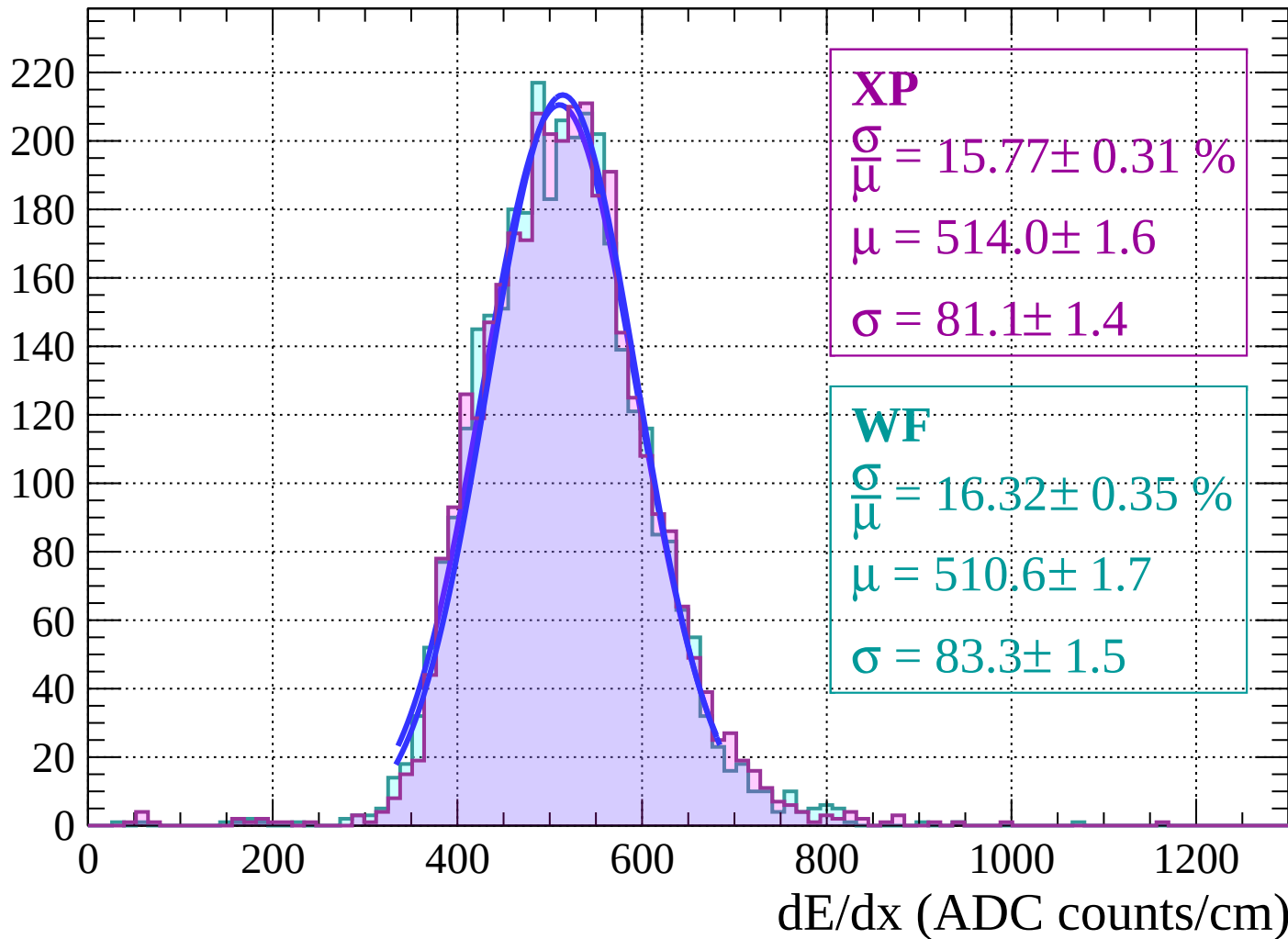


Energy loss | $32 < \theta < 34$



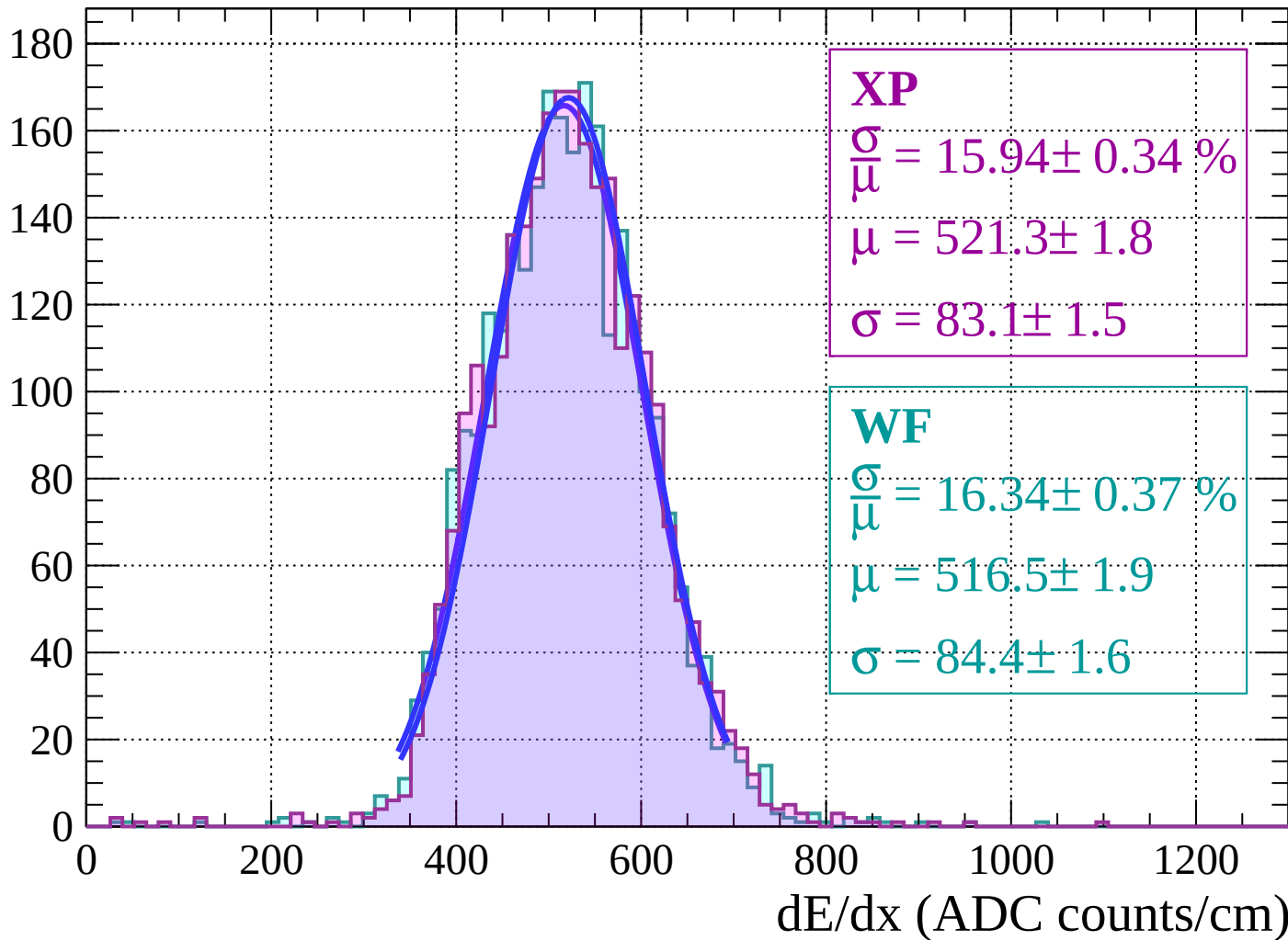
Energy loss | $34 < \theta < 36$

Count



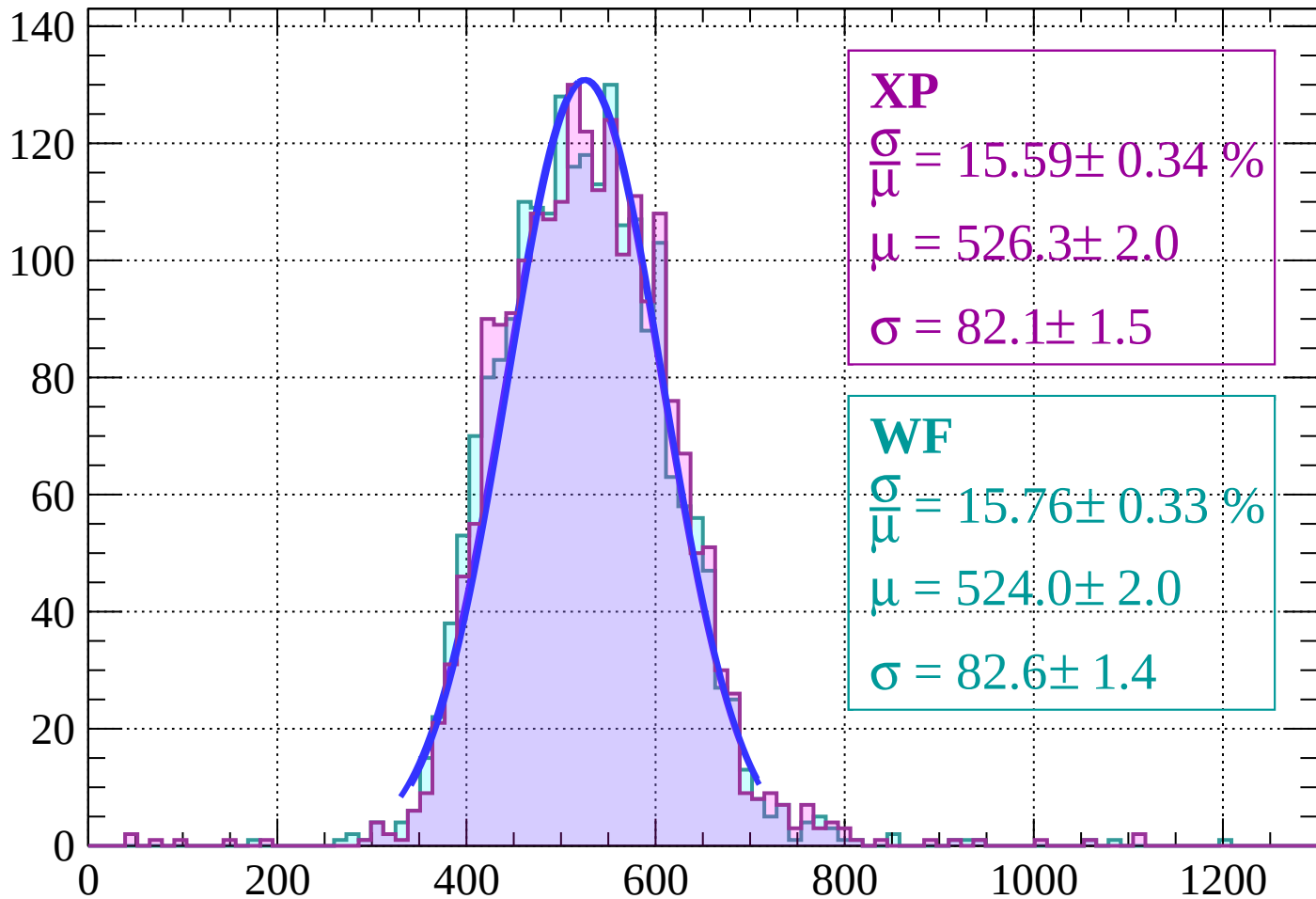
Energy loss | $36 < \theta < 38$

Count



Energy loss | $38 < \theta < 40$

Count



XP

$$\frac{\sigma}{\mu} = 15.59 \pm 0.34 \%$$

$$\mu = 526.3 \pm 2.0$$

$$\sigma = 82.1 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 15.76 \pm 0.33 \%$$

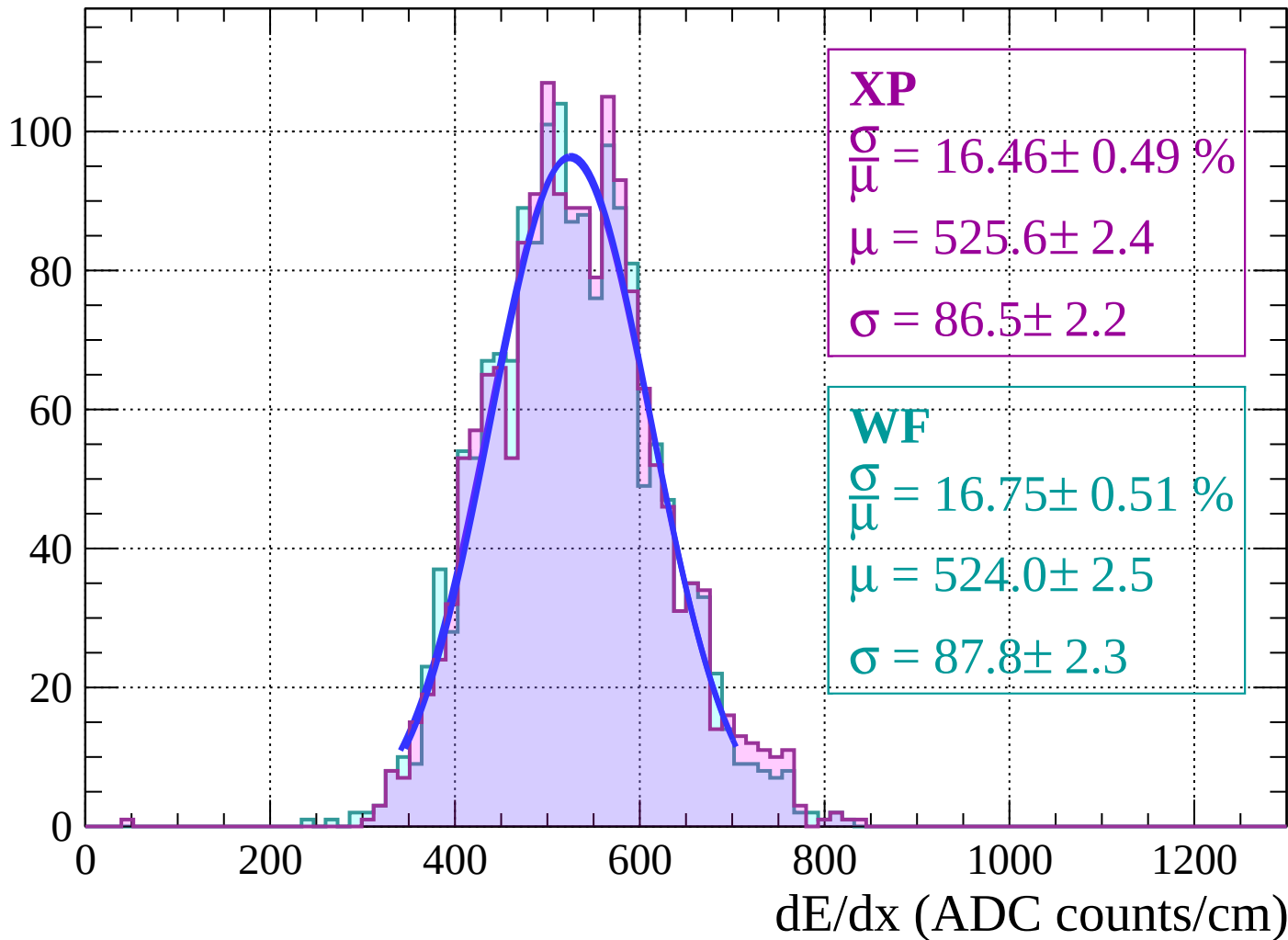
$$\mu = 524.0 \pm 2.0$$

$$\sigma = 82.6 \pm 1.4$$

dE/dx (ADC counts/cm)

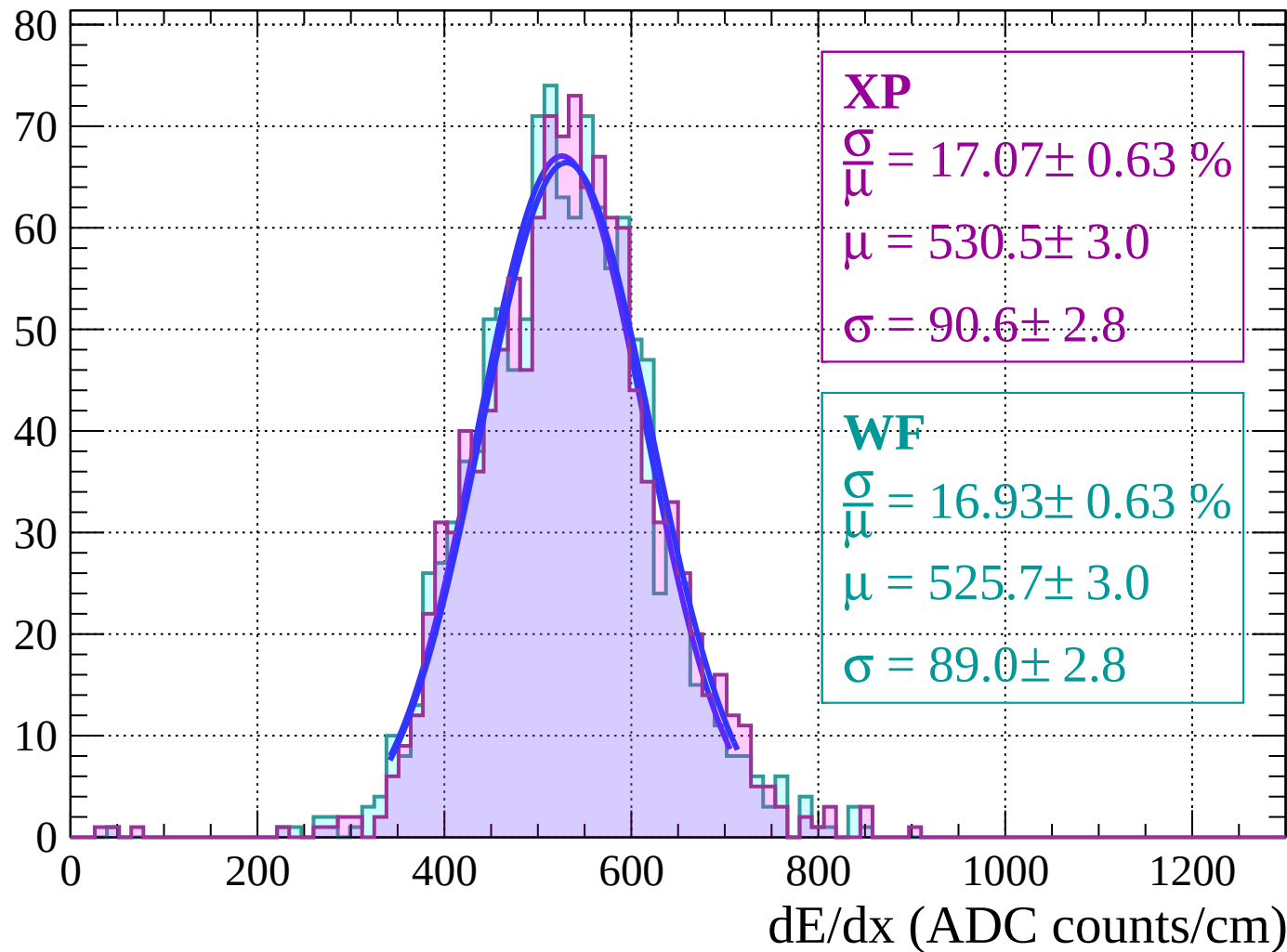
Energy loss | $40 < \theta < 42$

Count



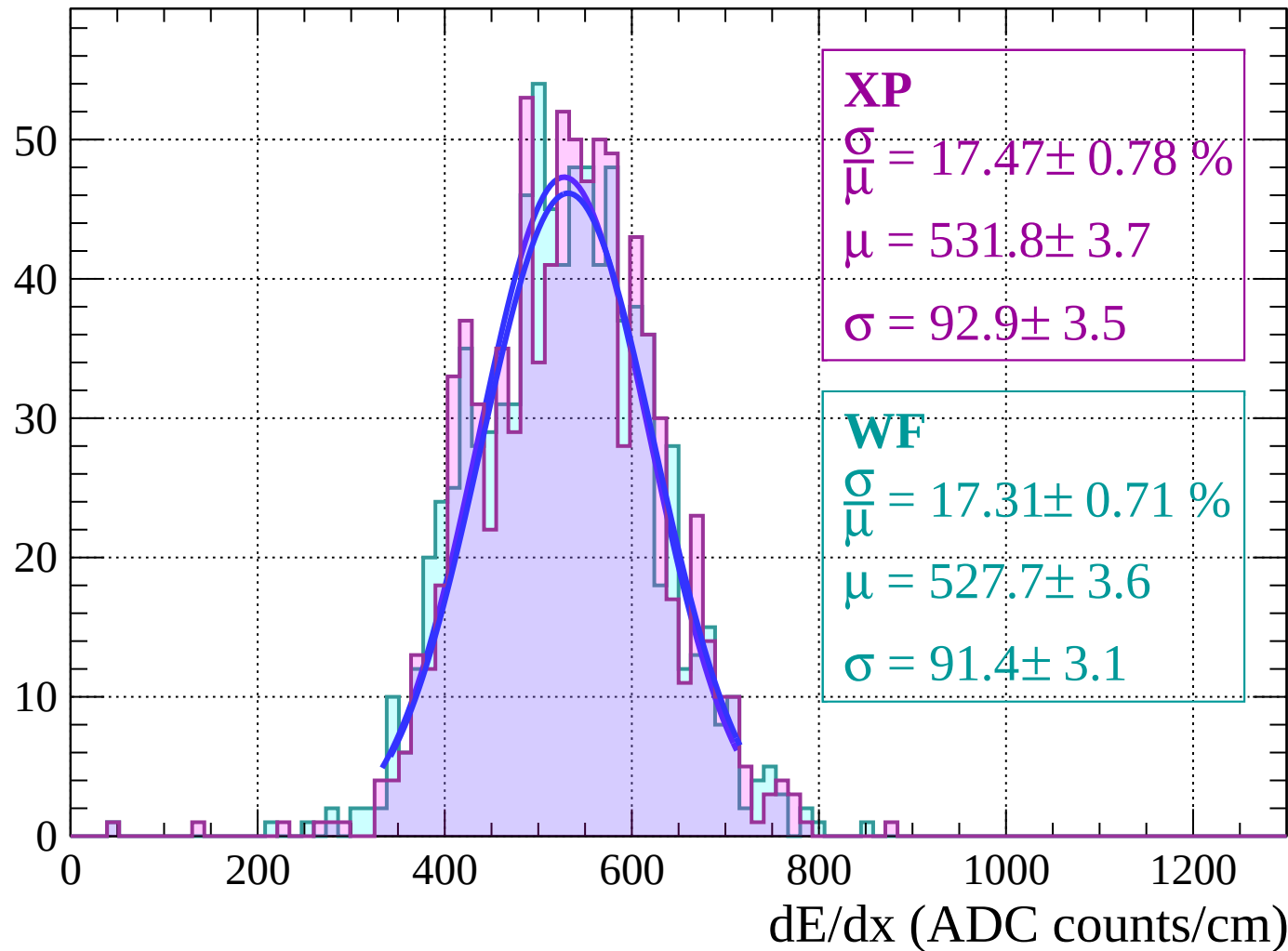
Energy loss | $42 < \theta < 44$

Count



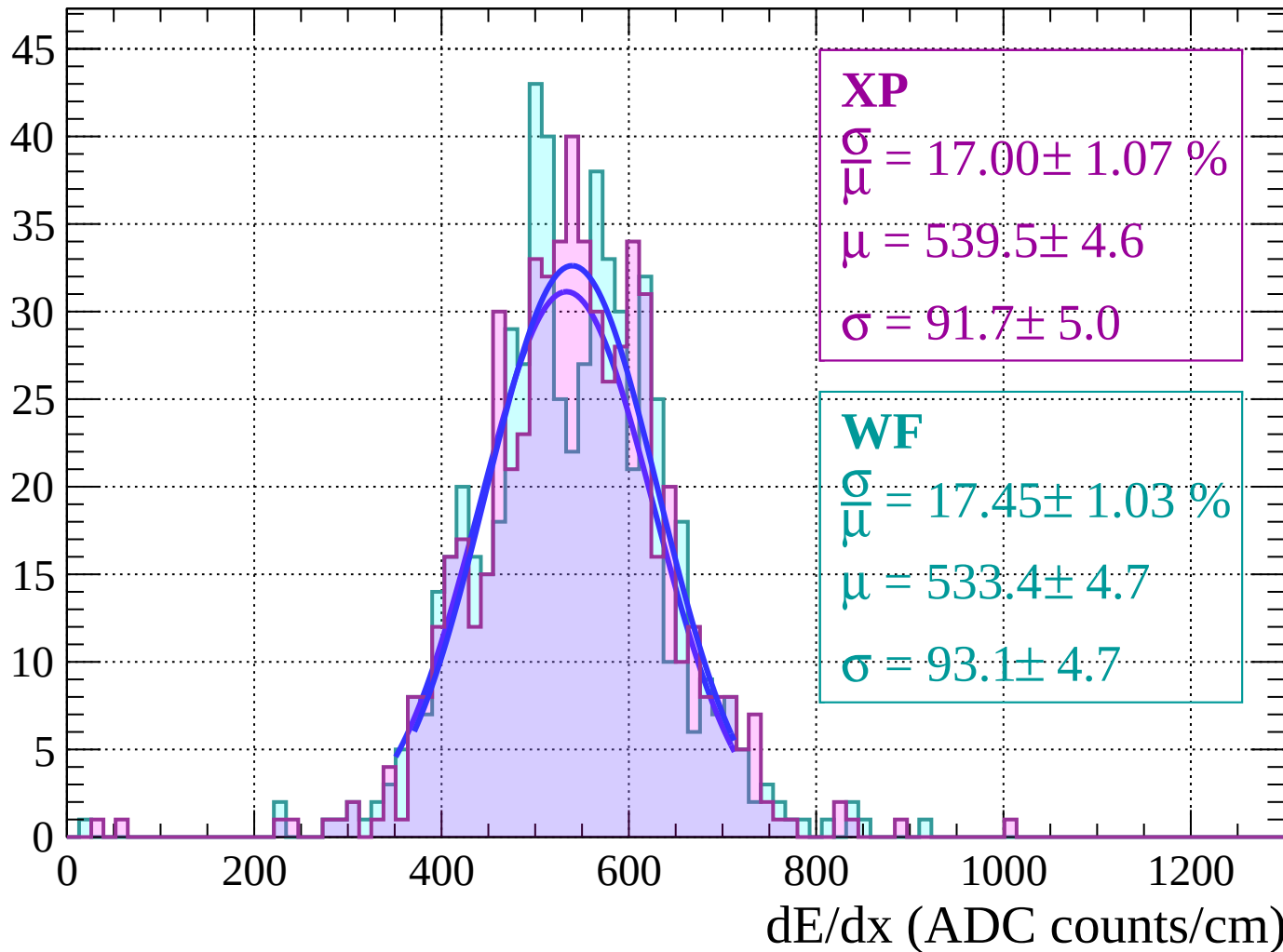
Energy loss | $44 < \theta < 46$

Count



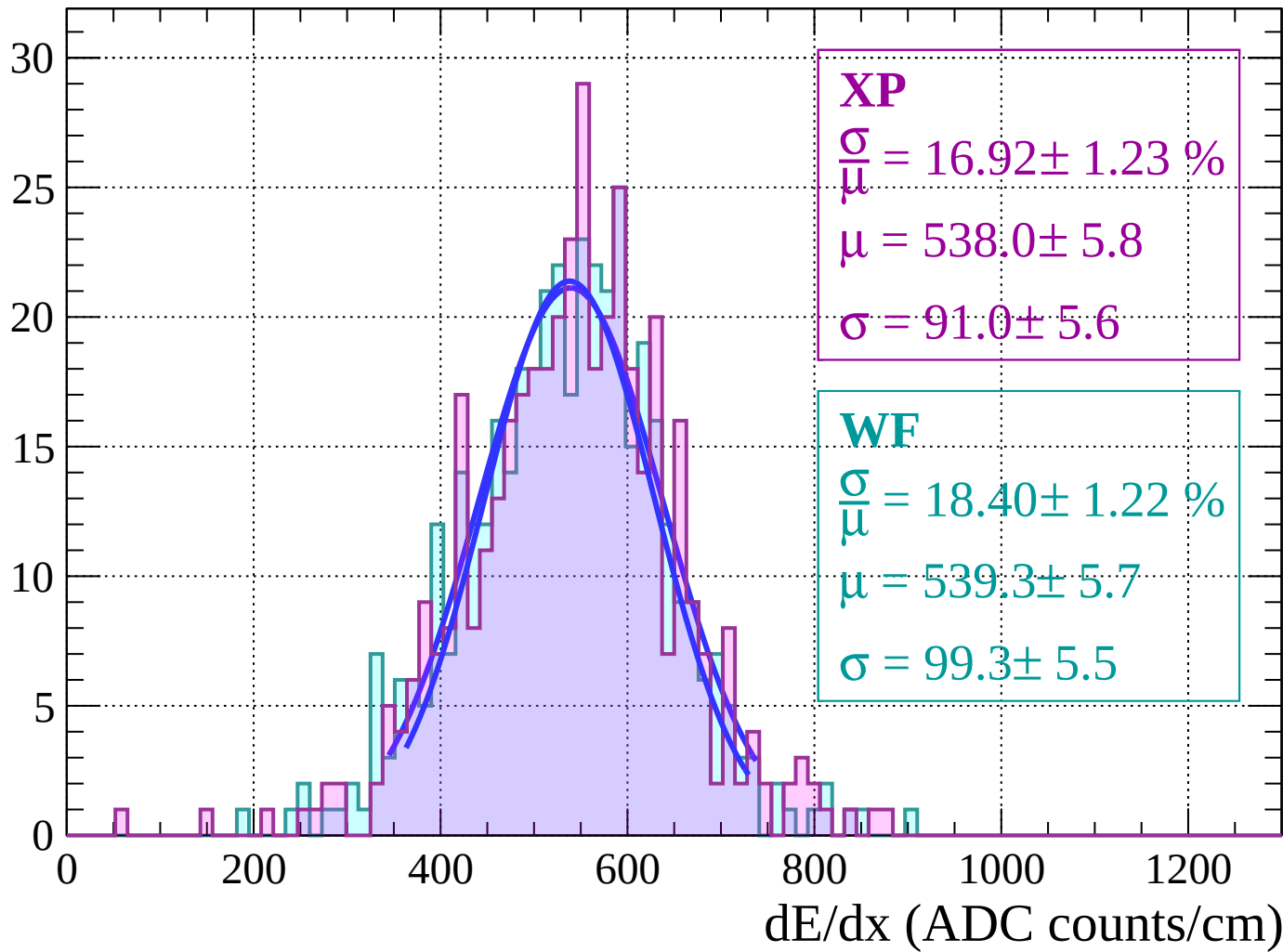
Energy loss | $46 < \theta < 48$

Count



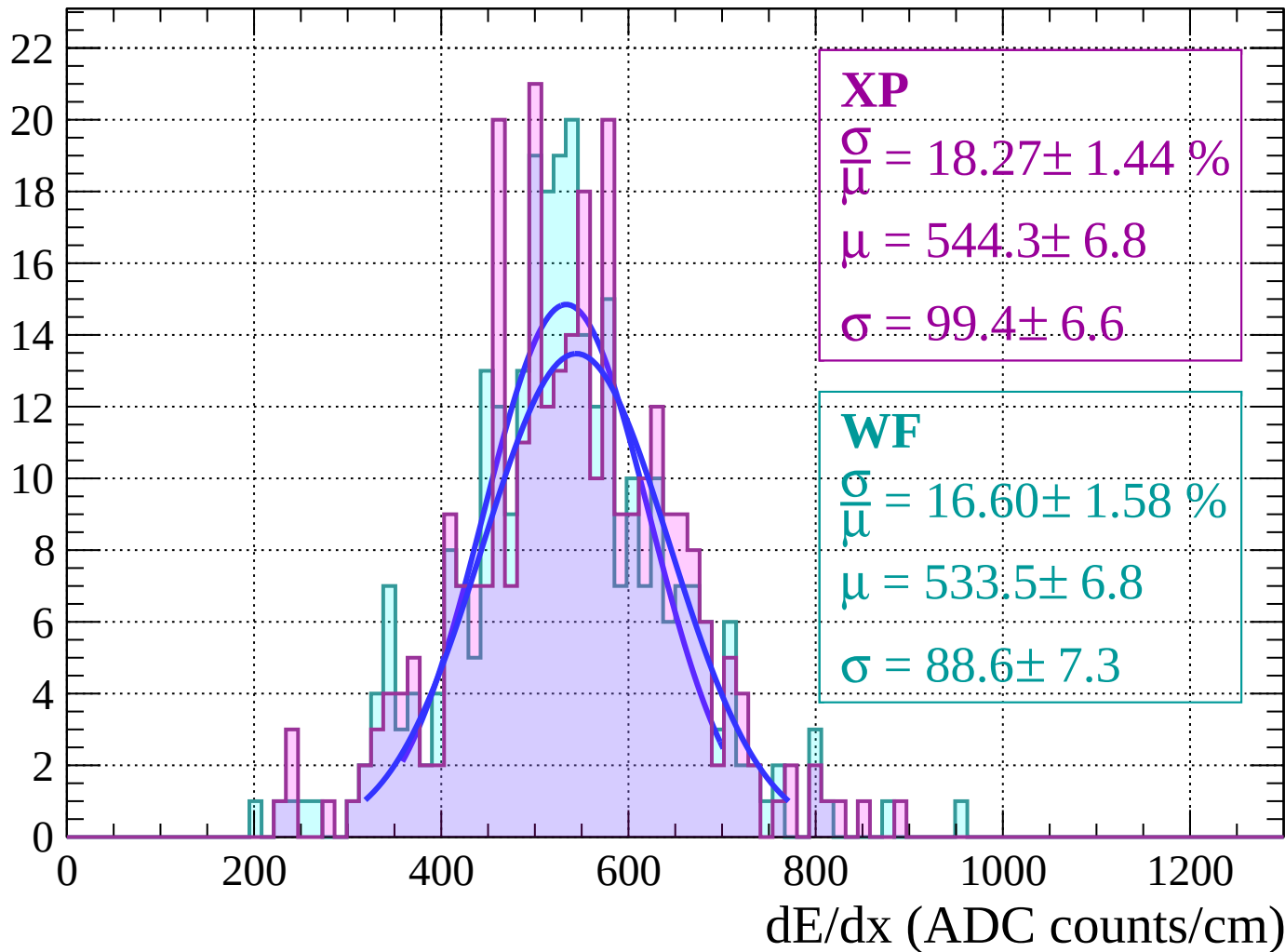
Energy loss | $48 < \theta < 50$

Count



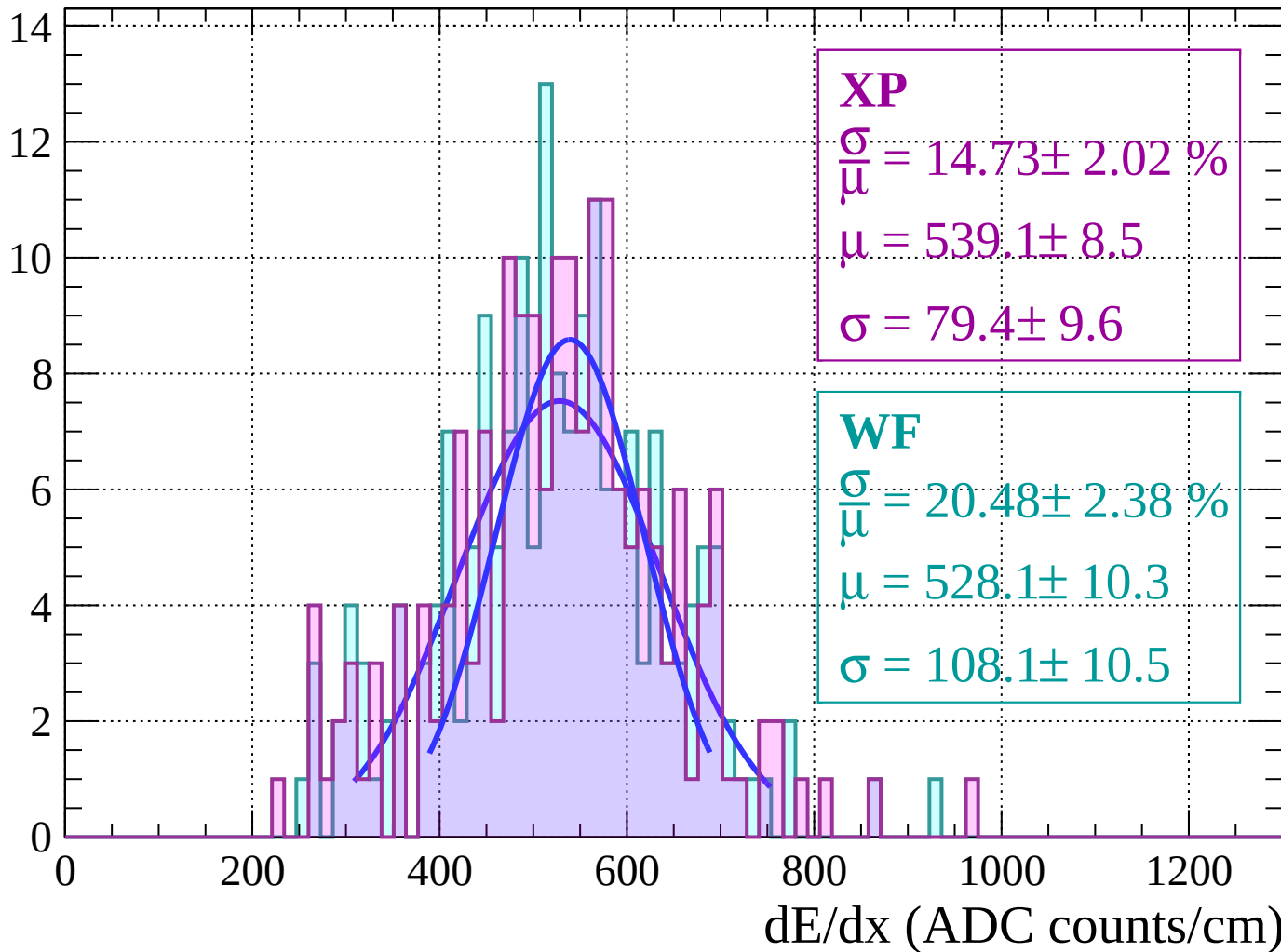
Energy loss | $50 < \theta < 52$

Count



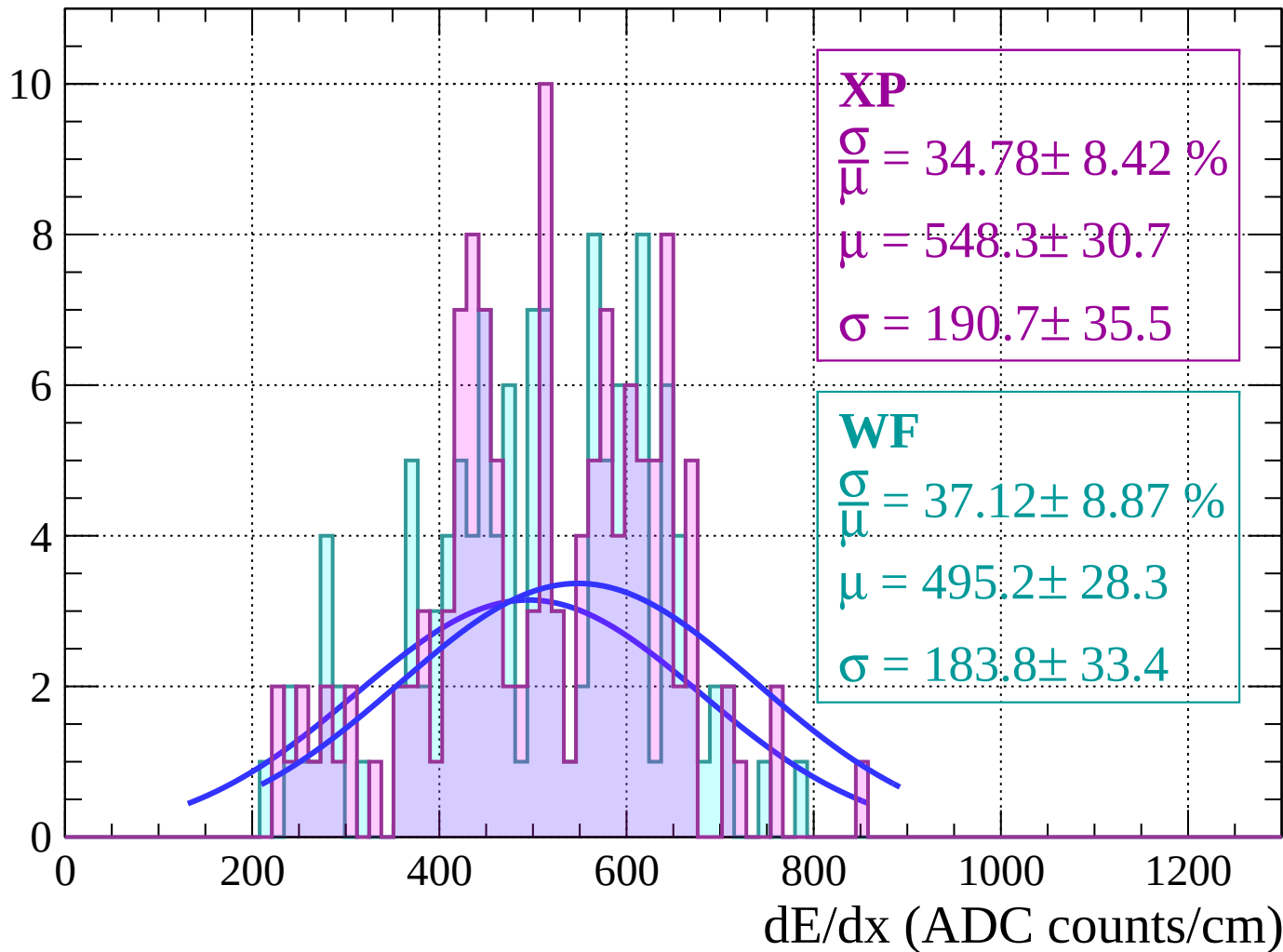
Energy loss | $52 < \theta < 54$

Count



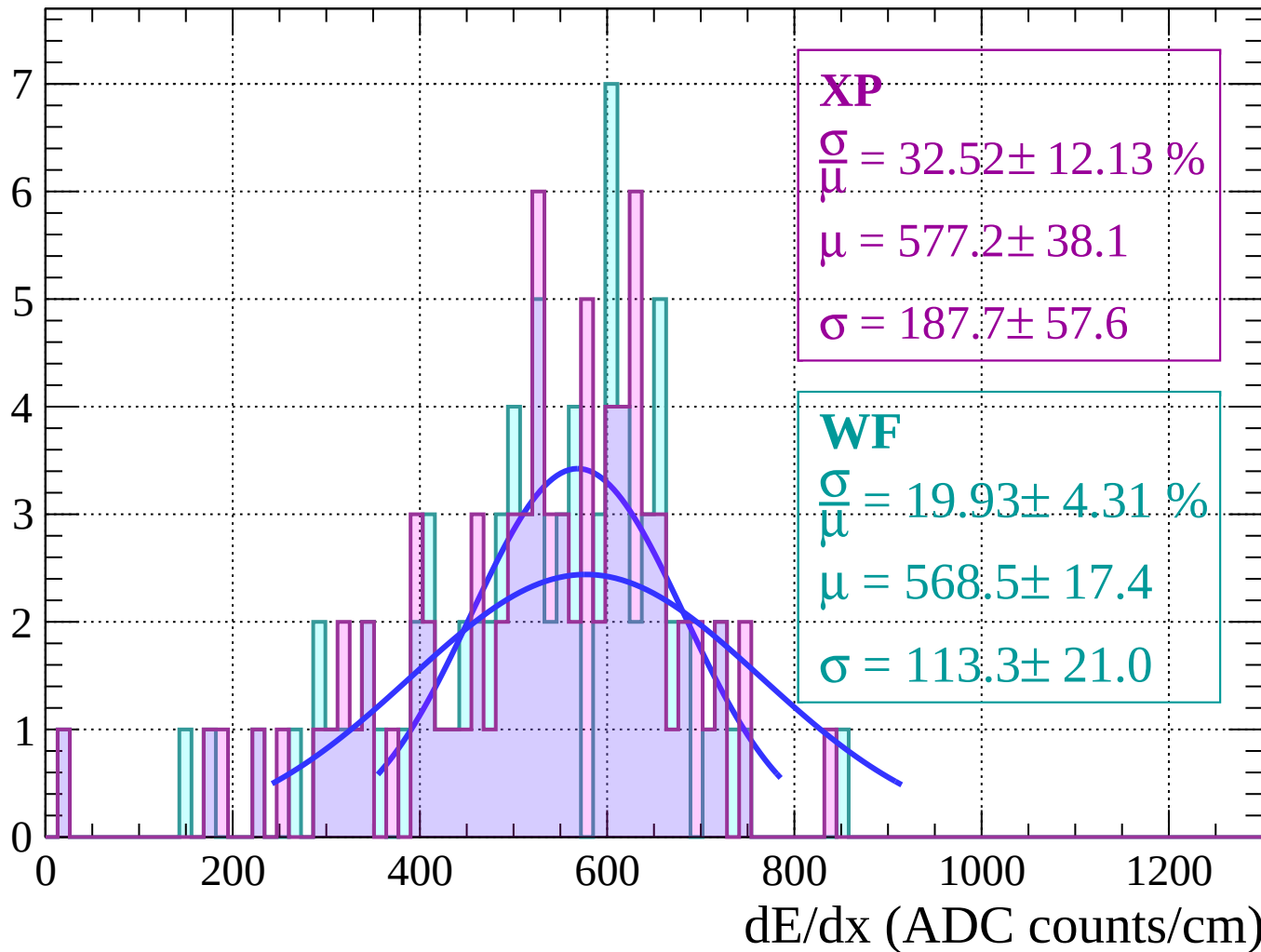
Energy loss | $54 < \theta < 56$

Count



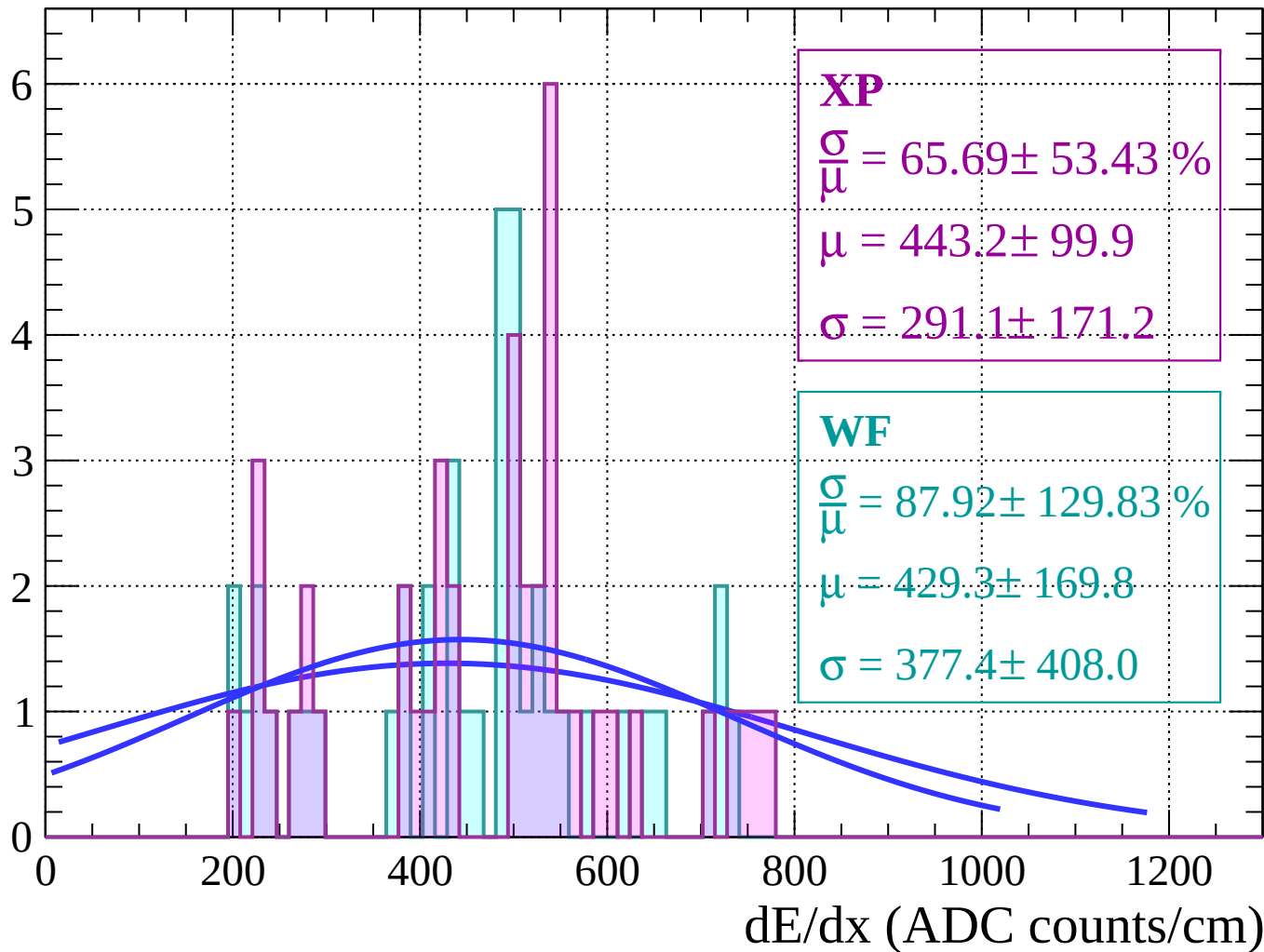
Energy loss | $56 < \theta < 58$

Count



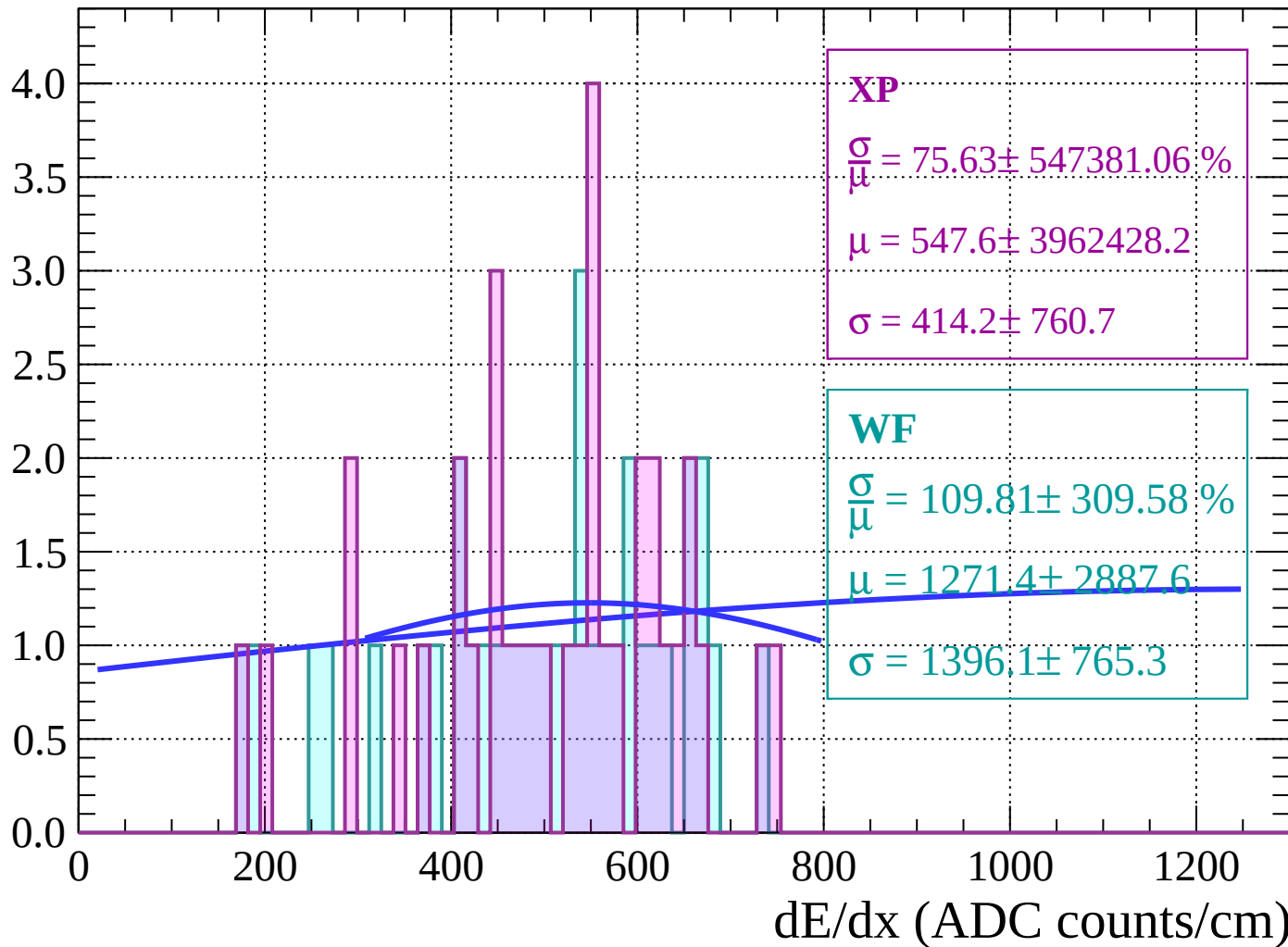
Energy loss | $58 < \theta < 60$

Count



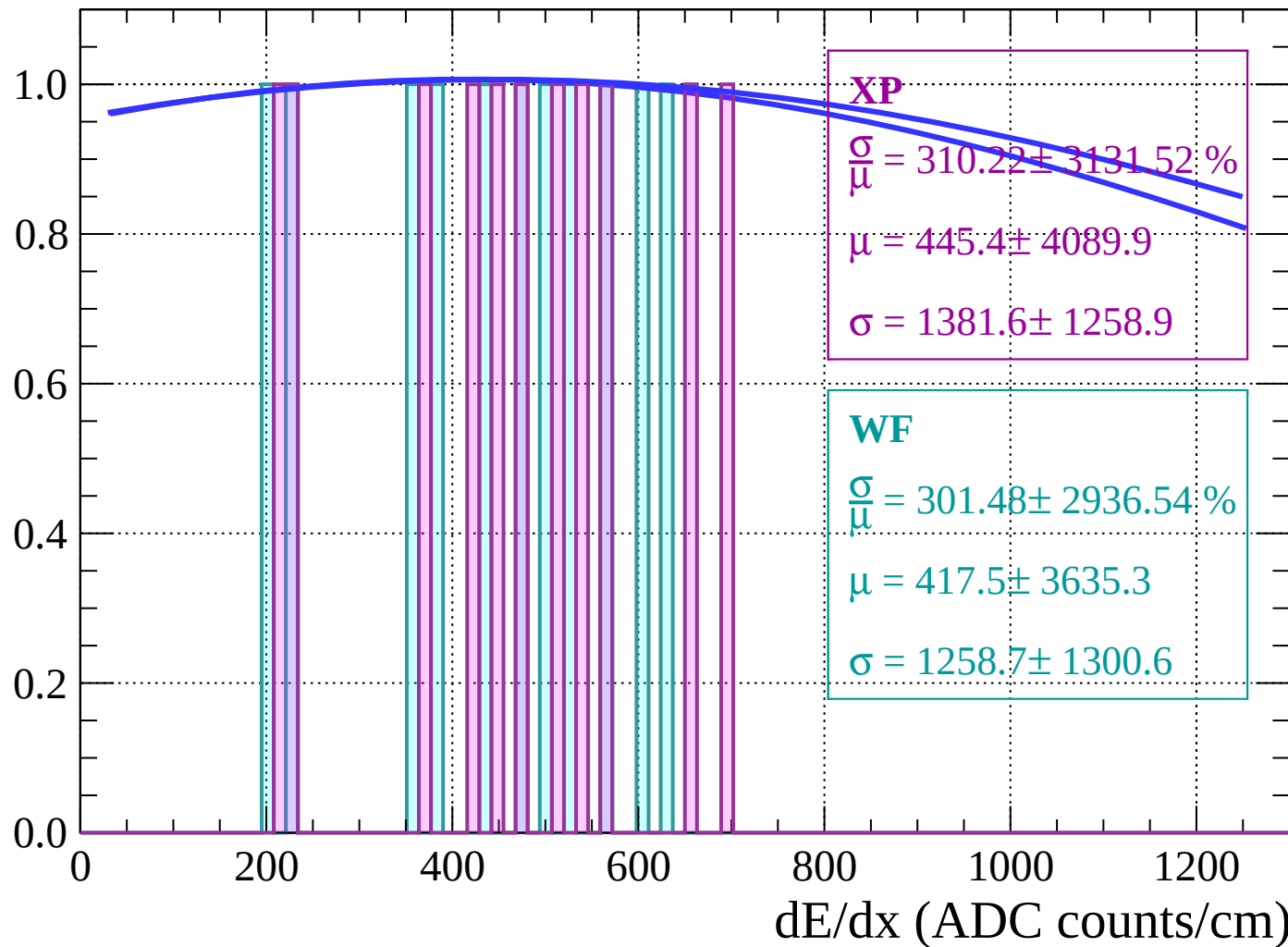
Energy loss | $60 < \theta < 62$

Count

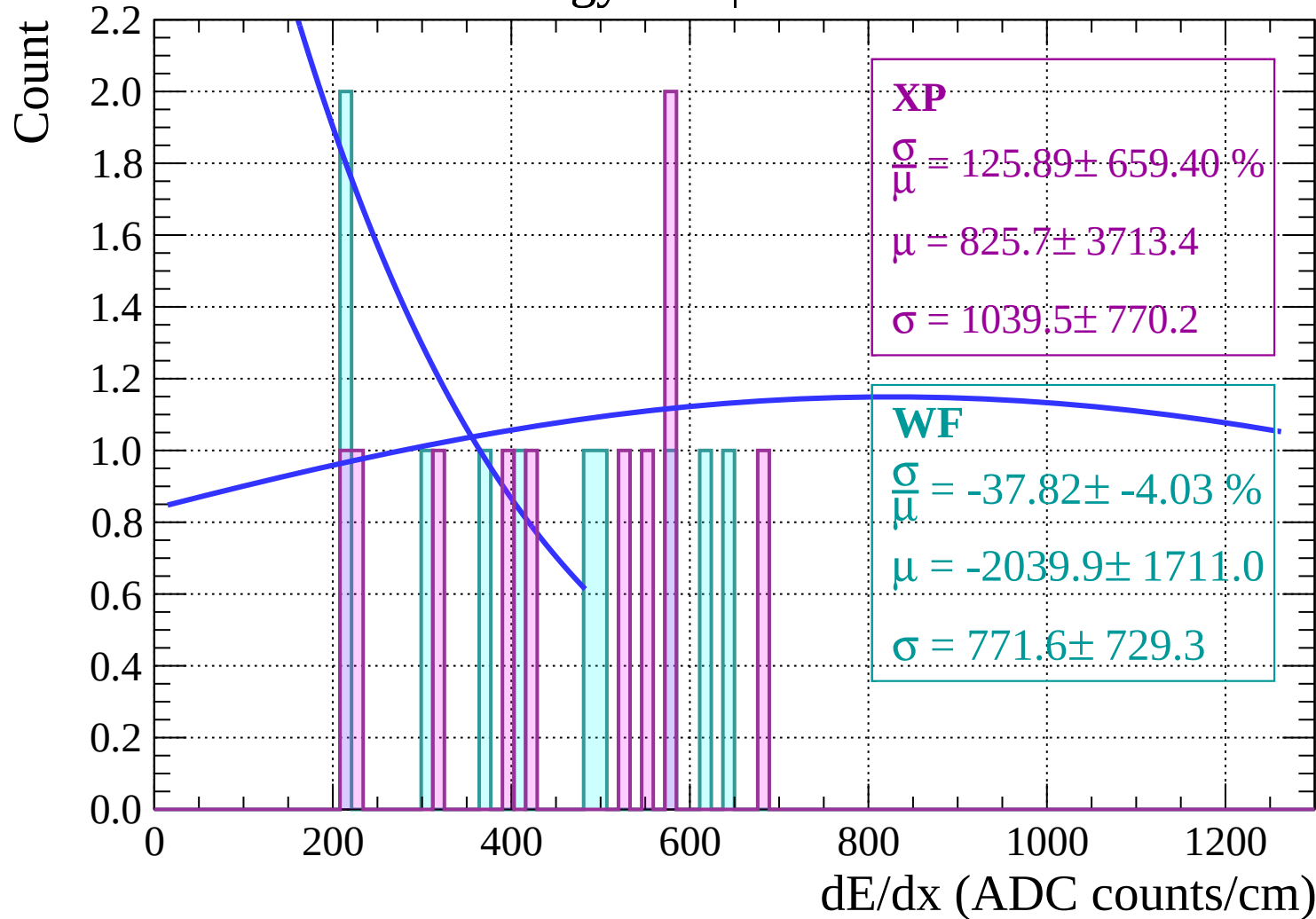


Energy loss | $62 < \theta < 64$

Count

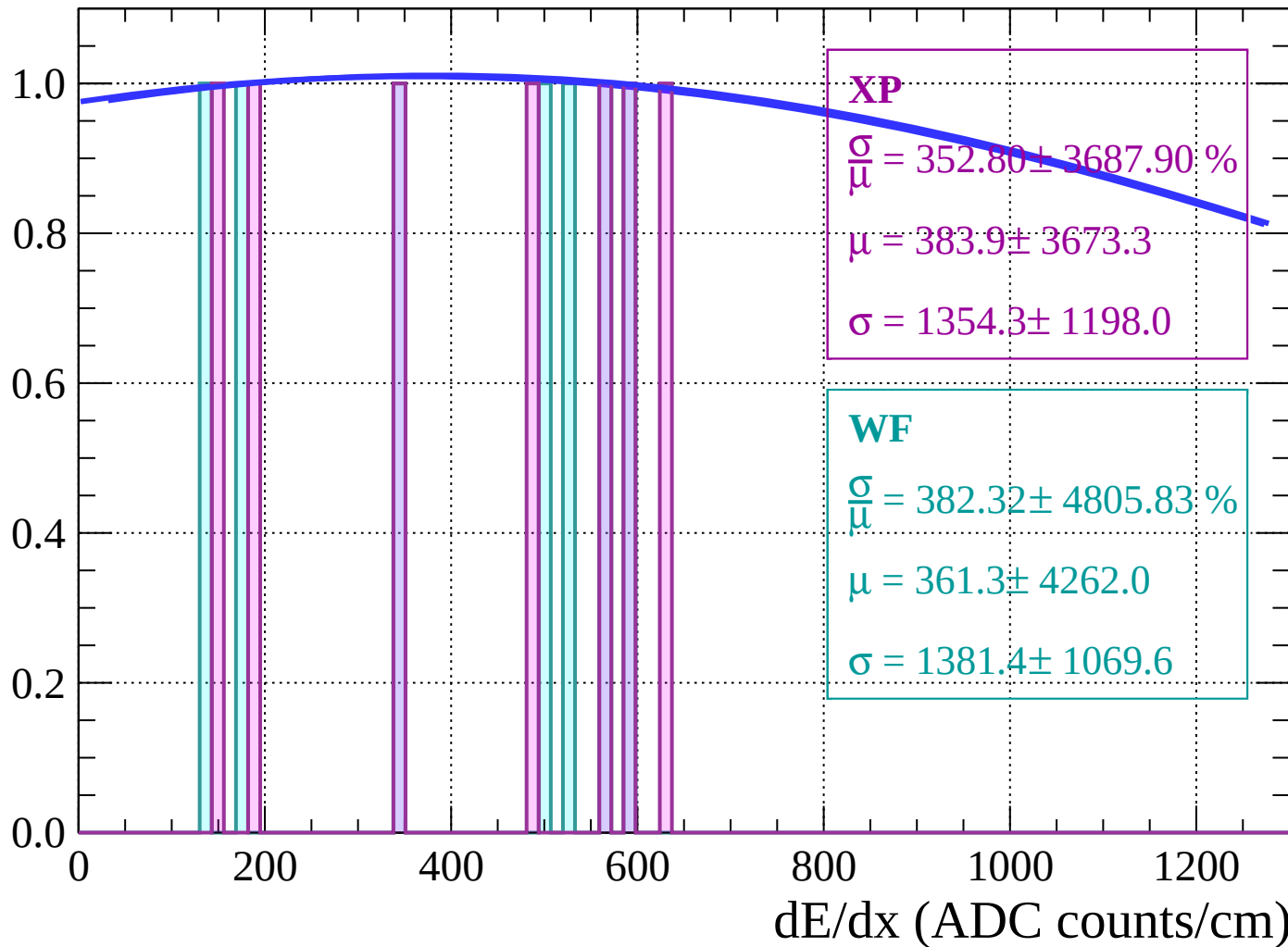


Energy loss | $64 < \theta < 66$



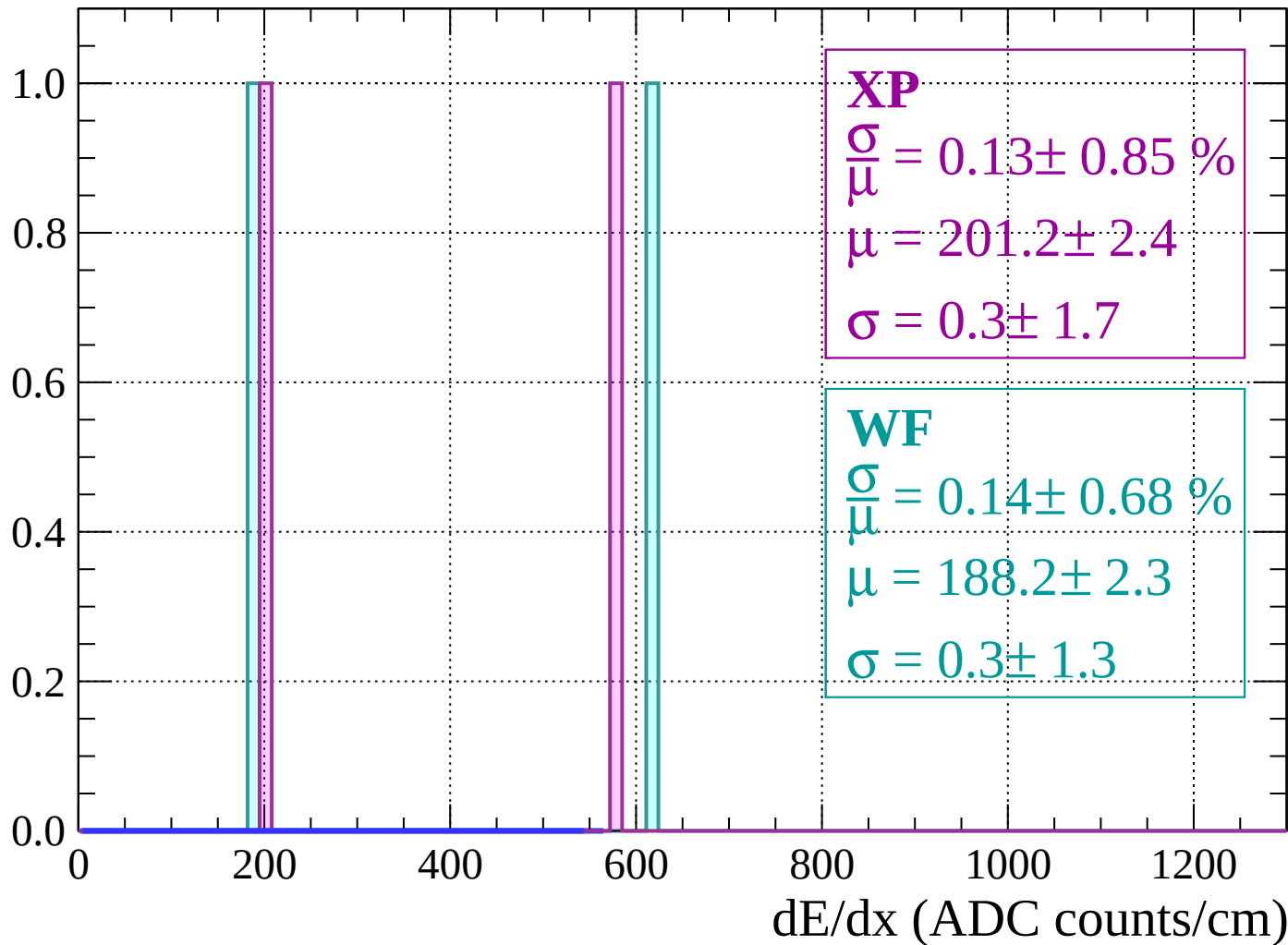
Energy loss | $66 < \theta < 68$

Count



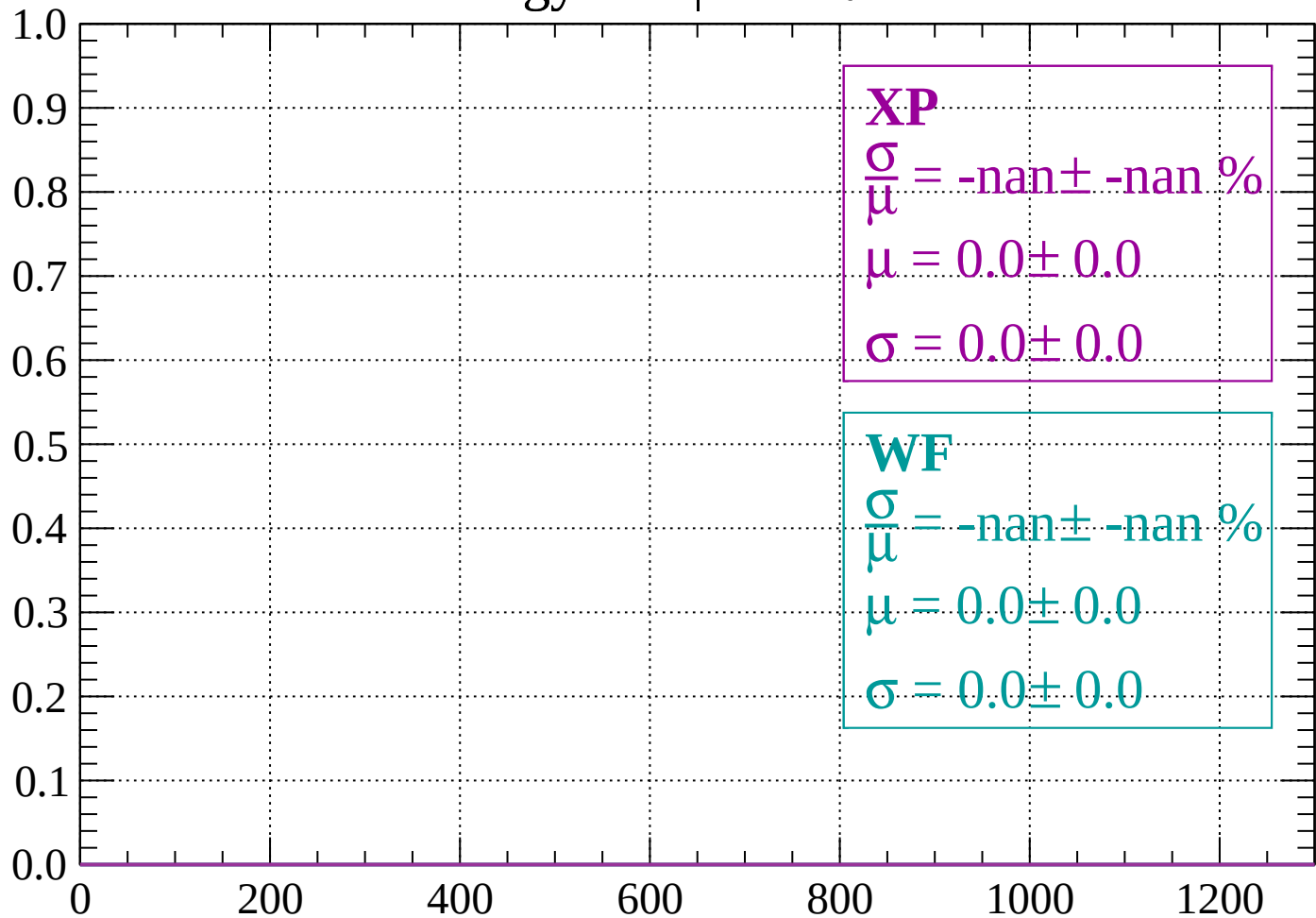
Energy loss | $68 < \theta < 70$

Count



Energy loss | $70 < \theta < 72$

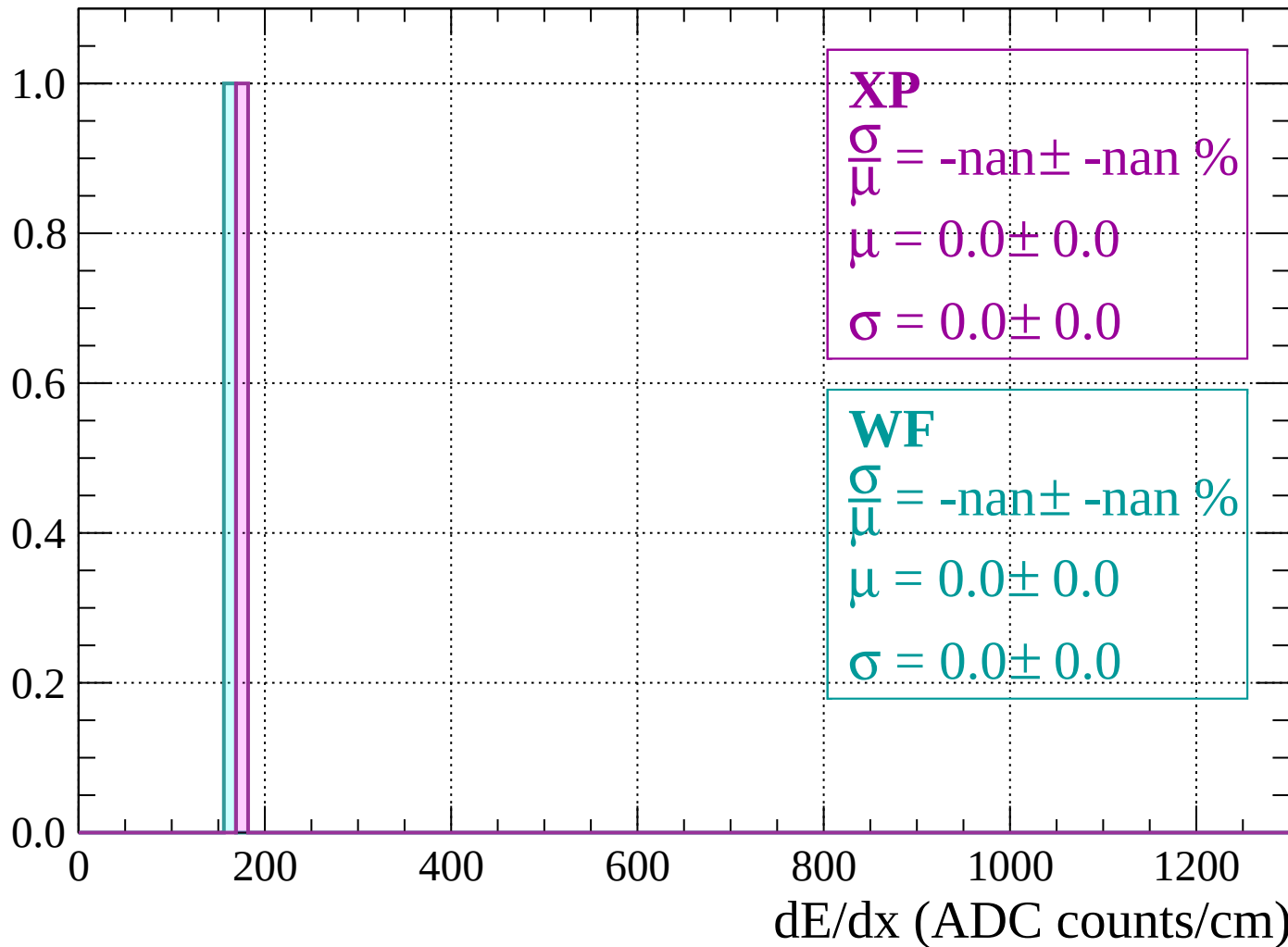
Count



dE/dx (ADC counts/cm)

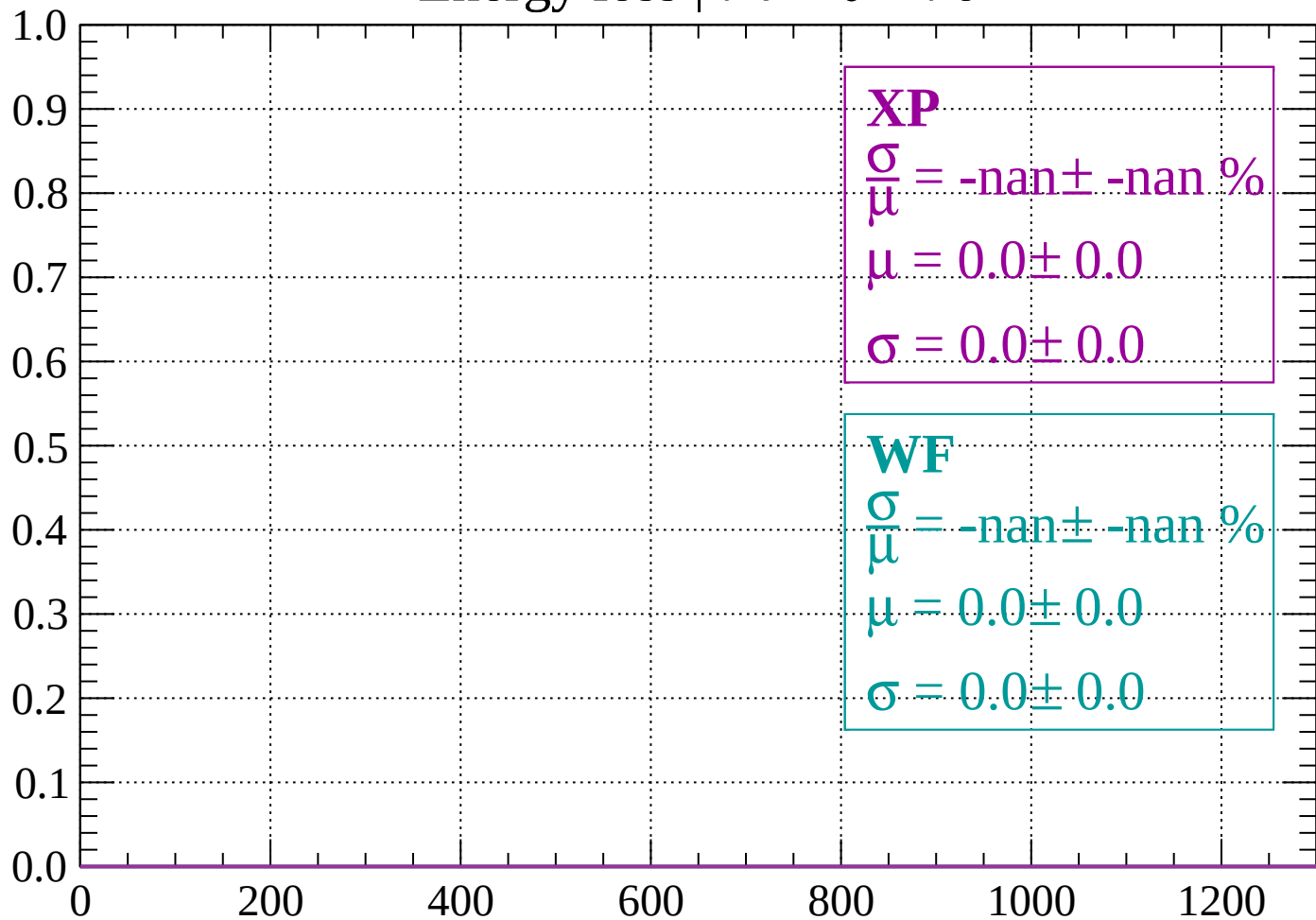
Energy loss | $72 < \theta < 74$

Count



Energy loss | $74 < \theta < 76$

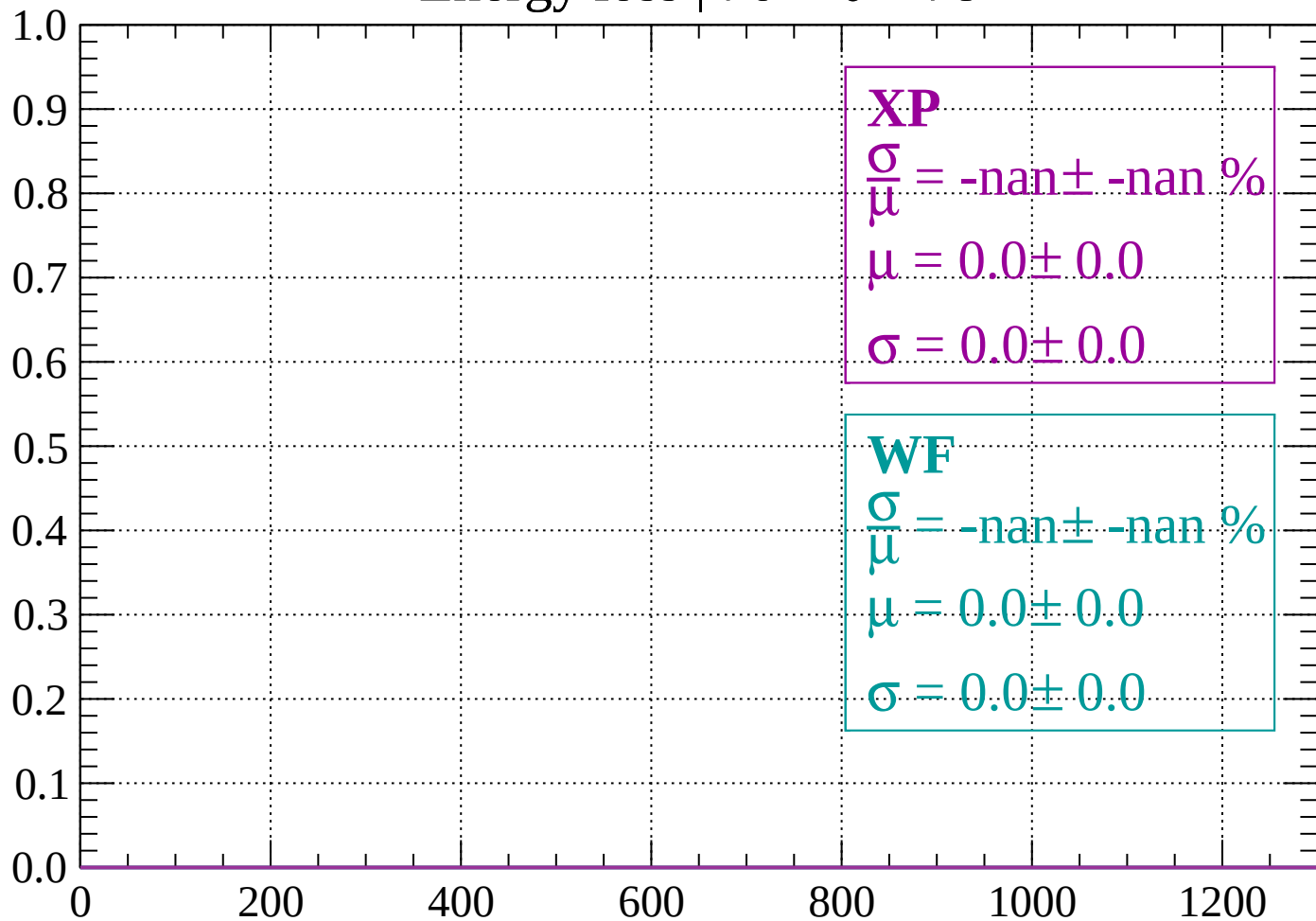
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

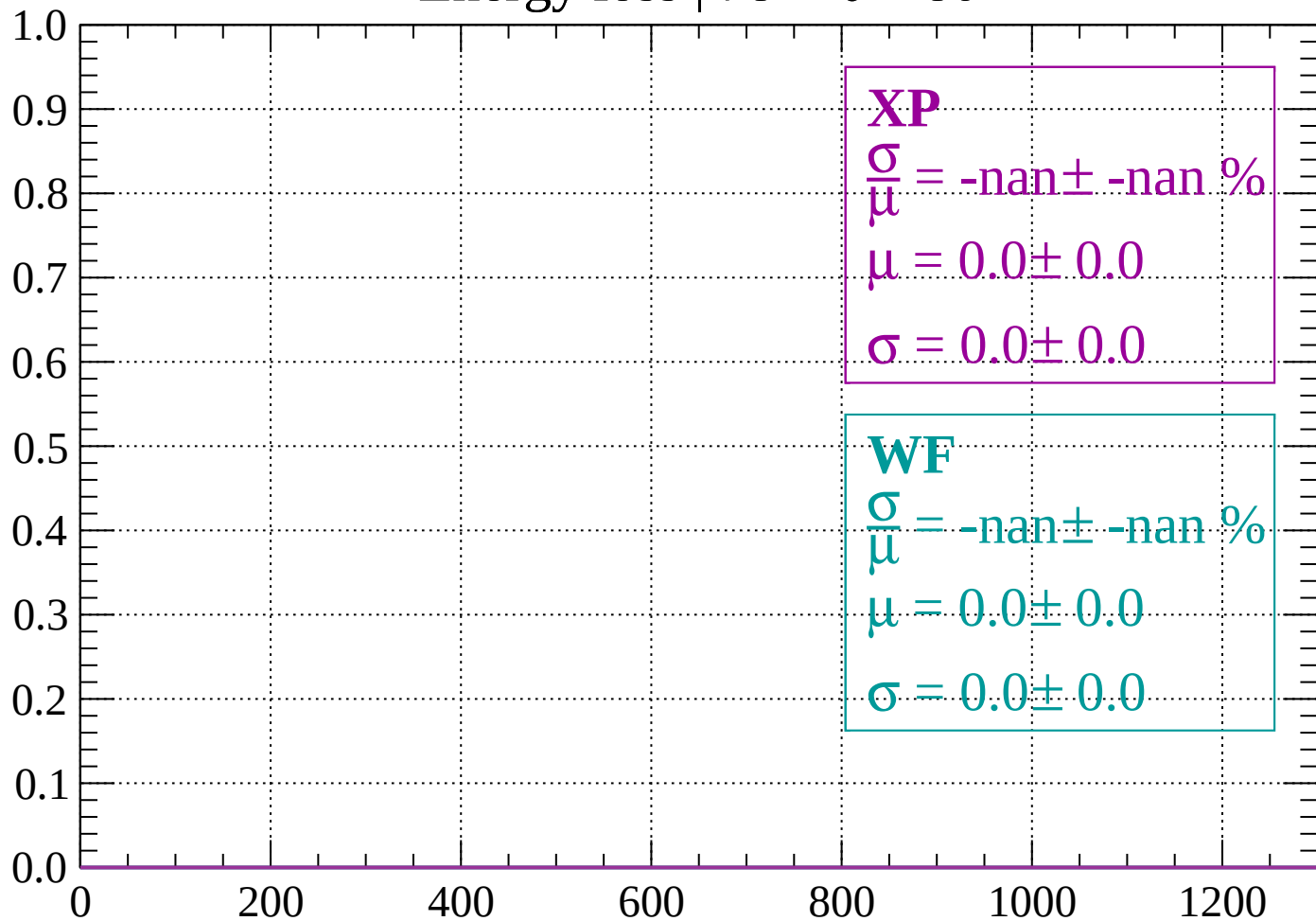
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

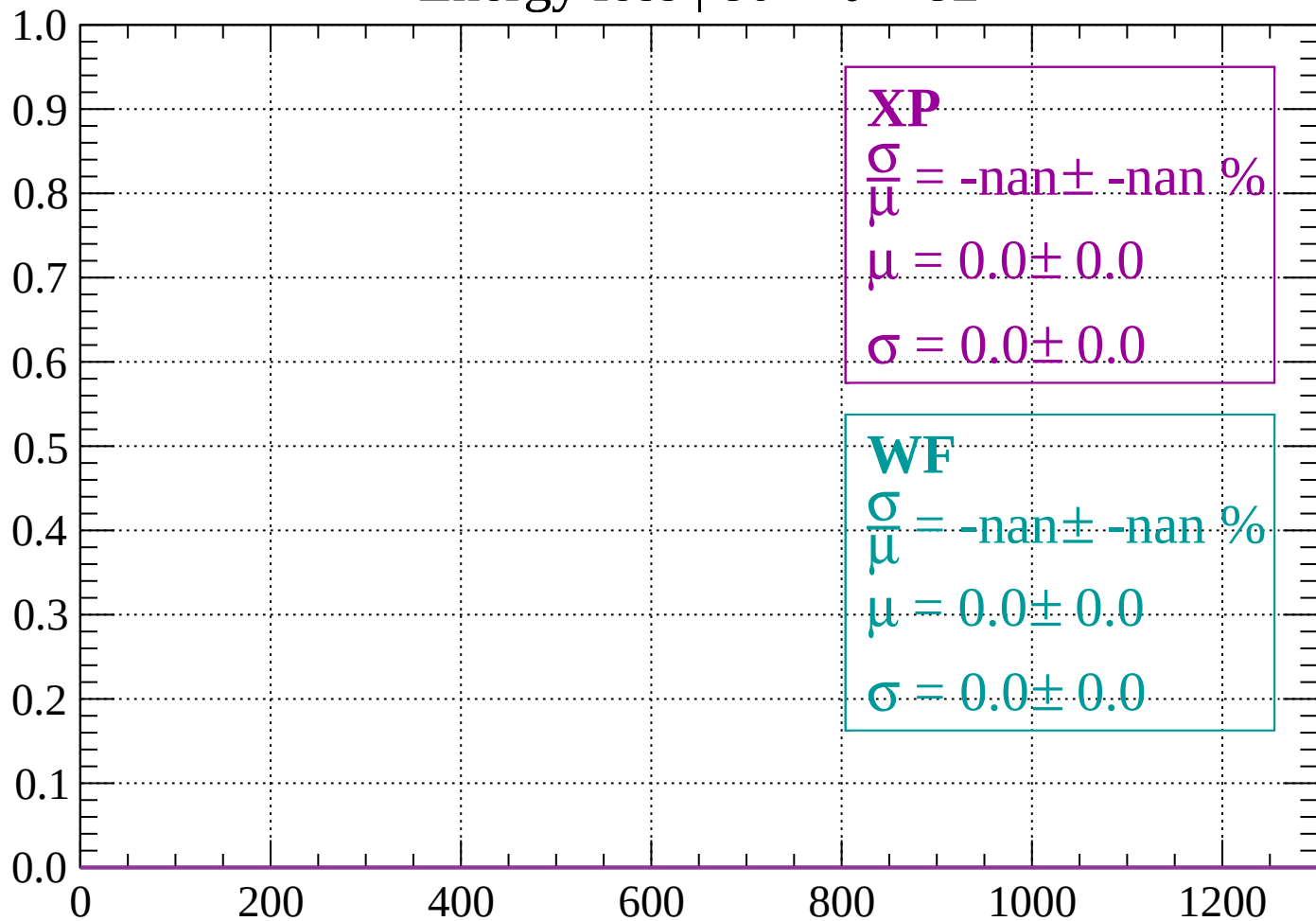
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

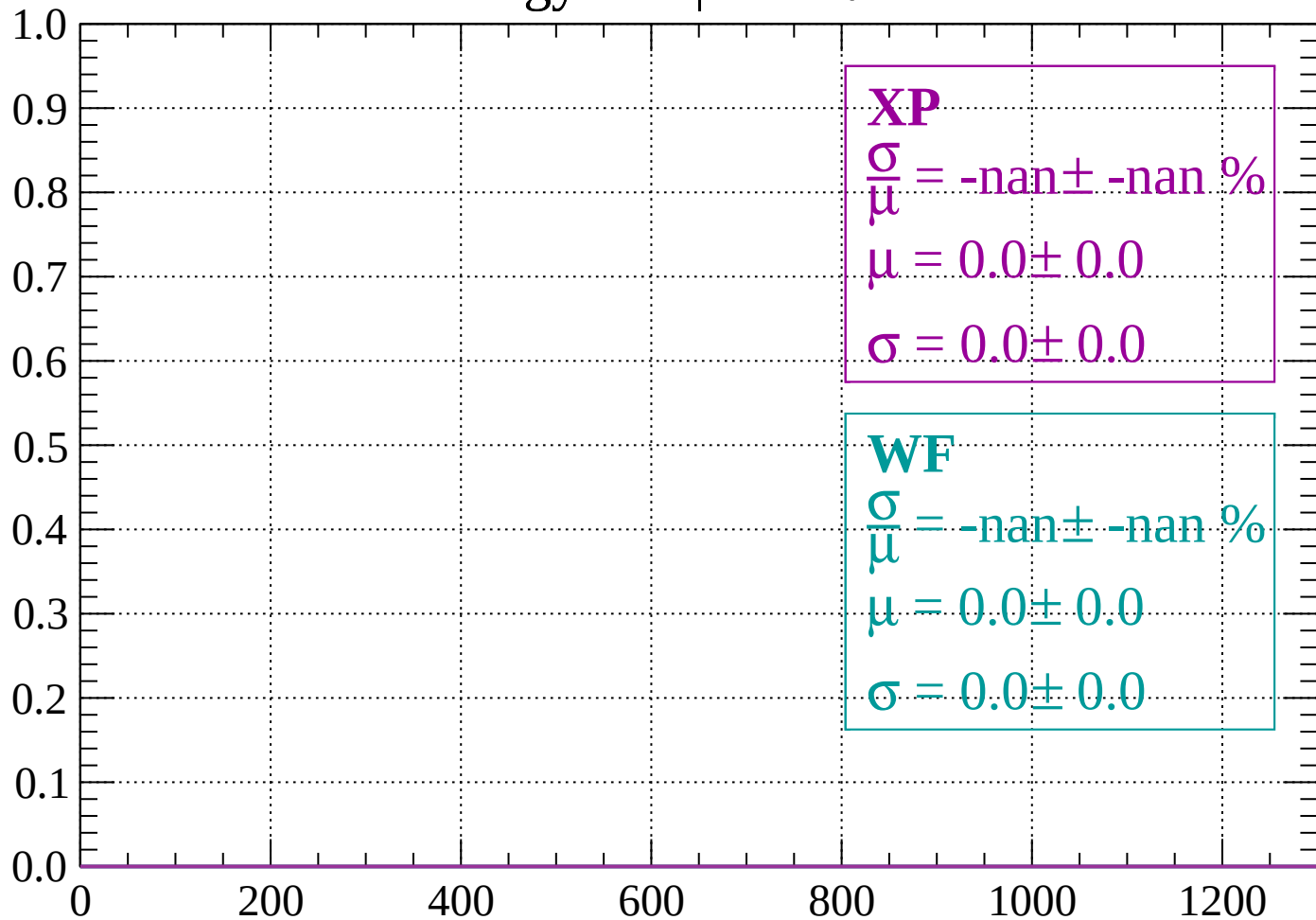
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

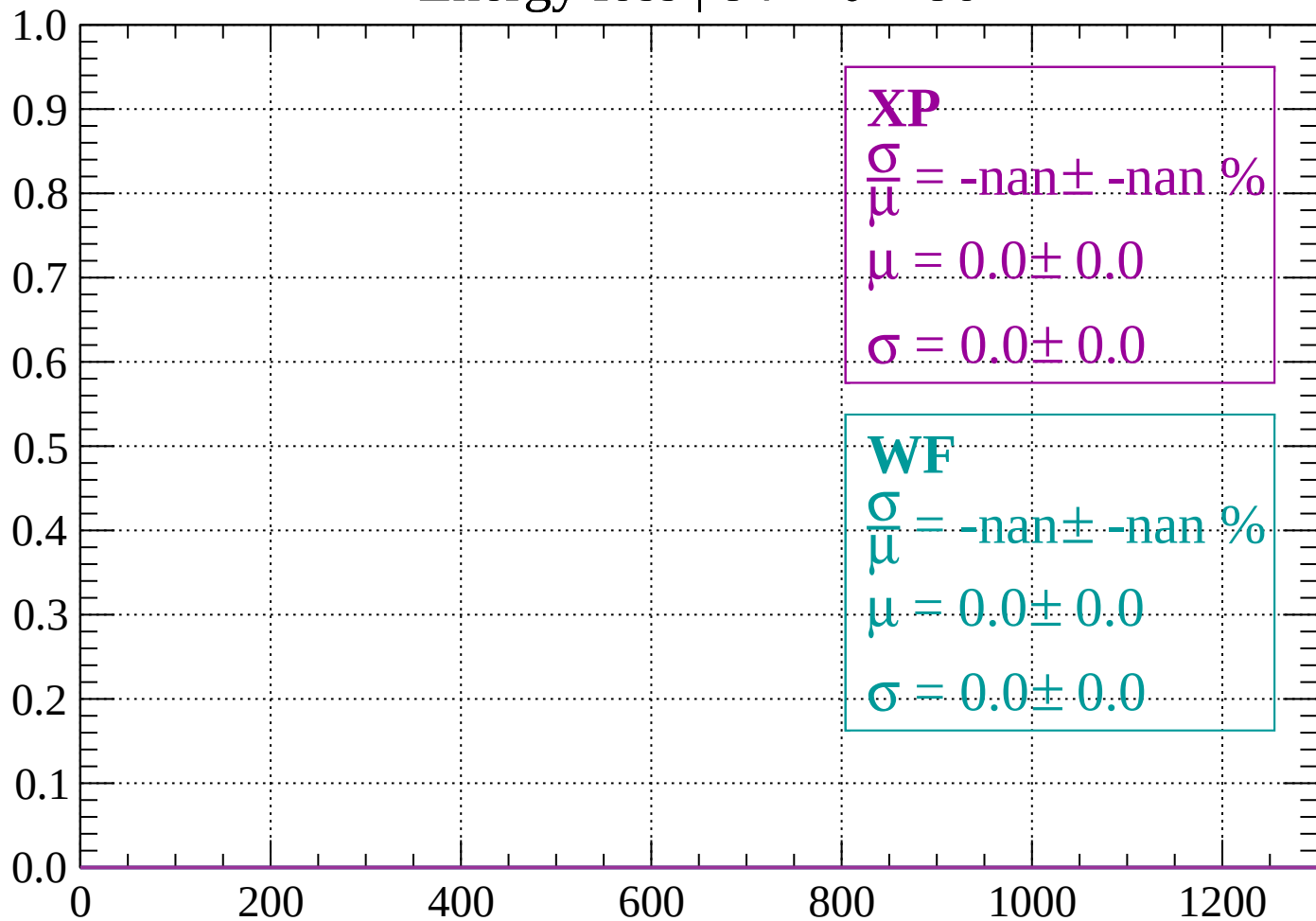
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

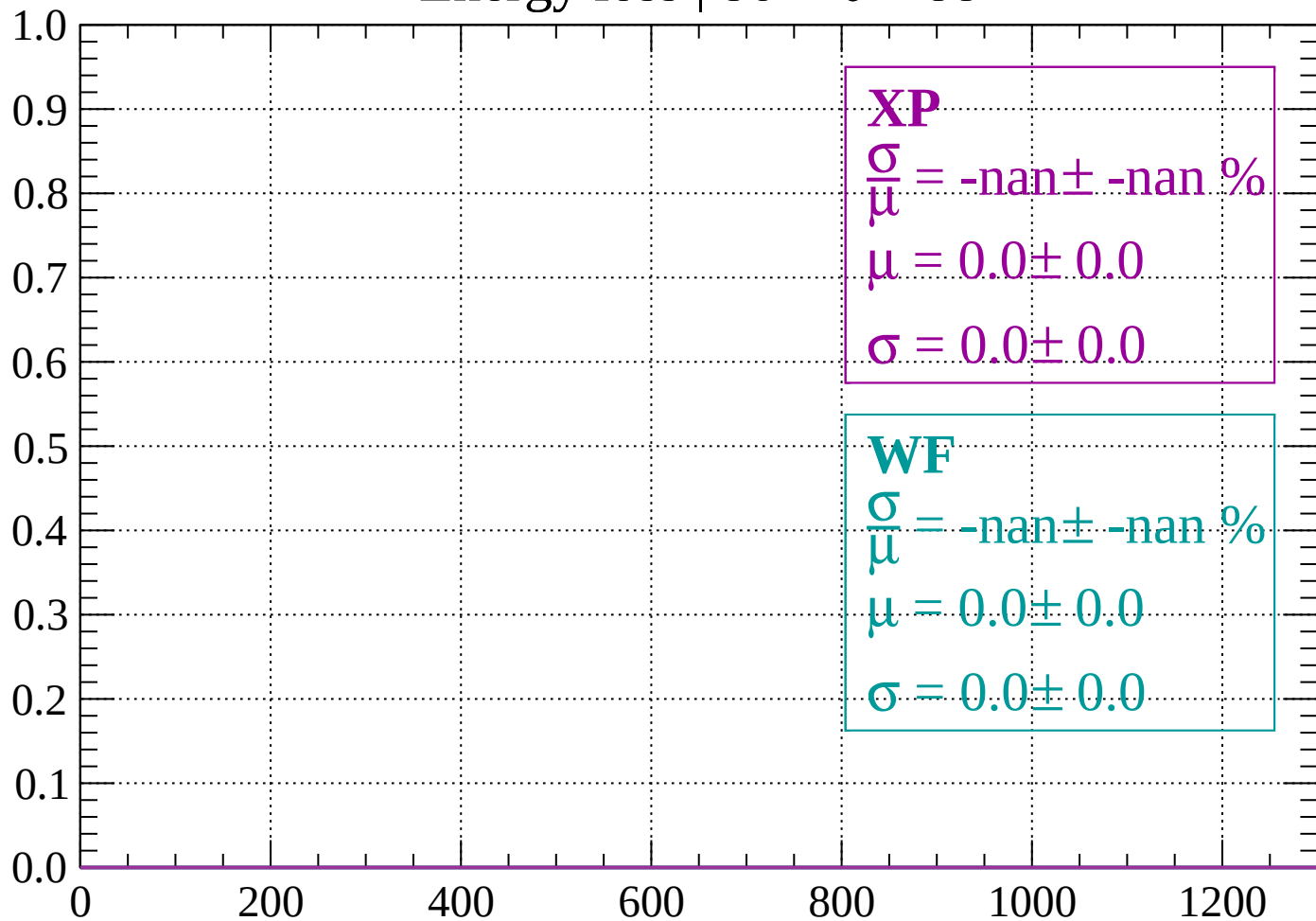
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

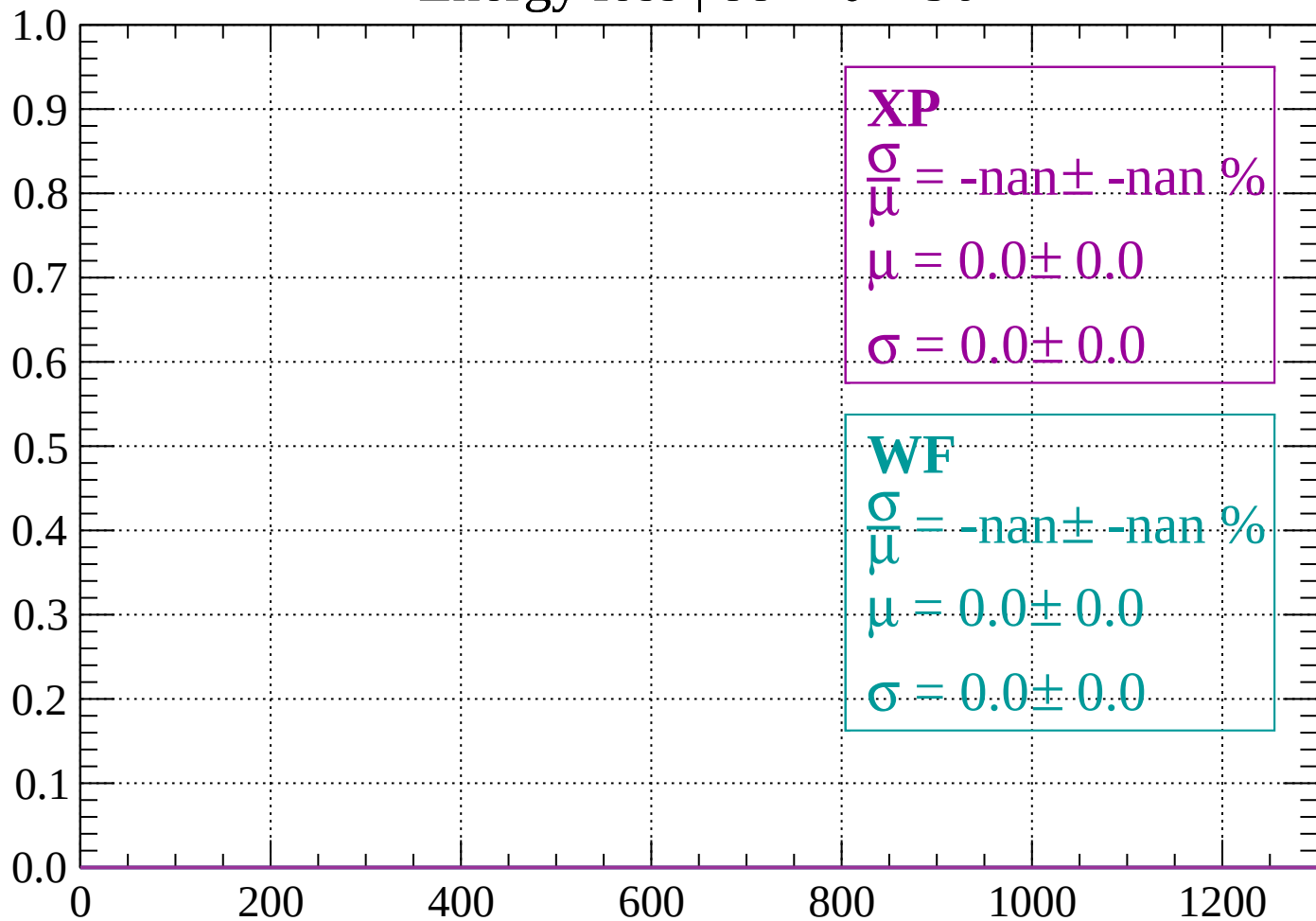
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

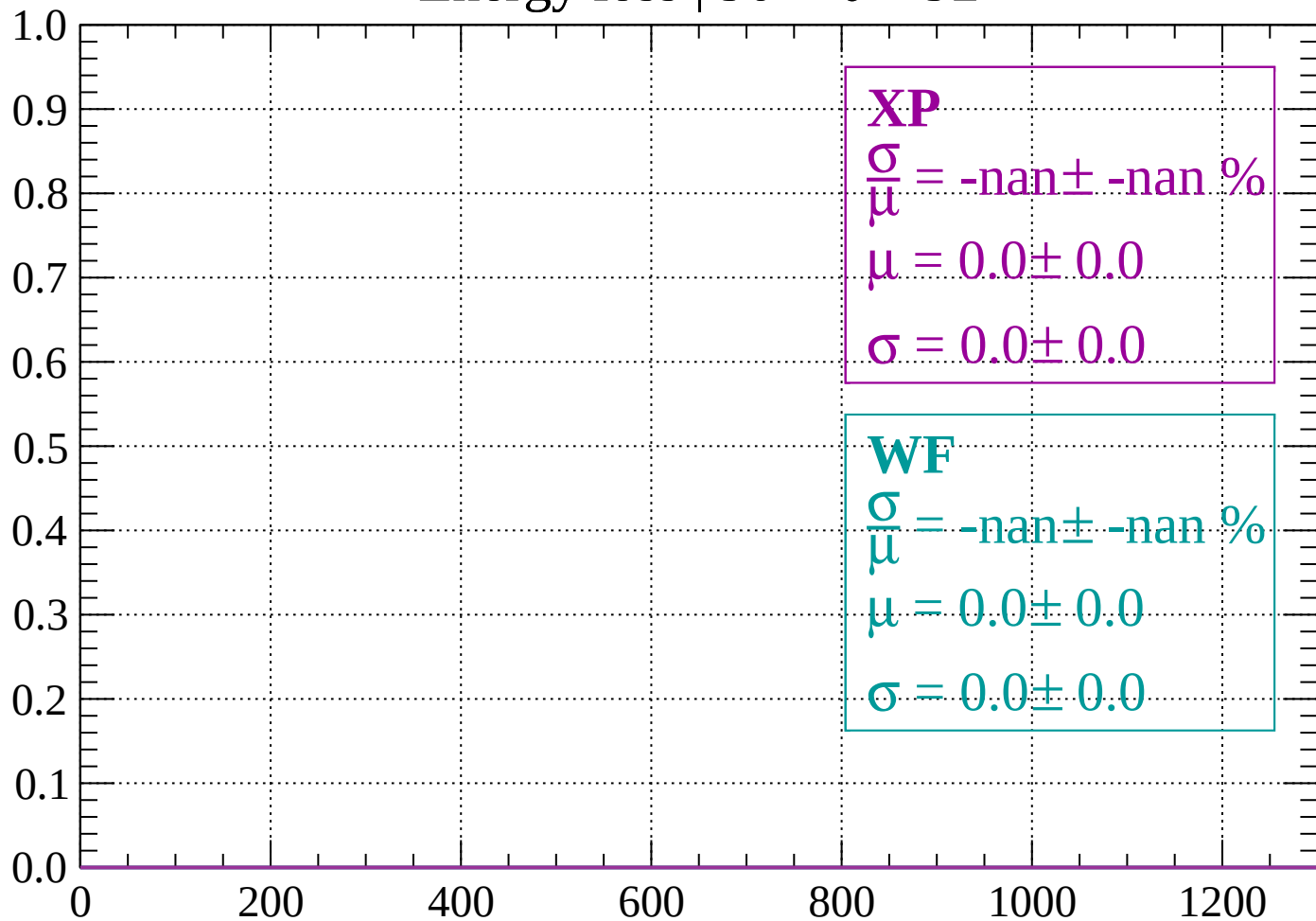
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

